Linear Algebra I/II

Lecture Notes & Transcript

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Contents

The Fibonacci Sequence (Chapter 1)

Introduction and Generalization

We begin our study with the classical Fibonacci sequence, a sequence of numbers that naturally emerges in various mathematical and biological contexts.

Definition 1.1 (*The Fibonacci Sequence*): The "*Fibonacci sequence*" is an infinite sequence of integers $a_0, a_1, a_2, a_3, \ldots, a_n, \ldots$ defined by the following recurrence relation:

$$\begin{cases} a_0 := 0 \\ a_1 := 1 \\ a_n := a_{n-1} + a_{n-2} \quad \text{for } n \geq 2 \end{cases} \tag{1.1}$$

Remark 1.1: This sequence was introduced to Western mathematics by Leonardo Fibonacci, a medieval mathematician from Pisa (1170–1250), to model the growth of rabbit populations. Computing the first few terms, we easily find:

$$a_0 = 0, \quad a_1 = 1, \quad a_2 = 1, \quad a_3 = 2, \quad a_4 = 3, \quad a_5 = 5, \quad a_6 = 8, \quad a_7 = 13, \quad a_8 = 21, \ldots$$

A natural, central question arises: Is there a closed, explicit formula for the general term a_n? To answer this, let us briefly look at other well-known mathematical sequences for comparison:

(a) **Arithmetic Sequence:** Defined by $a_0 := a$ and $a_n := a_{n-1} + d$ for $n \geq 1$. The explicit formula for the general term is given by $a_n = a + n \cdot d$.

(b) **Geometric Sequence:** Defined by $a_0 := a$ and $a_n := a_{n-1} \cdot q$ for $n \geq 1$. The explicit formula is given by $a_n = a \cdot q^n$.

Clearly, the recurrence relation for the Fibonacci sequence is fundamentally different. To find a formula for a_n, it is highly beneficial to step back and generalize the problem.

Definition 1.2 (*Generalized Fibonacci Sequence*): Let $a_0, a_1 \in \mathbb{R}$ be given initial real numbers. We define a sequence $\mathcal{A}$ to be a "*Fibonacci sequence*" if it satisfies the relation:

$$\begin{cases} \alpha_0 := a_0 \\ \alpha_1 := a_1 \\ \alpha_n := \alpha_{n-1} + \alpha_{n-2} \quad \text{for } n \geq 2 \end{cases} \tag{1.2}$$

We denote such a sequence uniquely determined by its starting values as $\mathcal{F}_{a_0,a_1}$. Furthermore, we denote the collection (or space) of all such generalized Fibonacci sequences by Fib.

Notice that our original, classical Fibonacci sequence is precisely $\mathcal{F}_{0,1}$.

The Algebraic Structure of Fib

The collection Fib possesses a very rich internal algebraic structure, which we will exploit to find our explicit formula.

Proposition 1.3 (*Closure Properties of* Fib): The collection of Fibonacci sequences, Fib, is closed under term-wise addition and scalar multiplication. Specifically,

(a) If $\mathcal{A}_1, \mathcal{A}_2 \in \mathrm{Fib}$, then $\mathcal{A}_1 + \mathcal{A}_2 \in \mathrm{Fib}$.

(b) If $\mathcal{A} \in \mathrm{Fib}$ and $\alpha \in \mathbb{R}$, then $\alpha\mathcal{A} \in \mathrm{Fib}$.

Proof:

We address each closure property:

(a) Let $\mathcal{A}_1 = (a_0, a_1, a_2, \dots)$ and $\mathcal{A}_2 = (b_0, b_1, b_2, \dots)$ be two sequences in Fib. We define their sum term-wise as $\mathcal{A}_1 + \mathcal{A}_2 := (c_0, c_1, c_2, \dots)$, where $c_n := a_n + b_n$. We must verify that the sequence (c_n) obeys the Fibonacci recurrence relation, namely $c_n = c_{n-1} + c_{n-2}$ for $n \geq 2$. By algebraic manipulation, we find:

$$c_{n-1} + c_{n-2} = (a_{n-1} + b_{n-1}) + (a_{n-2} + b_{n-2}) = (a_{n-1} + a_{n-2}) + (b_{n-1} + b_{n-2})$$

Because both $\mathcal{A}_1$ and $\mathcal{A}_2$ are Fibonacci sequences, we can substitute $a_{n-1} + a_{n-2} = a_n$ and $b_{n-1} + b_{n-2} = b_n$. Therefore,

$$c_{n-1} + c_{n-2} = a_n + b_n = c_n$$

This shows that $\mathcal{F}_{a_0,a_1} + \mathcal{F}_{b_0,b_1} = \mathcal{F}_{a_0+b_0,a_1+b_1}$, and therefore, Fib is closed under addition.

(b) Let $\mathcal{A} = (a_0, a_1, a_2, \dots) \in \mathrm{Fib}$ and $\alpha \in \mathbb{R}$. We define $\alpha\mathcal{A} := (\alpha a_0, \alpha a_1, \alpha a_2, \dots)$. We need to check that $\alpha a_n = \alpha a_{n-1} + \alpha a_{n-2}$. This is true because $a_n = a_{n-1} + a_{n-2}$, and we simply multiply both sides by α. In fact, we see that $\alpha\mathcal{F}_{a_0,a_1} = \mathcal{F}_{\alpha a_0,\alpha a_1}$, so Fib is closed under scalar multiplication.

Q.E.D.

Corollary 1.4: By Proposition 1.3, linear combinations of Fibonacci sequences are also Fibonacci sequences. That is, for any scalars $\alpha, \beta \in \mathbb{R}$, we have:

$$\alpha\mathcal{F}_{a_0,a_1} + \beta\mathcal{F}_{b_0,b_1} = \mathcal{F}_{\alpha a_0+\beta b_0,\alpha a_1+\beta b_1}$$

Remark 1.2: As a direct consequence of this structure, in order to find a general formula for *any* sequence $\mathcal{F}_{a_0,a_1}$, it is enough to find explicit formulas for the two foundational sequences $\mathcal{F}_{1,0}$ and $\mathcal{F}_{0,1}$. This is because any sequence can be decomposed as:

$$\mathcal{F}_{a_0,a_1} = a_0 \cdot \mathcal{F}_{1,0} + a_1 \cdot \mathcal{F}_{0,1}$$

We will see later in the course that Fib represents a 2-dimensional vector space. Every element in this space can be expressed by a linear combination of some two base elements (but 1 element alone is not enough). What precisely constitutes a *"vector space"* and *"dimension"* will be formally defined later in the course. The corresponding picture is drawn in Figure 1.1.

Deriving the Explicit Formula

Perhaps one of the sequences in Fib matches a form we already know?

First, we can rule out arithmetic sequences. Except for the trivial zero sequence $\mathcal{F}_{0,0}$, no Fibonacci sequence can be arithmetic. The reason is straightforward: for an arithmetic sequence, the difference between consecutive terms must be constant, meaning $a_n - a_{n-1} = d$. However, for a Fibonacci sequence, $a_n - a_{n-1} = a_{n-2}$. This difference grows and cannot remain constant. Thus, arithmetic sequences are unsuitable for this purpose. (Prof. Biran leaves the details of this as Exercise 1.2 **(d)**, and they are worth having, because the appeal to growth is not quite the argument: what the two displays give is that $a_{n-2} = d$ for every $n \geq 2$, that is, that the sequence is **constant**, equal to d throughout. A constant sequence obeys the Fibonacci recurrence exactly when $d = d + d$, which forces $d = 0$. No estimate of growth is needed, and the conclusion holds for negative and non-monotone sequences alike).

Next, we ask: perhaps for some choice of a_0, a_1, the sequence $\mathcal{F}_{a_0,a_1}$ coincides with a geometric sequence? Let us try a geometric sequence of the form $\mathcal{G} = (1, q, q^2, q^3, \dots, q^n, \dots)$.

Lemma 1.5: A geometric sequence $\mathcal{G} = (1, q, q^2, \dots)$ with $q \neq 0$ is a Fibonacci sequence if and only if $q = \frac{1\pm\sqrt{5}}{2}$.

Proof:

For $\mathcal{G}$ to be a Fibonacci sequence, it must satisfy the recurrence relation for all $n \geq 2$:

$$q^n = q^{n-1} + q^{n-2}$$

Since $q \neq 0$, we can divide the entire equation by q^{n-2}, which simplifies the condition to a quadratic equation:

$$q^2 = q + 1 \implies q^2 - q - 1 = 0$$

Applying the standard quadratic formula yields two possible solutions:

$$q = \frac{1 \pm \sqrt{5}}{2}$$

We denote these two specific roots as $\varphi := \frac{1+\sqrt{5}}{2} \approx 1.618$ (the golden ratio) and $\psi := \frac{1-\sqrt{5}}{2} \approx -0.618$. Consequently, we have discovered two purely geometric sequences that belong to Fib:

$$\mathcal{F}_{1,\varphi} = (1, \varphi, \varphi^2, \varphi^3, \dots, \varphi^n, \dots)$$

$$\mathcal{F}_{1,\psi} = (1, \psi, \psi^2, \psi^3, \dots, \psi^n, \dots)$$

Q.E.D.

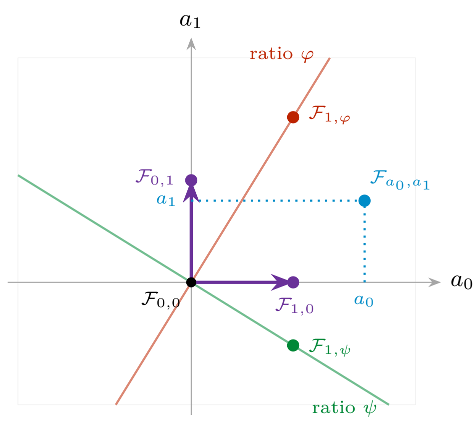


Figure 1.1: The space Fib drawn as a plane: a Fibonacci sequence is completely determined by its two initial values, so the point (a_0, a_1) *is* the sequence $\mathcal{F}_{a_0,a_1}$. Addition and rescaling of sequences become the usual addition and rescaling of points, which is the content of Proposition 1.3, and the decomposition $\mathcal{F}_{a_0,a_1} = a_0\mathcal{F}_{1,0} + a_1\mathcal{F}_{0,1}$ is just reading off coordinates. The two coloured lines carry the geometric Fibonacci sequences found in Lemma 1.5; they will turn out to be the natural axes of this plane.

Remark 1.3: The number φ is one of the most persistent constants in mathematics. The defining relation $\varphi^2 = \varphi + 1$ can be rewritten as $\varphi = 1 + \frac{1}{\varphi}$, which says that removing a unit square from a $\varphi \times 1$ rectangle leaves a rectangle of exactly the same proportions. Iterating this observation produces the classical subdivision of a rectangle into squares whose side lengths are consecutive Fibonacci numbers, drawn in Figure 1.2. The quarter circles inscribed into those squares fit together smoothly, and the resulting spiral grows by a factor of φ per quarter turn.

Theorem 1.6 (*Binet's Formula*): The explicit formula for the n-th term of the classical Fibonacci sequence $\mathcal{F}_{0,1}$ is given by "*Binet's Formula*":

$$a_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

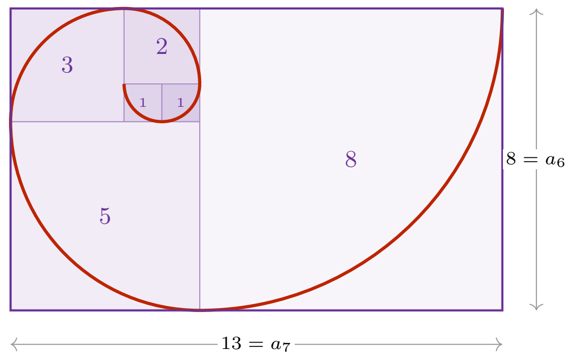


Figure 1.2: Squares with consecutive Fibonacci side lengths $1, 1, 2, 3, 5, 8$ tile a 13×8 rectangle, because $a_{n-1} + a_n = a_{n+1}$ is exactly the statement that a square of side a_n glued onto an $a_{n-1} \times a_n$ rectangle produces an $a_n \times a_{n+1}$ rectangle. Since $a_{n+1}/a_n \to \varphi$, these rectangles become ever closer to the golden rectangle.

Proof:

So far, by Lemma 1.5, we have found formulas for two specific sequences, $\mathcal{F}_{1,\varphi}$ and $\mathcal{F}_{1,\psi}$. Our original goal is to find a formula for the classical sequence $\mathcal{F}_{0,1}$.

We use the linearity principle established in Corollary 1.4. We seek scalars $\alpha, \beta \in \mathbb{R}$ such that:

$$\alpha \cdot \mathcal{F}_{1,\varphi} + \beta \cdot \mathcal{F}_{1,\psi} = \mathcal{F}_{0,1}$$

Comparing the first two terms ($n = 0$ and $n = 1$) of these sequences yields a system of linear equations:

$$\begin{cases} \alpha \cdot 1 + \beta \cdot 1 = 0 \\ \alpha \cdot \varphi + \beta \cdot \psi = 1 \end{cases} \tag{1.3}$$

From the first equation, we immediately deduce that $\beta = -\alpha$. Substituting this into the second equation gives:

$$\alpha \cdot \varphi - \alpha \cdot \psi = 1 \implies \alpha(\varphi - \psi) = 1$$

Notice that the difference $\varphi - \psi = \sqrt{5}$. Therefore, $\alpha = 1/\sqrt{5}$ and $\beta = -1/\sqrt{5}$. This leads directly to:

$$a_n = \frac{1}{\sqrt{5}}(\varphi^n - \psi^n) = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

Q.E.D.

Where the Idea Came From: Symmetries of Fib

The derivation of Theorem 1.6 is complete, but it rests on a step that looks like a lucky guess: why should one ever think of hunting for a **geometric** sequence inside Fib? The answer is not luck, and uncovering it introduces the single most important theme of this course.

Let us look at maps

$$T\colon \mathrm{Fib} \longrightarrow \mathrm{Fib},$$

also called functions, transformations or operators. Such a T takes as input an arbitrary Fibonacci sequence $\mathcal{A}$ and returns as output another Fibonacci sequence $T(\mathcal{A})$. We have just seen in Proposition 1.3 that Fib carries an algebraic structure: it is closed under addition and under multiplication by scalars. It is therefore natural to restrict attention to those maps that **respect** this structure.

Definition 1.7 (*Linear Map on* Fib): A map $T\colon \mathrm{Fib} \to \mathrm{Fib}$ is called a "*linear map*" if

$$\begin{cases} T(\mathcal{A}+\mathcal{B}) = T(\mathcal{A}) + T(\mathcal{B}) & \text{for all } \mathcal{A},\mathcal{B} \in \mathrm{Fib}, \\ T(\alpha\mathcal{A}) = \alpha\, T(\mathcal{A}) & \text{for all } \alpha \in \mathbb{R},\ \mathcal{A} \in \mathrm{Fib}\,. \end{cases} \tag{1.4}$$

There are many examples of such maps, though the first ones that come to mind are not very exciting:

(a) the identity $T := \mathrm{id}$, that is $T(\mathcal{A}) := \mathcal{A}$;
(b) the rescaling $T(\mathcal{A}) := \gamma \cdot \mathcal{A}$ for a fixed $\gamma \in \mathbb{R}$.

Neither of these takes any notice of the fact that the elements of Fib are *sequences*. Here is another one, which does:

Definition 1.8 (*The Shift Map*): The "*shift map*" $S\colon \mathrm{Fib} \to \mathrm{Fib}$ is defined by

$$S(a_0, a_1, a_2, \ldots, a_n, \ldots) := (a_1, a_2, \ldots, a_n, \ldots),$$

that is, it simply forgets the first term of the sequence.

Exercise 1.1: Check that S is linear.

Mathematicians like to probe a map $T\colon X \to X$ by looking for its special, or extreme, values. Two such notions will accompany us for the rest of the course.

Definition 1.9 (*Fixed Point and Eigenvector*): Let $T\colon X \to X$ be a map. An element $x_0 \in X$ is a "*fixed point*" of T if $T(x_0) = x_0$. An element $y_0 \in X$ is an "*eigenvector*" of T if $T(y_0)$ is proportional to y_0, that is, if

$$T(y_0) = \lambda \cdot y_0 \quad \text{for some } \lambda \in \mathbb{R}.$$

Let us test both notions on $X = \mathrm{Fib}$ with $T = S$ the shift map.

Proposition 1.10 (*Fixed Points of the Shift Map*): The only fixed point of the shift map S is the zero sequence $\mathcal{F}_{0,0}$.

Proof:

Suppose $\mathcal{A} = (a_0, a_1, a_2, \ldots)$ satisfies $S(\mathcal{A}) = \mathcal{A}$. Comparing the two sequences term by term gives

$$a_0 = a_1, \quad a_1 = a_2, \quad a_2 = a_3, \quad \ldots$$

and therefore $a_n = a_0$ for all n. Such a constant sequence is a Fibonacci sequence if and only if $a_0 = a_0 + a_0$, which forces $a_0 = 0$. Hence $\mathcal{A} = (0,0,0,\ldots) = \mathcal{F}_{0,0}$. **Q.E.D.**

So the fixed points of S are not interesting. Let us weaken the requirement and ask for eigenvectors instead. This is exactly where the geometric sequences appear.

Proposition 1.11 (*Eigenvectors of the Shift Map*): A sequence $\mathcal{A} \neq \mathcal{F}_{0,0}$ is an eigenvector of the shift map S if and only if $\mathcal{A}$ is a geometric sequence $\mathcal{A} = (a_0, a_0\lambda, a_0\lambda^2, \ldots)$ with $a_0 \neq 0$ whose ratio λ satisfies $\lambda^2 = \lambda + 1$. Consequently $\lambda \in \{\varphi, \psi\}$, while $a_0 \neq 0$ may be chosen freely.

Proof:

Assume $S(\mathcal{A}) = \lambda \cdot \mathcal{A}$ for some $\lambda \in \mathbb{R}$. Written out, the left-hand side is $(a_1, a_2, \ldots, a_{n+1}, \ldots)$ and the right-hand side is $(\lambda a_0, \lambda a_1, \ldots, \lambda a_n, \ldots)$. Comparing entries yields

$$a_n = a_{n-1} \cdot \lambda \quad \text{for all } n \geq 1,$$

and therefore

$$\mathcal{A} = (a_0,\, a_0\lambda,\, a_0\lambda^2,\, a_0\lambda^3,\, \ldots,\, a_0\lambda^n,\, \ldots),$$

which is precisely a geometric sequence. Note that $a_0 \neq 0$, since otherwise every term would vanish and $\mathcal{A}$ would be $\mathcal{F}_{0,0}$, which we excluded.

It remains to determine which ratios λ are admissible. Since $\mathcal{A}$ must in addition be a Fibonacci sequence, the relation $a_2 = a_1 + a_0$ has to hold, that is

$$\underbrace{a_0 \cdot \lambda^2}_{=a_2} = \underbrace{a_0 \cdot \lambda}_{=a_1} + a_0.$$

Dividing by $a_0 \neq 0$ gives $\lambda^2 = \lambda + 1$, the very same quadratic equation we met in Lemma 1.5. Its roots are φ and ψ, and a_0 is not constrained at all. Conversely, every geometric sequence with ratio φ or ψ satisfies $S(\mathcal{A}) = \lambda \cdot \mathcal{A}$ by the same computation read backwards. **Q.E.D.**

Conclusion 1.1: Taking $a_0 = 1$, the two geometric sequences

$$\mathcal{F}_{1,\varphi} = (1, \varphi, \varphi^2, \ldots, \varphi^n, \ldots) \qquad \text{and} \qquad \mathcal{F}_{1,\psi} = (1, \psi, \psi^2, \ldots, \psi^n, \ldots)$$

are Fibonacci sequences *and* they are eigenvectors of the shift operator S. The geometric sequences were therefore never a guess: they are forced upon us the moment we ask which sequences the shift map merely rescales.

It is worth restating this in the picture of Figure 1.1. Under the identification $\mathcal{F}_{a_0,a_1} \leftrightarrow (a_0, a_1)$, shifting a sequence discards a_0 and promotes a_1 and $a_2 = a_0 + a_1$ to the new initial pair, so

$$S\big(\mathcal{F}_{a_0,a_1}\big) = \mathcal{F}_{a_1,\, a_0+a_1}, \qquad \text{that is} \qquad \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \longmapsto \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}. \tag{1.5}$$

The shift map thus acts on the plane Fib through a single fixed matrix, and Proposition 1.11 identifies the two lines that this matrix maps onto themselves, as drawn in Figure 1.3.

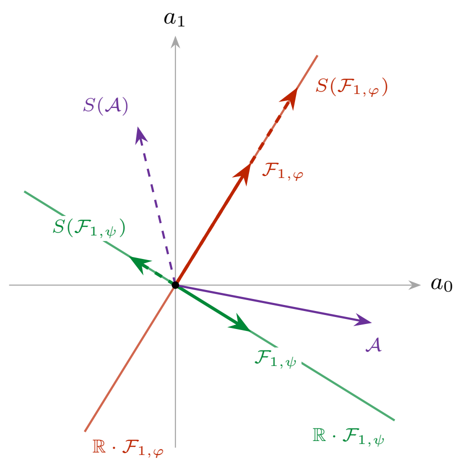


Figure 1.3: The shift map S acting on the plane Fib via (1.5); images are drawn dashed. A generic sequence $\mathcal{A}$ (purple) is thrown off its own direction. Only the two coloured lines are preserved: on $\mathbb{R} \cdot \mathcal{F}_{1,\varphi}$ the map stretches by $\varphi > 1$, while on $\mathbb{R} \cdot \mathcal{F}_{1,\psi}$ it reverses the direction and contracts by $|\psi| < 1$. These two lines are exactly the geometric Fibonacci sequences of Proposition 1.11.

Remark 1.4: The contraction along the second line is the geometric reason behind the asymptotics established below. Writing an arbitrary Fibonacci sequence in terms of the two eigen-directions, the ψ-component is crushed towards 0 under repeated shifting while the φ-component grows. Whatever sequence one starts from, its direction in the plane Fib converges to the line $\mathbb{R} \cdot \mathcal{F}_{1,\varphi}$.

The Upshot: Asymptotics and Matrix Form

AI-Note: This subsection has no counterpart in Prof. Biran's notes for Lecture 1A, which end with the exercises reproduced below. The closest-integer estimate, the matrix form and Cassini's identity are supplementary material added in this transcript. They are included because the matrix M reappears in the chapters on eigenvalues, but they should not be mistaken for lecture content.

The derivation above yields two profound observations regarding the growth and representation of the Fibonacci sequence.

Important remark 1.1: Clearly, as n grows, the term involving ψ becomes negligible. Since $|\psi| = |\frac{1-\sqrt{5}}{2}| \approx 0.618 < 1$, it follows that $\psi^n \to 0$ as $n \to \infty$. Therefore, for large n, we have the approximation $a_n \approx \frac{1}{\sqrt{5}}\varphi^n$. In fact, because $\left|\frac{\psi^n}{\sqrt{5}}\right| < \frac{1}{2}$ for all $n \geq 0$, we arrive at the following precise conclusion: the number a_n is the *closest integer* to

$$\frac{1}{\sqrt{5}} \cdot \left(\frac{1+\sqrt{5}}{2}\right)^n .$$

Furthermore, this implies that the ratio of consecutive terms satisfies:

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \varphi \tag{1.6}$$

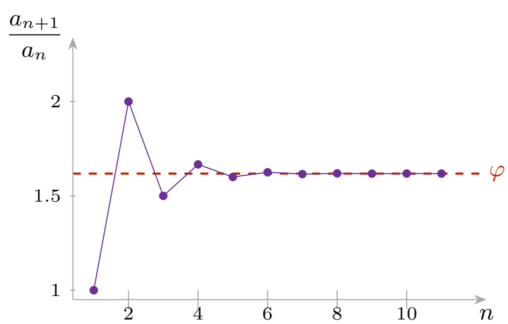


Figure 1.4: The quotients a_{n+1}/a_n of the classical Fibonacci sequence $\mathcal{F}_{0,1}$. They oscillate around the golden ratio φ, alternately overshooting and undershooting it, because the error term ψ^n changes sign at every step while its magnitude decays geometrically.

Finally, we can represent the recurrence relation using the language of matrices. Defining a vector $v_n := \binom{a_{n+1}}{a_n}$, the relation $a_{n+1} = a_n + a_{n-1}$ can be written as:

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n \\ a_{n-1} \end{pmatrix} \implies v_n = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a_1 \\ a_0 \end{pmatrix} \tag{1.7}$$

This formulation shows that the behavior of the sequence is entirely governed by the powers of the matrix $M := \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. It is the shift map of (1.5) written in the reversed coordinates $\binom{a_{n+1}}{a_n}$, so by Proposition 1.11 its eigenvalues are again φ and ψ.

Exercise 1.2 (*Interesting Exercises*)**:**

(a) **General Formula:** Find an explicit formula for the general Fibonacci sequence $\mathcal{F}_{a_0,a_1}$ that depends only on the initial values a_0, a_1 and the index n.

(b) **The Shift Operator:** Let $S : \text{Fib} \to \text{Fib}$ be the shift map defined by $S(a_0, a_1, \dots) = (a_1, a_2, \dots)$. Formally verify that S is a *linear map.*

(c) **Golden Ratio Continued Fraction:** Consider $\varphi = \frac{1+\sqrt{5}}{2}$.

$$\left.\varphi = 1 + \cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cdots}}}}\right\}$$

what is actually the meaning of this?

(d) **Non-existence of Arithmetic Fibonacci Sequences:** Complete the details to show that no Fibonacci sequence $\mathcal{F}_{a_0,a_1}$ (except for the trivial zero sequence) can be an arithmetic sequence.

(e) **Convergence:** For the classical Fibonacci sequence $\mathcal{F}_{0,1} = (a_0, a_1, a_2, \dots)$, calculate the limit

$$\lim_{n\to\infty} \frac{a_n}{a_{n-1}},$$

whose numerical behaviour is plotted in Figure 1.4.

(f) **Cassini's Identity:** Show that $a_{n+1}a_{n-1} - a_n^2 = (-1)^n$ for all $n \geq 1$. Hint: Use the determinant of the matrix M^n.

Solutions to the Exercises

Proof of Exercise 1.1, also Exercise 1.2 (b):

Let $\mathcal{A} = (a_0, a_1, a_2, \dots)$ and $\mathcal{B} = (b_0, b_1, b_2, \dots)$ be in Fib and let $\alpha \in \mathbb{R}$. Both operations on Fib are term-wise, and the shift acts on each term separately, so the two axioms of Definition 1.7 are read off directly:

$$S(\mathcal{A}+\mathcal{B}) = S(a_0+b_0,\, a_1+b_1,\, \dots) = (a_1+b_1,\, a_2+b_2,\, \dots) = S(\mathcal{A}) + S(\mathcal{B}),$$
$$S(\alpha\mathcal{A}) = S(\alpha a_0,\, \alpha a_1,\, \dots) = (\alpha a_1,\, \alpha a_2,\, \dots) = \alpha\, S(\mathcal{A}).$$

Both computations use Proposition 1.3 implicitly, in that the sequences $\mathcal{A}+\mathcal{B}$ and $\alpha\mathcal{A}$ are again in Fib, so that S may be applied to them at all. **Q.E.D.**

Solution of Exercise 1.2 (a):

We repeat the derivation of Theorem 1.6 with $\mathcal{F}_{0,1}$ replaced by an arbitrary $\mathcal{F}_{a_0,a_1}$. By Corollary 1.4 we look for $\alpha, \beta \in \mathbb{R}$ with $\alpha \cdot \mathcal{F}_{1,\varphi} + \beta \cdot \mathcal{F}_{1,\psi} = \mathcal{F}_{a_0,a_1}$, and comparing the terms with $n = 0$ and $n = 1$ gives

$$\begin{cases}\alpha + \beta = a_0, \;\; \alpha\varphi + \beta\psi = a_1.\end{cases}$$

Substituting $\beta = a_0 - \alpha$ into the second equation gives $\alpha(\varphi - \psi) = a_1 - a_0\psi$, and since $\varphi - \psi = \sqrt{5} \neq 0$,

$$\alpha = \frac{a_1 - a_0\psi}{\sqrt{5}}, \qquad \beta = a_0 - \alpha = \frac{a_0\varphi - a_1}{\sqrt{5}}.$$

Reading off the n-th term, the general Fibonacci sequence $\mathcal{F}_{a_0,a_1} = (\alpha_0, \alpha_1, \alpha_2, \dots)$ satisfies

$$\alpha_n = \frac{1}{\sqrt{5}}\left((a_1 - a_0\psi)\,\varphi^n - (a_1 - a_0\varphi)\,\psi^n\right) \qquad \text{for all } n \geq 0.$$

Two checks. For $a_0 = 0$ and $a_1 = 1$ this collapses to $\frac{1}{\sqrt{5}}(\varphi^n - \psi^n)$, which is Binet's formula. For $a_0 = 1$ and $a_1 = 0$ it gives $\frac{1}{\sqrt{5}}(\varphi\psi^n - \psi\varphi^n)$, whose values at $n = 0$ and $n = 1$ are $\frac{\varphi-\psi}{\sqrt{5}} = 1$ and $\frac{\varphi\psi-\psi\varphi}{\sqrt{5}} = 0$, as they must be. **Q.E.D.**

Solution of Exercise 1.2 (c):

The displayed expression is not a finite computation, so it has to be given a meaning, and the standard one is as a limit. Define a sequence by

$$x_1 := 1, \qquad x_{n+1} := 1 + \frac{1}{x_n},$$

so that x_n is the continued fraction truncated after n layers, and read the infinite fraction as $\lim_{n\to\infty} x_n$, provided that limit exists.

Two things then have to be said. First, **if** the limit x exists, it is φ: every $x_n \geq 1$, so $x \geq 1 > 0$, and passing to the limit in the recursion gives $x = 1 + \frac{1}{x}$, that is, $x^2 = x + 1$. That is exactly the equation of Lemma 1.5, whose roots are φ and ψ; since $\psi < 0$ and $x > 0$, we get $x = \varphi$.

Second, the limit does exist, and the reason is this chapter's own subject. Starting from $x_1 = 1 = a_2/a_1$, an induction gives

$$x_n = \frac{a_{n+1}}{a_n},$$

because $1 + \frac{a_n}{a_{n+1}} = \frac{a_{n+1}+a_n}{a_{n+1}} = \frac{a_{n+2}}{a_{n+1}}$ by the recurrence. So the convergents of the continued fraction are precisely the ratios of consecutive Fibonacci numbers, and part **(e)** below shows that these converge to φ.

The meaning of the picture, then, is that the golden ratio is the number whose continued fraction expansion consists of nothing but 1's, and that its convergents are the Fibonacci quotients plotted in Figure 1.4. **Q.E.D.**

Solution of Exercise 1.2 (d):

Let $\mathcal{F}_{a_0,a_1} = (\alpha_0, \alpha_1, \alpha_2, \dots)$ be a Fibonacci sequence which is also arithmetic, say with common difference d, so that $\alpha_n - \alpha_{n-1} = d$ for all $n \geq 1$. For $n \geq 2$ the Fibonacci recurrence gives $\alpha_n - \alpha_{n-1} = \alpha_{n-2}$, and comparing the two expressions yields

$$\alpha_{n-2} = d \qquad \text{for all } n \geq 2,$$

that is, $\alpha_m = d$ for every $m \geq 0$: the sequence is constant. Now impose the recurrence once on this constant sequence: $d = \alpha_2 = \alpha_1 + \alpha_0 = d + d$, so $d = 0$ and $\mathcal{F}_{a_0,a_1} = \mathcal{F}_{0,0}$.

Conversely $\mathcal{F}_{0,0}$ is arithmetic, with $d = 0$. So the zero sequence is the only arithmetic Fibonacci sequence, which is what the text asserts before Lemma 1.5. Note that no growth estimate entered: the argument is finished by the single equation $d = d + d$, and it applies to sequences with negative terms just as well. **Q.E.D.**

Solution of Exercise 1.2 (e):

By Theorem 1.6, for $n \geq 1$,

$$\frac{a_n}{a_{n-1}} = \frac{\varphi^n - \psi^n}{\varphi^{n-1} - \psi^{n-1}} = \varphi \cdot \frac{1 - (\psi/\varphi)^n}{1 - (\psi/\varphi)^{n-1}} \cdot \frac{1}{1},$$

where the second expression is obtained by dividing numerator and denominator by φ^n and φ^{n-1} respectively. Now $|\psi/\varphi| = \frac{|1-\sqrt{5}|}{1+\sqrt{5}} = \frac{\sqrt{5}-1}{\sqrt{5}+1} < 1$, so $(\psi/\varphi)^n \to 0$, and both the numerator and the denominator of the middle fraction tend to 1. Note also that $a_{n-1} \neq 0$ for $n \geq 2$, so the quotient is defined. Consequently

$$\lim_{n\to\infty} \frac{a_n}{a_{n-1}} = \varphi = \frac{1+\sqrt{5}}{2}.$$

The oscillation visible in Figure 1.4 is the sign of $(\psi/\varphi)^n$, which alternates, while its magnitude decays geometrically. **Q.E.D.**

Solution of Exercise 1.2 (f):

Write $M := \left(\begin{smallmatrix}1 & 1\\ 1 & 0\end{smallmatrix}\right)$ as in the text. We first claim that

$$M^n = \begin{pmatrix} a_{n+1} & a_n \\ a_n & a_{n-1} \end{pmatrix} \qquad \text{for all } n \geq 1,$$

where (a_n) is the classical Fibonacci sequence $\mathcal{F}_{0,1}$. For $n = 1$ the right-hand side is $\left(\begin{smallmatrix}a_2 & a_1\\ a_1 & a_0\end{smallmatrix}\right) = \left(\begin{smallmatrix}1 & 1\\ 1 & 0\end{smallmatrix}\right) = M$. Assuming the claim for n,

$$\begin{aligned} M^{n+1} = M^n \cdot M &= \begin{pmatrix} a_{n+1} & a_n \\ a_n & a_{n-1} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} a_{n+1} + a_n & a_{n+1} \\ a_n + a_{n-1} & a_n \end{pmatrix} = \begin{pmatrix} a_{n+2} & a_{n+1} \\ a_{n+1} & a_n \end{pmatrix}, \end{aligned}$$

using the recurrence twice, which is the claim for $n+1$.

Now take determinants. The determinant is multiplicative, so $\det(M^n) = (\det M)^n$, and $\det M = 1 \cdot 0 - 1 \cdot 1 = -1$. Reading the determinant off the claim instead gives $a_{n+1}a_{n-1} - a_n^2$. Equating the two,

$$a_{n+1}a_{n-1} - a_n^2 = (-1)^n \qquad \text{for all } n \geq 1,$$

which is Cassini's identity. For instance $n = 6$ gives $a_7 a_5 - a_6^2 = 13 \cdot 5 - 8^2 = 65 - 64 = 1 = (-1)^6$.

Q.E.D.

AI-Note: A word on what this chapter is doing, since it is the only one in the book that answers a question rather than building a theory.

Everything here is an instance of one move: replace a question about a single object, the number a_n, by a question about the whole set of objects like it, the space Fib. That set turns out to be closed under addition and scaling (Proposition 1.3), which is what makes Corollary 1.4 available, and from there a formula for two convenient sequences yields a formula for all of them. The two convenient sequences are the geometric ones, and Proposition 1.11 explains why: they are exactly the sequences the shift map rescales, so the search that produced them was forced rather than lucky. Every later chapter of the course repeats this move with the words *"vector space"*, *"basis"*, *"linear map"* and *"eigenvector"* in place of the ad-hoc ones used here.

The rigour pass over this chapter did three things. The two Definitions opening it, of the classical and of the generalized Fibonacci sequence, carried no `\label` at all and so could not be cited; they now can. The six exercises Prof. Biran leaves at the end, together with the linearity of S he asks for in passing, are written up in the section above, which is what every chapter from 14 on already does.

And the argument ruling out arithmetic Fibonacci sequences, which the text gives in one line, rests on the claim that the difference a_{n-2} *"grows and cannot remain constant"*. The conclusion is right but the reason is not the one that works: what the two displayed identities give is that the sequence is constant, and a constant sequence obeys the recurrence only when $d = d + d$, so $d = 0$. That is the whole argument, it needs no estimate of growth, and it covers sequences with negative terms, for which nothing grows. This is Exercise 1.2 **(d)**, so Prof. Biran evidently intends the details to be supplied; they now stand both here and in the solution.

Logic and Mathematics (Chapter 2)

Before advancing into the formal theory of vector spaces, we must establish the rigorous grammar of mathematical reasoning. This chapter provides the logical foundation necessary for constructing and understanding proofs.

Mathematical Statements and Basic Operators

Definition 2.1 (*Mathematical Statement*): A "*mathematical statement*" (or logical statement) is a declarative sentence that is definitively either **True (T)** or **False (F)**. It is a fundamental axiom of logic that a statement cannot be both true and false simultaneously.

Example 2.1 (*Examples of Logical Statements*): Consider the following declarations:

- **A:** *"For every real number x, $x^2 \geq 0$."*
- **B:** *"Every prime number $p \geq 3$ is odd."*
- **C:** *"Every odd number $p \geq 3$ is prime."*

Statements A and B are True. Statement C is False (e.g., 9 is odd but not prime).

Negation, Conjunction, and Disjunction

Definition 2.2 (*Negation*): The "*negation*" of a statement A, denoted $\neg A$ (and sometimes $\overline{A}$), asserts that A does not hold.

A	$\neg A$
T	F
F	T

Definition 2.3 (*Conjunction / AND*): The statement $A \wedge B$ (read *"A and B"*, and sometimes written $A \,\&\, B$) is True if and only if both A and B are True.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

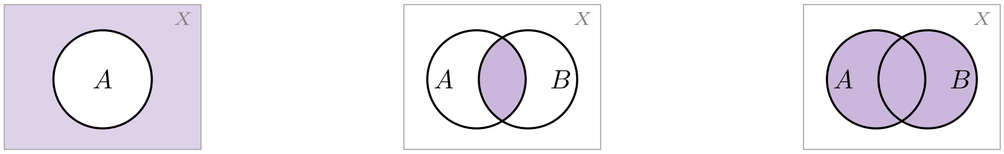
Example 2.2 (*Conjunction Example*): For $x \in \mathbb{R}$, the statement $(x^2 = 1) \wedge (x \geq 0)$ is logically equivalent to $x = 1$.

Definition 2.4 (*Disjunction / OR*): The statement $A \vee B$ (read *"A or B"*) is True if *at least one* of the statements A or B is True. This is an *inclusive* OR.

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Important remark 2.1: In everyday life, *"either A or B"* usually means one of the two but not both. In mathematics it means *"either A or B, or both"*: it could be only A, it could be only B, and it could be both A and B.

The three basic operators are conveniently visualised by shading the region of a Venn diagram on which the compound statement comes out True; see Figure 2.1.



(a) $\neg A$ (b) $A \wedge B$ (c) $A \vee B$

Figure 2.1: The three basic operators of Definitions 2.2 to 2.4, drawn on a universe X. The shaded region is where the compound statement is True. Note that in Figure 2.1c the overlap is shaded as well: this is the inclusive OR.

Example 2.3 (*Disjunction Example*): The statement $(x > 1) \vee (x < 2)$ is True for all $x \in \mathbb{R}$, since every real number satisfies at least one of the two conditions. Conversely, $(x < 1) \vee (x > 2)$ is precisely the negation of the statement $1 \leq x \leq 2$, as Figure 2.2 shows.

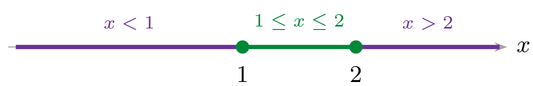


Figure 2.2: The statement $(x < 1) \vee (x > 2)$ holds exactly on the two purple rays, which together form the complement of the green interval. Hence it is the negation of $1 \leq x \leq 2$.

Implications and Equivalences

Definition 2.5 (*Implication*): The statement $A \implies B$ (*"A implies B"*) is defined to be False only when A is True and B is False. In all other cases, it is True.

A	B	$A \implies B$
T	T	T
T	F	F
F	T	T
F	F	T

Remark 2.1: Note that $A \implies B$ is logically equivalent to $\neg A \vee B$. Consequently, if the premise A is False, the implication is *vacuously true* regardless of B. As Prof. Biran noted:

$$(0 = 1) \implies \text{“The Earth is flat”}$$

is a mathematically **True** statement. Causality is irrelevant in formal logic.

Definition 2.6 (*Equivalence*): The statement $A \iff B$ (*"A if and only if B"*) holds when $A \implies B$ and $B \implies A$ are both True. This means A and B share the same truth value.

A	B	$A \iff B$
T	T	T
T	F	F
F	T	F
F	F	T

Example 2.4 (*Equivalences*): Both of the following statements are True:

$$(x^2 > 0) \iff (x \neq 0), \qquad (0 = 1) \iff \textit{“The Earth is flat”}.$$

The second one is True because both sides are False. Again, causality is irrelevant. (In the first one, x ranges over $\mathbb{R}$, as in the conjunction example above; the equivalence uses that a non-zero real has positive square, which is a fact about $\mathbb{R}$ and not a matter of logic).

Proposition 2.7 (*Contrapositive*): Let A, B be statements. Then

$$(A \implies B) \iff (\neg B \implies \neg A).$$

Exercise 2.1 (*Double Negation and De Morgan's Laws*): Prove Proposition 2.7 via a truth table, and show in the same way that

(a) **Double Negation:** $\neg(\neg A) = A$;

(b) **De Morgan's Laws:** $\neg(A \wedge B) = \neg A \vee \neg B$ and $\neg(A \vee B) = \neg A \wedge \neg B$.

Spell the two De Morgan laws out in words. (The sign $=$ between statements here is the equivalence $\iff$ of Definition 2.6: two statements are *“equal”* when they have the same truth value in every row of the table, which is exactly what a truth-table proof establishes).

Predicates and Quantifiers

A “*predicate*” $P(x)$ is a statement whose truth depends on a variable x. We use quantifiers to convert predicates into absolute statements.

Definition 2.8 (*Quantifiers*):

- **Universal ($\forall$):** *“For every x in X, $P(x)$ holds.”*
- **Existential ($\exists$):** *“There exists at least one x in X such that $P(x)$ holds.”*

Example 2.5 (*Quantifiers over Integers*): Let $P(n)$ be the predicate $n = n^2$ over $\mathbb{Z}$.

- $\forall n \in \mathbb{Z} : P(n)$ is False.
- $\exists n \in \mathbb{Z} : P(n)$ is True (take $n = 1$).
- $\forall n \in \mathbb{Z} : \big(n = n^2 \implies (n = 0) \vee (n = 1)\big)$ is True.

Quantifiers may of course be combined. For instance,

$$\forall y \in \mathbb{R} : \big(y \geq 0 \implies (\exists x \in \mathbb{R} : x^2 = y)\big)$$

is a True statement. In practice one writes this more compactly as

$$\forall\, 0 \leq y \in \mathbb{R},\ \exists x \in \mathbb{R} \ \text{ such that } \ x^2 = y. \tag{2.1}$$

Important remark 2.2 (*The Order of Quantifiers*): The order of quantifiers is non-commutative. Consider X (ETH students), Y (courses), and $A(x, y) :=$ *“student x takes course y”*.

- $\forall x \in X, \exists y \in Y : A(x, y)$ means every student takes at least one course (True).
- $\exists y \in Y, \forall x \in X : A(x, y)$ means there is one specific course taken by *every* student (False).

Figure 2.3 makes the asymmetry visible: reading the table row by row gives $\forall x\, \exists y$, whereas reading it column by column gives $\exists y\, \forall x$. Every row carries at least one mark, but no column is completely filled. The reader is encouraged to try the same interchange on (2.1) and to decide what the swapped statement asserts there.

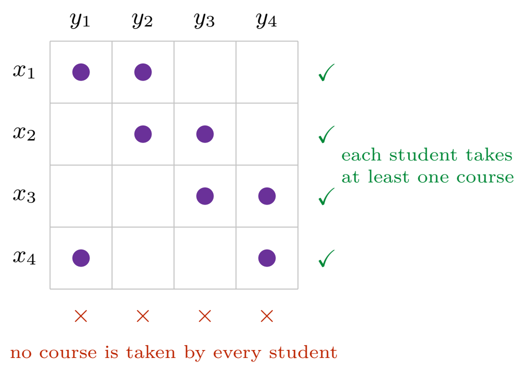


Figure 2.3: A mark in row x_i, column y_j records that student x_i takes course y_j. Every row contains a mark, so $\forall x \in X, \exists y \in Y : A(x, y)$ is True. No column is entirely filled, so $\exists y \in Y, \forall x \in X : A(x, y)$ is False. Swapping two quantifiers of different type therefore changes the meaning of a statement.

Negation and Unique Existence

Proposition 2.9 (*Negating Quantifiers*):

$$\neg(\forall x \in X : A(x)) \iff \exists x \in X : \neg A(x)$$
$$\neg(\exists x \in X : A(x)) \iff \forall x \in X : \neg A(x)$$

Remark 2.2 (*Shorthand Notation*): In practice, we often condense multiple quantifiers of the same type:

- $\exists x \in X, \exists y \in X \implies \exists x, y \in X$
- $\forall x \in X, \forall y \in X \implies \forall x, y \in X$

Definition 2.10 (*Unique Existence*): The notation $\exists! x \in X : P(x)$ means there exists exactly one x satisfying the predicate. Formally:

$$(\exists x \in X : P(x)) \wedge (\forall x, y \in X : (P(x) \wedge P(y) \implies x = y))$$

Example 2.6 (*Unique Existence of Roots*): $\exists! x \in \mathbb{R} : x^2 = 4$ is False (since $x = \pm 2$). However, $\exists! x \in \mathbb{R} : (x \geq 0) \wedge (x^2 = 4)$ is True.

Solutions to the Exercises

Proof of Proposition 2.7 and of Exercise 2.1:

Each of the four claims is settled by writing out the truth table and checking that the two columns in question agree row by row. That is what an equivalence of statements amounts to, by Definition 2.6.

The contrapositive. Comparing the third column with the sixth:

A	B	$A \implies B$	$\neg B$	$\neg A$	$\neg B \implies \neg A$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

The two columns are identical, which proves Proposition 2.7. The single row where they could have differed is the second, and there both are False for the same reason: a true premise with a false conclusion.

Double negation. Here two rows suffice:

A	$\neg A$	$\neg(\neg A)$
T	F	T
F	T	F

The first and third columns agree, so $\neg(\neg A) \iff A$.

The first De Morgan law. Comparing the fourth column with the seventh:

A	B	$A \wedge B$	$\neg(A \wedge B)$	$\neg A$	$\neg B$	$\neg A \vee \neg B$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

The second De Morgan law. Comparing the fourth column with the seventh:

A	B	$A \vee B$	$\neg(A \vee B)$	$\neg A$	$\neg B$	$\neg A \wedge \neg B$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

In words. The first law says that for *"A and B"* to fail it is necessary and sufficient that **at least one** of the two fails. The second says that for *"A or B"* to fail it is necessary and sufficient that **both** fail, which is where the inclusive reading of *"or"* recorded in Important remark 2.1 makes itself felt: were *"or"* exclusive, the second law would be false in the top row.

Two remarks worth carrying forward. First, the De Morgan laws are the statement-level version of Proposition 2.9: a universal quantifier is a long conjunction and an existential one a long disjunction, so negating either turns it into the other, exactly as here. Second, the contrapositive is the shape of every proof by contraposition in this book, of which the proof of Proposition 21.a.5 in ch. 21 is a typical example. **Q.E.D.**

AI-Note: This chapter is the only one in the book with no linear algebra in it, and it is easy to read past. It is worth saying what later chapters actually draw on.

Three things, mainly. The contrapositive of Proposition 2.7 is the shape of a large fraction of the proofs that follow: whenever a proof opens *"suppose the conclusion fails"*, it is this proposition being used. Proposition 2.9 is what makes a negated definition usable, and it is applied silently every time a proof shows that a family is **not** linearly independent by exhibiting one non-trivial relation. And Definition 2.10 is the pattern of nearly every uniqueness claim in the book: existence, then a proof that two candidates coincide. The proof of the uniqueness of the representation matrix in ch. 16 and the uniqueness half of Theorem 15.a.3 are both exactly this.

The one warning the chapter itself flags, in Important remark 2.1, is worth repeating: *"or"* is always inclusive here. The second De Morgan law depends on it, and so does the reading of *"A or B"* in every later statement.

The rigour pass added labels to the five Definitions and the Proposition that carried none, so that the results above can be cited at all, and wrote up the exercise, which is also the missing proof of Proposition 2.7: the chapter states the contrapositive and then defers its proof to the exercise, so without a solution the proposition stood unproved.

Set Theory (Chapter 3)

In this chapter, we formalize the language of sets, which provides the universal framework for all modern mathematical structures, including the vector spaces we will investigate later.

Basic Definitions and Notation

Definition 3.1 (*Set*): A "*set*" is a collection of distinct objects, which are referred to as the *elements* of the set.

In our notation, we denote sets typically by capital letters. If an object x is an element of a set M, we write $x \in M$. If it is not, we write $x \notin M$.

Remark 3.1: Crucially, within a set, the order of elements is irrelevant, and repetitions do not contribute to the set's identity. For example:

$$M = \{1, 3, \mathrm{moon}\} = \{3, \mathrm{moon}, 1\}$$

$$M = \{1, 1, 1\} = \{1\}$$

We frequently define sets using a property or condition $A(x)$ that the elements must satisfy. The standard notation for this *"set-builder"* form is:

$$M = \{x \mid A(x)\} \quad \text{or} \quad M = \{x : A(x)\}$$

Example 3.1 (*Set-Builder Notation*): The set of integers whose square is less than or equal to 9 is written as:

$$\{x \in \mathbb{Z} \mid x^2 \leq 9\} = \{-3, -2, -1, 0, 1, 2, 3\}$$

Definition 3.2 (*The Empty Set*): The "*empty set*", denoted by $\emptyset$ or $\{\}$, is the unique set containing no elements.

It is important to note that a set is allowed to contain another set as one of its elements. For instance, consider the set $M = \{1, \mathrm{elephant}, \{a, 5\}\}$. Here, the set $\{a, 5\}$ is a single element of M. Similarly, we can define a set containing only the empty set: $M = \{\emptyset\}$, which is a set containing exactly one element.

Subsets and Inclusion

We now define how sets relate to one another in terms of containment.

Definition 3.3 (*Subset and Proper Subset*): Let P and Q be sets.

(a) We say P is a "*subset*" of Q, denoted $P \subseteq Q$, if every element of P is also an element of Q. Formally: $\forall x \in P : x \in Q$.

(b) We say P is a "*proper subset*" of Q, denoted $P \subsetneq Q$, if $P \subseteq Q$ and $P \neq Q$.

(c) We write $P \nsubseteq Q$ to denote that P is not a subset of Q, meaning $\neg(P \subseteq Q)$.

Remark 3.2: Note that some authors use $P \subset Q$ to denote proper subsets, while others use it for generic subsets.

Example 3.2 (*Subsets and Non-Subsets*): We have

$$\{-1, 2, 5\} \subsetneq \{x \mid x \text{ is an integer and } x^2 \leq 30\},$$

the inclusion being proper because, for instance, 0 lies in the right-hand set but not in the left. On the other hand,

$$\{-1, 2, 5\} \not\subseteq \{x \mid x \text{ is an integer and } x^2 \leq 20\},$$

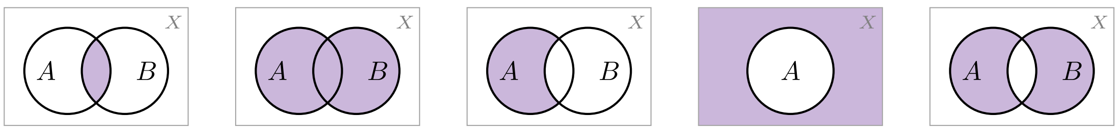
since $5^2 = 25 > 20$.

Claim: For any set Q, the empty set is a subset of Q (i.e., $\emptyset \subseteq Q$).

Set Operations

Just as we have logical operators for statements, we have operations to combine or modify sets. Let A and B be subsets of a universal set X.

(a) **Intersection:** $A \cap B = \{x \in X \mid x \in A \wedge x \in B\}$.
(b) **Union:** $A \cup B = \{x \in X \mid x \in A \vee x \in B\}$.
(c) **Difference:** $A \setminus B = \{x \in X \mid x \in A \wedge x \notin B\}$.
(d) **Complement:** The complement of A in X is $A^c = X \setminus A = \{x \in X \mid x \notin A\}$.



(a) $A \cap B$ (b) $A \cup B$ (c) $A \setminus B$ (d) A^c (e) $A \triangle B$

Figure 3.1: The set operations, drawn inside an ambient set X. In each panel the shaded region is the result of the operation. The last panel is the symmetric difference of Definition 3.4, which is exactly the union with the intersection removed.

Definition 3.4 (*Symmetric Difference*): The "*symmetric difference*" of A and B is defined by

$$A \triangle B := (A \cup B) \setminus (A \cap B).$$

Notation 3.1: The set X above is called the "*ambient set*", or the underlying set (in German, „*Grundmenge*“), of the discussion. The complement of A in X is sometimes also written $\overline{A}$ rather than A^c.

Unions and Intersections of Families of Sets

The operations $\cap$ and $\cup$ are not restricted to two sets; they extend to arbitrary collections.

Definition 3.5 (*Union and Intersection of a Family*): Let $\mathcal{A}$ be a family (that is, a set) of sets with $\mathcal{A} \neq \emptyset$. We define

$$\bigcup_{A \in \mathcal{A}} A := \{x \mid \exists A \in \mathcal{A} \text{ such that } x \in A\},$$

$$\bigcap_{A \in \mathcal{A}} A := \{x \mid \forall A \in \mathcal{A} : x \in A\}.$$

In practice, $\mathcal{A}$ sometimes merely parametrises, or indexes, a family of sets.

Example 3.3 (*An Indexed Family*): Let $\mathcal{A} = \{2, 3, 4, \dots\}$ and, for every $n \in \mathcal{A}$, set

$$A_n := \{x \mid x \in \mathbb{Z},\ x \geq 1, \text{ and } n^2 \text{ divides } x\}.$$

Thus $A_2 = \{4, 8, 12, \dots\}$, $A_3 = \{9, 18, 27, \dots\}$, and so on. Then

$$\bigcup_{n \in \mathcal{A}} A_n = \{a \in \mathbb{Z} \mid \exists k, r \in \mathbb{Z},\ k, r \geq 1, \text{ such that } a = k^2 \cdot r\}, \qquad \bigcap_{n \in \mathcal{A}} A_n = \emptyset.$$

The union is drawn in Figure 3.2. The intersection is empty because no integer $x \geq 1$ is divisible by n^2 for *every* n: any such x would exceed every bound.

AI-Note: As written in the notes, the condition describing the union allows $k = 1$, and with $k = 1$ the equation $a = k^2 \cdot r$ is satisfied by *every* $a \geq 1$. Taken literally the right-hand side would therefore be all of $\{a \in \mathbb{Z} \mid a \geq 1\}$, which is not the union of the A_n. Since the index set is $\mathcal{A} = \{2, 3, \dots\}$, the intended condition is $k \geq 2$, so that

$$\bigcup_{n \in \mathcal{A}} A_n = \{a \in \mathbb{Z} \mid a \geq 1 \text{ and } a \text{ is divisible by the square of some } k \geq 2\},$$

that is, the positive integers that are *not* squarefree. This is what Figure 3.2 shows.

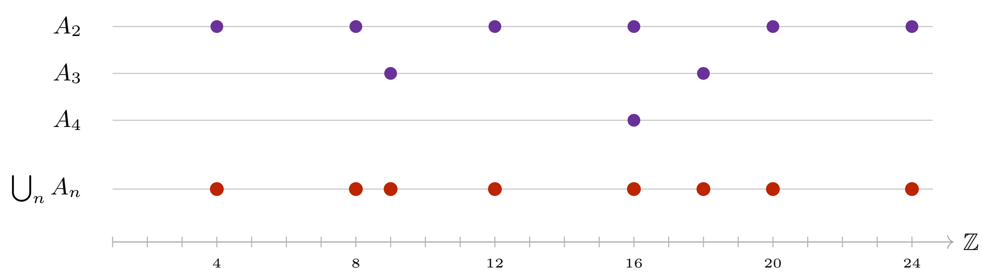


Figure 3.2: The family A_n of the preceding example, restricted to $1 \leq x \leq 24$. The set A_n collects the positive multiples of n^2, so $A_5, A_6, \dots$ contribute nothing in this range. The union (in red) consists of $4, 8, 9, 12, 16, 18, 20, 24$: precisely the integers divisible by a square greater than 1. Every row is eventually empty in any fixed range, which is why the intersection is empty.

Cartesian Products

To handle ordered pairs and higher-dimensional structures, we introduce the Cartesian product.

Definition 3.6 (*Cartesian Product*): The "*Cartesian product*" of two sets X and Y is the set of all ordered pairs (x, y) where $x \in X$ and $y \in Y$:

$$X \times Y := \{(x, y) \mid x \in X, y \in Y\}$$

Example 3.4 (*A Finite Cartesian Product*): Let $X = \{1, 2, \text{elephant}\}$ and $Y = \{4, \text{dog}\}$. Then

$$X \times Y = \{(1, 4), (1, \text{dog}), (2, 4), (2, \text{dog}), (\text{elephant}, 4), (\text{elephant}, \text{dog})\}.$$

The pairs are *ordered*, and this matters: $(1, 4) \in X \times Y$, whereas $(4, 1) \notin X \times Y$.

This construction generalizes to n sets. We denote the n-fold product of a set X with itself as X^n:

$$X^n := \underbrace{X \times \cdots \times X}_{n \text{ times}} = \{(x_1, \dots, x_n) \mid x_i \in X \text{ for } 1 \leq i \leq n\}$$

The elements of X^n are called n-tuples. For example, $X^2 := X \times X$, and $X^3 := X \times X \times X = \{(x_1, x_2, x_3) \mid x_1, x_2, x_3 \in X\}$, and so forth.

Example 3.5 (*Cartesian Products in the Plane*): If $X = \mathbb{R}$, then $X^2 = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ represents the standard Euclidean plane. If we take $X = [0, 2]$ and $Y = [1, 2]$, then $X \times Y$ represents a rectangle in the plane with corners at $(0, 1), (2, 1), (0, 2)$, and $(2, 2)$, as drawn in Figure 3.3.

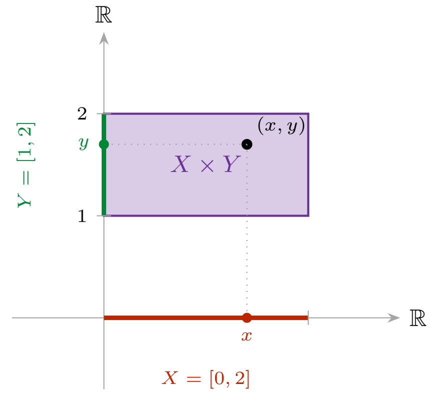


Figure 3.3: The Cartesian product $X \times Y$ for $X = [0, 2]$ and $Y = [1, 2]$. A point of the product is precisely a choice of one coordinate from each factor: dropping the point (x, y) onto the horizontal axis returns $x \in X$, and onto the vertical axis returns $y \in Y$.

Power Sets and Cardinality

Definition 3.7 (*Power Set*): Let X be a set. The "*power set*" of X, denoted by $\mathcal{P}(X)$ or 2^X, is the set of all subsets of X:

$$\mathcal{P}(X) := \{Q \mid Q \subseteq X\}$$

Example 3.6 (*Power Set of a Finite Set*): For $X = \{1, 2, 3\}$, the power set is:

$$\mathcal{P}(X) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Note that $\mathcal{P}(X)$ contains 8 elements.

Definition 3.8 (*Cardinality*): If X is a finite set, meaning a set with finitely many elements $X = \{x_1, \ldots, x_n\}$, we say that the "*cardinality*" of X is the number of elements in X, denoted by $|X| := n$ or $\operatorname{card}(X) := n$.

Exercise 3.1 (*Cardinality of the Power Set*): Let A be a finite set. Prove by induction that the cardinality of its power set is given by:

$$|\mathcal{P}(A)| = 2^{|A|}$$

A Note on the Foundations: Russell's Paradox

In the early development of set theory, it was assumed that any property $A(x)$ could define a set (Naive Set Theory). However, Bertrand Russell showed that this leads to a contradiction.

Suppose there exists a *"set of all sets"*, denoted by S. We then define a subset $R \subseteq S$ as follows:

$$R := \{P \in S \mid P \notin P\},$$

that is, R is the set of all sets that do not contain themselves. For example, S itself is a set, so $S \in S$, and therefore $S \notin R$.

Now, we ask: is R an element of itself?

- If $R \in R$, then by the definition of R, it must be that $R \notin R$. This is a contradiction.
- If $R \notin R$, then by the definition of R, it must be that $R \in R$. This is also a contradiction.

The two implications close into the cycle drawn in Figure 3.4: each of the only two possible answers forces the other one.

Claim: Neither S nor R can be regarded as a set.

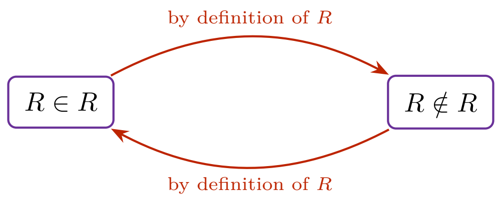


Figure 3.4: Russell's paradox as a cycle. A statement and its negation imply one another, so neither can be assigned a truth value. The fault lies not in the reasoning but in the assumption that S is a set.

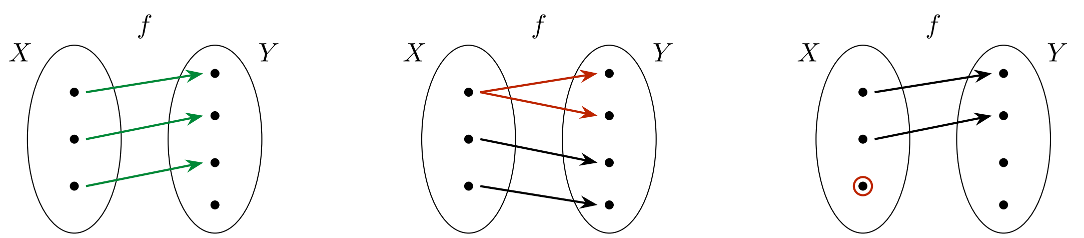
This paradox led to the development of Axiomatic Set Theory (e.g., Zermelo-Fraenkel), which places specific restrictions on how sets can be constructed to ensure mathematical consistency.

Maps (Chapter 4)

Functions and their Domains

Definition 4.1 (*Map / Function*): Let X and Y be sets. A map $f : X \to Y$ (also called a function or transformation) is an assignment that assigns to every element $x \in X$ a uniquely defined element $f(x) \in Y$.

In accordance with this definition, we can visually distinguish between valid and invalid assignments. A function must assign exactly one output to each input in its domain.



(a) O.K. (b) NOT O.K. (c) NOT O.K.

Figure 4.1: Which assignments are maps. In Figure 4.1a every element of X has exactly one arrow leaving it. In Figure 4.1b the top element of X is assigned two different values, so the value is not *uniquely defined*. In Figure 4.1c the circled element of X is assigned no value at all, so f is not defined on *every* element of X.

Informally, a map turns inputs into outputs:

$$\text{Input} \in X \xrightarrow{f} \text{output} \in Y.$$

X is called the *domain of definition* of f. Y is called the *target* (or target domain) of f.

Definition 4.2 (*Equality of Functions*): Two functions $f : X \to Y$ and $g : X' \to Y'$ are equal if they have the same domain of definition ($X' = X$), the same target ($Y' = Y$), and for all $x \in X$, we have $f(x) = g(x)$.

Sometimes a function is given by a formula, and sometimes it is not. Be careful: providing just a formula is often mathematically insufficient.

Example 4.1 (*Specifying Domain and Target*): **Bad notation:** $f(x) = x^2$. (What are the domain and target?)

O.K. notation: Let $X := \{x \in \mathbb{R} \mid 0 \le x\}$. Define $f : X \to \mathbb{R}$ by $f(x) = x^2$.

Example 4.2 (*Student Database Function*): Let $X = \{x \in \mathbb{Z} \mid x \ge 0\}$ and let Y be the set of family names. We might try to define a map $g : X \to Y$ by setting $g(x)$ to be the family name of the student at ETH whose student number is x.

This g is **not** defined! What is $g(0)$? What is $g(10^{10})$? There are no students with these IDs.

To fix this, we could define $Y' = Y \cup \{\text{ERROR!}\}$ and introduce a rigorous function $h : X \to Y'$ given by:

$$h(x) = \begin{cases} \text{family name} & \text{if a student with student no. } x \text{ exists} \\ \text{ERROR!} & \text{otherwise} \end{cases}$$

Remark 4.1: Notation: We frequently use the barred arrow ($\mapsto$) to denote how individual elements are mapped. For example, $f : X \to Y, X \ni x \mapsto f(x) \in Y$. Concretely: $f : \mathbb{R} \to \mathbb{R}, \mathbb{R} \ni x \mapsto x^5 \in \mathbb{R}$.

Images and Restrictions

Definition 4.3 (*Image of a Map*): Let $f : X \to Y$ be a map. The *image* of f is the set:

$$f(X) := \{y \in Y \mid \exists x \in X \text{ s.t. } f(x) = y\}$$

(sometimes written as $\text{image}(f)$). Clearly, $f(X) \subseteq Y$.

Definition 4.4 (*Restriction and Extension*): Let $f : X \to Y$ be a map, and let $A \subseteq X$ be a subset. We can define a new map $f|_A : A \to Y$, called the *restriction* of f to A, simply by $f|_A(x) := f(x)$ for all $x \in A$. Conversely, f is called an *extension* of $f|_A$.

Some Standard Maps

Example 4.3 (*Standard Maps*):

(a) **Identity Map:** $\text{id} : X \to X$ (sometimes denoted id_X), defined by $\text{id}(x) = x$ for all $x \in X$.
(b) **Constant Map:** $f : X \to Y$. Choose a fixed $y_0 \in Y$. Then $f(x) = y_0$ for all $x \in X$.
(c) **Characteristic Function:** Let $A \subseteq X$. The characteristic function on A, written $\mathbf{1}_A : X \to \mathbb{R}$ (and also χ_A), is defined as:

$$\mathbf{1}_A(x) := \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

(d) **Projections:** Given a Cartesian product $X_1 \times X_2$, the standard projection maps are:

$$\pi_1 : X_1 \times X_2 \to X_1, \quad (x_1, x_2) \mapsto x_1$$
$$\pi_2 : X_1 \times X_2 \to X_2, \quad (x_1, x_2) \mapsto x_2$$

Injectivity, Surjectivity, Bijectivity

Definition 4.5 (*Injectivity, Surjectivity, and Bijectivity*): Let $f : X \to Y$ be a map.

(a) We say f is **injective** if $\forall x_1, x_2 \in X$, $f(x_1) = f(x_2) \implies x_1 = x_2$. (Equivalently, by contrapositive: $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$).
(b) We say f is **surjective** if $\forall y \in Y, \exists x \in X$ such that $f(x) = y$. (Equivalently: $f(X) = Y$).
(c) We say f is **bijective** (or a *bijection*) if f is both injective and surjective. (Equivalently: $\forall y \in Y, \exists! x \in X : f(x) = y$).

Example 4.4 (*Injective and Surjective Squaring Functions*): Consider variations of the squaring function, which explicitly demonstrate the importance of the chosen domain and target:

- $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Not injective (e.g., $f(2) = f(-2)$). Not surjective (no real x maps to -1).
- $f : \mathbb{R}_{\geq 0} \to \mathbb{R}, f(x) = x^2$. Injective, but not surjective.
- $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}, f(x) = x^2$. Bijective.

Definition 4.6 (*Inverse Map*): Let $f : X \to Y$ be a bijection. The *inverse map* $f^{-1} : Y \to X$ is the map that assigns to every $y \in Y$ the unique $x \in X$ such that $f(x) = y$.

Example 4.5 (*Inverse of Squaring Function*): For the bijection $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f(x) = x^2$, the inverse map is $f^{-1} : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, strictly defined by $f^{-1}(y) = \sqrt{y}$.

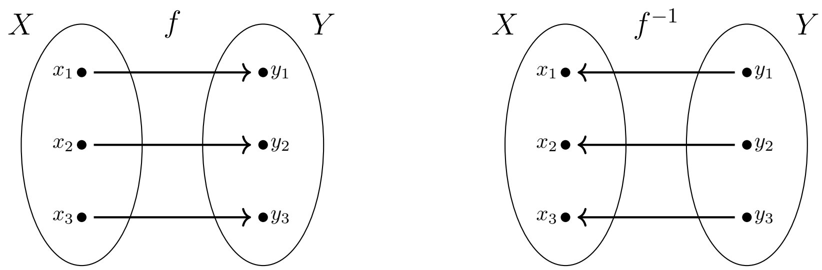


Figure 4.2: A bijection f and its inverse f^{-1}. Because f is bijective, every $y \in Y$ is hit by exactly one $x \in X$, so reversing every arrow again produces a map.

Composition of Maps

Definition 4.7 (*Composition*): Let $f : X \to Y$ and $g : Y \to Z$ be maps. The composition $g \circ f : X \to Z$ is defined by:

$$(g \circ f)(x) := g(f(x))$$

On elements, the composition is simply the two assignments performed one after the other:

$$x \xmapsto{f} f(x) \xmapsto{g} g(f(x))$$

Example 4.6 (*Composition with Inverse*): If $f : X \to Y$ is bijective and $f^{-1} : Y \to X$ is its inverse, then by definition:

$$f^{-1} \circ f = \mathrm{id}_X \quad \text{and} \quad f \circ f^{-1} = \mathrm{id}_Y$$

Remark 4.2: If $g \circ f$ is defined, $f \circ g$ is in general NOT defined (unless $Z = X$). But even when $Z = X$, in general $g \circ f \neq f \circ g$. Commutativity is not guaranteed.

Example 4.7 (*Non-Commutative Compositions*): Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$ and $g(x) = x + 1$.

$$\begin{aligned}(g \circ f)(x) &= g(x^2) = x^2 + 1\\ (f \circ g)(x) &= f(x+1) = (x+1)^2 = x^2 + 2x + 1\end{aligned}$$

Clearly, $g \circ f \neq f \circ g$.

Lemma 4.8 (*Composition of Injective and Surjective Maps*): Let $f : X \to Y$ and $g : Y \to Z$ be maps.

(a) If f and g are injective, then $g \circ f$ is injective.
(b) If f and g are surjective, then $g \circ f$ is surjective.
(c) If f and g are bijective, then $g \circ f$ is bijective.

Proof:

(a) Assume f and g are injective. Let $x_1, x_2 \in X$ such that $(g \circ f)(x_1) = (g \circ f)(x_2)$. This means $g(f(x_1)) = g(f(x_2))$. Because g is injective, this implies $f(x_1) = f(x_2)$. Because f is injective, this implies $x_1 = x_2$. Thus, $g \circ f$ is injective.
(b) Assume f and g are surjective. Let $z \in Z$. Because g is surjective, there exists $y \in Y$ such that $g(y) = z$. Because f is surjective, there exists $x \in X$ such that $f(x) = y$. Therefore, $g(f(x)) = g(y) = z$, which means $(g \circ f)(x) = z$. Thus, $g \circ f$ is surjective.
(c) This follows from **(a)** and **(b)**, because bijective means injective and surjective.

Q.E.D.

Graphs

Definition 4.9 (*Graph of a Map*): Let $f : X \to Y$ be a map. The "*graph*" of f is the following subset of $X \times Y$:

$$\mathrm{Graph}(f) := \{(x, y) \in X \times Y \mid y = f(x)\} \subseteq X \times Y.$$

For $X \subseteq \mathbb{R}$ and $Y = \mathbb{R}$ the graph is a curve in the plane, and the defining condition can be read off geometrically. Since f must assign to every $x \in X$ exactly one value, a subset of the plane is the graph of a map if and only if every vertical line meets it in exactly one point. This is the "*vertical line test*", illustrated in Figure 4.3.

(a) a graph (b) not a graph

Figure 4.3: The vertical line test. On the left every vertical line meets the curve exactly once, so the curve is the graph of a map. On the right the vertical line drawn meets the curve three times, so no single value can be assigned to that input and the curve is not the graph of a map.

We can often immediately determine the injectivity, surjectivity, and bijectivity of a function, or whether an assignment is a function at all, by examining its Cartesian graph using horizontal and vertical line tests.

Image and Inverse Image of Subsets

Definition 4.10 (*Image of a Subset*): Let $f : X \to Y$ be a map, and let $A \subseteq X$. The *image of A under f* is:

$$f(A) := \{y \in Y \mid \exists x \in A \text{ s.t. } f(x) = y\} = \{f(x) \mid x \in A\}$$

Definition 4.11 (*Inverse Image of a Subset*): Let $f : X \to Y$ be a map, and let $B \subseteq Y$. The *inverse image of B under f* is:

$$f^{-1}(B) := \{x \in X \mid f(x) \in B\} \subseteq X$$

Important remark 4.1: The notation $f^{-1}(B)$ is somewhat confusing, as this set exists and is perfectly well-defined even if the map f is not invertible!

Example 4.8 (*Preimages of Subsets*): Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ (which is neither injective nor surjective). We can still compute inverse images of subsets without issue:

$$\begin{aligned} f^{-1}(\{4\}) &= \{-2, 2\} \\ f^{-1}(\{1, 9\}) &= \{-1, 1, -3, 3\} \\ f^{-1}(\{0\}) &= \{0\} \\ f^{-1}(\{-1\}) &= \emptyset \\ f^{-1}(\{-1, 0\}) &= \{0\} \end{aligned}$$

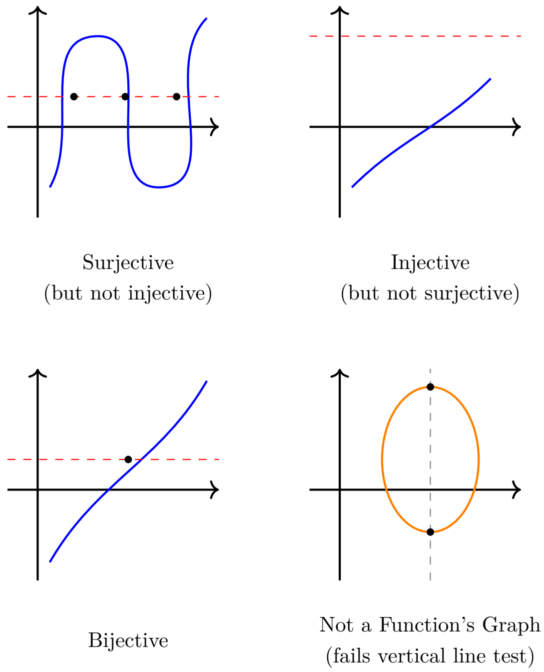


Figure 4.4: Reading off properties from a graph. A *horizontal* line at height y meets the graph once for each x with $f(x) = y$: meeting every horizontal line at least once means surjective, at most once means injective, and exactly once means bijective. A *vertical* line meeting the curve more than once means the curve is not a graph at all.

Partitions

Definition 4.12 (*Partition*): Let X be a set and $\mathcal{P}$ a family of subsets of X, none of which is empty. We say that $\mathcal{P}$ is a *partition* of X if the following hold:

(a) $X = \bigcup_{P \in \mathcal{P}} P$ (the subsets cover X).

(b) For any $P_1, P_2 \in \mathcal{P}$ with $P_1 \neq P_2$, we have $P_1 \cap P_2 = \emptyset$ (the subsets are mutually disjoint).

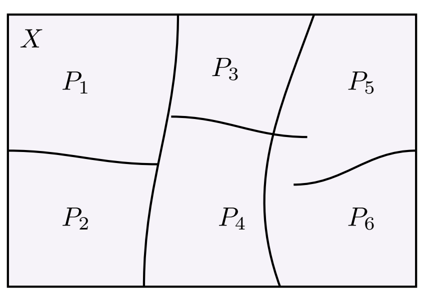


Figure 4.5: A partition of X. The pieces cover X and no two of them overlap, so every element of X lies in exactly one piece.

The last observation is what makes partitions useful, and it deserves to be stated on its own.

Lemma 4.13 (*Every Element Lies in Exactly One Piece*): Let $\mathcal{P}$ be a partition of X. Then for every $x \in X$ there exists exactly one $P \in \mathcal{P}$ with $x \in P$: at least one by condition **(a)** of Definition 4.12, and at most one by condition **(b)**.

Defining Functions by Cases

Partitions are precisely what licenses the case-by-case definitions we have already used, such as the characteristic function or the map h above.

Proposition 4.14 (*Gluing Functions along a Partition*): Let X and Y be sets and let $\mathcal{P}$ be a partition of X. Suppose that for every $P \in \mathcal{P}$ we are given a function $f_P : P \to Y$. Then there exists a unique function

$$f : X \to Y \quad \text{such that} \quad f|_P = f_P \;\text{ for all } P \in \mathcal{P}.$$

Proof:

Let $x \in X$. By Lemma 4.13, there exists exactly one $P \in \mathcal{P}$ with $x \in P$, so we may define

$$f(x) := f_P(x).$$

This assigns to every $x \in X$ a uniquely determined element of Y, so f is a well-defined map, and by construction $f|_P = f_P$ for every $P \in \mathcal{P}$.

For uniqueness, suppose $g : X \to Y$ also satisfies $g|_P = f_P$ for all $P \in \mathcal{P}$. Given $x \in X$, choose the piece P containing x; then $g(x) = g|_P(x) = f_P(x) = f(x)$. Hence $g = f$. Q.E.D.

Fields (Körper) (Chapter 5)

Before we can solve systems of linear equations or define abstract vector spaces, we must answer a more fundamental question: *Where do our numbers come from?*

In linear algebra, we do not restrict ourselves to the real numbers. We can perform our calculations over any algebraic structure where addition, subtraction, multiplication, and division behave exactly as we expect them to. Such a structure is called a **field** (in German: *Körper*).

The Field Axioms

To ensure that an algebraic structure behaves nicely, it must satisfy ten rigid rules. These are the Field Axioms.

Definition 5.1 (*Field*): A *field* is a tuple $(K, +, \cdot, 0, 1)$ consisting of a set K equipped with two binary operations:

$$+ : K \times K \to K, \quad (x, y) \mapsto x + y \quad \text{(Addition)}$$
$$\cdot : K \times K \to K, \quad (x, y) \mapsto x \cdot y \quad \text{(Multiplication)}$$

and two distinguished, unique elements $0, 1 \in K$, such that the following ten axioms hold:

Axioms of Addition:

- **(K1)** $\forall x, y, z \in K : x + (y + z) = (x + y) + z$ *(Associativity of Addition)*
- **(K2)** $\forall x, y \in K : x + y = y + x$ *(Commutativity of Addition)*
- **(K3)** $\forall x \in K : 0 + x = x$ *(Neutral Element of Addition)*
- **(K4)** $\forall x \in K, \exists x' \in K : x + x' = 0$ *(Inverse Element of Addition)*

Axioms of Multiplication:

- **(K5)** $\forall x, y, z \in K : x \cdot (y \cdot z) = (x \cdot y) \cdot z$ *(Associativity of Multiplication)*
- **(K6)** $\forall x \in K : 1 \cdot x = x$ *(Neutral Element of Multiplication)*
- **(K7)** $\forall x \in K \setminus \{0\}, \exists x' \in K : x' \cdot x = 1$ *(Inverse Element of Multiplication)*

Compatibility, Non-Triviality, and Commutativity:

- **(K8)** $\forall x, y, z \in K : x \cdot (y + z) = x \cdot y + x \cdot z$ and
  $\forall x, y, z \in K : (y + z) \cdot x = y \cdot x + z \cdot x$ *(Distributivity)*
- **(K9)** $1 \neq 0$ *(Non-triviality: the field has at least two elements)*
- **(K10)** $\forall x, y \in K : x \cdot y = y \cdot x$ *(Commutativity of Multiplication)*

Important remark 5.1: Axiom **(K10)** is not decorative. A structure satisfying **(K1)**–**(K9)** but not **(K10)** is called a "*skew field*" (or division ring); the quaternions are the standard example. Everything in this course is done over fields, so multiplication may always be reordered freely.

Remark 5.1 (*Notation*): We typically denote the additive inverse x' from (K4) as $-x$, allowing us to define subtraction: $x - y := x + (-y)$. Similarly, we denote the multiplicative inverse x' from (K7) as x^{-1} or $\frac{1}{x}$, allowing us to define division by non-zero elements: $x/y := x \cdot y^{-1}$.

AI-Note: The source material for this chapter is a single page, giving the definition above (and nothing else) in German, as *"Definition 1.3.9"*. Everything that follows — the examples, finite fields and modular arithmetic, $\mathbb{F}_p$, and the two proofs from the axioms, has no counterpart in the provided notes and was supplied by this transcript. It is standard material and consistent with the course, but it is not lecture content.

Infinite Fields

The most familiar fields are infinite. They form the backbone of calculus and classical geometry.

Example 5.1 (*Familiar Infinite Fields*):

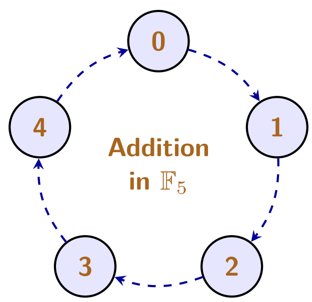
- $\mathbb{Q}$: The rational numbers (fractions). This is the smallest field containing the integers.
- $\mathbb{R}$: The real numbers.
- $\mathbb{C}$: The complex numbers, where we introduce $i^2 = -1$.

Important remark 5.2: The integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ do **not** form a field! While they satisfy addition and multiplication axioms perfectly, they completely fail (K7). For instance, there is no integer x' such that $2 \cdot x' = 1$. You cannot divide freely within $\mathbb{Z}$.

Finite Fields and Modular Arithmetic

A field does not have to be an infinite, continuous line of numbers. If we change our definitions of *"addition"* and *"multiplication"*, we can create perfectly valid fields that contain only a finite number of elements.

The secret to building a finite field is "*modular arithmetic*" (often called *"clock arithmetic"*). Instead of numbers growing infinitely on a number line, they wrap around a circle.



Definition 5.2 (*The Field $\mathbb{F}_p$*): Let p be a prime number. The set of integers modulo p, denoted as $\mathbb{F}_p$ (or sometimes $\mathbb{Z}/p\mathbb{Z}$), contains exactly p elements:

$$\mathbb{F}_p = \{0, 1, 2, \dots, p-1\}$$

Addition and multiplication are performed exactly as regular integers, but if the result exceeds $p-1$, we divide by p and keep only the **remainder**.

Theorem 5.3: $\mathbb{F}_p$ is a field if and only if p is a prime number.

Why must p be prime? Consider what happens if we try to build a field modulo 4 (which is not prime). In modulo 4, we have $2 \cdot 2 = 4 \equiv 0 \pmod 4$. We multiplied two non-zero numbers and got zero! In a true field, this is algebraically impossible, because it destroys the ability to divide (axiom **(K7)**). Prime numbers beautifully prevent this *"zero divisor"* problem, as Figure 5.1 makes visible.

Example 5.2 (*The Smallest Field: $\mathbb{F}_2$*): The absolute smallest field mathematically possible contains only two elements: $\{0, 1\}$. This is the field of Boolean logic, heavily used in computer science and cryptography. Operations are done modulo 2:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

$\cdot$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

(a) $\mathbb{F}_5$: no zero in the body

$\cdot$	1	2	3
1	1	2	3
2	2	$\boxed{0}$	2
3	3	2	1

(b) $\mathbb{Z}/4\mathbb{Z}$: $2 \cdot 2 = 0$

Figure 5.1: Multiplication tables of the non-zero elements. For the prime modulus 5 every entry is non-zero and every row is a rearrangement of $1, 2, 3, 4$, so each element has an inverse: the 1 in row x sits in column x^{-1}. For the composite modulus 4 the boxed entry $2 \cdot 2 = 0$ produces a zero divisor, and row 2 never reaches 1, so 2 has no inverse and **(K7)** fails.

Notice the fascinating property of $\mathbb{F}_2$: $1 + 1 = 0$. In this field, every element is its own additive inverse! Subtraction and addition are literally the same operation.

Example 5.3 (*Calculations in $\mathbb{F}_5$*): Let us explore $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$.

- **Addition:** $3 + 4 = 7$. But $7 \div 5 = 1$ with a remainder of 2. So, $3 + 4 = 2$ in $\mathbb{F}_5$.
- **Additive Inverses (K4):** What is -2? We need a number that adds to 2 to yield 0. Since $2 + 3 = 5 \equiv 0$, we know that $-2 = 3$.
- **Multiplicative Inverses (K7):** What is the inverse of 3 (i.e., $\frac{1}{3}$)? We need a number x such that $3 \cdot x = 1$. Let's test the options: $3 \cdot 2 = 6 \equiv 1$. Therefore, $3^{-1} = 2$.

Proving Basic Properties from the Axioms

The power of an axiomatic definition is that if we can prove a theorem using *only* axioms **(K1)** through **(K10)**, that theorem is universally true for **every field in existence**, whether it is the infinite space of real numbers $\mathbb{R}$, or the tiny binary world of $\mathbb{F}_2$.

Let us prove two seemingly *"obvious"* facts that actually require careful algebraic footwork to justify.

Lemma 5.4 (*Multiplication by Zero*): For any element x in a field K, we have $0 \cdot x = 0$.

Proof:

Let $x \in K$. By the neutral property of zero **(K3)**, we know that $0 = 0 + 0$. Let us multiply both sides of this equation by x:

$$(0 + 0) \cdot x = 0 \cdot x$$

By the distributivity axiom **(K8)**, we can expand the left side:

$$0 \cdot x + 0 \cdot x = 0 \cdot x$$

Now, by the additive inverse axiom **(K4)**, the element $(0 \cdot x)$ must have an inverse, $-(0 \cdot x)$. We add this inverse to both sides of our equation:

$$\big(0 \cdot x + 0 \cdot x\big) + \big(-(0 \cdot x)\big) = 0 \cdot x + \big(-(0 \cdot x)\big)$$

Using associativity of addition **(K1)**, we regroup the left side:

$$0 \cdot x + \Big(0 \cdot x + \big(-(0 \cdot x)\big)\Big) = 0$$

$$0 \cdot x + 0 = 0$$

Finally, applying **(K3)** once more, we drop the $+0$ to conclude:

$$0 \cdot x = 0$$

Q.E.D.

Lemma 5.5 (*Zero Divisors do not exist*): Let $a, b \in K$. If $a \cdot b = 0$, then either $a = 0$ or $b = 0$.

Proof:

Assume $a \cdot b = 0$. We want to show that if $a \neq 0$, then b must be exactly 0.

Suppose $a \neq 0$. Because a is a non-zero element in a field, axiom **(K7)** guarantees that it has a multiplicative inverse, a^{-1}. Let us multiply both sides of our equation $a \cdot b = 0$ by this inverse:

$$a^{-1} \cdot (a \cdot b) = a^{-1} \cdot 0$$

By Lemma 5.4, any element multiplied by 0 is 0, so the right side is 0:

$$a^{-1} \cdot (a \cdot b) = 0$$

By associativity of multiplication **(K5)**, we regroup the left side:

$$(a^{-1} \cdot a) \cdot b = 0$$

By the definition of the inverse **(K7)**, $a^{-1} \cdot a = 1$:

$$1 \cdot b = 0$$

Finally, by the neutral element of multiplication **(K6)**, $1 \cdot b = b$. Thus:

$$b = 0$$

We have proven that if the product is zero and a is not zero, b is forced to be zero. Q.E.D.

Systems of Linear Equations (Chapter 6)

Fields (Körper)

Before we solve systems of linear equations, we must fix the algebraic structure over which these equations are built. Throughout this course, we will work over a base "*field*" (in German: "*Körper*"), usually denoted by K.

A field is a set equipped with two operations (addition $+$ and multiplication $\cdot$) satisfying the ten axioms **(K1)–(K10)** of Definition 5.1: commutativity, associativity and distributivity, together with the existence of the identities 0 and 1 and of additive and multiplicative inverses. We briefly recall the examples we will use most often.

Example 6.1 (*Common Fields*):

- $\mathbb{Q} = \left\{a \mid a = \frac{m}{n},\ m, n \in \mathbb{Z},\ n \neq 0\right\}$ (the rational numbers)
- $\mathbb{R}$ (the real numbers)
- $\mathbb{C} = \{z \mid z = a + i \cdot b,\ a, b \in \mathbb{R}\}$, where $i \cdot i = -1$ (the complex numbers)

Example 6.2 (*Finite Fields*): Fields do not have to be infinite. We will frequently use finite fields, where arithmetic is performed modulo a prime number.

(a) $\mathbb{F}_2 = \{0, 1\}$. Operations are done modulo 2:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

(b) $\mathbb{F}_3 = \{0, 1, 2\}$. Operations are done modulo 3:

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

A Motivating Example

Let us solve a small system of linear equations over $\mathbb{R}$ by elimination:

$$\begin{cases} E_1: & x + 3y + 4z = -1 \\ E_2: & 2x - \ \ y + \ \ z = \ \ 5 \end{cases}$$

We operate on the equations so as to eliminate variables one at a time. First we remove x from the second equation by $-2 \cdot E_1 + E_2 \to E_2$:

$$\begin{cases} x + 3y + 4z = -1 \\ \quad\ - 7y - 7z = \ \ 7 \end{cases}$$

Next we normalise the second equation by $-\frac{1}{7} \cdot E_2 \to E_2$:

$$\begin{cases} x + 3y + 4z = -1 \\ \qquad\ \ y + \ \ z = -1 \end{cases}$$

Finally, instead of substituting backwards, we eliminate y **upwards** as well, using $-3 \cdot E_2 + E_1 \to E_1$:

$$\begin{cases} x \qquad\ + \ \ z = \ \ 2 \\ \qquad\ \ y + \ \ z = -1 \end{cases}$$

The system is now solved as far as it can be. There are three unknowns but only two equations, so one unknown remains free: let $z := c \in \mathbb{R}$ be an arbitrary real number. Then

$$y = -1 - c, \qquad x = 2 - c,$$

and the set of solutions is

$$\{(x, y, z) \mid x = 2 - c,\ y = -1 - c,\ z = c,\ c \in \mathbb{R}\}. \tag{6.1}$$

This is not a single point but an entire one-parameter family, geometrically, the line in which the two planes $x + 3y + 4z = -1$ and $2x - y + z = 5$ intersect, as sketched in Figure 6.1.

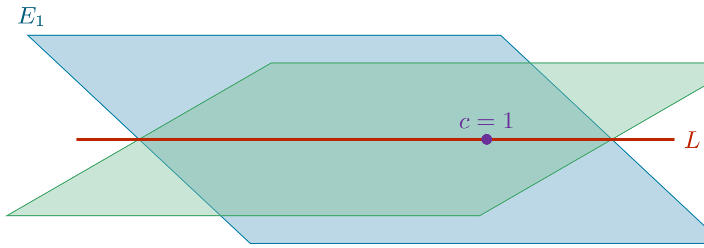


Figure 6.1: Schematic picture of the motivating example. Each equation cuts out a plane in $\mathbb{R}^3$, and the solution set L of the system is the line in which the two planes meet, one free parameter c, exactly as (6.1) records. Imposing a third independent equation adds a third plane, which meets L in a single point.

Example 6.3 (*One more equation removes the freedom*): Suppose we append a third equation $E_3 : 3x + 2y + z = 0$ to the system above. Substituting the parametrised solution (6.1) into it gives

$$3(2 - c) + 2(-1 - c) + c = 4 - 4c = 0,$$

so that $c = 1$ is forced. The freedom collapses and the system now has the unique solution

$$(x, y, z) = (1, -2, 1),$$

which is exactly the member of the family (6.1) with $c = 1$. A system may therefore have no solution, exactly one solution, or infinitely many; deciding which of these occurs is one of the goals of this chapter.

General Systems of Linear Equations

Definition 6.1 (*Linear Equation*): A "*linear equation*" in n variables $x_1, \dots, x_n$ over a field K is an equation of the form

$$a_1 x_1 + \cdots + a_n x_n = b,$$

where $a_1, \dots, a_n \in K$ are the "*coefficients*" and $b \in K$ is the "*constant term*".

- If $b = 0$, the equation is called "*homogeneous*".
- If $b \neq 0$, the equation is called "*inhomogeneous*".

Definition 6.2 (*System of Linear Equations*): A general system of m linear equations in n variables over a field K is written as

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases} \tag{6.2}$$

with $a_{i,j} \in K$ for $1 \le i \le m$, $1 \le j \le n$, and $b_i \in K$ for $1 \le i \le m$. Here i is the row index and j is the column index.

Definition 6.3 (*Solution*): A "*solution*" of the system is a tuple of scalars $s_1, \dots, s_n \in K$ that satisfies all m equations simultaneously when we substitute $x_j = s_j$.

Definition 6.4 (*Equivalence of Systems*): Two systems of linear equations in the same variables are called "*equivalent*" if they have exactly the same set of solutions.

Matrices

To study systems of linear equations systematically, we strip away the variables and isolate the coefficients into a rectangular array.

Definition 6.5 (*Matrix*): An $m \times n$ "*matrix*" over K is a rectangular array consisting of $m \cdot n$ elements of K, arranged in m rows and n columns.

Notation 6.1: We denote a matrix A by its entries,

$$A = (a_{i,j})_{1 \le i \le m,\, 1 \le j \le n} \quad \text{or simply} \quad (a_{ij}),$$

where i denotes the row index and j the column index.

We define the set of all such matrices as

$$\mathcal{M}_{m \times n}(K) := \text{the set of all } m \times n \text{ matrices over } K.$$

Definition 6.6 (*Row and Column Vectors*): Special cases of matrices occur when one of the dimensions is 1:

(a) An $n \times 1$ matrix is called a "*column vector*". The set of all column vectors of length n is denoted by $K^n := \mathcal{M}_{n \times 1}(K)$:

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in K^n.$$

(b) A $1 \times n$ matrix is called a "*row vector*". The set of all row vectors of length n is denoted by $K^n_{\text{row}} := \mathcal{M}_{1 \times n}(K)$:

$$(x_1, x_2, \dots, x_n) \in K^n_{\text{row}}.$$

Matrix-Vector Multiplication

Definition 6.7 (*Matrix-Vector Multiplication*): Let $A \in \mathcal{M}_{m \times n}(K)$ and $x \in K^n$. We define the product $A \cdot x$ to be the column vector in K^m obtained by pairing each row of A with the vector x:

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{1,1}x_1 + \dots + a_{1,n}x_n \\ a_{2,1}x_1 + \dots + a_{2,n}x_n \\ \vdots \\ a_{m,1}x_1 + \dots + a_{m,n}x_n \end{pmatrix} \in K^m.$$

With this notation, the whole system (6.2) collapses into a single matrix equation. We write

$$(S): \quad Ax = b, \tag{6.3}$$

where $A \in \mathcal{M}_{m \times n}(K)$ is the "*coefficient matrix*", $x \in K^n$ is the vector of variables, and $b \in K^m$ is the "*constant vector*".

Notation 6.2: For a system (S) as in (6.3) we write

$$L(S) := \{(x_1, \dots, x_n) \mid x_i \in K,\ Ax = b\} \subseteq K^n$$

for its "*set of solutions*". Two systems (S) and (S') in the same variables are equivalent (Definition 6.4) precisely when $L(S) = L(S')$.

Example 6.4 (*The motivating example in matrix form*): The system of the previous section can be rewritten first with indexed variables and then as a matrix equation:

$$\begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases} \iff \begin{cases} x_1 + 3x_2 + 4x_3 = -1 \\ 2x_1 - x_2 + x_3 = 5 \end{cases} \iff \underbrace{\begin{pmatrix} 1 & 3 & 4 \\ 2 & -1 & 1 \end{pmatrix}}_{2\times 3} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{3\times 1} = \underbrace{\begin{pmatrix} -1 \\ 5 \end{pmatrix}}_{2\times 1}.$$

Goals 6.1: Our aim for the rest of this chapter is to study the set of solutions $L(S)$ of a system $Ax = b$:

(a) Does the system have a solution at all, and if so, how many?

(b) How do we find **all** of the solutions, systematically?

(c) Does $L(S)$ carry a structure of its own?

Elementary Row Operations

Notation 6.3: We do not need to write down the unknowns $x_1, \dots, x_n$ at every step: the coefficient matrix A and the constant vector b already carry all the information. We therefore record the system as its "*augmented matrix*" (or "*extended matrix*")

$$(A \mid b) \in \mathcal{M}_{m\times(n+1)}(K),$$

obtained by appending b to A as an extra column.

Example 6.5 (*An augmented matrix*): The system

$$\begin{cases} 2x_1 - \ x_2 + 3x_3 + 2x_4 = 0 \\ \ x_1 + 4x_2 \qquad\ \ - \ x_4 = 3 \\ 2x_1 + 6x_2 - \ x_3 + 5x_4 = 9 \end{cases} \qquad \text{has the augmented matrix} \qquad \left(\begin{array}{cccc|c} 2 & -1 & 3 & 2 & 0 \\ 1 & 4 & 0 & -1 & 3 \\ 2 & 6 & -1 & 5 & 9 \end{array}\right).$$

The three operations below act on the equations of a system, and equally on the rows of its augmented matrix.

Definition 6.8 (*Elementary Row Operations*): Let E_i denote the i-th equation of a system, equivalently the i-th row of its augmented matrix. The three "*elementary row operations*" (EROs) are:

(a) **Type 1 (scaling).** Choose $\lambda \in K \setminus \{0\}$ and multiply row i by λ.
Notation: $\lambda \cdot E_i \to E_i$.

(b) **Type 2 (addition).** Choose $\lambda \in K$ and $1 \le i, j \le m$ with $i \neq j$, and replace row j by row j **plus** λ times row i.
Notation: $\lambda \cdot E_i + E_j \to E_j$.

(c) **Type 3 (interchange).** Swap row i and row j.
Notation: $E_i \leftrightarrow E_j$.

Definition 6.9 (*Row Equivalence*): Let C and C' be $r \times s$ matrices with entries in K. We say that C' is "*row equivalent*" to C if C' can be obtained from C by a finite sequence of elementary row operations,

$$C = C_0 \longrightarrow C_1 \longrightarrow \cdots \longrightarrow C_k = C'.$$

Theorem 6.10 (*Row Operations Preserve the Solution Set*): Let $(S): Ax = b$ and $(S'): A'x = b'$ be two systems of m linear equations in n unknowns. Suppose the augmented matrix $(A' \mid b')$ is row equivalent to $(A \mid b)$. Then

$$L(S') = L(S).$$

Proof:

We begin with three claims.

Claim (*Claim 1, One Operation Gives One Inclusion*): If (S') is obtained from (S) by one elementary row operation, then $L(S) \subseteq L(S')$.

Proof of Claim 1. Let $(x_1, \ldots, x_n) \in L(S)$. We must show that $(x_1, \ldots, x_n) \in L(S')$. We treat the three types separately.

(a) Operation $\lambda \cdot E_i \to E_i$ with $\lambda \neq 0$. All equations of (S') coincide with those of (S) except for equation i, which changes as

$$a_{i,1}x_1 + \cdots + a_{i,n}x_n = b_i \qquad \longrightarrow \qquad \lambda a_{i,1}x_1 + \cdots + \lambda a_{i,n}x_n = \lambda b_i.$$

If $(x_1, \ldots, x_n)$ satisfies the left equation, then multiplying both sides by λ shows that it satisfies the right one as well.

(b) Operation $\lambda \cdot E_i + E_j \to E_j$. Again all equations of (S) and (S') coincide except for equation j:

$$\begin{aligned} & \text{row } j \text{ in } (S): \quad a_{j,1}x_1 + \cdots + a_{j,n}x_n = b_j \\ \longrightarrow \qquad & \text{row } j \text{ in } (S'): \quad \big(a_{j,1} + \lambda a_{i,1}\big)x_1 + \cdots + \big(a_{j,n} + \lambda a_{i,n}\big)x_n = b_j + \lambda b_i \end{aligned}$$

If $(x_1, \ldots, x_n)$ satisfies rows i and j of (S), then adding λ times the i-th equation to the j-th shows that it satisfies row j of (S') too.

(c) Operation $E_i \leftrightarrow E_j$. Here we clearly have

$$(x_1, \ldots, x_n) \in L(S) \implies (x_1, \ldots, x_n) \in L(S'),$$

because the order in which the equations of a system are listed has no effect whatsoever on its set of solutions.

This concludes the proof of Claim 1. $\diamond$

Claim (*Claim 2, Every Operation is Reversible*): If (S') is obtained from (S) by one elementary row operation, then (S) can be obtained from (S') by one elementary row operation.

Proof of Claim 2. There are three cases, and in each of them we exhibit the inverse operation explicitly:

- **(a)** If $(S) \xrightarrow{\lambda \cdot E_i \to E_i} (S')$ with $\lambda \neq 0$, then $(S') \xrightarrow{\frac{1}{\lambda} \cdot E_i \to E_i} (S)$.
- **(b)** If $(S) \xrightarrow{\lambda \cdot E_i + E_j \to E_j} (S')$ with $i \neq j$, then $(S') \xrightarrow{-\lambda \cdot E_i + E_j \to E_j} (S)$.
- **(c)** If $(S) \xrightarrow{E_i \leftrightarrow E_j} (S')$, then $(S') \xrightarrow{E_j \leftrightarrow E_i} (S)$.

Note that case **(a)** is where the restriction $\lambda \neq 0$ in Definition 6.8 earns its keep: without it, $\frac{1}{\lambda}$ would not exist and the operation would not be reversible. This concludes the proof of Claim 2. $\diamond$

Claim (*Claim 3, One Operation Preserves the Solution Set*): If (S') is obtained from (S) by one elementary row operation, then $L(S) = L(S')$.

Proof of Claim 3. By Claim 1, we have $L(S) \subseteq L(S')$. By Claim 2, (S) is obtained from (S') by one elementary row operation, and hence Claim 1 applies again with the roles of (S) and (S') reversed, giving $L(S') \subseteq L(S)$. The two inclusions together give equality. This concludes the proof of Claim 3. $\diamond$

We are now in a position to prove the theorem. By assumption, there is a finite sequence of elementary row operations

$$(S) = (S_0) \longrightarrow (S_1) \longrightarrow \cdots \longrightarrow (S_k) = (S').$$

Applying Claim 3 to each single step gives

$$L(S) = L(S_0) = L(S_1) = \cdots = L(S_k) = L(S').$$

Q.E.D.

Row-Reduced Matrices

Theorem 6.10 tells us that we may row-reduce as much as we like without changing the answer. It remains to decide **how far** the reduction can be pushed. The next two definitions describe the two successive normal forms we aim for.

Definition 6.11 (*Row-Reduced Matrix*): An $m \times n$ matrix A is called "*row-reduced*" if the following two conditions hold:

- **(a)** The first non-zero entry in each non-zero row of A is 1. This entry is called the "*pivot*" (or "*leading term*") of that row.
- **(b)** Each column of A which contains the leading non-zero entry of some row has all its other entries equal to 0.

Example 6.6 (*Row-reduced and not row-reduced*): Consider the following four matrices over $\mathbb{Q}$:

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & -3 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 0 \end{pmatrix} \qquad D = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 5 \end{pmatrix}$$

- A **is** row-reduced. The pivots sit in columns 4 and 2, and each of those columns is zero apart from its pivot.
- B is **not** row-reduced. Its pivots sit in columns 1, 2 and 3; but column 3 contains the pivot of row 3 and also the entry -1 in row 2, so condition **(b)** fails.
- C is **not** row-reduced. The first non-zero entry of row 1 is 2 and not 1, so condition **(a)** already fails.
- D **is** row-reduced. Its pivots sit in columns 3, 1 and 2, and each of those columns is zero apart from its pivot. Note that the pivots do **not** march to the right as we go down the rows, being row-reduced does not require this.

Theorem 6.12 (*Reduction to a Row-Reduced Matrix*): Every $m \times n$ matrix A is row equivalent to a row-reduced matrix.

Proof:

Let $A = (a_{ij})$ with $1 \le i \le m$, $1 \le j \le n$ and $a_{ij} \in K$. We proceed row by row.

Row #1. If all entries in the first row of A are 0, then condition **(a)** of Definition 6.11 is satisfied for row #1 and there is nothing to do. Otherwise, let k be the smallest index j for which $a_{1j} \neq 0$, that is

$$a_{1j} = 0 \quad \forall\, 1 \le j < k, \qquad a_{1k} \neq 0,$$

so that the first row has the shape $(0, \dots, 0, a_{1k}, *, \dots, *)$. Multiply row $\#1$ by $\frac{1}{a_{1k}}$; its leading entry is now 1, and condition **(a)** holds for row $\#1$. Next, for each $2 \le i \le m$, add $(-a_{ik})$ times row $\#1$ to row i, that is, perform

$$-a_{ik} \cdot E_1 + E_i \to E_i.$$

Schematically:

$$\begin{pmatrix} 0 & \dots & 0 & 1 & \dots \\ & \vdots & & \vdots & \\ a_{i1} & \dots & \dots & a_{ik} & \dots \\ & \vdots & & \vdots & \end{pmatrix} \xrightarrow{-a_{ik}\cdot E_1 + E_i \to E_i} \begin{pmatrix} 0 & \dots & 0 & 1 & \dots \\ & \vdots & & \vdots & \\ a_{i1} & \dots & a_{i,k-1} & 0 & \dots \\ & \vdots & & \vdots & \end{pmatrix}$$

After these operations, column k is zero apart from the 1 in row $\#1$.

Row $\#2$. If all entries in row $\#2$ are 0, we do nothing to this row. If some entry in row $\#2$ is non-zero, we multiply row $\#2$ by a scalar so that its leading term becomes 1. Recall that row $\#1$ has its leading non-zero entry in column k. Since we have just made all other entries of column k vanish, the leading non-zero term of row $\#2$ can **not** lie in column k: it must appear in some other column, say column k' with $k' \neq k$. We now add suitable multiples of row $\#2$ to all the other rows until we obtain a matrix in which all entries of column k' are 0, except the one in row $\#2$. The two possible configurations are

$$\begin{pmatrix} 0 & \dots & 0 & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & 0 & 1 & *\cdots* & 0 & * & \dots & * \\ & & 0 & & & 0 & & & \\ & & \vdots & & & \vdots & & & \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & \dots & 0 & 1 & *\cdots* & 0 & * & \dots & * \\ 0 & \dots & 0 & \dots & 0 & 1 & * & \dots & * \\ & & 0 & & & 0 & & & \\ & & \vdots & & & \vdots & & & \end{pmatrix}$$

according to whether $k' < k$ or $k < k'$.

Important remark 6.1: In the operations just performed using row $\#2$ we did **not** change any of the entries of row $\#1$ lying in columns $1, \dots, k$; in particular, nothing was changed in column k. Also, in case row $\#1$ was zero, the operations using row $\#2$ do not affect row $\#1$ at all. This is what guarantees that the work already done is never undone by the later steps.

We continue now to row $\#3$ and repeat the same procedure, and so on. After a finite number of elementary row operations we arrive at a row-reduced matrix. **Q.E.D.**

Row-Reduced Echelon Matrices

The matrix D of the example above is row-reduced, and yet its pivots appear in the disorderly column order $3, 1, 2$. Sorting the rows costs nothing and yields a genuine normal form.

Definition 6.13 (*Row-Reduced Echelon Matrix*): An $m \times n$ matrix A is called "*row-reduced echelon*" if the following hold:

- **(a)** A is row-reduced.
- **(b)** Every row of A which has all its entries 0 appears below every row which has a non-zero entry.
- **(c)** If rows $1, \dots, r$ are the non-zero rows of A, and if the leading non-zero entry of row i occurs in column k_i for $i = 1, \dots, r$, then

$$k_1 < k_2 < \cdots < k_r.$$

Condition **(c)** is what produces the characteristic staircase shape shown in Figure 6.2.

$$\begin{array}{cc}
 & \begin{matrix} & k_1 & & k_2 & & k_3 & \end{matrix} \\
\begin{matrix} 1 \\ \\ r \\ r+1 \\ m \end{matrix} & \begin{bmatrix} 0 & 1 & * & 0 & * & 0 & * \\ 0 & 0 & 0 & 1 & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}
\end{array}$$

Figure 6.2: A row-reduced echelon matrix. The circled pivots are 1; everything below the green staircase is 0, and each pivot column is 0 apart from its pivot. Because $k_1 < k_2 < k_3$, the pivots move strictly to the right as we descend, and the $m-r$ zero rows are pushed to the bottom. The columns carrying a $*$ are the ones without a pivot; they will turn into the free variables when the matrix is read back as a system.

Theorem 6.14 (*Reduction to Row-Reduced Echelon Form*): Every $m \times n$ matrix A is row equivalent to a row-reduced echelon matrix.

Proof:

By Theorem 6.12 we already know that A is row equivalent to a row-reduced matrix A'. Performing a finite number of row interchanges on A' (an operation of Type 3, which preserves row equivalence), we may list the non-zero rows in increasing order of their pivot columns and move all zero rows to the bottom. The result satisfies conditions **(b)** and **(c)** of Definition 6.13, and interchanging rows does not disturb condition **(a)**. Hence we arrive at a row-reduced echelon matrix. **Q.E.D.**

Remark 6.1: Why is this useful for solving systems of equations? Let $Ax = b$ be a system of linear equations and consider its augmented matrix $(A \mid b)$. We may perform on $(A \mid b)$ the same elementary operations that we perform on A, and so obtain a row-equivalent matrix $(A' \mid b')$ in which A' is a row-reduced echelon matrix. By Theorem 6.10, the systems $Ax = b$ and $A'x = b'$ have the same solutions, but $A'x = b'$ is far easier to solve, since each pivot expresses one variable directly in terms of the pivot-free ones.

Worked Examples

Example 6.7 (*Solving a system by full reduction*): Consider the system over $\mathbb{R}$ whose augmented matrix is

$$\left(\begin{array}{cccc|c} 0 & -9 & 3 & 4 & 9 \\ 1 & 4 & 0 & -1 & 5 \\ 2 & 6 & -1 & 5 & -5 \end{array}\right).$$

We first normalise the leading entry of the first row, and then clear its column:

$$\xrightarrow{-\frac{1}{9}\cdot E_1 \to E_1} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{9} & -1 \\ 1 & 4 & 0 & -1 & 5 \\ 2 & 6 & -1 & 5 & -5 \end{array}\right) \xrightarrow[-6\cdot E_1 + E_3 \to E_3]{-4\cdot E_1 + E_2 \to E_2} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{9} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{9} & 9 \\ 2 & 0 & 1 & \frac{23}{3} & 1 \end{array}\right)$$

Next we clear the first column below row 2 and normalise the third row:

$$\xrightarrow{-2\cdot E_2 + E_3 \to E_3} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{9} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{9} & 9 \\ 0 & 0 & -\frac{5}{3} & \frac{55}{9} & -17 \end{array}\right) \xrightarrow{-\frac{3}{5}\cdot E_3 \to E_3} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{9} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{9} & 9 \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right)$$

Now we clear the third column upwards, and finally sort the rows:

$$\xrightarrow[\frac{1}{3}\cdot E_3+E_1\to E_1]{-\frac{4}{3}\cdot E_3+E_2\to E_2} \left(\begin{array}{cccc|c} 0 & 1 & 0 & -\frac{5}{3} & \frac{12}{5} \\ 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right) \xrightarrow{E_1\leftrightarrow E_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 1 & 0 & -\frac{5}{3} & \frac{12}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right)$$

The last matrix is in row-reduced echelon form. Reading it back as a system gives

$$\begin{cases} x_1 \qquad\qquad + \frac{17}{3}x_4 = -\frac{23}{5} \\ \qquad x_2 \qquad - \frac{5}{3}x_4 = \frac{12}{5} \\ \qquad\qquad x_3 - \frac{11}{3}x_4 = \frac{51}{5} \end{cases}$$

Column 4 carries no pivot, so x_4 is free. Take $x_4 := a \in \mathbb{R}$ arbitrary. Then

$$x_3 = \frac{11}{3}a + \frac{51}{5}, \qquad x_2 = \frac{5}{3}a + \frac{12}{5}, \qquad x_1 = -\frac{17}{3}a - \frac{23}{5},$$

and these are **all** the solutions of the original system, by Theorem 6.10.

Example 6.8 (*Conditions for consistency*): Let us now see how the solutions behave for a general choice of the constants $b_1, b_2, b_3 \in \mathbb{R}$ in the system

$$\begin{pmatrix} 1 & -2 & 1 \\ 2 & 1 & 1 \\ 0 & 5 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

We construct the augmented matrix and reduce it:

$$\left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 2 & 1 & 1 & b_2 \\ 0 & 5 & -1 & b_3 \end{array}\right) \xrightarrow{-2\cdot E_1+E_2\to E_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 0 & 5 & -1 & b_2-2b_1 \\ 0 & 5 & -1 & b_3 \end{array}\right)$$

$$\xrightarrow{-1\cdot E_2+E_3\to E_3} \left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 0 & 5 & -1 & b_2-2b_1 \\ 0 & 0 & 0 & b_3-b_2+2b_1 \end{array}\right) \xrightarrow{\frac{1}{5}\cdot E_2\to E_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 0 & 1 & -\frac{1}{5} & \frac{1}{5}(b_2-2b_1) \\ 0 & 0 & 0 & b_3-b_2+2b_1 \end{array}\right)$$

$$\xrightarrow{2\cdot E_2+E_1\to E_1} \left(\begin{array}{ccc|c} 1 & 0 & \frac{3}{5} & \frac{1}{5}(b_1+2b_2) \\ 0 & 1 & -\frac{1}{5} & \frac{1}{5}(b_2-2b_1) \\ 0 & 0 & 0 & b_3-b_2+2b_1 \end{array}\right)$$

The last matrix is in row-reduced echelon form, and the equivalent system reads

$$\begin{cases} x_1 \qquad + \frac{3}{5}x_3 = \frac{1}{5}(b_1+2b_2) \\ \qquad x_2 - \frac{1}{5}x_3 = \frac{1}{5}(b_2-2b_1) \\ \qquad\qquad\quad 0 = b_3-b_2+2b_1 \end{cases}$$

Two cases arise from the last row.

- If $b_3 - b_2 + 2b_1 \neq 0$, the third equation reads $0 = \text{non-zero}$, and the system has **no** solutions.
- If $b_3 - b_2 + 2b_1 = 0$, the third equation is vacuous and the system is equivalent to

$$\begin{cases} x_1 + \frac{3}{5}x_3 = \frac{1}{5}(b_1+2b_2) \\ x_2 - \frac{1}{5}x_3 = \frac{1}{5}(b_2-2b_1) \end{cases}$$

  Column 3 carries no pivot, so x_3 is free. Take $x_3 := c \in \mathbb{R}$ arbitrary; then

$$x_2 = \frac{1}{5}c + \frac{1}{5}(b_2-2b_1), \qquad x_1 = -\frac{3}{5}c + \frac{1}{5}(b_1+2b_2).$$

So the constants b_1, b_2, b_3 cannot be chosen freely: the system is consistent if and only if $b_3 - b_2 + 2b_1 = 0$, and when it is, it has infinitely many solutions.

Example 6.9 (*A system with infinitely many solutions*): As a final illustration, consider

$$\begin{cases} x_1 - 2x_2 + \ \ x_3 + \ \ x_4 = 1 \\ 2x_1 - 4x_2 + \ \ x_3 + 3x_4 = 2 \\ x_1 - 2x_2 + 2x_3 - \ \ x_4 = 1 \end{cases}$$

We write the augmented matrix and apply EROs:

$$\left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 2 & -4 & 1 & 3 & 2 \\ 1 & -2 & 2 & -1 & 1 \end{array}\right) \xrightarrow[-1\cdot E_1+E_3\to E_3]{-2\cdot E_1+E_2\to E_2} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \end{array}\right)$$

$$\xrightarrow{1\cdot E_2+E_3\to E_3} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right)$$

From the third row, $-x_4 = 0$, and therefore, $x_4 = 0$. From the second row, $-x_3 + x_4 = 0$, and therefore, $x_3 = 0$. Substituting into the first row gives $x_1 - 2x_2 = 1$, that is, $x_1 = 1 + 2x_2$.

Here x_2 is the free variable. Let $x_2 := \alpha \in \mathbb{R}$. The general solution is

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1+2\alpha \\ \alpha \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Observe the shape of the answer: one particular solution plus an arbitrary multiple of a fixed vector. That the solution set of a system should always look like this is precisely the kind of structure our **(c)** question asked about, and we will return to it once we have the language of vector spaces.

AI-Note: The lecture notes for this chapter use F for the base field and R_i for the rows; this transcript writes K and E_i throughout, in line with the conventions used in the rest of the document. The two matrices in the schematic step of the proof of Theorem 6.12, and the final example above, are expanded slightly beyond what the notes show; the geometric picture in Figure 6.1 and the staircase in Figure 6.2 were added here. Everything else follows the notes directly.

Vector Spaces (Chapter 7)

Definition and Axioms

Definition 7.1 (*Vector Space*): A "*vector space*" over a field K is a set V, endowed with two operations:

$$+ : V \times V \longrightarrow V, \quad (v_1, v_2) \mapsto v_1 + v_2 \qquad \text{(Addition of vectors)}$$
$$\cdot : K \times V \longrightarrow V, \quad (\alpha, v) \mapsto \alpha \cdot v \qquad \text{(Scalar multiplication)}$$

such that the following conditions (axioms) hold:

(V1) $\forall v_1, v_2, v_3 \in V\colon\ v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$.
(V2) $\exists$ an element $0 \in V$ such that $0 + v = v + 0 = v$ for all $v \in V$. (Sometimes denoted 0_V).
(V3) $\forall v \in V, \exists v' \in V$ such that $v + v' = 0$.
(V4) $\forall v_1, v_2 \in V\colon\ v_1 + v_2 = v_2 + v_1$.
(V5) $\forall a, b \in K, \forall v \in V\colon\ a \cdot (b \cdot v) = (a \cdot b) \cdot v$.
(V6) $\forall v \in V\colon\ 1 \cdot v = v$.
(V7) $\forall \alpha \in K, \forall v_1, v_2 \in V\colon\ \alpha \cdot (v_1 + v_2) = \alpha \cdot v_1 + \alpha \cdot v_2$.
(V8) $\forall \alpha_1, \alpha_2 \in K, \forall v \in V\colon\ (\alpha_1 + \alpha_2) \cdot v = \alpha_1 \cdot v + \alpha_2 \cdot v$.

The elements of K are called "*scalars*" and the elements of V are called "*vectors*".

Lemma 7.2 (*Elementary Consequences of the Axioms*): Let V be a vector space over a field K. Then the following properties hold:

(a) The 0 vector from (V2) is unique. To avoid confusion (e.g., with $0 \in K$), we will sometimes denote it by 0_V.
(b) The element v' (corresponding to v) from (V3) is unique. We will denote it by $-v$. We will also write $v - w := v + (-w)$ for all $v, w \in V$.
(c) For all $v \in V$, we have $0 \cdot v = 0_V$ (i.e., $0_K \cdot v = 0_V$).
(d) For all $\alpha \in K$, we have $\alpha \cdot 0_V = 0_V$.
(e) $(-1) \cdot v = -v$ for all $v \in V$.
(f) $-(-v) = v$ for all $v \in V$.
(g) If $a \cdot v = 0_V$ for some $a \in K$ and $v \in V$, then either $a = 0$ or $v = 0_V$.
(h) Associativity for $+$ holds also for sums of n elements. That is, $v_1 + \cdots + v_n$ makes sense no matter how we place the brackets:

$$v_1 + \big(v_2 + (\cdots + (v_{n-1} + v_n) \ldots)\big), \qquad (v_1 + v_2) + (\ldots), \qquad \text{and so on,}$$

all give the same element of V.

Proof:

(a) Assume there are two such zero elements, 0_1 and 0_2. Then, by applying (V2) to both, we get: $0_1 = 0_1 + 0_2 = 0_2 + 0_1 = 0_2$. Thus, $0_1 = 0_2$.
(b) Assume there are two inverses v_1' and v_2' for an element v. Then we can write:

$$v_1' = v_1' + 0_V = v_1' + (v + v_2') = (v_1' + v) + v_2' = 0_V + v_2' = v_2'$$

Thus, the inverse is unique.
(c) Consider $0 \cdot v = (0+0) \cdot v = 0 \cdot v + 0 \cdot v$, where the second equality is (V8). By adding $-(0 \cdot v)$ to both sides, we yield $0_V = 0 \cdot v$.
(d) Similarly, $\alpha \cdot 0_V = \alpha \cdot (0_V + 0_V) = \alpha \cdot 0_V + \alpha \cdot 0_V$ by (V7). Adding $-(\alpha \cdot 0_V)$ yields $0_V = \alpha \cdot 0_V$.
(e) Observe that $v + (-1) \cdot v = 1 \cdot v + (-1) \cdot v = (1 - 1) \cdot v = 0 \cdot v = 0_V$. Since the inverse is unique by part **(b)**, it must be that $(-1) \cdot v = -v$.

(f) By definition, $-(-v)$ is the unique element that added to $-v$ gives 0_V. But v itself satisfies $(-v)+v = v+(-v) = 0_V$ by (V4) and (V3), so uniqueness in part **(b)** forces $-(-v) = v$.

(g) Let $a \in K$ and $v \in V$ and assume that $a \cdot v = 0_V$. If $a = 0$ there is nothing to prove, so suppose $a \neq 0$. Because K is a field, there is an element $a^{-1} \in K$ with $a^{-1} \cdot a = 1$. Therefore,

$$v \overset{\text{(V6)}}{=} 1 \cdot v = (a^{-1} \cdot a) \cdot v \overset{\text{(V5)}}{=} a^{-1} \cdot (a \cdot v) = a^{-1} \cdot 0_V \overset{\textbf{(d)}}{=} 0_V.$$

So either $a = 0$ or $v = 0_V$, as claimed.

(h) This follows from (V1) by induction on n, exactly as for the associativity of addition in a field. From now on we will therefore write $v_1 + \cdots + v_n$ without brackets.

Q.E.D.

The Coordinate Space K^n

Example 7.1 (*The Coordinate Space K^n*): Let K be a field and $n \in \mathbb{N}$. Put

$$K^n := \{(a_1, \ldots, a_n) \mid a_i \in K,\ 1 \le i \le n\}.$$

The set K^n becomes a vector space over K if we endow it with the following operations. Let $v = (a_1, \ldots, a_n)$ and $w = (b_1, \ldots, b_n) \in K^n$. We define addition in K^n component-wise, using the addition of the field K in each slot:

$$v + w = (a_1, \ldots, a_n) + (b_1, \ldots, b_n) := (a_1 + b_1, \ldots, a_n + b_n)$$

For $\alpha \in K$, we define scalar multiplication as:

$$\alpha \cdot v = \alpha \cdot (a_1, \ldots, a_n) := (\alpha \cdot a_1, \ldots, \alpha \cdot a_n)$$

The zero vector is explicitly defined as $0 := (0, \ldots, 0) \in K^n$. With these operations K^n becomes a vector space over K; the proof rests entirely on the fact that K itself satisfies the field axioms. We call K^n also the "*n-dimensional coordinate space*" over K.

Exercise 7.1: Check the details: verify each of the eight axioms (V1)–(V8) for K^n.

Important remark 7.1: Sometimes we write the elements of K^n as row vectors and sometimes as column vectors, whichever is more convenient at the time. The two conventions describe the same vector space, and we will switch between them without further comment, but whenever a matrix product $A \cdot x$ is involved, the elements of K^n are always to be read as column vectors, as in Definition 6.7.

Linear Subspaces

Definition 7.3 (*Linear Subspace*): Let V be a vector space over K. A subset $U \subseteq V$ is called a "*linear subspace*" of V (or just a "*subspace*" of V) if the following holds:

- **(LSS1)** $U \neq \emptyset$.
- **(LSS2)** For all $u_1, u_2 \in U$, we have $u_1 + u_2 \in U$ (closed under addition).
- **(LSS3)** For all $\alpha \in K$ and all $u \in U$, we have $\alpha \cdot u \in U$ (closed under scalar multiplication).

Lemma 7.4 (*A Subspace is a Vector Space*): Let V be a vector space over K and let $U \subseteq V$ be a subspace. Then U is a vector space in its own right, when endowed with the operations inherited from V.

Proof:

Conditions (LSS2) and (LSS3) say precisely that the operations $+$ and $\cdot$ of V restrict to operations

$$+ \colon U \times U \longrightarrow U, \qquad \cdot \colon K \times U \longrightarrow U,$$

so that U carries the two operations required by Definition 7.1. That these operations satisfy the eight axioms of a vector space then follows by direct verification, since each axiom is an identity that already holds for all elements of V and therefore in particular for all elements of U.

The one axiom that deserves a word is (V2), because it asserts the **existence** of an element of U rather than an identity. Pick any element $u \in U$; this is possible because $U \neq \emptyset$ by (LSS1). By (LSS3),

$$0_V = 0_K \cdot u \in U,$$

where the first equality is Lemma 7.2 **(c)**. Now $0_V + w = w$ holds for every $w \in U$, because it already holds for every $w \in V$ by (V2) for V. Hence 0_V serves as the zero element of U.

Q.E.D.

Exercise 7.2: Carry out the direct verification of the remaining seven axioms for U.

In practice one rarely checks (LSS1)–(LSS3) separately. The following criterion compresses them into two conditions, one of which is a single membership test.

Lemma 7.5 (*Criterion for a Linear Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subset. Then U is a linear subspace of V if and only if the following holds:

- **(a)** $0_V \in U$.
- **(b)** For all $\alpha_1, \alpha_2 \in K$ and all $u_1, u_2 \in U$, we have $\alpha_1 u_1 + \alpha_2 u_2 \in U$.

Proof:

Direction 1 ($\Longrightarrow$): Suppose $U \subseteq V$ is a subspace. Then $0_V \in U$ was established in the proof of Lemma 7.4, which gives **(a)**. For **(b)**, given $\alpha_1, \alpha_2 \in K$ and $u_1, u_2 \in U$, condition (LSS3) gives $\alpha_1 u_1 \in U$ and $\alpha_2 u_2 \in U$, and then (LSS2) gives $\alpha_1 u_1 + \alpha_2 u_2 \in U$.

Direction 2 ($\Longleftarrow$): Suppose **(a)** and **(b)** hold. Then $U \neq \emptyset$ by **(a)**, which is (LSS1). Let $u_1, u_2 \in U$. By **(b)** applied with $\alpha_1 = \alpha_2 = 1$,

$$\underbrace{1 \cdot u_1 + 1 \cdot u_2}_{= u_1 + u_2} \in U,$$

which is (LSS2). Finally, let $\alpha \in K$ and $u \in U$. By **(b)** applied with $\alpha_1 = \alpha$, $\alpha_2 = 0$ and $u_1 = u_2 = u$,

$$\underbrace{\alpha \cdot u + 0 \cdot u}_{= \alpha \cdot u} \in U,$$

which is (LSS3). Hence U is a linear subspace.

Q.E.D.

Examples of Subspaces

Example 7.2 (*The Trivial Subspace*): Let V be a vector space over K. Then $\{0_V\} \subseteq V$ is a linear subspace: it is non-empty, and $0_V + 0_V = 0_V$ as well as $\alpha \cdot 0_V = 0_V$ for every $\alpha \in K$, by Lemma 7.2 **(d)**. At the other extreme, $V \subseteq V$ is of course a subspace of itself.

Example 7.3 (*A Plane Through the Origin, and One That Is Not*): Fix $b \in K$ and consider the subset

$$U_b := \{(x_1, x_2, x_3) \in K^3 \mid x_1 + x_2 + x_3 = b\} \subseteq K^3.$$

Claim (*U_b is a subspace exactly for $b = 0$*): $U_b \subseteq K^3$ is a linear subspace if and only if $b = 0$.

Proof of the direction $\Longrightarrow$. Suppose U_b is a subspace of K^3. Let (x_1, x_2, x_3) and $(y_1, y_2, y_3) \in U_b$. Then by definition,

$$x_1 + x_2 + x_3 = b, \qquad y_1 + y_2 + y_3 = b. \tag{7.1}$$

Since U_b is a subspace, it is closed under addition, and therefore, $(x_1+y_1, x_2+y_2, x_3+y_3) \in U_b$. By the definition of U_b, this implies

$$(x_1+y_1)+(x_2+y_2)+(x_3+y_3) = b.$$

But rearranging the terms and using (7.1), we also get

$$(x_1+y_1)+(x_2+y_2)+(x_3+y_3) = (x_1+x_2+x_3)+(y_1+y_2+y_3) = b+b.$$

Therefore, $b+b=b$, which implies $b=0$. $\diamond$

Exercise 7.3: Prove the direction $\Longleftarrow$ of the claim on Page 50 that U_b is a subspace exactly for $b=0$: show that if $b=0$, then U_0 satisfies (LSS1), (LSS2) and (LSS3), and is therefore a linear subspace of K^3.

The Solution Set of a Linear System

The previous example is a special case of something much more important. It answers the third of the questions we set ourselves in the previous chapter, in the list of goals on Page 41: the solution set of a system of linear equations does carry a structure of its own, provided the system is homogeneous.

Let $A \in \mathcal{M}_{m\times n}(K)$ and fix $b \in K^m$, viewed as a column vector. Consider

$$L := \{x \in K^n \mid A\cdot x = b\} \subseteq K^n,$$

where the elements of K^n are viewed as column vectors as well, following the remark on row and column notation on Page 49. This is exactly the set $L(S)$ of solutions of the system $(S): Ax=b$ from (6.3).

Lemma 7.6 (*Linearity of Matrix-Vector Multiplication*): Let $A \in \mathcal{M}_{m\times n}(K)$. For all $u,v \in K^n$ and all $a \in K$ we have:

(a) $A\cdot(u+v) = A\cdot u + A\cdot v$.

(b) $A\cdot(a\cdot u) = a\cdot(A\cdot u)$.

Proof:

Write $A=(a_{ij})$ with $1\le i\le m$, $1\le j\le n$, and let

$$u = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \qquad v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$

We prove **(a)** by comparing the i-th entries of both sides. By Definition 6.7, entry number i of $A\cdot u$ is $a_{i1}u_1+\cdots+a_{in}u_n$, and entry number i of $A\cdot v$ is $a_{i1}v_1+\cdots+a_{in}v_n$. Adding these in K and regrouping by distributivity gives

$$\begin{aligned}\big(A\cdot u + A\cdot v\big)_i &= \big(a_{i1}u_1+\cdots+a_{in}u_n\big)+\big(a_{i1}v_1+\cdots+a_{in}v_n\big)\\ &= a_{i1}(u_1+v_1)+a_{i2}(u_2+v_2)+\cdots+a_{in}(u_n+v_n)\\ &= \big(A\cdot(u+v)\big)_i,\end{aligned}$$

the last equality again by Definition 6.7, since the j-th entry of $u+v$ is u_j+v_j. As this holds for every $1\le i\le m$, the two column vectors are equal. **Q.E.D.**

Exercise 7.4: Prove statement **(b)** of Lemma 7.6.

Theorem 7.7 (*When the Solution Set is a Subspace*): $L \subseteq K^n$ is a linear subspace if and only if $b = 0 \in K^m$.

Proof:

Direction 1 ($\Longleftarrow$): We assume $b = 0$ and have to show that $L \subseteq K^n$ is a subspace.

First, $x = 0 \in L$, since indeed $A \cdot 0 = 0$. Next, if $x = (x_1, \dots, x_n) \in L$ and $y = (y_1, \dots, y_n) \in L$, then by Lemma 7.6 **(a)**,

$$A \cdot (x + y) = A \cdot x + A \cdot y = 0 + 0 = 0,$$

and therefore, $x + y \in L$. This shows that $x, y \in L$ implies $x + y \in L$. Finally, let $x \in L$ and $a \in K$. By Lemma 7.6 **(b)**,

$$A \cdot (a \cdot x) = a \cdot (A \cdot x) = a \cdot 0 = 0,$$

so $a \cdot x \in L$. This shows L is a subspace.

Direction 2 ($\Longrightarrow$): Suppose $L \subseteq K^n$ is a subspace. We need to show that $b = 0$. Since L is a subspace, $0 \in L$ by Lemma 7.5 **(a)**. By the definition of L, this means $A \cdot 0 = b$. But $A \cdot 0 = 0$, and therefore, $b = 0$. **Q.E.D.**

Important remark 7.2: So the solution set of a **homogeneous** system is a linear subspace of K^n, while the solution set of an inhomogeneous one is never a subspace; it does not even contain 0. It is not, however, shapeless: recall the last example of the previous chapter, whose general solution had the form *"one particular solution plus an arbitrary multiple of a fixed vector"*. We will make that observation precise once we have the notion of dimension.

More Examples of Vector Spaces

Spaces of Sequences K^∞

Example 7.4 (*Spaces of Sequences*): Let K be a field. Define

$$K^\infty := \{(a_0, a_1, a_2, \dots, a_k, \dots) \mid a_i \in K \ \forall i\},$$

the set of all infinite sequences with entries in K. We define $+$ and $\cdot$ on K^∞ entry-wise:

$$\begin{aligned}(a_0, a_1, \dots, a_k, \dots) + (b_0, b_1, \dots, b_k, \dots) &:= (a_0 + b_0, a_1 + b_1, \dots, a_k + b_k, \dots),\\ a \cdot (a_0, a_1, \dots, a_k, \dots) &:= (a \cdot a_0, a \cdot a_1, \dots, a \cdot a_k, \dots) \quad \text{for } a \in K.\end{aligned}$$

Exercise 7.5: With these operations, K^∞ is a vector space over K.

Example 7.5 (*Fibonacci Sequences as a Subspace*): Recall the Fibonacci sequences from Proposition 1.3:

$$\mathrm{Fib} = \{(a_0, a_1, \dots, a_k, \dots) \in \mathbb{R}^\infty \mid a_k = a_{k-1} + a_{k-2} \ \forall k \geq 2\}.$$

Then $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ is a linear subspace, which is precisely what we verified by hand in the very first chapter, before we had the word for it.

More generally, if we fix $\alpha, \beta \in \mathbb{R}$, then

$$U_{\alpha,\beta} := \{(a_0, a_1, \dots, a_k, \dots) \in \mathbb{R}^\infty \mid a_k = \alpha a_{k-1} + \beta a_{k-2} \ \forall k \geq 2\}$$

is also a linear subspace of $\mathbb{R}^\infty$, with $\mathrm{Fib} = U_{1,1}$.

Spaces of Functions K^S

Example 7.6 (*Function Spaces*): Fix a field K and a non-empty set S. We define the function space

$$K^S := \{f \colon S \to K \mid f \text{ is a function}\}.$$

We define addition $+$ and scalar multiplication $\cdot$ on K^S as follows. Let $f, g \in K^S$ and $a \in K$. We define $(f + g) \colon S \to K$ point-wise by $x \mapsto f(x) + g(x)$, where the addition on the right is that of K, and we define $(a \cdot f) \colon S \to K$ point-wise by $x \mapsto a \cdot f(x)$.

Exercise 7.6: K^S is a vector space over K.

Example 7.7 (*Continuous Functions*): Take $K = \mathbb{R}$ and let $S := [0,1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$. Define

$$C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0,1]\} \subseteq \mathbb{R}^S.$$

Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.

Exercise 7.7: Prove that $C(S)$ is a linear subspace of $\mathbb{R}^S$: check that the constant function 0 is continuous, and that a sum of two continuous functions, as well as a scalar multiple of a continuous function, is again continuous on $[0,1]$.

Polynomials $K[x]$

Example 7.8 (*Polynomial Spaces*): Let K be a field and let x be a **formal** variable; we may think of x as a *"free element of K"*. A "*polynomial*" over K (or polynomial with coefficients in K) is an expression of the form

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \qquad n \in \mathbb{N},$$

where $a_i \in K$ for all $0 \leq i \leq n$. The a_i are called the "*coefficients*" of f. The "*zero polynomial*" is the one with $a_0 = a_1 = \cdots = 0$. Two polynomials are equal if and only if all their corresponding coefficients are equal.

To add two polynomials

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n, \qquad g(x) = b_0 + b_1 x + \cdots + b_m x^m,$$

we first pad the shorter one with zeros: if $n < m$ we set $a_{n+1} = \cdots = a_m = 0$, and if $n > m$ we set $b_{m+1} = \cdots = b_n = 0$. Putting $r := \max\{n, m\}$, both are then written with the same number of terms and we may define

$$f(x) + g(x) := (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \cdots + (a_r + b_r)x^r.$$

If $c \in K$, we define

$$c \cdot f(x) := c \cdot a_0 + c \cdot a_1 x + c \cdot a_2 x^2 + \cdots + c \cdot a_n x^n.$$

Notation 7.1: We write $K[x]$ for the set of all polynomials over K.

Exercise 7.8: With the operations above, $K[x]$ is a vector space over K.

If we view polynomials as functions $K \to K$, then the polynomials over K give us a linear subspace of K^K. This second point of view is however genuinely coarser than the first, and the following warning explains why.

Important remark 7.3: For some fields (e.g., $K = \mathbb{F}_2$, or $\mathbb{F}_p$ with p prime) two **different** polynomials may give the same function. For example, over $K = \mathbb{F}_2$ the polynomials $f(x) = x$ and $g(x) = x^2$ produce the same output value for every input in $\mathbb{F}_2$. But **as polynomials** $f(x)$ and $g(x)$ are strictly different objects, since their coefficients differ.

The Degree of a Polynomial

Let $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$ be a polynomial with $a_n \neq 0$. We say that the "*degree*" of f is n, and we write $\deg(f) := n \in \mathbb{Z}_{\geq 0}$. If $a_0 = a_1 = \cdots = a_n = 0$, meaning f is the zero polynomial, we set $\deg(f) := -\infty$. Consequently, $\deg(f) = -\infty$ if and only if $f = 0$.

Remark 7.1: Every polynomial has a well-defined degree: if $f \neq 0$ and $f(x) = a_0 + a_1 x + \cdots + a_r x^r$, then

$$\deg(f) := \max\{k \in \mathbb{Z}_{\geq 0} \mid a_k \neq 0\},$$

and this maximum is taken over a non-empty finite set, hence exists.

Example 7.9 (*Polynomials of Bounded Degree*): Fix $d \in \mathbb{Z}_{\geq 0} \cup \{-\infty\}$ and define

$$K[x]_d := \{f \in K[x] \mid \deg(f) \leq d\},$$

where we agree that $-\infty \leq k$ for all $k \in \mathbb{Z}$.

Claim (*$K[x]_d$ is a subspace*): $K[x]_d \subseteq K[x]$ is a linear subspace of $K[x]$.

Later, we will see that $K[x]_d$ is a subspace of dimension $d+1$.

Question 7.1:

(a) Is the set $\{f \in K[x] \mid \deg(f) = d\}$ also a subspace of $K[x]$?

(b) Consider the subset

$$\left\{ p \in K[x]_5 \;\middle|\; p(x) = a_0 + a_1 x + \cdots + a_5 x^5,\ \sum_{i=0}^{5} a_i = 0 \right\} \subseteq K[x]_5,$$

that is, all polynomials of degree ≤ 5 for which the sum of the coefficients is 0. Is this subset a linear subspace of $K[x]_5$?

Spaces of Matrices

Example 7.10 (*Matrix Spaces*): Consider the set $\mathcal{M}_{m\times n}(K)$ of all $m \times n$ matrices with entries in K. We define addition of matrices component-wise: if $A = (a_{ij})$ and $B = (b_{ij})$, then $A + B := (a_{ij} + b_{ij})$. Similarly, for $\alpha \in K$ we define scalar multiplication component-wise by $\alpha \cdot A := (\alpha\, a_{ij})$.

Claim (*$\mathcal{M}_{m\times n}(K)$ is a vector space*): With these operations, $\mathcal{M}_{m\times n}(K)$ is a vector space over K. Its zero vector is the $m \times n$ matrix all of whose entries are 0.

AI-Note: The lecture notes index sequences from 1, writing $K^\infty = \{(a_1, a_2, \dots)\}$ and Fib with the recursion starting at $k \geq 3$. This transcript indexes from 0 instead, so that Fib here is literally the same object as the one built in Proposition 1.3; nothing else changes. The notes also use W for a subspace, where this transcript writes U, matching the later chapters. The proofs of Lemma 7.2 **(e)**, **(f)** and **(h)** are left as exercises in the notes and are supplied here; the exercises marked as such above are the ones the notes leave open.

Span and Linear Combinations (Chapter 8)

Preparation

Let V be a vector space over a field K. Let $S \subseteq V$ be a subset (which is not necessarily a linear subspace). We aim to answer two fundamental and closely related questions:

(a) What is the **smallest** linear subspace of V that contains S?

(b) What structure do we obtain if we consider the subset of V formed by taking all possible linear combinations of elements from S?

To formalize the concept of a *"smallest"* subspace, we must first prove that the intersection of any collection of subspaces remains a subspace.

Lemma 8.1 (*Intersections of Subspaces*): Let V be a vector space over K. Let $\{W_i\}_{i \in I}$ be a collection of linear subspaces $W_i \subseteq V$ of V, indexed by an arbitrary set I of indices. Then their intersection

$$W := \bigcap_{i \in I} W_i$$

is a linear subspace of V.

Proof:

First, since every W_i is a subspace, the zero vector 0_V is contained in every W_i by Lemma 7.5 **(a)**. Therefore, $0_V \in \bigcap_{i \in I} W_i = W$.

Next, let $a_1, a_2 \in K$ and $w_1, w_2 \in W$. We must show that

$$a_1 w_1 + a_2 w_2 \in W. \tag{8.1}$$

Choose an arbitrary index $j \in I$. Since $w_1, w_2 \in \bigcap_{i \in I} W_i$, we in particular have $w_1, w_2 \in W_j$. Because $W_j \subseteq V$ is itself a linear subspace, Lemma 7.5 **(b)** applies to it and gives

$$a_1 w_1 + a_2 w_2 \in W_j.$$

We have proved that $a_1 w_1 + a_2 w_2 \in W_j$ holds for **every** $j \in I$, and therefore,

$$a_1 w_1 + a_2 w_2 \in \bigcap_{i \in I} W_i = W,$$

which is what we wanted to prove in (8.1). By Lemma 7.5, W is a linear subspace. **Q.E.D.**

The Span

Definition 8.2 (*Span*): Let $S \subseteq V$ be a subset. Let

$$\mathcal{N} := \{W \mid W \subseteq V \text{ is a linear subspace, and } S \subseteq W\}$$

be the collection of all linear subspaces of V that contain S. We define the "*span*" of S (in V) as

$$\mathrm{Sp}(S) := \bigcap_{W \in \mathcal{N}} W.$$

Definition 8.3 (*Linear Combination*): Let $n \in \mathbb{N}$, let $a_1, \dots, a_n \in K$ be scalars and let $v_1, \dots, v_n \in V$ be vectors. A "*linear combination*" of the vectors $v_1, \dots, v_n$ with coefficients $a_1, \dots, a_n$ is the vector

$$a_1 v_1 + \dots + a_n v_n \in V \quad \left(\text{or equivalently, } \sum_{i=1}^{n} a_i v_i\right).$$

Important remark 8.1: In this course, we consider **only finitely many elements** in the sums of linear combinations. Infinite sums require topological notions of limits and convergence, which fall outside the scope of general vector spaces.

The span deserves its name: it is a subspace, and it is the smallest one containing S.

Lemma 8.4 (*The Span is the Smallest Subspace Containing S*): Let $S \subseteq V$ be a subset. Then:

(a) $\mathrm{Sp}(S) \subseteq V$ is a linear subspace.

(b) If $W \subseteq V$ is a linear subspace with $S \subseteq W$, then $\mathrm{Sp}(S) \subseteq W$.

In other words, $\mathrm{Sp}(S)$ is a linear subspace of V, and moreover, among those subspaces that contain S, it is the smallest one.

Proof:

Statement **(a)** is immediate from Lemma 8.1: by Definition 8.2, $\mathrm{Sp}(S)$ is an intersection of a collection of linear subspaces of V, and any such intersection is again a linear subspace. Note that the collection $\mathcal{N}$ is never empty, since V itself is a subspace of V containing S.

For **(b)**, let $W \subseteq V$ be a linear subspace with $S \subseteq W$. Then W is one of the members of the collection $\mathcal{N}$ over which the intersection in Definition 8.2 is taken. An intersection is contained in each of the sets being intersected, and therefore,

$$\mathrm{Sp}(S) = \bigcap_{W' \in \mathcal{N}} W' \subseteq W.$$

Q.E.D.

The next lemma answers the second of our two opening questions, and shows that the two answers agree.

Lemma 8.5 (*Span as the Set of Linear Combinations*): Let $\emptyset \neq S \subseteq V$. Then $\mathrm{Sp}(S)$ is exactly the set of all possible linear combinations of vectors from S:

$$\begin{aligned}\mathrm{Sp}(S) &= \{a_1 v_1 + \cdots + a_n v_n \mid n \in \mathbb{N},\ a_i \in K,\ v_i \in S \text{ for all } 1 \leq i \leq n\} \\ &= \text{the subset of } V \text{ obtained by taking all possible linear combinations of vectors from } S.\end{aligned}$$

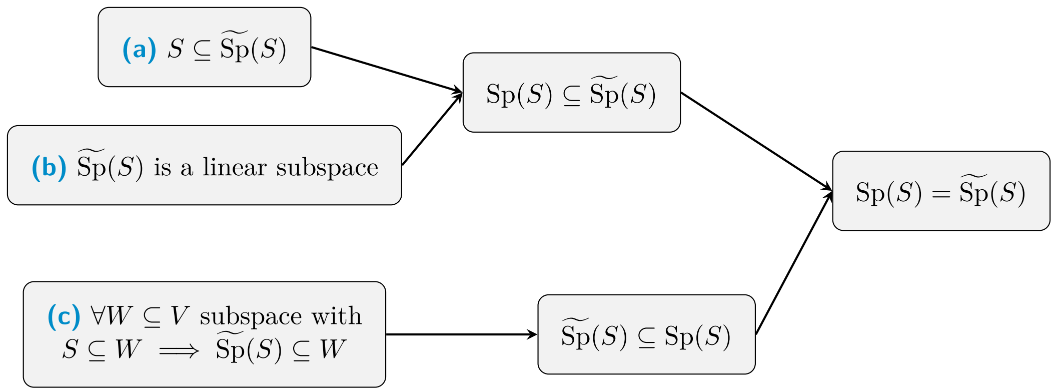
Remark 8.1: The set S itself can be finite or infinite. Regardless of the size of S, each individual linear combination uses only a finite number of elements from S.

Proof:

Let us denote the set of all linear combinations by

$$\widetilde{\mathrm{Sp}}(S) := \{a_1 v_1 + \cdots + a_n v_n \mid n \in \mathbb{N},\ a_i \in K,\ v_i \in S \text{ for all } 1 \leq i \leq n\}.$$

Before diving into the algebra, let us visualize the logical strategy of the proof. We will establish three facts, which together force $\mathrm{Sp}(S) = \widetilde{\mathrm{Sp}}(S)$:



Note that **(a)** and **(b)** imply that $\mathrm{Sp}(S) \subseteq \widetilde{\mathrm{Sp}}(S)$: together they say that $\widetilde{\mathrm{Sp}}(S)$ is one of the subspaces containing S, so Lemma 8.4 **(b)** applies to it. Furthermore, **(c)** implies that $\widetilde{\mathrm{Sp}}(S) \subseteq \mathrm{Sp}(S)$, because $\mathrm{Sp}(S)$ is itself a subspace W containing S by Lemma 8.4. Let us now prove the three statements.

(a) Let $v \in S$. Then we can write $v = 1 \cdot v \in \widetilde{\mathrm{Sp}}(S)$, which is a linear combination with $n = 1$. Thus $S \subseteq \widetilde{\mathrm{Sp}}(S)$.

(b) We verify the criterion of Lemma 7.5. First, $0_V \in \widetilde{\mathrm{Sp}}(S)$: since $S \neq \emptyset$ by assumption, we may pick some $u \in S$, and then $0_V = 0 \cdot u \in \widetilde{\mathrm{Sp}}(S)$ by Lemma 7.2 **(c)**.
Second, let $v, w \in \widetilde{\mathrm{Sp}}(S)$ and $\alpha, \beta \in K$. Write

$$v = a_1 v_1 + \cdots + a_n v_n, \qquad w = b_1 w_1 + \cdots + b_m w_m,$$

where $n, m \in \mathbb{N}$, $a_i, b_j \in K$ and $v_i, w_j \in S$ for all $1 \leq i \leq n$, $1 \leq j \leq m$. Then

$$\alpha v + \beta w = \alpha a_1 v_1 + \cdots + \alpha a_n v_n + \beta b_1 w_1 + \cdots + \beta b_m w_m,$$

which is a linear combination of the vectors $v_1, \ldots, v_n, w_1, \ldots, w_m \in S$ with coefficients $\alpha a_1, \ldots, \alpha a_n, \beta b_1, \ldots, \beta b_m \in K$. Hence $\alpha v + \beta w \in \widetilde{\mathrm{Sp}}(S)$, and $\widetilde{\mathrm{Sp}}(S)$ is a linear subspace.

(c) Let $W \subseteq V$ be a linear subspace such that $S \subseteq W$, and let $v \in \widetilde{\mathrm{Sp}}(S)$. We need to show that $v \in W$. Indeed, write $v = a_1 v_1 + \cdots + a_n v_n$ for some $n \in \mathbb{N}$, $v_i \in S$ and $a_i \in K$. By assumption $S \subseteq W$, and hence $v_i \in W$ for every i. Since W is a linear subspace it is closed under scalar multiplication and under finite sums, and therefore,

$$v = a_1 v_1 + \cdots + a_n v_n \in W.$$

Thus $\widetilde{\mathrm{Sp}}(S) \subseteq W$.

Q.E.D.

Summary 8.1: We now have two descriptions of the same object, and we may use whichever is more convenient:

$$\mathrm{Sp}(S) = \begin{cases} \bigcap_{W \in \mathcal{N}} W & \text{(by definition: the smallest subspace containing } S\text{)}, \\ \text{all linear combinations of vectors from } S & \text{(by Lemma 8.5)}. \end{cases}$$

They can be regarded as equivalent definitions of the span.

Notation 8.1: Let $n \in \mathbb{N}$ and $v_1, \ldots, v_n \in V$. We write

$$\mathrm{Sp}(v_1, \ldots, v_n) := \mathrm{Sp}(\{v_1, \ldots, v_n\}),$$

dropping the set brackets when the spanning set is given as an explicit finite list.

Lemma 8.6 (*The Span of a Finite List*): Let $n \in \mathbb{N}$ and $v_1, \ldots, v_n \in V$. Then

$$\mathrm{Sp}(v_1, \ldots, v_n) = \left\{ \sum_{i=1}^{n} \alpha_i v_i \;\middle|\; \alpha_i \in K \text{ for all } 1 \leq i \leq n \right\}.$$

Remark 8.2: Note that here the number of summands is fixed at n, whereas in Lemma 8.5 it was allowed to vary and the vectors were allowed to repeat. The point is that repetitions and omissions can always be absorbed into the coefficients: a repeated vector has its coefficients added together, and an omitted one is given the coefficient 0.

Exercise 8.1: Prove Lemma 8.6, using Lemma 8.5 together with the observation of the remark above about repetitions and omissions.

Remark 8.3: The span of the empty set is the zero subspace:

$$\mathrm{Sp}(\emptyset) = \{0_V\}.$$

This is not a convention but a consequence of Definition 8.2. Every subspace $W \subseteq V$ satisfies $\emptyset \subseteq W$, so the collection $\mathcal{N}$ consists of **all** subspaces of V; and since $\{0_V\}$ is one of them, the intersection of all of them is contained in $\{0_V\}$, while 0_V lies in every subspace. Using the other description, in terms of linear combinations, we just think of the sum of an empty collection of elements as 0.

Generating Sets and Finite-Dimensionality

Definition 8.7 (*Generating Set*): Let V be a vector space over K and let $S \subseteq V$. We say that S "*generates*" (or "*spans*") V if

$$\mathrm{Sp}(S) = V.$$

In this case S is called a "*generating set*" of V. More generally, if $\mathrm{Sp}(S) = U$, where $U \subseteq V$ is a subspace of V, we say that S generates (or spans) U.

Definition 8.8 (*Finite-Dimensional Vector Space*): A vector space V is called "*finite-dimensional*" if there exists a **finite** set $S \subseteq V$ such that $\mathrm{Sp}(S) = V$. Otherwise, V is called "*infinite-dimensional*".

Important remark 8.2: Note that this definition says nothing yet about **how many** vectors a generating set must contain, only that some finite generating set exists. Attaching an actual number to a finite-dimensional space requires the notion of a basis, and is the business of the next few chapters.

The Subspaces of the Plane

Before turning to computations, let us use the span to list **all** linear subspaces of a small vector space. Take $V := \mathbb{R}^2$ over $K := \mathbb{R}$.

Example 8.1 (*Lines Through the Origin*):

- The zero subspace: $\{0_{\mathbb{R}^2}\} = \{(0,0)\} = \mathrm{Sp}(\emptyset) = \mathrm{Sp}(0_{\mathbb{R}^2})$.
- Let $0 \neq v = (a,b) \in \mathbb{R}^2$. Then
$$\mathrm{Sp}(v) = \{\alpha \cdot v \mid \alpha \in \mathbb{R}\} = \{(\alpha a, \alpha b) \mid \alpha \in \mathbb{R}\}.$$
Geometrically, this is the line L_v that passes through $(0,0)$ and through $v = (a,b)$, as drawn in Figure 8.1.
- Let $0 \neq v, w \in \mathbb{R}^2$. Then
$$\mathrm{Sp}(v) = \mathrm{Sp}(w) \iff \exists\, \alpha \neq 0 \text{ such that } w = \alpha \cdot v \iff L_v = L_w.$$

Exercise 8.2: Prove the two equivalences stated on Page 58: for $0 \neq v, w \in \mathbb{R}^2$, show that $\mathrm{Sp}(v) = \mathrm{Sp}(w)$ if and only if $w = \alpha \cdot v$ for some $\alpha \neq 0$, and that this holds if and only if $L_v = L_w$.

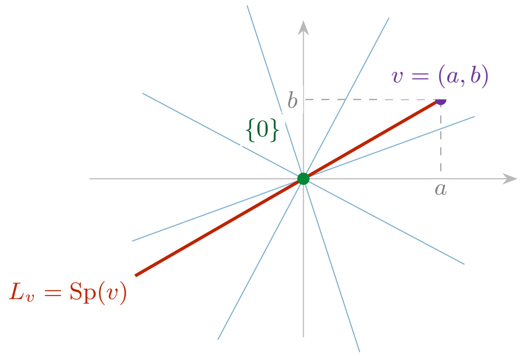


Figure 8.1: The linear subspaces of $\mathbb{R}^2$. There are exactly three kinds: the single point $\{0\}$ (green), the lines through the origin $L_v = \mathrm{Sp}(v)$ for $0 \neq v$ (one of them highlighted, a few others faint), and $\mathbb{R}^2$ itself. Two non-zero vectors span the same line precisely when each is a non-zero multiple of the other; as soon as w lies off L_v, the pair v, w already spans the whole plane.

Exercise 8.3 (*Important*): Let $W \subseteq \mathbb{R}^2$ be a linear subspace. Suppose $W = \mathrm{Sp}(v, w)$, where $v \neq 0$ and $w \notin \mathrm{Sp}(v)$. Show that $W = \mathbb{R}^2$.

Summary 8.2: $\mathbb{R}^2$ has the following linear subspaces, and no others:

(a) $\{0_{\mathbb{R}^2}\}$;
(b) $L_v = \mathrm{Sp}(v)$, where $0 \neq v \in \mathbb{R}^2$;
(c) $\mathbb{R}^2$ itself.

Checking for Span: The Bridge to Computations

Up to this point, our description of the span has been abstract. We now ask a practical, computational question.

Question 8.1: Let $v_1, \ldots, v_n \in K^m$ and $w \in K^m$. How can we determine whether or not $w \in \mathrm{Sp}(v_1, \ldots, v_n)$?

For example, take $v_1 := (1, 3)$ and $v_2 := (7, 73) \in \mathbb{R}^2$, together with $w := (11, 137)$. Here w is indeed in the span, because

$$-3 \cdot (1, 3) + 2 \cdot (7, 73) = (-3 + 14,\ -9 + 146) = (11, 137).$$

But how does one **find** the coefficients -3 and 2? Do they always exist? Are they unique?

Preparation: a matrix product is a linear combination of columns

Let

$$A = \begin{pmatrix} a_{1,1} & \ldots & a_{1,n} \\ \vdots & & \vdots \\ a_{m,1} & \ldots & a_{m,n} \end{pmatrix} \in \mathcal{M}_{m\times n}(K), \qquad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K^n,$$

and let v_j denote the j-th column of A, that is

$$v_1 := \begin{pmatrix} a_{1,1} \\ \vdots \\ a_{m,1} \end{pmatrix}, \qquad v_2 := \begin{pmatrix} a_{1,2} \\ \vdots \\ a_{m,2} \end{pmatrix}, \qquad \ldots \qquad v_n := \begin{pmatrix} a_{1,n} \\ \vdots \\ a_{m,n} \end{pmatrix} \quad \in K^m,$$

so that A may be written in column-block form as

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{pmatrix}.$$

Carrying out the product $A \cdot x$ of Definition 6.7 and regrouping the terms by columns instead of by rows, we obtain

$$A \cdot x = x_1 \begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + x_2 \begin{pmatrix} | \\ v_2 \\ | \end{pmatrix} + \dots + x_n \begin{pmatrix} | \\ v_n \\ | \end{pmatrix} = \sum_{i=1}^{n} x_i \cdot v_i. \tag{8.2}$$

This reveals one of the most useful insights in all of linear algebra: **a matrix-vector product $A \cdot x$ is simply a linear combination of the columns of A, weighted by the entries of x.** Conversely, we can write any linear combination as a multiplication of a matrix A by a vector x.

The consequence for membership in a span

So the problem *"find coefficients $x_1, \dots, x_n \in K$ such that $x_1 v_1 + \dots + x_n v_n = w$"* is the very same problem as *"find a solution $x \in K^n$ of the system $A \cdot x = w$"*, where A is the matrix whose columns are $v_1, \dots, v_n$. We record this as a theorem.

Theorem 8.9 (*Span and Consistency of a Linear System*): Let $w, v_1, \dots, v_n \in K^m$. Let $A \in \mathcal{M}_{m \times n}(K)$ be the matrix whose columns are exactly the vectors $v_1, \dots, v_n$. Then the target vector w lies in the span of $v_1, \dots, v_n$ if and only if the system of linear equations $A \cdot x = w$ has at least one solution $x = (x_1, \dots, x_n)^\mathsf{T} \in K^n$:

$$A \cdot x = w \text{ has a solution } x \in K^n \iff x_1 v_1 + \dots + x_n v_n = w \iff w \in \mathrm{Sp}(v_1, \dots, v_n).$$

Proof:

The first equivalence is exactly (8.2), read from right to left and from left to right. The second is Lemma 8.6, which says that the elements of $\mathrm{Sp}(v_1, \dots, v_n)$ are precisely the vectors of the form $\sum_{i=1}^{n} x_i v_i$ with $x_i \in K$. **Q.E.D.**

Remark 8.4 (*How to check this in practice*): Because of this equivalence, whenever you are asked *"is w in the span of these vectors?"*, you do not need to guess the coefficients. You simply construct the augmented matrix $(A \mid w)$ and perform Gauss elimination, bringing it to row-reduced echelon form as in Theorem 6.14. If the resulting system yields a contradiction, the vector is **not** in the span. If the system is solvable, the solutions $x_1, \dots, x_n$ are exactly the coefficients needed to build w, and by Theorem 6.10, row reduction has not changed which coefficients those are. Whether the coefficients are **unique** is a separate question, and it is precisely the question of linear independence, which occupies the next chapter.

Some Spaces and How to Generate Them

To conclude, let us look at some standard vector spaces and identify natural sets of *"building block"* vectors that span them entirely.

Example 8.2 (*The Coordinate Space K^n*): Let $V := K^n$ and let

$$e_1 := (1, 0, \dots, 0), \quad e_2 := (0, 1, 0, \dots, 0), \quad \dots, \quad e_n := (0, \dots, 0, 1)$$

be the standard vectors. Every vector $(x_1, \dots, x_n) \in K^n$ can be written as $x_1 e_1 + \dots + x_n e_n$, and therefore,

$$K^n = \mathrm{Sp}(e_1, \dots, e_n).$$

Exercise 8.4: Check the identity $K^n = \mathrm{Sp}(e_1, \dots, e_n)$ in detail.

Example 8.3 (*Polynomials*): Let $W := K[x]_d$ be the space of polynomials over K of degree $\leq d$, introduced as a subspace of $K[x]$ on Page 54. We can generate this space using the standard monomials:

$$W = \mathrm{Sp}(1, x, x^2, \dots, x^d).$$

Let $V := K[x]$ be the space of **all** polynomials over K. Generating this space requires an infinite set, since there is no maximum degree:

$$V = \mathrm{Sp}\big(\{1, x, x^2, \dots, x^n, \dots\}\big).$$

Example 8.4 (*Spaces of Matrices*): Let $\mathcal{M} := \mathcal{M}_{m\times n}(K)$ be the space of all $m \times n$ matrices with entries in K. For $1 \leq k \leq m$ and $1 \leq \ell \leq n$, let $E_{k\ell} \in \mathcal{M}$ be the matrix that has all its entries 0 except the entry in position (k, ℓ), which is 1. In other words, $E_{k\ell} = (a_{ij})$, where

$$a_{ij} = \begin{cases} 1 & \text{if } i = k \text{ and } j = \ell, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{so} \qquad E_{k\ell} = \begin{array}{c} \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix} \end{array} \begin{array}{l} \\ \\ \leftarrow \text{row } k \\ \\ \\ \end{array}$$

with the 1 sitting in column ℓ. There are exactly $m \cdot n$ such matrices.

For any matrix $A = (a_{ij}) \in \mathcal{M}_{m\times n}(K)$, we can pull it apart into a linear combination in which the scalar weights are simply the matrix entries themselves:

$$A = \sum_{k=1}^{m} \sum_{\ell=1}^{n} a_{k\ell} \cdot E_{k\ell},$$

and therefore,

$$\mathcal{M}_{m\times n}(K) = \mathrm{Sp}\big(\{E_{k\ell} \mid 1 \leq k \leq m,\ 1 \leq \ell \leq n\}\big).$$

For instance, in $\mathcal{M}_{2\times 2}(K)$ any matrix can be constructed like this:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

so that a matrix may be viewed as a *“vector”* built out of simpler matrices.

Exercise 8.5: Show that K^n, $\mathcal{M}_{m\times n}(K)$ and $K[x]_d$ are finite-dimensional in the sense of Definition 8.8. What about $K[x]$? Prove your answer.

AI-Note: The lecture notes write $\mathrm{Sp}(S)$ throughout, which this transcript keeps. The logical roadmap diagram inside the proof of Lemma 8.5, Figure 8.1, and the closing sentence about uniqueness in the practical remark on Page 60 were added here; everything else follows the notes. Lemma 8.4 is stated in the notes with the proof left as *“check the details”*, and is proved in full above.

Linear Independence (Chapter 9)

Linear Independence of a Finite List

Let V be a vector space over a field K. In the previous chapter, we studied how a set of vectors can *span* a subspace. We now ask whether a spanning set contains *"redundant"* vectors. This leads us to the core algebraic concept of linear independence.

Definition 9.1 (*Linear Independence*): Let $v_1, \ldots, v_n$ be a finite list of vectors in V. We say that the list $v_1, \ldots, v_n$ is **linearly independent** if the only linear combination that equals the zero vector 0_V is the trivial combination (where all coefficients are zero).

In formal notation, the vectors are linearly independent if, for all $a_1, \ldots, a_n \in K$:

$$a_1v_1 + a_2v_2 + \cdots + a_nv_n = 0_V \implies a_1 = 0, a_2 = 0, \ldots, a_n = 0$$

Remark 9.1 (*The Empty Set*): By mathematical convention, the empty list of vectors $\emptyset$ is defined to be linearly independent. But *why* does this make sense?

Recall from Chapter 8 that we defined the span of the empty set as the zero subspace: $\mathrm{Sp}(\emptyset) = \{0_V\}$. If the empty set were to be linearly *dependent*, there would need to exist a linear combination of its elements that non-trivially yields 0_V. However, since $\emptyset$ contains literally no vectors, we cannot even form a linear combination with a non-zero coefficient! Because the condition for dependence is impossible to satisfy, the empty set is vacuously, and logically, independent.

Before looking at examples, we can immediately deduce several structural facts directly from the definition.

Lemma 9.2 (*Basic Properties of Linear Independence*): Let V be a vector space. The following properties hold for any finite list of vectors:

- **(a)** The zero vector 0_V can always be written as a trivial linear combination of any finite list of vectors: $0 \cdot v_1 + \cdots + 0 \cdot v_n = 0_V$.
- **(b)** The order of the vectors in a list does not affect their linear independence.
- **(c)** If a list $v_1, \ldots, v_n$ contains two identical vectors (i.e., $v_k = v_l$ for some $k \neq l$), then the list is NOT linearly independent.
- **(d)** If the zero vector 0_V is an element of the list, then the list is NOT linearly independent.

Proof:

We address each property:

- **(a)** Follows trivially from the vector space axioms (specifically $0 \cdot v = 0_V$ for all $v \in V$).
- **(b)** Follows trivially from the commutativity of vector addition in V.
- **(c)** Assume $v_k = v_l$ for some $1 \leq k < l \leq n$. We can construct a specific linear combination where the coefficient for v_k is 1, the coefficient for v_l is -1, and all other coefficients are 0:

  $$0 \cdot v_1 + \cdots + 1 \cdot v_k + \cdots + (-1) \cdot v_l + \cdots + 0 \cdot v_n = v_k - v_l = 0_V$$

  Because we found a linear combination equal to 0_V where not all coefficients are zero (specifically, $a_k = 1 \neq 0$), the list fails the definition and is therefore not linearly independent.
- **(d)** Suppose $v_1 = 0_V$. We can assign a non-zero coefficient (e.g., 1) to v_1 and zero to all other vectors:

  $$1 \cdot 0_V + 0 \cdot v_2 + \cdots + 0 \cdot v_n = 0_V + 0_V + \cdots + 0_V = 0_V$$

  Again, since $a_1 = 1 \neq 0$, we have a non-trivial combination yielding the zero vector. The list is not linearly independent.

Q.E.D.

Property **(b)** above has a convenient consequence, which we will use constantly from here on.

Notation 9.1 (*From Lists to Sets*): Since the order of the vectors is irrelevant, whenever the list $v_1, \dots, v_n$ has **no repetitions** we can talk about the linear independence of the **set** $\{v_1, \dots, v_n\}$, and we shall do so freely. The proviso matters, because a set silently discards repetitions: $\{v, v\} = \{v\}$, whereas the two-term list v, v is never independent by Lemma 9.2 **(c)**. We return to this distinction on Page 64.

Examples of Linear Independence

Let us apply this definition to various vector spaces.

Example 9.1 (*In* $\mathbb{R}^2$): Consider the vectors $v_1 := \binom{1}{0}$ and $v_2 := \binom{0}{1}$ in $\mathbb{R}^2$.
Assume $a_1 v_1 + a_2 v_2 = 0_{\mathbb{R}^2}$. This means:

$$a_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus, $a_1 = 0$ and $a_2 = 0$. The vectors $\{v_1, v_2\}$ are linearly independent.

Example 9.2 (*Linearly dependent vectors in* $\mathbb{R}^2$): Consider $v_1 := \binom{1}{1}$ and $v_2 := \binom{2}{2}$.
We can easily spot that $2v_1 - v_2 = 0_{\mathbb{R}^2}$. Because we have coefficients $a_1 = 2$ and $a_2 = -1$ (which are non-zero) that yield the zero vector, these vectors are NOT linearly independent.

Example 9.3 (*The Standard Basis in* K^n): Consider the standard basis vectors in K^n:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots \quad , \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Setting $a_1 e_1 + a_2 e_2 + \cdots + a_n e_n = 0_{K^n}$ yields the column vector $(a_1, a_2, \dots, a_n)^\mathsf{T} = (0, 0, \dots, 0)^\mathsf{T}$. This immediately forces $a_i = 0$ for all i. Thus, the set $\{e_1, e_2, \dots, e_n\}$ is always linearly independent.

Example 9.4 (*Single Vectors*): Let V be a vector space and $v \in V$. The single-element set $\{v\}$ is linearly independent if and only if $v \neq 0_V$. (If $v = 0_V$, we know from Lemma 9.2 that it is not independent. If $v \neq 0_V$, then $a \cdot v = 0_V \implies a = 0$).

Arbitrary Families and Linear Dependence

As we advance in linear algebra, we frequently need to deal with infinite collections of vectors (such as the set of all polynomials). We must rigorously extend our definition to handle this. Before we do, we should say what a family actually is.

Question 9.1: What do we mean by a family of vectors in V, written $\mathcal{F} = \{v_i\}_{i \in I}$ and indexed by a set I?

Definition 9.3 (*Family*): The meaning of this is, by definition, that we have a set I (called the **set of indices**), and that we are given a function

$$\mathcal{F} \colon I \longrightarrow V, \qquad I \ni i \longmapsto v_i \in V.$$

So $\mathcal{F}$ assigns to each index $i \in I$ an element $v_i \in V$.

Example 9.5 (*Families and Their Index Sets*):

(1) $I = \{1, \ldots, n\}$ is a finite set. Then a family $\mathcal{F} = \{v_i\}_{i \in I}$ can be just written as $v_1, \ldots, v_n$; the finite lists of the previous subsections are thus a special case.

(2) $I = \mathbb{Z}_{\geq 1} = \{1, 2, 3, \ldots\}$. Then $\mathcal{F} = \{v_i\}_{i \in \mathbb{Z}_{\geq 1}}$ can be written as $v_1, v_2, v_3, \ldots$.

(3) We can have families $\mathcal{F} = \{v_i\}_{i \in I}$ indexed by other (in fact, any) sets I. For example $I = \mathbb{R}$, and so on.

Remark 9.2 (*Families of Elements of Any Set*): There is nothing special about V being a vector space as far as the concept of families goes. If X is a set and I is another set, we can talk about families $\mathcal{F} = \{x_i\}_{i \in I}$ of elements from X, indexed (or parametrized) by $i \in I$. Again, this means a function

$$\mathcal{F} \colon I \longrightarrow X, \qquad I \ni i \longmapsto x_i \in X.$$

Remark 9.3 (*Repetitions and Injectivity*): If this function $\mathcal{F}$ is **not** injective, then the family $\mathcal{F} = \{x_i\}_{i \in I}$ will have **repetitions**, i.e., there exist $i_1, i_2 \in I$ with $i_1 \neq i_2$ such that

$$x_{i_1} = x_{i_2}.$$

Conversely, an injective $\mathcal{F}$ is precisely a family without repetitions. This is the one feature of families that interacts with linear independence.

Back to linear independence.

Definition 9.4 (*Linear Independence of an Arbitrary Family*): Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a *family of vectors* in V, indexed by an arbitrary set I (which may be infinite). We say that the family $\mathcal{F}$ is linearly independent if **every finite subset** of $\mathcal{F}$ is linearly independent.

Spelled out in terms of the indexing, this says: for every $n \in \mathbb{N}$ and any sequence of **distinct** indices $i_1, \ldots, i_n \in I$ (i.e., $i_k \neq i_\ell$ for all $k \neq \ell$), the list of vectors

$$v_{i_1},\ v_{i_2},\ \ldots,\ v_{i_n}$$

is linearly independent in the sense of Definition 9.1.

Remark 9.4 (*Repetitions Rule Out Independence*): Of course, if in $\mathcal{F}$ there is a repetition of some vector (i.e., there exist $i, j \in I$ with $i \neq j$ such that $v_i = v_j$), then the family $\mathcal{F}$ can **not** be linearly independent. Indeed, i, j is a choice of two distinct indices, so Definition 9.4 demands that the two-term list v_i, v_j be independent; but that list consists of the same vector twice, and is not independent by Lemma 9.2 **(c)**.

Since a family carries more information than the set of its values, it is worth asking how much of that extra information linear independence can actually see.

Question 9.2: What is the difference between a family $\mathcal{F} = \{v_i\}_{i \in I}$ and the set $S = \{v_i \mid i \in I\}$?

Answer 9.1:

(a) The set S does **not** contain repetitions (in case there are any in $\mathcal{F}$).

(b) The set S does **not** *"remember"* the indexing $i \in I$ of the elements v_i of S.

Now, **(b)** has no effect on linear independence, since by Lemma 9.2 **(b)** the order and the names of the indices are irrelevant. The effect of **(a)** is clear and has been discussed on Page 64: a repetition makes the family dependent while leaving the set unchanged.

Sometimes we consider a set of vectors $\emptyset \neq S \subseteq V$ directly, and it costs nothing to define independence for it.

Definition 9.5 (*Linear Independence of a Set*)**:** Let $\emptyset \neq S \subseteq V$ be a set of vectors. We say that S is **linearly independent** if every finite list $v_1, \dots, v_n \in S$, $n \geq 1$, that has no repetitions (i.e., $v_i \neq v_j$ for all $i \neq j$) is linearly independent.

Example 9.6 (*Families of Polynomials*)**:** In the space of bounded polynomials $K[x]_d$, the set of standard monomials $\{1, x, x^2, \dots, x^d\}$ is linearly independent. This remains true even for the **infinite family** of all monomials $\mathcal{F} = \{1, x, x^2, \dots\} \subseteq K[x]$. Any finite subset of this family will just be a finite collection of distinct monomials, which can only sum to the zero polynomial if all their coefficients are strictly zero.

Exercise 9.1: For each $k \geq 0$ let $e_k \in K^\infty$ be the sequence whose k-th entry is 1 and all of whose other entries are 0, where K^∞ is the space of sequences from Page 52. Show that the set

$$\{e_0, e_1, e_2, \dots, e_k, \dots\} \subseteq K^\infty$$

is linearly independent. (Note that this set does **not** span K^∞: a linear combination is a finite sum, so it can only produce sequences with finitely many non-zero entries.)

We now formally define the opposite of linear independence, specifically focusing on how it applies to an arbitrary family.

Definition 9.6 (*Linear Dependence*)**:** A family $\mathcal{F} = \{v_i\}_{i \in I}$ of vectors in V is called **linearly dependent** if it is NOT linearly independent.

In other words, this means there exists some finite integer $n \in \mathbb{N}$, a selection of n distinct indices $i_1, \dots, i_n \in I$, and a set of scalars $a_1, \dots, a_n \in K$ **where not all a_j are zero**, such that:

$$a_1 v_{i_1} + a_2 v_{i_2} + \cdots + a_n v_{i_n} = 0_V$$

Such a combination, in which at least one coefficient is non-zero, is called a **non-trivial linear combination** of $v_{i_1}, \dots, v_{i_n}$. In other words, there is a non-trivial linear combination of elements from $\mathcal{F}$ that gives 0_V.

Remark 9.5 (*Two Ways to Be Dependent*)**:** Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a family of vectors in V.

(1) If there exists an $i \in I$ such that $v_i = 0_V$, then $\mathcal{F}$ is linearly dependent. Just take $n = 1$, the single index $i_1 := i$ with $v_{i_1} = 0_V$, and any $a_1 \neq 0$; then

$$a_1 \cdot v_{i_1} = a_1 \cdot 0_V = 0_V.$$

(2) If $\mathcal{F}$ has a repetition of some vector, then $\mathcal{F}$ is linearly dependent. This is the remark on Page 64, restated in the language of dependence.

Remark 9.6 (*Dependent Sets*)**:** We can define a linearly dependent set S in a similar way to the above: $S \subseteq V$ is linearly dependent if it is not linearly independent in the sense of Definition 9.5, which spelled out gives the same description as in Definition 9.6, with the n distinct indices replaced by n pairwise distinct vectors $v_1, \dots, v_n \in S$.

Back to Linear Independence: Redundancy

This definition leads to an incredibly useful geometric intuition: a set of vectors is linearly dependent if and only if at least one vector is *"redundant"*, meaning it can be built entirely out of the other vectors in the set.

Algebraic Definition
$\{v_1, \dots, v_n\}$ is
Linearly Dependent
$(\sum a_i v_i = 0_V,\ a_i \neq 0)$

$\xleftrightarrow{\textbf{Proposition 9.7}}$

Geometric Redundancy
At least one vector v_k
is a linear combination
of the others.
$(v_k \in \mathrm{Sp}(v_1, \dots, \widehat{v_k}, \dots, v_n))$

Proposition 9.7 (*Equivalence of Dependence and Redundancy*): A list of vectors $v_1, \dots, v_n$ is linearly dependent if and only if at least one vector in the list can be written as a linear combination of the other vectors.

Proof:

We must prove both directions.

Direction 1 ($\implies$): Assume the list $v_1, \dots, v_n$ is linearly dependent. By definition, there exist scalars $a_1, \dots, a_n \in K$, not all zero, such that:

$$a_1v_1 + a_2v_2 + \cdots + a_nv_n = 0_V$$

Since not all coefficients are zero, we can choose an index k such that $a_k \neq 0$. We can isolate a_kv_k on one side of the equation by subtracting all other terms:

$$a_kv_k = -\sum_{\substack{i=1 \\ i \neq k}}^{n} a_iv_i$$

Because $a_k \neq 0$, its multiplicative inverse a_k^{-1} exists in the field K. Multiplying both sides by a_k^{-1} yields:

$$v_k = \sum_{\substack{i=1 \\ i \neq k}}^{n} \left(-\frac{a_i}{a_k}\right) v_i$$

Thus, v_k is a linear combination of the other vectors.

Direction 2 ($\impliedby$): Assume that at least one vector, say v_k, can be written as a linear combination of the others. This means there exist scalars b_i such that:

$$v_k = \sum_{\substack{i=1 \\ i \neq k}}^{n} b_iv_i$$

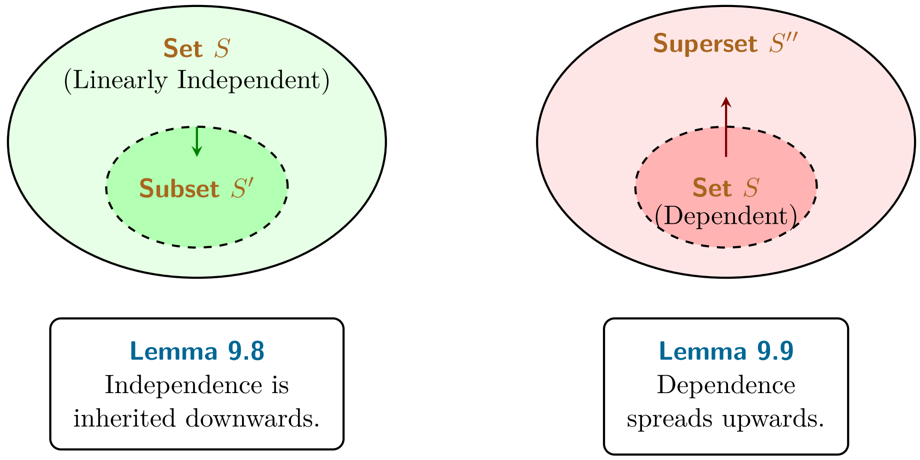
We can subtract v_k from both sides to obtain:

$$0_V = -1 \cdot v_k + \sum_{\substack{i=1 \\ i \neq k}}^{n} b_iv_i$$

This equation is a linear combination of the full list $v_1, \dots, v_n$ that equals 0_V. Furthermore, the coefficient of v_k is -1, which is non-zero. Since we have found a non-trivial linear combination equaling 0_V, the list is linearly dependent by definition. **Q.E.D.**

Inheritance in Subsets and Supersets

How does linear independence behave when we add or remove vectors from a set? There is a strict, logical hierarchy to how these properties are inherited.



Lemma 9.8 (*Subsets of Independent Sets*): Let $S \subseteq V$ be a linearly independent set. If $S' \subseteq S$ is a subset, then S' is also linearly independent.

Proof:

We prove this by contradiction. Suppose S' is NOT linearly independent. Then S' must be linearly dependent. By definition, there exist vectors $v_1, \dots, v_k \in S'$ and scalars $a_1, \dots, a_k \in K$, not all zero, such that:

$$a_1 v_1 + \dots + a_k v_k = 0_V$$

However, since $S' \subseteq S$, every vector v_i is also an element of the larger set S. Therefore, this exact same equation represents a non-trivial linear combination of elements in S that equals 0_V. This would mean S is linearly dependent, which directly contradicts our initial assumption that S is linearly independent. Thus, S' must be linearly independent. **Q.E.D.**

Lemma 9.9 (*Supersets of Dependent Sets*): Let $S \subseteq V$ be a linearly dependent set. If S'' is a superset containing S (i.e., $S \subseteq S'' \subseteq V$), then S'' is also linearly dependent.

Proof:

Because S is linearly dependent, there exist vectors $v_1, \dots, v_k \in S$ and scalars $a_1, \dots, a_k \in K$, not all zero, such that:

$$a_1 v_1 + \dots + a_k v_k = 0_V$$

Since $S \subseteq S''$, the vectors $v_1, \dots, v_k$ also belong to S''. Therefore, this exact same non-trivial linear combination exists within S'', immediately satisfying the definition of linear dependence for S''. **Q.E.D.**

AI-Note: Definition 9.3 and the surrounding discussion of families, the index-set examples, the remark that nothing here needs V to be a vector space, and the link between repetitions and injectivity, are in the notes and were missing from this transcript; the same holds for Definition 9.5, for the comparison of a family with the set of its values on Page 64, for the exercise on Page 65, and for the remarks on Page 65. The notes write the two non-zero coefficients in the proof of Lemma 9.2 **(c)** as 1 and 1, which does not give $v_k - v_l$; the transcript already had the corrected 1 and -1, which is kept. Because this transcript indexes sequences from 0 (Page 52), the standard vectors of K^∞ in that exercise are $e_0, e_1, e_2, \dots$ rather than $e_1, e_2, \dots$ as in the notes. Proposition 9.7 and the two diagrams are not in the notes at this point; the proposition is kept here because the basis-extraction argument in the chapters on bases relies on it.

Bases of Vector Spaces (Part A) (Chapter 10)

Foundations of Bases and the Exchange Theorem (Section 10.a)

Definition and Unique Representation

We have previously explored two fundamental properties a set of vectors can possess: the ability to *span* the entire space, and the property of being *linearly independent* (having no redundancies). A basis is simply a set that achieves both simultaneously.

Definition 10.a.1 (*Basis*): Let V be a vector space over a field K. A subset $S \subseteq V$ is called a **basis** of V (plural: "*bases*") if the following two conditions hold:

(a) S is linearly independent.

(b) S generates (or spans) V, meaning $\mathrm{Sp}(S) = V$.

Why is a basis so mathematically desirable? Because it provides a perfect *"coordinate system"* for our abstract vector space.

Proposition 10.a.2 (*Unique Representation*): A subset $S \subseteq V$ is a basis of V if and only if every vector $v \in V$ can be written in a **unique way** as a linear combination of vectors in S.

To prove this rigorously, especially since S might be an infinite set, like the set of all polynomials, we must first formalize what we mean by *"a unique way as a linear combination."* In a vector space, we can only ever add a *finite* number of vectors together.

We formalize this using **finitely supported functions**.

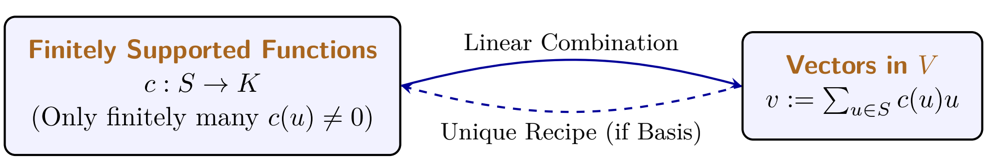
Definition (*Finitely Supported Function and Linear Combination over a Set*): Let V be a vector space over K and let $S \subseteq V$. A function $c : S \to K$, which assigns a scalar coefficient to every vector in S, is called "*finitely supported*" if it is non-zero on only finitely many elements, that is, if the *"support"* of the function,

$$\mathrm{supp}(c) := \{u \in S \mid c(u) \neq 0\},$$

is a finite set. For such a c, the associated **linear combination of elements of S** is the sum

$$\sum_{u \in S} c(u) \cdot u \in V.$$

Because $\mathrm{supp}(c)$ is finite, the number of non-zero terms in this sum is strictly finite, making it a perfectly valid vector operation. (Furthermore, because vector addition is commutative, the order in which we sum these elements does not matter).



To say that every $v \in V$ can be written in a unique way means: if we have two finitely supported functions $c, c' : S \to K$ such that:

$$\sum_{u \in S} c(u) \cdot u = \sum_{u \in S} c'(u) \cdot u$$

then it must be true that $c(u) = c'(u)$ for all $u \in S$.

In case S is a finite set this has a very simple meaning. Write $S = \{v_1, \dots, v_n\}$. Then if

$$v = a_1v_1 + \cdots + a_nv_n \quad \text{with } a_i \in K, \qquad \text{and} \qquad v = b_1v_1 + \cdots + b_nv_n \quad \text{with } b_i \in K,$$

then $a_1 = b_1,\ \dots,\ a_n = b_n$.

Proof of Proposition 10.a.2:

We must prove both directions.

Direction 1 ($\Longrightarrow$): Assume S is a basis. Because S spans V, every $v \in V$ can be written as a linear combination of elements in S. We must show this combination is unique. Suppose there are two combinations yielding the same vector: $\sum_{u\in S} c(u)u = \sum_{u\in S} c'(u)u$. Subtracting one from the other yields:

$$\sum_{u\in S} \big(c(u) - c'(u)\big)u = 0_V$$

Because S is a basis, it is linearly independent. The only way a linear combination of independent vectors can equal 0_V is if all coefficients are zero (this is precisely Definition 9.5, applied to the finitely many $u \in S$ at which c and c' differ). Thus, $c(u) - c'(u) = 0 \implies c(u) = c'(u)$ for all $u \in S$. The representation is unique.

Direction 2 ($\Longleftarrow$): Assume every $v \in V$ has a unique representation. Since every vector can be represented, S clearly spans V. We must show S is independent. Consider the linear combination $\sum_{u\in S} c(u)u = 0_V$. We already know one valid representation for the zero vector: assigning 0 to every coefficient. Since the hypothesis guarantees representations are *unique*, the all-zero assignment must be the *only* representation. Thus, S is linearly independent. **Q.E.D.**

Standard Examples of Bases

Let us identify the standard bases for the families of vector spaces we encounter most frequently.

Example 10.1 (*The Coordinate Space*): In $V = K^n$, the standard basis is $S = \{e_1, e_2, \dots, e_n\}$, where e_i is the column vector with a 1 in the i-th position and 0 elsewhere.

Example 10.2 (*Bounded Polynomials*): In $V = K[x]_d$, the space of polynomials of degree $\le d$, the standard basis is $S = \{1, x, x^2, \dots, x^d\}$.

Example 10.3 (*All Polynomials*): In $V = K[x]$, the space of all polynomials, the standard basis is the infinite set $S = \{1, x, x^2, \dots, x^n, \dots\}$. (Notice how beautifully our *"finitely supported function"* handles this: any specific polynomial has a finite degree, so it requires only a finite number of non-zero coefficients from this infinite basis).

Example 10.4 (*Matrices*): In $V = \mathcal{M}_{m\times n}(K)$, the standard basis is $S = \{E_{i,j} \mid 1 \le i \le m, 1 \le j \le n\}$, where $E_{i,j}$ is the matrix with a 1 in the (i,j) entry and 0 everywhere else. Written out, the 1 sits at the crossing of row i and column j:

$$E_{i,j} = \begin{array}{cl} \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} & \begin{array}{l} \\ \\ \leftarrow \text{row } i \\ \\ \\ \end{array} \end{array}$$

For example, $\mathcal{M}_{2\times 2}(K)$ has the following set as a basis:

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

The Size of a Basis and Exchange Lemmas

Looking at K^n, the standard basis has exactly n elements. Looking at $K[x]_d$, the standard basis has $d+1$ elements. This raises a monumental question: **Are all bases of a given vector space V exactly the same size?**

Yes, they are. Our next major goal is to prove this fact. To do so, we must understand how vectors can be swapped in and out of spanning sets without breaking the span. We develop this through a sequence of four powerful lemmas.

Lemma 10.a.3 (*Lemma A: Simple Exchange*): Let $v_1, \ldots, v_n$ be a basis of V. Let $u = a_1v_1 + \cdots + a_nv_n \in V$ be a vector such that $a_1 \neq 0$. Then the modified list $u, v_2, \ldots, v_n$ is also a basis of V.

Proof:

We must show the new list is independent and spans V. **Independence:** Assume $b_1u + b_2v_2 + \cdots + b_nv_n = 0_V$. Substitute the definition of u:

$$b_1(a_1v_1 + a_2v_2 + \cdots + a_nv_n) + b_2v_2 + \cdots + b_nv_n = 0_V$$

Regrouping the terms by the basis vectors v_i:

$$(b_1a_1)v_1 + (b_1a_2 + b_2)v_2 + \cdots + (b_1a_n + b_n)v_n = 0_V$$

Since $v_1, \ldots, v_n$ are linearly independent, all coefficients must be zero. In particular, $b_1a_1 = 0$. Because we assumed $a_1 \neq 0$, we are forced to conclude $b_1 = 0$. Once $b_1 = 0$, the equation reduces to $b_2v_2 + \cdots + b_nv_n = 0_V$, which forces $b_2 = \cdots = b_n = 0$. Thus, the new list is independent.

Spanning: We can rearrange $u = a_1v_1 + a_2v_2 + \cdots + a_nv_n$ to solve for v_1:

$$v_1 = a_1^{-1}u - a_1^{-1}a_2v_2 - \cdots - a_1^{-1}a_nv_n$$

This shows $v_1 \in \mathrm{Sp}(u, v_2, \ldots, v_n)$. Since all the other basis vectors $v_2, \ldots, v_n$ are trivially in this span, all vectors of the original basis are contained within it. Thus $V = \mathrm{Sp}(v_1, \ldots, v_n) \subseteq \mathrm{Sp}(u, v_2, \ldots, v_n)$. The reverse inclusion is immediate, since $u, v_2, \ldots, v_n$ all lie in V and V is a subspace of itself, so the two spans are equal and the new list spans V. **Q.E.D.**

Lemma 10.a.4 (*Lemma B: Targeted Redundancy*): Let $v_1, \ldots, v_n$ be a list of vectors. Suppose it is linearly dependent. Then there exists an index $1 \leq j \leq n$ such that $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$. Moreover, if we drop v_j from the list, the span of the remaining vectors is exactly the same as the original span.

Proof:

Because the list is dependent, there exist coefficients not all zero such that $\sum_{i=1}^{n} a_iv_i = 0_V$. Let j be the *largest* index such that $a_j \neq 0$. Then our equation is actually:

$$a_1v_1 + a_2v_2 + \cdots + a_jv_j = 0_V$$

Because $a_j \neq 0$, we can solve for v_j:

$$v_j = -\frac{a_1}{a_j}v_1 - \cdots - \frac{a_{j-1}}{a_j}v_{j-1}$$

This proves $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$. Because v_j can be synthesized entirely from the vectors strictly preceding it, dropping v_j does not shrink the span.

Let us spell that last step out as the two inclusions it really is. Let $v \in \mathrm{Sp}(v_1, \ldots, v_n)$, so that

$$v = c_1v_1 + \cdots + c_nv_n \qquad \text{for some } c_1, \ldots, c_n \in K.$$

Substituting the expression for v_j obtained above into this equation writes v as a linear combination of $v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n$, and therefore,

$$\mathrm{Sp}(v_1, \ldots, v_n) \subseteq \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n).$$

The reverse inclusion

$$\mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n) \subseteq \mathrm{Sp}(v_1, \ldots, v_n)$$

is obvious, since every vector of the shorter list occurs in the longer one. The two inclusions give

$$\mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n) = \mathrm{Sp}(v_1, \ldots, v_n).$$

Q.E.D.

Exercise 10.1: What happens if $j = 1$ in Lemma 10.a.4?

The next lemma refines the previous one. It pins down **where** in the list the droppable vector can sit: if an initial segment of the list is independent, the droppable vector must lie strictly beyond it. This is what will later stop the exchange process from ever ejecting one of the vectors we are trying to insert.

Lemma 10.a.5 (*Lemma B′: The Drop Index Lies Beyond Any Independent Head*): Same assumptions as in Lemma 10.a.4, so let $v_1, \ldots, v_n$ be a linearly dependent list and let j be the index it provides. Assume in addition that $v_1, \ldots, v_k$ are linearly independent, where $1 \leq k \leq n$ is some index. Then $k < j$.

Proof:

Assume by contradiction that $j \leq k$. By Lemma 10.a.4 we have $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$, that is,

$$v_j = a_1 v_1 + \cdots + a_{j-1} v_{j-1} \qquad \text{for some } a_1, \ldots, a_{j-1} \in K.$$

Moving v_j to the other side gives

$$0_V = a_1 v_1 + \cdots + a_{j-1} v_{j-1} + (-1) v_j.$$

The coefficient of v_j is $-1 \neq 0$, so this is a non-trivial linear combination of $v_1, \ldots, v_j$ that gives 0_V. Hence $v_1, \ldots, v_j$ are linearly dependent. Since $j \leq k$, the list $v_1, \ldots, v_j$ sits inside the list $v_1, \ldots, v_k$, so $v_1, \ldots, v_k$ are linearly dependent as well. This contradicts the assumption that $v_1, \ldots, v_k$ are linearly independent, and therefore, $k < j$. **Q.E.D.**

Lemma 10.a.6 (*Lemma C: Span Overcrowding*): If $\mathrm{Sp}(v_1, \ldots, v_n) = V$ and $u \in V$, then the list $u, v_1, \ldots, v_n$ is linearly dependent.

Proof:

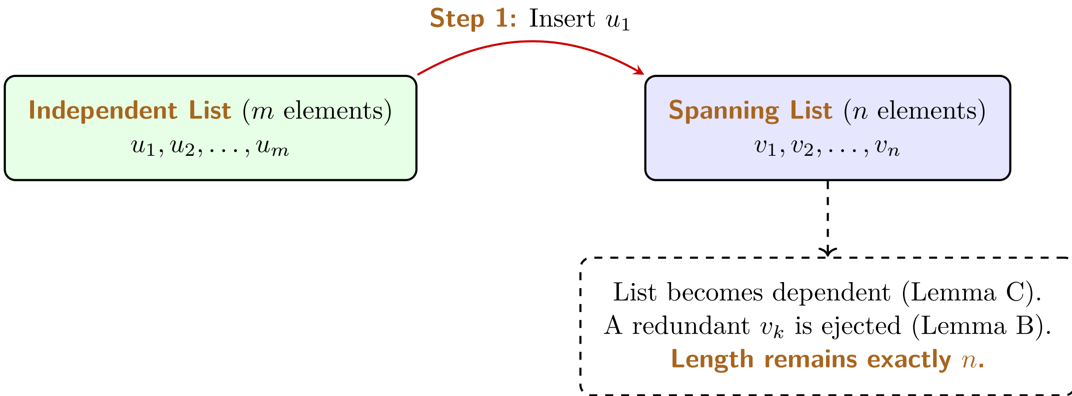
Because the v_i's span V, we can write $u = a_1 v_1 + \cdots + a_n v_n$. Rearranging this gives:

$$1 \cdot u - a_1 v_1 - \cdots - a_n v_n = 0_V$$

Since the coefficient of u is 1 (which is non-zero), this is a non-trivial linear combination that equals zero. The list is dependent. **Q.E.D.**

We now arrive at the crescendo of this chapter: The Steinitz Exchange Lemma. This theorem rigorously proves that an independent set can never outnumber a spanning set.

Theorem 10.a.7 (*Lemma D: Steinitz Exchange Theorem*): Let $u_1, \ldots, u_m$ be a linearly independent list in V. Let $v_1, \ldots, v_n$ be a list that spans V (i.e., $\mathrm{Sp}(v_1, \ldots, v_n) = V$). Then $m \leq n$.



Proof:

We proceed by injecting the independent u vectors into the spanning list one by one, forcefully replacing v vectors to maintain a constant list length of n.

Step 1: Add u_1 to the front of the spanning list: $u_1, v_1, \ldots, v_n$. Because the v_i's span V, Lemma C (Lemma 10.a.6) dictates this new list is dependent. By Lemma B (Lemma 10.a.4), there is a redundant vector in this list that can be dropped without altering the span. Could the redundant vector be u_1? No. Lemma B specifies the redundant vector is in the span of the vectors *preceding* it. u_1 is at the very front, so that span is $\mathrm{Sp}(\emptyset) = \{0_V\}$ (see Remark 8.3), and $u_1 \neq 0_V$ because it is part of an independent list (Lemma 9.2 **(d)**). Thus, the redundant vector must be one of the v_i's, say v_k. We drop v_k. Our list is now $u_1, v_1, \ldots, \widehat{v_k}, \ldots, v_n$. It still spans V, and it has exactly n elements.

Step 2: Add u_2 directly behind u_1: $u_1, u_2, v_1, \ldots, \widehat{v_k}, \ldots, v_n$. By Lemma C, this is dependent. By Lemma B, a redundant vector exists. It can be neither u_1 nor u_2: those two head the list and are linearly independent, so Lemma B' (Lemma 10.a.5) forces the drop index to lie strictly beyond them. Therefore, the redundant vector *must* be another v. We drop it. Length remains n.

The Contradiction: Assume, for the sake of contradiction, that $m > n$. We can continue the above process exactly n times. After step n, we will have systematically ejected every single v_i from the list, leaving us with a list consisting purely of u-vectors: $u_1, u_2, \ldots, u_n$. Crucially, because we only ever used Lemma B to drop vectors, this list of u-vectors **still spans V**.

Because $m > n$, we still have at least one vector left in our independent set: u_{n+1}. Let us add it to our current spanning list:

$$u_1, u_2, \ldots, u_n, u_{n+1}$$

Because $u_1, \ldots, u_n$ spans V, Lemma C states that $u_{n+1}, u_1, \ldots, u_n$ is linearly dependent, and hence so is the list above, the order being immaterial by Lemma 9.2 **(b)**. However, this list is just a subset of our original $u_1, \ldots, u_m$, which was explicitly assumed to be linearly independent, and so it is independent itself by Lemma 9.8. This is a fatal mathematical contradiction.

Therefore, our assumption that $m > n$ must be false. We conclude that $m \leq n$. **Q.E.D.**

Remark 10.1 (*The General Step*): At step j of the argument above, the list reads $u_1, \ldots, u_{j-1}, w_1, \ldots, w_{n-(j-1)}$, where the w's are the survivors of the original spanning list, and we insert u_j after u_{j-1}. Lemma 10.a.6 makes the resulting list of $n+1$ vectors dependent, and Lemma 10.a.4 supplies a vector that may be dropped without changing the span. That vector is never one of $u_1, \ldots, u_j$: those are linearly independent, being part of an independent list (Lemma 9.8), so Lemma 10.a.5 forces the drop index to lie strictly beyond them. The ejected vector is therefore always one of the w's, which is what keeps the length at exactly n while the u's accumulate at the front.

Remark 10.2 (*We Have Proved More Than $m \le n$*): The argument gives more than the inequality. It shows that $m \le n$ **and** that one can remove m vectors from the list $v_1, \dots, v_n$ in such a way that the remaining list $v_{i_1}, \dots, v_{i_{n-m}}$, where $1 \le i_1 < \cdots < i_{n-m} \le n$, spans V when put together with $u_1, \dots, u_m$:

$$\mathrm{Sp}(u_1, \dots, u_m, v_{i_1}, \dots, v_{i_{n-m}}) = V.$$

Indeed, we already know that $m \le n$. Apply the procedure from the proof for the steps $j = 1, \dots, m$. In each step we add another u_j to the list and drop one of the v_i's, while keeping the span equal to V. After $j = m$ steps we obtain the claimed list $u_1, \dots, u_m, v_{i_1}, \dots, v_{i_{n-m}}$.

Solutions to the Exercises

Solution to Exercise 10.1:

The index j supplied by Lemma 10.a.4 may indeed be 1, and in that case the conclusion degenerates in a way that is worth reading carefully. The vectors strictly preceding v_1 form the empty list, so

$$\mathrm{Sp}(v_1, \dots, v_{j-1}) = \mathrm{Sp}(\emptyset) = \{0_V\}$$

by Remark 8.3. The conclusion $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$ therefore says nothing weaker or stronger than

$$v_1 = 0_V.$$

This case occurs exactly when $v_1 = 0_V$. In one direction, the proof takes j to be the largest index with $a_j \ne 0$, so $j = 1$ means $a_1 \ne 0$ while $a_2 = \cdots = a_n = 0$; the dependency relation then reads $a_1 v_1 = 0_V$, and multiplying by a_1^{-1} gives $v_1 = 0_V$. In the other direction, if $v_1 = 0_V$ then

$$1 \cdot v_1 + 0 \cdot v_2 + \cdots + 0 \cdot v_n = 0_V$$

is a non-trivial dependency whose largest non-zero index is 1, so the proof may return $j = 1$. (This is Lemma 9.2 **(d)** seen from the other side: a list containing 0_V is never independent, and here the redundant vector is the zero vector itself.)

The second assertion of the lemma survives unchanged: dropping $v_1 = 0_V$ leaves $\mathrm{Sp}(v_2, \dots, v_n) = \mathrm{Sp}(v_1, \dots, v_n)$, since the zero vector contributes nothing to any linear combination.

This degenerate case is precisely what Step 1 in the proof of Theorem 10.a.7 has to rule out. There u_1 heads the list, so $j = 1$ would say $u_1 = 0_V$; but u_1 belongs to an independent list and is therefore non-zero, so the drop index cannot be 1 and one of the v_i's must be ejected instead. From Step 2 on, the head is no longer a single vector and this role is taken over by Lemma 10.a.5.

Q.E.D.

AI-Note: The notes name their four lemmas A–D differently from this transcript: what is called Lemma 10.a.4 here is Lemma A in the notes, and the notes' Lemma B is Lemma 10.a.5, which was missing from the transcript altogether. The existing letters are left alone to keep the references in the later chapters valid, and the restored lemma is lettered B'. Lemma 10.a.3 does not appear in this part of the notes. Also restored: the finite-set reading of uniqueness before the proof of Proposition 10.a.2, the explicit form of $E_{i,j}$ and the basis of $\mathcal{M}_{2\times 2}(K)$ on Page 69, the second half of the proof of Lemma 10.a.4 (the notes prove the equality of spans by two inclusions; the transcript asserted it), Exercise 10.1, and Remark 10.2, which the later chapters rely on. The proof of Proposition 10.a.2 is given in the notes under the simplifying assumption that S is finite, with the infinite case noted as similar; the transcript proves it for general S and that is kept.

The rigour pass over this chapter corrected two things and named several reasons. In the proof of Theorem 10.a.7, the span of the vectors preceding u_1 was given as $\emptyset$; it is $\mathrm{Sp}(\emptyset) = \{0_V\}$, which is what makes $u_1 \ne 0_V$ the relevant fact there. And the same proof ran its exchange backwards: it inserted each u_j at the front, so the head accumulated as $u_j, \dots, u_1$ and the proof ended at $u_{n+1}, u_n, \dots, u_1$. The notes insert u_j behind u_{j-1} and end at $u_1, \dots, u_n, u_{n+1}$, which is also what both remarks below the proof say, Remark 10.1 for the general step and Remark 10.2 for the stronger form. Nothing was false either way, since independence and span do not depend on the order, but the proof disagreed with the two remarks that explain it; the proof now follows the notes. Steps that were taken silently now cite Remark 8.3, Lemma 9.2 **(b)** and **(d)**, Lemma 9.8 and Definition 9.5, and the proof of Lemma 10.a.3 now closes the spanning half with the reverse inclusion instead of stopping at $V \subseteq \mathrm{Sp}(u, v_2, \ldots, v_n)$. The notes introduce finitely supported functions in running prose, under no heading, on a page where they do write *"Def."* for the basis; that passage is now the unnumbered Definition on Page 68. The solution to Exercise 10.1 is supplied.

Bases of a Vector Space (Part B) (Chapter 11)

The Dimension Theorem (Section 11.b)

The Strong Exchange Lemma

Before we establish the main theorem of this chapter, we formalize a stronger version of the Steinitz Exchange Theorem (Lemma D from the previous section). We proved that an independent list cannot be larger than a spanning list. However, the exact mechanism of the proof actually guarantees something even more powerful: we can precisely substitute the independent vectors into the spanning set.

Lemma 11.b.1 (*Lemma D': Strong Steinitz Exchange*): Let V be a vector space over K, and suppose that $V = \mathrm{Sp}(v_1, \ldots, v_n)$ for some list of vectors $v_1, \ldots, v_n \in V$. Let $u_1, \ldots, u_m \in V$ be a linearly independent list.

Then $m \leq n$, and moreover, one can delete exactly m vectors from the spanning list $v_1, \ldots, v_n$ such that the remaining $n - m$ vectors, together with all the vectors $u_1, \ldots, u_m$, still span V.

$$\mathrm{Sp}(u_1, \ldots, u_m, v'_1, \ldots, v'_{n-m}) = V$$

(where the v' vectors are the survivors of the original spanning list).

Proof:

The fact that $m \leq n$ was already proven in Lemma D. To prove the *"moreover"* part, we simply apply the exact step-by-step substitution procedure described in the proof of Lemma D. For steps $j = 1, 2, \ldots, m$, we sequentially add u_j to the list, behind the vectors $u_1, \ldots, u_{j-1}$ already inserted. At each step the list has $n+1$ vectors and still spans V, so it is dependent (Lemma C), and we forcefully eject a redundant v_k vector (Lemma B), which preserves the span and brings the length back to n. The ejected vector is never one of the u's, by Lemma 10.a.5; this is the general step recorded on Page 72. After m steps, we have injected all m independent vectors and deleted exactly m of the original spanning vectors. The resulting mixed list still spans V. See also Page 73, where the same statement is drawn out of the proof of Lemma D directly. **Q.E.D.**

The Existence and Uniqueness of Dimension

We are now ready to state and prove the foundational theorem of finite-dimensional vector spaces. It guarantees that bases always exist and that they provide an absolute, rigid measure of the *"size"* of a vector space.

Theorem 11.b.2 (*The Basis Theorem*): Let V be a finite-dimensional vector space over K. Then V has a basis with finitely many elements.

Moreover, **every** basis of V is finite, and any two bases of V have exactly the same number of elements.

Remark 11.1 (*Infinite Dimensional Spaces*): This theorem is actually true even if V is not finite-dimensional (this is known as the *"Hamel Basis Theorem"* and requires Zorn's Lemma). However, in this course, we will only prove the theorem for finite-dimensional spaces. The infinite-dimensional case is rigorously covered in advanced algebra courses (e.g., *Grundstrukturen* in the 2nd semester).

Preparation: Two Lemmas

For the proof of the theorem we need some preparation. The first lemma is the converse reading of the redundancy lemma of the previous chapter: there, a dependent list was shown to contain a vector lying in the span of its predecessors; here, a list in which **no** vector lies in the span of its predecessors is shown to be independent.

Lemma 11.b.3 (*Lemma E: No Predecessor Span Means Independent*): Let $w_1, \dots, w_\ell \in V$ and assume that

$$w_j \notin \mathrm{Sp}(w_1, \dots, w_{j-1}) \qquad \text{for all } 1 \le j \le \ell.$$

Then the list $w_1, \dots, w_\ell$ is linearly independent.

Proof:

If $w_1, \dots, w_\ell$ were linearly dependent, then by Lemma 10.a.4 there would exist an index $1 \le j \le \ell$ with $w_j \in \mathrm{Sp}(w_1, \dots, w_{j-1})$, contradicting the assumption. **Q.E.D.**

The second lemma is the one that actually produces a basis. Note that it extracts the basis from **within** the given spanning list, rather than merely asserting that one exists somewhere.

Lemma 11.b.4 (*Lemma F: Every Finite Spanning List Contains a Basis*): Suppose that $\mathrm{Sp}(v_1, \dots, v_n) = V$. Then there exists a subset of $\{v_1, \dots, v_n\}$ which is a basis for V.

Proof:

The proof is algorithmic and is based on n steps. At each step we consider another vector from the list $v_1, \dots, v_n$ and decide whether or not to drop it from the list. At step number j we consider v_j:

Step 1. If $v_1 = 0_V$ we drop v_1 from the list. If $v_1 \neq 0_V$ we keep it.

Step j, for $2 \le j \le n$. If $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$ we drop v_j. Otherwise we keep it.

(The two steps apply one and the same test. For $j = 1$ the vectors preceding v_1 form the empty list, so $\mathrm{Sp}(v_1, \dots, v_{j-1}) = \mathrm{Sp}(\emptyset) = \{0_V\}$ by Remark 8.3, and "$v_1 \in \mathrm{Sp}(\emptyset)$" says exactly "$v_1 = 0_V$". This is the degenerate case treated in the solution to Exercise 10.1.)

After performing the n steps above we obtain a new, possibly shorter, list

$$v_{i_1}, v_{i_2}, \dots, v_{i_m}, \qquad \text{where } 1 \le i_1 < i_2 < \cdots < i_m \le n.$$

We claim that $\mathrm{Sp}(v_{i_1}, \dots, v_{i_m}) = V$. Indeed, in any of the steps $1 \le j \le n$ we dropped the vector v_j only if $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$, and by Lemma 10.a.4 such a vector may be dropped without changing the span. (That lemma produces one particular index j, whereas here the index is whichever one the step is at; but the two-inclusion argument proving its second assertion uses only the property $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$ and never how j was found, so it applies to every index having that property.) Therefore,

$$\mathrm{Sp}(v_{i_1}, \dots, v_{i_m}) = \mathrm{Sp}(v_1, \dots, v_n) = V.$$

We also claim that $v_{i_1}, \dots, v_{i_m}$ is linearly independent. Indeed, for every $1 \le k \le m$ the vector v_{i_k} survived its step, which means

$$v_{i_k} \notin \mathrm{Sp}(v_1, v_2, \dots, v_{i_k - 1}),$$

and since $v_{i_1}, \dots, v_{i_{k-1}}$ are all taken from among $v_1, \dots, v_{i_k - 1}$, so that $\mathrm{Sp}(v_{i_1}, \dots, v_{i_{k-1}}) \subseteq \mathrm{Sp}(v_1, \dots, v_{i_k - 1})$ by Lemma 8.4 **(b)**, this gives in particular

$$v_{i_k} \notin \mathrm{Sp}(v_{i_1}, v_{i_2}, \dots, v_{i_{k-1}}).$$

By Lemma 11.b.3, the list $v_{i_1}, v_{i_2}, \dots, v_{i_m}$ is linearly independent.

Together with the fact that $\mathrm{Sp}(v_{i_1}, \dots, v_{i_m}) = V$, we get that $v_{i_1}, \dots, v_{i_m}$ is a basis for V.

Q.E.D.

Proof of the Basis Theorem

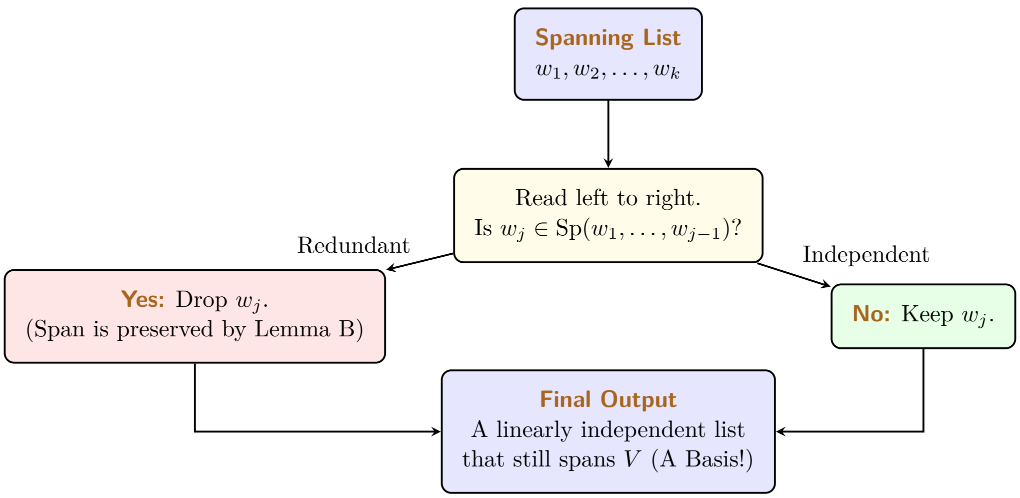
Because this theorem is so monumental, we will break its proof down into three distinct, manageable statements.

Statement 1: V has a finite basis.

Proof:

Because we assumed V is finite-dimensional, there must exist some finite list of vectors that spans V. Let us call this spanning set $S = \{w_1, \dots, w_k\}$. If S is already linearly independent, it is a basis, and we are done.

If S is linearly dependent, we can extract a basis from it using the following systematic reduction algorithm (often called a *"Redundancy Filter"*):



Specifically:

(a) If the zero vector 0_V is in the list, drop it.

(b) Read the list from left to right. If a vector w_j is a linear combination of its strictly preceding vectors (i.e., $w_j \in \mathrm{Sp}(w_1, \dots, w_{j-1})$), drop it.

By Lemma B (Lemma 10.a.4), dropping this redundant vector does not change the span of the list. We repeat this until we have checked every vector. The remaining list still spans V. Furthermore, by Lemma 11.b.3, it must be linearly independent, because we systematically deleted every vector that could be written as a combination of the vectors preceding it. Thus, the remaining list is a finite basis. This is precisely the algorithm of Lemma 11.b.4, applied to the spanning list $w_1, \dots, w_k$.

Q.E.D.

Statement 2: Every basis of V is finite.

Proof:

We prove this by contradiction. Suppose there exists a basis $\mathcal{A}$ of V that is *not* a finite set.

From Statement 1, we know that V definitely possesses at least one finite basis. Let us call this finite basis $S = \{v_1, \dots, v_n\}$. The size of this finite basis is exactly n.

Because $\mathcal{A}$ is infinite, it contains infinitely many linearly independent vectors. Therefore, we can arbitrarily select $n+100$ distinct vectors from $\mathcal{A}$. Let us call them $u_1, \ldots, u_{n+100}$. Because they belong to a basis, this chosen list of $n+100$ vectors is strictly linearly independent (a subset of an independent set is independent, by Lemma 9.8).

We now have two lists:

- A spanning list S containing exactly n elements.
- An independent list containing $n+100$ elements.

By the Steinitz Exchange Theorem (Lemma D, Theorem 10.a.7), the number of elements in any independent list must be less than or equal to the number of elements in any spanning list. Therefore, we must have:

$$n + 100 \leq n$$

This implies $100 \leq 0$, which is a blatant mathematical contradiction! Thus, our initial assumption must be wrong. Every basis of V must be finite. **Q.E.D.**

Statement 3: Any two bases have the same cardinality.

Proof:

Let $\mathcal{A}$ and $\mathcal{B}$ be two arbitrary bases of V. By Statement 2, we know that both $\mathcal{A}$ and $\mathcal{B}$ must be finite sets. Let $|\mathcal{A}|$ denote the number of elements in $\mathcal{A}$, and $|\mathcal{B}|$ denote the number of elements in $\mathcal{B}$.

Because they are bases, both sets are simultaneously independent and spanning. This allows us to apply Lemma D twice in alternating directions:

Independent in Spanning

$$\begin{tikzcd} \textbf{Basis } \mathcal{A} \arrow[r, bend left, "|\mathcal{A}| \leq |\mathcal{B}|"] & \textbf{Basis } \mathcal{B} \arrow[l, bend left, "|\mathcal{B}| \leq |\mathcal{A}|"] \end{tikzcd}$$

Independent in Spanning

(a) Because $\mathcal{A}$ is linearly independent and $\mathcal{B}$ spans V, Lemma D (Theorem 10.a.7) dictates that $|\mathcal{A}| \leq |\mathcal{B}|$.

(b) Reversing their roles, because $\mathcal{B}$ is linearly independent and $\mathcal{A}$ spans V, Lemma D dictates that $|\mathcal{B}| \leq |\mathcal{A}|$.

Since $|\mathcal{A}| \leq |\mathcal{B}|$ and $|\mathcal{B}| \leq |\mathcal{A}|$, it must be that $|\mathcal{A}| = |\mathcal{B}|$. All bases contain exactly the same number of elements. **Q.E.D.**

The Concept of Dimension

Because the number of elements in a basis is a rigid, absolute invariant of the vector space, we can finally define the space's dimension.

Definition 11.b.5 (*Dimension*): Let V be a finite-dimensional vector space over K. We define the **dimension** of V to be the number of elements in any basis of V: if $\mathcal{B}$ is a basis of V and n is the number of its elements, we set

$$\dim V := n \in \mathbb{Z}_{\geq 0}.$$

If V is not finite-dimensional, we write $\dim V = \infty$.

Important remark 11.1: $\dim V$ is **well defined**: the definition does not depend on the specific choice of a basis for V. This is exactly what Statement 3 of Theorem 11.b.2 buys us. Without it, the phrase *"the number of elements in any basis"* would not name a single number, and the definition above would be empty.

Remark 11.2 (*The Zero Space*): Consider the zero vector space $V = \{0_V\}$. It has exactly two subsets, $\emptyset$ and $\{0_V\}$, and both of them span it: $\mathrm{Sp}(\{0_V\}) = \{0_V\} = V$, and $\mathrm{Sp}(\emptyset) = \{0_V\} = V$ as well, by Remark 8.3. Of the two, however, $\{0_V\}$ is linearly dependent, since a list containing the zero vector never is (Lemma 9.2 **(d)**). The only linearly independent subset that spans V is therefore the empty set $\emptyset$, which is independent by Remark 9.1. Therefore, the basis of the zero space is $\emptyset$. Because the empty set contains 0 elements, we have:

$$\dim\{0_V\} = 0$$

Examples of Dimensions

Now that we have formally defined dimension, we can readily determine the dimensions of our familiar vector spaces by simply counting the elements in their standard bases.

Example 11.1 (*The Coordinate Space*): The standard basis for K^n is $\{e_1, \dots, e_n\}$. Because this set contains exactly n elements, we have:

$$\dim K^n = n$$

Example 11.2 (*Bounded Polynomials*): The standard basis for the space of polynomials of degree $\leq d$ is $\{1, x, x^2, \dots, x^d\}$. Be careful when counting here! Because we start at $x^0 = 1$, this set contains $d+1$ elements. Thus:

$$\dim K[x]_d = d + 1$$

Example 11.3 (*Matrices*): The standard basis for the matrix space $\mathcal{M}_{m\times n}(K)$ consists of the individual single-entry matrices $E_{i,j}$. Since there are m rows and n columns, there are exactly $m \cdot n$ such matrices. Thus:

$$\dim \mathcal{M}_{m\times n}(K) = m \cdot n$$

Example 11.4 (*Function Spaces*): Let S be a **finite** set. Recall from Example 7.6 that the space of all functions from S to K, denoted as $K^S := \{f \mid f : S \to K\}$, forms a vector space over K.

The dimension of this space matches the cardinality of the underlying set S:

$$\dim K^S = |S|$$

Exercise 11.1 (*Basis for the Space of Functions on a Finite Set*): If S is a finite set, say $S = \{s_1, \dots, s_n\}$, can you construct an explicit basis for K^S?
(Hint: Try to define a family of functions f_i where each function outputs 1 for exactly one specific element s_i, and 0 for all other elements).

Remark 11.3: When the domain S is an infinite set, the space K^S is infinite-dimensional. In this case, we unfortunately do not have a *"nice"* or easily describable basis for it, and the formula above no longer applies: by Definition 11.b.5 we simply write $\dim K^S = \infty$. It is worth seeing why the obvious candidate fails. The functions f_s of the exercise above are still linearly independent for infinite S, but they no longer span K^S, because a linear combination is a **finite** sum and so can only produce functions that vanish outside a finite set. Exercise 9.1 is exactly this phenomenon for $S = \mathbb{Z}_{\geq 0}$.

Solutions to the Exercises

Solution to Exercise 11.1:

Let $S = \{s_1, \dots, s_n\}$ be finite, and follow the hint: for each $1 \le i \le n$ define $f_i \in K^S$ by

$$f_i(s_j) := \begin{cases} 1 & \text{if } j = i, \\ 0 & \text{if } j \neq i. \end{cases}$$

We claim that $\mathcal{B} := \{f_1, \dots, f_n\}$ is a basis of K^S.

Spanning. Let $f \in K^S$ be any function and put $a_i := f(s_i) \in K$. For every $1 \le j \le n$,

$$\left(\sum_{i=1}^{n} a_i f_i\right)(s_j) = \sum_{i=1}^{n} a_i f_i(s_j) = a_j = f(s_j),$$

because every term with $i \neq j$ vanishes. The two functions therefore agree at every element of S, and so they are equal by Definition 4.2. Hence $f = \sum_{i=1}^{n} a_i f_i \in \mathrm{Sp}(\mathcal{B})$.

Independence. Suppose $a_1 f_1 + \cdots + a_n f_n$ is the zero function. Evaluating at s_j and using the same computation gives $a_j = 0$, and this for every $1 \le j \le n$. So the only linear combination producing the zero function is the trivial one.

Note also that the f_i are pairwise distinct, since $f_i(s_i) = 1 \neq 0 = f_j(s_i)$ whenever $i \neq j$, so $\mathcal{B}$ really has n elements and not fewer. By Definition 11.b.5,

$$\dim K^S = n = |S|,$$

which is the formula stated in the example. The construction is the same one that gives the standard basis $e_1, \dots, e_n$ of K^n: taking $S = \{1, 2, \dots, n\}$ turns f_i into e_i. **Q.E.D.**

AI-Note: The notes prepare the Basis Theorem with two lemmas that the transcript had dropped: Lemma 11.b.3 and Lemma 11.b.4, both restored above with their proofs. Statement 1 of the theorem is proved in the notes by citing Lemma 11.b.4; the transcript inlines the algorithm instead, which is kept, with the citation corrected. That citation had pointed at Proposition 9.7, which is about a vector lying in the span of **all** the others, whereas the filter only ever tests a vector against the ones preceding it. Lemma 11.b.3 is the statement that closes that gap. Also restored: Important remark 11.1, which the notes flag explicitly. The notes state Lemma 11.b.1 here and refer back to the closing remark of the previous lecture for its proof; both are kept, and the cross-reference is made explicit.

The rigour pass corrected three things. The proof of Lemma 11.b.1 inserted each u_j at the front of the list, the same reversed bookkeeping that the proof of Theorem 10.a.7 carried before this pass, and it described the ejection of a v_k as restoring independence. It does not: dropping a redundant vector preserves the span and returns the length to n, and says nothing at all about independence, which may perfectly well still fail, as it does whenever the original spanning list was itself redundant. What the procedure maintains is that the u's standing at the head are independent, and that is what lets Lemma 10.a.5 keep the ejected vector among the v's.

Definition 11.b.5 set $\dim V := n$ with n never introduced; the basis whose elements n counts is now named. The remark on the zero space asserted that $\{0_V\}$ is its only spanning set, and then, in the next sentence, that $\emptyset$ spans it too. Both $\emptyset$ and $\{0_V\}$ span it; only the former is independent, which is the point being made.

The example on function spaces stated $\dim K^S = |S|$ for an arbitrary set S. That holds only when S is finite. For infinite S the space K^S is not finite-dimensional, so Definition 11.b.5 assigns it ∞ and the equation does not typecheck; the functions f_s stay independent but stop spanning, since a linear combination is a finite sum. The example is now stated for finite S and the following remark carries the infinite case. Its exercise was unlabelled and is now written up, giving the basis that the example's formula counts.

Steps taken silently now cite Remark 8.3, Lemma 8.4 **(b)** for the monotonicity of the span, Lemma 9.8, Lemma 9.2 **(d)**, Remark 9.1 and Theorem 10.a.7, which Statements 2 and 3 of Theorem 11.b.2 had invoked as *"Lemma D"* in words only. One gap is flagged rather than closed: Lemma 10.a.4 attaches its second assertion to the one index its proof produces, whereas Lemma 11.b.4 drops whichever index its step is at. The two-inclusion argument there uses only the predecessor property, never how the index was found, and a parenthetical now says so.

Dimension and Subspaces (Chapter 12)

In this chapter, we synthesize the fundamental results regarding finite-dimensional vector spaces and explore the relationship between a space and its subspaces through the lens of dimensionality.

Summary 12.1 (*Properties of Finite-Dimensional Spaces*): Let V be a finite-dimensional vector space over a field K. The following structural principles hold:

- **(a)** Every finite list of vectors that spans V contains a sublist which constitutes a "*basis*" of V. Formally, if $(v_1, \ldots, v_n)$ spans V, then there exists a sublist $(v_{i_1}, \ldots, v_{i_d})$ with $1 \leq i_1 < i_2 < \cdots < i_d \leq n$ that is a basis for V, where $d = \dim V$.
- **(b)** Every linearly independent list of vectors in V can be extended to a basis of V. If the list $(u_1, \ldots, u_m)$ is linearly independent, there exist vectors $w_1, \ldots, w_{d-m}$ such that
$$(u_1, \ldots, u_m, w_1, \ldots, w_{d-m})$$
is a basis of V, where $d = \dim V$.

The notes record these two principles as a summary of the previous lecture and do not prove them. Everything in this chapter rests on them, and item **(2)**, the "*basis extension property*", is used repeatedly in the chapters that follow, so we supply the derivation here. It is short, and it uses nothing beyond the two lemmas of the previous chapter.

Proof of Summary 12.1:

(1) This is Lemma 11.b.4, which extracts a basis from within the list $(v_1, \ldots, v_n)$ itself. Reading its elements in increasing order of index gives the sublist $(v_{i_1}, \ldots, v_{i_d})$, and $d = \dim V$ by Definition 11.b.5, since every basis of V has that many elements.

(2) Let $(u_1, \ldots, u_m)$ be linearly independent, and let $\mathcal{B} = (b_1, \ldots, b_d)$ be a basis of V, which exists and has $d = \dim V$ elements by Theorem 11.b.2. In particular $\mathrm{Sp}(b_1, \ldots, b_d) = V$, so Lemma 11.b.1 applies to the independent list $(u_1, \ldots, u_m)$ and the spanning list $\mathcal{B}$. It yields $m \leq d$ together with $d - m$ survivors $w_1, \ldots, w_{d-m}$ of $\mathcal{B}$ such that
$$\mathrm{Sp}(u_1, \ldots, u_m, w_1, \ldots, w_{d-m}) = V.$$
This list spans V and has exactly d entries. By Lemma 11.b.4 it contains a sublist which is a basis of V, and that sublist has d elements, again because every basis of V does. A sublist of a list of length d that itself has length d is the whole list. Hence $(u_1, \ldots, u_m, w_1, \ldots, w_{d-m})$ is a basis of V, and it extends the given list. **Q.E.D.**

Theorem 12.1 (*Equivalent Characterizations of a Basis*): Let V be a vector space with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Let $v_1, \ldots, v_n \in V$ be a list of n vectors. Then the following statements are equivalent:

- **(a)** The vectors $v_1, \ldots, v_n$ are linearly independent.
- **(b)** The vectors $v_1, \ldots, v_n$ generate V, i.e., $\mathrm{Sp}(v_1, \ldots, v_n) = V$.
- **(c)** The vectors $v_1, \ldots, v_n$ constitute a basis for V.

Proof:

The equivalence is established through the following implications:

(a) $\implies$ (c): Suppose that **(a)** holds, so the list $v_1, \ldots, v_n$ is linearly independent. According to the property that every linearly independent list can be extended to a basis, and observing that the list already possesses length $n = \dim V$, the extension requires no additional vectors. Thus, the list is already a basis for V, satisfying **(c)**.

(c) $\implies$ (a) & (b): This follows directly from the definition of a basis, which by requirement must satisfy both the condition of linear independence in **(a)** and the spanning property in **(b)**.

(b) $\implies$ (c): If the spanning property in **(b)** holds, the list $v_1, \dots, v_n$ generates V. By the properties of finite-dimensional spaces, this list must contain a sublist that is a basis. Since $\dim V = n$, any basis must have exactly length n. Therefore, the sublist must be the entire list itself, making $v_1, \dots, v_n$ a basis as required by **(c)**. **Q.E.D.**

The previous theorem settles what happens for a list whose length is exactly $\dim V$. The next one records what happens when the length is wrong in either direction: too short a list cannot reach every vector, and too long a list must contain a redundancy.

Theorem 12.2 (*Lists of the Wrong Length*): Suppose that $n = \dim V < \infty$, and let $v_1, \dots, v_k$ be a list of vectors in V. Then:

(1) If $k < n$, the list $v_1, \dots, v_k$ can **not** span V.
(2) If $k > n$, the list $v_1, \dots, v_k$ can **not** be linearly independent.

Proof:

(1) Assume that $k < n$, and suppose by contradiction that $\mathrm{Sp}(v_1, \dots, v_k) = V$. By item **(1)** of the summary on Page 82 we can delete some of the vectors from the list $v_1, \dots, v_k$ and obtain a basis $v_{i_1}, \dots, v_{i_{k'}}$ of V. Then

$$\dim V = k' \le k < n = \dim V,$$

a contradiction. (The first equality is Definition 11.b.5, applied to the basis just produced, and the second step holds because a sublist is no longer than the list it comes from.)

(2) Assume that $k > n$, and suppose by contradiction that $v_1, \dots, v_k$ are linearly independent. By item **(2)** of the summary on Page 82 we can extend this list to a basis of V. Then

$$\dim V \ge k > n = \dim V,$$

a contradiction. (The first inequality holds because that basis contains all k vectors of the list, and has $\dim V$ elements by Definition 11.b.5.) **Q.E.D.**

Proposition 12.3 (*Dimensional Constraints on Subspaces*): Let V be a finite-dimensional vector space over K. Then every linear subspace $U \subseteq V$ is also finite-dimensional. Moreover, $\dim U \le \dim V$, with the equality $\dim U = \dim V$ holding if and only if $U = V$.

Proof:

Let $n := \dim V$. We first establish that U possesses a finite basis. If $U = \{0\}$, the empty set serves as the basis and we are done. Assume $U \neq \{0\}$. We select a non-zero vector $v_1 \in U$. If $\mathrm{Sp}(v_1) = U$, we have found our basis.

If $\mathrm{Sp}(v_1) \neq U$, we choose a second vector $v_2 \in U \setminus \mathrm{Sp}(v_1)$. It follows from the properties of linear independence (Lemma 11.b.3) that the list $\{v_1, v_2\}$ is linearly independent. We proceed iteratively: after j steps, we obtain a linearly independent list $\{v_1, \dots, v_j\} \subseteq U$. If $\mathrm{Sp}(v_1, \dots, v_j) = U$, the process terminates. Otherwise, we choose $v_{j+1} \in U \setminus \mathrm{Sp}(v_1, \dots, v_j)$. By the principle that adding a vector outside the span of a linearly independent set preserves independence (again Lemma 11.b.3), the expanded list $\{v_1, \dots, v_{j+1}\}$ remains linearly independent.

Crucially, in a space V of dimension n, no linearly independent list can exceed length n; this is Theorem 12.2 **(2)**. Consequently, this algorithmic procedure must terminate in k steps, where $k \le n$. At this stage, we have a list $\{v_1, \dots, v_k\}$ that is linearly independent and spans U, thereby constituting a basis for U. This implies U is finite-dimensional and $\dim U = k \le n = \dim V$.

It remains to settle when equality holds. One direction is immediate: if $U = V$, then clearly $\dim U = \dim V$. The converse is the substantial one.

Conversely, if $\dim U = \dim V = n$, then U contains a linearly independent list of length n, namely a basis $v_1, \dots, v_n$ of U, which stays independent when read as a list in V, independence being a condition on linear combinations and not on the ambient space. By Theorem 12.1, such a list satisfies property **(a)** and is therefore also a basis for V. This implies $\mathrm{Sp}(v_1, \dots, v_n) = V$. Since $U = \mathrm{Sp}(v_1, \dots, v_n)$ by the construction of the basis, we conclude $U = V$. **Q.E.D.**

AI-Note: Theorem 12.2 is stated and proved in the notes and was missing from this transcript; it is restored above. The notes justify the two independence steps in the proof of Proposition 12.3 by citing Lemma E of the previous chapter (Lemma 11.b.3), which the transcript appealed to without naming; the citations are put back, leaving the surrounding wording as it was. The trivial direction of the equality claim in Proposition 12.3, that $U = V$ gives $\dim U = \dim V$, was also missing and is supplied. The notes restate Lemma 11.b.3 in full at this point as a recollection from the earlier lecture; it is only cross-referenced here.

The rigour pass supplied the proof of Summary 12.1. The notes give the two principles as a recollection of the previous lecture and prove neither, and nothing else in the book proves them either, although item **(2)**, the basis extension property, is used some ten times from ch. 14 onwards, occasionally credited in passing to *"the Steinitz Exchange Lemma"*. Both follow from Lemma 11.b.4 and Lemma 11.b.1 in a few lines. The derivation avoids Theorem 12.1, which stands below it and whose own proof appeals to both items; going through it would be circular.

Also: the converse half of the equality claim in Proposition 12.3 opened with *"Furthermore"*, which announces a further fact rather than the converse that the sentence above it had just set up, and the independent list it produces is a basis of U, which is what makes it independent in V as well. Four silent steps now name their reason, among them Theorem 12.2 **(2)**, which the proof of Proposition 12.3 restates in words although it stands two results above.

Row and Column Spaces (Chapter 13)

In this chapter, we bridge the abstract theory of vector spaces with the algorithmic power of matrix theory. We use Gauss elimination as a scalpel to dissect the structure of coordinate spaces.

Linear Combinations and Matrix Equations

Consider the vector space K^m over a field K. Let $w_1, \dots, w_n \in K^m$ be a collection of vectors. We begin by addressing three fundamental questions that govern our understanding of these spaces.

Question 13.1 (*The Three Fundamental Questions*):

(a) How can we systematically determine if a given vector $w \in K^m$ belongs to the span $\mathrm{Sp}(w_1, \dots, w_n)$?

(b) Can we describe the subspace $\mathrm{Sp}(w_1, \dots, w_n)$ precisely using algebraic conditions?

(c) How can we determine whether the vectors $w_1, \dots, w_n$ are linearly independent?

Answer 13.1 (*To Questions 1 and 2*): By definition, $w \in \mathrm{Sp}(w_1, \dots, w_n)$ if and only if there exist scalars $x_1, \dots, x_n \in K$ such that $w = x_1 w_1 + \cdots + x_n w_n$. We can assemble the vectors w_j as columns of an $m \times n$ matrix A. This abstract linear combination is equivalent to the matrix equation:

$$\underbrace{\begin{pmatrix} | & & | \\ w_1 & \dots & w_n \\ | & & | \end{pmatrix}}_{A} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} | \\ w \\ | \end{pmatrix}$$

Thus, w is in the span if and only if the linear system $Ax = w$ is *"consistent"* (i.e., it possesses at least one valid solution vector x). We analyze this by applying Gauss elimination on the *"augmented matrix"* $(A \mid w)$.

To answer the second question and describe the entire span precisely, we replace the specific target w with a general, variable vector $b = (b_1, \dots, b_m)^\mathsf{T}$ and reduce the augmented matrix $(A \mid b)$ to a row-reduced echelon form $(A' \mid b')$.

Two cases have to be distinguished. Suppose first that A' has no zero rows at all, so that every row of A' carries a pivot. Then the system $A'x = b'$ can be solved no matter what the right-hand side b' is, and by Theorem 6.10 the original system $Ax = b$ is then solvable as well. Since this argument applies to every $b \in K^m$, we conclude that in this case

$$\mathrm{Sp}(w_1, \dots, w_n) = K^m.$$

In the remaining case, A' does have zero rows. This row operation procedure transforms our original problem into a much simpler, equivalent linear system $A'x = b'$. As shown in Prof. Biran's notes, the resulting matrix generally has the following characteristic block structure:

$$(A' \mid b') = \left(\begin{array}{ccc|c} * & \dots & * & b'_1 \\ \vdots & \text{Rows with pivots} & \vdots & \vdots \\ * & \dots & * & b'_r \\ \hline 0 & \dots & 0 & b'_{r+1} \\ \vdots & \text{Zero rows} & \vdots & \vdots \\ 0 & \dots & 0 & b'_m \end{array}\right) \tag{13.1}$$

In this reduced block structure, any row of A' consisting entirely of zeros translates to an algebraic equation of the form $0 = b'_i$. For the overall system to be consistent (meaning a solution actually exists), it is an absolute requirement that the right-hand side of these specific equations also evaluates to zero.

In other words, the system is consistent if and only if the entries in b' corresponding to the zero-rows of A' strictly vanish. Therefore, we can concisely describe the span as the set of all vectors satisfying these constraints:

$$\mathrm{Sp}(w_1, \dots, w_n) = \left\{ b \in K^m \;\middle|\; b'_{r+1} = \cdots = b'_m = 0 \right\}.$$

Example 13.1 (*Calculating a Specific Span*): Let $v_1 = (1,1,1)^\intercal$ and $v_2 = (1,0,-1)^\intercal \in K^3$. To describe $\mathrm{Sp}(v_1, v_2)$, we set up the augmented matrix with a general vector b and row-reduce:

$$\left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 1 & 0 & b_2 \\ 1 & -1 & b_3 \end{array}\right) \xrightarrow[-1\cdot E_1 + E_3 \to E_3]{-1\cdot E_1 + E_2 \to E_2} \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & -1 & b_2 - b_1 \\ 0 & -2 & b_3 - b_1 \end{array}\right)$$

$$\xrightarrow{-1\cdot E_2 \to E_2} \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & 1 & b_1 - b_2 \\ 0 & -2 & b_3 - b_1 \end{array}\right)$$

$$\xrightarrow{2\cdot E_2 + E_3 \to E_3} \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & 1 & b_1 - b_2 \\ \hline 0 & 0 & b_1 - 2b_2 + b_3 \end{array}\right)$$

$$\xrightarrow{-1\cdot E_2 + E_1 \to E_1} \left(\begin{array}{cc|c} 1 & 0 & b_2 \\ 0 & 1 & b_1 - b_2 \\ \hline 0 & 0 & b_1 - 2b_2 + b_3 \end{array}\right)$$

The last step clears the entry above the second pivot, which is what makes the left block row-reduced and not merely echelon; it does not affect the zero row, and it has the pleasant side effect that the solution can now be read off directly, as $x_1 = b_2$ and $x_2 = b_1 - b_2$.

The system is consistent if and only if the expression corresponding to the zero-row evaluates to zero. Thus, the span is perfectly described by that single plane equation:

$$\mathrm{Sp}(v_1, v_2) = \{(b_1, b_2, b_3)^\intercal \in K^3 \mid b_1 - 2b_2 + b_3 = 0\}.$$

Answer 13.2 (*To Question 3*): The vectors are linearly independent if and only if the homogeneous equation $Ax = 0$ has only the trivial solution (i.e., $x_1 = \cdots = x_n = 0$).

This occurs if and only if the row-reduced echelon form A' has a pivot in every column. If even a single column lacks a pivot, the corresponding variable becomes *"free"* (e.g., it can be parameterized to take on any scalar value). The existence of a free variable immediately allows for non-trivial solutions, which by definition means the vectors are linearly dependent.

The machinery just developed gives a second, entirely computational proof of the two counting statements of Theorem 12.2, now specialised to the concrete space K^m. It is instructive to compare the two arguments: the earlier one was extracted from the abstract theory of bases, whereas the one below simply counts pivots.

Theorem 13.1 (*Fundamental Constraints on Coordinate Vectors*): Let $v_1, \dots, v_n \in K^m$.

(a) If $n > m$, then the vectors $v_1, \dots, v_n$ are linearly dependent.

(b) If $n < m$, then the vectors $v_1, \dots, v_n$ cannot span K^m.

Proof:

(a) Let $A = (v_1 \mid \cdots \mid v_n)$. The number of pivots in A' is bounded by the number of rows m. If $n > m$, at least $n - m$ columns are forced to be without pivots. These free variables guarantee non-trivial solutions to $Ax = 0$, hence dependence.

(b) The number of pivots is bounded by the number of columns n. If $n < m$, there are at least $m - n$ zero-rows in A'. By the logic of equation (13.1), we can easily choose a vector b such that the resulting b' has non-zero entries in those bottom rows, making the system inconsistent.

It is worth recording what the elimination produces here in slightly more detail. Applying Gauss elimination to the extended matrix $(v_1 \mid \cdots \mid v_n \mid b)$ yields a matrix whose left block is in row-reduced echelon form and whose last column splits into two parts: an upper part b', sitting next to the rows with pivots, and a lower part b'', sitting next to the zero rows. Every entry of b'' is an expression of the form $\alpha_1 b_1 + \cdots + \alpha_m b_m$ with $\alpha_1, \ldots, \alpha_m \in K$, where the coefficients α_i may of course differ from entry to entry. The span is therefore

$$\mathrm{Sp}(v_1, \ldots, v_n) = \{b \in K^m \mid b'' = 0\} \subsetneq K^m,$$

The inclusion is strict: b'' has at least one entry, and the linear form $\alpha_1 b_1 + \cdots + \alpha_m b_m$ computing it cannot have all its coefficients equal to zero, since the row operations leading to it are reversible and hence cannot annihilate a whole row. (Spelled out: those operations act on the last column alone as a map $b \mapsto \binom{b'}{b''}$ of K^m to itself, and each of the three types of Definition 6.8 can be undone by another one, so the map is a bijection. Were some coordinate of the image identically zero, the image would lie in a proper subspace of K^m and the map could not be onto.) Choosing b with that entry of b'' non-zero therefore gives an inconsistent system, and such a b exists.

Q.E.D.

Row and Column Spaces

Definition 13.2 (*Subspaces of a Matrix*): Let $A \in \mathcal{M}_{m\times n}(K)$, and denote by $u_1, \ldots, u_m \in K^n$ the rows of A and by $v_1, \ldots, v_n \in K^m$ the columns of A, so that

$$A = \begin{pmatrix} — & u_1 & — \\ & \vdots & \\ — & u_m & — \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \ldots & v_n \\ | & & | \end{pmatrix}.$$

(a) The "*column space*" $\mathrm{Cols}(A) \subseteq K^m$ is the span of the columns of A, that is, $\mathrm{Cols}(A) := \mathrm{Sp}(v_1, \ldots, v_n)$.

(b) The "*row space*" $\mathrm{Rows}(A) \subseteq K^n$ is the span of the rows of A, that is, $\mathrm{Rows}(A) := \mathrm{Sp}(u_1, \ldots, u_m)$.

Note the difference in the ambient spaces: the rows of A have n entries and are viewed as row vectors, so that $\mathrm{Rows}(A)$ is a subspace of K^n, while the columns have m entries and are viewed as column vectors, so that $\mathrm{Cols}(A)$ is a subspace of K^m.

Proposition 13.3 (*Invariance of Row Space*): Elementary row operations do not change the row space: $\mathrm{Rows}(A) = \mathrm{Rows}(A')$. More generally, if $A, B \in \mathcal{M}_{m\times n}(K)$ are row-equivalent matrices in the sense of Definition 6.9, then $\mathrm{Rows}(A) = \mathrm{Rows}(B)$.

Proof:

New rows are, simply, linear combinations of the old rows, and because elementary operations are reversible, the spans remain identical. We now carry this out for each of the three types of Definition 6.8. Let $M \in \mathcal{M}_{m\times n}(K)$ be a matrix with rows $w_1, \ldots, w_m \in K^n$.

Type 3 ($E_i \leftrightarrow E_j$). Exchanging two rows of M merely reorders the list $w_1, \ldots, w_m$, and the span of a list does not depend on the order of its members. Hence $\mathrm{Rows}(M)$ is unchanged by such an operation.

Type 1 ($\lambda \cdot E_i \to E_i$, with $\lambda \in K \setminus \{0\}$). Replacing w_i by λw_i does not change the span of the rows either: the new row lies in $\mathrm{Sp}(w_1, \dots, w_m)$, and conversely $w_i = \lambda^{-1}(\lambda w_i)$ lies in the span of the new list, which is where the hypothesis $\lambda \neq 0$ enters. So $\mathrm{Rows}(M)$ is left unchanged by such an operation as well.

Type 2 ($\lambda \cdot E_j + E_i \to E_i$, with $i \neq j$ and $\lambda \in K$). Here we claim that

$$\mathrm{Sp}(w_1, \dots, w_i, \dots, w_m) = \mathrm{Sp}(w_1, \dots, w_i + \lambda w_j, \dots, w_m),$$

where on the right-hand side the i-th member of the list has been replaced by $w_i + \lambda w_j$ and all other members are unchanged. Indeed, the inclusion

$$\mathrm{Sp}(w_1, \dots, w_i + \lambda w_j, \dots, w_m) \subseteq \mathrm{Sp}(w_1, \dots, w_i, \dots, w_m)$$

holds because every linear combination of the vectors on the left is obviously also a linear combination of $w_1, \dots, w_m$. But we also have the reverse inclusion

$$\mathrm{Sp}(w_1, \dots, w_i, \dots, w_m) \subseteq \mathrm{Sp}(w_1, \dots, w_i + \lambda w_j, \dots, w_m),$$

because $w_i = (w_i + \lambda w_j) - \lambda w_j$, and since $i \neq j$, the vector w_j itself is still one of the members of the list on the right. Consequently, whenever we write a linear combination of $w_1, \dots, w_m$, we may substitute this expression for w_i and obtain a linear combination of $w_1, \dots, w_i + \lambda w_j, \dots, w_m$. This proves the claim, and with it that an operation of Type 2 does not change $\mathrm{Rows}(M)$ either.

Finally, let A and B be row-equivalent. By Definition 6.9, there exists a finite sequence of matrices

$$A = A_0 \longrightarrow A_1 \longrightarrow \cdots \longrightarrow A_{r-1} \longrightarrow A_r = B$$

such that A_{i+1} is obtained from A_i by a single elementary row operation for every $0 \leq i \leq r-1$. By what we have just proved, each of these steps leaves the row space unchanged, and therefore

$$\mathrm{Rows}(A) = \mathrm{Rows}(A_0) = \mathrm{Rows}(A_1) = \cdots = \mathrm{Rows}(A_r) = \mathrm{Rows}(B).$$

Q.E.D.

Important remark 13.1: Row operations **do** change the column space, as seen by $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. However, they **preserve** the linear relations between columns: $\sum \alpha_i w_i = 0 \iff \sum \alpha_i w_i' = 0$.

The second assertion of Important remark 13.1 carries the entire weight of Theorem 13.5 below, which transfers a statement about the echelon form back to the original matrix. The notes do not prove it, so we record it here, together with the two consequences in which it is actually used.

AI-Lemma 13.1 (*Row Operations Preserve the Relations Between Columns*): Let $A, A' \in \mathcal{M}_{m \times n}(K)$ be row-equivalent matrices, and denote by $w_1, \dots, w_n$ the columns of A and by $w_1', \dots, w_n'$ the columns of A'. Then, for all scalars $\alpha_1, \dots, \alpha_n \in K$,

$$\alpha_1 w_1 + \cdots + \alpha_n w_n = 0 \quad \text{if and only if} \quad \alpha_1 w_1' + \cdots + \alpha_n w_n' = 0.$$

Consequently, for every subset of indices $S \subseteq \{1, \dots, n\}$:

- **(a)** the columns $(w_i)_{i \in S}$ are linearly independent if and only if the columns $(w_i')_{i \in S}$ are linearly independent;
- **(b)** for every index j, we have $w_j \in \mathrm{Sp}\left((w_i)_{i \in S}\right)$ if and only if $w_j' \in \mathrm{Sp}\left((w_i')_{i \in S}\right)$.

Proof:

Writing $x := (\alpha_1, \dots, \alpha_n)^\mathsf{T}$, the equation $\alpha_1 w_1 + \cdots + \alpha_n w_n = 0$ says precisely that $Ax = 0$, and likewise $\alpha_1 w_1' + \cdots + \alpha_n w_n' = 0$ says precisely that $A'x = 0$. Performing on $(A \mid 0)$ the very sequence of elementary row operations that leads from A to A' produces $(A' \mid 0)$, since a zero column stays a zero column under any row operation. So the two augmented matrices are row-equivalent, and Theorem 6.10 gives the two homogeneous systems the same solution set. This is the asserted equivalence.

For **(a)**, a linear relation among the columns indexed by S is the same thing as a linear relation among all n columns whose coefficients vanish outside S. The equivalence just proved matches such relations for A with such relations for A' coefficient by coefficient, so one family admits a non-trivial relation exactly when the other does.

For **(b)**, an expression $w_j = \sum_{i\in S} \gamma_i w_i$ is the relation $\sum_{i\in S} \gamma_i w_i - w_j = 0$, which again has vanishing coefficients outside $S \cup \{j\}$. Applying the equivalence to it, and reading the resulting relation in the other direction, gives $w'_j = \sum_{i\in S} \gamma_i w'_i$, with the same coefficients γ_i; and symmetrically.

Q.E.D.

Both spaces now have a dimension, and these two numbers deserve names of their own.

Definition 13.4 (*Row Rank and Column Rank*): Let $A \in \mathcal{M}_{m\times n}(K)$. We define the "*row rank*" and the "*column rank*" of A as

$$\operatorname{rank}_{\text{row}}(A) := \dim \operatorname{Rows}(A), \qquad \operatorname{rank}_{\text{col}}(A) := \dim \operatorname{Cols}(A).$$

Remark 13.1: The notes flag right here that these two numbers always coincide, $\operatorname{rank}_{\text{row}}(A) = \operatorname{rank}_{\text{col}}(A)$, and postpone the proof. In this chapter the equality already falls out of the two basis theorems below, in Theorem 13.9; we return to it in Theorem 17.c.1, where it is proved once more, this time from the theory of linear maps and without any appeal to Gauss elimination.

Constructing Bases

Theorem 13.5 (*Basis for the Column Space*): The columns of A that correspond to the *"pivot columns"* of its row-reduced echelon form B constitute a basis for $\operatorname{Cols}(A)$.

Proof:

By the principle that row operations preserve the linear dependence relations between columns (AI-Lemma 13.1), we have:

$$\sum_{i=1}^{n} \alpha_i w_i = 0 \iff \sum_{i=1}^{n} \alpha_i w'_i = 0.$$

Thus, a set of columns in A is linearly independent (or spans the column space) if and only if the corresponding set of columns in B possesses the same property in $\operatorname{Cols}(B)$.

Consider the row-reduced echelon matrix $B \in \mathcal{M}_{m\times n}(K)$ with r pivots. We visualize its block structure and the specific indices of its pivot columns $j_1, j_2, \ldots, j_r$:

$$B = \begin{array}{c} \begin{array}{cccccccc} & \text{col } j_1 & & \text{col } j_2 & & \text{col } j_r & & \\ & \downarrow & & \downarrow & & \downarrow & & \end{array} \\ \left(\begin{array}{cccccccc} \cdots & \mathbf{1} & * & \mathbf{0} & * & \mathbf{0} & * & \cdots \\ \cdots & \mathbf{0} & \cdots & \mathbf{1} & \cdots & \mathbf{0} & * & \cdots \\ & \vdots & & \vdots & \ddots & \vdots & \vdots & \\ \cdots & \mathbf{0} & \cdots & \mathbf{0} & \cdots & \mathbf{1} & * & \cdots \\ \hline & 0 & & 0 & & 0 & 0 & \cdots \\ & \vdots & & \vdots & & \vdots & \vdots & \ddots \\ & 0 & & 0 & & 0 & 0 & \cdots \end{array}\right) \end{array} \begin{array}{l} \\ \\ \leftarrow 1 \\ \leftarrow 2 \\ \vdots \\ \leftarrow r \\ \\ \\ \\ \end{array} \tag{13.2}$$

In this echelon form, the pivot columns are precisely the first r standard basis vectors of K^m:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots, \quad e_r = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

where for each e_i, the entry 1 is located at the i-th row. Because B is in row-reduced echelon form, every entry in every column w'_j that lies below the r-th row must be zero. This implies that:

$$w'_j \in K^r \times \{0\}^{m-r} = \left\{(x_1, \ldots, x_r, 0, \ldots, 0)^\mathsf{T} \in K^m\right\}.$$

Since the standard basis vectors $e_1, \ldots, e_r$ are clearly linearly independent and they span the subspace $K^r \times \{0\}^{m-r}$, we establish the following chain of inclusions:

$$K^r \times \{0\}^{m-r} = \operatorname{Sp}(e_1, \ldots, e_r) \subseteq \operatorname{Cols}(B) \subseteq K^r \times \{0\}^{m-r}.$$

This double inclusion forces the equality $\operatorname{Cols}(B) = \operatorname{Sp}(e_1, \ldots, e_r) = K^r \times \{0\}^{m-r}$. Consequently, the pivot columns of B form a basis for its column space. By the preservation of linear relations, the corresponding columns in the original matrix A form a basis for $\operatorname{Cols}(A)$: they are linearly independent by part **(a)** of AI-Lemma 13.1, and every other column of A lies in their span by part **(b)**, since this is so for B. **Q.E.D.**

Theorem 13.6 (*Basis for the Row Space*): The non-zero rows of the row-reduced echelon matrix B form a basis for $\operatorname{Rows}(A)$.

Proof:

By Proposition 13.3, $\operatorname{Rows}(A) = \operatorname{Rows}(B)$. Let r be the number of pivots in B. The non-zero rows of B are $w'_1, \ldots, w'_r$, the zero rows of a row-reduced echelon matrix standing below them. They span $\operatorname{Rows}(B)$: by Definition 13.2 that space is spanned by **all** rows of B, and discarding the zero ones changes no linear combination.

To show linear independence, consider the indices of the pivot columns $j_1, \ldots, j_r$. For each row $i \in \{1, \ldots, r\}$, the entry in the j_i-th column is 1, while in all other rows $k \neq i$, the entry in the j_i-th column is 0. Suppose $\sum_{i=1}^r \alpha_i w'_i = 0$. Looking at the j_k-th component of this vector sum, we have:

$$\sum_{i=1}^r \alpha_i B_{i,j_k} = \alpha_k \cdot 1 + \sum_{i \neq k} \alpha_i \cdot 0 = \alpha_k = 0.$$

Since this holds for all k, all coefficients must be zero. Thus, the non-zero rows are linearly independent and form a basis. **Q.E.D.**

Corollary 13.7 (*Rank Invariance*): The number of pivots is independent of the sequence of row operations, as it always equals $\dim \operatorname{Rows}(A)$.

Proof:

By Theorem 13.6, the non-zero rows of any row-reduced echelon form B of A form a basis for $\operatorname{Rows}(A)$. The number of such rows is exactly the number of pivots. Since the dimension of a vector space is a fixed property, the number of pivots must be the same regardless of the specific row-reduction process used to reach an echelon form. **Q.E.D.**

The row basis theorem is not merely a structural statement; read in the right direction, it is a recipe. It answers the practical question of how to produce a basis for a subspace that has been handed to us as a span.

Corollary 13.8 (*Algorithm for Finding a Basis*): Let $U := \mathrm{Sp}(u_1, \dots, u_m) \subseteq K^n$ be the subspace spanned by a list of vectors $u_1, \dots, u_m \in K^n$, which we regard as row vectors. Form the matrix

$$A := \begin{pmatrix} — & u_1 & — \\ & \vdots & \\ — & u_m & — \end{pmatrix} \in \mathcal{M}_{m \times n}(K),$$

and apply Gauss elimination to A until a row-reduced echelon matrix M is reached. Then the non-zero rows of M form a basis for U.

Proof:

By construction, $U = \mathrm{Rows}(A)$. The matrix M is row-equivalent to A, so $\mathrm{Rows}(A) = \mathrm{Rows}(M)$ by Proposition 13.3, and the non-zero rows of M form a basis for that space by Theorem 13.6.

Q.E.D.

Theorem 13.9 (*The Rank Equality*): For any $A \in \mathcal{M}_{m \times n}(K)$, $\dim \mathrm{Rows}(A) = \dim \mathrm{Cols}(A)$, or equivalently, in the terminology of Definition 13.4, $\mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{col}}(A)$. This shared dimension is the "*rank*" of A, denoted $\mathrm{rank}(A)$.

Proof:

Let r be the number of pivots in the RREF of A. By Theorem 13.5, $\dim \mathrm{Cols}(A) = r$. By Theorem 13.6, $\dim \mathrm{Rows}(A) = r$. Thus, the dimensions are exactly equal. **Q.E.D.**

The Transpose of a Matrix

The row space and the column space of a matrix are built from two different readings of the same array of scalars. The operation that turns one reading into the other is the transpose, which we introduce here and will use constantly from now on.

Definition 13.10 (*Transpose of a Matrix*): Let $A = (a_{ij}) \in \mathcal{M}_{m \times n}(K)$. The "*transposed matrix*" of A is the matrix $A^\mathsf{T} := (b_{ij}) \in \mathcal{M}_{n \times m}(K)$ whose entries are given by

$$b_{ij} := a_{ji} \qquad \text{for all } 1 \le i \le n,\ 1 \le j \le m.$$

Notation 13.1: The transposed matrix is sometimes also denoted by A^t.

Example 13.2 (*Transposing a Rectangular Matrix*): Transposing turns the rows of a matrix into columns and its columns into rows:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}^\mathsf{T} = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}.$$

Proposition 13.11 (*Simple Properties of the Transpose*): For all matrices $A, B \in \mathcal{M}_{m \times n}(K)$ and every scalar $\lambda \in K$, the following hold:

(a) $(A + B)^\mathsf{T} = A^\mathsf{T} + B^\mathsf{T}$.
(b) $(\lambda A)^\mathsf{T} = \lambda \cdot A^\mathsf{T}$.
(c) $(A^\mathsf{T})^\mathsf{T} = A$.
(d) $\mathrm{Rows}(A^\mathsf{T}) = \mathrm{Cols}(A)$ and $\mathrm{Cols}(A^\mathsf{T}) = \mathrm{Rows}(A)$.

Proof:

(a) The entry of $(A+B)^\mathsf{T}$ in position (i,j) is the entry of $A+B$ in position (j,i), namely $a_{ji}+b_{ji}$, which is exactly the entry of $A^\mathsf{T}+B^\mathsf{T}$ in position (i,j).

(b) Likewise, the entry of $(\lambda A)^\mathsf{T}$ in position (i,j) is λa_{ji}, which is the entry of $\lambda \cdot A^\mathsf{T}$ in position (i,j).

(c) Transposing twice returns the entry in position (i,j) to a_{ij}.

(d) By Definition 13.10, the i-th row of A^T consists of the entries $a_{1i},\dots,a_{mi}$, that is, it is the i-th column of A written as a row vector. The rows of A^T are therefore precisely the columns of A, so the two spans agree: $\mathrm{Rows}(A^\mathsf{T}) = \mathrm{Cols}(A)$. The second identity follows by applying the first one to A^T and using **(c)**.

Q.E.D.

Remark 13.2: Statement **(d)** is the reason the transpose is the natural bookkeeping device for the two ranks. Taking dimensions in Proposition 13.11 **(d)** gives

$$\mathrm{rank}_{\mathrm{row}}(A^\mathsf{T}) = \mathrm{rank}_{\mathrm{col}}(A), \qquad \mathrm{rank}_{\mathrm{col}}(A^\mathsf{T}) = \mathrm{rank}_{\mathrm{row}}(A),$$

so every statement about row rank is a statement about the column rank of the transposed matrix, and conversely.

AI-Note: Four pieces of the notes were missing from this transcript and are restored above: the case distinction in the answer to Questions 1 and 2, where an A' without zero rows forces $\mathrm{Sp}(w_1,\dots,w_n) = K^m$; the definition of row rank and column rank (Definition 13.4), used from Theorem 17.c.1 onwards but never introduced; Corollary 13.8, which the notes state as an algorithm; and the whole closing section on the transpose, whose notation A^T the transcript already used in this chapter and in fourteen later ones without ever defining it. The proof of Proposition 13.3 had been compressed to a single sentence; the notes treat the three types of row operation separately and prove the Type 2 case by double inclusion, so that argument is restored around the sentence that was there. Theorem 13.1 is the computational counterpart of Theorem 12.2, which is how the notes present it; the description of the span by the vanishing of b'' is put back into its proof.

Two things go beyond the notes. AI-Lemma 13.1 is not in Biran's text: the transcript asserts the preservation of linear relations between columns in an important remark and then rests Theorem 13.5 on it, so it is stated and proved here as an AI-Lemma, on the strength of Theorem 6.10. And the notes defer $\mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{col}}(A)$ to a later lecture, while this chapter already reaches it in Theorem 13.9; the argument is sound and is kept, with the relation to Theorem 17.c.1 recorded in a remark. In the worked span computation on Page 86 the elimination now carries Prof. Biran's arrow labels and includes the normalisation step $-1 \cdot E_2 \to E_2$, without which the displayed chain never reaches a row-reduced echelon matrix.

The rigour pass found nothing false in this chapter. What it did was finish three arguments that stopped just short. The elimination in Page 86 still ended one operation before a row-reduced echelon form, since the entry above the second pivot was never cleared, although the paragraph setting up the method says the reduction goes that far; the step is added, and it lets the solution be read off. In the proof of Theorem 13.1 **(b)**, the strictness of the inclusion rested on row operations being *"reversible and hence unable to annihilate a whole row"*, which is right but leaves the reader to reconstruct why: those operations act on the last column as a bijection of K^m, and a coordinate that vanished identically would confine the image to a proper subspace. And in the proof of Theorem 13.6, that the non-zero rows of B span $\mathrm{Rows}(B)$ was credited to the definition, whereas Definition 13.2 spans **all** the rows; what is needed is that discarding the zero ones changes no linear combination.

The important remark on row operations and columns is now labelled Important remark 13.1, since the sentence introducing AI-Lemma 13.1 refers back to its second assertion.

One thing was left alone deliberately. The rank of a matrix, $\operatorname{rank}(A)$, is named inside the statement of Theorem 13.9 rather than in a definition of its own. That is unusual, but the notion is not thereby uncitable, Theorem 13.9 being numbered, and nothing later in the book appeals to a definition of matrix rank that does not exist; the two places that say *"by definition,* $\operatorname{rank}(\cdot)$*"* in ch. 17c mean the rank of a linear map, which Definition 15.b.13 supplies.

Sums of Vector Spaces (Chapter 14)

Definition 14.1 (*Sum of Subspaces*): Let V be a vector space over K, and let $U, W \subseteq V$ be linear subspaces. We define the "*sum*" of U and W, denoted by $U + W$, as the set:

$$U + W := \{u + w \mid u \in U, w \in W\}.$$

In a similar way, we can define sums of more than two subspaces. If $U_1, \ldots, U_r \subseteq V$ are subspaces, we define their total sum as:

$$U_1 + \cdots + U_r := \{u_1 + \cdots + u_r \mid u_1 \in U_1, \ldots, u_r \in U_r\}.$$

Proposition 14.2 (*Properties of Sums*): Let $U, W \subseteq V$ be subspaces. Then, the following properties hold:

(a) $U + W = \operatorname{Sp}(U \cup W)$. In particular, $U + W \subseteq V$ is a subspace, and U and W are themselves subspaces of $U + W$.

(b) If U and W are finite-dimensional, then so is $U + W$. Moreover, one can systematically find a basis for $U + W$ by following these sequential steps: first, choose a basis $(v_1, \ldots, v_k)$ for $U \cap W$ (where $k = \dim(U \cap W)$). Second, extend this to a basis $(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k})$ for U (where $\ell = \dim U$). Third, extend the initial intersection basis to a basis $(v_1, \ldots, v_k, w_1, \ldots, w_{m-k})$ for W (where $m = \dim W$). Finally, the combined elements

$$(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k}, w_1, \ldots, w_{m-k})$$

form a basis for $U + W$.

(c) If U and W are finite-dimensional, the dimensions satisfy the "*dimension formula*"

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof:

(a) Clearly $U + W \subseteq \operatorname{Sp}(U \cup W)$, by the characterisation of $\operatorname{Sp}(U \cup W)$ as the set of all linear combinations of elements from $U \cup W$ (Lemma 8.5, applicable since $U \cup W$ contains 0_V and is in particular non-empty). But we also have $\operatorname{Sp}(U \cup W) \subseteq U + W$. Indeed, if $v \in \operatorname{Sp}(U \cup W)$, then

$$v = \underbrace{\sum_{i=1}^{s} a_i u_i}_{\in U} + \underbrace{\sum_{j=1}^{r} b_j w_j}_{\in W},$$

with $a_i, b_j \in K$, $u_i \in U$ and $w_j \in W$, which implies that $v \in U + W$. This proves $\operatorname{Sp}(U \cup W) \subseteq U + W$. Putting all together, we have $U + W = \operatorname{Sp}(U \cup W)$. The rest of the statements in **(a)** follow immediately.

(b) That the subspace $U + W$ is generated by the combined vectors

$$(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k}, w_1, \ldots, w_{m-k})$$

is not quite immediate from the definitions, and the notes leave it to the reader, so it is set as Exercise 14.1 below and solved at the end of the chapter. It remains to show that these vectors are linearly independent.

Suppose there exists a linear combination equal to the zero vector:

$$\alpha_1 v_1 + \cdots + \alpha_k v_k + \beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k} + \gamma_1 w_1 + \cdots + \gamma_{m-k} w_{m-k} = 0. \tag{14.1}$$

We can rearrange this equation to isolate the terms belonging to U and W:

$$\underbrace{\beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k}}_{\in U} = \underbrace{-\alpha_1 v_1 - \cdots - \alpha_k v_k - \gamma_1 w_1 - \cdots - \gamma_{m-k} w_{m-k}}_{\in W}.$$

Let us denote this common vector by x. Since the left side is entirely in U and the right side is entirely in W, we have $x \in U$ and $x \in W$, which implies $x \in U \cap W$.

Because $(v_1, \dots, v_k)$ is a basis for $U \cap W$, there must exist scalars $\delta_1, \dots, \delta_k \in K$ such that:

$$x = \delta_1 v_1 + \cdots + \delta_k v_k.$$

Equating this with our original expression for x in U, we obtain:

$$\beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k} = \delta_1 v_1 + \cdots + \delta_k v_k,$$

or, after moving everything to one side,

$$\sum_{i=1}^{\ell-k} \beta_i u_i - \sum_{j=1}^{k} \delta_j v_j = 0.$$

However, the vectors $(v_1, \dots, v_k, u_1, \dots, u_{\ell-k})$ form a basis for U, meaning they are strictly linearly independent. Consequently, all coefficients in this sum must be zero. In particular, $\beta_1 = \cdots = \beta_{\ell-k} = 0$.

Substituting these zero values back into our initial equation (14.1), the terms involving u_i vanish, leaving:

$$\alpha_1 v_1 + \cdots + \alpha_k v_k + \gamma_1 w_1 + \cdots + \gamma_{m-k} w_{m-k} = 0.$$

Since the vectors $(v_1, \dots, v_k, w_1, \dots, w_{m-k})$ form a basis for W, they are linearly independent. This immediately implies that $\alpha_1 = \cdots = \alpha_k = 0$ and $\gamma_1 = \cdots = \gamma_{m-k} = 0$.

Because all coefficients α_i, β_i, and γ_j have been forced to zero, the combined family of vectors is linearly independent, and therefore constitutes a valid basis for $U + W$.

Statement **(c)** now follows by simply counting the basis vectors produced in **(b)**. The basis consists of the k vectors v_i, the $\ell - k$ vectors u_i, and the $m - k$ vectors w_j, and therefore,

$$\begin{aligned}
\dim(U + W) &= k + (\ell - k) + (m - k) \\
&= \ell + m - k \\
&= \dim U + \dim W - \dim(U \cap W).
\end{aligned}$$

Q.E.D.

Exercise 14.1 (*The Combined Family Spans the Sum*): The proof above verifies that the family

$$(v_1, \dots, v_k, u_1, \dots, u_{\ell-k}, w_1, \dots, w_{m-k})$$

is linearly independent. Check that it also spans $U + W$, which the notes leave to the reader.

Corollary 14.3 (*Equivalent Conditions for Trivial Intersection*): Let V be a vector space, and let $U, W \subseteq V$ be two finite-dimensional linear subspaces. The following statements are equivalent:

- **(a)** $\dim(U + W) = \dim U + \dim W$.
- **(b)** $\dim(U \cap W) = 0$.
- **(c)** $U \cap W = \{0_V\}$.
- **(d)** For every $v \in U + W$, there exist unique vectors $u \in U$ and $w \in W$ such that $v = u + w$.
- **(e)** The equality $0 = u + w$ with $u \in U$ and $w \in W$ implies that $u = 0$ and $w = 0$.

Proof:

The equivalence **(a)** $\iff$ **(b)** follows from the previous Proposition 14.2 **(c)**, whose dimension formula reads $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$. And **(b)** $\iff$ **(c)** holds because there is only one linear subspace of dimension 0, namely the space $\{0_V\}$: a subspace of dimension 0 has the empty set as a basis, and $\mathrm{Sp}(\emptyset) = \{0_V\}$ by Remark 8.3, which is the reading of Remark 11.2 in the other direction.

The remaining three statements are tied to **(c)** by the cycle of implications **(c)** $\implies$ **(d)** $\implies$ **(e)** $\implies$ **(c)**.

(c) $\implies$ (d): Suppose that $v \in U + W$ can be written in two ways: $v = u + w = u' + w'$, where $u, u' \in U$ and $w, w' \in W$. Rearranging this equation yields $u - u' = w' - w$. The left side is in U and the right side is in W, which means the vector $u - u'$ resides in the intersection $U \cap W$. Since we assume $U \cap W = \{0_V\}$, it follows that $u - u' = 0$ and $w' - w = 0$. Therefore, $u = u'$ and $w = w'$, proving uniqueness.

(d) $\implies$ (e): Consider the zero vector, which can obviously be written as $0 = 0_U + 0_W$. If we also have $0 = u + w$ for some $u \in U$ and $w \in W$, the assumed uniqueness in statement **(d)** immediately forces $u = 0$ and $w = 0$.

(e) $\implies$ (c): Let $x \in U \cap W$. We can express the zero vector as $0 = x + (-x)$. Since $x \in U \cap W$, we know $x \in U$ and $-x \in W$. If statement **(e)** holds, this equality strictly implies $x = 0$. Consequently, the only element in the intersection is the zero vector, so $U \cap W = \{0_V\}$.

Q.E.D.

Definition 14.4 (*Complement of a Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. A subspace $W \subseteq V$ is called a "*complement*" of U if it satisfies the following two conditions:

$$U + W = V \quad \text{and} \quad U \cap W = \{0_V\}.$$

Remark 14.1: Note that if W is a complement of U, then U and W satisfy all of the five equivalent statements derived from Corollary 14.3.

Proposition 14.5 (*Existence of a Complement*): Let V be a finite-dimensional vector space, and let $U \subseteq V$ be a subspace. Then, there exists a subspace $W \subseteq V$ which is a complement to U.

Proof:

Note first that U is finite-dimensional, by Proposition 12.3. Choose a basis $\mathcal{B}_U = (u_1, \dots, u_\ell)$ for U, where $\ell = \dim U$. By the basis extension property, item **(2)** of Summary 12.1, we can extend this to a full basis for V:

$$\mathcal{B}_V = (u_1, \dots, u_\ell, w_1, \dots, w_m),$$

where $\ell + m = \dim V$. We define $W := \mathrm{Sp}(w_1, \dots, w_m)$.

We claim that W is a valid complement of U. Indeed, since the combined basis spans the entirety of V, it is clear that $U + W = V$: by Proposition 14.2 **(a)**, $U + W = \mathrm{Sp}(U \cup W)$, and this span contains all of $u_1, \dots, u_\ell, w_1, \dots, w_m$, hence contains $\mathrm{Sp}(\mathcal{B}_V) = V$. To see that $U \cap W = \{0\}$, let $x \in U \cap W$. Because $x \in U$, we can write $x = \sum_{i=1}^{\ell} a_i u_i$. Because $x \in W$, we can also write $x = \sum_{j=1}^{m} b_j w_j$. Equating these expressions yields:

$$\sum_{i=1}^{\ell} a_i u_i = \sum_{j=1}^{m} b_j w_j \implies \sum_{i=1}^{\ell} a_i u_i - \sum_{j=1}^{m} b_j w_j = 0.$$

Since $\mathcal{B}_V$ is a basis for V, all of its constituent vectors are strictly linearly independent. This forces all scalar coefficients to be zero: $a_i = 0$ for all i and $b_j = 0$ for all j. Thus, $x = 0$, proving that $U \cap W = \{0\}$.

Q.E.D.

Important remark 14.1: If $\{0_V\} \subsetneq U \subsetneq V$, then there are many different subspaces $W \subseteq V$ which are complements of U. In general, there is no canonical choice for a complement W. (Both strict inclusions are needed. The subspace $U = \{0_V\}$ has the single complement $W = V$, since $\{0_V\} + W = W$, and $U = V$ has the single complement $W = \{0_V\}$; in either case there is nothing to choose.)

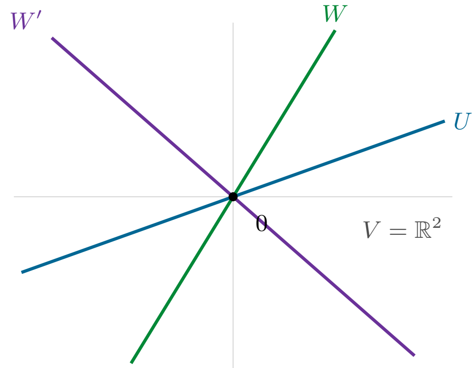


Figure 14.1: Two complements of the same subspace. Inside $V = \mathbb{R}^2$, take for U any line through the origin. Every other line through the origin meets U only in 0 and together with U spans the whole plane, so both W and W' are complements of U, and so is any of the infinitely many further lines through the origin.

Solutions to the Exercises

Proof that the Combined Family Spans $U + W$ (on Page 95):

Write $\mathcal{S} := (v_1, \dots, v_k, u_1, \dots, u_{\ell-k}, w_1, \dots, w_{m-k})$ for the combined family.

Every member of $\mathcal{S}$ lies in $U + W$: the vectors v_i and u_i lie in U, the vectors w_j lie in W, and both U and W are contained in $U + W$ by Proposition 14.2 **(a)**. Since $U + W$ is a subspace, it follows that $\mathrm{Sp}(\mathcal{S}) \subseteq U + W$.

For the reverse inclusion, let $x \in U + W$ and write $x = u + w$ with $u \in U$ and $w \in W$. Because $(v_1, \dots, v_k, u_1, \dots, u_{\ell-k})$ is a basis for U, there are scalars with

$$u = \sum_{i=1}^{k} \alpha_i v_i + \sum_{i=1}^{\ell-k} \beta_i u_i,$$

and because $(v_1, \dots, v_k, w_1, \dots, w_{m-k})$ is a basis for W, there are scalars with

$$w = \sum_{i=1}^{k} \alpha'_i v_i + \sum_{j=1}^{m-k} \gamma_j w_j.$$

Adding the two expressions writes x as a linear combination of the members of $\mathcal{S}$, the coefficient of v_i being $\alpha_i + \alpha'_i$. Hence $U + W \subseteq \mathrm{Sp}(\mathcal{S})$, and the two inclusions give $\mathrm{Sp}(\mathcal{S}) = U + W$. **Q.E.D.**

AI-Note: Restored from the notes: the proof of statement **(a)** of Proposition 14.2, which the transcript skipped with the words *"We will explicitly prove statement (b)"*; the two dimension criteria in Corollary 14.3; and the closing remark on Page 96, with the sketch that accompanies it in the notes redrawn as Figure 14.1.

Corollary 14.3 deserves a word. The notes state **five** equivalent conditions, and the transcript kept only four of them: the two dimension criteria, $\dim(U+W) = \dim U + \dim W$ and $\dim(U \cap W) = 0$, were lost, and in their place stood a statement with no counterpart in the notes, namely that every $x \in U \cap W$ can be written uniquely as $x = u + w$. That statement is true, but it is a garbled duplicate of statement **(e)**, and it has been dropped in favour of Biran's list. That something was lost here is visible in the transcript itself: the remark following Definition 14.4 already spoke of *"the five equivalent statements"*. The hypothesis that U and W be finite-dimensional is the notes' own and is needed for the two dimension criteria to mean anything; the transcript had dropped it as well.

The notes prove the independence claim in Proposition 14.2 **(b)** by eliminating the W-coefficients first, whereas the transcript isolates the U-part first. The two routes are mirror images of one another and equally correct, so the transcript's version is kept. The notes also mark the spanning half of that statement as an exercise, *"exc. check this!"*; it is stated as an exercise on Page 95 and solved at the end of the chapter.

The rigour pass corrected one statement and named five reasons. Important remark 14.1 said that a subspace $U \subsetneq V$ has many complements. It needs $U \neq \{0_V\}$ as well: the zero subspace has $W = V$ as its only complement, since $\{0_V\} + W = W$, and there is nothing to choose. The figure beside the remark already assumed as much, taking for U a line rather than a point. Both strict inclusions are now in the statement.

The proof of Proposition 14.2 **(b)** opened by saying that $U + W$ is generated by the combined family *"by definition"*. It is not: that is precisely what Exercise 14.1 asks the reader to check, as the notes do, and the sentence now says so instead of claiming the point.

Proposition 14.5 extends a basis of U to one of V *"by the Steinitz Exchange Lemma"*. The result being used is the basis extension property, item **(2)** of Summary 12.1, which is what the citation now names; that item is proved in ch. 12. The same proof also needs U to be finite-dimensional before it can speak of a basis of U, which is Proposition 12.3. Also cited now: Lemma 8.5 in **(a)**, Proposition 14.2 **(a)** for the step $U + W = V$, and Remark 8.3 for there being only one subspace of dimension zero. One index clash was tidied: the closing sentence of the proof of **(b)** listed the coefficients as $\alpha_i, \beta_j, \gamma_k$, and k is the dimension of $U \cap W$ throughout the chapter.

Linear Maps (Chapter 15)

Introduction to Linear Maps (Section 15.a)

Linear maps, also known as linear transformations, are homomorphisms of vector spaces. They are the primary objects of study in linear algebra, as they preserve the underlying algebraic structure of vector spaces.

Definition 15.a.1 (*Linear Map*): Let V and W be vector spaces over a field K. A map $T : V \to W$ is called a "*linear map*" if it satisfies the following two conditions:

(a) For all $v_1, v_2 \in V$, $T(v_1 + v_2) = T(v_1) + T(v_2)$.

(b) For all $\alpha \in K$ and $v \in V$, $T(\alpha \cdot v) = \alpha \cdot T(v)$.

Remark 15.1 (*Notation and Terminology*): In many contexts, we omit the parentheses and write Tv instead of $T(v)$. The conditions outlined in Definition 15.a.1 essentially state that $T : V \to W$ is a map that respects the operations defined on the vector spaces; specifically, the addition and scalar multiplication in V are mapped perfectly to the corresponding operations in W.

We denote the set of all linear maps from V to W by $\operatorname{Hom}_K(V, W)$, or simply $\operatorname{Hom}(V, W)$. While some authors use alternative notations such as $\operatorname{Lin}(V, W)$ or $\hom_K(V, W)$, $\operatorname{Hom}_K(V, W)$ remains the standard academic convention.

Exercise 15.1 (*Linearity Criterion*): Prove that a map $T : V \to W$ is linear if and only if for all $\alpha_1, \alpha_2 \in K$ and $v_1, v_2 \in V$:

$$T(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 T(v_1) + \alpha_2 T(v_2).$$

Example 15.1 (*Fundamental Examples of Linear Maps*):

(a) The identity map $\mathrm{id}_V : V \to V$, defined by $\mathrm{id}_V(v) := v$ for all $v \in V$.

(b) The zero map $0 : V \to W$, defined by $0(v) := 0_W$ for all $v \in V$.

Clearly, these initial examples are elementary, yet they serve as vital benchmarks.

(c) The derivative map $D : K[x] \to K[x]$. For any polynomial $p(x) = \sum_{k=0}^{n} a_k x^k$, we define its image as:

$$D(p(x)) := \sum_{k=1}^{n} k a_k x^{k-1}.$$

This map is linear, as the derivative of a linear combination of polynomials is simply the same linear combination of their derivatives.

(d) Consider the vector space $C([a, b])$ consisting of continuous functions $f : [a, b] \to \mathbb{R}$. The definite integral defines a linear map $T : C([a, b]) \to \mathbb{R}$, given by:

$$T(f) := \int_a^b f(x)\, dx.$$

(e) The shift map $S : K^\infty \to K^\infty$, defined on infinite sequences as:

$$S((a_1, a_2, a_3, \dots)) := (a_2, a_3, a_4, \dots).$$

Example 15.2 (*Matrix Transformation*): This is a central example in finite-dimensional linear algebra. Let $A \in \mathcal{M}_{m\times n}(K)$ be a matrix. We define a map $T_A : K^n \to K^m$ via matrix-vector multiplication:

$$T_A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} := A \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Recall from matrix arithmetic that $A \cdot (\alpha \cdot x) = \alpha \cdot (A \cdot x)$ and $A \cdot (x + y) = A \cdot x + A \cdot y$. Consequently, T_A is a linear map.

Proposition 15.a.2 (*Basic Properties of Linear Maps*): Let V and W be vector spaces over a field K, and let $T : V \to W$ be a linear map. Then, the following properties hold:

(a) For all $n \in \mathbb{N}$, $\alpha_1, \ldots, \alpha_n \in K$, and $v_1, \ldots, v_n \in V$, we have:
$$T\left(\sum_{i=1}^{n} \alpha_i v_i\right) = \sum_{i=1}^{n} \alpha_i T(v_i).$$

(b) $T(0_V) = 0_W$.

Proof:

(a) The first statement follows directly from the definition of linearity, Definition 15.a.1, by applying induction on the number of vectors n.

(b) To prove the second statement, we evaluate the image of the zero vector:
$$0_W = T(0_V) - T(0_V) = T(0_V - 0_V) = T(0_V).$$
Therefore, any linear map must map the zero vector of the domain directly to the zero vector of the codomain.

Q.E.D.

Theorem 15.a.3 (*Existence and Uniqueness of Linear Maps on a Basis*): Let V and W be vector spaces over K with V finite-dimensional, and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V. Given any arbitrary sequence of vectors $w_1, \ldots, w_n \in W$, there exists a unique linear map $T : V \to W$ such that:
$$T(v_i) = w_i, \quad \forall\, 1 \le i \le n.$$

Proof:

Existence: Let $v \in V$ be an arbitrary vector. Since $\mathcal{B}$ is a basis, there exist unique scalars $a_1, \ldots, a_n \in K$ such that $v = \sum_{i=1}^{n} a_i v_i$. We define the map T by setting:
$$T(v) := \sum_{i=1}^{n} a_i w_i. \tag{15.1}$$
Because the coordinates a_i are uniquely determined for each v, the map T is well-defined.

We claim that T is linear. Indeed, let $v, v' \in V$. There exist $a_1, \ldots, a_n \in K$ with $v = a_1 v_1 + \cdots + a_n v_n$ and $a'_1, \ldots, a'_n \in K$ with $v' = a'_1 v_1 + \cdots + a'_n v_n$, so that
$$v + v' = (a_1 + a'_1)v_1 + \cdots + (a_n + a'_n)v_n.$$
By our definition of T, this gives
$$T(v+v') = (a_1+a'_1)w_1+\cdots+(a_n+a'_n)w_n = (a_1w_1+\cdots+a_nw_n)+(a'_1w_1+\cdots+a'_nw_n) = T(v)+T(v').$$
Also, if $\alpha \in K$, then $\alpha v = \alpha a_1 v_1 + \cdots + \alpha a_n v_n$, and therefore,
$$T(\alpha v) = \alpha a_1 w_1 + \cdots + \alpha a_n w_n = \alpha(a_1 w_1 + \cdots + a_n w_n) = \alpha T(v).$$
This proves that $T : V \to W$ is a linear map.

We claim next that $T(v_i) = w_i$ for all $1 \le i \le n$. Indeed, writing $v_i = 0 \cdot v_1 + \cdots + 1 \cdot v_i + \cdots + 0 \cdot v_n$ exhibits the coordinates of v_i, whence
$$T(v_i) = 0 \cdot w_1 + \cdots + 1 \cdot w_i + \cdots + 0 \cdot w_n = w_i.$$
This concludes the proof of the existence of T.

Uniqueness: Suppose there exists another linear map $S : V \to W$ such that $S(v_i) = w_i$ for all $1 \le i \le n$. For any $v = \sum_{i=1}^{n} a_i v_i \in V$, the linearity of S dictates:
$$S(v) = S\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i S(v_i) = \sum_{i=1}^{n} a_i w_i = T(v).$$

Since $S(v) = T(v)$ for all $v \in V$, the maps are functionally identical, establishing that $S = T$.
Q.E.D.

Example 15.3 (*Linear Maps on Coordinate Spaces*): Consider the coordinate space K^n equipped with the standard basis $\mathcal{E} = (e_1, \dots, e_n)$. Let $w_1, \dots, w_n \in K^m$ be any vectors. Theorem 15.a.3 guarantees a unique linear map $T : K^n \to K^m$ such that $T(e_i) = w_i$ for all $1 \le i \le n$.

Question 15.1: Can we provide an explicit matrix description for this map T?

Answer 15.1: Let $A \in \mathcal{M}_{m\times n}(K)$ be the matrix whose j-th column is precisely the vector w_j:

$$A = \begin{pmatrix} | & & | \\ w_1 & \dots & w_n \\ | & & | \end{pmatrix}.$$

The linear transformation $T_A : K^n \to K^m$ defined by $v \mapsto A \cdot v$ satisfies $T_A(e_j) = w_j$ for all j. Indeed, multiplying A by e_i picks out the i-th column of A:

$$T_A(e_i) = \begin{pmatrix} | & & | & & | \\ w_1 & \dots & w_i & \dots & w_n \\ | & & | & & | \end{pmatrix} \cdot \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} | \\ w_i \\ | \end{pmatrix},$$

where the 1 on the right sits in entry i. Due to the absolute uniqueness property established in Theorem 15.a.3, we immediately conclude that $T = T_A$.

Lemma 15.a.5 (*Matrix Representation of Linear Maps*): For every linear transformation $T : K^n \to K^m$, there exists a unique matrix $A \in \mathcal{M}_{m\times n}(K)$ such that $T = T_A$.

Proof:

Let $w_i := T(e_i)$ for each $i \in \{1, \dots, n\}$. As demonstrated in the preceding example, the matrix A formed by using these specific vectors as columns defines a linear map T_A that perfectly agrees with T on the standard basis $\mathcal{E}$. Because $\mathcal{E}$ is a basis for K^n, Theorem 15.a.3 strictly implies that T and T_A must be the exact same map.

The matrix is moreover unique, which the notes do not claim but which costs one line: if $B \in \mathcal{M}_{m\times n}(K)$ also satisfies $T = T_B$, then the i-th column of B equals $B \cdot e_i = T_B(e_i) = T(e_i) = w_i$ for every i, so B and A have the same columns and $B = A$.
Q.E.D.

Kernel, Image, and the Rank-Nullity Theorem (Section 15.b)

In this section, we explore two fundamental subspaces associated with every linear map: the kernel and the image. These subspaces provide deep insight into the structure of the transformation and ultimately lead us to one of the most celebrated results in linear algebra, the Rank-Nullity Theorem.

Proposition 15.b.1 (*Kernel and Image*): Let V and W be vector spaces over a field K, and let $T : V \to W$ be a linear map.

(a) The set $\ker(T) := \{v \in V \mid T(v) = 0_W\} \subseteq V$ is a linear subspace of V. It is called the "*kernel*" of T.

(b) The set $\operatorname{Im}(T) := \{T(v) \mid v \in V\} \subseteq W$ is a linear subspace of W. It is called the "*image*" of T.

Remark 15.2 (*Set-Theoretic Perspective*): Using standard notation from set theory, we can concisely write the kernel as the fiber over the zero vector, $\ker(T) = T^{-1}(\{0_W\})$, and the image as the range of the entire domain, $\operatorname{Im}(T) = T(V)$. (Note: The image is sometimes denoted as $\operatorname{Image}(T)$).

Proof of Proposition 15.b.1:

(a) To show that $\ker(T) \subseteq V$ is a subspace, let $a, b \in K$, and let $v_1, v_2 \in \ker(T)$. By the linearity of T, we have:

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = a \cdot 0_W + b \cdot 0_W = 0_W.$$

This implies that $av_1 + bv_2 \in \ker(T)$. Furthermore, since $T(0_V) = 0_W$, the zero vector $0_V \in \ker(T)$. Thus, the kernel is a valid subspace.

(b) To show that $\operatorname{Im}(T) \subseteq W$ is a subspace, let $a, b \in K$, and let $w_1, w_2 \in \operatorname{Im}(T)$. By definition, there exist vectors $v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. Therefore, we can write:

$$aw_1 + bw_2 = aT(v_1) + bT(v_2) = T(av_1 + bv_2).$$

Since $av_1 + bv_2 \in V$, it follows immediately that $aw_1 + bw_2 \in \operatorname{Im}(T)$. Moreover, since $T(0_V) = 0_W$, we have $0_W \in \operatorname{Im}(T)$. Thus, the image is a subspace.

Q.E.D.

Example 15.4 (*The Kernel of a Matrix Transformation*): Let $A \in \mathcal{M}_{m\times n}(K)$ and consider the linear map $T_A : K^n \to K^m$, recalling that $T_A(x) = A \cdot x$ for all $x \in K^n$. Then

$$\ker(T_A) = \{x \in K^n \mid A \cdot x = 0\},$$

which is precisely the solution set of the homogeneous linear system with coefficient matrix A. The kernel is therefore not a new object at all: it is the solution set of a homogeneous system, seen from the point of view of linear maps.

Proposition 15.b.2 (*Injectivity, Surjectivity, Kernel and Image*): Let $T : V \to W$ be a linear map. Then:

(a) T is injective if and only if $\ker(T) = \{0_V\}$.

(b) T is surjective if and only if $\operatorname{Im}(T) = W$.

Recall that a map is called "*bijective*" if it is both injective and surjective.

Proof:

Statement **(b)** is a restatement of the definition of surjectivity, since $\operatorname{Im}(T) = T(V)$, so there is nothing to prove. We turn to **(a)**.

First, suppose T is injective. If $v \in \ker(T)$, then $T(v) = 0_W$. Since we also know that $T(0_V) = 0_W$, the injectivity of T forces $v = 0_V$. Thus, $\ker(T) = \{0_V\}$.

Conversely, suppose that $\ker(T) = \{0_V\}$. Let $v_1, v_2 \in V$ such that $T(v_1) = T(v_2)$. By linearity, we obtain:

$$T(v_1 - v_2) = T(v_1) - T(v_2) = 0_W.$$

This implies that $v_1 - v_2 \in \ker(T)$. By our assumption, the only element in the kernel is the zero vector, and therefore, $v_1 - v_2 = 0_V$, which yields $v_1 = v_2$. Consequently, T is injective.

Q.E.D.

Exercise 15.2 (*Images and Preimages of Subspaces*): Let $T : V \to W$ be a linear map. Show that:

(a) if $V' \subseteq V$ is a linear subspace, then $T(V') \subseteq W$ is a linear subspace;

(b) if $W' \subseteq W$ is a linear subspace, then $T^{-1}(W') \subseteq V$ is a linear subspace, where $T^{-1}(W') := \{v \in V \mid T(v) \in W'\}$;

(c) both statements are generalisations of Proposition 15.b.1. Explain why.

Isomorphisms

Recall from set theory that a map $f : X \to Y$ is called bijective if it is both injective and surjective, and that in this case there exists a unique inverse map $g : Y \to X$, denoted by f^{-1}, meaning that

$$g \circ f = \mathrm{id}_X \quad \text{and} \quad f \circ g = \mathrm{id}_Y .$$

Conversely, if $f : X \to Y$ is a map for which some $g : Y \to X$ satisfies $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$, then f is bijective. What is the right analogue of this for linear maps?

Definition 15.b.3 (*Isomorphism*): A linear map $T : V \to W$ is called an "*isomorphism*" if there exists a **linear** map $S : W \to V$ such that

$$S \circ T = \mathrm{id}_V \quad \text{and} \quad T \circ S = \mathrm{id}_W .$$

Two vector spaces V and W are defined to be *"isomorphic"*, denoted as $V \cong W$, if there exists an isomorphism between them.

$$\begin{tikzcd} V \arrow[r, shift left, "T"] & W \arrow[l, shift left, "S"] \end{tikzcd}$$

Remark 15.3: An isomorphism $T : V \to W$ is obviously bijective: the linear map S of Definition 15.b.3 is in particular an inverse map of T in the set-theoretic sense recalled above. The converse is the substantial question, and it is not a matter of definition, because the definition of an isomorphism demands that the inverse be **linear**: is every bijective linear map an isomorphism?

Lemma 15.b.4 (*Bijective Linear Maps are Isomorphisms*): Let $T : V \to W$ be a linear map which is bijective. Then the inverse map $T^{-1} : W \to V$ is linear. In other words, a bijective linear map is an isomorphism.

Proof:

Let $a, b \in K$ and $w_1, w_2 \in W$. Put $v_1 := T^{-1}(w_1)$ and $v_2 := T^{-1}(w_2)$, so that $T(v_1) = w_1$ and $T(v_2) = w_2$. Since T is linear, we have

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = aw_1 + bw_2.$$

Applying T^{-1} to both sides gives

$$T^{-1}(aw_1 + bw_2) = av_1 + bv_2 = a \cdot T^{-1}(w_1) + b \cdot T^{-1}(w_2),$$

so T^{-1} is a linear map. **Q.E.D.**

Summary 15.1: For linear maps, bijective = isomorphism.

Lemma 15.b.5 (*Compositions of Linear Maps are Linear*): Let $T : V \to W$ and $S : W \to U$ be linear maps. Then the composition $S \circ T : V \to U$ is also linear.

Proof:

Let $a, b \in K$ and $v_1, v_2 \in V$. Then

$$\begin{aligned}(S \circ T)(av_1 + bv_2) &= S\big(T(av_1 + bv_2)\big) = S\big(aT(v_1) + bT(v_2)\big) \\ &= a \cdot S(T(v_1)) + b \cdot S(T(v_2)) = a \cdot (S \circ T)(v_1) + b \cdot (S \circ T)(v_2).\end{aligned}$$

Q.E.D.

Exercise 15.3 (*Isomorphism is an Equivalence Relation*): Show that $\cong$, that is, being isomorphic, defines an equivalence relation on the set of vector spaces over K.

Definition 15.b.6 (*Endomorphisms and Automorphisms*): A linear map $T : V \to V$, whose domain and target are the same space V, is sometimes called an "*endomorphism*" of V. We denote the set of all endomorphisms of V by $\operatorname{End}(V)$, so that $\operatorname{End}(V) = \operatorname{Hom}(V, V)$. A linear map $T : V \to V$ which is an isomorphism is called an "*automorphism*" of V; in slogan form, endomorphism + isomorphism = automorphism.

Definition 15.b.7 (*Monomorphisms and Epimorphisms*): If a linear map T is injective, one says that T is a "*monomorphism*"; if T is surjective, one says that T is an "*epimorphism*".

Remark 15.4: We will not use the terminology of Definition 15.b.7 in this course, and will simply say injective and surjective instead.

The Rank Theorem

Lemma 15.b.8 (*The Image is Spanned by the Images of a Basis*): Let $T : V \to W$ be a linear map, and let $v_1, \ldots, v_n$ be a basis of V. Then

$$\operatorname{Im}(T) = \operatorname{Sp}\big(T(v_1), \ldots, T(v_n)\big).$$

Proof:

We first show that $\operatorname{Im}(T) \supseteq \operatorname{Sp}(T(v_1), \ldots, T(v_n))$. Let $a_1, \ldots, a_n \in K$. Then

$$\sum_{i=1}^{n} a_i T(v_i) = T\left(\sum_{i=1}^{n} a_i v_i\right) \in \operatorname{Im}(T),$$

which shows that $\operatorname{Sp}(T(v_1), \ldots, T(v_n)) \subseteq \operatorname{Im}(T)$.

We show now that $\operatorname{Im}(T) \subseteq \operatorname{Sp}(T(v_1), \ldots, T(v_n))$. Let $w \in \operatorname{Im}(T)$. By definition, there exists $v \in V$ such that $T(v) = w$. Since $v_1, \ldots, v_n$ is a basis for V, there exist $a_1, \ldots, a_n \in K$ such that $v = \sum_{i=1}^{n} a_i v_i$, and therefore,

$$w = T(v) = T\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i T(v_i).$$

So $w \in \operatorname{Sp}(T(v_1), \ldots, T(v_n))$, which proves the second inclusion. **Q.E.D.**

The dimension of the image is the quantity the next theorem is about; it is given the name $\operatorname{rank}(T)$ at the end of this section, in Definition 15.b.13.

Theorem 15.b.9 (*Rank-Nullity Theorem*): Let V be a finite-dimensional vector space, and let $T : V \to W$ be a linear map. Then, the following dimension formula holds:

$$\dim V = \dim \ker(T) + \dim \operatorname{Im}(T).$$

Proof:

Being a subspace of the finite-dimensional space V, the kernel $\ker(T)$ is itself finite-dimensional by Proposition 12.3, so it has a basis. Let $k := \dim \ker(T)$ and $n := \dim V$, and let $(u_1, \ldots, u_k)$ be a basis for $\ker(T)$. By the basis extension property, item **(2)** of Summary 12.1, we can extend this linearly independent set to a full basis for V, denoted by $\mathcal{B} = (u_1, \ldots, u_k, v_1, \ldots, v_{n-k})$. We claim that the family $\mathcal{C} = (T(v_1), \ldots, T(v_{n-k}))$ forms a basis for $\operatorname{Im}(T)$.

(a) **Spanning:** Let $w \in \mathrm{Im}(T)$. By definition, there exists some $v \in V$ such that $w = T(v)$. We can express v uniquely as a linear combination of the basis vectors in $\mathcal{B}$:

$$v = \sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{n-k} b_j v_j.$$

Applying the linear map T, and recalling that $T(u_i) = 0_W$ for all i, we find:

$$\begin{aligned} w &= T\left(\sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{n-k} b_j v_j\right) \\ &= \sum_{i=1}^{k} a_i T(u_i) + \sum_{j=1}^{n-k} b_j T(v_j) \\ &= 0_W + \sum_{j=1}^{n-k} b_j T(v_j). \end{aligned}$$

Thus, the vectors in $\mathcal{C}$ span $\mathrm{Im}(T)$.

(b) **Linear Independence:** Suppose there exist scalars $b_j \in K$ such that:

$$\sum_{j=1}^{n-k} b_j T(v_j) = 0_W.$$

By the linearity of T, this is equivalent to $T\left(\sum_{j=1}^{n-k} b_j v_j\right) = 0_W$, which implies that the vector $\sum_{j=1}^{n-k} b_j v_j$ lies in $\ker(T)$. Because $(u_1, \ldots, u_k)$ is a basis for the kernel, there must exist scalars $a_i \in K$ such that:

$$\sum_{j=1}^{n-k} b_j v_j = \sum_{i=1}^{k} a_i u_i,$$

or, after moving everything to one side,

$$\sum_{j=1}^{n-k} b_j v_j - \sum_{i=1}^{k} a_i u_i = 0_V.$$

However, since $\mathcal{B}$ is a basis for V, all of its constituent vectors are linearly independent. This forces all coefficients to be zero, specifically $b_j = 0$ for all $1 \leq j \leq n-k$.

Therefore, $\mathcal{C}$ is linearly independent and spans the image, making it a basis for $\mathrm{Im}(T)$. Consequently, $\dim \mathrm{Im}(T) = n - k = \dim V - \dim \ker(T)$.

Q.E.D.

Remark 15.5: The spanning step above can be shortened: applying Lemma 15.b.8 to the basis $\mathcal{B}$ gives

$$\mathrm{Im}(T) = \mathrm{Sp}\left(\underbrace{T(u_1)}_{=0}, \ldots, \underbrace{T(u_k)}_{=0}, T(v_1), \ldots, T(v_{n-k})\right) = \mathrm{Sp}\left(T(v_1), \ldots, T(v_{n-k})\right),$$

which is how the notes obtain it. Only the independence of $\mathcal{C}$ then remains to be checked.

The theorem also admits a second, more structural proof, which replaces the manipulation of coordinates by a single well-chosen decomposition of V. It uses the existence of a complement from Proposition 14.5 and the fact that isomorphisms carry bases to bases, Theorem 15.b.12 below; neither of these depends on the rank theorem, so no circularity arises.

Second proof of Theorem 15.b.9:

By Proposition 14.5, the subspace $\ker(T) \subseteq V$ admits a complement, that is, a subspace $U \subseteq V$ with

$$\ker(T) + U = V \quad \text{and} \quad \ker(T) \cap U = \{0_V\}.$$

We claim that the restriction $T|_U : U \to \operatorname{Im}(T)$ is an isomorphism.

It is injective: if $u \in U$ satisfies $T(u) = 0_W$, then $u \in \ker(T) \cap U = \{0_V\}$, so $u = 0_V$, and Proposition 15.b.2 **(a)** applies. It is surjective onto $\operatorname{Im}(T)$: given $w \in \operatorname{Im}(T)$, write $w = T(v)$ for some $v \in V$ and decompose $v = x + u$ with $x \in \ker(T)$ and $u \in U$; then

$$w = T(x+u) = T(x) + T(u) = 0_W + T(u) = T(u).$$

Being a bijective linear map, $T|_U$ is an isomorphism by Lemma 15.b.4. By Theorem 15.b.12, it carries a basis of U to a basis of $\operatorname{Im}(T)$, and therefore, $\dim U = \dim \operatorname{Im}(T)$.

Finally, the dimension formula of Proposition 14.2 **(c)**, applied to the subspaces $\ker(T)$ and U, gives

$$\dim V = \dim\big(\ker(T) + U\big) = \dim\ker(T) + \dim U - \underbrace{\dim\big(\ker(T) \cap U\big)}_{=0} = \dim\ker(T) + \dim\operatorname{Im}(T).$$

Q.E.D.

Corollary 15.b.10 (*Dimension Obstructions to Injectivity and Surjectivity*): Let $T : V \to W$ be a linear map, where V and W are finite-dimensional vector spaces. Then:

(a) If $\dim W < \dim V$, then T is **not** injective.

(b) If $\dim W > \dim V$, then T is **not** surjective.

(c) If $\dim W = \dim V$, then T is bijective if and only if T is injective, if and only if T is surjective.

Proof:

(a) Since $\operatorname{Im}(T)$ is a subspace of W, we have $\dim\operatorname{Im}(T) \le \dim W < \dim V$ by Proposition 12.3. By Theorem 15.b.9,

$$\dim\ker(T) = \dim V - \dim\operatorname{Im}(T) > 0,$$

so $\ker(T) \neq \{0_V\}$, and T is not injective by Proposition 15.b.2 **(a)**.

(b) By Theorem 15.b.9 and the assumption,

$$\dim\operatorname{Im}(T) = \dim V - \dim\ker(T) \le \dim V < \dim W,$$

so $\operatorname{Im}(T) \subsetneq W$, and T is not surjective by Proposition 15.b.2 **(b)**.

(c) Here the chain of equivalences reads

$$T \text{ injective} \iff \ker(T) = \{0_V\} \iff \dim\ker(T) = 0 \iff \dim V = \dim\operatorname{Im}(T) \iff \dim W = \dim\operatorname{Im}(T),$$

the third step by Theorem 15.b.9 and the last by the assumption $\dim V = \dim W$. Since $\operatorname{Im}(T)$ is a subspace of W, the final equality holds if and only if $\operatorname{Im}(T) = W$, this being the equality case of Proposition 12.3, that is, if and only if T is surjective. As each of the two properties implies the other, either one already makes T bijective.

Q.E.D.

Remark 15.6: By Lemma 15.b.4, the word *"bijective"* in statement **(c)** may equivalently be read as *"is an isomorphism"*. So for maps between spaces of equal finite dimension, injectivity alone already forces T to be an isomorphism.

Corollary 15.b.11 (*Isomorphism is Detected by Dimension*): Two finite-dimensional vector spaces V and W over K are isomorphic if and only if $\dim V = \dim W$.

Proof:

Suppose first that $T : V \to W$ is an isomorphism. By Definition 15.b.3, it admits a linear inverse and is in particular bijective, so $\ker(T) = \{0_V\}$ and $\operatorname{Im}(T) = W$ by Proposition 15.b.2. The rank theorem, Theorem 15.b.9, then gives

$$\dim W = \dim \operatorname{Im}(T) = \dim V - \dim \ker(T) = \dim V.$$

Conversely, assume that $\dim V = \dim W$, and denote this common dimension by n. Let $v_1, \ldots, v_n$ be a basis for V and $w_1, \ldots, w_n$ a basis for W. By Theorem 15.a.3, there is a linear map $T : V \to W$ with $T(v_i) = w_i$ for all $1 \leq i \leq n$. By Lemma 15.b.8,

$$\operatorname{Im}(T) = \operatorname{Sp}\big(T(v_1), \ldots, T(v_n)\big) = \operatorname{Sp}(w_1, \ldots, w_n) = W,$$

so T is surjective and $\dim \operatorname{Im}(T) = n$. By Theorem 15.b.9, $\dim \ker(T) = n - n = 0$, so T is injective as well. Being a bijective linear map, T is an isomorphism by Lemma 15.b.4, and hence $V \cong W$. **Q.E.D.**

Isomorphisms and Vector Space Equivalence

Isomorphisms allow us to classify vector spaces and identify when two ostensibly different spaces are essentially the *"same"* in terms of their linear algebraic structure.

Theorem 15.b.12 (*Isomorphisms Preserve Bases*): Let $T : V \to W$ be an isomorphism between two vector spaces V and W. Let $\mathcal{S} = \{v_i\}$ be a family of vectors in V. We define its image family as $T(\mathcal{S}) = \{T(v_i)\}_{v_i \in \mathcal{S}} \subseteq W$. Then, the following hold:

- **(a)** $\mathcal{S}$ is linearly independent in V if and only if $T(\mathcal{S})$ is linearly independent in W.
- **(b)** $\mathcal{S}$ spans V if and only if $T(\mathcal{S})$ spans W.
- **(c)** $\mathcal{S}$ is a basis of V if and only if $T(\mathcal{S})$ is a basis of W.

The notes state this theorem without proof. Since the second proof of Theorem 15.b.9 above appeals to it, we supply one; it is short, and it uses nothing beyond the definition of an isomorphism.

Proof:

Write $S := T^{-1} : W \to V$ for the inverse of T, which is linear because T is an isomorphism. Each statement is an equivalence, and in each case the implication from right to left follows by applying the already-proved implication from left to right to the isomorphism S and the family $T(\mathcal{S})$, since $S(T(\mathcal{S})) = \mathcal{S}$. It therefore suffices to prove the three implications from left to right.

- **(a)** Let $\mathcal{S}$ be linearly independent, and suppose that a finite subfamily satisfies $\sum_i a_i T(v_i) = 0_W$. By linearity, $T\left(\sum_i a_i v_i\right) = 0_W$, and applying S gives $\sum_i a_i v_i = 0_V$. The independence of $\mathcal{S}$ forces every a_i to vanish, so $T(\mathcal{S})$ is linearly independent.
- **(b)** Let $\mathcal{S}$ span V, and let $w \in W$. Then $S(w) \in V$ is a linear combination $S(w) = \sum_i a_i v_i$ of members of $\mathcal{S}$, and applying T gives
$$w = T(S(w)) = \sum_i a_i T(v_i),$$
so $T(\mathcal{S})$ spans W.
- **(c)** Immediate from the first two statements, since a basis is precisely a linearly independent spanning family.

Q.E.D.

Remark 15.7 (*The Significance of Isomorphism*): If two vector spaces are isomorphic ($V \cong W$), they possess identical linear algebraic properties, such as having the exact same dimension. We can conceptually think of V and W as the exact same space, just represented in two different ways.

However, it is crucial to note that identifying V with W requires the explicit choice of an isomorphism $T : V \to W$. Because there is usually no preferred (or *"canonical"*) choice for this map, it is generally better practice not to forcefully identify different vector spaces simply because they are isomorphic. We will explore the nuances of canonical choices later in the course.

Definition 15.b.13 (*Rank of a Linear Map*): Let $T \in \mathrm{Hom}(V, W)$ be a linear map such that $\dim \mathrm{Im}(T) < \infty$. We define the "*rank*" of T as the dimension of its image:

$$\mathrm{rank}(T) := \dim \mathrm{Im}(T).$$

Exercise 15.4 (*Rank of a Composition*): Let $T : V \to W$ and $S : W \to U$ be linear maps, where U, V and W are finite-dimensional. Show that:

(a) $\mathrm{rank}(S \circ T) \leq \min\{\mathrm{rank}(S), \mathrm{rank}(T)\}$;
(b) if S is injective, then $\mathrm{rank}(S \circ T) = \mathrm{rank}(T)$;
(c) if T is surjective, then $\mathrm{rank}(S \circ T) = \mathrm{rank}(S)$.

Solutions to the Exercises

Linearity Criterion (on Page 99):

Suppose first that T is linear. Then, for all $\alpha_1, \alpha_2 \in K$ and $v_1, v_2 \in V$,

$$T(\alpha_1 v_1 + \alpha_2 v_2) = T(\alpha_1 v_1) + T(\alpha_2 v_2) = \alpha_1 T(v_1) + \alpha_2 T(v_2),$$

using additivity for the first equality and homogeneity for the second.

Conversely, assume the displayed identity for all scalars and vectors. Taking $\alpha_1 = \alpha_2 = 1$ gives $T(v_1 + v_2) = T(v_1) + T(v_2)$, which is additivity. Taking $\alpha_2 = 0$ and $v_2 = 0_V$ gives

$$T(\alpha_1 v_1) = \alpha_1 T(v_1) + 0 \cdot T(0_V) = \alpha_1 T(v_1),$$

which is homogeneity. So T satisfies both conditions of Definition 15.a.1. **Q.E.D.**

Images and Preimages of Subspaces (on Page 102):

(a) Since $0_V \in V'$ and $T(0_V) = 0_W$, we have $0_W \in T(V')$. Let $a, b \in K$ and $w_1, w_2 \in T(V')$, say $w_1 = T(v_1)$ and $w_2 = T(v_2)$ with $v_1, v_2 \in V'$. Then $av_1 + bv_2 \in V'$, because V' is a subspace, and

$$aw_1 + bw_2 = aT(v_1) + bT(v_2) = T(av_1 + bv_2) \in T(V').$$

(b) Since $T(0_V) = 0_W \in W'$, we have $0_V \in T^{-1}(W')$. Let $a, b \in K$ and $v_1, v_2 \in T^{-1}(W')$, so that $T(v_1), T(v_2) \in W'$. Then

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) \in W',$$

because W' is a subspace, and therefore $av_1 + bv_2 \in T^{-1}(W')$.

(c) Both parts of Proposition 15.b.1 are special cases. Taking $V' := V$ in **(a)** gives that $\mathrm{Im}(T) = T(V)$ is a subspace of W, and taking $W' := \{0_W\}$ in **(b)** gives that $\ker(T) = T^{-1}(\{0_W\})$ is a subspace of V, exactly the two set-theoretic descriptions recorded in the remark on Page 102.

Q.E.D.

Isomorphism is an Equivalence Relation (on Page 104):

Reflexivity. The identity $\mathrm{id}_V : V \to V$ is linear and satisfies $\mathrm{id}_V \circ \mathrm{id}_V = \mathrm{id}_V$, so it is an isomorphism, and $V \cong V$.

Symmetry. Let $V \cong W$, witnessed by an isomorphism $T : V \to W$ with linear inverse $S : W \to V$. Then S is itself linear with linear inverse T, since $T \circ S = \mathrm{id}_W$ and $S \circ T = \mathrm{id}_V$. So S is an isomorphism and $W \cong V$.

Transitivity. Let $T : V \to W$ and $T' : W \to U$ be isomorphisms with inverses S and S'. By Lemma 15.b.5, both $T' \circ T : V \to U$ and $S \circ S' : U \to V$ are linear, and

$$(S \circ S') \circ (T' \circ T) = S \circ (S' \circ T') \circ T = S \circ \mathrm{id}_W \circ T = S \circ T = \mathrm{id}_V,$$

and symmetrically $(T' \circ T) \circ (S \circ S') = \mathrm{id}_U$. So $T' \circ T$ is an isomorphism and $V \cong U$. **Q.E.D.**

Rank of a Composition (on Page 108):

Throughout, we use the identity

$$\mathrm{Im}(S \circ T) = \{S(T(v)) \mid v \in V\} = \{S(w) \mid w \in \mathrm{Im}(T)\} = S\big(\mathrm{Im}(T)\big),$$

so that $\mathrm{Im}(S \circ T)$ is the image of the restricted linear map $S|_{\mathrm{Im}(T)} : \mathrm{Im}(T) \to U$.

(a) Since $\mathrm{Im}(T) \subseteq W$, we have $S(\mathrm{Im}(T)) \subseteq S(W) = \mathrm{Im}(S)$, and both are subspaces of U by Exercise 15.2 **(a)**, so Proposition 12.3 gives $\mathrm{rank}(S \circ T) \leq \mathrm{rank}(S)$. Applying Theorem 15.b.9 to $S|_{\mathrm{Im}(T)}$ gives

$$\mathrm{rank}(S \circ T) = \dim \mathrm{Im}\left(S|_{\mathrm{Im}(T)}\right) = \dim \mathrm{Im}(T) - \dim \ker\left(S|_{\mathrm{Im}(T)}\right) \leq \mathrm{rank}(T),$$

and the two bounds together give $\mathrm{rank}(S \circ T) \leq \min\{\mathrm{rank}(S), \mathrm{rank}(T)\}$.

(b) If S is injective, then so is its restriction $S|_{\mathrm{Im}(T)}$, so $\ker(S|_{\mathrm{Im}(T)}) = \{0_W\}$ and the displayed computation becomes an equality: $\mathrm{rank}(S \circ T) = \dim \mathrm{Im}(T) = \mathrm{rank}(T)$.

(c) If T is surjective, then $\mathrm{Im}(T) = W$, and therefore $\mathrm{Im}(S \circ T) = S(W) = \mathrm{Im}(S)$, so that $\mathrm{rank}(S \circ T) = \mathrm{rank}(S)$.

Q.E.D.

AI-Note: This was the most heavily damaged chapter so far. Of the thirteen numbered statements in the notes for Section 15.b, the transcript kept five, and it hid the loss with a counter jump: a `\setcounter{theorem}{11}` placed just before Theorem 15.b.12, annotated as aligning the numbering with the notes. Restored above, in the notes' own order, so that the printed numbers now agree with the handwritten ones throughout: Definition 15.b.3, Lemma 15.b.4, Lemma 15.b.5, Definition 15.b.6, Definition 15.b.7, Lemma 15.b.8 and Corollary 15.b.11, along with the example of $\ker(T_A)$, the recollection on bijections that motivates Definition 15.b.3, and three of the notes' exercises. In Proposition 15.b.2 the surjectivity criterion was missing, and in Corollary 15.b.10 the two dimension obstructions, the statements that really do the work in practice, had been dropped in favour of the third part alone.

The gap that mattered most was Definition 15.b.3 itself. The transcript carried a definition of **isomorphic spaces** at the very end of the chapter, numbered 15.b.13, while the word *"isomorphism"* was already used in Corollary 15.b.10 and Theorem 15.b.12 above it. Worse, the old proof of Corollary 15.b.10 concluded from bijectivity that the map is an isomorphism, which is exactly Lemma 15.b.4, the one result in the notes that was missing. That definition has been moved to the notes' position, 15.b.3, and merged with the notes' formulation.

Since Prof. Biran leaves Theorem 15.b.12 unproved, a proof is supplied here. It is also used for a second proof of Theorem 15.b.9, added above: rather than manipulating coordinates, one takes a complement U of $\ker(T)$ as in Proposition 14.5 and observes that $T|_U$ is an isomorphism onto $\mathrm{Im}(T)$, after which the dimension formula of Proposition 14.2 finishes the argument. Neither ingredient depends on the rank theorem.

In Section 15.a, the notes verify in full that the map constructed in Theorem 15.a.3 is linear and that it sends v_i to w_i; the transcript replaced both verifications with the phrase *"it is straightforward to show"*. They are restored, as is the computation showing that $A \cdot e_i$ picks out the i-th column of A. The uniqueness of the matrix in Lemma 15.a.5 is asserted by the transcript but not proved, and not claimed at all in the notes; a one-line proof is added. Two cross-references used `\eqref` on a definition and on a theorem, which typesets as a bare equation number; they are now `\cref`.

The rigour pass found one defect of statement and five silent steps. Theorem 15.a.3 introduced V and a basis of it, and then spoke of vectors $w_1, \ldots, w_n \in W$ and of a map $T : V \to W$ without W ever having been named; it is now introduced with V, as a vector space over the same field.

The proof of Theorem 15.b.9 extends a basis of $\ker(T)$ to one of V *"by the Steinitz Exchange Lemma"*. What it uses is the basis extension property, item **(2)** of Summary 12.1, which is proved in ch. 12; and before a basis of $\ker(T)$ can be spoken of at all, that subspace has to be known to be finite-dimensional, which is Proposition 12.3. That same proposition is what justifies three further steps in Corollary 15.b.10 and in the solution to Exercise 15.4: that a subspace has no larger dimension than the space containing it, and that equality of the two forces them to coincide. The last of these is the equality case, and it is what turns "$\dim \operatorname{Im}(T) = \dim W$" into surjectivity in part **(c)**.

Linear Maps and Bases (Chapter 16)

Writing Linear Maps Using Bases and Coordinates

In this lecture, we will assume all vector spaces to be *"finite-dimensional"* unless otherwise stated. We will write bases of a vector space V as an ordered list (or tuple) $\mathcal{B} = (v_1, \ldots, v_n)$.

Definition 16.1 (*Coordinate Vector*): Let V be a vector space over K, and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis of V. Let $v \in V$. We define the "*coordinate vector*" of v with respect to the basis $\mathcal{B}$ as

$$[v]_{\mathcal{B}} := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \ldots, a_n \in K$ are the unique scalars in K such that $v = a_1 v_1 + \cdots + a_n v_n$.

Remark 16.1: Note that $[v]_{\mathcal{B}}$ is uniquely defined, since every $v \in V$ has a unique representation as a linear combination of the basis vectors $v_1, \ldots, v_n$.

Definition (*Coordinate Map*): Let V be a vector space over K, and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis of V. Define a map $\Phi_{\mathcal{B}} : V \to K^n$ given by:

$$\Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \in K^n.$$

We call $\Phi_{\mathcal{B}}$ the "*coordinate map*" of the basis $\mathcal{B}$.

Proposition 16.2 (*Isomorphism of the Coordinate Map*): The map $\Phi_{\mathcal{B}}$ is an isomorphism.

Proof:

We first show that $\Phi_{\mathcal{B}}$ is a linear map. Let $a, b \in K$, and let $v, v' \in V$. Write v and v' as linear combinations of the elements of $\mathcal{B}$:

$$v = a_1 v_1 + \cdots + a_n v_n, \quad \text{and} \quad v' = b_1 v_1 + \cdots + b_n v_n,$$

with $a_i, b_i \in K$. By definition, we have:

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad \text{and} \quad [v']_{\mathcal{B}} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

Now, $av + bv' = (aa_1 + bb_1)v_1 + \cdots + (aa_n + bb_n)v_n$, and hence:

$$\Phi_{\mathcal{B}}(av + bv') = [av + bv']_{\mathcal{B}} = \begin{pmatrix} aa_1 + bb_1 \\ \vdots \\ aa_n + bb_n \end{pmatrix} = a \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + b \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a\Phi_{\mathcal{B}}(v) + b\Phi_{\mathcal{B}}(v').$$

This proves that $\Phi_{\mathcal{B}}$ is linear.

We now claim that $\Phi_{\mathcal{B}}$ is an isomorphism. By previous results (Lemma 15.b.4), it is enough to show $\Phi_{\mathcal{B}}$ is bijective. Indeed, if $v \in \ker(\Phi_{\mathcal{B}})$, then $v = 0v_1 + \cdots + 0v_n = 0$, which implies $\ker(\Phi_{\mathcal{B}}) = \{0\}$. This shows that $\Phi_{\mathcal{B}}$ is injective.

For the surjectivity of $\Phi_{\mathcal{B}}$, let $(a_1, \ldots, a_n)^\mathsf{T} \in K^n$. Put $v := a_1 v_1 + \cdots + a_n v_n \in V$. Then, $\Phi_{\mathcal{B}}(v) = (a_1, \ldots, a_n)^\mathsf{T}$, which implies $\operatorname{Im}(\Phi_{\mathcal{B}}) = K^n$. **Q.E.D.**

Remark 16.2: The inverse map to $\Phi_{\mathcal{B}}$ is given by:

$$\Phi_{\mathcal{B}}^{-1} : K^n \to V, \quad \Phi_{\mathcal{B}}^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} := a_1 v_1 + \cdots + a_n v_n.$$

Exhibiting this map is in fact a second, shorter proof that $\Phi_{\mathcal{B}}$ is bijective: both composites are the identity, since $\mathcal{B}$ is a basis, and hence every $v \in V$ is recovered from its own coordinates by $a_1 v_1 + \cdots + a_n v_n$, while every tuple $(a_1, \dots, a_n)^{\mathsf{T}} \in K^n$ is by Definition 16.1 the coordinate vector of that same combination.

Remark 16.3: The previous Proposition 16.2 gives yet another proof for the following Corollary.

Corollary 16.3 (*Classification of Finite-Dimensional Vector Spaces*): Let V be a vector space over K with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Then, $V \cong K^n$.

Proof:

Choose a basis $\mathcal{B}$ for V. Then, $\Phi_{\mathcal{B}} : V \to K^n$ is an isomorphism. **Q.E.D.**

Remark 16.4:

- The Corollary does NOT hold for $\dim V = \infty$, at least not as stated.
- We have seen that $V \cong K^n$ where $n = \dim V < \infty$, but in general, there does not exist a *"canonical"* (or preferred) isomorphism $V \to K^n$. For example, $\Phi_{\mathcal{B}}$ depends on a choice of a basis $\mathcal{B}$.

Example 16.1 (*Coordinate Vector Calculation*): Consider $V = \mathbb{R}^2$, let $\mathcal{E} = \left(\binom{1}{0}, \binom{0}{1}\right)$ be the standard basis, and let $\mathcal{B} = \left(\binom{1}{1}, \binom{1}{-1}\right)$ be another basis (check that $\mathcal{B}$ is indeed a basis!). We have

$$\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{E}} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \text{and we want to find} \quad \left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} a \\ b \end{pmatrix} =?$$

Let us first try to determine $\left[\binom{1}{0}\right]_{\mathcal{B}}$ and $\left[\binom{0}{1}\right]_{\mathcal{B}}$. We set $\left[\binom{1}{0}\right]_{\mathcal{B}} = \binom{a_1}{a_2}$, where

$$a_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + a_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

This leads to the system of linear equations:

$$\begin{cases} a_1 + a_2 = 1 \\ a_1 - a_2 = 0 \end{cases}$$

Solving this system yields $a_1 = a_2 = \frac{1}{2}$, and therefore, $\left[\binom{1}{0}\right]_{\mathcal{B}} = \binom{1/2}{1/2}$.
Similarly, for the second standard basis vector, we set

$$\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad \text{where} \quad b_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The resulting system is:

$$\begin{cases} b_1 + b_2 = 0 \\ b_1 - b_2 = 1 \end{cases}$$

Solving this yields $b_1 = \frac{1}{2}$ and $b_2 = -\frac{1}{2}$, so $\left[\binom{0}{1}\right]_{\mathcal{B}} = \binom{1/2}{-1/2}$.
Finally, using the fact that $\Phi_{\mathcal{B}}$ is linear, we calculate:

$$\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{B}} = x \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right]_{\mathcal{B}} + y \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} x/2 \\ x/2 \end{pmatrix} + \begin{pmatrix} y/2 \\ -y/2 \end{pmatrix} = \begin{pmatrix} \frac{x+y}{2} \\ \frac{x-y}{2} \end{pmatrix}.$$

Representation Matrices

Now we arrive at the main question. Let V and W be vector spaces over K, let $n := \dim V$ and $m := \dim W$, and let $T : V \to W$ be a linear map. Fix a basis $\mathcal{B}$ for V and a basis $\mathcal{C}$ for W. Consider the following diagram, in which the two paths from V to K^m agree, so that the diagram is commutative (the lower arrow is defined to be that very composition, so that commutativity holds by construction):

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\Phi_{\mathcal{B}}"'] & W \arrow[d, "\Phi_{\mathcal{C}}"] \\ K^n \arrow[r, "\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}"'] & K^m \end{tikzcd}$$

Since $\Phi_{\mathcal{B}}^{-1}$, T, and $\Phi_{\mathcal{C}}$ are all linear maps, the map $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} : K^n \to K^m$ is also linear. By previous results (Lemma 15.a.5), any linear map $K^n \to K^m$ can be written using a matrix $A \in \mathcal{M}_{m\times n}(K)$. In other words, there exists a unique matrix $A \in \mathcal{M}_{m\times n}(K)$ such that $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} = T_A$.

Definition (*Representation Matrix*): Let V and W be finite-dimensional vector spaces over K, put $n := \dim V$ and $m := \dim W$, let $T : V \to W$ be a linear map, and fix a basis $\mathcal{B}$ for V and a basis $\mathcal{C}$ for W. The unique matrix $A \in \mathcal{M}_{m\times n}(K)$ satisfying

$$T_A = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$$

is called the "*representation matrix*" of T with respect to the bases $\mathcal{B}$ and $\mathcal{C}$. We also say that A represents T in the bases $\mathcal{B}$ and $\mathcal{C}$. We denote this matrix by $[T]_{\mathcal{C}}^{\mathcal{B}} \in \mathcal{M}_{m\times n}(K)$.

Question 16.1:

(a) How can we calculate the entries of A?

(b) How does A depend on different choices of the bases $\mathcal{B}$ and $\mathcal{C}$?

(c) If we have three vector spaces V, W, U with bases $\mathcal{B}, \mathcal{C}, \mathcal{D}$ respectively, and $T : V \to W$ and $S : W \to U$ are linear maps where T is represented by the matrix M and S by the matrix N, then what is the matrix representing $S \circ T : V \to U$ with respect to the bases $\mathcal{B}$ and $\mathcal{D}$?

Let $T : V \to W$ be a linear map, let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V, and let $\mathcal{C} = (w_1, \dots, w_m)$ be a basis for W. We wish to determine the entries of its representation matrix $A := [T]_{\mathcal{C}}^{\mathcal{B}} \in \mathcal{M}_{m\times n}(K)$. By definition (see Page 113), A is the unique matrix such that the linear map $T_A : K^n \to K^m$, given by left-multiplication by A, satisfies $T_A = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$.

Proposition (*Columns of the Representation Matrix*): Let $T : V \to W$ be a linear map, let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V, let $\mathcal{C} = (w_1, \dots, w_m)$ be a basis for W, and put $A := [T]_{\mathcal{C}}^{\mathcal{B}}$. Then, the j-th column of the representation matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ is precisely the coordinate vector of $T(v_j)$ with respect to the basis $\mathcal{C}$. Thus, we can write A in block-column form as:

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \in \mathcal{M}_{m\times n}(K).$$

Equivalently, if we write $A = (a_{ij}) \in \mathcal{M}_{m\times n}(K)$, the entries a_{ij} of the j-th column are determined by expressing $T(v_j) \in W$ as a linear combination of the basis vectors $\mathcal{C} = (w_1, \dots, w_m)$:

$$T(v_j) = a_{1j}w_1 + a_{2j}w_2 + \dots + a_{mj}w_m = \sum_{i=1}^{m} a_{ij}w_i.$$

Proof:

Recall that the j-th column of A is given by $T_A(e_j)$, where $e_j \in K^n$ is the j-th standard basis vector. We can evaluate $T_A(e_j)$ using the commutative diagram:

$$T_A(e_j) = \left(\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}\right)(e_j) = \Phi_{\mathcal{C}}\left(T\left(\Phi_{\mathcal{B}}^{-1}(e_j)\right)\right).$$

Recall that the coordinate vector of the basis vector $v_j \in V$ with respect to its own basis $\mathcal{B}$ is the j-th standard basis vector $e_j \in K^n$, meaning $\Phi_{\mathcal{B}}(v_j) = [v_j]_{\mathcal{B}} = e_j$. Applying the inverse coordinate map gives $\Phi_{\mathcal{B}}^{-1}(e_j) = v_j$. Substituting this into our expression, we have:

$$T_A(e_j) = \Phi_{\mathcal{C}}(T(v_j)) = [T(v_j)]_{\mathcal{C}}.$$

Since the j-th column of A is $T_A(e_j)$, this is exactly the asserted block-column form. The description by entries is the same statement read coordinate by coordinate: writing $[T(v_j)]_{\mathcal{C}} = (a_{1j}, \dots, a_{mj})^{\mathsf{T}}$ says, by Definition 16.1, that $T(v_j) = \sum_{i=1}^{m} a_{ij} w_i$. **Q.E.D.**

Proposition 16.4 (*Evaluation via Representation Matrix*): Let $T : V \to W$ be a linear map between two finite-dimensional vector spaces V and W. Let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W, respectively. Then, for all $v \in V$:

$$[T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T(v)]_{\mathcal{C}}.$$

Proof:

First Proof (Computational): Let $v \in V$ be an arbitrary vector. We can express v uniquely as a linear combination of the basis vectors in $\mathcal{B} = (v_1, \dots, v_n)$:

$$v = \sum_{j=1}^{n} x_j v_j.$$

By definition, the coordinate vector of v with respect to $\mathcal{B}$ is $[v]_{\mathcal{B}} = (x_1, \dots, x_n)^{\mathsf{T}} \in K^n$. Since T is linear, we can apply it to v:

$$T(v) = T\left(\sum_{j=1}^{n} x_j v_j\right) = \sum_{j=1}^{n} x_j T(v_j).$$

Now, we take the coordinate vector of $T(v)$ with respect to the basis $\mathcal{C}$. Since the coordinate map $\Phi_{\mathcal{C}}$ is an isomorphism, it is linear, and we obtain:

$$[T(v)]_{\mathcal{C}} = \Phi_{\mathcal{C}}(T(v)) = \Phi_{\mathcal{C}}\left(\sum_{j=1}^{n} x_j T(v_j)\right) = \sum_{j=1}^{n} x_j [T(v_j)]_{\mathcal{C}}.$$

Observe that this last expression is exactly the definition of the matrix-vector product of the matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ (whose columns are $[T(v_j)]_{\mathcal{C}}$, see Page 113) and the column vector $[v]_{\mathcal{B}}$ (whose entries are x_j):

$$\sum_{j=1}^{n} x_j [T(v_j)]_{\mathcal{C}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}}.$$

Therefore, we conclude that $[T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T(v)]_{\mathcal{C}}$.

Second Proof (Structural): Recall that the representation matrix $A := [T]_{\mathcal{C}}^{\mathcal{B}}$ is uniquely defined by the relation $T_A = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$. Evaluating the linear transformation T_A at the coordinate vector $[v]_{\mathcal{B}} = \Phi_{\mathcal{B}}(v) \in K^n$ yields:

$$T_A([v]_{\mathcal{B}}) = \left(\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}\right)(\Phi_{\mathcal{B}}(v)) = \Phi_{\mathcal{C}}\left(T\left(\Phi_{\mathcal{B}}^{-1}(\Phi_{\mathcal{B}}(v))\right)\right) = \Phi_{\mathcal{C}}(T(v)) = [T(v)]_{\mathcal{C}}.$$

On the other hand, the map T_A acts on vectors in K^n by left-multiplication by the matrix A. Thus, we also have:

$$T_A([v]_{\mathcal{B}}) = A \cdot [v]_{\mathcal{B}} = [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}}.$$

Equating both expressions gives $[T]^{\mathcal{B}}_{\mathcal{C}} \cdot [v]_{\mathcal{B}} = [T(v)]_{\mathcal{C}}$, which completes the proof. **Q.E.D.**

Example 16.2 (*Examples of Representation Matrices*):

(a) Let $V = K^n$ and $W = K^m$, and let $T : K^n \to K^m$ be given by $T = T_A$ for $A \in \mathcal{M}_{m\times n}(K)$, so $T(v) = A \cdot v$. Take $\mathcal{B} := (e_1, \dots, e_n)$ and $\mathcal{C} := (e_1, \dots, e_m)$ to be the standard bases. Then, $[T]^{\mathcal{B}}_{\mathcal{C}} = A$. *(Check the details!)*

(b) Let $V = K[x]_3$ and $W = K[x]_2$, and let $D : V \to W$ be the derivative map defined by $D(p(x)) = p'(x)$. Let $\mathcal{B} := (1, x, x^2, x^3)$ and $\mathcal{C} := (1, x, x^2)$. We calculate:

$$\begin{aligned} D(1) &= 0 = 0\cdot 1 + 0\cdot x + 0\cdot x^2 \\ D(x) &= 1 = 1\cdot 1 + 0\cdot x + 0\cdot x^2 \\ D(x^2) &= 2x = 0\cdot 1 + 2\cdot x + 0\cdot x^2 \\ D(x^3) &= 3x^2 = 0\cdot 1 + 0\cdot x + 3\cdot x^2 \end{aligned}$$

Therefore, the representation matrix is:

$$[D]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

(c) Revisiting the previous item, suppose we view D as a linear map $D : K[x]_3 \to K[x]_3$, so that the same map is now given the same basis on both sides (strictly, by Definition 4.2 this is a **different** map, since the target has changed; only the rule $p \mapsto p'$ is unchanged, and the effect of enlarging the target is exactly the extra zero row below). Then,

$$[D]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Representing Composition of Linear Maps by Matrices

Let $T : V \to W$ and $S : W \to U$ be linear maps. Let $\mathcal{A} = (v_1, \dots, v_n)$ be a basis for V, let $\mathcal{B} = (w_1, \dots, w_m)$ be a basis for W, and let $\mathcal{C} = (u_1, \dots, u_p)$ be a basis for U. The dimensions are thus $n = \dim V$, $m = \dim W$, and $p = \dim U$.

Consider the composition $S \circ T : V \longrightarrow U$.

Question 16.2: What is the matrix $[S \circ T]^{\mathcal{A}}_{\mathcal{C}}$? How is it related to the matrices $[T]^{\mathcal{A}}_{\mathcal{B}}$ and $[S]^{\mathcal{B}}_{\mathcal{C}}$?

To answer this, let us write the representation matrices as follows:

$$[T]^{\mathcal{A}}_{\mathcal{B}} = (a_{ij}), \quad [S]^{\mathcal{B}}_{\mathcal{C}} = (b_{ij}), \quad [S\circ T]^{\mathcal{A}}_{\mathcal{C}} = (c_{ij}).$$

By definition (recall the column description of a representation matrix on Page 113), the images of the basis vectors are given by:

$$T(v_j) = \sum_{k=1}^{m} a_{kj} w_k, \quad \forall\, 1 \le j \le n, \tag{16.1}$$

$$S(w_k) = \sum_{i=1}^{p} b_{ik} u_i, \quad \forall\, 1 \le k \le m, \tag{16.2}$$

$$(S\circ T)(v_j) = \sum_{i=1}^{p} c_{ij} u_i, \quad \forall\, 1 \le j \le n. \tag{16.3}$$

But we can also calculate $(S \circ T)(v_j)$ using (16.1) and (16.2):

$$\begin{aligned}(S \circ T)(v_j) &= S\left(\sum_{k=1}^{m} a_{kj} w_k\right) \\ &= \sum_{k=1}^{m} a_{kj} S(w_k) \\ &= \sum_{k=1}^{m} a_{kj}\left(\sum_{i=1}^{p} b_{ik} u_i\right) \\ &= \sum_{k=1}^{m}\sum_{i=1}^{p} a_{kj} b_{ik} u_i = \sum_{i=1}^{p}\left(\sum_{k=1}^{m} b_{ik} a_{kj}\right) \cdot u_i.\end{aligned}$$

Comparing this with (16.3) (the coefficients of a vector with respect to the basis $\mathcal{C} = (u_1, \ldots, u_p)$ are unique, so the two representations must agree term by term), we find that $c_{ij} = \sum_{k=1}^{m} b_{ik} a_{kj}$. In other words, row i of $[S]_{\mathcal{C}}^{\mathcal{B}}$ multiplied by column j of $[T]_{\mathcal{B}}^{\mathcal{A}}$ yields the entry c_{ij}:

$$\begin{pmatrix} b_{i1} & \ldots & b_{im} \end{pmatrix} \cdot \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} = c_{ij}.$$

This reveals the following.

Proposition (*Representation Matrix of a Composition*): Let $T : V \to W$ and $S : W \to U$ be linear maps between finite-dimensional vector spaces, and let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ be bases for V, W, and U, respectively, with $n := \dim V$, $m := \dim W$, and $p := \dim U$. Write $[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij})$, $[S]_{\mathcal{C}}^{\mathcal{B}} = (b_{ij})$, and $[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = (c_{ij})$. Then,

$$c_{ij} = \sum_{k=1}^{m} b_{ik} a_{kj} \qquad \text{for all } 1 \le i \le p \text{ and } 1 \le j \le n.$$

In the notation introduced in the next chapter, this reads

$$[S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}} = [S \circ T]_{\mathcal{C}}^{\mathcal{A}},$$

where the matrix dimensions follow $(p \times m) \cdot (m \times n) = (p \times n)$.

Remark 16.5: There is a shorter route to the same statement, in the spirit of the second proof of Proposition 16.4. Put $A := [T]_{\mathcal{B}}^{\mathcal{A}}$, $B := [S]_{\mathcal{C}}^{\mathcal{B}}$, and $C := [S \circ T]_{\mathcal{C}}^{\mathcal{A}}$. Each of the three matrices is defined by the corresponding commutative diagram, and therefore,

$$T_C = \Phi_{\mathcal{C}} \circ (S \circ T) \circ \Phi_{\mathcal{A}}^{-1} = \left(\Phi_{\mathcal{C}} \circ S \circ \Phi_{\mathcal{B}}^{-1}\right) \circ \left(\Phi_{\mathcal{B}} \circ T \circ \Phi_{\mathcal{A}}^{-1}\right) = T_B \circ T_A,$$

where the middle equality holds because $\Phi_{\mathcal{B}}^{-1} \circ \Phi_{\mathcal{B}} = \mathrm{id}_W$. So C is the unique matrix whose associated map is the composition $T_B \circ T_A$, and the computation above merely reads off its entries. This is also the reason why the new operation, once it is defined, has to be defined the way it is.

In order to obtain $[S \circ T]_{\mathcal{C}}^{\mathcal{A}}$ from $[T]_{\mathcal{B}}^{\mathcal{A}}$ and $[S]_{\mathcal{C}}^{\mathcal{B}}$, we need to use a new operation on matrices. It is called *"matrix multiplication"*. But is this operation really new?

— Yes, but not entirely...

AI-Note: This chapter came through the transcription essentially intact: every statement of the notes is present, in the notes' order, and the printed numbers 16.1–16.4 already agreed with the handwritten ones. The abrupt final line is Prof. Biran's own; his notes end there, in the middle of the thought that chapter 17 picks up.

What needed repair was small and mostly typographic. The third example carried a `\setcounter{enumi}{1}`, annotated as matching the professor's numbering, which made the list print two items labelled **(b)**; the notes label that item $2'$ because it revisits the previous one, so the counter manipulation is dropped and the revisiting is said in words instead. The sentence introducing the commutative diagram ended in the word *"test"*, evidently a leftover. A `\usetikzlibrary{babel}` sat in the middle of the document, although the preamble already loads that library. Prof. Biran's aside *"check the details!"* on the first representation-matrix example is restored, and his closing line is now set as the reply it is rather than as a stray hyphen.

The proof that $\Phi_{\mathcal{B}}$ is an isomorphism cited the definition of isomorphic spaces for the step that bijectivity suffices. That step is Lemma 15.b.4, which the transcript of the previous chapter had dropped altogether and which is now restored; the citation points there.

One addition of the transcript is worth keeping and worth flagging: the notes prove Proposition 16.4 only structurally, by observing that the asserted identity is exactly the commutative diagram defining $[T]^{\mathcal{B}}_{\mathcal{C}}$. The computational proof that precedes it in the text above is not Prof. Biran's.

A second pass, this one looking only for gaps in rigour, found that the chapter states four of its central facts in running prose and never in an environment: the coordinate map $\Phi_{\mathcal{B}}$, which Proposition 16.2 is named after and which chapter 18 uses throughout; the representation matrix itself; the column description of $[T]^{\mathcal{B}}_{\mathcal{C}}$, which chapters 17b, 18, 22a and 25 all cite as *"the definition of the representation matrix"* although it is a derived fact; and the composition rule that is the punchline of the chapter. Each of the four now has an environment of its own. All four are deliberately unnumbered, so that Prof. Biran's numbering 16.1–16.4 is unchanged and nothing downstream shifts; they are cited with `\cpageref` rather than with `\cref`. The prose inside them is the transcript's, with only the hypotheses restated so that each block stands on its own.

Three assertions were given their reason in a parenthesis: the diagram above commutes by construction, its lower arrow being defined as that very composition; the comparison of coefficients in the composition computation rests on the uniqueness of a representation in a basis; and the third example called $D\colon K[x]_3 \to K[x]_3$ *"the same map"* as $D\colon K[x]_3 \to K[x]_2$, which Definition 4.2 expressly denies, the target being part of the data of a map. The extra zero row in that example is exactly what the enlarged target records. Finally, two short second proofs were added, both marked as mine: the explicit inverse of $\Phi_{\mathcal{B}}$, which the notes already write down, proves bijectivity by itself, and the composition rule drops out of the three defining diagrams in a single line.

Matrices (Chapter 17)

Matrix Multiplication (Section 17.a)

Definition 17.a.1 (*Matrix Multiplication*): Let $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$. We define a new matrix, which is called the "*product*" (or "*multiplication*") of A and B, denoted by $A \cdot B \in \mathcal{M}_{m\times p}(K)$.
Write $A = (a_{ij})_{\substack{1\le i\le m\\ 1\le j\le n}}$ and $B = (b_{ij})_{\substack{1\le i\le n\\ 1\le j\le p}}$. The matrix $A \cdot B$ has entries $(c_{ij})_{\substack{1\le i\le m\\ 1\le j\le p}}$, where

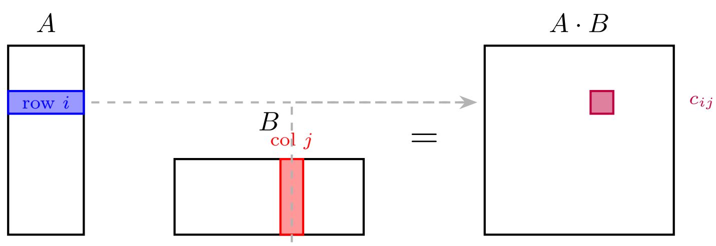
$$c_{ij} := a_{i1}b_{1j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}$$

We often write AB instead of $A \cdot B$.

Important remark 17.1: The product AB is defined only when the number of columns of A equals the number of rows of B.

An alternative description of $A \cdot B$ can be visualized by taking the dot product of the i-th row of A with the j-th column of B:

$$c_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B) = a_{i1}b_{1j} + \cdots + a_{ik}b_{kj} + \cdots + a_{in}b_{nj}$$



Another way to describe it is by considering the columns of B, denoted as $w_1, \ldots, w_p$. The matrix $A \cdot B$ is then formed by multiplying A with each column vector w_j:

$$A \cdot \begin{pmatrix} | & & | \\ w_1 & \ldots & w_p \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ Aw_1 & \ldots & Aw_p \\ | & & | \end{pmatrix}$$

Example 17.1 (*Matrix Multiplication Example*): We compute the product of a 2×3 matrix and a 3×2 matrix:

$$\begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 3\cdot 1 + 2\cdot 0 + 1\cdot 4 & 3\cdot 2 + 2\cdot 1 + 1\cdot 0 \\ 1\cdot 1 + 0\cdot 0 + 2\cdot 4 & 1\cdot 2 + 0\cdot 1 + 2\cdot 0 \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 9 & 2 \end{pmatrix}$$

In the other order, multiplying the 3×2 matrix by the 2×3 matrix yields a 3×3 matrix:

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 5 \\ 1 & 0 & 2 \\ 12 & 8 & 4 \end{pmatrix}$$

If $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$, then $A \cdot B$ is defined, but $B \cdot A$ is not defined, unless $p = m$.

Some Special Cases

(a) If $A \in \mathcal{M}_{m\times n}(K)$ and $v \in K^n_{\text{col}}$, we have previously defined the matrix-vector product $A \cdot v \in K^m_{\text{col}}$. However, we can also identify $K^n_{\text{col}} = \mathcal{M}_{n\times 1}(K)$. If we view v as an $n \times 1$ matrix, then the matrix multiplication of A and v yields the exact same result as applying A to the vector v.

(b) If $A \in \mathcal{M}_{m\times n}(K)$ and $w \in K^m_{\text{row}} = \mathcal{M}_{1\times m}(K)$, then we can define the left-multiplication $w \cdot A \in \mathcal{M}_{1\times n}(K) = K^n_{\text{row}}$. If A is represented by its row vectors $v_1, \dots, v_m$, this operation scales and sums the rows of A according to the entries of w. This gives the product a row-wise description matching the column-wise one above. If A has rows $v_1, \dots, v_m$ and $B \in \mathcal{M}_{n\times p}(K)$ has columns $w_1, \dots, w_p$, then

$$A \cdot B = \begin{pmatrix} | & & | \\ Aw_1 & \dots & Aw_p \\ | & & | \end{pmatrix} = \begin{pmatrix} — & v_1 \cdot B & — \\ & \vdots & \\ — & v_m \cdot B & — \end{pmatrix}.$$

(c) Let $v = (a_1, \dots, a_n) \in K^n_{\text{row}} = \mathcal{M}_{1\times n}(K)$, and let $w = (b_1, \dots, b_n)^{\mathsf{T}} \in K^n_{\text{col}} = \mathcal{M}_{n\times 1}(K)$. Then the product $v \cdot w \in \mathcal{M}_{1\times 1}(K) = K$ is simply the standard scalar product:

$$v \cdot w = a_1 b_1 + \cdots + a_n b_n$$

Connections to Linear Maps

As a conclusion from our previous discussions on linear maps (the computation on Page 116), let $T : V \to W$ and $S : W \to U$ be linear maps between finite-dimensional vector spaces. Let $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{E}$ be bases for V, W, and U, respectively. The matrix representation of the composition $S \circ T$ is given by the product of their respective representation matrices:

$$[S \circ T]^{\mathcal{B}}_{\mathcal{E}} = [S]^{\mathcal{C}}_{\mathcal{E}} \cdot [T]^{\mathcal{B}}_{\mathcal{C}}$$

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[rr, bend left, "S\circ T"] & W \arrow[r, "S"] & U \end{tikzcd}$$

Here $\mathcal{B}$ is a basis of the source V, $\mathcal{C}$ a basis of the space W in the middle, and $\mathcal{E}$ a basis of the target U; each matrix in the identity above carries the basis of its own source as a superscript and the basis of its own target as a subscript, and the two occurrences of $\mathcal{C}$ meet in the middle exactly where T and S meet.

Notation and The Identity Matrix

Let $\mathcal{I}$ be an index set. For $i, j \in \mathcal{I}$, we define the "*Kronecker-delta*" as:

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Using this, we define the "*identity matrix*" $I_n \in \mathcal{M}_{n\times n}(K)$ as $I_n = (\delta_{ij})_{\substack{1\le i\le n \\ 1\le j\le n}}$. Explicitly, this is:

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

Proposition (*The Identity Map is Represented by I_n*)**:** Let V be a finite-dimensional vector space, and let $n := \dim V$. Let $\mathrm{id}_V : V \to V$ be the identity map, and let $\mathcal{B}$ be any basis for V.

Then, $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n$. In other words, the matrix representing the identity map with respect to the basis $\mathcal{B}$ is precisely the identity matrix I_n.

Exercise 17.1 (*The Identity Map is Represented by I_n*): Prove the Proposition on Page 119.

Properties of Matrix Multiplication

Proposition 17.a.2 (*Algebraic Properties of Matrix Multiplication*): Matrix multiplication satisfies the following algebraic properties:

(a) Associativity: For all $A \in \mathcal{M}_{m\times n}(K)$, $B \in \mathcal{M}_{n\times p}(K)$, and $C \in \mathcal{M}_{p\times q}(K)$, we have:
$$(AB)C = A(BC) \in \mathcal{M}_{m\times q}(K)$$

(b) Distributivity: For all $A, B \in \mathcal{M}_{m\times n}(K)$, $C \in \mathcal{M}_{n\times p}(K)$, and $C' \in \mathcal{M}_{q\times m}(K)$, we have:
$$(A+B)C = AC + BC \in \mathcal{M}_{m\times p}(K)$$
$$C'(A+B) = C'A + C'B \in \mathcal{M}_{q\times n}(K)$$

(c) Identity: For all $A \in \mathcal{M}_{m\times n}(K)$, we have $I_m A = A$ and $AI_n = A$.

(d) Scalar Multiplication: For all $\alpha \in K$, $A \in \mathcal{M}_{m\times n}(K)$, and $B \in \mathcal{M}_{n\times p}(K)$, we have:
$$\alpha(AB) = (\alpha A)B = A(\alpha B)$$

Proof:

Let $T_A : K^n \to K^m$, $T_B : K^p \to K^n$, and $T_C : K^q \to K^p$ be the linear maps defined by left-multiplication with the matrices A, B, and C, respectively (i.e., $T_A(v) = A \cdot v$ for all $v \in K^n$, $T_B(w) = B \cdot w$ for all $w \in K^p$, and $T_C(u) = C \cdot u$ for all $u \in K^q$). For any dimension r, we denote the standard basis of K^r by $\mathcal{E}_r$. We have already established that (Example 16.2, first item):
$$[T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} = A, \quad [T_B]_{\mathcal{E}_n}^{\mathcal{E}_p} = B, \quad \text{and} \quad [T_C]_{\mathcal{E}_p}^{\mathcal{E}_q} = C$$

(a) We rely on the associativity of function composition, which states that $(T_A \circ T_B) \circ T_C = T_A \circ (T_B \circ T_C)$. These are both maps from K^q to K^m. We now pass to the matrix representation of these maps with respect to the standard bases $\mathcal{E}_q$ and $\mathcal{E}_m$:
$$[(T_A \circ T_B) \circ T_C]_{\mathcal{E}_m}^{\mathcal{E}_q} = [T_A \circ (T_B \circ T_C)]_{\mathcal{E}_m}^{\mathcal{E}_q}$$
By applying the rule for the matrix representation of composed maps (on Page 116), we get:
$$[T_A \circ T_B]_{\mathcal{E}_m}^{\mathcal{E}_p} \cdot [T_C]_{\mathcal{E}_p}^{\mathcal{E}_q} = [T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} \cdot [T_B \circ T_C]_{\mathcal{E}_n}^{\mathcal{E}_q}$$
Expanding further, we obtain:
$$\left([T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} \cdot [T_B]_{\mathcal{E}_n}^{\mathcal{E}_p}\right) \cdot [T_C]_{\mathcal{E}_p}^{\mathcal{E}_q} = [T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} \cdot \left([T_B]_{\mathcal{E}_n}^{\mathcal{E}_p} \cdot [T_C]_{\mathcal{E}_p}^{\mathcal{E}_q}\right)$$
Substituting the matrices back in yields $(A \cdot B) \cdot C = A \cdot (B \cdot C)$.

(b) The shapes here are the ones of statement **(b)** and not the ones fixed above, since A and B must now be summable: we have $A, B \in \mathcal{M}_{m\times n}(K)$ and $C \in \mathcal{M}_{n\times p}(K)$, so that $T_A, T_B : K^n \to K^m$ and $T_C : K^p \to K^n$. From the properties of linear maps, we know that $(T_A + T_B) \circ T_C = T_A \circ T_C + T_B \circ T_C$. Passing to the matrix representations, we have:
$$[(T_A + T_B) \circ T_C]_{\mathcal{E}_m}^{\mathcal{E}_p} = [T_A \circ T_C + T_B \circ T_C]_{\mathcal{E}_m}^{\mathcal{E}_p}$$
This implies that:
$$[T_A + T_B]_{\mathcal{E}_m}^{\mathcal{E}_n} \cdot [T_C]_{\mathcal{E}_n}^{\mathcal{E}_p} = [T_A \circ T_C]_{\mathcal{E}_m}^{\mathcal{E}_p} + [T_B \circ T_C]_{\mathcal{E}_m}^{\mathcal{E}_p}$$
Which directly gives us $(A+B) \cdot C = A \cdot C + B \cdot C$ (both steps use that a sum of maps given by matrices is again given by a matrix, namely $T_M + T_N = T_{M+N}$ for M, N of equal shape, since both sides send v to $Mv + Nv = (M+N)v$; in particular $[T_A + T_B]_{\mathcal{E}_m}^{\mathcal{E}_n} = A + B$). The other distributive identity in **(b)** is proven similarly.

(c) The identity matrix I_m represents the identity map: $I_m = [\mathrm{id}_{K^m}]^{\mathcal{E}_m}_{\mathcal{E}_m}$. Since composing with the identity map does not change a function, we have $\mathrm{id}_{K^m} \circ T_A = T_A$. Therefore:

$$[\mathrm{id}_{K^m} \circ T_A]^{\mathcal{E}_n}_{\mathcal{E}_m} = [T_A]^{\mathcal{E}_n}_{\mathcal{E}_m} = A$$

This implies that $I_m \cdot A = A$ (the composition rule turns the left-hand side into $[\mathrm{id}_{K^m}]^{\mathcal{E}_m}_{\mathcal{E}_m} \cdot [T_A]^{\mathcal{E}_n}_{\mathcal{E}_m} = I_m \cdot A$). The proof for $AI_n = A$ follows the same logic.

(d) Let $T_A : K^n \to K^m$ and $T_B : K^p \to K^n$ be the linear maps associated with A and B, respectively. We want to show that $\alpha(AB) = (\alpha A)B = A(\alpha B)$. This is equivalent to showing that the corresponding linear maps are equal.

First, consider the map $Q : K^n \to K^n$ defined by $Q(v) := \alpha v$. This map is linear. If $\mathcal{E}_n$ is the standard basis for K^n, then for any $e_j \in \mathcal{E}_n$, $Q(e_j) = \alpha e_j$. Thus, the matrix representation of Q with respect to $\mathcal{E}_n$ is $[Q]^{\mathcal{E}_n}_{\mathcal{E}_n} = \alpha I_n$.

Now, let's look at the compositions:

- The linear map corresponding to $(\alpha A)B$ is $T_{\alpha A} \circ T_B$. We know that $T_{\alpha A}(v) = (\alpha A)v = \alpha(Av) = \alpha T_A(v)$, so $T_{\alpha A} = \alpha T_A$. Thus, $T_{(\alpha A)B} = (\alpha T_A) \circ T_B$.
- The linear map corresponding to $A(\alpha B)$ is $T_A \circ T_{\alpha B}$. Similarly, $T_{\alpha B} = \alpha T_B$. Thus, $T_{A(\alpha B)} = T_A \circ (\alpha T_B)$.

We need to show that $(\alpha T_A) \circ T_B = T_A \circ (\alpha T_B)$. For any $v \in K^p$:

$$\begin{aligned} ((\alpha T_A) \circ T_B)(v) &= (\alpha T_A)(T_B(v)) \\ &= \alpha(T_A(T_B(v))) \quad \text{(by definition of scalar multiplication for maps)} \end{aligned}$$

And for the other side:

$$\begin{aligned} (T_A \circ (\alpha T_B))(v) &= T_A((\alpha T_B)(v)) \\ &= T_A(\alpha(T_B(v))) \quad \text{(by definition of scalar multiplication for maps)} \\ &= \alpha(T_A(T_B(v))) \quad \text{(by linearity of } T_A) \end{aligned}$$

Since both compositions yield the same result, the corresponding matrices are equal: $(\alpha A)B = A(\alpha B)$.

Finally, to show $\alpha(AB) = (\alpha A)B$: The matrix $\alpha(AB)$ corresponds to the linear map $\alpha(T_A \circ T_B)$. The matrix $(\alpha A)B$ corresponds to the linear map $(\alpha T_A) \circ T_B$. We have already shown that $(\alpha T_A) \circ T_B = \alpha(T_A \circ T_B)$ (from the first part of the derivation above). Therefore, $\alpha(AB) = (\alpha A)B$.

Q.E.D.

The notes prepare statement **(d)** differently, through the following observation about the map that scales every vector by a fixed factor.

Claim 17.1 (*The Scaling Map is Represented by αI_n*): Let V be a finite-dimensional vector space over K, let $\alpha \in K$, and let $\mathcal{B}$ be a basis for V. Consider the map $Q : V \to V$ defined by $Q(v) := \alpha v$. Then Q is linear, and moreover

$$[Q]^{\mathcal{B}}_{\mathcal{B}} = \alpha \cdot I_n = \begin{pmatrix} \alpha & & 0 \\ & \ddots & \\ 0 & & \alpha \end{pmatrix}, \qquad \text{where } n = \dim V.$$

Exercise 17.2 (*Preparation for Statement (d)*): Prove Claim 17.1, and use it to prove statement **(d)** of Proposition 17.a.2.

Remark 17.1: One can prove Proposition 17.a.2 also by direct calculation with the entries. It is worth trying once, say for $(AB)C = A(BC)$, to see that it is not very pleasant, and to appreciate how much work the passage to linear maps saves.

Remark 17.2: In the space $\mathcal{M}_{n\times n}(K)$, we can add and subtract matrices, and we can also multiply matrices. There is additionally a multiplicative unit I_n. Because these operations satisfy associativity, distributivity, and other relevant axioms, it follows that $\mathcal{M}_{n\times n}(K)$ forms a (non-commutative) ring with a unity.

Important remark 17.2: Matrix multiplication is, in general, **not commutative**. There exist matrices A and B such that $A \cdot B \neq B \cdot A$. For example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$
$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Invertible Matrices

Definition 17.a.3 (*Invertibility*): An $n \times n$ matrix $A \in \mathcal{M}_{n\times n}(K)$ is called "*invertible*" if there exists a matrix $B \in \mathcal{M}_{n\times n}(K)$ such that $A \cdot B = B \cdot A = I_n$.

Proposition (*Uniqueness of the Inverse Matrix*): If A is invertible, then there exists **only one** matrix B such that $A \cdot B = B \cdot A = I_n$.

Proof:

Assume there are two matrices B and C such that $AB = BA = I_n$ and $AC = CA = I_n$. Then, we can calculate:

$$B = B \cdot I_n = B \cdot (A \cdot C) = (B \cdot A) \cdot C = I_n \cdot C = C$$

This proves the uniqueness. **Q.E.D.**

Definition 17.a.4 (*Inverse Matrix and the General Linear Group*): If A is invertible, the unique matrix B such that $AB = BA = I_n$ (unique by the Proposition on Page 122) is called the "*inverse*" of A and is denoted by A^{-1}. We have:

$$A \cdot A^{-1} = A^{-1} \cdot A = I_n$$

The set of all invertible $n \times n$ matrices over K is denoted by:

$$\mathrm{GL}_n(K) = \mathrm{GL}(n, K) := \{A \in \mathcal{M}_{n\times n}(K) \mid A \text{ is invertible}\}$$

Definition 17.a.5 (*Group*): Let G be a set and $\cdot : G \times G \to G$, $(a, b) \mapsto a \cdot b \in G$, be a binary operation. We say that $(G, \cdot)$ is a "*group*" if the following axioms hold:

- **(a) Associativity:** For all $g_1, g_2, g_3 \in G$, we have $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3$.
- **(b) Identity Element:** There exists an element $e \in G$ such that $e \cdot g = g \cdot e = g$ for all $g \in G$. The element e is called the "*unit*" (or "*identity*") of G.
- **(c) Inverse Element:** For every $g \in G$, there exists an element $h \in G$ such that $g \cdot h = h \cdot g = e$. The element h is called the "*inverse*" of g.

Furthermore, if G satisfies $g_1 \cdot g_2 = g_2 \cdot g_1$ for all $g_1, g_2 \in G$, we say that G is a "*commutative group*" (or "*Abelian group*").

Proposition 17.a.6 (*Group Properties*): Let G be a group. Then:

- **(a)** The unit $e \in G$ is unique. (The unit e is sometimes denoted as $1 \in G$.)
- **(b)** For every $g \in G$, its inverse is uniquely defined. It is denoted by g^{-1} (so that $g \cdot g^{-1} = g^{-1} \cdot g = e$).
- **(c)** For every $g \in G$, we have $(g^{-1})^{-1} = g$.

(d) For all $g_1, g_2 \in G$, we have $(g_1 \cdot g_2)^{-1} = g_2^{-1} \cdot g_1^{-1}$.

Proof:

(a) Let e and e' both be units of G. Then, $e = e \cdot e' = e'$, where the first equality holds because e' is a unit, and the second because e is one.

(b) Let h and h' both be inverses of g. Then,

$$h = h \cdot e = h \cdot (g \cdot h') = (h \cdot g) \cdot h' = e \cdot h' = h',$$

using associativity in the middle step.

(c) The defining equations $g \cdot g^{-1} = g^{-1} \cdot g = e$ say, read the other way round, that g is an inverse of g^{-1}. By **(b)** the inverse is unique, and therefore, $(g^{-1})^{-1} = g$.

(d) We check that $g_2^{-1} \cdot g_1^{-1}$ is a two-sided inverse of $g_1 \cdot g_2$:

$$(g_1 g_2)(g_2^{-1} g_1^{-1}) = g_1 (g_2 g_2^{-1}) g_1^{-1} = g_1 \cdot e \cdot g_1^{-1} = g_1 g_1^{-1} = e,$$

and symmetrically $(g_2^{-1} g_1^{-1})(g_1 g_2) = e$. By **(b)** again, $(g_1 g_2)^{-1} = g_2^{-1} g_1^{-1}$.

Q.E.D.

Proposition 17.a.7 (*The General Linear Group*): The set $\mathrm{GL}_n(K)$, endowed with matrix multiplication $(A, B) \mapsto A \cdot B$, is a group. The unity is the identity matrix I_n. The inverse of a matrix $A \in \mathrm{GL}_n(K)$ is A^{-1}. Moreover, for all $A, B \in \mathrm{GL}_n(K)$, we have:

$$(A^{-1})^{-1} = A \quad \text{and} \quad (AB)^{-1} = B^{-1} A^{-1}$$

Proof:

Let $A, B \in \mathrm{GL}_n(K)$. To establish that $\mathrm{GL}_n(K)$ forms a group, we first need to show closure, namely that $AB \in \mathrm{GL}_n(K)$ as well. Indeed, we can verify this by checking the inverse property:

$$(B^{-1} \cdot A^{-1})(AB) = B^{-1} A^{-1} A B = B^{-1} I_n B = B^{-1} B = I_n$$

and, multiplying from the right,

$$(AB)(B^{-1} A^{-1}) = A B B^{-1} A^{-1} = A I_n A^{-1} = A A^{-1} = I_n$$

This proves that AB is invertible with inverse $B^{-1} A^{-1}$, and therefore, $AB \in \mathrm{GL}_n(K)$. The remaining group axioms are already available: associativity is Proposition 17.a.2 **(a)**, the identity element is I_n by Proposition 17.a.2 **(c)**, and inverses exist by the very definition of $\mathrm{GL}_n(K)$. The other statements in the proposition follow immediately from what we have proved earlier regarding the uniqueness of inverses, namely Proposition 17.a.6 **(c)** and **(d)**. **Q.E.D.**

Corollary 17.a.8 (*Unique Solution via Matrix Inverse*): Let $A \in \mathrm{GL}_n(K)$. Then, for all $b \in K^n$, the linear system of equations $A \cdot x = b$ has a unique solution, namely $x = A^{-1} \cdot b$.

Proof:

If $Ax = b$, then multiplying both sides from the left by A^{-1} yields:

$$x = I_n x = (A^{-1} A) x = A^{-1}(Ax) = A^{-1} b.$$

Conversely, if we substitute $x = A^{-1} b$ into the equation, we obtain:

$$A(A^{-1} b) = (A A^{-1}) b = I_n b = b$$

This confirms both the existence and uniqueness of the solution. **Q.E.D.**

Proposition 17.a.9 (*Equivalence of Invertibility and Isomorphism*): Let $A \in \mathcal{M}_{n \times n}(K)$. Then, $A \in \mathrm{GL}_n(K)$ (i.e., A is invertible) if and only if the associated linear map $T_A : K^n \to K^n$ is an isomorphism. Moreover, in this case, $(T_A)^{-1} = T_{A^{-1}}$.

Proof:

First, assume that A is invertible. We claim that $T_{A^{-1}} \circ T_A = \mathrm{id}_{K^n}$ and $T_A \circ T_{A^{-1}} = \mathrm{id}_{K^n}$, which will show that T_A is an isomorphism. Indeed, for all $v \in K^n$:

$$T_{A^{-1}} \circ T_A(v) = A^{-1} \cdot (T_A(v)) = A^{-1} \cdot A \cdot v = I_n \cdot v = v$$

Similarly, one shows that for all $v \in K^n$:

$$T_A \circ T_{A^{-1}}(v) = I_n \cdot v = v$$

This proves that T_A is an isomorphism.

Conversely, assume that $T_A : K^n \to K^n$ is an isomorphism. Let $S := (T_A)^{-1}$. By Lemma 15.a.5, we know that any linear map from K^n to K^n is given by left-multiplication with a matrix, so $S = T_B$ for some $B \in \mathcal{M}_{n\times n}(K)$. Now, for all $w \in K^n$:

$$w = S \circ T_A(w) = T_B(A \cdot w) = B \cdot A \cdot w = (BA) \cdot w$$

We apply this equality for the standard basis vectors $w = e_1, w = e_2, \ldots, w = e_n$ and obtain that $BA = I_n$. Similarly, using $T_A \circ S = \mathrm{id}_{K^n}$, one shows that $AB = I_n$ too. Thus, A is invertible.

Q.E.D.

Exercise 17.3 (*Properties of the Transpose Map*):

(a) Let $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$. Prove that $(AB)^\mathsf{T} = B^\mathsf{T} \cdot A^\mathsf{T}$.
(*Hint: Proving this by direct calculation is not so bad.*)

(b) Let $A \in \mathrm{GL}_n(K)$. Prove that $A^\mathsf{T} \in \mathrm{GL}_n(K)$ and that $(A^\mathsf{T})^{-1} = (A^{-1})^\mathsf{T}$.

Definition 17.a.10 (*Triangular and Diagonal Matrices*): A matrix $A \in \mathcal{M}_{n\times n}(K)$, with entries $A = (a_{ij})_{\substack{1\le i\le n\\ 1\le j\le n}}$, is called:

- "*Upper triangular*" if $a_{ij} = 0$ for all $i > j$.
- "*Lower triangular*" if $a_{ij} = 0$ for all $i < j$.
- "*Diagonal*" if $a_{ij} = 0$ for all $i \neq j$. A diagonal matrix has the form:

$$\begin{pmatrix} * & & 0 \\ & \ddots & \\ 0 & & * \end{pmatrix}$$

Lemma 17.a.11 (*Closure of Triangular Matrices*): Let A and B be two diagonal matrices (respectively, both upper triangular, or both lower triangular). Then, their product $A \cdot B$ is of the same type.

The notes leave the proof as an exercise; it is carried out among the solutions at the end of this section, on Page 124.

Solutions to the Exercises

Proof that the Identity Map is Represented by I_n (on Page 119):

Let $\mathcal{B} = (v_1, \ldots, v_n)$. The j-th column of $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}}$ is the coordinate vector of $\mathrm{id}_V(v_j) = v_j$ with respect to $\mathcal{B}$. Writing $v_j = 0 \cdot v_1 + \cdots + 1 \cdot v_j + \cdots + 0 \cdot v_n$ shows that this coordinate vector is e_j, whose i-th entry is δ_{ij}. So the (i,j)-entry of $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}}$ is δ_{ij}, which is exactly I_n. **Q.E.D.**

Proof of Claim 17.1 and of Proposition 17.a.2 (d):

The map $Q(v) := \alpha v$ is linear, since $Q(av+bv') = \alpha(av+bv') = a\alpha v + b\alpha v' = aQ(v) + bQ(v')$. For its matrix, the j-th column of $[Q]^{\mathcal{B}}_{\mathcal{B}}$ is $[Q(v_j)]_{\mathcal{B}} = [\alpha v_j]_{\mathcal{B}} = \alpha e_j$, and a matrix whose j-th column is αe_j for every j is precisely αI_n.

Now let $\alpha \in K$, $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$. Applying the claim in K^m with respect to the standard basis, multiplication by α on K^m is represented by αI_m, and therefore $\alpha C = (\alpha I_m)\cdot C$ for every $C \in \mathcal{M}_{m\times q}(K)$; likewise $\alpha A = (\alpha I_m)A$. Using associativity, Proposition 17.a.2 **(a)**, twice:

$$\alpha(AB) = (\alpha I_m)(AB) = \big((\alpha I_m)A\big)B = (\alpha A)B.$$

For the remaining identity, note that $(\alpha I_m)A = A(\alpha I_n)$, since both sides have (i,j)-entry αa_{ij}. Hence

$$(\alpha A)B = \big(A(\alpha I_n)\big)B = A\big((\alpha I_n)B\big) = A(\alpha B),$$

again by associativity. Together these give $\alpha(AB) = (\alpha A)B = A(\alpha B)$. **Q.E.D.**

Proof of Exercise 17.3:

(a) Write $A = (a_{ij}) \in \mathcal{M}_{m\times n}(K)$ and $B = (b_{jk}) \in \mathcal{M}_{n\times p}(K)$, and put $C := AB$, so that $c_{ik} = \sum_{\ell=1}^{n} a_{i\ell}b_{\ell k}$. Both $(AB)^{\mathsf{T}}$ and $B^{\mathsf{T}}A^{\mathsf{T}}$ lie in $\mathcal{M}_{p\times m}(K)$, so it suffices to compare entries. The (k,i)-entry of $(AB)^{\mathsf{T}}$ is

$$c_{ik} = \sum_{\ell=1}^{n} a_{i\ell}b_{\ell k}.$$

On the other hand, the (k,i)-entry of $B^{\mathsf{T}}A^{\mathsf{T}}$ is the k-th row of B^{T} times the i-th column of A^{T}, that is,

$$\sum_{\ell=1}^{n} (B^{\mathsf{T}})_{k\ell}(A^{\mathsf{T}})_{\ell i} = \sum_{\ell=1}^{n} b_{\ell k}a_{i\ell},$$

which is the same sum, since the entries are scalars and K is commutative. Hence $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$.

(b) Let $A \in \mathrm{GL}_n(K)$. Using **(a)** and $I_n^{\mathsf{T}} = I_n$,

$$A^{\mathsf{T}}\cdot(A^{-1})^{\mathsf{T}} = (A^{-1}A)^{\mathsf{T}} = I_n^{\mathsf{T}} = I_n, \qquad (A^{-1})^{\mathsf{T}}\cdot A^{\mathsf{T}} = (AA^{-1})^{\mathsf{T}} = I_n^{\mathsf{T}} = I_n.$$

So A^{T} has a two-sided inverse, which means $A^{\mathsf{T}} \in \mathrm{GL}_n(K)$ and $(A^{\mathsf{T}})^{-1} = (A^{-1})^{\mathsf{T}}$. Note that no determinants are needed: part **(a)** alone produces the inverse explicitly.

Q.E.D.

Proof of Lemma 17.a.11:

We give the upper triangular case; the other two follow from it.

Let $A = (a_{ij})$ and $B = (b_{ij})$ be upper triangular, so that $a_{ik} = 0$ whenever $i > k$, and $b_{kj} = 0$ whenever $k > j$. Write $C := AB = (c_{ij})$, so that $c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$.

Fix a pair of indices with $i > j$, and consider a single term $a_{ik}b_{kj}$ of that sum. Exactly one of the following two cases occurs:

- $k < i$. Then $a_{ik} = 0$, because A is upper triangular.
- $k \geq i$. Then $k \geq i > j$, so $k > j$, and hence $b_{kj} = 0$, because B is upper triangular.

In either case the term vanishes, so $c_{ij} = 0$ for all $i > j$, which says precisely that C is upper triangular.

For the lower triangular case, note that A is lower triangular if and only if A^{T} is upper triangular. So if A and B are lower triangular, then A^{T} and B^{T} are upper triangular, and by Exercise 17.3 **(a)** together with the case already proved,

$$(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}} \quad \text{is upper triangular,}$$

which makes AB lower triangular. Finally, a matrix is diagonal exactly when it is both upper and lower triangular, so the diagonal case follows from the two cases just proved. **Q.E.D.**

AI-Note: Every numbered statement of the notes for this section is present in the transcript, and 17.a.1–17.a.11 already matched the handwritten numbering. What was missing sits around them.

Restored: the preparation the notes give for statement **(d)** of Proposition 17.a.2, namely Claim 17.1, that scaling by α is represented by αI_n, which is how the notes ask for that statement to be proved; the remark that the proposition can also be proved by direct calculation, with Prof. Biran's warning that it is *"not very pleasant"*; and the row-wise description of a product in the second special case, which the notes give alongside the column-wise one.

Removed: a second, verbatim copy of the remark that $\mathcal{M}_{n\times n}(K)$ is a non-commutative ring. It appeared twice in the transcript, a few lines apart, and once in the notes.

Two things were wrong rather than missing. The diagram under *"Connections to Linear Maps"* placed $\mathrm{Sp}(\mathcal{B})$ below V joined by a rotated "$\in$", asserting $\mathrm{Sp}(\mathcal{B}) \in V$, which is not a relation between a span and the space containing it; the notes have no such diagram, and it is replaced by the composition diagram with the bookkeeping said in words. And the proof of Lemma 17.a.11 opened with the word *"Exercise"* and then argued, in the middle of the paragraph, that *"for a_{ik} to be non-zero, k must be less than or equal to i"*, the condition runs the other way. The conclusion was right and the argument was not.

The notes mark four things *"Exc."*: the identity matrix remark, the preparation claim and statement **(d)**, the two transpose identities, and Lemma 17.a.11. All four are now stated as exercises and solved on Page 124. The transcript had instead embedded *"proof sketches"* inside the exercise environment, and one of them derived $A^{\mathsf{T}} \in \mathrm{GL}_n(K)$ from $\det(A^{\mathsf{T}}) = \det(A)$, determinants are three chapters away, and the transpose identity of part **(a)** produces the inverse directly.

Finally, terms being defined for the first time, the product, the Kronecker-delta, the identity matrix, invertible, the inverse, upper and lower triangular, diagonal, were marked with `\qt` rather than `\newterm`, which the house style reserves for exactly this.

A later pass looked only for gaps in rigour, and turned up one real error. The whole proof of Proposition 17.a.2 carried its representation matrices the wrong way round: it wrote $[T_A]^{\mathcal{E}_m}_{\mathcal{E}_n}$ for the map $T_A\colon K^n \to K^m$, putting the basis of the target on top. The convention of this book, and the one used in Section 17.b and everywhere after, is that the superscript carries the basis of the source. Since every occurrence in that proof was swapped, the argument was internally consistent and read perfectly well, which is exactly why it survived; but each matrix in it named the wrong map. All of them are now the right way round.

Statement **(b)** of that proposition has different shapes from **(a)** and **(d)**, because A and B have to be summable there. Its proof nonetheless inherited the maps fixed for **(a)**, so that T_C quietly changed from $K^q \to K^p$ into $K^q \to K^n$ halfway through. The shapes for **(b)** are now stated where they change, and the displays use them. The step from maps back to matrices in **(b)** also needs $T_M + T_N = T_{M+N}$, which is one line and is now given.

Proposition 17.a.6 stood with neither a proof nor an exercise, although the notes mark four other statements of this section *"Exc."* and this one is not among them. Its four items are one line each and are proved above. The proof of Proposition 17.a.7 appealed to *"what we have proved earlier"*; it now names the results, including the two axioms it borrows from Proposition 17.a.2. Three pointers were added where a fact was used without one, and the uniqueness of the inverse matrix, which the definition immediately after it relies on for the word *"the"*, was given a label and is cited there.

Change of Basis (Section 17.b)

Question 17.1: How does the representation matrix $[T]$ of a linear map $T : V \to W$ depend on the choices of bases $\mathcal{B}$ and $\mathcal{C}$?

Corollary 17.b.1 (*Change of Basis Formula*): Let V and W be finite-dimensional vector spaces over K. Let $n := \dim V$ and $m := \dim W$. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V, and let $\mathcal{C}$ and $\mathcal{C}'$ be two bases for W. Then, $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \in \mathrm{GL}_n(K)$ and $[\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'} \in \mathrm{GL}_m(K)$, with inverses given by:

$$[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} = ([\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1} \quad \text{and} \quad [\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'} = ([\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}})^{-1}$$

Let $T : V \to W$ be a linear map. Then, its representation matrix changes according to the formula:

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Proof:

We can write T as the composition $T = \mathrm{id}_W \circ T \circ \mathrm{id}_V = \mathrm{id}_W \circ (T \circ \mathrm{id}_V)$. Passing to the matrix representations, this yields:

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T \circ \mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Also, we have the identity $\mathrm{id}_V = \mathrm{id}_V \circ \mathrm{id}_V$. This implies that:

$$I_n = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \cdot [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}}$$

and similarly:

$$I_n = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

This demonstrates that $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$ is invertible and that its inverse is indeed $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}}$, which is the asserted identity $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} = ([\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}$. The proof for $[\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'}$ follows the exact same logic.

Q.E.D.

Definition 17.b.2 (*Change of Basis Matrix*): Let V be a finite-dimensional vector space over K, and let $n := \dim V$. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases of V. The matrix $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \in \mathrm{GL}_n(K)$ is called the "*change of basis matrix*" between $\mathcal{B}$ and $\mathcal{B}'$, or the "*transition matrix*" from basis $\mathcal{B}$ to basis $\mathcal{B}'$.

Remark 17.3: If $\mathcal{B} = (w_1, \dots, w_n)$ and $\mathcal{B}' = (w_1', \dots, w_n')$, then the transition matrix is explicitly constructed by placing the coordinate vectors of the old basis elements with respect to the new basis into the columns:

$$[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [w_1]_{\mathcal{B}'} & \dots & [w_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}$$

Proposition 17.b.3 (*Representation of Isomorphisms*): Let $T : V \to W$ be an isomorphism between two finite-dimensional vector spaces. Let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W, respectively. Then, the matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ is an invertible matrix, and furthermore:

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^{-1}$$

Proof:

Let $n := \dim V = \dim W$, the two dimensions agreeing because an isomorphism carries a basis to a basis. By the properties of composed linear maps and representation matrices (the composition rule on Page 116, together with the Proposition on Page 119 for the last equality), we have:

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n$$

and likewise:

$$[T]_{\mathcal{C}}^{\mathcal{B}} \cdot [T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = [\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}} = I_n$$

This proves the claim.

Q.E.D.

Equivalence and Similarity of Matrices

Question 17.2: Suppose $T : V \to V$ is an endomorphism. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V. What is the relation between $[T]_{\mathcal{B}'}^{\mathcal{B}'}$ and $[T]_{\mathcal{B}}^{\mathcal{B}}$?

Corollary 17.b.4 (*Similarity of Representation Matrices*): Let V be a finite-dimensional vector space, and let $T : V \to V$ be a linear map. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V. Then:

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} = ([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Definition 17.b.5 (*Equivalence and Similarity*): We define two types of relations between matrices:

(a) Let $A, B \in \mathcal{M}_{m\times n}(K)$. We say that A and B are "*equivalent*" if there exist matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that $B = PAQ$.

(b) Let $A, B \in \mathcal{M}_{n\times n}(K)$. We say that A and B are "*similar*" if there exists a matrix $P \in \mathrm{GL}_n(K)$ such that $B = P^{-1}AP$.

Exercise 17.4 (*Properties and Characterization of Matrix Equivalence*):

(a) Show that "*equivalence*" between $m \times n$ matrices defines an equivalence relation on $\mathcal{M}_{m\times n}(K)$.

(b) Show that two matrices $A, B \in \mathcal{M}_{m\times n}(K)$ are equivalent if and only if there exist two vector spaces V and W with $\dim V = n$ and $\dim W = m$, two bases $\mathcal{B}$ and $\mathcal{B}'$ of V, and two bases $\mathcal{C}$ and $\mathcal{C}'$ of W such that:

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} \quad \text{and} \quad B = [T]_{\mathcal{C}'}^{\mathcal{B}'}$$

In other words, A is equivalent to B if and only if A and B are matrix representations of the exact same linear map T, but with respect to different choices of bases for V and W.

(c) Show that "*similarity*" between $n \times n$ matrices defines an equivalence relation on $\mathcal{M}_{n\times n}(K)$.

(d) Show that two square matrices $M, N \in \mathcal{M}_{n\times n}(K)$ are similar if and only if there exists an n-dimensional space V, an endomorphism $T : V \to V$, and two bases $\mathcal{B}$ and $\mathcal{B}'$ of V such that $M = [T]_{\mathcal{B}}^{\mathcal{B}}$ and $N = [T]_{\mathcal{B}'}^{\mathcal{B}'}$.

Proposition 17.b.6 (*Normal Form for Linear Maps*): Let V and W be finite-dimensional vector spaces over K, with dimensions $n = \dim V$ and $m = \dim W$. Let $T : V \to W$ be a linear map with $\mathrm{rank}(T) = r$. (Recall that $\mathrm{rank}(T) := \dim \mathrm{Im}(T)$). Then, there exists a basis $\mathcal{B} = (v_1, \ldots, v_n)$ of V and a basis $\mathcal{C} = (w_1, \ldots, w_m)$ of W such that:

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & O_{r,n-r} \\ O_{m-r,r} & O_{m-r,n-r} \end{pmatrix}$$

where $O_{p,q}$ stands for the zero matrix in $\mathcal{M}_{p\times q}(K)$.

Proof:

Put $\ell := \dim \ker(T)$. By the rank theorem, we know that $r = n - \ell$. Let $v_1, \ldots, v_\ell$ be a basis of $\ker(T)$, and extend it to a full basis $(v_1, \ldots, v_\ell, v_{\ell+1}, \ldots, v_n)$ of V. It will be useful to work below with the rearranged basis:

$$\mathcal{B} := (v_{\ell+1}, \ldots, v_n, v_1, \ldots, v_\ell)$$

In the proof of the rank theorem, we have seen that $T(v_{\ell+1}), \ldots, T(v_n)$ form a basis for $\mathrm{Im}(T)$. Define $w_i := T(v_{\ell+i})$ for $i = 1, \ldots, r$ (i.e., $w_1 = T(v_{\ell+1}), \ldots, w_r = T(v_{\ell+r})$), and extend this list to a basis of W:

$$\mathcal{C} := (w_1, \ldots, w_r, w_{r+1}, \ldots, w_m)$$

It now directly follows from these definitions that the representation matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ takes the exact block form described in the proposition. Indeed, the j-th column of $[T]_{\mathcal{C}}^{\mathcal{B}}$ is the coordinate vector of the image of the j-th member of $\mathcal{B}$. For $1 \le j \le r$ that member is $v_{\ell+j}$, whose image is w_j by construction, so the column is $[w_j]_{\mathcal{C}} = e_j \in K^m$. For $r < j \le n$ the member is one of $v_1, \dots, v_\ell$, which lie in $\ker(T)$, so the column is $[0_W]_{\mathcal{C}} = 0$. A matrix whose first r columns are $e_1, \dots, e_r$ and whose remaining columns vanish is exactly the asserted block matrix. **Q.E.D.**

Corollary 17.b.7 (*Matrix Equivalence and Normal Form*): Let $A \in \mathcal{M}_{m\times n}(K)$. Then, there exist invertible matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that:

$$PAQ = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

where $r = \mathrm{rank}_{\mathrm{col}}(A)$. In particular, $\mathrm{rank}_{\mathrm{col}}(A) \le \min\{m, n\}$.

Proof:

Consider the linear map $T_A : K^n \to K^m$ defined by left-multiplication, $T_A(v) = A \cdot v$. Then, $r = \mathrm{rank}_{\mathrm{col}}(A) = \dim \mathrm{Im}(T_A)$, because we have previously seen that

$$\mathrm{Im}(T_A) = \mathrm{Sp}(T_A(e_1), \dots, T_A(e_n)) = \mathrm{Sp}(\text{columns of } A).$$

Now, we apply the previous proposition to T_A, yielding that:

$$[T_A]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

for some basis $\mathcal{B}$ of K^n and some basis $\mathcal{C}$ of K^m. Using the change of basis formula, we can rewrite this as:

$$\underbrace{[\mathrm{id}_{K^m}]_{\mathcal{C}}^{\mathcal{E}_m}}_{=:P} \cdot \underbrace{[T_A]_{\mathcal{E}_m}^{\mathcal{E}_n}}_{=A} \cdot \underbrace{[\mathrm{id}_{K^n}]_{\mathcal{E}_n}^{\mathcal{B}}}_{=:Q} = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

where $\mathcal{E}_n$ and $\mathcal{E}_m$ are the standard bases of K^n and K^m, respectively, and where the bases meet diagonally as they must: the superscript $\mathcal{E}_m$ of the left factor matches the subscript $\mathcal{E}_m$ of the middle one, and the subscript $\mathcal{E}_n$ of the right factor matches the superscript $\mathcal{E}_n$ of the middle one. Setting $P := [\mathrm{id}_{K^m}]_{\mathcal{C}}^{\mathcal{E}_m}$ and $Q := [\mathrm{id}_{K^n}]_{\mathcal{E}_n}^{\mathcal{B}}$ concludes the proof. **Q.E.D.**

Remark 17.4: The fact that $\mathrm{rank}_{\mathrm{col}}(A) \le \min\{m, n\}$ also follows from observing the linear map $T_A : K^n \to K^m$. We have:

$$\mathrm{rank}_{\mathrm{col}}(A) = \dim \mathrm{Im}(T_A) = n - \dim \ker(T_A) \le n$$

and simultaneously,

$$\mathrm{rank}_{\mathrm{col}}(A) = \dim \mathrm{Im}(T_A) \le \dim K^m = m$$

Question 17.3: Recall that we have an equivalence relation on $\mathcal{M}_{m\times n}(K)$: $A \sim B$ if there exist $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that $B = PAQ$. Can we describe the equivalence classes of this equivalence relation?

Corollary 17.b.8 (*Classification of Matrices by Rank*): Let $A, B \in \mathcal{M}_{m\times n}(K)$. Then, $A \sim B$ if and only if $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{col}}(B)$. Moreover, for every $0 \le r \le \min\{m, n\}$, there is precisely one equivalence class of matrices A with $\mathrm{rank}_{\mathrm{col}} = r$.

Proof:

Let $A, B \in \mathcal{M}_{m\times n}(K)$. First, assume that $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{col}}(B)$, and denote this common column rank by r. By Corollary 17.b.7, we know that $A \sim \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ and $B \sim \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$. Since $\sim$ is an equivalence relation, it naturally follows that $A \sim B$.

Conversely, assume that $A \sim B$. This implies that there exist matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that $B = PAQ$. Transitioning to the associated linear maps, this means $T_B = T_P \circ T_A \circ T_Q$. We analyze the image of T_B:

$$\mathrm{Im}(T_B) = T_B(K^n) = T_P \circ T_A(T_Q(K^n)) = T_P \circ T_A(K^n) = T_P(\mathrm{Im}(T_A))$$

Note that $T_Q(K^n) = K^n$ because T_Q is an isomorphism. Consequently, the dimension of the image is:

$$\mathrm{rank}_{\mathrm{col}}(B) = \dim \mathrm{Im}(T_B) = \dim T_P(\mathrm{Im}(T_A)) = \dim \mathrm{Im}(T_A) = \mathrm{rank}_{\mathrm{col}}(A)$$

where the second to last equality holds because T_P is an isomorphism, preserving the dimension of subspaces.

The last statement of the corollary now follows. The two implications just proved say that the equivalence classes are exactly the sets of matrices of a common column rank, so distinct ranks give distinct classes, and every value $0 \le r \le \min\{m, n\}$ actually occurs, since

$$\begin{pmatrix} I_r & O \\ O & O \end{pmatrix} \in \mathcal{M}_{m\times n}(K)$$

has columns $e_1, \dots, e_r, 0, \dots, 0$ and hence column rank r. There is therefore precisely one class for each such r. **Q.E.D.**

Solutions to the Exercises

Proof of Exercise 17.4:

Throughout, write $A \sim B$ for equivalence and $A \approx B$ for similarity.

(a) **Equivalence is an equivalence relation.** It is reflexive, since $A = I_m A I_n$ with $I_m \in \mathrm{GL}_m(K)$ and $I_n \in \mathrm{GL}_n(K)$. It is symmetric: if $B = PAQ$ with $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$, then multiplying by P^{-1} on the left and Q^{-1} on the right gives $A = P^{-1}BQ^{-1}$, and P^{-1}, Q^{-1} are again invertible by Proposition 17.a.7. It is transitive: if $B = PAQ$ and $C = P'BQ'$, then

$$C = P'(PAQ)Q' = (P'P)A(QQ'),$$

and $P'P \in \mathrm{GL}_m(K)$, $QQ' \in \mathrm{GL}_n(K)$, again by Proposition 17.a.7.

(b) **Equivalent matrices represent one map in two pairs of bases.** Suppose first that $A = [T]^{\mathcal{B}}_{\mathcal{C}}$ and $B = [T]^{\mathcal{B}'}_{\mathcal{C}'}$ for a single linear map $T : V \to W$. By Corollary 17.b.1,

$$B = \underbrace{[\mathrm{id}_W]^{\mathcal{C}}_{\mathcal{C}'}}_{\in \mathrm{GL}_m(K)} \cdot A \cdot \underbrace{[\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}}_{\in \mathrm{GL}_n(K)},$$

so $A \sim B$.

Conversely, suppose $B = PAQ$ with $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$. Take $V := K^n$, $W := K^m$ and $T := T_A$, and let $\mathcal{E}_n$, $\mathcal{E}_m$ be the standard bases, so that $A = [T]^{\mathcal{E}_n}_{\mathcal{E}_m}$. Let $\mathcal{B}'$ be the list of columns of Q and let $\mathcal{C}'$ be the list of columns of P^{-1}; both are bases, since a square matrix is invertible precisely when its columns form a basis. The matrix $[\mathrm{id}]^{\mathcal{B}'}_{\mathcal{E}_n}$ has as its j-th column the coordinate vector of the j-th member of $\mathcal{B}'$ with respect to the standard basis, that is, the j-th column of Q; hence $[\mathrm{id}_{K^n}]^{\mathcal{B}'}_{\mathcal{E}_n} = Q$, and likewise $[\mathrm{id}_{K^m}]^{\mathcal{C}'}_{\mathcal{E}_m} = P^{-1}$, so that $[\mathrm{id}_{K^m}]^{\mathcal{E}_m}_{\mathcal{C}'} = P$ by Corollary 17.b.1. That same corollary now gives

$$[T]^{\mathcal{B}'}_{\mathcal{C}'} = [\mathrm{id}_{K^m}]^{\mathcal{E}_m}_{\mathcal{C}'} \cdot [T]^{\mathcal{E}_n}_{\mathcal{E}_m} \cdot [\mathrm{id}_{K^n}]^{\mathcal{B}'}_{\mathcal{E}_n} = PAQ = B,$$

so A and B do represent the same map in two pairs of bases.

(c) **Similarity is an equivalence relation.** Reflexivity: $A = I_n^{-1} A I_n$. Symmetry: if $B = P^{-1}AP$, then $A = PBP^{-1} = (P^{-1})^{-1}B(P^{-1})$. Transitivity: if $B = P^{-1}AP$ and $C = R^{-1}BR$, then

$$C = R^{-1}P^{-1}APR = (PR)^{-1}A(PR),$$

using $(PR)^{-1} = R^{-1}P^{-1}$ from Proposition 17.a.7.

(d) **Similar matrices represent one endomorphism in two bases.** If $M = [T]_{\mathcal{B}}^{\mathcal{B}}$ and $N = [T]_{\mathcal{B}'}^{\mathcal{B}'}$ for an endomorphism $T : V \to V$, then Corollary 17.b.4 gives $N = P^{-1}MP$ with $P := [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \in \mathrm{GL}_n(K)$, so $M \approx N$.
Conversely, let $N = P^{-1}MP$ with $P \in \mathrm{GL}_n(K)$. Take $V := K^n$, $T := T_M$ and $\mathcal{B} := \mathcal{E}_n$, so that $M = [T]_{\mathcal{B}}^{\mathcal{B}}$. Let $\mathcal{B}'$ be the list of columns of P, a basis of K^n, so that $[\mathrm{id}_{K^n}]_{\mathcal{E}_n}^{\mathcal{B}'} = P$ as in **(b)**. Then Corollary 17.b.4 gives

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \left([\mathrm{id}_{K^n}]_{\mathcal{E}_n}^{\mathcal{B}'}\right)^{-1} \cdot [T]_{\mathcal{E}_n}^{\mathcal{E}_n} \cdot [\mathrm{id}_{K^n}]_{\mathcal{E}_n}^{\mathcal{B}'} = P^{-1}MP = N.$$

Q.E.D.

AI-Note: Every numbered statement of the notes for this section is present in the transcript, in the notes' order, and 17.b.1–17.b.8 already matched the handwritten numbering; the proofs follow Prof. Biran's.

The four parts of Exercise 17.4 are his, marked *"Exc."* in the notes and left unanswered by the transcript; they are solved above. Two of them are the substantial ones: equivalence of matrices means representing the same linear map in two pairs of bases, and similarity means representing the same endomorphism in two bases. Both directions need the observation that for an invertible P, the columns of P form a basis whose transition matrix from the standard basis is P itself.

Also here: the block form asserted at the end of the proof of Proposition 17.b.6 is now verified column by column rather than declared to follow from the definitions, and the notation was brought to the house macros in three places, where $\mathcal{M}$ and GL had been typeset as plain M and GL.

A later pass, looking only for gaps in rigour, found two slips in the bookkeeping of bases. The proof of Corollary 17.b.1 closed by saying that the inverse of $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$ is $([\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}$, one inverse too many: the two displays just above it show that the inverse is $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}}$, which is exactly the identity the corollary asserts.

More seriously, in the proof of Corollary 17.b.7 the left factor was written $[\mathrm{id}_{K^m}]_{\mathcal{E}_m}^{\mathcal{C}}$. Its bases do not meet those of the middle factor $[T_A]_{\mathcal{E}_m}^{\mathcal{E}_n}$ the way the composition rule requires, so the product as printed does not compose, and the matrix named P was the inverse of the one meant. It has to be $[\mathrm{id}_{K^m}]_{\mathcal{C}}^{\mathcal{E}_m}$, which is also the matrix that the solution to part **(b)** of Exercise 17.4 arrives at independently. The correction is made and the diagonal matching is now said in words.

Two smaller things: the final sentence of Corollary 17.b.8, that there is precisely one class for each rank r, was stated but not argued, and now is; and the proof of Proposition 17.b.3 said *"conversely"* where it merely composes in the other order, and left unsaid why $\dim V = \dim W$ and which two earlier results its display uses.

Column Rank and Row Rank (Section 17.c)

Theorem 17.c.1 (*Equality of Column and Row Rank*): Let $A \in \mathcal{M}_{m\times n}(K)$. Then, $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{row}}(A)$.

Remark 17.5 (*Notation and Properties of Rank*):

(a) Because of this theorem, we will from now on denote $\mathrm{rank}(A) := \mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{row}}(A)$.

(b) Suppose A' is a row-reduced echelon matrix that is row-equivalent to A. Then, we have:

$$\mathrm{rank}(A) = \mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{row}}(A') = \text{number of pivots in } A'$$

(the middle equality because row-equivalent matrices have the same row space, Proposition 13.3, and the last because the non-zero rows of a row-reduced echelon matrix form a basis of that space, Theorem 13.6, and there is one such row per pivot).

(c) Recall that for a linear map $T : V \to W$, we defined $\mathrm{rank}(T) := \dim \mathrm{Im}(T)$. Moreover, if $A \in \mathcal{M}_{m\times n}(K)$, then $\mathrm{rank}(T_A) = \dim \mathrm{Im}(T_A) = \mathrm{rank}_{\mathrm{col}}(A)$.

Preparation for the Proof

Lemma 17.c.2 (*Rank Invariance under Composition with Isomorphisms*): Let $T : V \to W$ be a linear map, and let $S : W \to W'$ and $L : P \to V$ be isomorphisms. Then:

(a) $\operatorname{rank}(S \circ T) = \operatorname{rank}(T)$.

(b) $\operatorname{rank}(T \circ L) = \operatorname{rank}(T)$.

Proof:

(a) By definition, $\operatorname{rank}(S \circ T) = \dim(S(T(V)))$. Since S is an isomorphism, its restriction to the subspace $T(V) \subseteq W$ is injective. An injective linear map preserves the dimension of a vector space, and therefore:

$$\dim(S(T(V))) = \dim(T(V)) = \operatorname{rank}(T)$$

(b) By definition, $\operatorname{rank}(T \circ L) = \dim(T(L(P)))$. Since L is an isomorphism, it is surjective, which implies that $L(P) = V$. Consequently:

$$\dim(T(L(P))) = \dim(T(V)) = \operatorname{rank}(T)$$

Q.E.D.

Lemma 17.c.3 (*Invertibility and Isomorphism*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then A is invertible if and only if $T_A : K^n \to K^n$ is an isomorphism.

Proof:

Suppose first that A is invertible. Then $T_{A^{-1}} = (T_A)^{-1}$, so T_A is an isomorphism.

Conversely, suppose that T_A is an isomorphism. We know that $A = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_n}$, and by a previous result the representation matrix of an isomorphism is invertible (Proposition 17.b.3), so A is invertible.

This is the same statement as Proposition 17.a.9, which was proved directly from the definition of invertibility; the two lines here instead read it off the change of basis machinery of Section 17.b. Prof. Biran states it in both places, and both proofs are worth having. **Q.E.D.**

Remark 17.6 (*Invertibility of the Transpose*): It is worth recording alongside this that $A \in \mathrm{GL}_n(K)$ if and only if $A^\mathsf{T} \in \mathrm{GL}_n(K)$, with $(A^\mathsf{T})^{-1} = (A^{-1})^\mathsf{T}$. This is Exercise 17.3 **(b)** of Section 17.a, and it needs nothing from the present section: applying $(XY)^\mathsf{T} = Y^\mathsf{T} X^\mathsf{T}$ to $A^{-1}A = AA^{-1} = I_n$ exhibits $(A^{-1})^\mathsf{T}$ as a two-sided inverse of A^T, and the converse follows by applying the same argument to A^T and using $(A^\mathsf{T})^\mathsf{T} = A$.

Lemma 17.c.4 (*Rank Invariance under Matrix Equivalence*): Let $A, B \in \mathcal{M}_{m\times n}(K)$ such that $B = PAQ$, where $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$. Then:

(a) $\operatorname{rank}_{\mathrm{col}}(A) = \operatorname{rank}_{\mathrm{col}}(B)$.

(b) $\operatorname{rank}_{\mathrm{row}}(A) = \operatorname{rank}_{\mathrm{row}}(B)$.

Proof:

(a) Transitioning to the associated linear maps, we have $T_B = T_P \circ T_A \circ T_Q$. Since P and Q are invertible matrices, both T_P and T_Q are isomorphisms. Applying Lemma 17.c.2, we deduce that $\operatorname{rank}(T_B) = \operatorname{rank}(T_A)$. Recalling that the rank of a linear map given by a matrix matches the column rank of that matrix, we obtain $\operatorname{rank}_{\mathrm{col}}(B) = \operatorname{rank}_{\mathrm{col}}(A)$.

(b) Taking the transpose of the equation $B = PAQ$ yields $B^\mathsf{T} = Q^\mathsf{T} A^\mathsf{T} P^\mathsf{T}$. Since P and Q are invertible matrices, their transposes P^T and Q^T are also invertible (we have previously seen that $(P^\mathsf{T})^{-1} = (P^{-1})^\mathsf{T}$). We can now apply statement **(a)** of this Lemma 17.c.4, which we have already proven, to the matrices A^T and B^T. This gives us $\operatorname{rank}_{\mathrm{col}}(B^\mathsf{T}) = \operatorname{rank}_{\mathrm{col}}(A^\mathsf{T})$.

By the definition of the transpose, this immediately translates to $\text{rank}_{\text{row}}(B) = \text{rank}_{\text{row}}(A)$, since the columns of A^{T} are the rows of A, so that $\text{rank}_{\text{col}}(A^{\mathsf{T}}) = \text{rank}_{\text{row}}(A)$ by Definition 13.4.

Q.E.D.

Proof of the Main Theorem

We are now ready for the proof of Theorem 17.c.1.

Proof of Theorem 17.c.1:

Let $r := \text{rank}_{\text{col}}(A)$. By Corollary 17.b.7, we know that there exist invertible matrices $P \in \text{GL}_m(K)$ and $Q \in \text{GL}_n(K)$ such that:

$$PAQ = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix} =: B$$

By Lemma 17.c.4, we know that $\text{rank}_{\text{row}}(A) = \text{rank}_{\text{row}}(B)$. Thus, it suffices to determine the row rank of B.

The matrix B is of the block form $\left(\begin{smallmatrix} I_r & O \\ O & O \end{smallmatrix}\right)$. The non-zero rows of B are precisely the first r standard basis vectors $e_1^{\mathsf{T}}, e_2^{\mathsf{T}}, \ldots, e_r^{\mathsf{T}} \in K^n_{\text{row}}$. Since these vectors form a linearly independent set, the dimension of the row space of B is exactly r. Consequently, $\text{rank}_{\text{row}}(B) = r$.

Therefore, we conclude that $\text{rank}_{\text{row}}(A) = r = \text{rank}_{\text{col}}(A)$, completing the proof. **Q.E.D.**

Important remark 17.3 (*Column Space versus Row Space*): Although $\text{rank}_{\text{col}}(A) = \text{rank}_{\text{row}}(A)$, the column space and the row space are, in general, different vector spaces. First of all, if $A \in \mathcal{M}_{m\times n}(K)$, then the column space is a subspace of K^m, whereas the row space is a subspace of K^n_{row} (which is isomorphic to K^n). But even in the case where $m = n$ (i.e., $A \in \mathcal{M}_{n\times n}(K)$), these two spaces are generally distinct! They share the exact same dimension, but they are not the same space.

AI-Note: The transcript has all four numbered statements of this section, but 17.c.3 is not the one in the notes. Prof. Biran's Lemma 2 records that A is invertible if and only if T_A is an isomorphism, in two lines; the transcript put in its place a lemma on the invertibility of A^{T}, with a proof that runs through row-reduced echelon forms and pivots. That lemma is true, and its first sentence is precisely Prof. Biran's Lemma 2 used as a step, but the statement is not his and nothing in this section uses it. The notes' lemma is restored as Lemma 17.c.3, and the transpose statement is kept as Remark 17.6, where it belongs: it is Exercise 17.3 **(b)** of Section 17.a, needs nothing from this section, and comes with the inverse written out.

Two citations were vague, *"a previous proposition regarding the equivalence of matrices"* is Corollary 17.b.7, and GL was typeset as plain GL in the main proof.

A later pass looking only for gaps in rigour found nothing wrong here, only steps taken silently. Three of them now name their reason: the chain in the notation remark rests on the invariance of the row space under row operations and on the fact that the non-zero rows of a row-reduced echelon matrix form a basis of it; the passage from $\text{rank}_{\text{col}}(A^{\mathsf{T}})$ to $\text{rank}_{\text{row}}(A)$ is the definition of the two ranks; and Lemma 17.c.3 is the same statement as Proposition 17.a.9, which is now said, since a reader meeting it twice should know it is not a new result.

One thing was left alone deliberately: Lemma 17.c.2 calls a vector space P, while P is an invertible matrix two statements later. Nothing is wrong, but the collision is a trap for the eye.

More on Matrices and Linear Equations (Section 17.d)

Let $A \in \mathcal{M}_{m\times n}(K)$, and consider the homogeneous system of linear equations $A \cdot x = 0$. We denote the space of solutions of this system by:

$$\mathrm{Sol}(A,0) := \{x \in K^n \mid Ax = 0\} = \ker(T_A)$$

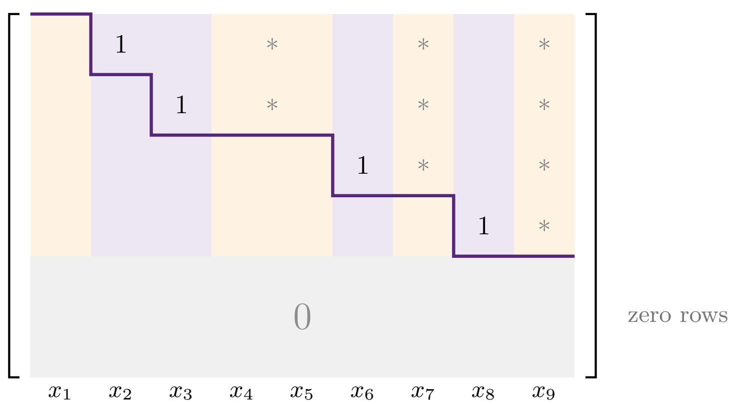
Note that by the Rank-Nullity Theorem, Theorem 15.b.9, we have

$$\dim \mathrm{Sol}(A,0) = n - \mathrm{rank}(A),$$

because $\dim\ker(T_A) = n - \dim\mathrm{Im}(T_A)$ (and $\dim\mathrm{Im}(T_A) = \mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}(A)$ by Remark 17.5). Let A' be a row-reduced echelon matrix that is row-equivalent to A (for example, A' is obtained from A by Gauss elimination). We then know that the number of pivots of A' equals $\mathrm{rank}(A)$ (Remark 17.5 again), and the number of free variables is $n - \mathrm{rank}(A)$.

We can visualize the structure of A' and its corresponding variables $x_1, \dots, x_n$ as in Figure 17.1. The columns containing the leading 1's (pivots) correspond to the pivot variables, while the remaining columns correspond to the free variables. Clearly, the solution spaces are identical (this is Theorem 6.10: row-equivalent systems have the same solutions, applied to the homogeneous system):

$$\mathrm{Sol}(A,0) = \mathrm{Sol}(A',0)$$



pivot columns: x_2, x_3, x_6, x_8 **(**$r = \mathrm{rank}(A) = 4$ **of them)**
free columns: x_1, x_4, x_5, x_7, x_9 **(**$\ell = n - \mathrm{rank}(A) = 5$ **of them)**

Figure 17.1: The staircase shape of a row-reduced echelon matrix A', here with $n = 9$ columns and $\mathrm{rank}(A) = 4$. Every entry below the purple staircase vanishes. Each of the r non-zero rows contributes exactly one leading 1, whose column is a **pivot** column; the remaining $\ell = n-r$ columns are the **free** columns. Choosing the ℓ free variables arbitrarily forces the r pivot variables, which is precisely the map Φ below.

Denote by $x_{i_1}, \dots, x_{i_\ell}$ the free variables, where $1 \le i_1 < \dots < i_\ell \le n$ and $\ell = n - \mathrm{rank}(A)$. Similarly, denote by $x_{j_1}, \dots, x_{j_r}$ the pivot variables, where $1 \le j_1 < \dots < j_r \le n$ and $r = \mathrm{rank}(A)$.

Definition (*The Free-Variable Parametrisation of the Solution Space*): For each choice of values for the free variables $x_{i_1}, \dots, x_{i_\ell} \in K$, the pivot variables $x_{j_1}, \dots, x_{j_r}$ are uniquely determined. So, we obtain a map $\Phi : K^\ell \to \mathrm{Sol}(A,0)$, where $\ell = n - \mathrm{rank}(A)$, defined by:

$$\Phi(\lambda_1, \dots, \lambda_\ell) := \text{the unique solution of } Ax = 0 \text{ (which is the solution of } A'x = 0)$$

with $x_{i_1} = \lambda_1, \dots, x_{i_k} = \lambda_k, \dots, x_{i_\ell} = \lambda_\ell$.

Proposition 17.d.1 (*Linearity and Isomorphism of the Solution Map*): The map Φ of the Definition on Page 134 is linear. Moreover, it is an isomorphism.

Proof:

Linearity of Φ: The k-th non-zero row of A' gives the following equation:

$$x_{j_k} + (\text{linear combination of the free variables that come after } j_k) = 0$$

This allows us to isolate the pivot variable x_{j_k}:

$$x_{j_k} = -\sum_{\{q \mid 1 \le q \le \ell,\, i_q > j_k\}} a'_{k,i_q} \cdot x_{i_q}$$

This explicitly shows that the pivot variables $x_{j_1}, \ldots, x_{j_r}$ depend linearly on the free variables $x_{i_1}, \ldots, x_{i_\ell}$. This proves the linearity of Φ.

Isomorphism: To prove that Φ is an isomorphism, it is enough to show that $\ker(\Phi) = \{0\}$, because $\Phi : K^\ell \to \mathrm{Sol}(A, 0)$ is a linear map between two spaces of the exact same dimension ℓ (the target has dimension $\ell = n - \mathrm{rank}(A)$ by the count at the start of this section; for maps between spaces of equal finite dimension, injectivity already gives bijectivity by Corollary 15.b.10 **(c)**, and a bijective linear map is an isomorphism by Lemma 15.b.4).

But this clearly holds: if $\lambda = (\lambda_1, \ldots, \lambda_\ell) \in \ker(\Phi)$, which means $\Phi(\lambda_1, \ldots, \lambda_\ell) = 0$, then by the definition of Φ, we must have $\lambda_1 = \cdots = \lambda_\ell = 0$. This is simply because each λ_k is an actual entry in the output vector $\Phi(\lambda_1, \ldots, \lambda_\ell)$.

Furthermore, we can explicitly state the inverse map Φ^{-1}. For any $x \in \mathrm{Sol}(A, 0)$, we extract the entries corresponding to the free variables:

$$\Phi^{-1}(x) = \begin{pmatrix} x_{i_1} \\ \vdots \\ x_{i_\ell} \end{pmatrix}$$

This completes the proof. **Q.E.D.**

Corollary 17.d.2 (*Structure of Non-Homogeneous Solutions*): Let $b \in K^m$, and $A \in \mathcal{M}_{m \times n}(K)$. Then:

(a) $\mathrm{Sol}(A, b) := \{x \in K^n \mid A \cdot x = b\} \neq \emptyset$ if and only if $b \in \mathrm{Cols}(A)$.

(b) If $\mathrm{Sol}(A, b) \neq \emptyset$ and $y \in \mathrm{Sol}(A, b)$, then:

$$\mathrm{Sol}(A, b) = y + \mathrm{Sol}(A, 0) := \{y + x \mid x \in \mathrm{Sol}(A, 0)\}$$

Exercise 17.5 (*Proof of Corollary 17.d.2*): The notes leave this proof as an exercise, with the following hints.

(a) Write the matrix A in terms of its column vectors $v_1, \ldots, v_n$:

$$A = \begin{pmatrix} | & & | \\ v_1 & \ldots & v_n \\ | & & | \end{pmatrix}$$

For any vector $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K^n$, the matrix-vector product is a linear combination of the columns:

$$A \cdot x = x_1 \begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + \cdots + x_n \begin{pmatrix} | \\ v_n \\ | \end{pmatrix}$$

(b) Let $y \in \mathrm{Sol}(A, b)$. First, assume $x \in \mathrm{Sol}(A, 0)$. Then $A \cdot x = 0$ and $A \cdot y = b$. Adding these together yields:

$$A(y + x) = A \cdot y + A \cdot x = b + 0 = b$$

Conversely, if $z \in \mathrm{Sol}(A, b)$, meaning $A \cdot z = b$, then define $x := z - y$. We obviously have $z = y + x$, and checking the multiplication gives:

$$A \cdot x = A \cdot (z - y) = A \cdot z - A \cdot y = b - b = 0$$

This shows that any solution z can be written as $y + x$ for some $x \in \mathrm{Sol}(A, 0)$.

Proposition 17.d.3 (*Rank Criterion for Solvability*): Let $A \in \mathcal{M}_{m\times n}(K)$ and $b \in K^m$. The following statements are equivalent:

(a) $\mathrm{rank}(A \mid b) = \mathrm{rank}(A)$.

(b) The system of equations $Ax = b$ has a solution.

Proof:

Write A in terms of its columns,

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

We have the following chain of equivalences:

$$\begin{aligned} \mathrm{rank}(A \mid b) = \mathrm{rank}(A) &\iff \mathrm{rank}_{\mathrm{col}}(A \mid b) = \mathrm{rank}_{\mathrm{col}}(A) \\ &\iff \dim \mathrm{Sp}(v_1, \dots, v_n, b) = \dim \mathrm{Sp}(v_1, \dots, v_n) \\ &\iff b \in \mathrm{Sp}(v_1, \dots, v_n) \\ &\iff b \in \mathrm{Cols}(A) \\ &\iff \mathrm{Sol}(A, b) \neq \emptyset \end{aligned}$$

The middle step deserves a word: $\mathrm{Sp}(v_1, \dots, v_n)$ is contained in $\mathrm{Sp}(v_1, \dots, v_n, b)$ whatever b is, and two nested subspaces of equal finite dimension coincide, so the two spans have the same dimension if and only if they are equal, which happens if and only if b already lies in the smaller one. The last step holds directly by Corollary 17.d.2, thus finishing the proof. **Q.E.D.**

Proposition 17.d.4 (*Criterion for a Unique Solution*): Let $A \in \mathcal{M}_{m\times n}(K)$ and $b \in K^m$. The following statements are equivalent:

(a) $\mathrm{rank}(A) = \mathrm{rank}(A \mid b) = n$.

(b) The system $Ax = b$ has a unique solution.

Exercise 17.6 (*Proof of Proposition 17.d.4*): The notes leave this proof as an exercise, with the following hint. Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

Note that:

$$\mathrm{rank}(A) = n \iff \dim \mathrm{Sp}(v_1, \dots, v_n) = n$$

Since there are n column vectors, this means that $v_1, \dots, v_n$ are linearly independent, and therefore, they form a basis for $\mathrm{Cols}(A)$.

Solutions to the Exercises

Proof of Corollary 17.d.2:

(a) As the hint records, for $x = (x_1, \dots, x_n)^{\mathsf{T}} \in K^n$ the product $A \cdot x$ is the linear combination $x_1 v_1 + \cdots + x_n v_n$ of the columns of A. Hence

$$\mathrm{Sol}(A, b) \neq \emptyset \iff \exists\, x \in K^n : x_1 v_1 + \cdots + x_n v_n = b \iff b \in \mathrm{Sp}(v_1, \dots, v_n) = \mathrm{Cols}(A),$$

the last equality being the definition of the column space.

(b) Let $y \in \mathrm{Sol}(A,b)$. If $x \in \mathrm{Sol}(A,0)$, then $A(y+x) = Ay+Ax = b+0 = b$, so $y+x \in \mathrm{Sol}(A,b)$, which gives the inclusion $y+\mathrm{Sol}(A,0) \subseteq \mathrm{Sol}(A,b)$. Conversely, if $z \in \mathrm{Sol}(A,b)$, put $x := z-y$; then $z = y+x$ and

$$Ax = A(z-y) = Az - Ay = b - b = 0,$$

so $x \in \mathrm{Sol}(A,0)$ and $z \in y + \mathrm{Sol}(A,0)$. The two inclusions give the asserted equality.

Q.E.D.

Proof of Proposition 17.d.4:

Suppose first that $\mathrm{rank}(A) = \mathrm{rank}(A \mid b) = n$. Since the two ranks agree, Proposition 17.d.3 gives a solution $y \in \mathrm{Sol}(A,b)$. Since $\mathrm{rank}(A) = n$, the computation at the start of this section gives

$$\dim \mathrm{Sol}(A,0) = n - \mathrm{rank}(A) = 0,$$

so $\mathrm{Sol}(A,0) = \{0\}$. By Corollary 17.d.2 **(b)**, $\mathrm{Sol}(A,b) = y + \{0\} = \{y\}$, so the solution is unique.

Conversely, suppose $Ax = b$ has a unique solution y. In particular a solution exists, so $\mathrm{rank}(A \mid b) = \mathrm{rank}(A)$ by Proposition 17.d.3. If now $x \in \mathrm{Sol}(A,0)$, then $y+x$ is again a solution of $Ax = b$ by Corollary 17.d.2 **(b)**, and uniqueness forces $y+x = y$, that is, $x = 0$. So $\mathrm{Sol}(A,0) = \{0\}$, and therefore

$$0 = \dim \mathrm{Sol}(A,0) = n - \mathrm{rank}(A),$$

which gives $\mathrm{rank}(A) = n$. In the language of the hint: the columns $v_1, \dots, v_n$ are then linearly independent and form a basis of $\mathrm{Cols}(A)$, which is exactly the statement that the coefficients in $b = x_1 v_1 + \cdots + x_n v_n$ are uniquely determined. **Q.E.D.**

AI-Note: This section is complete in the transcript: all four numbered statements, in the notes' order, with the notes' proofs and with Prof. Biran's hints for the two he leaves as exercises. The staircase picture that the notes draw by hand is rendered as Figure 17.1.

The two proofs the notes mark *"Exc."* stood in the text as `\begin{proof}` blocks reading *"This is left as an exercise"*, followed by the hints. They are now exercises carrying those hints, and both are solved above. The second one needs a little more than the hint gives: the uniqueness criterion follows by combining Proposition 17.d.3 with the dimension count $\dim \mathrm{Sol}(A,0) = n - \mathrm{rank}(A)$ from the opening of the section.

A later pass looking only for gaps in rigour found nothing wrong here. What it found was the same pattern as in Section 17.a: an object defined in running prose with a numbered statement about it standing directly underneath. The map Φ, which sends a choice of free variables to the solution it determines, now has a definition of its own, unnumbered so that 17.d.1–17.d.4 are untouched, and Proposition 17.d.1 points at it.

Five steps were taken silently and now say why they hold: the dimension of the solution space rests on the rank theorem together with $\dim \mathrm{Im}(T_A) = \mathrm{rank}(A)$; the pivot count is Remark 17.5; the word *"clearly"* in front of $\mathrm{Sol}(A,0) = \mathrm{Sol}(A',0)$ covers Theorem 6.10, that row-equivalent systems have the same solutions; the passage from a trivial kernel to an isomorphism is Corollary 15.b.10 **(c)** followed by Lemma 15.b.4; and the middle link of the chain of equivalences in Proposition 17.d.3 needs the observation that two nested subspaces of equal finite dimension coincide, without which equal dimensions of the two spans would not put b in the smaller one.

Finally, the augmented matrix is written $(A \mid b)$ in the chapter on systems of linear equations, where it is introduced, and in the solutions at the end of this section, but was written $(A|b)$ in the two propositions. The three occurrences are unified.

Elementary Row Operations Revisited (Section 17.e)

Let $A \in \mathcal{M}_{n\times p}(K)$. We have previously learned about elementary row operations, which are utilized as an integral part of the Gaussian elimination algorithm. The main point of this section is to demonstrate that all of these operations can be expressed through matrix multiplications.

Notation 17.1: Let $n \in \mathbb{Z}_{\geq 1}$, and let $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by $E_{ij} \in \mathcal{M}_{n\times n}(K)$ the matrix whose entries are all zero, except for the entry in the i-th row and j-th column, which is equal to 1.

We will now define three types of elementary matrices in $\mathcal{M}_{n\times n}(K)$. The notes introduce them without a number of their own, so that the numbering of this section starts at Lemma 17.e.1 below.

Definition (*Elementary Matrices*): We define the following three types of elementary matrices:

(a) **Type I:** Let $1 \leq i \leq n$, $1 \leq j \leq n$ with $i \neq j$, and let $\alpha \in K$. We define $Q_{ij}(\alpha)$ to be the matrix obtained from I_n by applying the row operation $R_i + \alpha R_j \longrightarrow R_i$ (i.e., replacing the i-th row by the sum of the i-th row and α times the j-th row). Explicitly, we have:
$$Q_{ij}(\alpha) = I_n + \alpha \cdot E_{ij}$$

(b) **Type II:** Let $1 \leq i \leq n$, $1 \leq j \leq n$ with $i \neq j$. We define P_{ij} to be the matrix obtained from I_n by permuting (exchanging) the i-th row and the j-th row, denoted by $R_i \leftrightarrow R_j$.

(c) **Type III:** Let $1 \leq i \leq n$, and let $0 \neq \alpha \in K$. We denote by $S_i(\alpha)$ the matrix obtained from I_n by applying the row operation $\alpha R_i \longrightarrow R_i$ (i.e., multiplying the i-th row by the non-zero scalar α).

Exercise 17.7 (*Explicit Form of the Permutation Matrix*): Show that the Type II elementary matrix can be written explicitly as $P_{ij} = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$.

Example 17.2 (*Examples of Elementary Matrices*): Consider the space $\mathcal{M}_{4\times 4}(K)$. Examples of the three types of elementary matrices are:
$$Q_{1,3}(\alpha) = \begin{pmatrix} 1 & 0 & \alpha & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$
$$P_{2,4} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$
$$S_3(\alpha) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Lemma 17.e.1 (*Matrix Multiplication and Row Operations*): Let $A \in \mathcal{M}_{n\times p}(K)$. Multiplying A from the left by an elementary matrix performs the corresponding elementary row operation on A:

(a) Multiplying A from the left with $Q_{ij}(\alpha)$ results in the row operation $R_i + \alpha R_j \longrightarrow R_i$:
$$A \xrightarrow{R_i + \alpha R_j \longrightarrow R_i} Q_{ij}(\alpha) \cdot A$$

(b) Multiplying A from the left with P_{ij} results in the row operation $R_i \leftrightarrow R_j$:
$$A \xrightarrow{R_i \leftrightarrow R_j} P_{ij} \cdot A$$

(c) Multiplying A from the left with $S_i(\alpha)$ results in the row operation $\alpha R_i \longrightarrow R_i$:

$$A \xrightarrow{\alpha R_i \longrightarrow R_i} S_i(\alpha) \cdot A$$

Important remark 17.4 (*Column Operations*): The same principle holds true for elementary column operations, but via right-multiplication. Here the elementary matrices are the ones of $\mathcal{M}_{p\times p}(K)$, that is, their indices run over $1 \le i \le p$, $1 \le j \le p$: it is the p columns of $A \in \mathcal{M}_{n\times p}(K)$ that are being operated on, and a product such as $A \cdot Q_{ij}(\alpha)$ is defined only for a $p \times p$ factor on the right.

(a') Multiplying A from the right with $Q_{ij}(\alpha)$ results in the column operation $C_j + \alpha C_i \longrightarrow C_j$:

$$A \xrightarrow{C_j + \alpha C_i \longrightarrow C_j} A \cdot Q_{ij}(\alpha)$$

(b') Multiplying A from the right with P_{ij} results in the column operation $C_i \leftrightarrow C_j$:

$$A \xrightarrow{C_i \leftrightarrow C_j} A \cdot P_{ij}$$

(c') Multiplying A from the right with $S_i(\alpha)$ results in the column operation $\alpha C_i \longrightarrow C_i$:

$$A \xrightarrow{\alpha C_i \longrightarrow C_i} A \cdot S_i(\alpha)$$

Proof of Lemma 17.e.1 and of Important remark 17.4:

Statements **(a)**, **(b)**, and **(c)** can be proven directly by calculating the matrix multiplications; the notes leave this calculation to the reader, and it is carried out for **(a)** among the solutions at the end of the section.

Statements **(a')**, **(b')**, and **(c')** can be deduced from the row operation versions by passing to the transposed matrix. For example, to prove **(a')**:

$$(A \cdot Q_{ij}(\alpha))^{\mathsf{T}} = Q_{ij}(\alpha)^{\mathsf{T}} \cdot A^{\mathsf{T}} = Q_{ji}(\alpha) \cdot A^{\mathsf{T}}$$

By part **(a)**, applied to $A^{\mathsf{T}} \in \mathcal{M}_{p\times n}(K)$ with the roles of n and p interchanged, this is exactly the matrix obtained from A^{T} after applying to it the row operation $R_j + \alpha R_i \longrightarrow R_j$. Now the rows of A^{T} are the columns of A, so performing that row operation on A^{T} is the same as performing the column operation $C_j + \alpha C_i \longrightarrow C_j$ on A and transposing the result. Hence the matrix above is equal to:

$$(\text{matrix obtained from } A \text{ after applying } C_j + \alpha C_i \longrightarrow C_j)^{\mathsf{T}}$$

Taking the transpose of both sides, it follows that $A \cdot Q_{ij}(\alpha)$ is the matrix obtained from A after the column operation $C_j + \alpha C_i \longrightarrow C_j$. The proofs for **(b')** and **(c')** follow a similar logic, using that $P_{ij}^{\mathsf{T}} = P_{ij}$ and $S_i(\alpha)^{\mathsf{T}} = S_i(\alpha)$ in place of $Q_{ij}(\alpha)^{\mathsf{T}} = Q_{ji}(\alpha)$. **Q.E.D.**

Lemma 17.e.2 (*Inverses of Elementary Matrices*): Every elementary matrix is invertible, and the inverse of each is an elementary matrix of the exact same type:

$$Q_{ij}(\alpha)^{-1} = Q_{ij}(-\alpha), \quad P_{ij}^{-1} = P_{ij}, \quad \text{and} \quad S_i(\alpha)^{-1} = S_i(\alpha^{-1})$$

Exercise 17.8 (*Proof of Lemma 17.e.2*): The notes leave this proof as an exercise. *(Hint:* Note that $P_{ij} \cdot P_{ij}$ is the matrix obtained from P_{ij} by exchanging rows i and j, which brings us immediately back to I_n. Apply similar logical steps for the other types.*)*

AI-Proposition 17.1 (*Invertibility and Full Rank*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then A is invertible if and only if $\operatorname{rank}(A) = n$.

Proof:

By Lemma 17.c.3, A is invertible if and only if the associated map $T_A : K^n \to K^n$ is an isomorphism. Source and target here have the same dimension n, so by Corollary 15.b.10 **(c)** this happens if and only if T_A is surjective, that is, if and only if $\operatorname{Im}(T_A) = K^n$. Since $\operatorname{Im}(T_A)$

is a subspace of K^n, Proposition 12.3 turns that last condition into $\dim \mathrm{Im}(T_A) = n$. Finally, $\dim \mathrm{Im}(T_A) = \mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}(A)$ by Remark 17.5 **(c)** and **(a)**. **Q.E.D.**

Theorem 17.e.3 (*Decomposition into Elementary Matrices*): For every invertible matrix $A \in \mathrm{GL}_n(K)$, there exists an integer $k \geq 1$ and a sequence of elementary matrices $T_1, \dots, T_k$ such that:

$$T_k \cdot T_{k-1} \dots T_1 \cdot A = I_n$$

Furthermore, we have $A = T_1^{-1} \dots T_k^{-1}$ and $A^{-1} = T_k \dots T_1$. In particular, any invertible matrix A can be written as a finite product of elementary matrices (recall from Lemma 17.e.2 that each T_ℓ^{-1} is again an elementary matrix).

Proof:

We already know that every matrix $A \in \mathcal{M}_{n\times n}(K)$ can be brought to a row-reduced echelon matrix, which we will call A', using the Gaussian elimination algorithm (Theorem 6.14).

In our case, A is invertible, and hence $\mathrm{rank}(A) = n$ by AI-Proposition 17.1. Because row operations do not change the rank (row-equivalent matrices have the same row space by Proposition 13.3, hence the same row rank, which is the rank by Remark 17.5 **(a)**), we have $\mathrm{rank}(A') = \mathrm{rank}(A)$, which implies that $\mathrm{rank}(A') = n$. The only $n \times n$ row-reduced echelon matrix with rank n is the identity matrix, meaning $A' = I_n$. Indeed, the rank of A' is its number of pivots by Remark 17.5 **(b)**, so A' carries n pivots in n columns, one in each; since the pivots move strictly to the right as one descends the rows, the pivot of the ℓ-th row lies in the ℓ-th column, and since A' is row-reduced, that pivot is the only non-zero entry of its column.

Since each step in the Gaussian elimination algorithm corresponds to an elementary operation, we can deduce from Lemma 17.e.1 that there exist elementary matrices $T_1, \dots, T_k$ (of types I, II, and III) such that:

$$T_k \cdot T_{k-1} \dots T_1 \cdot A = I_n$$

(If $A = I_n$ to begin with, the algorithm performs no operation at all and this argument produces no matrices; the statement still holds as written, with $k := 1$ and $T_1 := S_1(1) = I_n$, which is an elementary matrix of Type III because $1 \neq 0$.) By left-multiplying successively with the inverses of these elementary matrices, which exist by Lemma 17.e.2, we isolate A and deduce that $A = T_1^{-1} \dots T_k^{-1}$. Taking the inverse of A yields $A^{-1} = T_k \dots T_1$, since inverting a product reverses its order and undoes each factor, by Proposition 17.a.7. **Q.E.D.**

Theorem 17.e.4 (*Matrix Inversion Algorithm*): Let $A \in \mathcal{M}_{n\times n}(K)$. Construct the augmented matrix by placing A alongside I_n, denoted as $(A \mid I_n) \in \mathcal{M}_{n\times 2n}(K)$, that is, the matrix whose first n columns are those of A and whose last n columns are those of I_n. Then, A is invertible if and only if $(A \mid I_n)$ can be brought, via a sequence of elementary row operations, to the form $(I_n \mid B)$ for some matrix $B \in \mathcal{M}_{n\times n}(K)$. Furthermore, if this is the case, we have $A^{-1} = B$.

Before proving this theorem, we require an intermediate technical result.

Lemma 17.e.5 (*Block Matrix Multiplication*): Let $B, C, D \in \mathcal{M}_{n\times n}(K)$. Then:

$$B \cdot (C \mid D) = (B \cdot C \mid B \cdot D)$$

Exercise 17.9 (*Proof of Lemma 17.e.5*): The notes leave this proof as an exercise: it follows directly from the definition of matrix multiplication. *(Hint:* Consider the (k,ℓ)-th entry of $B \cdot (C \mid D)$ and show that it matches the (k,ℓ)-th entry of $(B \cdot C \mid B \cdot D)$.*)*

Proof of Theorem 17.e.4:

Assume that A is invertible. By Theorem 17.e.3, there exist elementary matrices $T_1, \dots, T_k$ such that $T_k \dots T_1 \cdot A = I_n$. Using Lemma 17.e.5, we can multiply the augmented matrix $(A \mid I_n)$ from the left by this chain of elementary matrices:

$$T_k \dots T_1 \cdot (A \mid I_n) = (T_k \dots T_1 \cdot A \mid T_k \dots T_1 \cdot I_n) = (I_n \mid T_k \dots T_1)$$

Left-multiplying by T_1, then by T_2, and so on up to T_k is, by Lemma 17.e.1, nothing other than performing the corresponding k elementary row operations in that order. So $(A \mid I_n)$ can indeed be brought to the required form by a sequence of elementary row operations.

Let us define $B := T_k \dots T_1$. We can clearly see that:

$$B \cdot A = (T_k \dots T_1) \cdot A = I_n$$

and using the decomposition of A provided by Theorem 17.e.3,

$$A \cdot B = (T_1^{-1} \dots T_k^{-1}) \cdot (T_k \dots T_1) = I_n$$

This explicitly implies that $A^{-1} = B$.

Conversely, assume that $(A \mid I_n)$ can be brought to $(I_n \mid B)$ by a sequence of elementary row operations, for some matrix $B \in \mathcal{M}_{n\times n}(K)$. This directly implies that the matrix A itself is row-equivalent to I_n: each of those operations is left-multiplication by an elementary matrix (Lemma 17.e.1), and by Lemma 17.e.5 such a multiplication acts on the two blocks separately, so the left-hand block undergoes on its own exactly the same sequence of row operations, ending at I_n. Consequently, A must have rank n, again because row operations leave the row space, and with it the rank, unchanged (Proposition 13.3 and Remark 17.5 **(a)**), which inherently means that A is invertible by AI-Proposition 17.1. This concludes the proof. **Q.E.D.**

Example 17.3 (*Computing the Inverse of a Two-by-Two Matrix*): Let us find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathcal{M}_{2\times 2}(\mathbb{Q})$. We place A and I_2 together in an augmented matrix and apply Gaussian elimination:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right) \xrightarrow{R_2 - 3R_1 \longrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right)$$

Next, we scale the second row to create a pivot of 1:

$$\xrightarrow{-\frac{1}{2}R_2 \longrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right)$$

Finally, we eliminate the entry above the second pivot:

$$\xrightarrow{R_1 - 2R_2 \longrightarrow R_1} \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right)$$

The left side is now the identity matrix I_2, which means the right side is our inverse matrix. We conclude that:

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

Solutions to the Exercises

Proof of Exercise 17.7:

Both sides are $n\times n$ matrices, so it suffices to compare entries. Write $M := I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$, and recall that the (k,ℓ)-entry of E_{pq} is 1 if $(k,\ell) = (p,q)$ and 0 otherwise, while the (k,ℓ)-entry of I_n is $\delta_{k\ell}$. Since $i \neq j$, the four matrix units involved sit in four distinct positions, and the entries of M are:

- $M_{kk} = 1$ for $k \notin \{i,j\}$, from I_n alone;
- $M_{ii} = 1 - 1 = 0$ and $M_{jj} = 1 - 1 = 0$;
- $M_{ij} = 0 + 1 = 1$ and $M_{ji} = 0 + 1 = 1$;

- $M_{k\ell} = 0$ in every other position.

That is exactly the matrix obtained from I_n by exchanging rows i and j: row i of M has its single 1 in column j, row j has its single 1 in column i, and every other row is unchanged. Hence $M = P_{ij}$. **Q.E.D.**

Proof of Lemma 17.e.1 (a):

Let $A = (a_{k\ell}) \in \mathcal{M}_{n\times p}(K)$. Using $Q_{ij}(\alpha) = I_n + \alpha E_{ij}$ and distributivity,

$$Q_{ij}(\alpha)\cdot A = (I_n + \alpha E_{ij})A = A + \alpha\, E_{ij}A.$$

Now $E_{ij}A$ has (k,ℓ)-entry $\sum_q (E_{ij})_{kq} a_{q\ell}$, and $(E_{ij})_{kq}$ vanishes unless $k = i$ and $q = j$. So $E_{ij}A$ has the j-th row of A sitting in its i-th row and zeros everywhere else. Adding α times this to A therefore leaves every row of A untouched except the i-th, which becomes

$$(\text{row } i \text{ of } A) + \alpha\cdot(\text{row } j \text{ of } A),$$

which is precisely the operation $R_i + \alpha R_j \longrightarrow R_i$. The calculations for P_{ij} and $S_i(\alpha)$ are of the same kind. **Q.E.D.**

Proof of Lemma 17.e.2:

For Type II, the hint does it: $P_{ij}\cdot P_{ij}$ is, by Lemma 17.e.1 **(b)**, the matrix obtained from P_{ij} by exchanging its rows i and j, and exchanging them a second time restores I_n. So $P_{ij}^{-1} = P_{ij}$.

For Type I, apply Lemma 17.e.1 **(a)** twice, or compute directly using $i \neq j$, which gives $E_{ij}E_{ij} = 0$:

$$Q_{ij}(\alpha)\,Q_{ij}(-\alpha) = (I_n + \alpha E_{ij})(I_n - \alpha E_{ij}) = I_n - \alpha E_{ij} + \alpha E_{ij} - \alpha^2 E_{ij}E_{ij} = I_n,$$

and the same computation with α and $-\alpha$ interchanged gives $Q_{ij}(-\alpha)Q_{ij}(\alpha) = I_n$. Hence $Q_{ij}(\alpha)^{-1} = Q_{ij}(-\alpha)$.

For Type III, $S_i(\alpha)S_i(\alpha^{-1})$ scales the i-th row of $S_i(\alpha^{-1})$ by α, turning its (i,i)-entry α^{-1} into 1 and leaving the other rows alone, so the product is I_n; symmetrically for the other order. This uses $\alpha \neq 0$, which is part of the definition of $S_i(\alpha)$. In every case the inverse is an elementary matrix of the same type. **Q.E.D.**

Proof of Lemma 17.e.5:

Write $(C \mid D) \in \mathcal{M}_{n\times 2n}(K)$ for the matrix whose first n columns are those of C and whose last n columns are those of D, and call these columns $u_1, \dots, u_{2n}$. By the column-wise description of a product (Section 17.a), multiplying on the left by B acts on each column separately:

$$B\cdot\begin{pmatrix} | & & | \\ u_1 & \dots & u_{2n} \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ Bu_1 & \dots & Bu_{2n} \\ | & & | \end{pmatrix}.$$

The first n columns of $(C \mid D)$ are the columns $c_1, \dots, c_n$ of C, and the last n are the columns $d_1, \dots, d_n$ of D. So the first n columns of $B\cdot(C \mid D)$ are $Bc_1, \dots, Bc_n$, that is, the columns of $B\cdot C$, and the last n are $Bd_1, \dots, Bd_n$, the columns of $B\cdot D$. Hence $B\cdot(C \mid D) = (B\cdot C \mid B\cdot D)$. **Q.E.D.**

AI-Note: All five numbered statements of the notes are present in the transcript, in the notes' order and with the notes' proofs, but the printed numbers were each one too large: the three types of elementary matrix are introduced in the notes under *"Notation"*, without a number, while the transcript wrapped them in a numbered `definition`, which consumed 17.e.1 and pushed everything after it. That definition is now unnumbered, and 17.e.1 through 17.e.5 line up with the handwritten ones again.

The notes call the matrix units E_{ij}; the transcript renamed them R_{ij}, which collides with the R_i that this very section uses for rows — R_{ij} and $R_i + \alpha R_j \to R_i$ appear within two lines of each other. The notes' E_{ij} is restored. Note that the notes switch from the E_i of the chapter on linear systems to R_i here for rows; that switch is Prof. Biran's own and is left alone.

The notes mark four things for the reader: the explicit form of P_{ij}, the direct calculation behind Lemma 17.e.1, Lemma 17.e.2, and Lemma 17.e.5. All four are now exercises with solutions above. The proof of Lemma 17.e.5 in the transcript also carried its opening sentence twice, once with an exercise prompt and once with a hint.

Finally, the transcript's example of a Type II matrix was $P_{3,4}$, correctly computed but not the notes' example; it is back to Prof. Biran's $P_{2,4}$.

A later pass looking only for gaps in rigour found one product that does not compose. The elementary matrices are defined in $\mathcal{M}_{n\times n}(K)$, which is what Lemma 17.e.1 needs, since it multiplies $A \in \mathcal{M}_{n\times p}(K)$ from the left. But Important remark 17.4 multiplies the same A from the right by the same matrices, and $A \cdot Q_{ij}(\alpha)$ is defined only for a $p \times p$ factor. The remark now says that its elementary matrices are the ones of $\mathcal{M}_{p\times p}(K)$; the mathematics of all three statements is otherwise untouched, and the proof gains the note that it applies **(a)** to A^T with the roles of n and p interchanged.

The chapter also carries the same pattern seen in the chapter on linear maps and bases: two of its steps rest on facts the book never states. Theorem 17.e.3 opens with *"A is invertible, and hence* $\operatorname{rank}(A) = n$*"*, and the proof of Theorem 17.e.4 closes with the converse implication; neither direction is stated anywhere in the notes, each being assembled on the spot from Lemma 17.c.3, Corollary 15.b.10 and Proposition 12.3. Since the equivalence is nowhere in the lecture material, it is stated as AI-Proposition 17.1 rather than as an unnumbered Proposition: the ai-family carries the robot marker that tells a reader this is not Prof. Biran's, and runs on its own counter, so 17.e.1–17.e.5 stay where they are.

Seven further steps now name the reason they hold: the existence of the row-reduced echelon form, the invariance of the rank under row operations (twice), the identification of the only $n \times n$ row-reduced echelon matrix of rank n, the passage from left-multiplication by a chain of elementary matrices to a sequence of row operations (twice, once in each direction of Theorem 17.e.4), and the reversal of order when a product is inverted. The statement of Theorem 17.e.3 asks for $k \geq 1$ while its proof produces no elementary matrix at all when $A = I_n$; the proof now closes that gap with $T_1 := S_1(1) = I_n$.

Two smaller things. The proof following Important remark 17.4 argues both that remark and the lemma above it, so its title now says which; and where it read *"But taking the transpose again"*, nothing was in fact being transposed, the step being the observation that the rows of A^T are the columns of A. The logic was right either way.

Constructing New Vector Spaces Out of Old Ones (Chapter 18)

The Homomorphism Space, The Dual Space And The Dual Map (Section 18.a)

The Homomorphism Space

Proposition 18.a.1 (*The Vector Space Structure of the Homomorphism Space*): Let V and W be two vector spaces over K. Then, $\mathrm{Hom}(V,W)$ has the structure of a vector space over K, if we endow it with the following operations:

- For all $T_1, T_2 \in \mathrm{Hom}(V,W)$, $(T_1+T_2)(v) := T_1(v) + T_2(v)$ for all $v \in V$.
- For all $T \in \mathrm{Hom}(V,W)$ and $\alpha \in K$, $(\alpha \cdot T)(v) := \alpha \cdot T(v)$ for all $v \in V$.

Proof:

We claim that $T_1 + T_2 \in \mathrm{Hom}(V,W)$, meaning $T_1 + T_2$ is a linear map. Indeed, for a scalar a and a vector v, we have

$$\begin{aligned}(T_1+T_2)(a\cdot v) &= T_1(a\cdot v) + T_2(a\cdot v) = a\cdot T_1(v) + a\cdot T_2(v)\\ &= a\cdot (T_1(v)+T_2(v)) = a\cdot (T_1+T_2)(v),\end{aligned}$$

and for vectors v_1, v_2, we have

$$\begin{aligned}(T_1+T_2)(v_1+v_2) &= T_1(v_1+v_2) + T_2(v_1+v_2) = T_1(v_1)+T_1(v_2)+T_2(v_1)+T_2(v_2)\\ &= (T_1+T_2)(v_1) + (T_1+T_2)(v_2).\end{aligned}$$

Consequently, $T_1 + T_2$ is a linear map. The proof that $\alpha \cdot T$ is linear is analogous.

So far, we have defined $+\colon \mathrm{Hom}(V,W)\times \mathrm{Hom}(V,W) \to \mathrm{Hom}(V,W)$ and $\cdot\colon K \times \mathrm{Hom}(V,W) \to \mathrm{Hom}(V,W)$.

Regarding the neutral element, the map $0\colon V \to W$ defined by $v \mapsto 0$ for all $v \in V$ is linear. We claim that $0 + T = T$. Indeed,

$$(0+T)(v) = 0(v) + T(v) = 0 + T(v) = T(v)$$

for all $v \in V$. Similarly, one shows that $T + 0 = T$. **Q.E.D.**

Exercise 18.1 (*The Axioms for* $\mathrm{Hom}(V,W)$): Complete the proof that $\mathrm{Hom}(V,W)$ is a vector space by checking that it satisfies all the axioms of a vector space, using the zero map $0\colon V \to W$ as the additive identity. Show also that $\alpha \cdot T$ is linear for all $\alpha \in K$ and $T \in \mathrm{Hom}(V,W)$, and that $\Psi^{\mathcal{B}}_{\mathcal{C}}(T_1+T_2) = \Psi^{\mathcal{B}}_{\mathcal{C}}(T_1) + \Psi^{\mathcal{B}}_{\mathcal{C}}(T_2)$ in Theorem 18.a.2.

For example, consider the distributivity axiom. For all $\alpha \in K$ and $T_1, T_2 \in \mathrm{Hom}(V,W)$, we have $\alpha\cdot(T_1+T_2) = \alpha\cdot T_1 + \alpha \cdot T_2$. Indeed, for $v \in V$,

$$\begin{aligned}(\alpha\cdot(T_1+T_2))(v) &= \alpha\cdot(T_1+T_2)(v) = \alpha(T_1(v)+T_2(v))\\ &= \alpha\cdot T_1(v) + \alpha\cdot T_2(v) = (\alpha\cdot T_1)(v) + (\alpha\cdot T_2)(v)\\ &= (\alpha\cdot T_1 + \alpha\cdot T_2)(v).\end{aligned}$$

Therefore, $\alpha\cdot(T_1+T_2) = \alpha\cdot T_1 + \alpha\cdot T_2$.

Theorem 18.a.2 (*The Canonical Isomorphism from the Homomorphism Space* $\mathrm{Hom}(V,W)$ *to the Matrix Space* $\mathcal{M}_{m\times n}(K)$): Let V and W be finite-dimensional vector spaces over K, let $\mathcal{B} =$

$(v_1, \dots, v_n)$ be a basis for V, and let $\mathcal{C} = (w_1, \dots, w_m)$ be a basis for W. Then, $\Psi^{\mathcal{B}}_{\mathcal{C}} \colon \operatorname{Hom}(V,W) \to \mathcal{M}_{m\times n}(K)$, defined by

$$\Psi^{\mathcal{B}}_{\mathcal{C}}(T) := [T]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix},$$

is a linear map, and moreover, an isomorphism.

Proof:

Recall that $T_{[T]^{\mathcal{B}}_{\mathcal{C}}} = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$. Recall also that $\Phi_{\mathcal{B}}(v) = [v]_{\mathcal{B}} \in K^n$ for all $v \in V$. Furthermore, $\Phi_{\mathcal{C}}(w) = [w]_{\mathcal{C}} \in K^m$ for all $w \in W$. We have

$$\Phi_{\mathcal{B}}^{-1}\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 v_1 + \dots + a_n v_n, \quad \Phi_{\mathcal{C}}^{-1}\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = b_1 w_1 + \dots + b_m w_m.$$

We claim that the map $\Psi^{\mathcal{B}}_{\mathcal{C}}$ is bijective. Indeed, define $\Theta \colon \mathcal{M}_{m\times n}(K) \to \operatorname{Hom}(V,W)$ by $\Theta(A) := \Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}} \in \operatorname{Hom}(V,W)$ for $A \in \mathcal{M}_{m\times n}(K)$. We have:

$$\begin{aligned}
\Psi^{\mathcal{B}}_{\mathcal{C}} \circ \Theta(A) &= \Psi^{\mathcal{B}}_{\mathcal{C}}(\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}) \\
&= \big([\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}(v_1)]_{\mathcal{C}} \dots [\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}(v_n)]_{\mathcal{C}}\big) \\
&= \begin{pmatrix} | & & | \\ T_A\Phi_{\mathcal{B}}(v_1) & \dots & T_A\Phi_{\mathcal{B}}(v_n) \\ | & & | \end{pmatrix} \\
&= \begin{pmatrix} | & & | \\ T_A(e_1) & \dots & T_A(e_n) \\ | & & | \end{pmatrix} = A.
\end{aligned}$$

This uses the facts that $[\Phi_{\mathcal{C}}^{-1}(b)]_{\mathcal{C}} = b$ for all $b \in K^m$, and $\Phi_{\mathcal{B}}(v_k) = e_k$. This proves that $\Psi^{\mathcal{B}}_{\mathcal{C}} \circ \Theta = \mathrm{id}_{\mathcal{M}_{m\times n}(K)}$.

Now, by definition, $\Theta \circ \Psi^{\mathcal{B}}_{\mathcal{C}}(T) = \Phi_{\mathcal{C}}^{-1} \circ T_{\Psi^{\mathcal{B}}_{\mathcal{C}}(T)} \circ \Phi_{\mathcal{B}} = \Phi_{\mathcal{C}}^{-1} \circ T_{[T]^{\mathcal{B}}_{\mathcal{C}}} \circ \Phi_{\mathcal{B}} = T$. This shows that $\Theta \circ \Psi^{\mathcal{B}}_{\mathcal{C}} = \mathrm{id}_{\operatorname{Hom}(V,W)}$. Therefore, $\Psi^{\mathcal{B}}_{\mathcal{C}}$ is bijective.

It remains to prove that $\Psi^{\mathcal{B}}_{\mathcal{C}}$ is linear. Indeed, let $\alpha \in K$, and let $T \in \operatorname{Hom}(V,W)$. Then,

$$\begin{aligned}
\Psi^{\mathcal{B}}_{\mathcal{C}}(\alpha \cdot T) &= ([(\alpha T)(v_1)]_{\mathcal{C}} \dots [(\alpha T)(v_n)]_{\mathcal{C}}) \\
&= ([\alpha T(v_1)]_{\mathcal{C}} \dots [\alpha T(v_n)]_{\mathcal{C}}) \\
&= (\alpha[T(v_1)]_{\mathcal{C}} \dots \alpha[T(v_n)]_{\mathcal{C}}) \\
&= \alpha \cdot \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \\
&= \alpha \cdot \Psi^{\mathcal{B}}_{\mathcal{C}}(T).
\end{aligned}$$

Similarly, one shows that $\Psi^{\mathcal{B}}_{\mathcal{C}}(T_1 + T_2) = \Psi^{\mathcal{B}}_{\mathcal{C}}(T_1) + \Psi^{\mathcal{B}}_{\mathcal{C}}(T_2)$. **Q.E.D.**

Corollary 18.a.3 (*Dimensionality of the Homomorphism Space*): If V and W are finite-dimensional vector spaces, then the homomorphism space $\operatorname{Hom}(V,W)$ is also finite-dimensional, and $\dim \operatorname{Hom}(V,W) = \dim V \cdot \dim W$.

Proof:

Let $n := \dim V$ and $m := \dim W$. By Theorem 18.a.2, we have an isomorphism
$$\mathrm{Hom}(V, W) \cong \mathcal{M}_{m\times n}(K).$$
This means $\mathrm{Hom}(V, W)$ is a finite-dimensional space. Furthermore, isomorphic spaces have the same dimension by Corollary 15.b.11, so
$$\dim \mathrm{Hom}(V, W) = \dim \mathcal{M}_{m\times n}(K) = m \cdot n,$$
the last equality because the $m \cdot n$ matrix units E_{ij} form a basis of $\mathcal{M}_{m\times n}(K)$. **Q.E.D.**

Rings of Endomorphisms

The notes say at this point only *"explain briefly the structure of a ring"* and *"explain what is a homomorphism of rings"*, since both were given orally. The book uses both notions from here on, and already used the word *"ring"* on Page 122, so we record them.

AI-Definition 18.1 (*Ring*): A "*ring*" is a set R together with two operations, an addition $+\colon R \times R \to R$ and a multiplication $\cdot\colon R \times R \to R$, such that:

(a) $(R, +)$ is a commutative group, whose neutral element is denoted by 0;

(b) the multiplication is associative: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$;

(c) the two operations are connected by the distributive laws
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (b + c) \cdot a = b \cdot a + c \cdot a \qquad \text{for all } a, b, c \in R.$$

If there is an element $1 \in R$ with $1 \cdot a = a \cdot 1 = a$ for all $a \in R$, we call R a "*ring with a unity*", and 1 its "*unity*". If moreover $a \cdot b = b \cdot a$ for all $a, b \in R$, the ring is called "*commutative*".

AI-Definition 18.2 (*Homomorphism of Rings*): Let R and S be rings with unity. A map $\varphi\colon R \to S$ is a "*homomorphism of rings*" if for all $a, b \in R$,
$$\varphi(a + b) = \varphi(a) + \varphi(b), \qquad \varphi(a \cdot b) = \varphi(a) \cdot \varphi(b), \qquad \varphi(1_R) = 1_S.$$
A bijective homomorphism of rings is called an "*isomorphism of rings*"; its inverse is then automatically a homomorphism of rings as well.

Example 18.1 (*Examples of Rings*):

(a) Every field K is also a commutative ring.

(b) $\mathbb{Z}$ is a commutative ring.

(c) $\mathcal{M}_{n\times n}(K)$ is a ring, where $A \cdot B$ is matrix multiplication, and the unity is I_n. Clearly, it is not commutative.

Corollary 18.a.4 (*The Ring of Endomorphisms*): Let V be a vector space over K. Then, $\mathrm{End}(V) := \mathrm{Hom}(V, V)$ is a ring with a unity. This ring is, in general, not commutative. The multiplication in $\mathrm{End}(V)$ is given by composition $T_1 \cdot T_2 := T_1 \circ T_2$ for all $T_1, T_2 \in \mathrm{Hom}(V, V)$, and the unity is id_V. Moreover, if V is finite-dimensional with $\dim V = n$, and $\mathcal{B} = (w_1, \ldots, w_n)$ is a basis for V, then $\Psi^{\mathcal{B}}_{\mathcal{B}}\colon \mathrm{Hom}(V, V) \to \mathcal{M}_{n\times n}(K)$ is an isomorphism of rings.

Proof:

First, we verify that $(\mathrm{End}(V), +, \circ)$ satisfies the ring axioms. We already know that $(\mathrm{End}(V), +)$ is an abelian group from the proposition regarding the vector space structure of $\mathrm{Hom}(V, W)$, Proposition 18.a.1 with $W = V$. We now verify the properties related to multiplication (composition):

(a) **Associativity of Multiplication**: For any $T_1, T_2, T_3 \in \mathrm{End}(V)$, function composition is inherently associative. That is, $(T_1 \circ T_2) \circ T_3 = T_1 \circ (T_2 \circ T_3)$.

(b) Existence of Multiplicative Identity: The identity map $\mathrm{id}_V \colon V \to V$ is an endomorphism. For any $T \in \mathrm{End}(V)$, we have $T \circ \mathrm{id}_V = T$ and $\mathrm{id}_V \circ T = T$. Thus, id_V is the multiplicative identity.

(c) Distributivity: For any $T, T_1, T_2 \in \mathrm{End}(V)$, we must show both left and right distributivity.

- **Left Distributivity**: For any $v \in V$,
$$(T \circ (T_1 + T_2))(v) = T((T_1 + T_2)(v)) = T(T_1(v) + T_2(v)).$$
By the linearity of T, this equals $T(T_1(v)) + T(T_2(v)) = (T \circ T_1)(v) + (T \circ T_2)(v) = (T \circ T_1 + T \circ T_2)(v)$. Hence, $T \circ (T_1 + T_2) = T \circ T_1 + T \circ T_2$.
- **Right Distributivity**: For any $v \in V$,
$$\begin{aligned}((T_1 + T_2) \circ T)(v) &= (T_1 + T_2)(T(v)) \\ &= T_1(T(v)) + T_2(T(v)) \\ &= (T_1 \circ T)(v) + (T_2 \circ T)(v) \\ &= (T_1 \circ T + T_2 \circ T)(v).\end{aligned}$$
Hence, $(T_1 + T_2) \circ T = T_1 \circ T + T_2 \circ T$.

Thus, $\mathrm{End}(V)$ is a ring with unity id_V. It is generally not commutative, as demonstrated by matrix multiplication.

Now, assume V is finite-dimensional with $\dim V = n$, and let $\mathcal{B}$ be a basis for V. We have already established that $\Psi^{\mathcal{B}}_{\mathcal{B}} \colon \mathrm{End}(V) \to \mathcal{M}_{n\times n}(K)$ is a linear isomorphism, this being Theorem 18.a.2 with $W = V$ and $\mathcal{C} = \mathcal{B}$. To prove it is a ring isomorphism, we must show it preserves the multiplicative structure and the multiplicative identity.

(a) Preservation of Multiplication: For any $T_1, T_2 \in \mathrm{End}(V)$, a fundamental result from Linear Algebra I states that the matrix representation of a composition of linear maps is the product of their matrix representations (the composition rule, on Page 116). That is,
$$\Psi^{\mathcal{B}}_{\mathcal{B}}(T_1 \circ T_2) = [T_1 \circ T_2]^{\mathcal{B}}_{\mathcal{B}} = [T_1]^{\mathcal{B}}_{\mathcal{B}} \cdot [T_2]^{\mathcal{B}}_{\mathcal{B}} = \Psi^{\mathcal{B}}_{\mathcal{B}}(T_1) \cdot \Psi^{\mathcal{B}}_{\mathcal{B}}(T_2).$$

(b) Preservation of Multiplicative Identity: The matrix representation of the identity endomorphism with respect to any basis $\mathcal{B}$ is the identity matrix I_n (the Proposition on Page 119).
$$\Psi^{\mathcal{B}}_{\mathcal{B}}(\mathrm{id}_V) = [\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = I_n.$$

Since $\Psi^{\mathcal{B}}_{\mathcal{B}}$ is a bijective linear map that preserves both multiplication and the multiplicative identity, it is an isomorphism of rings. **Q.E.D.**

The Dual Space

An important special case of $\mathrm{Hom}(V, W)$ is when $W = K$.

Definition 18.a.5 (*The Dual Space and Linear Functionals*): Let V be a vector space over K. The "*dual space*" of V is defined to be the vector space $V^* := \mathrm{Hom}(V, K)$. The elements of V^* are linear maps $\ell \colon V \to K$, and are sometimes called "*functionals*".

Corollary 18.a.6 (*Dimensionality of the Dual Space*): Let V be a finite-dimensional vector space over K. Then, $\dim V^* = \dim V$.

Proof:

We have $\dim V^* = \dim \mathrm{Hom}(V, K) = \dim V \cdot \dim K = \dim V \cdot 1 = \dim V$, using Corollary 18.a.3 for the middle equality and, for the one after it, that K is one-dimensional as a vector space over itself, with basis (1). **Q.E.D.**

Let $V = K^n_{\text{col}}$. Then, we can identify V^* with K^n_{row} as follows: If $\ell \in K^n_{\text{row}}$, then ℓ defines a functional $f_\ell \colon K^n_{\text{col}} \to K$ by $f_\ell(v) := \ell \cdot v \in \mathcal{M}_{1\times 1}(K) \cong K$. The map $D\colon K^n_{\text{row}} \to (K^n_{\text{col}})^*$ defined by $D(\ell) := f_\ell$ is a linear map, and it is in fact an isomorphism; the notes leave the verification to the reader, and it is carried out among the solutions at the end of this section.

Exercise 18.2 (*The Row Space as the Dual of the Column Space*): Show that the map $D\colon K^n_{\text{row}} \to (K^n_{\text{col}})^*$, $D(\ell) := f_\ell$, is a linear map and an isomorphism.

Definition (*The Functionals Dual to a Basis*): Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V. For $1 \le i \le n$, define a functional $v_i^* \in V^*$ as follows: $v_i^*(v_j) := \delta_{ij}$. This means that $v_i^*(v_i) = 1$, and $v_i^*(v_j) = 0$ for all $j \neq i$. Since $v_1, \dots, v_n$ form a basis for V, our definition determines uniquely a well-defined functional $v_i^*\colon V \to K$ (this is Theorem 15.a.3: prescribing the values on a basis arbitrarily determines one and only one linear map).

Lemma 18.a.7 (*Construction of the Dual Basis*): The elements $v_1^*, \dots, v_n^* \in V^*$ form a basis for V^*. This basis is called the "*dual basis*" to $\mathcal{B}$. We denote this basis by $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$. Moreover, for all $\ell \in V^*$, $\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*$.

Proof:

Let $\ell \in V^*$. We claim that $\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*$, or in other words, we claim that $\ell(v) = \sum_{i=1}^n \ell(v_i) \cdot v_i^*(v)$. Indeed, both sides of the equation are linear maps $V \to K$, and therefore, it is enough to check that it holds for $v = v_1, v = v_2, \dots, v = v_n$, because $v_1, \dots, v_n$ form a basis for V (two linear maps that agree on a basis are equal, by the uniqueness half of Theorem 15.a.3). And indeed, for $v = v_k$, we have:

$$\sum_{i=1}^n \ell(v_i) \cdot v_i^*(v_k) = \sum_{i=1}^n \ell(v_i) \cdot \delta_{ik} = \ell(v_k).$$

The claim just proven shows that $v_1^*, \dots, v_n^*$ span V^*. Since $\dim V^* = \dim V = n$ by Corollary 18.a.6, it follows that $v_1^*, \dots, v_n^*$ form a basis for V^*: a list of n vectors spanning an n-dimensional space is a basis, by Theorem 12.1.

Q.E.D.

Question 18.1: Let V be a finite-dimensional vector space over K, and let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V. $\mathcal{B}^*$ and $\mathcal{C}^*$ are two bases for V^*. What is the relation between $[\mathrm{id}_{V^*}]^{\mathcal{C}^*}_{\mathcal{B}^*}$ and $[\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{B}}$?

Proposition 18.a.8 (*Dual Change of Basis and Transposition*): The relationship between transition matrices in the dual space and the primal space is characterized by transposition and inversion. Specifically, the transition matrix between dual bases $\mathcal{C}^*$ and $\mathcal{B}^*$ is the transpose of the inverse transition matrix between the original bases $\mathcal{C}$ and $\mathcal{B}$:

$$[\mathrm{id}_{V^*}]^{\mathcal{C}^*}_{\mathcal{B}^*} = \left([\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}}\right)^{\mathsf{T}} = \left(\left([\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{B}}\right)^{-1}\right)^{\mathsf{T}}.$$

Thus, the coordinate transformation in the dual space corresponds to the transpose of the inverse transformation in the primal space. The second equality is Corollary 17.b.1, which gives $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}} = ([\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{B}})^{-1}$.

Proof:

Let $\mathcal{B} = (v_1, \dots, v_n)$, and let $\mathcal{C} = (w_1, \dots, w_n)$. We have

$$[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{C}} & \dots & [v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}.$$

Let $A := [\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}}$ and write $A = (a_{ij})$. By definition, $v_i = \sum_{k=1}^n a_{ki} w_k$, its i-th column being $[v_i]_{\mathcal{C}}$ (the column description of a representation matrix, on Page 113). Let $\mathcal{C}^* = (w_1^*, \dots, w_n^*)$ be the dual basis of $\mathcal{C}$. We apply the previous lemma to $\ell = w_j^*$ and the basis $\mathcal{B}$:

$$w_j^* = \sum_{i=1}^n w_j^*(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^* \left(\sum_{k=1}^n a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*.$$

Here, only $k = j$ remains from the inner sum. This says that $[w_j^*]_{\mathcal{B}^*} = (a_{j1}, \dots, a_{jn})^{\mathsf{T}}$, which is the j-th column of A^{T}; reading those columns off by the same column description, this implies that $[\mathrm{id}_{V^*}]^{\mathcal{C}^*}_{\mathcal{B}^*} = A^{\mathsf{T}} = ([\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}})^{\mathsf{T}}$. **Q.E.D.**

The Dual Map

Lemma 18.a.9 (*Definition of the Dual Map*): Let V and W be vector spaces over K, and let $T\colon V \to W$ be a linear map. Define a new map $T^*\colon W^* \to V^*$ by $T^*(\ell) := \ell \circ T$ for all $\ell \in W^*$. Then, T^* is a linear map. It is called the "*dual map*" to T.

Proof:

Let $\ell \in W^*$, meaning $\ell\colon W \to K$ is linear. Clearly, $\ell \circ T\colon V \to K$ is a linear map because both T and ℓ are linear, so $T^*(\ell) := \ell \circ T \in V^*$. This shows that $T^*\colon W^* \to V^*$ has its target indeed in V^*, as claimed.

We will show now that T^* is a linear map. Let $\ell \in W^*$, and let $\alpha \in K$. We have that $T^*(\alpha\ell) \in V^*$ is the map:

$$T^*(\alpha\ell)(v) = ((\alpha\ell) \circ T)(v) = (\alpha\ell)(T(v)) = \alpha\ell(T(v)) = \alpha \cdot (\ell \circ T)(v) = \alpha(T^*\ell)(v).$$

So, $T^*(\alpha\ell) = \alpha \cdot T^*(\ell)$. Similarly, one shows that $T^*(\ell_1 + \ell_2) = T^*(\ell_1) + T^*(\ell_2)$. **Q.E.D.**

Proposition 18.a.10 (*Functoriality and Composition of Duals*): Let U, V, W be vector spaces over K, and let $T\colon V \to W$ and $S\colon W \to U$ be linear maps. Consider the dual maps $T^*\colon W^* \to V^*$ and $S^*\colon U^* \to W^*$. Then, $(S \circ T)^* = T^* \circ S^*$.

The notes mark this proof as an exercise; it is carried out among the solutions at the end of the section, together with the following one.

Exercise 18.3 (*Duals of Compositions, of the Identity and of the Zero Map*): Let V be a vector space over K. Prove Proposition 18.a.10, and show that:

(a) $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

(b) $(0\colon V \to V)^* = (0\colon V^* \to V^*)$.

Summary 18.1: In summary, applying $(-)^*$, or $\mathrm{Hom}(-, K)$, takes every vector space V and gives us a new vector space V^*. It also takes any linear map $T\colon V \to W$ and gives us a new map T^* between the duals of V and W, but the arrow is reversed: $T^*\colon W^* \to V^*$. This is an example of a "*contravariant functor*".

Lemma 18.a.11 (*Transposition of the Representation Matrix*): Let $S\colon V \to W$ be a linear map between two finite-dimensional vector spaces V and W over K. Let $\mathcal{B}$ be a basis for V, and let $\mathcal{C}$ be a basis for W. Consider $S^*\colon W^* \to V^*$ and the bases $\mathcal{C}^*, \mathcal{B}^*$ for W^*, V^*, respectively. Then, $[S^*]^{\mathcal{C}^*}_{\mathcal{B}^*} = ([S]^{\mathcal{B}}_{\mathcal{C}})^{\mathsf{T}}$.

Proof:

Write $\mathcal{B} = (v_1, \ldots, v_n)$, $\mathcal{C} = (w_1, \ldots, w_m)$, and $A := [S]^{\mathcal{B}}_{\mathcal{C}} = (a_{ij})$. The matrix A satisfies $S(v_j) = \sum_{i=1}^m a_{ij} w_i$. Now, for $w_j^* \in \mathcal{C}^*$, we have

$$S^*(w_j^*) = w_j^* \circ S = \sum_{i=1}^n (w_j^* \circ S)(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^*(S(v_i)) \cdot v_i^*,$$

where the second equality follows from the Lemma on the Dual Basis, Lemma 18.a.7, applied to the functional $w_j^* \circ S \in V^*$ and the basis $\mathcal{B}$. Substituting $S(v_i)$, we get

$$\sum_{i=1}^n w_j^* \left(\sum_{k=1}^m a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*.$$

The coordinates of $S^*(w_j^*)$ in the basis $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ are $\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix}$, which corresponds to row j in A, represented as a column (i.e., column j in A^T). These coordinates are, for $1 \le j \le m$, precisely the columns of $[S^*]^{\mathcal{C}^*}_{\mathcal{B}^*}$ (the column description of a representation matrix, on Page 113), and they agree column by column with those of A^T. Hence $[S^*]^{\mathcal{C}^*}_{\mathcal{B}^*} = A^\mathsf{T} = ([S]^{\mathcal{B}}_{\mathcal{C}})^\mathsf{T}$, as claimed.

Q.E.D.

Reflexivity

Let V be a vector space over K. We have V^*, but also $V^{**} := (V^*)^*$, obtained by applying the dual operation one more time to V^*. V^{**} is sometimes called the "*bidual space*" of V. Is there any relation between V^{**} and V?

Theorem 18.a.12 (*The Canonical Isomorphism to the Bidual Space*): Let V be a finite-dimensional vector space over K. Then, there is a canonical isomorphism $\tau \colon V \to (V^*)^*$ given by: for $v \in V$, $\tau(v) \colon V^* \to K$ is the map $\tau(v)(\ell) := \ell(v)$ for all $\ell \in V^*$.

Remark 18.1: The map $\tau \colon V \to (V^*)^*$ exists and is defined in the same way as above regardless of V being finite- or infinite-dimensional. Moreover, τ is always injective. However, when V is infinite-dimensional, τ fails to be surjective.

Proof:

We claim that τ is a linear map. Indeed, if $v \in V$ and $\alpha \in K$, we have

$$\begin{aligned} \tau(\alpha v) &= (V^* \ni \ell \mapsto \ell(\alpha v) \in K) \\ &= (V^* \ni \ell \mapsto \alpha \ell(v) \in K) \\ &= \alpha \cdot (V^* \ni \ell \mapsto \ell(v) \in K) = \alpha \tau(v). \end{aligned}$$

Similarly, one shows that $\tau(v' + v'') = \tau(v') + \tau(v'')$.

We now claim that τ is injective. To see this, we will show that $\ker(\tau) = 0$, which suffices by Proposition 15.b.2 **(a)**. Indeed, let $v \in \ker(\tau)$, meaning $\tau(v) = 0$. Then, $\ell(v) = 0$ for all $\ell \in V^*$. Choose a basis $\mathcal{B} = (v_1, \ldots, v_n)$ for V. Write $v = c_1 v_1 + \cdots + c_n v_n$. Apply v_k^* to v, and we obtain $v_k^*(c_1 v_1 + \cdots + c_n v_n) = 0$. But $v_k^*(c_1 v_1 + \cdots + c_n v_n) = c_k$. Therefore, $c_k = 0$, and this holds for $1 \le k \le n$, which implies that $v = 0$. This proves that τ is injective.

To finish the proof, note that $\dim(V^*)^* = \dim V^* = \dim V$, by two applications of Corollary 18.a.6. Therefore, since $\tau \colon V \to (V^*)^*$ is injective, it must be an isomorphism: between two spaces of the same finite dimension, an injective linear map is already bijective, by Corollary 15.b.10 **(c)**.

Q.E.D.

Remark 18.2: Theorem 18.a.12 is not true for infinite-dimensional vector spaces. More specifically, $\tau\colon V \to (V^*)^*$ is injective, but not surjective. In fact, when V is infinite-dimensional, $(V^*)^*$ is not isomorphic to V.

Naturality

The map $\tau\colon V \to (V^*)^*$ exists for every space V, and it is defined in the same way. To distinguish τ's for different spaces, we denote it by $\tau_V\colon V \to (V^*)^*$.

Proposition (*Naturality of the Canonical Map to the Bidual*): Let V and W be two vector spaces over K. Then, for any linear map $T\colon V \to W$, we have the following naturality condition, represented by a commutative diagram:

$$\begin{tikzcd} V \arrow[r, "\tau_V"] \arrow[d, "T"'] & (V^*)^* \arrow[d, "(T^*)^*"] \\ W \arrow[r, "\tau_W"'] & (W^*)^* \end{tikzcd}$$

In other words, $(T^*)^* \circ \tau_V = \tau_W \circ T$.

Proof:

Both sides are maps $V \to (W^*)^*$, so it is enough to compare their values at an arbitrary $v \in V$; those values are functionals on W^*, so it is enough in turn to compare them at an arbitrary $\ell \in W^*$. Unwinding the definition of the dual map, Lemma 18.a.9, twice, and that of τ, which is the same formula for every space,

$$\begin{aligned} \Big((T^*)^*\big(\tau_V(v)\big)\Big)(\ell) &= \big(\tau_V(v) \circ T^*\big)(\ell) = \tau_V(v)\big(T^*(\ell)\big) = \big(T^*(\ell)\big)(v) \\ &= (\ell \circ T)(v) = \ell\big(T(v)\big) = \tau_W\big(T(v)\big)(\ell). \end{aligned}$$

Hence $(T^*)^*(\tau_V(v)) = \tau_W(T(v))$ for every $v \in V$, which is precisely the commutativity of the square. Note that nothing here requires V or W to be finite-dimensional. **Q.E.D.**

Exercise 18.4 (*Canonical Evaluation on Dual Bases*): Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V. Let $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ be the dual basis, and let $(\mathcal{B}^*)^* = ((v_1^*)^*, \ldots, (v_n^*)^*)$ be the basis for $(V^*)^*$ which is dual to $\mathcal{B}^*$. Show that $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$, meaning $\tau(v_i) = (v_i^*)^*$ for all $1 \le i \le n$.

Solutions to the Exercises

Proof of Exercise 18.1:

For the linearity of $\alpha \cdot T$: for $a \in K$ and $v, v' \in V$,

$$(\alpha T)(av + v') = \alpha\, T(av + v') = \alpha\big(aT(v) + T(v')\big) = a \cdot \alpha T(v) + \alpha T(v') = a\,(\alpha T)(v) + (\alpha T)(v').$$

The vector space axioms for $\mathrm{Hom}(V, W)$ all reduce to the corresponding axioms in W, because both operations are defined pointwise. For instance, associativity of addition: for all $v \in V$,

$$\big((T_1 + T_2) + T_3\big)(v) = \big(T_1(v) + T_2(v)\big) + T_3(v) = T_1(v) + \big(T_2(v) + T_3(v)\big) = \big(T_1 + (T_2 + T_3)\big)(v),$$

using associativity in W in the middle; since two maps agreeing at every v are equal, $(T_1+T_2)+T_3 = T_1+(T_2+T_3)$. The same pattern gives commutativity, $1 \cdot T = T$, $(\alpha\beta) \cdot T = \alpha \cdot (\beta \cdot T)$, and the two distributive laws, the one for $\alpha \cdot (T_1 + T_2)$ is written out in the text above. The additive inverse of T is $(-1) \cdot T$, since $(T + (-1)T)(v) = T(v) - T(v) = 0$ for all v.

For the additivity of $\Psi^{\mathcal{B}}_{\mathcal{C}}$: its j-th column at $T_1 + T_2$ is

$$[(T_1 + T_2)(v_j)]_{\mathcal{C}} = [T_1(v_j) + T_2(v_j)]_{\mathcal{C}} = [T_1(v_j)]_{\mathcal{C}} + [T_2(v_j)]_{\mathcal{C}},$$

because the coordinate map $\Phi_{\mathcal{C}}$ is linear. Columnwise addition of matrices is addition of matrices, so $\Psi^{\mathcal{B}}_{\mathcal{C}}(T_1 + T_2) = \Psi^{\mathcal{B}}_{\mathcal{C}}(T_1) + \Psi^{\mathcal{B}}_{\mathcal{C}}(T_2)$. **Q.E.D.**

Proof of Exercise 18.2:

Linearity of D: for $\ell, \ell' \in K^n_{\text{row}}$, $\alpha \in K$ and every $v \in K^n_{\text{col}}$,

$$f_{\alpha\ell+\ell'}(v) = (\alpha\ell + \ell') \cdot v = \alpha(\ell \cdot v) + \ell' \cdot v = \alpha f_\ell(v) + f_{\ell'}(v),$$

by the distributivity of matrix multiplication, so $D(\alpha\ell + \ell') = \alpha D(\ell) + D(\ell')$.

Injectivity: if $D(\ell) = 0$, then $\ell \cdot v = 0$ for every $v \in K^n_{\text{col}}$; taking $v = e_i$ picks out the i-th entry of ℓ, so every entry of ℓ vanishes and $\ell = 0$. Since $\dim K^n_{\text{row}} = n = \dim K^n_{\text{col}} = \dim(K^n_{\text{col}})^*$ by Corollary 18.a.6, an injective linear map between these two spaces is an isomorphism by Corollary 15.b.10. **Q.E.D.**

Proof of Proposition 18.a.10 and of Exercise 18.3:

For the composition rule, both $(S \circ T)^*$ and $T^* \circ S^*$ are maps $U^* \to V^*$, so it suffices to evaluate them at an arbitrary $\ell \in U^*$:

$$(S \circ T)^*(\ell) = \ell \circ (S \circ T) = (\ell \circ S) \circ T = T^*(\ell \circ S) = T^*\big(S^*(\ell)\big) = (T^* \circ S^*)(\ell),$$

using only the associativity of composition. Hence $(S \circ T)^* = T^* \circ S^*$.

(a) For $\ell \in V^*$ we have $(\mathrm{id}_V)^*(\ell) = \ell \circ \mathrm{id}_V = \ell$, so $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

(b) For $\ell \in V^*$ and $v \in V$, $\big(0^*(\ell)\big)(v) = \ell(0(v)) = \ell(0_V) = 0$, so $0^*(\ell)$ is the zero functional for every ℓ, that is, 0^* is the zero map $V^* \to V^*$.

Q.E.D.

Proof of Exercise 18.4:

Both $\tau(v_i)$ and $(v_i^*)^*$ are elements of $(V^*)^*$, that is, functionals on V^*. Since $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ is a basis of V^*, it suffices to check that they agree on each v_j^*. By the definition of τ,

$$\tau(v_i)(v_j^*) = v_j^*(v_i) = \delta_{ji} = \delta_{ij},$$

while by the definition of the basis dual to $\mathcal{B}^*$,

$$(v_i^*)^*(v_j^*) = \delta_{ij}.$$

The two agree on every member of a basis of V^*, hence $\tau(v_i) = (v_i^*)^*$ for all i, that is, $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$. **Q.E.D.**

Direct Sums (Section 18.b)

The External Direct Sum

Proposition 18.b.1 (*Vector Space Structure of the Cartesian Product*): Let V and W be two vector spaces over K. Then, the Cartesian product $V \times W$ becomes a vector space over K, provided we endow it with the following operations:

- $(v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2)$ for all $(v_1, w_1), (v_2, w_2) \in V \times W$.

- $\alpha \cdot (v, w) := (\alpha v, \alpha w)$ for all $\alpha \in K$ and $(v, w) \in V \times W$.

The zero element of this space is given by $0_{V\times W} := (0_V, 0_W)$. This new vector space $V \times W$ is usually denoted by $V \oplus W$ and is called the direct sum of V and W.

Exercise 18.5 (*The Axioms for $V \oplus W$, and the Canonical Maps*): Verify that all vector space axioms are satisfied by $V \times W$ under the componentwise operations above, so that Proposition 18.b.1 holds. Check also that the four canonical maps of the following remark are linear, that p_V and p_W are surjective, and that i_V and i_W are injective.

Remark 18.3 (*Canonical Maps associated with Direct Sums*): The direct sum $V \oplus W$ is naturally equipped with several canonical linear maps:

(a) Canonical Embeddings:
- $i_V : V \to V \oplus W$ defined by $i_V(v) := (v, 0)$.
- $i_W : W \to V \oplus W$ defined by $i_W(w) := (0, w)$.

(b) Canonical Projections:
- $p_V : V \oplus W \to V$ defined by $p_V(v, w) := v$.
- $p_W : V \oplus W \to W$ defined by $p_W(v, w) := w$.

Clearly, all these maps are linear. Furthermore, the projections p_V and p_W are surjective, while the embeddings i_V and i_W are injective. By virtue of these embeddings, we can often identify V and W as subspaces of $V \oplus W$ through the identifications $v \leftrightarrow (v, 0)$ and $w \leftrightarrow (0, w)$.

Internal Direct Sums and Complements

Proposition 18.b.2 (*Isomorphism to a Complementary Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. Suppose $U' \subseteq V$ is another subspace that acts as a complement of U in V, which by Definition 14.4 means that $U + U' = V$ and $U \cap U' = \{0\}$. Then, the map

$$\varphi : U \oplus U' \to V, \quad \varphi(u, u') := u + u'$$

is a linear isomorphism.

Proof:

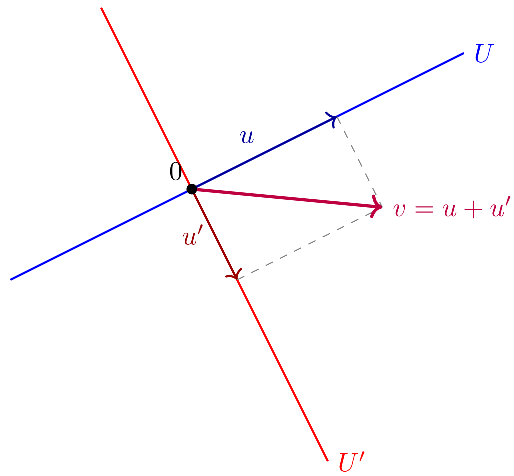
The linearity of φ follows immediately from the definitions of the operations in $U \oplus U'$ and the linearity of addition in V.

To establish injectivity, suppose $\varphi(u, u') = 0$, which implies $u + u' = 0$, and therefore $u = -u'$. Since $u \in U$ and $u' \in U'$, it follows that $u = -u' \in U \cap U'$. However, as U' is a complement of U, we have $U \cap U' = \{0\}$ by definition. Consequently, $u = u' = 0$, which shows that $\ker(\varphi) = \{0_{U\oplus U'}\}$. Thus, φ is injective.

Regarding surjectivity, we observe that since U' is a complement of U in V, we have $U + U' = V$ by definition. Thus, for any $v \in V$, there exist $u \in U$ and $u' \in U'$ such that $u + u' = v$. This implies $\varphi(u, u') = v$, and therefore $\mathrm{Im}(\varphi) = V$. This proves that φ is surjective, and consequently, an isomorphism. Q.E.D.

Remark 18.4: While $U \oplus U'$ is formally a different vector space from V, the existence of the canonical isomorphism φ allows us to treat V as the direct sum of its internal subspaces whenever a complement is specified.

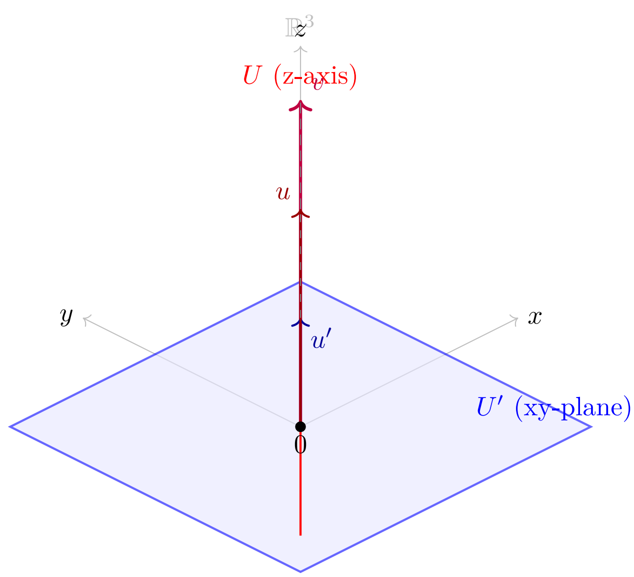
Geometrically, in $\mathbb{R}^2$, this represents the decomposition of any vector v uniquely into a sum of a vector $u \in U$ and a vector $u' \in U'$, constructed via a parallelogram parallel to the complementary subspaces.



Similarly, we can extend this intuition to $\mathbb{R}^3$. Consider U to be the z-axis (a line through the origin) and U' to be the xy-plane (a plane through the origin). Then U and U' are subspaces of $\mathbb{R}^3$.

- $U = \mathrm{Sp}(e_3) = \{(0,0,z) \mid z \in \mathbb{R}\}$
- $U' = \mathrm{Sp}(e_1, e_2) = \{(x,y,0) \mid x, y \in \mathbb{R}\}$

Any vector $v = (x,y,z) \in \mathbb{R}^3$ can be uniquely written as a sum of a vector $u \in U$ and a vector $u' \in U'$: $v = (0,0,z) + (x,y,0)$. Here, $u = (0,0,z) \in U$ and $u' = (x,y,0) \in U'$. Furthermore, $U \cap U' = \{(0,0,0)\}$, which is the zero vector. Thus, $\mathbb{R}^3 = U \oplus U'$. Geometrically, this means any point in 3D space can be reached by first moving along the z-axis and then moving in the xy-plane.



Generalization to Multiple Summands

The concept of a direct sum can be extended to any finite collection of vector spaces $U_1, \ldots, U_m$ over K. We define their direct sum as:

$$U_1 \oplus \cdots \oplus U_m := U_1 \times \cdots \times U_m,$$

endowed with coordinatewise operations.

For an infinite family of vector spaces $\{U_i\}_{i\in I}$, we define the direct sum $\bigoplus_{i\in I} U_i$ as the set of functions $t\colon I \to \bigcup_{i\in I} U_i$ such that $t(i)\in U_i$ for all $i\in I$, and $t(i)\neq 0$ for at most finitely many indices $i\in I$. This ensures that any element of the direct sum can be represented as a finite formal sum of components.

Solutions to the Exercises

Proof of Exercise 18.5:

Every axiom for $V\times W$ is checked coordinatewise and reduces to the same axiom in V and in W. For instance, for $(v_1,w_1),(v_2,w_2),(v_3,w_3)\in V\times W$,

$$\big((v_1,w_1)+(v_2,w_2)\big)+(v_3,w_3)=(v_1+v_2+v_3,\ w_1+w_2+w_3)=(v_1,w_1)+\big((v_2,w_2)+(v_3,w_3)\big),$$

using associativity in V in the first coordinate and in W in the second. The same pattern gives commutativity, the neutral element $(0_V,0_W)$, the additive inverse $-(v,w)=(-v,-w)$, and the four axioms involving scalars, for instance $\alpha\cdot\big((v_1,w_1)+(v_2,w_2)\big)=(\alpha v_1+\alpha v_2,\alpha w_1+\alpha w_2)=\alpha(v_1,w_1)+\alpha(v_2,w_2)$.

The canonical maps are linear for the same reason: $i_V(\alpha v+v')=(\alpha v+v',0)=\alpha(v,0)+(v',0)=\alpha\, i_V(v)+i_V(v')$, and

$$p_V\big(\alpha(v,w)+(v',w')\big)=p_V(\alpha v+v',\ \alpha w+w')=\alpha v+v'=\alpha\, p_V(v,w)+p_V(v',w'),$$

and symmetrically for i_W and p_W.

The projections are surjective, since $p_V(v,0)=v$ for every $v\in V$ and $p_W(0,w)=w$ for every $w\in W$. The embeddings are injective, since $i_V(v)=(v,0)=(0,0)$ forces $v=0$, so $\ker(i_V)=\{0\}$, and likewise for i_W. **Q.E.D.**

Quotient Spaces (Section 18.c)

Equivalence Relation and the Quotient Space

Let V be a vector space over K, and let $U\subseteq V$ be a subspace. We will define a new vector space over K, denoted by V/U, together with a linear map $\pi\colon V\to V/U$ such that π is surjective and $\ker(\pi)=U$. The space V/U will have some other good properties. The space V/U will be called the "*quotient*" of V by U, or V modulo U, and the map π the "*quotient map*", or the "*projection onto the quotient space*".

We begin by defining a relation on V. For $v_1,v_2\in V$, we define $v_1\sim v_2$ if and only if $v_1-v_2\in U$.

Lemma (*Equivalence Relation on V*): The relation $\sim$ defines an equivalence relation on V.

Proof:

Reflexivity: For all $v\in V$, $v\sim v$ because $v-v=0\in U$.

Symmetry: If $v_1\sim v_2$, then $v_1-v_2\in U$. Since U is a subspace, $-(v_1-v_2)=v_2-v_1\in U$, which implies $v_2\sim v_1$.

Transitivity: If $v_1\sim v_2$ and $v_2\sim v_3$, then $v_1-v_2\in U$ and $v_2-v_3\in U$. Because U is a subspace, their sum is in U:

$$(v_1-v_2)+(v_2-v_3)=v_1-v_3\in U,$$

which means $v_1\sim v_3$. **Q.E.D.**

We denote the equivalence class of $v\in V$ by $[v]$. Thus,

$$[v]=v+U=\{v+u\mid u\in U\}.$$

Definition 18.c.1 (*The Quotient Space*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. We define $V/U := V/\sim$ to be the set of equivalence classes of the equivalence relation $\sim$, which is one by the Lemma on Page 155. So, the elements of V/U are $[v]$, where $v \in V$.
Alternatively, each element in V/U is a set of the form $v + U$, where $v \in V$. The space V/U is called the quotient of V by U (or modulo U).

Geometrically, the nature of the quotient space $\mathbb{R}^2/U$ depends on the dimension of U:

- If $U = \{0\}$ is the trivial subspace, then each equivalence class $[v] = v + \{0\} = \{v\}$ is simply a single **point**. In this case, no points are *"collapsed"* together, and therefore $\mathbb{R}^2/\{0\} \cong \mathbb{R}^2$.
- If U is a **line** through the origin, the equivalence classes $[v]$ forming the space $\mathbb{R}^2/U$ are the lines parallel to U, as seen in Figure 18.1.

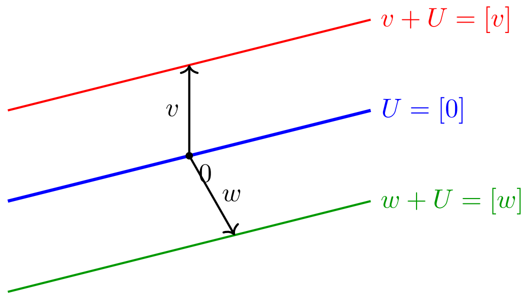


Figure 18.1: Geometric interpretation of the quotient space $\mathbb{R}^2/U$.

Similarly, we can extend this intuition to $\mathbb{R}^3$, as illustrated in Figure 18.2:

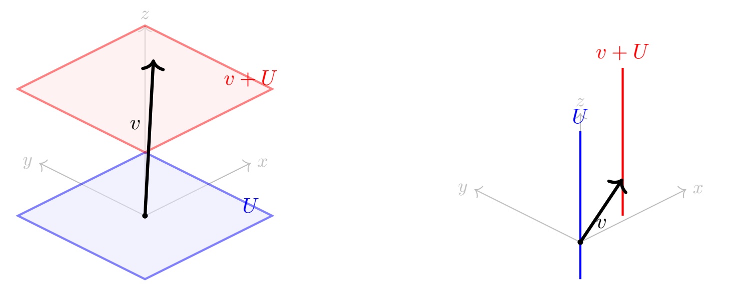
- If $U = \{0\}$ is the trivial subspace, then each equivalence class $[v] = \{v\}$ is simply a **point**. Since no two distinct vectors are identified, the quotient space $\mathbb{R}^3/\{0\}$ is isomorphic to $\mathbb{R}^3$ itself.
- If U is a **plane** through the origin, then $\mathbb{R}^3/U$ is 1-dimensional (by Proposition 18.c.4) and therefore isomorphic to a line. Each element $[v]$ is a plane parallel to U. The quotient space can be viewed as the set of all such parallel planes *"stacked"* along a transversal direction (see Figure 18.2a and Proposition 18.c.7).
- If U is a **line** through the origin, then $\mathbb{R}^3/U$ is 2-dimensional (by Proposition 18.c.4) and therefore isomorphic to a plane. Each element $[v]$ is a line parallel to U (see Figure 18.2b and Proposition 18.c.7).
- If $U = \mathbb{R}^3$ is the entire space, then there is only one equivalence class, $[v] = \mathbb{R}^3$. The quotient space $\mathbb{R}^3/\mathbb{R}^3$ is the zero vector space, with dimension 0.

Remark 18.5: The crucial insight is that an equivalence class $[v] = v + U = \{v + u \mid u \in U\}$ is not merely a single vector, but a translate of the subspace U. Consequently, the "shape" or dimension of $[v]$ is identical to the dimension of U. Specifically:

- If U is a line, then every $[v]$ is a line parallel to U.
- If U is a plane, then every $[v]$ is a plane parallel to U.

The dimension of the quotient space V/U represents the number of independent directions available to shift these objects. For instance, in $\mathbb{R}^3$, if U is a line (1-dimensional), there are 2 independent directions in which one can move this line to obtain a distinct parallel line (effectively moving the line across a 2-dimensional plane).

Thus, the space of all such lines is 2-dimensional. This geometric intuition aligns perfectly with the dimension formula established in Proposition 18.c.4. In the case where $U = \{0\}$, the equivalence class is a point (0-dimensional), and we retain all 3 degrees of freedom of the ambient space.



(a) Plane case ($1D$-quotient) (b) Line case ($2D$-quotient)

Figure 18.2: Geometric interpretation of the quotient space $\mathbb{R}^3/U$.

The Vector Space Structure of V/U

We want to turn V/U into a vector space over K. We define addition as $[v_1] + [v_2] := [v_1 + v_2]$ for all $[v_1], [v_2] \in V/U$. Alternatively, this can be written as $(v_1 + U) + (v_2 + U) := (v_1 + v_2) + U$. For scalar multiplication, let $\alpha \in K$ and $[v] \in V/U$. We define $\alpha \cdot [v] := [\alpha v]$. The zero element is defined as $0_{V/U} := [0_V] = U$.

Proposition 18.c.2 (*Well-definedness of Operations in V/U*): The operations defined above are well-defined. Moreover, they give V/U the structure of a vector space over K.

Proof:

To establish well-definedness, we must show that the operations do not depend on the choice of representatives. For scalar multiplication, suppose $[v] = [v']$, and let $\alpha \in K$. We need to show that $[\alpha v] = [\alpha v']$. Indeed, since $[v] = [v']$, we have $v - v' \in U$. Because U is a subspace, $\alpha(v - v') = \alpha v - \alpha v' \in U$. Consequently, $[\alpha v] = [\alpha v']$.

Similarly, for addition, suppose $[v_1] = [v_1']$ and $[v_2] = [v_2']$. We need to show that $[v_1+v_2] = [v_1'+v_2']$. By assumption, $v_1 - v_1' \in U$ and $v_2 - v_2' \in U$. Adding these yields:

$$(v_1 + v_2) - (v_1' + v_2') = (v_1 - v_1') + (v_2 - v_2') \in U.$$

Therefore, $v_1 + v_2 \sim v_1' + v_2'$, which implies $[v_1 + v_2] = [v_1' + v_2']$. This proves that the operations are well-defined.

It remains to show that they satisfy the axioms of a vector space. These follow directly from the corresponding axioms in V; the notes leave that verification to the reader, and it is carried out among the solutions at the end of this section. **Q.E.D.**

Exercise 18.6 (*The Axioms for V/U*): Complete the proof of Proposition 18.c.2 by checking that the operations on V/U satisfy the axioms of a vector space.

We now define a map $\pi : V \to V/U$ by $\pi(v) := [v] \in V/U$ for all $v \in V$.

Proposition 18.c.3 (*The Quotient Map*): The map $\pi : V \to V/U$ is a linear map. Moreover, it is surjective (i.e., $\mathrm{Im}(\pi) = V/U$), and $\ker(\pi) = U$.

Proof:

Linearity: For $v_1, v_2 \in V$, $\pi(v_1 + v_2) = [v_1 + v_2] = [v_1] + [v_2] = \pi(v_1) + \pi(v_2)$. For $\alpha \in K$ and $v \in V$, $\pi(\alpha v) = [\alpha v] = \alpha[v] = \alpha\pi(v)$.

Surjectivity: Let $x \in V/U$. By definition, x is an equivalence class $[v]$ of some element $v \in V$. Thus, $x = \pi(v)$.

Kernel: We claim that $\ker(\pi) = U$. First, let $v \in \ker(\pi)$, meaning $\pi(v) = 0$. Then, $[v] = [0]$, which implies $v = v - 0 \in U$. Hence, $\ker(\pi) \subseteq U$. Conversely, let $u \in U$. Then, $u \sim 0$ because $u - 0 \in U$. Therefore, $\pi(u) = [u] = [0] = 0$, so $u \in \ker(\pi)$. This shows that $U \subseteq \ker(\pi)$, completing the proof. **Q.E.D.**

Proposition 18.c.4 (*Dimension of the Quotient Space*): Suppose that V is a finite-dimensional vector space. Then, V/U is also finite-dimensional, and $\dim V/U = \dim V - \dim U$.

Proof:

We have seen that $\pi\colon V \to V/U$ is a surjective linear map, this being Proposition 18.c.3. This guarantees that V/U is finite-dimensional. (Generally, if $T\colon W' \to W''$ is a surjective linear map and W' is finite-dimensional, then W'' is also finite-dimensional, because the images of a basis of W' span W'').

We now apply the Rank-Nullity Theorem, Theorem 15.b.9, to π:

$$\dim V = \dim \ker(\pi) + \dim \mathrm{Im}(\pi).$$

Since $\ker(\pi) = U$ and $\mathrm{Im}(\pi) = V/U$, again by Proposition 18.c.3, we obtain $\dim V/U = \dim V - \dim U$. **Q.E.D.**

The Isomorphism Theorem and Universal Property

Theorem 18.c.5 (*The First Isomorphism Theorem*): Let V and W be vector spaces over K, and let $T\colon V \to W$ be a linear map. We define a new map $\overline{T}\colon V/\ker(T) \to \mathrm{Im}(T)$ as follows: For $x \in V/\ker(T)$, choose a representative $v \in V$ such that $x = [v]$. Define $\overline{T}(x) := T(v)$. Then:

(a) $\overline{T}$ is well-defined and is a linear map.

(b) $\overline{T}\colon V/\ker(T) \to \mathrm{Im}(T)$ is an isomorphism.

(c) The following diagram commutes: $T = i \circ \overline{T} \circ \pi$, where $i\colon \mathrm{Im}(T) \to W$ is the inclusion map.

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "\pi"'] & W \\
V/\ker(T) \arrow[r, "\overline{T}\ (\cong)"] & \mathrm{Im}(T) \arrow[u, "i"']
\end{tikzcd}$$

Proof:

(a) To show that $\overline{T}$ is well-defined, suppose $x = [v] = [v']$. We must show that $T(v) = T(v')$. Indeed, $[v] = [v']$ implies that $v - v' \in \ker(T)$, meaning $T(v - v') = 0$. By linearity, $T(v) = T(v')$.

The linearity of $\overline{T}$ is straightforward: $\overline{T}(\alpha[v]) = \overline{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha \overline{T}([v])$. Also, $\overline{T}([v_1] + [v_2]) = \overline{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \overline{T}([v_1]) + \overline{T}([v_2])$.

(b) To show that $\overline{T}$ is injective, let $x \in \ker(\overline{T})$, and write $x = [v]$ for some $v \in V$. Then, $\overline{T}(x) = T(v) = 0$, meaning $v \in \ker(T)$. This implies that $x = [v] = 0_{V/\ker(T)}$.

To show that $\overline{T}$ is surjective, let $w \in \mathrm{Im}(T)$. Then, there exists $v \in V$ such that $T(v) = w$. Consequently, $\overline{T}([v]) = T(v) = w$, proving surjectivity.

(c) The commutativity of the diagram follows immediately from the definitions; the notes leave this to the reader, and it is checked among the solutions at the end of this section.

Q.E.D.

Exercise 18.7 (*Commutativity of the Diagram in Theorem 18.c.5*): Check that $T = i \circ \overline{T} \circ \pi$, where $i\colon \operatorname{Im}(T) \to W$ is the inclusion.

Theorem 18.c.6 (*Universal Property of Quotient Spaces*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. The quotient space V/U has the following universal property: For every vector space W and every linear map $T\colon V \to W$ for which $T(U) = 0$ (i.e., for which $\ker(T) \supseteq U$), there exists a unique linear map $T'\colon V/U \to W$ such that $T = T' \circ \pi$. Moreover, $\ker(T') = \ker(T)/U$.

Remark 18.6: In algebraic jargon, every linear map $T\colon V \to W$ that maps U to 0 factors through V/U via $T = T' \circ \pi$, and moreover, in a unique way. We say that T induces the map T', or that T descends to a map $T'\colon V/U \to W$.

Proof:

We define $T'\colon V/U \to W$ as follows: let $x \in V/U$. Choose $v \in V$ such that $x = [v]$. Define $T'(x) := T(v)$.

Exercise 18.8 (*Well-definedness and Linearity of the Induced Map*):

(a) Show that T' is well-defined (i.e., if $[v] = [v']$, then $T(v) = T(v')$).

(b) Show that T' is linear.

The diagram commutes by construction, since for all $v \in V$, $T' \circ \pi(v) = T'([v]) = T(v)$.

For uniqueness, suppose $T = T'' \circ \pi$ is another factorization. We will show that $T'' = T'$. Let $x \in V/U$. Since π is surjective, there exists $v \in V$ such that $x = \pi(v)$. We then have $T(v) = T'(\pi(v))$ and $T(v) = T''(\pi(v))$, which immediately yields $T'(x) = T''(x)$. Q.E.D.

Exercise 18.9 (*Kernel of the Induced Quotient Map*): Prove that $\ker(T') = \ker(T)/U$.

Quotients, Complements, and Dual Spaces

Let V be a vector space over K, and let $U \subseteq V$ be a subspace. Let $W \subseteq V$ be a complement of U in the sense of Definition 14.4 (i.e., W is a subspace such that $U \cap W = \{0\}$ and $U + W = V$).

Proposition 18.c.7 (*Canonical Isomorphism from a Complement*): Under the assumptions above, there is a canonical isomorphism $Q\colon W \to V/U$, defined by $Q(w) := [w] \in V/U$ for all $w \in W$.

Remark 18.7: The isomorphism Q is canonical only once the complement W is specified. In general, there are many choices of complements to U, and there is no preferred choice.

Proof:

Consider the inclusion map $i\colon W \to V$ and the projection map $\pi\colon V \to V/U$. Since $Q = \pi \circ i$, Q is a linear map.

We claim that $\ker(Q) = \{0\}$. Indeed, if $w \in \ker(Q)$, then $[w] = 0$, implying that $w \in U$. However, since $w \in W$, we must have $w \in U \cap W = \{0\}$, and therefore $w = 0$. This shows that $\ker(Q) = \{0\}$.

We also claim that Q is surjective. Let $x \in V/U$ and choose $v \in V$ such that $x = [v]$. Since $V = U + W$, there exist $u \in U$ and $w \in W$ such that $v = w + u$. It follows that $[v] = [w+u] = [w]$, so $Q(w) = x$. This proves that Q is surjective, and consequently, an isomorphism. Q.E.D.

Exercise 18.10 (*Inverse Isomorphism from a Complement*): Describe the inverse of Q, $Q^{-1}\colon V/U \xrightarrow{\cong} W$.

Quotients, Complements, and Annihilators

Proposition 18.c.8 (*Annihilators and Dual Quotients*): Let $U \subseteq V$ be a subspace, and define the "*annihilator*" $U^{\perp} := \{\ell \in V^* \mid \ell|_U = 0\}$. Then:

(a) $U^{\perp} \subseteq V^*$ is a linear subspace.
(b) There is a canonical isomorphism $(V/U)^* \cong U^{\perp}$.
(c) If V is finite-dimensional, then $U^* \cong V^*/U^{\perp}$.

Proof:

(a) Clearly, the zero functional vanishes on U. For $\ell_1, \ell_2 \in U^{\perp}$ and $\alpha, \beta \in K$, we have $(\alpha\ell_1 + \beta\ell_2)(u) = \alpha\ell_1(u) + \beta\ell_2(u) = 0$ for all $u \in U$, establishing that $U^{\perp}$ is a subspace.

(b) We define a map $L\colon (V/U)^* \to U^{\perp}$ as follows: Let $S \in (V/U)^*$, i.e., $S\colon V/U \to K$. Define $L(S) \in V^*$ to be $L(S) := S \circ \pi$.

Exercise 18.11 (*The Map L Lands in $U^{\perp}$ and is Linear*):
(a) Show that for all $S \in (V/U)^*$, $S \circ \pi \in U^{\perp}$.
(b) Show that L is a linear map.

We claim that L is an isomorphism. To see this, we define a map $P\colon U^{\perp} \to (V/U)^*$ as follows. Let $T \in U^{\perp}$, i.e., $T\colon V \to K$ such that $T|_U = 0$. By the Universal Property of Quotients, Theorem 18.c.6 applied with $W = K$, there exists a unique $T'\colon V/U \to K$ such that $T = T' \circ \pi$. Define $P(T) := T'$.
We claim that $L \circ P = \mathrm{id}_{U^{\perp}}$ and $P \circ L = \mathrm{id}_{(V/U)^*}$. Indeed, for all $S\colon V/U \to K$, we have $P \circ L(S) = P(S \circ \pi) = S$, because of the uniqueness statement in Theorem 18.c.6. Also, for all $T \in U^{\perp}$, $L \circ P(T) = L(T') = T' \circ \pi = T$. This proves that L is bijective, and since we know L is linear, it is an isomorphism. It follows that P is also a linear isomorphism, being the inverse map of the bijective linear map L (Lemma 15.b.4).

(c) Assume V is finite-dimensional. Consider the restriction map $R\colon V^* \to U^*$ defined by $R(\varphi) := \varphi|_U$ for all $\varphi \in V^*$.

Exercise 18.12 (*Linearity of the Restriction Map*): Show that R is a linear map.

We claim that R is surjective. Indeed, let $\psi \in U^*$. We need to show that there exists $\varphi \in V^*$ such that $\varphi|_U = \psi$. The finite-dimensionality of V is important here. Pick a basis $u_1, \ldots, u_k$ for U, and extend it to a basis $u_1, \ldots, u_k, v_{k+1}, \ldots, v_n$ of V, which is possible by the Steinitz Exchange Lemma in the form of Lemma 11.b.1. Define a linear map $\varphi\colon V \to K$ by $\varphi(u_i) := \psi(u_i)$ for all $1 \le i \le k$, and $\varphi(v_j) := 0$ for all $k < j \le n$; prescribing the values on a basis in this way does define one and only one linear map, by Theorem 15.a.3. Clearly, $\varphi|_U = \psi$ because φ and ψ coincide on the elements of a basis of U, and two linear maps on U agreeing on a basis of U are equal (the uniqueness half of the same theorem). This proves $\mathrm{Im}(R) = U^*$.
We claim that $\ker(R) = U^{\perp}$. Indeed, this follows directly from definitions: $\ker(R) = \{\varphi \in V^* \mid \varphi|_U = 0\} = U^{\perp}$. By the First Isomorphism Theorem, Theorem 18.c.5 applied to R, we get an induced isomorphism $\overline{R}\colon V^*/\ker(R) \to \mathrm{Im}(R)$, which is to say $\overline{R}\colon V^*/U^{\perp} \xrightarrow{\cong} U^*$.

Q.E.D.

Exercise 18.13 (*Relationship between Annihilator of Image and Kernel of Dual Map*): Let $T\colon V \to W$ be a linear map between vector spaces. Prove that the annihilator of the image of T is equal to the kernel of its dual map:

$$(\operatorname{Im} T)^{\perp} = \ker(T^{*}).$$

Solutions to the Exercises

Proof of Exercise 18.6:

Every axiom is inherited from V by passing to classes. For instance, associativity: for $v_1, v_2, v_3 \in V$,

$$\big([v_1]+[v_2]\big)+[v_3] = [v_1+v_2]+[v_3] = [(v_1+v_2)+v_3] = [v_1+(v_2+v_3)] = [v_1]+\big([v_2]+[v_3]\big),$$

where the middle step is associativity in V. Commutativity is identical. The class $[0_V] = U$ is neutral, since $[v]+[0_V] = [v+0_V] = [v]$, and $[-v]$ is an additive inverse of $[v]$, since $[v]+[-v] = [0_V]$. For scalars, $\alpha([v_1]+[v_2]) = [\alpha(v_1+v_2)] = [\alpha v_1 + \alpha v_2] = \alpha[v_1]+\alpha[v_2]$, and similarly $(\alpha+\beta)[v] = \alpha[v]+\beta[v]$, $(\alpha\beta)[v] = \alpha(\beta[v])$ and $1\cdot[v] = [v]$. Each of these is the corresponding axiom in V with brackets placed around it, which is legitimate precisely because the operations were shown to be well defined. **Q.E.D.**

Proof of Exercise 18.7:

Both T and $i \circ \overline{T} \circ \pi$ are maps $V \to W$, so we evaluate at an arbitrary $v \in V$:

$$\big(i \circ \overline{T} \circ \pi\big)(v) = i\big(\overline{T}([v])\big) = i\big(T(v)\big) = T(v),$$

using the definition of π, then the definition of $\overline{T}$ on the class $[v]$ with representative v, and finally that i is the inclusion of $\operatorname{Im}(T)$ into W. **Q.E.D.**

Proof of Exercise 18.8:

(a) Suppose $[v] = [v']$. Then $v - v' \in U \subseteq \ker(T)$ by the hypothesis $T(U) = 0$, so $T(v-v') = 0$ and hence $T(v) = T(v')$. So the value $T'([v]) := T(v)$ does not depend on the representative chosen.

(b) For $\alpha \in K$ and $[v],[v'] \in V/U$,

$$T'\big(\alpha[v]+[v']\big) = T'\big([\alpha v + v']\big) = T(\alpha v + v') = \alpha T(v) + T(v') = \alpha\, T'([v]) + T'([v']),$$

using the definition of the operations on V/U, then the definition of T', then the linearity of T.

Q.E.D.

Proof of Exercise 18.9:

First note that $\ker(T) \supseteq U$, so the quotient $\ker(T)/U$ makes sense and is a subspace of V/U: its elements are the classes $[v]$ with $v \in \ker(T)$.

Let $x \in V/U$ and write $x = [v]$. Then

$$x \in \ker(T') \iff T'([v]) = 0 \iff T(v) = 0 \iff v \in \ker(T) \iff x \in \ker(T)/U,$$

where the second step is the definition of T'. The middle equivalence is legitimate for either representative of x, since T' is well defined. Hence $\ker(T') = \ker(T)/U$. **Q.E.D.**

Proof of Exercise 18.10:

Let $x \in V/U$. Since $V = U + W$, any representative v of x can be written as $v = u + w$ with $u \in U$ and $w \in W$, and then $x = [v] = [w]$. The map

$$Q^{-1}\colon V/U \to W, \qquad Q^{-1}(x) := w, \quad \text{where } x = [u+w] \text{ with } u \in U,\, w \in W,$$

is the inverse of Q. It is well defined: if $u+w=u'+w'$ with $u,u'\in U$ and $w,w'\in W$, then $w-w'=u'-u\in U\cap W=\{0\}$, so $w=w'$. And indeed $Q^{-1}(Q(w))=Q^{-1}([w])=w$ for $w\in W$, while $Q(Q^{-1}([v]))=Q(w)=[w]=[v]$.

Equivalently, and more compactly: $Q^{-1}=p_W\circ(\text{choice of representative})$, where p_W is the projection of $V=U\oplus W$ onto W furnished by Proposition 18.b.2; the content of the verification above is exactly that this projection is well defined on classes. **Q.E.D.**

Proof of Exercise 18.11:

(a) Let $S\in(V/U)^*$ and $u\in U$. Then $\pi(u)=[u]=0_{V/U}$, because $u\in U=\ker(\pi)$, and therefore

$$(S\circ\pi)(u)=S(0_{V/U})=0.$$

So $S\circ\pi$ vanishes on U, that is, $S\circ\pi\in U^\perp$.

(b) For $S,S'\in(V/U)^*$ and $\alpha\in K$, and every $v\in V$,

$$L(\alpha S+S')(v)=\big((\alpha S+S')\circ\pi\big)(v)=(\alpha S+S')\big(\pi(v)\big)=\alpha\,S(\pi(v))+S'(\pi(v))=\big(\alpha L(S)+L(S')\big)(v),$$

so $L(\alpha S+S')=\alpha L(S)+L(S')$.

Q.E.D.

Proof of Exercise 18.12:

For $\varphi,\varphi'\in V^*$, $\alpha\in K$ and every $u\in U$,

$$R(\alpha\varphi+\varphi')(u)=(\alpha\varphi+\varphi')|_U(u)=\alpha\varphi(u)+\varphi'(u)=\big(\alpha R(\varphi)+R(\varphi')\big)(u),$$

since the operations on V^* are defined pointwise. Two functionals on U agreeing at every $u\in U$ are equal, so $R(\alpha\varphi+\varphi')=\alpha R(\varphi)+R(\varphi')$. **Q.E.D.**

Proof of Exercise 18.13:

Both sides are subsets of W^*. For $\ell\in W^*$, the notes' chain of equivalences reads

$$\ell\in(\operatorname{Im}T)^\perp\iff\ell|_{\operatorname{Im}(T)}=0\iff\ell\circ T=0\iff T^*(\ell)=0\iff\ell\in\ker(T^*).$$

The middle equivalence is the only one with content: $\ell\circ T$ is the zero map on V exactly when $\ell(T(v))=0$ for every $v\in V$, and the values $T(v)$ are precisely the elements of $\operatorname{Im}(T)$, so this says exactly that ℓ vanishes on $\operatorname{Im}(T)$. The remaining steps are the definitions of $U^\perp$, of T^*, and of the kernel. **Q.E.D.**

AI-Note: The three lectures behind this chapter, 18.a on Hom, duals and biduals, 18.b on direct sums, 18.c on quotients, are all present in the transcript, statement for statement, and the printed numbers 18.a.1–18.a.12, 18.b.1–18.b.2 and 18.c.1–18.c.8 already agreed with the handwritten ones. What the audit found here was damage of a different kind: repetitions, one broken inference, and a great many exercises left unanswered.

Repetitions and misplacements. The paragraph introducing the bidual appeared twice, once inside a `summary` environment that had swallowed a `\subsection` heading along with it, and once again under *"Reflexivity"*; the first copy is removed and the summary now ends where the notes end it. Corollary 18.a.4 carried two proofs in a row, a sketch and a full verification; the sketch was contained in the verification and is dropped. A `definition*` naming the quotient and the quotient map stood before either had been constructed, duplicating the naming that comes with Definition 18.c.1; it is folded into the sentence that announces the construction. Four cross-references read *"Figure Figure 18.1"*, which typesets the word *"Figure"* twice.

One broken inference. Of the map $D\colon K^n_{\mathrm{row}} \to (K^n_{\mathrm{col}})^*$ the transcript said it *"is a linear map, and therefore, it is an isomorphism"*, linearity does not give that, and the notes do not claim it does: they say the map is an iso and mark the verification *"exc. Prove this!"*. It is now Exercise 18.2, solved on Page 151. Also repaired: the proof of Proposition 18.a.8 set $A := [\mathrm{id}_V]$ with no bases on it, and the matrix it means is $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}}$.

Rings. At the point where the notes say only *"explain briefly the structure of a ring"* and *"explain what is a homomorphism of rings"*, both given orally, the book had nothing, although it uses the word from Page 122 onwards and states Corollary 18.a.4 as an isomorphism of rings. AI-Definition 18.1 and AI-Definition 18.2 fill that gap; they are not Prof. Biran's text and are marked as AI-Definitions.

Exercises. The notes mark fourteen things *"Exc."* across the three lectures, from the vector space axioms for $\mathrm{Hom}(V, W)$ to $(\mathrm{Im}\, T)^{\perp} = \ker(T^*)$. Several stood in the transcript as `\begin{proof}` blocks reading *"Exercise"*, which leaves a stated result with no proof anywhere. All of them are now exercises, and each of the three sections ends with the solutions.

A later pass looking only for gaps in rigour found nothing false in this chapter, but five numbered results of Prof. Biran's carrying no `\label` at all, so that nothing could cite them: 18.a.3, 18.a.5, 18.a.11, 18.c.6 and 18.c.8. Two of them were being cited already, in words rather than by reference: the proof of Corollary 18.a.6 runs $\dim V^* = \dim \mathrm{Hom}(V, K) = \dim V \cdot \dim K$ on Corollary 18.a.3 without saying so, and the proof of Proposition 18.c.8 appeals twice to *"the theorem about the universal property of quotient spaces"*, which is Theorem 18.c.6 directly above it. All five are labelled now, with the handwritten numbers recorded above them as everywhere else, and the printed 18.a.1–18.a.12, 18.b.1–18.b.2 and 18.c.1–18.c.8 are unchanged.

A proof that stops one line early. The proof of Lemma 18.a.11 computes the coordinates of $S^*(w_j^*)$ in $\mathcal{B}^*$, observes that they form column j of A^{T}, and ends there, without ever saying that this identifies $[S^*]^{\mathcal{C}^*}_{\mathcal{B}^*}$ with A^{T}, which is what was to be shown. The concluding sentence is supplied.

Two things stated in running prose. The same pattern as in the chapter on linear maps and bases. The functionals v_i^*, on which Lemma 18.a.7 standing directly underneath is a statement, were defined in a paragraph of prose; they have an unnumbered Definition now, on Page 148. And the naturality of τ was asserted as a commutative square with no statement attached to it and no proof: it is the Proposition on Page 151, with the four-line verification that unwinds the definitions. Both are unnumbered, so nothing moved.

Silent steps. Some fifteen now name the result they rest on, across all three sections. The ones with real content: that isomorphic spaces have equal dimension, which is what carries Corollary 18.a.3; that two linear maps agreeing on a basis are equal and that prescribing values on a basis determines a linear map, which are the two halves of Theorem 15.a.3 and are used three times between the dual basis and the surjectivity of the restriction map; that n spanning vectors in an n-dimensional space are a basis; that an injective map between spaces of equal finite dimension is already an isomorphism, which finishes Theorem 18.a.12; the column description of a representation matrix, twice; and the basis extension behind Proposition 18.c.8 **(c)**. Corollary 18.a.4 appealed to *"a fundamental result from Linear Algebra I"* for the matrix of a composition; that result is the composition rule earlier in this book, and is now pointed at.

Two smaller things. Proposition 18.b.2 said that U' being a complement *"implies"* $U + U' = V$ and $U \cap U' = \{0\}$, where those two conditions are the definition of a complement, Definition 14.4, not a consequence of it. And one `\cpageref` was introduced by *"in"* rather than *"on"*, which typesets as *"in page 136"*; the same slip survives in the chapters on bases and on miscellaneous topics, which this pass has not yet reached.

Miscellaneous Topics and Motivation (Chapter 19)

Geometric Motivation: Why Abstraction Matters (Section 19.a)

It is a common temptation to simply identify every n-dimensional vector space over K with the standard coordinate space K^n. However, while they are algebraically isomorphic for any fixed parameter, they may not be *"continuously"* isomorphic when the vector spaces depend dynamically on parameters.

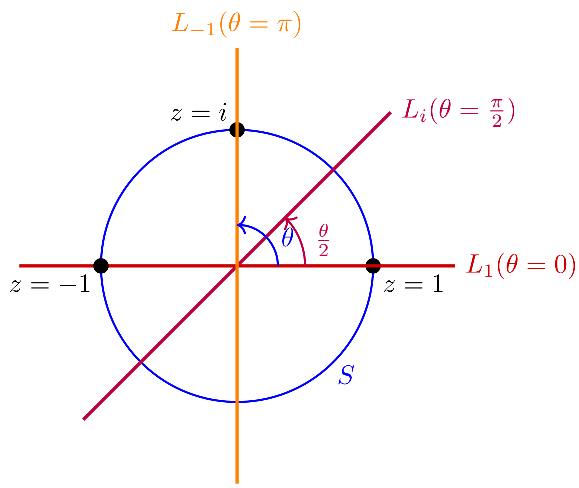
The Möbius Vector Bundle

Definition 19.1 (*A Continuous Family of Subspaces*): Let $S \subseteq \mathbb{C}$ be the unit circle defined by $S = \{e^{i\theta} \mid \theta \in [0, 2\pi]\}$. For each $z = e^{i\theta} \in S$, we define a 1-dimensional subspace $L_z \subseteq \mathbb{R}^2$ as follows:

$$L_z := \mathrm{Sp}\left(\left(\cos\tfrac{\theta}{2}, \sin\tfrac{\theta}{2}\right)\right).$$

The inner parentheses are those of the vector: L_z is the span of the **single** vector $(\cos\frac{\theta}{2}, \sin\frac{\theta}{2}) \in \mathbb{R}^2$, not of the two scalars.

Notice a topological anomaly: $z = 1$ corresponds to both $\theta = 0$ and $\theta = 2\pi$. In the first case ($\theta = 0$), the spanning vector is $(1,0)$, so $L_1 = \mathrm{Sp}\left((1,0)\right)$. In the second case ($\theta = 2\pi$), the spanning vector is $(-1,0)$, which gives the exact same subspace $L_1 = \mathrm{Sp}\left((-1,0)\right)$. The line L_z rotates at exactly half the speed of the point z.



Proposition 19.2 (*Non-triviality of the Möbius Bundle*): There does not exist a family of linear isomorphisms $T_z \colon L_z \to \mathbb{R}$ that depends continuously on the parameter $z \in S$.

Proof Sketch:

Consider the disjoint union of all these lines parameterized by z, which forms a space $M = \{(z, v) \mid z \in S, v \in L_z\}$. Topologically, this forms a vector bundle over S^1 known as the Möbius strip.

If such a continuous family of isomorphisms T_z existed, it would allow us to construct a global homeomorphism $\Phi \colon M \to S \times \mathbb{R}$. Removing the zero-section (the center circle where $v = 0$) from both spaces would imply that $M \setminus \{0\} \cong S \times (\mathbb{R} \setminus \{0\})$.

However, $S \times (\mathbb{R} \setminus \{0\})$ consists of two disconnected cylinders, an *"upper"* and a *"lower"* half. In contrast, if you cut a Möbius strip down its centre line, it remains a single connected piece. This topological contradiction demonstrates that we cannot consistently and continuously identify all L_z with $\mathbb{R}$. Abstraction is therefore necessary to handle *"twisted"* families of spaces. This belongs to a branch of mathematics called topology. The conclusion for us is the one the notes draw: each L_z is a one-dimensional vector space over $\mathbb{R}$, so $L_z \cong \mathbb{R}$ for every single z, and yet the whole family cannot be identified with $\mathbb{R}$ in a continuous way. **Q.E.D.**

Affine Geometry (Section 19.b)

In our strict definition of a vector space, there is always a distinguished origin 0. Affine geometry studies spaces that *"look like"* vector spaces locally, but have forgotten their origin.

Definition 19.3 (*Affine Space*): A subset $X \subseteq V$ is called an "*affine space*" if there exists a linear subspace $U \subseteq V$ and a translation vector $x_0 \in V$ such that $X = x_0 + U$.

We say that X is modeled on the subspace U, and we write the "*direction space*" as $\vec{X} := U$. The elements of X are called "*points*", while the elements of $\vec{X}$ are called "*vectors*". The "*dimension*" of X is defined as $\dim X := \dim U$.

Remark 19.1: The point x_0 in Definition 19.3 is not uniquely determined by X, unless $\dim X = 0$: any point of X will serve. The subspace U, on the other hand, **is** uniquely determined by X, which is what makes the notation $\vec{X}$ legitimate. This is left as an exercise.

Exercise 19.1 (*The Direction Space is Determined by the Affine Space*): Let $X \subseteq V$ be an affine space. Show that the subspace U of Definition 19.3 is uniquely determined by X, and that any point of X may be taken as the base point x_0. Deduce that x_0 is unique precisely when $\dim X = 0$.

Proposition 19.4 (*Algebra of Affine Spaces*): Let X be an affine space modeled on $\vec{X}$. Then:

- **(a)** We cannot inherently add two points $p, q \in X$, as $p + q \notin X$ in general.
- **(b)** We can translate a point $p \in X$ by a vector $v \in \vec{X}$ to obtain a new point $p + v \in X$.
- **(c)** For any two points $p, q \in X$, their difference defines a unique vector $\vec{pq} := q - p \in \vec{X}$.

Proof:

For **(b)**, write $p = x_0 + u$ with $u \in \vec{X}$. Then $p + v = x_0 + (u + v) \in x_0 + \vec{X} = X$, since $u + v \in \vec{X}$. For **(c)**, write $p = x_0 + u$ and $q = x_0 + v$ with $u, v \in \vec{X}$; then $q - p = v - u \in \vec{X}$. Statement **(a)** is the observation that on X there is no longer a zero: an affine space has forgotten its origin.

Q.E.D.

Note also that the two operations fit together as one expects: $p + \vec{pq} = q$ for all $p, q \in X$.

Example 19.1 (*Affine Spaces*):

- **(a)** An "*affine point*" is an affine space $X = \{x\} \subseteq V$ with $\vec{X} = \{0\}$, so $\dim X = 0$.
- **(b)** An "*affine line*": let $x, y \in V$ be distinct and put $v := \vec{xy}$. Then
$$\ell_{xy} := \{x + tv \mid t \in K\}$$
is an affine line, modeled on the one-dimensional subspace $\mathrm{Sp}(v)$.
- **(c)** In the same way one considers affine 2-dimensional planes, and generally affine k-dimensional planes, modeled on k-dimensional subspaces.
- **(d)** Let $A \in \mathcal{M}_{m\times n}(K)$ and $b \in K^m$, and let $x_0 \in K^n$ be a solution of $Ax = b$. Then, by Corollary 17.d.2,
$$\underbrace{\mathrm{Sol}(A, b)}_{\text{affine space}} = x_0 + \underbrace{\mathrm{Sol}(A, 0)}_{\text{vector space}},$$

so the solution set of an inhomogeneous linear system is exactly an affine space modeled on the solution space of the associated homogeneous system.

Remark 19.2: One can go further and define "*affine maps*" $f\colon X \to Y$ between affine spaces, and, given an affine space X and a subspace $U \subseteq \vec{X}$, a new affine space X/U modeled on $\vec{X}/U$.

Definition 19.5 (*Parallel Affine Spaces*): Let $X, Y \subseteq V$ be affine spaces. We say that X and Y are "*parallel*", written $X \parallel Y$, if $\vec{X} = \vec{Y}$; equivalently, if $X = p + U$ and $Y = q + U$ for one and the same subspace U. We say that X is "*weakly parallel*" to Y if $\vec{X} \subseteq \vec{Y}$.

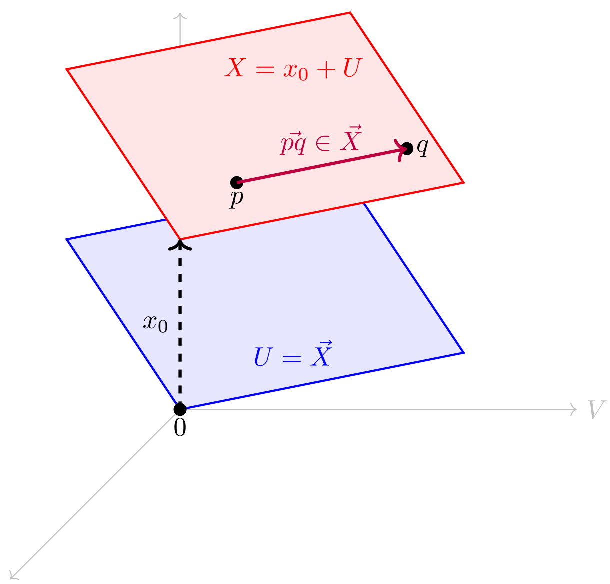
Important remark 19.1 (*Ratios of Vectors on a Line*): If L is a **one-dimensional** vector space over K, then vectors in L can be divided by one another. Indeed, let $u, v \in L$ with $u \neq 0$. Since $\dim L = 1$, the vector u alone is a basis of L, so there is a unique scalar $\lambda \in K$ with $v = \lambda \cdot u$, and we define

$$\frac{v}{u} := \lambda \in K.$$

Consequently, if X is an affine line and $a, b, c \in X$ with $a \neq b$, then $\vec{ab}$ and $\vec{ac}$ both lie in the one-dimensional space $\vec{X}$ and $\vec{ab} \neq 0$, so the ratio

$$\frac{\vec{ac}}{\vec{ab}} \in K$$

is defined. These are proportions, not distances: no notion of length is involved, and none is available, since K need not be $\mathbb{R}$.



Theorem 19.6 (*Thales' Theorem in Affine Geometry*): Let V be a 3-dimensional vector space over K, and let H, H', H'' be three distinct parallel 2-dimensional affine planes in V. Let ℓ_1, ℓ_2 be two affine lines, neither of them weakly parallel to H, and put

$$p_1 := H \cap \ell_1, \quad p_1' := H' \cap \ell_1, \quad p_1'' := H'' \cap \ell_1, \qquad p_2 := H \cap \ell_2, \quad p_2' := H' \cap \ell_2, \quad p_2'' := H'' \cap \ell_2.$$

Then

$$\frac{\vec{p_1p_1''}}{\vec{p_1p_1'}} = \frac{\vec{p_2p_2''}}{\vec{p_2p_2'}}.$$

In other words, this ratio does not depend on the line ℓ chosen, but only on the three planes H, H', H''.

Proof:

Put $U := \vec{H} = \vec{H}' = \vec{H}''$, which is a 2-dimensional subspace of V, the three planes being parallel. Consider the quotient V/U and the projection $\pi\colon V \to V/U$. By Proposition 18.c.4,

$$\dim V/U = \dim V - \dim U = 3 - 2 = 1,$$

so V/U is a line, and ratios of its non-zero vectors are defined as in Important remark 19.1.

Both ratios in the statement are defined: p_1, p_1', p_1'' lie on the affine line ℓ_1, whose direction space is one-dimensional, and $\vec{p_1p_1'} \neq 0$ because $p_1 \in H$ and $p_1' \in H'$ are points of distinct parallel planes. Put

$$\lambda_1 := \frac{\vec{p_1p_1''}}{\vec{p_1p_1'}}, \qquad \lambda_2 := \frac{\vec{p_2p_2''}}{\vec{p_2p_2'}},$$

so that $\vec{p_1p_1''} = \lambda_1 \vec{p_1p_1'}$ and $\vec{p_2p_2''} = \lambda_2 \vec{p_2p_2'}$. Our task is to show $\lambda_1 = \lambda_2$.

Now $\pi(\vec{p_1p_1'}) = \pi(\vec{p_2p_2'})$. Indeed,

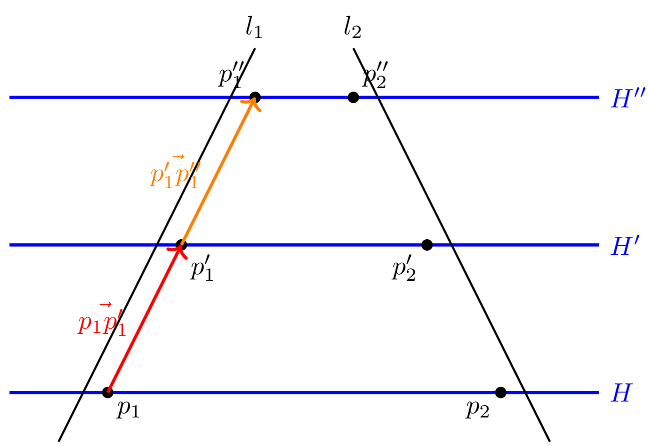
$$\vec{p_1p_1'} - \vec{p_2p_2'} = \vec{p_1p_2} + \vec{p_2'p_1'} \in U,$$

because $p_1, p_2 \in H$ gives $\vec{p_1p_2} \in \vec{H} = U$, and $p_2', p_1' \in H'$ gives $\vec{p_2'p_1'} \in \vec{H}' = U$; and $\ker(\pi) = U$ by Proposition 18.c.3. The same argument with H'' in place of H' gives $\pi(\vec{p_1p_1''}) = \pi(\vec{p_2p_2''})$. Combining, and using the linearity of π,

$$\lambda_1\,\pi(\vec{p_1p_1'}) = \pi(\lambda_1\vec{p_1p_1'}) = \pi(\vec{p_1p_1''}) = \pi(\vec{p_2p_2''}) = \pi(\lambda_2\vec{p_2p_2'}) = \lambda_2\,\pi(\vec{p_2p_2'}) = \lambda_2\,\pi(\vec{p_1p_1'}).$$

Finally $\pi(\vec{p_1p_1'}) \neq 0$: otherwise $\vec{p_1p_1'} \in \ker(\pi) = U$, and since $\vec{p_1p_1'}$ spans the direction space of ℓ_1, this would make ℓ_1 weakly parallel to H, which we excluded. Cancelling the non-zero vector $\pi(\vec{p_1p_1'})$ in the one-dimensional space V/U yields $\lambda_1 = \lambda_2$. **Q.E.D.**

Remark 19.3: Observe what this proof did **not** use: no coordinates, no lengths, no angles, and no assumption on K beyond its being a field. The whole argument is the observation that collapsing the direction U of the three planes turns them into three points of a line, where the ratio is visible at once. The four points p_1, p_2, p_1', p_2' need not lie in a common plane, and the same goes for p_1, p_2, p_1'', p_2''.



Introduction to Eigenvalues: Fibonacci Sequences (Section 19.c)

The Vector Space of Fibonacci Sequences

Consider the set $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ of all infinite real sequences $(a_0, a_1, a_2, \dots)$ that satisfy the linear recurrence relation $a_n = a_{n-1} + a_{n-2}$ for all $n \geq 2$.

Proposition 19.7 (*Structure of* Fib): The set Fib is a 2-dimensional subspace of $\mathbb{R}^\infty$. The evaluation map $F\colon \mathrm{Fib} \to \mathbb{R}^2$ sending a sequence to its first two terms, $(a_n) \mapsto (a_0, a_1)$, is a linear isomorphism.

The notes state this without proof, and the computation at the end of this section uses both of its halves, so we supply one.

Proof:

That $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ is a linear subspace is Example 7.5. The map F is linear, since both of its coordinates are, addition and scalar multiplication in $\mathbb{R}^\infty$ being defined entry by entry.

F is injective: if $(a_n) \in \mathrm{Fib}$ has $a_0 = a_1 = 0$, then $a_n = a_{n-1} + a_{n-2}$ gives $a_n = 0$ for every $n \geq 2$ by induction, so the sequence is the zero sequence and $\ker(F) = \{0\}$; now apply Proposition 15.b.2 **(a)**. It is surjective: given $(c_0, c_1) \in \mathbb{R}^2$, set $a_0 := c_0$, $a_1 := c_1$ and define $a_n := a_{n-1} + a_{n-2}$ for $n \geq 2$. This recursion defines a sequence, it satisfies the Fibonacci relation by construction, so it lies in Fib, and F sends it to (c_0, c_1).

Being a bijective linear map, F is an isomorphism by Lemma 15.b.4, and therefore $\dim \mathrm{Fib} = \dim \mathbb{R}^2 = 2$ by Corollary 15.b.11. **Q.E.D.**

To find a closed-form formula for the Fibonacci numbers (like Binet's formula), we study the **shift map** $S\colon \mathbb{R}^\infty \to \mathbb{R}^\infty$ defined by shifting the sequence one position to the left:

$$S(a_0, a_1, a_2, \dots) := (a_1, a_2, a_3, \dots).$$

Since the shifted sequence also satisfies the Fibonacci recurrence, this map restricts to a linear endomorphism $S'\colon \mathrm{Fib} \to \mathrm{Fib}$.

By utilizing the isomorphism F, we can translate the abstract shift operator S' into concrete matrix multiplication on $\mathbb{R}^2$. The sequence (a_0, a_1) shifts to $(a_1, a_2) = (a_1, a_0 + a_1)$, which corresponds to multiplication by the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

$$\begin{tikzcd}
\mathrm{Fib} \arrow[r, "S'"] \arrow[d, "F"', "\cong"] & \mathrm{Fib} \arrow[d, "\cong"', "F"] \\
\mathbb{R}^2 \arrow[r, "T_A"'] & \mathbb{R}^2
\end{tikzcd}$$

Eigenvalues and Eigenvectors

The problem of finding a geometric sequence $(1, \lambda, \lambda^2, \dots)$ inside Fib is equivalent to finding sequences that only scale by λ when shifted. That is, we seek vectors $\mathcal{F}$ such that $S'(\mathcal{F}) = \lambda \mathcal{F}$.

Definition 19.8 (*Eigenvalues and Eigenvectors*): Let V be a vector space and $T \in \mathrm{End}(V)$. A scalar $\lambda \in K$ is called an "*eigenvalue*" of T if there exists a non-zero vector $v \in V$ such that:

$$T(v) = \lambda v.$$

The vector v is called an "*eigenvector*" of T associated with the eigenvalue λ.

Remark 19.4 (*Reformulations of the Eigenvalue Condition*): Unwinding the definition, λ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot \mathrm{id}_V\colon V \to V$ has a non-trivial kernel, since $T(v) = \lambda v$ says exactly that $(T - \lambda\, \mathrm{id}_V)(v) = 0$.

If we assume in addition that V is finite-dimensional, then by Corollary 15.b.10 **(c)** the last condition is equivalent to

$$T - \lambda \cdot \mathrm{id}_V : V \to V \quad \text{is **not** an isomorphism,}$$

and, passing to a basis $\mathcal{B}$ of V and writing $n := \dim V$, to

$$[T]_{\mathcal{B}}^{\mathcal{B}} - \lambda \cdot I_n \in \mathcal{M}_{n\times n}(K) \quad \text{is **not** invertible.}$$

The last equivalence uses Proposition 17.b.3 together with $[\lambda \cdot \mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = \lambda I_n$ from Claim 17.1. This is the form that will let us actually compute eigenvalues, once we have determinants: it turns the question into a question about a single matrix.

Proposition 19.9 (*The Golden Ratio Diagonalises the Shift*): Put $\varphi := \frac{1+\sqrt{5}}{2}$ and $\psi := \frac{1-\sqrt{5}}{2}$. Then the two geometric sequences

$$x_{1,\varphi} := (1, \varphi, \varphi^2, \dots), \qquad x_{1,\psi} := (1, \psi, \psi^2, \dots)$$

belong to Fib, and they form a basis of Fib. Moreover φ and ψ are eigenvalues of the shift $S' \colon \mathrm{Fib} \to \mathrm{Fib}$, with $x_{1,\varphi}$ and $x_{1,\psi}$ as associated eigenvectors.

Proof:

A geometric sequence $(1, \lambda, \lambda^2, \dots)$ lies in Fib precisely when $\lambda^n = \lambda^{n-1} + \lambda^{n-2}$ for all $n \geq 2$, that is, when $\lambda^2 = \lambda + 1$. (The case $n = 2$ already gives that equation, and conversely $\lambda^2 = \lambda + 1$ yields every other case on multiplying by λ^{n-2}. Note that $\lambda = 0$ is not a solution, so no division by zero is hidden here.) The roots of $\lambda^2 - \lambda - 1$ are exactly φ and ψ, so both sequences lie in Fib.

They are linearly independent: applying the isomorphism F of Proposition 19.7 sends them to $(1, \varphi)$ and $(1, \psi)$ in $\mathbb{R}^2$, which are independent because $\varphi \neq \psi$. Since $\dim \mathrm{Fib} = 2$, they form a basis.

Finally, shifting a geometric sequence multiplies it by its ratio:

$$S'(1, \lambda, \lambda^2, \dots) = (\lambda, \lambda^2, \lambda^3, \dots) = \lambda \cdot (1, \lambda, \lambda^2, \dots),$$

so $x_{1,\varphi}$ and $x_{1,\psi}$ are eigenvectors of S' with eigenvalues φ and ψ. **Q.E.D.**

All that is left in order to recover the closed formula is to express the Fibonacci sequence itself in this basis. The Fibonacci sequence is $x_{0,1} = (0, 1, 1, 2, 3, 5, \dots)$, and we seek $\alpha, \beta \in \mathbb{R}$ with $\alpha\, x_{1,\varphi} + \beta\, x_{1,\psi} = x_{0,1}$. Comparing the first two entries gives the system

$$\begin{cases} \alpha \cdot 1 + \beta \cdot 1 = 0, \\ \alpha \cdot \varphi + \beta \cdot \psi = 1, \end{cases}$$

whose solution is $\alpha = \frac{1}{\sqrt{5}}$ and $\beta = -\frac{1}{\sqrt{5}}$. Reading off the n-th entry recovers the formula from the first chapter,

$$f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right) \qquad \text{for all } n \geq 0.$$

This is the whole method in miniature, and it is the motivation for the chapters to come: find the eigenvalues of an endomorphism, use the associated eigenvectors as a basis, and in that basis the map becomes multiplication by scalars, one coordinate at a time. What is still missing is a way to find the eigenvalues, and that is what determinants will provide.

Solutions to the Exercises

Solution to Exercise 19.1:

Write $X = x_0 + U$ as in Definition 19.3. The key observation is that U can be recovered from X alone, without reference to x_0, as the set of differences of points of X:

$$U = \{\, q - p \mid p, q \in X \,\}.$$

Indeed, if $p = x_0 + u$ and $q = x_0 + u'$ with $u, u' \in U$, then $q - p = u' - u \in U$, which gives the inclusion $\supseteq$ read from right to left; and conversely every $u \in U$ arises as $u = (x_0 + u) - x_0$, a difference of two points of X. The right-hand side mentions only X, so if $X = x_0 + U = x_1 + U'$ for subspaces U, U', both are equal to that same set of differences, and hence $U = U'$. This is what makes the notation $\vec{X} := U$ legitimate.

For the base point, let $p \in X$ be arbitrary and write $p = x_0 + u$ with $u \in U$. Then

$$p + U = x_0 + u + U = x_0 + U = X,$$

the middle equality because $u + U = U$ for $u \in U$, a subspace being closed under addition and containing $-u$. So every point of X serves as a base point.

Finally, the base point is unique exactly when X has a single element. If $\dim X = \dim U = 0$ then $U = \{0_V\}$ and $X = \{x_0\}$, so there is nothing to choose. If $\dim X > 0$ then U contains some $u \neq 0_V$, and x_0 and $x_0 + u$ are two distinct points of X, each of which is a valid base point by the paragraph above. **Q.E.D.**

AI-Note: This chapter is the one place in the book where the notes are mostly pictures and asides, and the transcript kept the pictures but lost several of the asides. Restored above:

In the affine section, the dimension of an affine space; the observation that the base point x_0 is not determined by X while the direction space is (Page 165); the identity $p + \vec{pq} = q$; Prof. Biran's four examples, of which the fourth is the pleasant one, the solution set of $Ax = b$ **is** an affine space, modeled on the solution space of $Ax = 0$; the mention of affine maps and affine quotients; the definition of parallel and weakly parallel; and, above all, Page 166: in a one-dimensional space one vector may be divided by another. Without that observation the fraction in the statement of Theorem 19.6 has no meaning at all, and the transcript stated Thales without it.

Theorem 19.6 itself stood without a proof and with a hypothesis too loose to be true as stated, the ambient space must be three-dimensional, the planes two-dimensional, and the two lines must not be weakly parallel to H, or the intersections need not be single points. Prof. Biran's proof is restored: collapse the common direction U of the three planes, so that V/U becomes a line, and the equality of ratios falls out. He annotates it *"a truly coordinate-free proof"*, and it is a striking use of the quotient construction from the previous chapter.

In the closing section, the reformulations of the eigenvalue condition, non-trivial kernel, failure to be an isomorphism, non-invertibility of $[T]^{\mathcal{B}}_{\mathcal{B}} - \lambda I_n$, which are what makes the notion computable later. And the payoff the chapter was building towards: that φ and ψ are the eigenvalues of the shift on Fib, that the two geometric sequences form a basis, and that expanding the Fibonacci sequence in that basis returns the formula from chapter 1. In the transcript this last part existed only as two commented-out lines at the end of the file, one of them a corrected copy of the other.

Also: the proposition on Fib had `$Fib$` in its title, typeset as a product of three italic letters rather than with the `\Fib` macro, and five raw quotation marks became `\qt`.

The rigour pass closed two gaps and corrected one notation. The lines of the Möbius family were written $\mathrm{Sp}(\cos\frac{\theta}{2}, \sin\frac{\theta}{2})$, and likewise $\mathrm{Sp}(1, 0)$ and $\mathrm{Sp}(-1, 0)$ further down. Under this book's own convention, $\mathrm{Sp}(v_1, \ldots, v_n)$ is the span of a **list**, so those read as spans of two scalars rather than of one vector of $\mathbb{R}^2$. The vector's own parentheses are now written.

Proposition 19.7 stood without a proof, although the computation closing the chapter uses both of its halves: the isomorphism F to test the independence of the two geometric sequences, and $\dim \mathrm{Fib} = 2$ to conclude from that independence that they form a basis. A proof is supplied; it is four lines, the substance being that a Fibonacci sequence is determined by its first two terms and that any pair of reals starts one.

The uniqueness of the direction space U, on which the notation $\vec{X}$ of Definition 19.3 depends for its meaning, was asserted in Remark 19.1 and left to the reader. It is now Exercise 19.1, with its solution at the end of the chapter; the point is that U is the set of differences of points of X, a description in which the base point does not appear.

Determinants (Chapter 20)

Introduction and Motivation (Section 20.a)

The special case of 2×2 matrices

In linear algebra, determining whether a matrix is invertible is a fundamental problem. Let us begin by analyzing the familiar 2×2 case. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(K)$. There is a very simple, well-known algebraic criterion to check whether A is invertible.

Proposition 20.a.1 (*Invertibility of 2×2 Matrices*): The matrix A is invertible if and only if $ad - bc \neq 0$. Moreover, if A is invertible, then its inverse is given by the explicit formula:

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The scalar quantity $ad - bc$ is called the "*determinant*" of A, denoted $\det(A)$.

Proof:

First, A is not invertible if and only if its rank is strictly less than 2; this is AI-Proposition 17.1, read in the contrapositive. Because A is a 2×2 matrix, having a rank less than 2 implies that its column vectors are linearly dependent: the column rank equals the rank by Theorem 13.9, so two columns spanning a space of dimension at most 1 cannot be independent, by Theorem 12.2 **(2)**.

Exercise 20.1 (*Dependence of Two Columns*): Check that the columns of A being linearly dependent means that either the first column is a scalar multiple of the second, $\binom{a}{c} = \lambda\binom{b}{d}$ for some $\lambda \in K$, or the second is a scalar multiple of the first, $\binom{b}{d} = \tau\binom{a}{c}$ for some $\tau \in K$.

If $\binom{a}{c} = \lambda\binom{b}{d}$, then $a = \lambda b$ and $c = \lambda d$. Consequently, $ad - bc = (\lambda b)d - b(\lambda d) = 0$. If, on the other hand, $\binom{b}{d} = \tau\binom{a}{c}$, then $b = \tau a$ and $d = \tau c$. Consequently, $ad - bc = a(\tau c) - (\tau a)c = 0$. Thus, we have proved that if A is not invertible, then $ad - bc = 0$.

Conversely, assume $ad - bc = 0$. If $a = 0$, then $bc = 0$, so either $b = 0$ or $c = 0$. This means either $A = \left(\begin{smallmatrix} 0 & 0 \\ c & d \end{smallmatrix}\right)$ or $A = \left(\begin{smallmatrix} 0 & b \\ 0 & d \end{smallmatrix}\right)$. In both of these cases, A, clearly, has linearly dependent columns and is therefore not invertible.

Now, assume $a \neq 0$. This implies $d = \frac{bc}{a}$, and thus we can write $A = \begin{pmatrix} a & b \\ c & \frac{bc}{a} \end{pmatrix}$. We can observe that the second column of A is exactly a multiple of the first column because $\binom{b}{\frac{bc}{a}} = \frac{b}{a}\binom{a}{c}$. This implies that the columns of A are linearly dependent. Hence, $\operatorname{rank}(A) < 2$, which implies A is not invertible. **Q.E.D.**

To confirm the sufficiency of this condition and to find the explicit inverse, the formula for A^{-1} can be verified by direct matrix multiplication:

$$\frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad - bc}\begin{pmatrix} ad - bc & bd - bd \\ -ac + ac & -bc + ad \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

An inverse has to satisfy the identity in both orders, and the product taken the other way round is the same computation:

$$\frac{1}{ad - bc}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc}\begin{pmatrix} ad - bc & -ab + ab \\ cd - dc & -cb + ad \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Goals 20.1: While the 2×2 formula is easy to memorize, it does not immediately suggest how to handle larger matrices. Our main goals now are to make the determinant conceptually rigorous and to generalize the determinant function to arbitrary $n \times n$ matrices.

Multilinear and Alternating Functions

We wish to generalize the determinant to arbitrary dimensions. To this end, we should view an $n \times n$ matrix not just as a grid of numbers, but as a collection of n row vectors.

Definition 20.a.2 (*n-Linearity*): Let $D : \mathcal{M}_{n\times n}(K) \to K$ be a function. We say that D is "*n-linear*" if for every $1 \le i \le n$, D is linear as a function of the i-th row when all the other $n-1$ rows are held fixed.

We can think of the matrix space $\mathcal{M}_{n\times n}(K)$ as a Cartesian product of row vectors: $K^n_{\text{row}} \times \cdots \times K^n_{\text{row}}$. If A has rows $\alpha_1, \ldots, \alpha_n$, we write $D(A) = D(\alpha_1, \ldots, \alpha_n)$. Saying that D is n-linear means that for any chosen row index i, the following two linearity conditions hold:

(a) $D(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) = D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + D(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n)$.

(b) $D(\alpha_1, \ldots, c \cdot \alpha_i, \ldots, \alpha_n) = c \cdot D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n)$ for all scalars $c \in K$.

Example 20.1 (*Product of Row Entries*): For a matrix $A \in \mathcal{M}_{n\times n}(K)$, write $A(i,j)$ for its entry at place i,j. Let $k_1, \ldots, k_n$ be a fixed sequence of integers such that $1 \le k_i \le n$ for all i. Fix also a scalar $c \in K$. Consider $D : \mathcal{M}_{n\times n}(K) \to K$ defined by the product of one entry from each row:

$$D(A) := c \cdot A(1,k_1) \cdot A(2,k_2) \cdots A(n,k_n).$$

We claim that D is an n-linear function.

To see this, consider $D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) = b \cdot A(i,k_i)$, where $b \in K$ is the product of all the fixed terms. Notice that b depends only on the rows $\alpha_1, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_n$ and does not depend on row i. Let $\alpha_i' = (\alpha_{i,1}', \ldots, \alpha_{i,n}')$ be another possible i-th row. Then, by standard distribution:

$$D(\ldots, \alpha_i + \alpha_i', \ldots) = b \cdot (A(i,k_i) + \alpha_{i,k_i}') = D(\ldots, \alpha_i, \ldots) + D(\ldots, \alpha_i', \ldots).$$

Similarly, for any scalar $\lambda \in K$:

$$D(\ldots, \lambda \cdot \alpha_i, \ldots) = b \cdot (\lambda A(i,k_i)) = \lambda \cdot D(\ldots, \alpha_i, \ldots).$$

Therefore, D is indeed n-linear.

Example 20.2 (*Diagonal Product*): A more concrete example: $D(A) := A_{11} \cdot A_{22} \cdot \cdots \cdot A_{nn}$ (i.e., the product of the terms on the main diagonal) is an n-linear function.

Now, let us try to find all 2-linear functions $D : \mathcal{M}_{2\times 2}(K) \to K$. Write the identity matrix as $I = \left(\begin{smallmatrix} -\epsilon_1- \\ -\epsilon_2- \end{smallmatrix}\right)$, where $\epsilon_1 = (1,0)$ and $\epsilon_2 = (0,1)$.

If D is 2-linear, then we have:

$$\begin{aligned} D(A) &= D(A_{11}\epsilon_1 + A_{12}\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}D(\epsilon_1, A_{21}\epsilon_1 + A_{22}\epsilon_2) + A_{12}D(\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}A_{21}D(\epsilon_1,\epsilon_1) + A_{11}A_{22}D(\epsilon_1,\epsilon_2) + A_{12}A_{21}D(\epsilon_2,\epsilon_1) + A_{12}A_{22}D(\epsilon_2,\epsilon_2). \end{aligned}$$

Therefore, D is completely determined by its values on the following four pairings: $D(\epsilon_1,\epsilon_1)$, $D(\epsilon_1,\epsilon_2)$, $D(\epsilon_2,\epsilon_1)$, and $D(\epsilon_2,\epsilon_2)$.

Lemma 20.a.3 (*Closure of n-Linear Functions*): Any linear combination of n-linear functions is again an n-linear function.

Proof:

It is enough to prove this for a linear combination of two n-linear functions. Let $D, E : \mathcal{M}_{n\times n}(K) \to K$ be n-linear functions and $a, b \in K$. We define the combined function as $(aD + bE)(A) :=$

$aD(A)+bE(A)$. By expanding $(aD+bE)(\dots,\alpha_i+\alpha_i',\dots)$ and $(aD+bE)(\dots,\lambda\alpha_i,\dots)$ using the individual linearities of D and E, we see the result naturally holds. **Q.E.D.**

Example 20.3 (*Two-by-Two Determinant Function*): Consider $D:\mathcal{M}_{2\times 2}(K)\to K$ defined by $D(A):=A_{11}A_{22}-A_{12}A_{21}$. The function D is 2-linear because it can be written as $D=D_1+D_2$, where $D_1(A)=A_{11}A_{22}$ and $D_2(A)=-A_{12}A_{21}$, both of which are 2-linear by our previous example. Furthermore, $D(I)=1$, and if A' is obtained from A by interchanging the rows, then $D(A')=-D(A)$.

Indeed, $D(A')=D\left(\begin{smallmatrix}A_{21} & A_{22}\\ A_{11} & A_{12}\end{smallmatrix}\right)=A_{21}\cdot A_{12}-A_{22}A_{11}=-(A_{11}A_{22}-A_{12}A_{21})=-D(A)$.

Definition 20.a.4 (*Alternating Function*): Let $D:\mathcal{M}_{n\times n}(K)\to K$ be an n-linear function. We say that D is "*alternating*" if for every matrix A in which some two rows are equal, we have $D(A)=0$.

Proposition 20.a.5 (*Adjacent Swap Property*): Let $D:\mathcal{M}_{n\times n}(K)\to K$ be an n-linear function, and assume that D has the property that whenever A has two equal **adjacent** rows, then $D(A)=0$. Then both of the following hold:

(a) D is alternating, that is, $D(A)=0$ whenever **any** two rows of A are equal, adjacent or not.

(b) For any matrix B, if B' is obtained from B by interchanging two of its rows, then $D(B')=-D(B)$.

Proof:

We begin with the proof of **(b)**. Assume first that B' is obtained from B by interchanging two adjacent rows (say, row k and row $k+1$). Consider evaluating D on a matrix where both the k-th and $(k+1)$-th rows are equal to $\beta_k+\beta_{k+1}$. By our assumption, this yields 0. Expanding this via n-linearity gives:

$$\begin{aligned}D(\dots,\beta_k+\beta_{k+1},\beta_k+\beta_{k+1},\dots)&=D(\dots,\beta_k,\beta_k,\dots)+D(\dots,\beta_k,\beta_{k+1},\dots)\\&+D(\dots,\beta_{k+1},\beta_k,\dots)+D(\dots,\beta_{k+1},\beta_{k+1},\dots)=0.\end{aligned}$$

By assumption, the terms with adjacent equal rows are zero, leaving $D(B)+D(B')=0$, which naturally implies $D(B')=-D(B)$.

Now suppose B' is obtained from B by interchanging rows k and ℓ where $k<\ell$. We can obtain B' by doing a sequence of adjacent row interchanges. Moving row ℓ to position k requires $r=\ell-k$ adjacent interchanges. Next, moving the original row k down to position ℓ requires $r-1$ adjacent interchanges. The total number of interchanges is $2r-1$. Since this is always an odd number, we flip the sign an odd number of times. Thus, $D(B')=(-1)^{2r-1}D(B)=-D(B)$.

We now prove **(a)**. Let A be a matrix in which row i equals row j where $i<j$. If $j=i+1$, then by assumption $D(A)=0$. If $j>i+1$, we interchange row j with row $i+1$ to obtain a matrix A' with two equal adjacent rows. By assumption $D(A')=0$, but by **(b)**, $D(A')=-D(A)$ (since we performed an odd number of adjacent swaps), which implies $D(A)=0$. **Q.E.D.**

The Determinant Function

Definition 20.a.6 (*Determinant Function*): A function $D:\mathcal{M}_{n\times n}(K)\to K$ is called a "*determinant function*" if it is n-linear, it is alternating, and it normalizes the identity matrix such that $D(I)=1$.

Example 20.4 (*Warmup*): Let us find all determinant functions on $M_{2\times 2}(K)$. Write $I=\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right)=\left(\begin{smallmatrix}-\epsilon_1-\\-\epsilon_2-\end{smallmatrix}\right)$, meaning $\epsilon_1=(1,0)$ and $\epsilon_2=(0,1)$. If D is a 2-linear function, we can expand it algebraically:

$$\begin{aligned} D(A) &= D(A_{11}\epsilon_1 + A_{12}\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}D(\epsilon_1, A_{21}\epsilon_1 + A_{22}\epsilon_2) + A_{12}D(\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}A_{21}D(\epsilon_1,\epsilon_1) + A_{11}A_{22}D(\epsilon_1,\epsilon_2) \\ &\quad + A_{12}A_{21}D(\epsilon_2,\epsilon_1) + A_{12}A_{22}D(\epsilon_2,\epsilon_2). \end{aligned}$$

Consequently, D is completely determined by its values on these standard basis row pairings.

If D is alternating, then evaluating equal rows gives zero, meaning $D(\epsilon_1,\epsilon_1) = D(\epsilon_2,\epsilon_2) = 0$. Additionally, swapping rows flips the sign, so $D(\epsilon_2,\epsilon_1) = -D(\epsilon_1,\epsilon_2) = -D(I)$. Finally, if D is a determinant function, $D(I) = 1$. Substituting these values back into our expansion, we perfectly recover the familiar formula:

$$D(A) = A_{11}A_{22}(1) + A_{12}A_{21}(-1) = A_{11}A_{22} - A_{12}A_{21}.$$

Therefore, the 2×2 determinant function is completely unique!

Definition 20.a.7 (*Submatrix*): Let $A \in \mathcal{M}_{n\times n}(K)$, $n \geq 2$, and $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by $A(i|j) \in \mathcal{M}_{(n-1)\times(n-1)}(K)$ the *"submatrix"* obtained from A by deleting row i and column j. If D is an $(n-1)$-linear function, we intuitively define $D_{ij}(A) := D(A(i|j))$.

Theorem 20.a.8 (*Recursive Construction of Determinants*): Let $n \geq 2$, and let D be an $(n-1)$-linear function which is alternating. Let $1 \leq j \leq n$. Define $E_j : \mathcal{M}_{n\times n}(K) \to K$ by

$$E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij} D_{ij}(A).$$

Then, E_j is an n-linear function and it is alternating. Furthermore, if D is a determinant function, then so is E_j.

Proof:

Clearly, $D_{ij}(A)$ is independent of the i-th row of A because that row is deleted in the construction of $A(i|j)$. Since D is $(n-1)$-linear, the map $A \mapsto D_{ij}(A)$ is linear when viewed as a function of each of its remaining rows. The function $A \mapsto A_{ij}D_{ij}(A)$ is therefore n-linear. Since linear combinations of n-linear functions are n-linear, $E_j(A)$ is n-linear too.

We show that E_j is alternating. By Proposition 20.a.5, it is enough to show $E_j(A) = 0$ whenever A has two equal adjacent rows. Assume $\alpha_k = \alpha_{k+1}$ for some index k. For any $i \neq k, k+1$, the submatrix $A(i|j)$ still has two equal rows, hence $D_{ij}(A) = 0$. This leaves only two non-zero terms in the sum:

$$E_j(A) = (-1)^{k+j} A_{kj} D_{kj}(A) + (-1)^{k+1+j} A_{(k+1)j} D_{(k+1)j}(A).$$

But, since $\alpha_k = \alpha_{k+1}$, we have $A_{kj} = A_{(k+1)j}$ and $A(k|j) = A(k+1|j)$, meaning $D_{kj}(A) = D_{(k+1)j}(A)$. Thus, the two terms perfectly cancel out, yielding $E_j(A) = 0$. This proves E_j is alternating.

Finally, assume D is a determinant function, meaning $D(I_{n-1}) = 1$. Note that $I_n(j|j) = I_{n-1}$ for all j. When computing $E_j(I_n)$, the term $(I_n)_{ij}$ is 1 if $i = j$, and 0 if $i \neq j$. Therefore, the sum collapses to a single term:

$$E_j(I_n) = (-1)^{j+j} D_{jj}(I_n) = D(I_n(j|j)) = 1.$$

This shows that E_j is a valid determinant function. **Q.E.D.**

Corollary 20.a.9 (*Existence of Determinants*): For all $n \geq 1$, at least one determinant function exists.

Proof:

By induction on n. For $n = 1$, the function $D(A) := A_{11}$ is 1-linear, and it satisfies $D(I_1) = 1$; it is alternating for the empty reason that a 1×1 matrix has only one row and so cannot have two equal ones. For the step, suppose a determinant function D on $\mathcal{M}_{(n-1)\times(n-1)}(K)$ exists, with $n \geq 2$. Then Theorem 20.a.8, applied with any fixed j, produces the determinant function E_j on $\mathcal{M}_{n\times n}(K)$. **Q.E.D.**

Example 20.5 (*Three-by-Three Determinant Expansion*): Consider $A \in \mathcal{M}_{3\times 3}(K)$. Using the uniquely defined 2×2 determinant function $|B| := B_{11}B_{22} - B_{12}B_{21}$, we can explicitly write out $E_1(A)$ (expanding along the first column) as:

$$E_1(A) = A_{11}\begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{21}\begin{vmatrix} A_{12} & A_{13} \\ A_{32} & A_{33} \end{vmatrix} + A_{31}\begin{vmatrix} A_{12} & A_{13} \\ A_{22} & A_{23} \end{vmatrix}.$$

We can similarly construct $E_2(A)$ and $E_3(A)$ along the other columns, and all of these form valid determinant functions on $M_{3\times 3}(K)$.

Uniqueness of the Determinant (Section 20.b)

We have shown that a determinant function exists. Now, we will prove that it is absolutely unique. Let $D : \mathcal{M}_{n\times n}(K) \to K$ be an n-linear alternating function. Let $A \in \mathcal{M}_{n\times n}(K)$ be a matrix with rows $\alpha_1, \ldots, \alpha_n$. We denote the identity matrix as I, whose rows are the standard basis vectors $\epsilon_1, \ldots, \epsilon_n$ of K^n_{row}.

Clearly, we can express each row of A in terms of the standard basis: $\alpha_i = \sum_{j=1}^n A(i,j)\epsilon_j$ for $1 \leq i \leq n$. Using the n-linearity of D, we can expand $D(A)$ as follows:

$$\begin{aligned} D(A) &= D(\alpha_1, \ldots, \alpha_n) \\ &= D\left(\sum_{j=1}^n A(1,j)\epsilon_j, \alpha_2, \ldots, \alpha_n\right) \\ &= \sum_{j=1}^n A(1,j) \cdot D(\epsilon_j, \alpha_2, \ldots, \alpha_n) \\ &= \sum_{j=1}^n A(1,j) \cdot D\left(\epsilon_j, \sum_{k=1}^n A(2,k)\epsilon_k, \ldots, \alpha_n\right) \\ &= \sum_{j,k} A(1,j) \cdot A(2,k) \cdot D(\epsilon_j, \epsilon_k, \ldots, \alpha_n) \end{aligned}$$

where both j and k run between 1 and n. Continuing this expansion for all n rows yields an enormous sum:

$$D(A) = \sum_{k_1,\ldots,k_n} A(1,k_1) \cdot A(2,k_2) \cdots A(n,k_n) \cdot D(\epsilon_{k_1}, \epsilon_{k_2}, \ldots, \epsilon_{k_n})$$

where each of the indices $k_1, \ldots, k_n$ runs between 1 and n.

Thus far, we have not used the fact that D is alternating. Because D is alternating, the term $D(\epsilon_{k_1}, \ldots, \epsilon_{k_n}) = 0$ whenever two of the indices $k_1, \ldots, k_n$ are equal. Thus, we are interested only in ordered sequences $(k_1, \ldots, k_n)$ in which $k_i \in \{1, \ldots, n\}$ for all i, and $k_i \neq k_j$ for all $i \neq j$.

Such a sequence is called a "*permutation*" of $\{1, \ldots, n\}$, or a permutation of degree n. Alternatively, we can think of a permutation of degree n as a bijective function $\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$. Then, the sequence $(k_1, \ldots, k_n) = (\sigma(1), \sigma(2), \ldots, \sigma(n))$ is a permutation as defined earlier. In other words, a permutation of degree n is simply a re-ordering of a list of n elements: $(1, \ldots, n) \mapsto (\sigma(1), \ldots, \sigma(n))$.

We can nicely rewrite our expansion of $D(A)$ as:

$$D(A) = \sum_{\sigma} A(1,\sigma(1)) \cdot A(2,\sigma(2)) \cdots A(n,\sigma(n)) \cdot D(\epsilon_{\sigma(1)}, \epsilon_{\sigma(2)}, \dots, \epsilon_{\sigma(n)})$$

where σ runs over all possible permutations of degree n.

Properties of Permutations

An important property about permutations is that if σ is a permutation of degree n, one can pass from the sequence $(1,\dots,n)$ to $(\sigma(1),\dots,\sigma(n))$ by performing a finite sequence of interchanges of pairs.

More precisely, a "*transposition*" is a permutation $\theta : \{1,\dots,n\} \to \{1,\dots,n\}$ for which there exist $i,j \in \{1,\dots,n\}$ with $i \neq j$ such that $\theta(i) = j$, $\theta(j) = i$, and $\theta(k) = k$ for all $k \neq i,j$.

Every permutation σ can be written as a composition of transpositions:

$$\sigma = \theta_1 \circ \cdots \circ \theta_m \tag{20.1}$$

where $\theta_1,\dots,\theta_m$ are transpositions. However, these θ_i's and the total number m are not unique. The same permutation σ can be written as $\sigma = \theta_1 \circ \cdots \circ \theta_m = \theta'_1 \circ \cdots \circ \theta'_{m'}$.

Remark 20.1: Why is the decomposition (20.1) always possible? Start with the sorted list $(1,\dots,n)$. If $\sigma(1) \neq 1$, then interchange 1 and $\sigma(1)$. We now have $(\sigma(1),\dots,1,\dots,n)$. Next, look at the second position. If it is not $\sigma(2)$, interchange it with $\sigma(2)$, and proceed similarly down the line. Of course, there are other ways to do it. For example, the permutation $(3,2,1)$ can be obtained in 3 steps:

$$(1,2,3) \to (2,1,3) \to (2,3,1) \to (3,2,1)$$

But, it can also be obtained in just one step: $(1,2,3) \to (3,2,1)$.

Lemma 20.b.1 (*Parity of Transpositions*): If $(\sigma(1),\dots,\sigma(n))$ is obtained from $(1,\dots,n)$ in two different ways—one time by a sequence of m transpositions, and another time by a sequence of m' transpositions—then m and m' must have the same parity. That is, m' is even if and only if m is even, and m' is odd if and only if m is odd.

In other words, $(-1)^{m'} = (-1)^m$. The parity depends only on the permutation σ, and not on the particular sequence of transpositions used.

A permutation σ is called "*even*" if m is even, and "*odd*" if m is odd. The number $\operatorname{sgn}(\sigma) := (-1)^m \in \{-1,1\}$ is called the "*sign*" of σ.

Proof of Lemma 20.b.1:

We know from previous results (Corollary 20.a.9) that determinant functions $E : \mathcal{M}_{n\times n}(K) \to K$ exist for $n \geq 1$. Let us fix one such determinant function E (we denote it by E to distinguish it from D in the previous discussion).

Consider the evaluation $E(\epsilon_{\sigma(1)},\dots,\epsilon_{\sigma(n)})$. If $(\sigma(1),\dots,\sigma(n))$ can be obtained from $(1,\dots,n)$ by a sequence of m transpositions, then the matrix with rows $\epsilon_{\sigma(1)},\dots,\epsilon_{\sigma(n)}$ can be obtained from the identity matrix I by a sequence of m interchanges of pairs of rows.

Because E is alternating, swapping two rows flips the sign of the output. Therefore:

$$E(\epsilon_{\sigma(1)},\dots,\epsilon_{\sigma(n)}) = (-1)^m E(\epsilon_1,\dots,\epsilon_n) = (-1)^m E(I) = (-1)^m.$$

But, this exact same logic applies to the sequence of m' transpositions. Thus:

$$E(\epsilon_{\sigma(1)},\dots,\epsilon_{\sigma(n)}) = (-1)^{m'} E(\epsilon_1,\dots,\epsilon_n) = (-1)^{m'} E(I) = (-1)^{m'}.$$

This identically implies that $(-1)^m = (-1)^{m'}$, proving the lemma. **Q.E.D.**

Remark 20.2 (*No Circularity Here*): The argument just given proves a fact about permutations by using determinants, and the Leibniz formula below will express the determinant using sgn. That is not a circle. What the proof needs is only Corollary 20.a.9, the **existence** of some determinant function, and that was obtained from Theorem 20.a.8 by a recursion on n in which the sign of a permutation never appears: the signs there are the explicit factors $(-1)^{i+j}$ attached to positions, not to permutations. Only after sgn has been shown to be well defined is it used, in Theorem 20.b.2, to write down the one determinant function in closed form.

Returning to our original function $D(A)$, we have seen that:

$$D(A) = \sum_{\sigma} A(1,\sigma(1)) \cdots A(n,\sigma(n)) \cdot D(\epsilon_{\sigma(1)}, \dots, \epsilon_{\sigma(n)}).$$

By the alternating property of D, performing the transpositions necessary to sort $(\sigma(1), \dots, \sigma(n))$ back to $(1, \dots, n)$ yields:

$$D(\epsilon_{\sigma(1)}, \dots, \epsilon_{\sigma(n)}) = \operatorname{sgn}(\sigma) \cdot D(\epsilon_1, \dots, \epsilon_n) = \operatorname{sgn}(\sigma) \cdot D(I). \tag{20.2}$$

Theorem 20.b.2 (*Leibniz Formula & Uniqueness*): For all $n \geq 1$, there exists a unique determinant function on $\mathcal{M}_{n\times n}(K)$. We will denote it by det. It is mathematically given by the formula:

$$\det(A) = \sum_{\sigma} \operatorname{sgn}(\sigma) \cdot A(1,\sigma(1)) \cdot A(2,\sigma(2)) \cdots A(n,\sigma(n)).$$

Proof:

Existence is Corollary 20.a.9, so only uniqueness is at issue, and that follows immediately from (20.2). Indeed, if D is a determinant function, then $D(I) = 1$. Substituting this into (20.2) yields the exact formula for $D(A)$ stated in the theorem, proving that the function is entirely unique.

Q.E.D.

Corollary 20.b.3 (*Scaling of Alternating Functions*): For any n-linear alternating function $D : \mathcal{M}_{n\times n}(K) \to K$, we have $D(A) = \det(A) \cdot D(I)$ for all $A \in \mathcal{M}_{n\times n}(K)$.

Proof:

Note that the expansion leading to (20.2) used only that D is n-linear and alternating, never that $D(I) = 1$. Substituting (20.2) into that expansion gives

$$D(A) = \sum_{\sigma} A(1,\sigma(1)) \cdots A(n,\sigma(n)) \cdot \operatorname{sgn}(\sigma) \cdot D(I) = \left(\sum_{\sigma} \operatorname{sgn}(\sigma) A(1,\sigma(1)) \cdots A(n,\sigma(n)) \right) D(I),$$

and the bracket is $\det(A)$ by Theorem 20.b.2. **Q.E.D.**

The Symmetric Group

Denote by S_n the set of permutations $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$. The set S_n has the algebraic structure of a group under composition: if $\sigma, \tau \in S_n$, their composition is $(\sigma \circ \tau)(x) = \sigma(\tau(x))$. The identity function id is the neutral element. The inverse σ^{-1} exists because σ is bijective. Furthermore, the size of the symmetric group is $|S_n| = n!$.

Exercise 20.2 (*The Order of the Symmetric Group*): Show that $|S_n| = n!$.

Lemma 20.b.4 (*Multiplicativity of Signum*): For all $\sigma, \tau \in S_n$, we have $\operatorname{sgn}(\sigma \circ \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau)$.

Proof:

Suppose $\sigma = \theta_1 \circ \cdots \circ \theta_m$ and $\tau = \lambda_1 \circ \cdots \circ \lambda_\ell$, where the θ_i's and λ_j's are all transpositions. Then, their composition is simply $\sigma \circ \tau = \theta_1 \circ \cdots \circ \theta_m \circ \lambda_1 \circ \cdots \circ \lambda_\ell$, which explicitly consists of $m + \ell$ transpositions. Therefore:

$$\operatorname{sgn}(\sigma \circ \tau) = (-1)^{m+\ell} = (-1)^m \cdot (-1)^\ell = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau).$$

Q.E.D.

Multiplicativity of the Determinant

Theorem 20.b.5 (*Multiplicativity of the Determinant*): For any matrices $A, B \in M_{n\times n}(K)$, we have $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Proof:

Fix a matrix $B \in \mathcal{M}_{n\times n}(K)$, and consider a function $D : \mathcal{M}_{n\times n}(K) \to K$ defined by $D(A) := \det(A \cdot B)$.

Claim: D is n-linear and alternating.

Proof of claim: Let the rows of A be $\alpha_1, \ldots, \alpha_n$. If we view α_i as a $1 \times n$ matrix, then the matrix product $\alpha_i \cdot B$ is also $1 \times n$. In fact, the rows of the product matrix $A \cdot B$ are exactly $\alpha_1 \cdot B, \ldots, \alpha_n \cdot B$. So, $D(A) = \det(\alpha_1 \cdot B, \ldots, \alpha_n \cdot B)$.

Clearly, for two row vectors α_i and α_i', we have $(\alpha_i + \alpha_i') \cdot B = \alpha_i \cdot B + \alpha_i' \cdot B$. Also, for any scalar $c \in K$, $(c\alpha_i) \cdot B = c(\alpha_i \cdot B)$. Since the determinant function det is n-linear with respect to its rows, it heavily follows that D is also n-linear.

Furthermore, if two rows of A are equal (i.e., $\alpha_i = \alpha_j$), then the corresponding rows of $A \cdot B$ are also equal ($\alpha_i \cdot B = \alpha_j \cdot B$). Because det is alternating, $D(A) = \det(\alpha_1 \cdot B, \ldots, \alpha_n \cdot B) = 0$. This rigorously proves the claim that D is alternating.

We continue with the proof of Theorem 20.b.5. By Corollary 20.b.3, any n-linear alternating function satisfies $D(A) = \det(A) \cdot D(I)$. We compute $D(I)$ using our definition: $D(I) = \det(I \cdot B) = \det(B)$. Substituting this back yields $D(A) = \det(A) \cdot \det(B)$, completing the proof. Q.E.D.

Corollary 20.b.6 (*Determinant of an Inverse*): If A is invertible, then $\det(A) \neq 0$. Moreover, $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof:

Assume A is invertible, and denote its inverse by A^{-1}. Since $A \cdot A^{-1} = I$, taking the determinant of both sides and applying Theorem 20.b.5 yields $\det(A) \cdot \det(A^{-1}) = \det(I) = 1$. This automatically implies that $\det(A)$ cannot be zero, and dividing by $\det(A)$ yields the expected formula.

Q.E.D.

More on Determinants: Properties of Determinants and the Transpose (Section 20.c)

Recall that for any matrix $M \in M_{m\times n}(K)$, we have the transposed matrix $M^\mathsf{T} \in M_{n\times m}(K)$, where the entries are given by $M^\mathsf{T}(i, j) = M(j, i)$.

Theorem 20.c.1 (*Determinant of the Transpose*): For all $A \in \mathcal{M}_{n\times n}(K)$, we have $\det(A^\mathsf{T}) = \det(A)$.

Proof:

If σ is a permutation of degree n, then by definition of the transpose, $A^\mathsf{T}(i,\sigma(i)) = A(\sigma(i),i)$. Therefore, expanding the determinant definition gives:

$$\det(A^\mathsf{T}) = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\cdot A(\sigma(1),1)\cdots A(\sigma(n),n).$$

If we let $j=\sigma(i)$, then $i=\sigma^{-1}(j)$, and consequently $A(\sigma(i),i) = A(j,\sigma^{-1}(j))$. Hence, we can rearrange the product completely:

$$A(\sigma(1),1)\cdots A(\sigma(n),n) = A(1,\sigma^{-1}(1))\cdots A(n,\sigma^{-1}(n)) \tag{20.3}$$

This holds because every factor on the left-hand side of (20.3) will appear exactly once on the right-hand side, and vice-versa.

Now, since $\sigma\circ\sigma^{-1} = \mathrm{id}$, we have $\operatorname{sgn}(\sigma)\cdot\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\mathrm{id}) = 1$, which implies $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\sigma^{-1})$. When σ runs over S_n (all possible permutations of degree n), its inverse σ^{-1} also naturally runs over the entirety of S_n.

This implies:

$$\begin{aligned}
\det(A^\mathsf{T}) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)A(\sigma(1),1)\cdots A(\sigma(n),n)\\
&= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma^{-1})A(1,\sigma^{-1}(1))\cdots A(n,\sigma^{-1}(n))\\
&= \sum_{\tau\in S_n} \operatorname{sgn}(\tau)A(1,\tau(1))\cdots A(n,\tau(n))\\
&= \det(A).
\end{aligned}$$

Q.E.D.

Example 20.6 (*Determinant of a Transposed Matrix*): Let $A = \left(\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right)$. Its transpose is $A^\mathsf{T} = \left(\begin{smallmatrix}1&3\\2&4\end{smallmatrix}\right)$. We have $\det(A) = (1)(4)-(2)(3) = -2$, and $\det(A^\mathsf{T}) = (1)(4)-(3)(2) = -2$. Thus, the theorem holds.

Corollary 20.c.2 (*Column Alternating Property*): If $D : \mathcal{M}_{n\times n}(K)\to K$ is an alternating n-linear function of the rows of the matrix, then D is also an alternating n-linear function of the columns of the matrix. In fact, $D(A^\mathsf{T}) = D(A)$ for all $A\in\mathcal{M}_{n\times n}(K)$. In particular, this holds for $D=\det$.

Proof:

For any $A\in M_{n\times n}(K)$, we have previously established $D(A) = \det(A)\cdot D(I)$. Applying this to A^T gives $D(A^\mathsf{T}) = \det(A^\mathsf{T})\cdot D(I)$. Since $\det(A^\mathsf{T}) = \det(A)$, this evaluates exactly to $\det(A)\cdot D(I) = D(A)$. Q.E.D.

Because the determinant acts symmetrically on rows and columns, we can perform column operations just as we do row operations.

Theorem 20.c.3 (*Invariance Under Row Operations*): Let $A\in\mathcal{M}_{n\times n}(K)$.

(a) If B is obtained from A by adding a multiple of one row of A to another row (i.e., $c\cdot E_j + E_i \to E_i$ for $i\neq j$), or by doing the analogous operation on columns, then $\det(B) = \det(A)$.

(b) If B is obtained from A by interchanging two rows (i.e., $E_i \leftrightarrow E_j$ for $i\neq j$), or columns, then $\det(B) = -\det(A)$.

(c) If B is obtained from A by multiplying a row, or a column, by a scalar (i.e., $c\cdot E_i\to E_i$), then $\det(B) = c\det(A)$.

Proof:

Statement **(b)** follows directly from the fact that the determinant is an alternating function. Statement **(c)** follows from the n-linearity of the determinant.

To prove **(a)**, let A have rows $\alpha_1, \ldots, \alpha_n$. Assume $j < i$. Expanding by linearity on the i-th row gives:

$$\begin{aligned} \det(B) &= \det(\alpha_1, \ldots, \alpha_i + c\alpha_j, \ldots, \alpha_j, \ldots, \alpha_n) \\ &= \det(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + c\det(\alpha_1, \ldots, \alpha_j, \ldots, \alpha_j, \ldots, \alpha_n) \\ &= \det(A) + 0 = \det(A). \end{aligned}$$

The second term identically vanishes because two rows are equal (α_j appears twice). If $j > i$, the proof is similar.

In each of the three statements, the assertion about columns follows from the one about rows: a column operation on A is the corresponding row operation on A^T, and $\det(A^\mathsf{T}) = \det(A)$ by Theorem 20.c.1. **Q.E.D.**

Determinants of Block Matrices

Calculating the determinant of a large matrix by expanding all permutations is computationally expensive. Block matrices provide a powerful shortcut. Consider a block matrix M of the form $M = \left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{r\times s}(K)$.

Proposition 20.c.4 (*Determinant of Block Matrices*): For a block matrix M defined as above, $\det(M) = \det(A)\cdot\det(C)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{r\times s}(K)$.

Proof:

Define a helper function $D(A,B,C) := \det\left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$ as a function $D : \mathcal{M}_{r\times r}(K) \times \mathcal{M}_{r\times s}(K) \times \mathcal{M}_{s\times s}(K) \to K$.

If we fix the first two matrices A and B, then the function mapping $C \mapsto D(A,B,C)$ is s-linear and alternating. By Corollary 20.b.3, any such function satisfies:

$$D(A,B,C) = \det(C)\cdot D(A,B,I_s). \tag{20.4}$$

Let us calculate $D(A,B,I_s) = \det\left(\begin{smallmatrix} A & B \\ 0 & I_s \end{smallmatrix}\right)$. By doing row operations of the type $R_i + cR_j \to R_i$ (where $j \neq i$), we can use the last s rows of the identity matrix I_s to eliminate the entries in B entirely. This transforms the matrix into $\left(\begin{smallmatrix} A & 0 \\ 0 & I_s \end{smallmatrix}\right)$.

Since these operations do not change the determinant, we have:

$$\det\begin{pmatrix} A & B \\ 0 & I_s \end{pmatrix} = \det\begin{pmatrix} A & 0 \\ 0 & I_s \end{pmatrix}. \tag{20.5}$$

The function $A \mapsto \det\left(\begin{smallmatrix} A & 0 \\ 0 & I_s \end{smallmatrix}\right)$ is r-linear and alternating. By Corollary 20.b.3 again, it evaluates to $\det(A)\cdot\det\left(\begin{smallmatrix} I_r & 0 \\ 0 & I_s \end{smallmatrix}\right) = \det(A)\cdot\det(I_{r+s}) = \det(A)\cdot 1 = \det(A)$.

Going back through (20.5) and then to (20.4), we finally get that $D(A,B,C) = \det(A)\cdot\det(C)$. **Q.E.D.**

Exercise 20.3 (*The Other Block Shape*): Show that $\det\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right) = \det(A)\cdot\det(C)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{s\times r}(K)$. (Hint: Use column operations to reduce B to zero, or consider the transpose of the matrix and apply Proposition 20.c.4.)

Example 20.7 (*Calculation of a 4×4 Determinant*): Let $A = \left(\begin{smallmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & -1 & -1 \\ 1 & 2 & 3 & 0 \end{smallmatrix}\right) \in \mathcal{M}_{4\times 4}(\mathbb{R})$. We begin by performing row operations $R_2 - 2R_1 \to R_2$, $R_3 - 4R_1 \to R_3$, and $R_4 - R_1 \to R_4$. This transforms A into:

$$\begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix}$$

Now, we clean the second column below the diagonal using $R_3 - \frac{5}{4}R_2 \to R_3$ and $R_4 - \frac{3}{4}R_2 \to R_4$. This reveals a beautiful block structure:

$$\left(\begin{array}{cc|cc} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ \hline 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{array}\right)$$

By applying Proposition 20.c.4 to this partitioned form, we can see that the determinant of the 4×4 matrix is simply the product of the determinants of the 2×2 diagonal blocks:

$$\det(A) = \det \begin{pmatrix} 1 & -1 \\ 0 & 4 \end{pmatrix} \cdot \det \begin{pmatrix} -4 & -8 \\ 4 & 0 \end{pmatrix}$$

Consequently, we evaluate the individual pieces:

$$\det(A) = (4) \cdot (32) = 128.$$

Cofactor Expansion and the Classical Adjoint

Recall that if D is a determinant function on $\mathcal{M}_{(n-1)\times(n-1)}(K)$, then for any fixed $j \in \{1, \dots, n\}$, the function $E_j(A) := \sum_{i=1}^{n}(-1)^{i+j}A_{ij}D(A(i|j))$ is a determinant function as well. But, we have seen that on $\mathcal{M}_{n\times n}(K)$ there is a strictly unique determinant function, $\det$.

Corollary 20.c.5 (*Laplace Expansion*): For all $A \in \mathcal{M}_{n\times n}(K)$, we have the following n different formulas for $\det(A)$ (expanding along column j):

$$\det(A) = \sum_{i=1}^{n}(-1)^{i+j}A_{ij} \det A(i|j), \quad \text{for any } j = 1, \dots, n.$$

The scalars $C_{ij} := (-1)^{i+j} \det A(i|j)$ are called the "*i, j cofactor*" of A. We can succinctly write the expansion of $\det(A)$ by the cofactors of the j-th column as $\det(A) = \sum_{i=1}^{n} A_{ij} \cdot C_{ij}$.

Lemma (*Orthogonality of Cofactors*): If we replace A_{ij} in the formula by A_{ik} (where k is fixed), we always get 0. In other words, for all $A \in \mathcal{M}_{n\times n}(K)$ and for $j \neq k$, we have $\sum_{i=1}^{n} A_{ik} \cdot C_{ij} = 0$.

Proof:

Let $B \in \mathcal{M}_{n\times n}(K)$ be the matrix obtained from A by replacing (not exchanging!) column j of A by column k of A. So, in B, column j equals column k. Clearly, $\det(B) = 0$ since B has two equal columns.

Expanding $\det(B)$ along the j-th column gives:

$$0 = \det(B) = \sum_{i=1}^{n}(-1)^{i+j}B_{ij} \det B(i|j).$$

By construction, $B_{ij} = A_{ik}$ for all i. Furthermore, since we only modified column j, deleting column j from both matrices leaves identical submatrices: $B(i|j) = A(i|j)$ for all i. Thus:

$$0 = \sum_{i=1}^{n}(-1)^{i+j}A_{ik} \det A(i|j) = \sum_{i=1}^{n} A_{ik} \cdot C_{ij}.$$

Q.E.D.

Summary 20.1: For $A \in \mathcal{M}_{n\times n}(K)$, we have $\sum_{i=1}^{n} A_{ik} \cdot C_{ij} = \delta_{jk} \cdot \det(A)$, where δ_{jk} is the Kronecker delta (which is 1 if $j = k$, and 0 otherwise).

Definition (*Classical Adjoint*): The transpose of the matrix of cofactors, $(C_{ij})^{\mathsf{T}}$, is called the "*classical adjoint*" (or "*adjugate*") of A, denoted $\operatorname{adj} A \in \mathcal{M}_{n\times n}(K)$. Explicitly, $(\operatorname{adj} A)_{ij} = C_{ji} = (-1)^{i+j} \det A(j|i)$.

Summary 20.1 immediately implies the matrix equation:

$$(\operatorname{adj} A) \cdot A = (\det A) \cdot I \tag{20.6}$$

The bookkeeping is worth doing once. The entry of $(\operatorname{adj} A) \cdot A$ in place (j, k) is

$$\big((\operatorname{adj} A) \cdot A\big)_{jk} = \sum_{i=1}^{n} (\operatorname{adj} A)_{ji}\, A_{ik} = \sum_{i=1}^{n} C_{ij}\, A_{ik} = \delta_{jk} \cdot \det A,$$

using $(\operatorname{adj} A)_{ji} = C_{ij}$ from the Definition on Page 183 for the middle equality and Summary 20.1 for the last, and $\delta_{jk} \det A$ is exactly the entry of $(\det A) \cdot I$ in that place.

Lemma (*Left Adjugate Identity*): It is also true that $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$.

Proof:

Since $A^{\mathsf{T}}(i|j) = (A(j|i))^{\mathsf{T}}$, we have

$$(\operatorname{adj} A^{\mathsf{T}})_{ij} = (-1)^{i+j} \det A^{\mathsf{T}}(j|i) = (-1)^{i+j} \det(A(i|j)^{\mathsf{T}}) = (-1)^{i+j} \det A(i|j).$$

This perfectly matches $(\operatorname{adj} A)_{ji}$, which is exactly $((\operatorname{adj} A)^{\mathsf{T}})_{ij}$. Thus, $\operatorname{adj} A^{\mathsf{T}} = (\operatorname{adj} A)^{\mathsf{T}}$. Applying (20.6) to the matrix A^{T} gives $(\operatorname{adj} A^{\mathsf{T}}) \cdot A^{\mathsf{T}} = (\det A^{\mathsf{T}}) \cdot I$. Using our transpose identities, this becomes $(\operatorname{adj} A)^{\mathsf{T}} \cdot A^{\mathsf{T}} = (\det A) \cdot I$. Taking the transpose of both sides finally yields $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$. **Q.E.D.**

Theorem 20.c.6 (*Inverse via Adjugate*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then, A is invertible if and only if $\det A \neq 0$. When A is invertible, its inverse is given by $A^{-1} = \frac{1}{\det A} \cdot (\operatorname{adj} A)$.

Proof:

One direction is Corollary 20.b.6: if A is invertible then $\det A \neq 0$.

Conversely, assume $\det A \neq 0$. Then the scalar $\frac{1}{\det A} \in K$ exists, and the two adjugate identities (20.6) and Page 183 give

$$A \cdot \left(\frac{1}{\det A} \operatorname{adj} A\right) = \frac{1}{\det A}(A \cdot \operatorname{adj} A) = \frac{1}{\det A}(\det A) \cdot I = I,$$

and in the same way $\left(\frac{1}{\det A} \operatorname{adj} A\right) \cdot A = I$. So $\frac{1}{\det A} \operatorname{adj} A$ is a two-sided inverse of A, which makes A invertible with $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$. **Q.E.D.**

Cramer's Rule

Let $A \in \mathcal{M}_{n\times n}(K)$. We would like an explicit formula to solve the system of linear equations $A \cdot x = b$, where $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ are the unknowns and $b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n$ is a given vector.

Assume $A \cdot x = b$. Multiplying both sides by the adjoint gives $(\operatorname{adj} A) \cdot A \cdot x = (\operatorname{adj} A) \cdot b$. This directly implies $(\det A) \cdot x = (\operatorname{adj} A) \cdot b$. Looking at the j-th row of this vector equation, we get:

$$\begin{aligned}(\det A) \cdot x_j &= \sum_{i=1}^{n} (\operatorname{adj} A)_{ji} \cdot b_i \\ &= \sum_{i=1}^{n} (-1)^{i+j} b_i \det A(i|j) \\ &= \det(A(j,b))\end{aligned}$$

where $A(j,b)$ is the matrix obtained from A by replacing column j with the vector b.

Theorem 20.c.7 (*Cramer's Rule*): Let $A \in \mathcal{M}_{n\times n}(K)$ with $\det A \neq 0$. Let $b = (b_1, \dots, b_n)^\mathsf{T} \in K^n$. Then, the system $Ax = b$, where $x = (x_1, \dots, x_n)^\mathsf{T}$, has a unique solution, and it is given by the formula:

$$x_j = \frac{\det A(j,b)}{\det A} \quad \text{for all } 1 \leq j \leq n,$$

where $A(j,b)$ is the matrix obtained from A by replacing column j with b.

Proof of Theorem 20.c.7:

Since $\det A \neq 0$, the matrix A is invertible by Theorem 20.c.6. Then $x := A^{-1}b$ solves the system, because $A(A^{-1}b) = b$, and it is the only solution: any x with $Ax = b$ satisfies $x = A^{-1}(Ax) = A^{-1}b$. So the system has a unique solution, and there is no need to appeal to Linear Algebra I for it. We have just shown above that $(\det A) \cdot x_j = \det A(j,b)$. Since $\det A \neq 0$, we can safely divide to obtain $x_j = \frac{\det A(j,b)}{\det A}$. **Q.E.D.**

Example 20.8 (*Solving a System using Cramer's Rule*): Consider the system $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 5 \\ 5 \end{smallmatrix}\right)$. We have $\det(A) = 2(3) - 1(1) = 5$. Using Cramer's rule:

$$x_1 = \frac{\det \begin{pmatrix} 5 & 1 \\ 5 & 3 \end{pmatrix}}{5} = \frac{15-5}{5} = 2, \quad x_2 = \frac{\det \begin{pmatrix} 2 & 5 \\ 1 & 5 \end{pmatrix}}{5} = \frac{10-5}{5} = 1.$$

Thus, the solution is $x = \binom{2}{1}$.

Determinants of Endomorphisms

Warm up 20.1: Let $A \in \mathcal{M}_{n\times n}(K)$, and let $P \in \mathrm{GL}(n,K)$ (meaning P is an invertible $n \times n$ matrix). Then, by multiplicativity (Theorem 20.b.5):

$$\det(P^{-1} \cdot A \cdot P) = (\det P^{-1}) \cdot (\det A) \cdot (\det P) = \frac{1}{\det P} \cdot \det A \cdot \det P = \det A.$$

In other words, similar matrices mathematically have the exact same determinant.

Let V be a finite-dimensional vector space over K, and put $n := \dim V$. Let $T : V \to V$ be a linear map (an endomorphism). Pick a basis $\mathcal{B}$ for V. We have a matrix representation $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_{n\times n}(K)$, and we can consider its determinant $\det[T]_{\mathcal{B}}^{\mathcal{B}} \in K$.

Question 20.1: Does $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ depend on the specific choice of the basis $\mathcal{B}$?

Answer 20.1: Let $\mathcal{B}'$ be another arbitrary basis for V. Denote by $P := [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$ the transition matrix between the basis $\mathcal{B}'$ and $\mathcal{B}$. Recall that $[T]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$. We also formally have $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = ([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} = P^{-1}$. So, $[T]_{\mathcal{B}'}^{\mathcal{B}'} = P^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot P$. This implies $\det[T]_{\mathcal{B}'}^{\mathcal{B}'} = \det[T]_{\mathcal{B}}^{\mathcal{B}}$.

Recall also that if $T, S : V \to V$ are two linear maps, then $[S \circ T]_{\mathcal{B}}^{\mathcal{B}} = [S]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}$. If T is an isomorphism, then $[T^{-1}]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^{-1}$. Finally, $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I$.

Corollary 20.c.8 (*Determinant of an Endomorphism*)**:** Let V be a finite-dimensional vector space over K, and let $T: V \to V$ be a linear map. Then, $\det([T]^{\mathcal{B}}_{\mathcal{B}})$ does NOT depend on the choice of the basis $\mathcal{B}$ of V. We denote this intrinsic scalar simply by $\det(T) \in K$.

Proposition (*Properties of the Determinant of an Endomorphism*)**:** Let V be a vector space over K with $1 \le \dim V < \infty$. The function $\det : \mathrm{End}(V) \to K$, $T \mapsto \det(T)$, has the following elegant properties:

(a) $\det(S \circ T) = \det(S) \cdot \det(T)$ for all $S, T \in \mathrm{End}(V)$.

(b) T is an isomorphism if and only if $\det T \neq 0$. If T is an isomorphism, then $\det(T^{-1}) = \frac{1}{\det T}$.

(c) $\det(\mathrm{id}_V) = 1$.

(d) $\det(0) = 0 \in K$ (where 0 is the zero linear map).

Proof:

Fix a basis $\mathcal{B}$ of V and write $n := \dim V$, so that $\det(T) = \det[T]^{\mathcal{B}}_{\mathcal{B}}$ by Corollary 20.c.8. Each statement is then the corresponding statement about matrices.

(a) The composition rule gives $[S \circ T]^{\mathcal{B}}_{\mathcal{B}} = [S]^{\mathcal{B}}_{\mathcal{B}} \cdot [T]^{\mathcal{B}}_{\mathcal{B}}$, and Theorem 20.b.5 turns the product of matrices into the product of determinants.

(b) By Proposition 17.b.3, T is an isomorphism precisely when $[T]^{\mathcal{B}}_{\mathcal{B}}$ is invertible, which by Theorem 20.c.6 happens precisely when $\det[T]^{\mathcal{B}}_{\mathcal{B}} \neq 0$. The formula for $\det(T^{-1})$ is then Corollary 20.b.6.

(c) $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = I_n$ and $\det(I_n) = 1$, the latter being the normalisation in Definition 20.a.6.

(d) $[0]^{\mathcal{B}}_{\mathcal{B}}$ is the zero matrix, and every summand of the Leibniz formula of Theorem 20.b.2 then carries a factor 0. This is where $\dim V \ge 1$ is needed: for $n \ge 1$ each summand has at least one factor. **Q.E.D.**

Important remark 20.1: In contrast to an endomorphism $T : V \to V$, if we take a linear map $T : V \to W$ between two vector spaces of the same dimension and bases $\mathcal{B}, \mathcal{C}$ of V, W, then $\det[T]^{\mathcal{B}}_{\mathcal{C}}$ generally depends on the choice of $\mathcal{B}$ and $\mathcal{C}$.

Solutions to the Exercises

Proof of Exercise 20.1:

Write $u := \binom{a}{c}$ and $v := \binom{b}{d}$ for the two columns. By Proposition 9.7, the list (u, v) is linearly dependent exactly when one of the two vectors is a linear combination of the other. Explicitly: dependence gives scalars $\mu, \nu \in K$, not both zero, with $\mu u + \nu v = 0$. If $\mu \neq 0$ then $u = -\frac{\nu}{\mu} v$, which is the first alternative with $\lambda := -\frac{\nu}{\mu}$; and if $\mu = 0$ then $\nu \neq 0$, so $v = 0 = 0 \cdot u$, which is the second alternative with $\tau := 0$. Conversely, either alternative exhibits a non-trivial vanishing combination, so the columns are dependent. Note that the case $u = 0$ is contained in the first alternative, taking $\lambda := 0$. **Q.E.D.**

Proof of Exercise 20.2:

A permutation $\sigma \in S_n$ is determined by the list $(\sigma(1), \ldots, \sigma(n))$, which is an arrangement of $1, \ldots, n$ with no repetitions. Building such a list, there are n choices for $\sigma(1)$; once it is fixed, $\sigma(2)$ may be any of the remaining $n-1$ values; then $n-2$ for $\sigma(3)$, and so on, with a single choice left for $\sigma(n)$. Distinct sequences of choices give distinct permutations and every permutation arises exactly once, so

$$|S_n| = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1 = n!.$$

Q.E.D.

Proof of Exercise 20.3:

Take the transpose. By Theorem 20.c.1, $\det\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right) = \det\left(\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right)^{\mathsf{T}}\right)$, and transposing a block matrix transposes each block and reflects the arrangement, giving

$$\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right)^{\mathsf{T}} = \begin{pmatrix} A^{\mathsf{T}} & B^{\mathsf{T}} \\ 0 & C^{\mathsf{T}} \end{pmatrix},$$

where now $A^{\mathsf{T}} \in \mathcal{M}_{r\times r}(K)$, $C^{\mathsf{T}} \in \mathcal{M}_{s\times s}(K)$ and $B^{\mathsf{T}} \in \mathcal{M}_{r\times s}(K)$. This is exactly the shape of Proposition 20.c.4, so

$$\det\begin{pmatrix} A & 0 \\ B & C \end{pmatrix} = \det\begin{pmatrix} A^{\mathsf{T}} & B^{\mathsf{T}} \\ 0 & C^{\mathsf{T}} \end{pmatrix} = \det(A^{\mathsf{T}})\cdot\det(C^{\mathsf{T}}) = \det(A)\cdot\det(C),$$

using Theorem 20.c.1 once more on each factor. **Q.E.D.**

AI-Note: This chapter came through in good order. All of 20.a.1–20.a.9, 20.b.1–20.b.6 and 20.c.1–20.c.8 are present, in the notes' order, with the notes' proofs, and the printed numbers already matched, including the deliberate use of unnumbered `lemma*` and `definition*` around the cofactor identity and the classical adjoint, which is what keeps 20.c.6 landing on the theorem about the inverse.

Three repairs. Theorem 20.c.6 was stated without any proof; it is the point the whole cofactor discussion has been building to, and the two lines it needs are supplied. The exercise inside the proof of Proposition 20.a.1 stated the dichotomy as *"either the first column is the zero vector or the second column is a scalar multiple of the first"*, while the proof two lines below it uses the notes' version, either column is a multiple of the other — and the zero case is subsumed there; the statement now matches what is used. And a sentence in the proof of Page 183 was left behind a second time as a commented-out copy.

Prof. Biran marks three things *"exc."*: the column dichotomy just mentioned, $|S_n| = n!$, and the lower-triangular block shape. Two of them stood in the text as `remark[Exercise]` and one as a parenthesis; all three are now exercises, solved above.

The rigour pass supplied three missing proofs and repaired one statement. Proposition 20.a.5 opened *"let D be an alternating function"* and then offered *"D is alternating"* as one of two equivalent conditions, assuming what it concludes; its hypothesis is that D is n-linear and vanishes on matrices with two equal **adjacent** rows, and both listed statements are conclusions drawn from that. Its list, and that of Theorem 20.c.3, also printed as $1, 2, 3$ while both proofs refer to **(a)**, **(b)** and **(c)**.

Corollary 20.a.9 and Corollary 20.b.3 were both stated without proof. The first carries Lemma 20.b.1, and the second is the engine of Theorem 20.b.5, Proposition 20.c.4 and Corollary 20.c.2; each proof is three lines. Theorem 20.b.2 asserts existence as well as uniqueness and its proof addressed only uniqueness, which is now said. The order of these results looks circular and is not, so Remark 20.2 says once why: existence comes from a recursion in which the sign of a permutation never appears.

The four properties of $\det : \mathrm{End}(V) \to K$ stood in running prose with no environment, although Remark 19.4 in the previous chapter turns on the second of them; they are now the unnumbered Proposition on Page 185, with proofs, so 20.c.1–20.c.8 do not move. Statement **(d)** needs $\dim V \geq 1$, which is now a hypothesis.

Smaller things: the inverse in Proposition 20.a.1 was verified in one order only, and an inverse must satisfy the identity in both; the rank criterion the same proof opens with is AI-Proposition 17.1 from ch. 17e; the passage from Summary 20.1 to (20.6), which the text calls immediate, is one line of index bookkeeping and is now written out; and the proof of Theorem 20.c.7 appealed to *"Linear Algebra I"* for the unique solubility of $Ax = b$, which Theorem 20.c.6 two results above already gives.

Eigenvalues & Eigenvectors (Chapter 21)

Motivation (Section 21.a)

Fibonacci revisited

Consider the sequence $f_0, f_1, f_2, f_3, \dots, f_n, \dots$ defined by $f_0 = 0$, $f_1 = 1$, and $f_n := f_{n-1} + f_{n-2}$ for $n \geq 2$. This yields the sequence: $0, 1, 1, 2, 3, 5, 8, 13, 21, \dots$

We found a formula for f_n:

$$f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right) \quad \forall n \geq 0.$$

We defined $\mathrm{Fib} :=$ the set of all sequences $(a_0, a_1, a_2, \dots, a_n, \dots)$ that satisfy $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$.

Claim: $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ is a 2-dimensional vector subspace of the space of all infinite sequences

$$(x_0, x_1, x_2, \dots, x_n, \dots).$$

In fact, the map $F : \mathrm{Fib} \longrightarrow \mathbb{R}^2$ given by $F(a_0, a_1, \dots, a_n, \dots) := (a_0, a_1) \in \mathbb{R}^2$ is an isomorphism.

Claim: Put $\varphi := \frac{1+\sqrt{5}}{2}$ and $\psi := \frac{1-\sqrt{5}}{2}$. Then the two geometric sequences $(1, \varphi, \varphi^2, \dots, \varphi^n, \dots)$ and $(1, \psi, \psi^2, \dots, \psi^n, \dots)$ belong to Fib and, moreover, they form a basis for Fib.

We also saw that using these two sequences, we could easily find an explicit formula for all Fibonacci sequences. But how did we find these two special sequences?

The shift map $S : \mathrm{Fib} \longrightarrow \mathrm{Fib}$ defined by $S(a_0, a_1, \dots, a_n, \dots) := (a_1, a_2, \dots, a_{n+1}, \dots)$ is a linear map. Let us try to find a Fibonacci sequence $\mathcal{F} \neq 0$ such that $S(\mathcal{F}) = \lambda \cdot \mathcal{F}$ for some λ.

$$S(\mathcal{F}) = \lambda \cdot \mathcal{F} \iff \lambda = \frac{1 \pm \sqrt{5}}{2} \text{ and } \mathcal{F} = (a, a\lambda, a\lambda^2, \dots, a\lambda^n, \dots)$$

Definition 21.a.1 (*Eigenvalue and Eigenvector*): Let V be a vector space over K and $T : V \to V$ a linear map (an endomorphism of V). We say that $\lambda \in K$ is an "*eigenvalue*" (a.k.a. "*characteristic value*") of T if there exists a vector $v \neq 0$ such that $Tv = \lambda v$. Any vector $v \neq 0$ with the property that $Tv = \lambda v$ is called an "*eigenvector*" (of T) for the eigenvalue λ.

Definition 21.a.2 (*Diagonalizable Endomorphism*): Let V be a finite-dimensional vector space over K, put $n := \dim V$, and let $T \in \mathrm{End}(V)$. We say that T is "*diagonalizable*" if there exists a basis $\mathcal{B} := (v_1, \dots, v_n)$ for V such that the representation matrix of T with respect to $\mathcal{B}$ is diagonal, that is,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} \tag{21.1}$$

for some $\lambda_1, \dots, \lambda_n \in K$.

Lemma 21.a.3 (*Diagonalizable Matrix Criterion*): T is diagonalizable if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ consisting of eigenvectors of T (i.e., $\forall i$, $Tv_i = \lambda_i \cdot v_i$ for some $\lambda_i \in K$). Moreover, if for some basis $\mathcal{B} = (v_1, \dots, v_n)$ the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape (21.1), then for all i, λ_i is an eigenvalue of T and v_i is an eigenvector for λ_i.

Proof:

Recall that $[T]_{\mathcal{B}}^{\mathcal{B}} = ([Tv_1]_{\mathcal{B}} \dots [Tv_n]_{\mathcal{B}})$ and $[Tv_j]_{\mathcal{B}} = \begin{pmatrix} \alpha_{1j} \\ \vdots \\ \alpha_{nj} \end{pmatrix}$, where $Tv_j = \alpha_{1j}v_1 + \dots + \alpha_{nj}v_n$.

The matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape (21.1) if and only if $\alpha_{ij} = 0$ for all $i \neq j$, and $\alpha_{kk} = \lambda_k \iff Tv_k = \lambda_k \cdot v_k$. **Q.E.D.**

Remark 21.1: We saw that for all $A \in \mathcal{M}_{n\times n}(K)$, we have a linear map $T_A : K^n \to K^n$, $T_A(v) := A \cdot v$. We say that a matrix $A \in \mathcal{M}_{n\times n}(K)$ is diagonalizable if the linear map T_A is diagonalizable.

Example 21.1 (*Stretching Transformation in $\mathbb{R}^2$*): Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $K = \mathbb{R}$. $T_A\binom{1}{1} = \binom{3}{3} = 3\binom{1}{1}$. Thus, $\lambda = 3$ is an eigenvalue and $v = \binom{1}{1}$ is a corresponding eigenvector. Furthermore, $T_A\binom{1}{-1} = \binom{1}{-1}$. This implies that $\lambda = 1$ is an eigenvalue and $v = \binom{1}{-1}$ is a corresponding eigenvector. Geometrically, this represents a stretching by a factor of 3 in $\mathbb{R}^2$ along one axis and the identity map along the other axis.

Example 21.2 (*Finding Eigenvalues via Quadratic Equations*): Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Let us find all possible eigenvalues and eigenvectors. We are looking for $A \cdot v = \lambda v$ with $\lambda \in \mathbb{R}$ and $0 \neq v \in \mathbb{R}^2$. This gives $(A - \lambda I)v = 0$. This implies that $A - \lambda I$ is not invertible, which in turn means $\det(A - \lambda I) = 0$ (which leads to a quadratic equation).

Proposition 21.a.4 (*Linear Independence of Eigenvectors*): Let V be a vector space over K and $T : V \to V$ a linear map. Suppose that $\lambda_1, \dots, \lambda_n \in K$ are eigenvalues of T which are pairwise distinct (i.e., $\lambda_i \neq \lambda_j$ for all $i \neq j$). Let $v_1, \dots, v_n$ be eigenvectors for $\lambda_1, \dots, \lambda_n$ respectively. Then $v_1, \dots, v_n$ are linearly independent. (Put shortly: eigenvectors corresponding to a list of distinct eigenvalues are always linearly independent).

Proof:

We proceed by induction on n. Base case $n = 1$: v_1 is an eigenvector for λ_1. By Definition 21.a.1, $v_1 \neq 0$, which implies that v_1 is linearly independent.

Inductive step for $n \geq 1$: Suppose the proposition holds for any list of n eigenvalues and eigenvectors. We will prove now that the proposition holds also for a list of $n+1$ eigenvalues and eigenvectors. Let $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ be pairwise distinct eigenvalues of T, and $v_1, \dots, v_n, v_{n+1}$ a given list of eigenvectors of $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ respectively. We need to prove that $v_1, \dots, v_n, v_{n+1}$ are linearly independent. Assume that

$$a_1v_1 + \dots + a_nv_n + a_{n+1}v_{n+1} = 0. \tag{21.2}$$

We need to show that $a_1 = \dots = a_n = a_{n+1} = 0$. Apply the transformation T to (21.2):

$$a_1T(v_1) + \dots + a_nT(v_n) + a_{n+1}T(v_{n+1}) = 0.$$

But since $T(v_i) = \lambda_i v_i$, we have:

$$a_1\lambda_1v_1 + \dots + a_n\lambda_nv_n + a_{n+1}\lambda_{n+1}v_{n+1} = 0. \tag{21.3}$$

Consider now the combination $(21.3) - \lambda_{n+1} \cdot (21.2)$:

$$a_1(\lambda_1 - \lambda_{n+1})v_1 + \dots + a_n(\lambda_n - \lambda_{n+1})v_n + a_{n+1}(\lambda_{n+1} - \lambda_{n+1})v_{n+1} = 0.$$

By the induction hypothesis, $v_1, \dots, v_n$ are linearly independent. Therefore, $a_1(\lambda_1 - \lambda_{n+1}) = \dots = a_n(\lambda_n - \lambda_{n+1}) = 0$. By assumption, the eigenvalues are distinct, so $\lambda_i \neq \lambda_{n+1}$. Consequently, $a_1 = \dots = a_n = 0$. Going back to (21.2), we get $a_{n+1}v_{n+1} = 0$. But since $v_{n+1} \neq 0$ (because v_{n+1} is an eigenvector), this implies that $a_{n+1} = 0$. **Q.E.D.**

The Characteristic Polynomial

Proposition 21.a.5 (*Eigenvalue Injective Criterion*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Then λ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot \mathrm{id}_V$ is not injective.

Proof:

Every step below is an equivalence, which is what the statement requires. The scalar λ is an eigenvalue if and only if there exists a non-zero vector $v \in V$ such that $Tv - \lambda v = 0$, by Definition 21.a.1. This says exactly that $(T - \lambda \cdot \mathrm{id}_V)v = 0$ for some $v \neq 0$, that is, that $\ker(T - \lambda \cdot \mathrm{id}_V) \neq \{0_V\}$. And by Proposition 15.b.2 **(a)**, a linear map is injective precisely when its kernel is trivial, so the last condition holds if and only if $T - \lambda \cdot \mathrm{id}_V$ fails to be injective. **Q.E.D.**

Definition 21.a.6 (*Characteristic Polynomial*): Let V be a finite-dimensional vector space over K and $T \in \operatorname{End}(V)$. We define the "*characteristic polynomial*" of T as $P_T(x) := \det(T - x \cdot \mathrm{id}_V)$, where x is a formal variable. We define the characteristic polynomial of a matrix $A \in \mathcal{M}_{n\times n}(K)$ in the same way: $P_A(x) := \det(A - x \cdot I_n)$.

Remark 21.2 (*How to Read* $\det(A - xI_n)$): Two words are in order about what Definition 21.a.6 means.

First, the entries of $A - x \cdot I_n$ are not scalars but elements of $K[x]$, whereas det was defined in ch. 20 for matrices over a field. What is meant is the Leibniz formula of Theorem 20.b.2,

$$\det B = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot B_{1,\sigma 1} \cdots B_{n,\sigma n},$$

which is a polynomial expression in the entries and therefore makes sense verbatim over any commutative ring, $K[x]$ included. Read that way, $P_A(x)$ is an element of $K[x]$. Evaluation at a scalar $\mu \in K$ is a ring homomorphism $K[x] \to K$, so it may be carried inside that sum; consequently

$$P_A(\mu) = \det(A - \mu \cdot I_n),$$

the determinant on the right now being an ordinary one, over K. We use this repeatedly and it is the only bridge between the formal polynomial and the scalars.

Second, $P_T(x)$ is defined through a basis, since det of an endomorphism is. It does not depend on which one: two bases give similar matrices, by the computation in Corollary 20.c.8, and similar matrices have equal characteristic polynomials by Lemma 21.a.8 **(e)**. So $P_T(x) = P_{[T]_{\mathcal{B}}^{\mathcal{B}}}(x)$ for every basis $\mathcal{B}$, which is what the proof of Remark 21.3 uses.

AI-Proposition 21.1 (*Eigenvalues are the Roots of the Characteristic Polynomial*): Let V be a finite-dimensional vector space over K with $\dim V \geq 1$, let $T \in \operatorname{End}(V)$ and let $\lambda \in K$. Then λ is an eigenvalue of T if and only if $P_T(\lambda) = 0$.

Proof:

Each step is an equivalence. By Proposition 21.a.5, λ is an eigenvalue of T if and only if $T - \lambda \cdot \mathrm{id}_V$ is not injective. Its domain and target are both V, so Corollary 15.b.10 **(c)** applies and injectivity is the same as bijectivity here; by Lemma 15.b.4, that is the same as being an isomorphism. So λ is an eigenvalue if and only if $T - \lambda \cdot \mathrm{id}_V$ is **not** an isomorphism, which by part **(b)** of the Proposition on Page 185 happens exactly when

$$\det(T - \lambda \cdot \mathrm{id}_V) = 0.$$

Finally, that scalar determinant is the value $P_T(\lambda)$ of the formal polynomial, by the evaluation identity of Remark 21.2. **Q.E.D.**

Definition 21.a.7 (*Eigenspace*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let λ be an eigenvalue of T. The "*eigenspace*" corresponding to λ is defined as:

$$\mathrm{Eig}_T(\lambda) := \{v \in V \mid Tv = \lambda v\} = \ker(T - \lambda \cdot \mathrm{id}_V) = \{v \in V \mid v \text{ is an eigenvector of } \lambda\} \cup \{0\}.$$

Clearly (by this definition), $\mathrm{Eig}_T(\lambda) \subseteq V$ is a linear subspace.

Lemma 21.a.8 (*Properties of the Characteristic Polynomial*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then:

(a) $P_A(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$ (i.e., $P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + \cdots + c_0$, with $c_i \in K$ for all i, and $c_n = (-1)^n$).

(b) $P_{A^\intercal}(x) = P_A(x)$.

(c) If $A = \begin{pmatrix} \lambda_1 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}$ has upper triangular form, then $P_A(x) = (\lambda_1 - x)\cdot\dots\cdot(\lambda_n - x)$.

(d) If $A = \left(\begin{array}{c|c} B_1 & * \\ \hline 0 & B_2 \end{array}\right)$ is a block matrix, then $P_A(x) = P_{B_1}(x) \cdot P_{B_2}(x)$.

(e) If A and B are similar matrices (i.e., $B = P^{-1}AP$ for some invertible matrix P), then $P_A(x) = P_B(x)$.

Proof:

(a) Let

$$P_A(x) = \det(A - x \cdot I_n) = \det \begin{pmatrix} A_{11} - x & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} - x & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} - x \end{pmatrix}.$$

Denote this matrix (inside the determinant) by B. This determinant equals

$$\sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \cdot B_{1,\sigma 1} \dots B_{n,\sigma n} = (A_{11} - x) \cdot (A_{22} - x) \dots (A_{nn} - x)$$
$$+ \text{(terms from the permutations } \sigma \neq \mathrm{id}).$$

If $\sigma \neq \mathrm{id} = \left(\begin{smallmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{smallmatrix}\right)$, then at least one factor in the product $B_{1,\sigma 1} \dots B_{n,\sigma n}$ does not contain x, so each such term has degree $\leq n-1$. Consequently, the whole determinant equals

$$(-1)^n x^n + (\text{terms of degree} \leq n-1).$$

(b) $P_{A^\intercal}(x) = \det(A^\intercal - x \cdot I_n) = \det((A - x \cdot I_n)^\intercal) = \det(A - x \cdot I_n) = P_A(x)$.

(c)+(d) These properties follow directly from the definition of the determinant and its behavior with block matrices, as established in Chapter 20 (Determinants). For example, for (c), the determinant of an upper triangular matrix is the product of its diagonal entries. For (d), the determinant of a block triangular matrix is the product of the determinants of its diagonal blocks.

(e) $P_B(x) = \det(B - x \cdot I_n) = \det(P^{-1}AP - x \cdot I_n) = \det(P^{-1}(A - x \cdot I_n)P) = \det(A - x \cdot I_n) = P_A(x)$.

Q.E.D.

Remark 21.3: If V is an n-dimensional vector space over K and $T \in \mathrm{End}(V)$, then $P_T(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$.

Proof:

Let $\mathcal{B}$ be a basis for V. Then $[T - x \cdot \mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = [T]^{\mathcal{B}}_{\mathcal{B}} - x \cdot I_n$. This implies that $P_T(x) = \det([T - x \cdot \mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}}) = \det([T]^{\mathcal{B}}_{\mathcal{B}} - x \cdot I_n) = P_{[T]^{\mathcal{B}}_{\mathcal{B}}}(x)$. The claim of the remark now follows from **(a)** of Lemma 21.a.8.

Q.E.D.

Definition 21.a.9 (*Trace*): Let $A \in \mathcal{M}_{n\times n}(K)$. Define the "*trace*" (in German: „*Spur*") of A, denoted $\mathrm{Tr}(A) := \sum_{i=1}^{n} A_{ii} \in K$ (which is the sum of the entries along the diagonal of A). If V is a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$, define $\mathrm{Tr}(T) := \mathrm{Tr}([T]_{\mathcal{B}}^{\mathcal{B}})$, where $\mathcal{B}$ is some basis for V. We will see soon that $\mathrm{Tr}(T)$ is well defined (i.e., it does not depend on the choice of $\mathcal{B}$).

Lemma 21.a.10 (*Properties of Trace*): For all $A, B \in \mathcal{M}_{n\times n}(K)$, we have:

(a) $\mathrm{Tr}(A \cdot B) = \mathrm{Tr}(B \cdot A)$.

(b) If A and B are similar matrices, then $\mathrm{Tr}(A) = \mathrm{Tr}(B)$.

In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are two bases for V, then $\mathrm{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \mathrm{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'})$. Thus, $\mathrm{Tr}(T)$ is well defined.

Proof:

(a) $\mathrm{Tr}(A \cdot B) = \sum_{i=1}^{n}(A \cdot B)_{ii} = \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij} \cdot B_{ji} = \sum_{i,j} A_{ij}B_{ji} = \sum_{i,j} A_{ji}B_{ij} = \sum_{i=1}^{n}\sum_{j=1}^{n} B_{ij}A_{ji} = \sum_{i=1}^{n}(B \cdot A)_{ii} = \mathrm{Tr}(B \cdot A)$.

(b) If $B = P^{-1}AP$, then $\mathrm{Tr}(B) = \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}(P^{-1} \cdot (AP)) = \mathrm{Tr}(AP \cdot P^{-1}) = \mathrm{Tr}(A)$.

Regarding the claim about $\mathrm{Tr}(T)$, recall that $[T]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$, and $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = ([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1}$. Use **(b)** to conclude the proof. **Q.E.D.**

Why is the trace interesting? Let $A \in \mathcal{M}_{n\times n}(K)$ and $P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$ its characteristic polynomial. We have seen that $c_n = (-1)^n$.

Lemma 21.a.11 (*Coefficients of the Characteristic Polynomial*): $c_{n-1} = (-1)^{n-1} \cdot \mathrm{Tr}(A)$ and $c_0 = \det(A)$. In other words, $P_A(x) = (-1)^n x^n + (-1)^{n-1}\,\mathrm{Tr}(A) \cdot x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1 x + \det(A)$.

Proof:

$$P_A(x) = \det \begin{pmatrix} A_{11} - x & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} - x & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} - x \end{pmatrix}.$$ Denote this matrix by B.

$$P_A(x) = (A_{11} - x) \cdot (A_{22} - x) \dots (A_{nn} - x) + \sum_{\sigma \in S_n, \sigma \neq \mathrm{id}} \mathrm{sgn}(\sigma) B_{1,\sigma 1} \dots B_{n,\sigma n}. \tag{21.4}$$

If $\sigma \neq \mathrm{id}$, then there exist at least two indices $i \neq j$ such that $i \neq \sigma i$ and $j \neq \sigma j$. Thus, the product contains at least two factors ($B_{i,\sigma i}$ and $B_{j,\sigma j}$) that are not from the diagonal of B, and so do not contain x. Therefore, the degree of the sum in (21.4) is $\leq n-2$. $P_A(x) = (A_{11} - x)(A_{22} - x) \dots (A_{nn} - x) + (\text{another polynomial of degree} \leq n-2)$. $= (-1)^n x^n + \left[A_{11}(-x)^{n-1} + \cdots + A_{nn}(-x)^{n-1}\right] + \dots$ Expanding the product $(A_{11} - x)(A_{22} - x) \dots (A_{nn} - x)$ yields $(-1)^n x^n + (-1)^{n-1}(A_{11} + \cdots + A_{nn})x^{n-1} + \dots$. $= (-1)^n x^n + (-1)^{n-1} \cdot \mathrm{Tr}(A) x^{n-1} + \cdots + c_1 x + c_0$.

Finally, $c_0 = P_A(0) = \det(A - 0 \cdot I_n) = \det(A)$ (obtained by substituting $x = 0$). **Q.E.D.**

Definition 21.a.12 (*Direct Sum*): Let V be a vector space over K and $U_1, \dots, U_r \subseteq V$ subspaces. Denote $W := U_1 + \cdots + U_r \subseteq V$. We say that W is a "*direct sum*" of $U_1, \dots, U_r$ (or that $U_1 + \cdots + U_r$ is a direct sum) if every $w \in W$ has a unique representation as $w = u_1 + \cdots + u_r$, with $u_i \in U_i$ for all i.

Exercise 21.1 (*Equivalent Conditions for Direct Sums*): $U_1 + \cdots + U_r$ is a direct sum if and only if any of the following conditions hold:

(a) The only way to write $0 = u_1 + \cdots + u_r$, with $u_i \in U_i$ for all i, is with $u_1 = \cdots = u_r = 0$.

(b) For all $2 \leq j \leq r$, we have $U_j \cap (U_1 + \cdots + U_{j-1}) = \{0\}$.

Under the additional assumption that $U_1, \ldots, U_r$ are all finite-dimensional, $U_1 + \cdots + U_r$ is a direct sum if and only if whenever $\mathcal{B}_1, \ldots, \mathcal{B}_r$ are bases for $U_1, \ldots, U_r$ respectively, then $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ forms a basis for $U_1 + \cdots + U_r$.

Exercise 21.2 (*Isomorphism between External and Internal Direct Sums*): If $U_1 + \cdots + U_r$ is a direct sum, then there exists a canonical isomorphism $U_1 \oplus \cdots \oplus U_r \xrightarrow{\sim} U_1 + \cdots + U_r$, given by $(u_1, \ldots, u_r) \mapsto u_1 + \cdots + u_r$. So when $U_1 + \cdots + U_r$ is a direct sum, we can identify $U_1 + \cdots + U_r$ with $U_1 \oplus \cdots \oplus U_r$.

Proposition 21.a.13 (*Direct Sum of Eigenspaces*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Let $\lambda_1, \ldots, \lambda_r \in K$ be pairwise distinct eigenvalues of T. Then the sum $\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_r)$ is a direct sum.

Proof:

Assume $u_1 + \cdots + u_r = u_1' + \cdots + u_r'$, where $u_i, u_i' \in \operatorname{Eig}_T(\lambda_i)$ for all $1 \leq i \leq r$. We need to show that $u_i = u_i'$ for all $i \iff (u_i - u_i' = 0 \text{ for all } i)$. We have:

$$(u_1 - u_1') + \cdots + (u_r - u_r') = 0. \tag{21.5}$$

Assume by contradiction that there exists an i such that $u_i - u_i' \neq 0$. Remove from the list $u_1 - u_1', \ldots, u_r - u_r'$ all those vectors that are 0. We get a non-empty list $u_{i_1} - u_{i_1}', \ldots, u_{i_\ell} - u_{i_\ell}'$ of non-zero vectors, and by (21.5), we still have

$$(u_{i_1} - u_{i_1}') + \cdots + (u_{i_\ell} - u_{i_\ell}') = 0. \tag{21.6}$$

where $u_{i_k} - u_{i_k}' \in \operatorname{Eig}_T(\lambda_{i_k})$ Now for all k, $u_{i_k} - u_{i_k}'$ is either an eigenvector of λ_{i_k} or 0, but by our assumptions, $u_{i_k} - u_{i_k}' \neq 0$. Recall from Proposition 21.a.4 that eigenvectors corresponding to pairwise distinct eigenvalues are linearly independent. $u_{i_1} - u_{i_1}', \ldots, u_{i_\ell} - u_{i_\ell}'$ are linearly independent vectors, and by (21.6), their sum is 0. This is a contradiction. **Q.E.D.**

Corollary 21.a.14 (*Diagonalizability via Eigenspaces*): Let V be finite-dimensional (over K) and $T \in \operatorname{End}(V)$ with distinct eigenvalues $\lambda_1, \ldots, \lambda_r \in K$. Then, T is diagonalizable if and only if the dimensions of its eigenspaces sum to $\dim V$:

$$\sum_{i=1}^{r} \dim \operatorname{Eig}_T(\lambda_i) = \dim V.$$

Proof:

Choose a basis $\mathcal{B}_i$ for $\operatorname{Eig}_T(\lambda_i)$. Recall that $W := \operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_r)$ is a direct sum. This implies that $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for W. Now $W \subseteq V$ is a linear subspace, and $\dim W = \sum_{i=1}^{r} |\mathcal{B}_i| = \sum_{i=1}^{r} \dim \operatorname{Eig}_T(\lambda_i) \leq \dim V$. Since $W \subseteq V$ is a subspace of the finite-dimensional space V, we have $W = V$ if and only if $\dim W = \dim V$, that is, if and only if $\sum_{i=1}^{r} \dim \operatorname{Eig}_T(\lambda_i) = \dim V$.

It remains to connect this to diagonalizability. If the dimension equality holds, then $W = V$, so $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for V consisting of eigenvectors of T, and therefore, T is diagonalizable by Lemma 21.a.3. Conversely, if T is diagonalizable, pick a basis $\mathcal{C} = (w_1, \ldots, w_n)$ of V consisting of eigenvectors. Every w_j lies in $\operatorname{Eig}_T(\lambda_i)$ for some i (the eigenvalues of T are exactly $\lambda_1, \ldots, \lambda_r$), hence $w_j \in W$ for all j. Consequently, $V = \operatorname{Sp}(w_1, \ldots, w_n) \subseteq W$, so $W = V$ and the dimension equality holds. **Q.E.D.**

Exercise 21.3: If V is finite-dimensional, then T can have only finitely many eigenvalues. (Of course, it is implicitly assumed $\lambda_j \neq \lambda_k$ for all $j \neq k$).

Solutions to the Exercises

Proof of Exercise 21.1:

Throughout, write $W := U_1 + \cdots + U_r$.

Directness implies (a). The zero vector always admits the representation $0 = 0 + \cdots + 0$ with $0 \in U_i$ for every i. If the sum is direct, this representation is the only one, which is precisely **(a)**.

(a) implies directness. Let $w \in W$ and suppose $w = u_1 + \cdots + u_r = u_1' + \cdots + u_r'$ with $u_i, u_i' \in U_i$. Subtracting the two representations gives $0 = (u_1 - u_1') + \cdots + (u_r - u_r')$, and $u_i - u_i' \in U_i$ because U_i is a subspace. By **(a)**, every summand vanishes, so $u_i = u_i'$ for all i; the representation of w is therefore unique.

(a) implies (b). Fix $2 \le j \le r$ and let $u \in U_j \cap (U_1 + \cdots + U_{j-1})$. Write $u = u_1 + \cdots + u_{j-1}$ with $u_i \in U_i$. Then

$$u_1 + \cdots + u_{j-1} + (-u) + 0 + \cdots + 0 = 0$$

is a representation of 0 whose i-th entry lies in U_i for every i (the j-th entry being $-u \in U_j$). By **(a)** all entries vanish, and in particular $u = 0$.

(b) implies (a). Suppose $0 = u_1 + \cdots + u_r$ with $u_i \in U_i$ and not all u_i equal to 0. Let j be the largest index with $u_j \neq 0$. If $j = 1$, then $0 = u_1$ contradicts $u_1 \neq 0$. If $j \ge 2$, then

$$u_j = -(u_1 + \cdots + u_{j-1}) \in U_j \cap (U_1 + \cdots + U_{j-1}) = \{0\},$$

again contradicting $u_j \neq 0$. Hence all the u_i vanish.

The basis criterion. Assume in addition that every U_i is finite-dimensional, and let $\mathcal{B}_i := (b_1^{(i)}, \ldots, b_{d_i}^{(i)})$ be a basis for U_i. The concatenated list $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ always spans W, since every $u_i \in U_i$ is a combination of $\mathcal{B}_i$. So the only question is linear independence.

If the sum is direct, consider a vanishing combination of the concatenated list and collect, for each i, the terms coming from $\mathcal{B}_i$ into a single vector $u_i \in U_i$. Then $u_1 + \cdots + u_r = 0$, so $u_i = 0$ for every i by **(a)**, and the linear independence of $\mathcal{B}_i$ forces all the coefficients in the i-th group to vanish. Thus $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for W, for every choice of the bases $\mathcal{B}_i$.

Conversely, suppose $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for W for some choice of bases $\mathcal{B}_i$, and let $0 = u_1 + \cdots + u_r$ with $u_i \in U_i$. Expanding each u_i in the basis $\mathcal{B}_i$ exhibits a vanishing combination of the concatenated list; by its linear independence all coefficients are 0, hence $u_i = 0$ for every i. This is **(a)**, so the sum is direct. **Q.E.D.**

Proof of Exercise 21.2:

Consider the map

$$\Phi : U_1 \oplus \cdots \oplus U_r \longrightarrow U_1 + \cdots + U_r, \qquad \Phi(u_1, \ldots, u_r) := u_1 + \cdots + u_r.$$

Addition and scalar multiplication in the external direct sum $U_1 \oplus \cdots \oplus U_r$ are performed componentwise, so

$$\Phi\big((u_1, \ldots, u_r) + \mu(u_1', \ldots, u_r')\big) = \sum_{i=1}^{r} (u_i + \mu u_i') = \Phi(u_1, \ldots, u_r) + \mu \Phi(u_1', \ldots, u_r'),$$

and Φ is linear. It is surjective, because by the very definition of $U_1 + \cdots + U_r$ every element of the target is of the form $u_1 + \cdots + u_r$ with $u_i \in U_i$.

For injectivity, let $(u_1, \ldots, u_r) \in \ker \Phi$, that is, $u_1 + \cdots + u_r = 0$ with $u_i \in U_i$. Since the sum is direct, part **(a)** of Exercise 21.1 gives $u_1 = \cdots = u_r = 0$, so $\ker \Phi = \{0\}$. Hence Φ is an isomorphism.

The isomorphism is canonical in the sense that its definition involves no choice whatsoever, no bases, no complements, only the addition of V. This is what licenses the identification of $U_1 + \cdots + U_r$ with $U_1 \oplus \cdots \oplus U_r$ whenever the sum is direct. **Q.E.D.**

Proof of Exercise 21.3:

Put $n := \dim V$ and let $\lambda_1, \ldots, \lambda_m \in K$ be pairwise distinct eigenvalues of T. Choose, for each i, an eigenvector $v_i \neq 0$ for λ_i. By Proposition 21.a.4, the vectors $v_1, \ldots, v_m$ are linearly independent, and a linearly independent list in an n-dimensional space has at most n members. Hence $m \leq n$.

Since m was the number of **any** list of pairwise distinct eigenvalues, T has at most $n = \dim V$ distinct eigenvalues; in particular, only finitely many. **Q.E.D.**

Multiplicities of Eigenvalues (Section 21.b)

Example 21.3 (*Identical Characteristic Polynomials*): Consider the matrices $A := \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $B := \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. We have that their characteristic polynomials are identical: $P_A(x) = P_B(x) = (1-x)^2 = (x-1)^2$.

Example 21.4 (*Jordan Block*): Consider the matrix $J_{\lambda,n} \in \mathcal{M}_{n\times n}(K)$, for $\lambda \in K$.

$$J := J_{\lambda,n} := \begin{pmatrix} \lambda & 1 & 0 & \ldots & 0 \\ 0 & \lambda & 1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & 0 & \lambda \end{pmatrix} \tag{21.7}$$

This is called a "*Jordan block*". It features λ's along the main diagonal, 1's along the "*diagonal*" just above the main diagonal, and 0's everywhere else. The characteristic polynomial is $P_{J_{\lambda,n}}(x) = (\lambda - x)^n$. This implies that J has only one eigenvalue, namely λ.

Let us find the eigenvectors for the eigenvalue λ:

$$J - \lambda \cdot I_n = \begin{pmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

This matrix clearly has $n-1$ linearly independent columns. Therefore, the rank is $\operatorname{rank}(J - \lambda \cdot I_n) = n-1$. By the rank-nullity theorem, $\dim\ker(J - \lambda \cdot I_n) = 1$. Consequently, the eigenspace $\operatorname{Eig}_J(\lambda)$ is 1-dimensional. Clearly, $e_1 = (1, 0, \ldots, 0)^{\mathsf{T}}$ is an eigenvector for λ. Thus, $\operatorname{Eig}_J(\lambda) = \operatorname{Sp}(e_1)$.

Important remark 21.1: There is an important difference between polynomials $p(x) \in K[x]$ and the functions they define $K \to K$. For example, if $K = \mathbb{Z}_2$, then $p(x) = x$ and $q(x) = x^2$ are different polynomials, but they define the same function $\mathbb{Z}_2 \to \mathbb{Z}_2$. (What happens for $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}, \ldots$?)

Exercise 21.4 (*Polynomials versus Polynomial Functions*): Settle the question raised at the end of Important remark 21.1: for which fields K do two distinct polynomials $p(x), q(x) \in K[x]$ always define distinct functions $K \to K$? In particular, what happens for $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$?

Lemma 21.b.1 (*Decomposition of Diagonalizable Matrices*): Let $A \in \mathcal{M}_{n\times n}(K)$ (or $T \in \operatorname{End}(V)$) be diagonalizable. Then $P_A(x)$ decomposes as a product of linear (i.e., degree 1) factors:

$$P_A(x) = (\lambda_1 - x) \cdot \cdots \cdot (\lambda_n - x),$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A (but there might be repetitions among the λ_i's).

Proof:

By assumption, there exists an invertible matrix P such that:

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} =: D. \tag{21.8}$$

(i.e., A is similar to D). Since similar matrices have the same characteristic polynomial, we have:

$$P_A(x) = P_D(x) = \det \begin{pmatrix} \lambda_1 - x & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n - x \end{pmatrix} = (\lambda_1 - x)\dots(\lambda_n - x).$$

Q.E.D.

Example 21.5 (*Rotation Matrix without Real Eigenvalues*): Let $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ over $K = \mathbb{R}$. The associated linear map $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ represents a rotation by 90° in the counter-clockwise direction. The characteristic polynomial is $P_A(x) = \det \left(\begin{smallmatrix} -x & -1 \\ 1 & -x \end{smallmatrix}\right) = x^2 + 1$. This polynomial has no roots in $\mathbb{R}$. And, of course, A has no real eigenvalues.

A quick digression about polynomials...

Let $0 \neq p(x) \in K[x]$ and $\lambda \in K$. By dividing $p(x)$ by $(x - \lambda)$, we can write:

$$p(x) = (x - \lambda) \cdot q(x) + R, \tag{21.9}$$

where $\deg q(x) = \deg p(x) - 1$, and $R \in K$ is the remainder. If we substitute $x = \lambda$ into (21.9), we obtain $p(\lambda) = 0 \cdot q(\lambda) + R = R$. Thus, $p(x) = (x - \lambda) \cdot q(x) + p(\lambda)$. Conclusion: $p(\lambda) = 0 \iff p(x) = (x - \lambda) \cdot q(x)$ for some $q(x) \in K[x]$. We say that λ is a "*zero*" of $p(x)$ (in German: „*Nullstelle*“), or a "*root*" of $p(x)$. Sometimes, λ is also a zero of $q(x)$, so $q(x) = (x - \lambda) \cdot h(x)$, which implies $p(x) = (x - \lambda)^2 \cdot h(x)$, etc.

Definition 21.b.2 (*Multiplicity of a Zero*): Let $0 \neq p(x) \in K[x]$ and $\lambda \in K$. We say that λ is a zero of "*multiplicity*" $m \in \mathbb{N}_{>0}$ if $p(x) = (x - \lambda)^m \cdot g(x)$, where $g(x) \in K[x]$ and $g(\lambda) \neq 0$ (i.e., λ is not a zero of $g(x)$). If $m = 1$, we say λ is a "*simple zero*" of $p(x)$. If $m \geq 2$, we say λ is a "*multiple zero*" of $p(x)$.

Geometric & Algebraic Multiplicity of Eigenvalues

Definition 21.b.3 (*Geometric and Algebraic Multiplicity*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Let λ be an eigenvalue of T. We define the "*geometric multiplicity*" of λ to be:

$$m_g(T; \lambda) := \dim \operatorname{Eig}_T(\lambda).$$

Assume now, in addition, that V is finite-dimensional. We define the "*algebraic multiplicity*" of λ to be the multiplicity of the zero λ in the characteristic polynomial $P_T(x)$ of T. We denote it by $m_a(T; \lambda)$.

Proposition 21.b.4 (*Multiplicity Inequality*): For any eigenvalue λ, the geometric multiplicity is less than or equal to the algebraic multiplicity:

$$m_g(T; \lambda) \leq m_a(T; \lambda).$$

Proof:

Let $v_1, \dots, v_k$ be a basis for the eigenspace $\operatorname{Eig}_T(\lambda)$. We extend this basis to a basis $\mathcal{B} := (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$ of V. Then the matrix of T with respect to the basis $\mathcal{B}$ has the block form:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = ([Tv_1]_{\mathcal{B}} \dots [Tv_k]_{\mathcal{B}} \mid [Tw_{k+1}]_{\mathcal{B}} \dots [Tw_n]_{\mathcal{B}})$$

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|c} \lambda I_k & C_1 \\ \hline 0 & C \end{array}\right) \tag{21.10}$$

where $C \in \mathcal{M}_{(n-k)\times(n-k)}(K)$. We compute the characteristic polynomial by utilizing the determinant property for block matrices (Proposition 20.c.4):

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = (\lambda - x)^k \cdot \det(C - xI_{n-k})$$
$$P_T(x) = (\lambda - x)^k \cdot P_C(x) = (-1)^k (x-\lambda)^k \cdot P_C(x).$$

This factorization directly shows that the multiplicity of λ as a zero of $P_T(x)$ is at least k. Therefore, we conclude that $m_a(T;\lambda) \geq k = m_g(T;\lambda)$. **Q.E.D.**

Example 21.6 (*Multiplicities of a Jordan Block*): Revisiting the Jordan block $J := J_{\lambda,n}$ defined in (21.7): We previously saw that $P_J(x) = (\lambda - x)^n$, which directly implies that $m_a(J;\lambda) = n$. We also found that $\mathrm{Eig}_J(\lambda)$ is a 1-dimensional space, which means $m_g(J;\lambda) = 1$.

Solutions to the Exercises

Solution of Exercise 21.4:

The assignment $p(x) \mapsto \hat{p}$, where $\hat{p} : K \to K$ denotes the function $\hat{p}(a) := p(a)$, is K-linear, so two polynomials $p(x), q(x) \in K[x]$ define the same function if and only if their difference $r(x) := p(x) - q(x)$ vanishes at **every** point of K. The question is therefore: which non-zero polynomials vanish identically as functions?

A non-zero polynomial has at most deg many zeros. We claim that a polynomial $0 \neq r(x) \in K[x]$ of degree d has at most d distinct zeros in K. We induct on d. If $d = 0$, then r is a non-zero constant and has no zeros at all. Let $d \geq 1$. If r has no zero, there is nothing to prove; otherwise pick a zero $\lambda \in K$. By the Conclusion of the digression above, $r(x) = (x - \lambda) \cdot q(x)$ with $\deg q(x) = d - 1$. If $\mu \neq \lambda$ is another zero of r, then $(\mu - \lambda) \cdot q(\mu) = 0$ with $\mu - \lambda \neq 0$, and since a field has no zero divisors, $q(\mu) = 0$. So every zero of r other than λ is a zero of q, of which there are at most $d - 1$ by the inductive hypothesis. Hence r has at most d zeros.

Infinite fields. If K is infinite, then no non-zero polynomial can vanish at every point of K, since it has only finitely many zeros. Consequently $r(x) = 0$, that is, $p(x) = q(x)$: over an infinite field, distinct polynomials always define distinct functions. This settles $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$, for these fields the phenomenon of Important remark 21.1 cannot occur, and one may safely identify a polynomial with the function it defines.

Finite fields. If K is finite, the phenomenon always occurs. Indeed, put

$$r(x) := \prod_{a \in K} (x - a) \in K[x].$$

This polynomial is non-zero (it is monic of degree $|K|$), yet $\hat{r}(a) = 0$ for every $a \in K$. Thus $p(x)$ and $p(x) + r(x)$ are distinct polynomials defining the same function, for any $p(x)$ whatsoever. For $K = \mathbb{Z}_2$ this gives $r(x) = x \cdot (x - 1) = x^2 - x$, and adding it to $p(x) = x$ returns exactly Prof. Biran's $q(x) = x^2$.

There is also a proof by counting, which shows that a collision must occur without exhibiting one: if $|K| = m < \infty$, then there are exactly m^m functions $K \to K$, whereas $K[x]$ is infinite, so the map $p(x) \mapsto \hat{p}$ cannot possibly be injective. **Q.E.D.**

Diagonalizability (Section 21.c)

Theorem 21.c.1 (*Equivalent Conditions for Diagonalizability*): Let V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$. The following statements are equivalent:

(a) T is diagonalizable.

(b) There exists a basis for V whose elements are eigenvectors of T.

(c) The characteristic polynomial $P_T(x)$ splits into a product of linear factors and, for every eigenvalue of T, the geometric and algebraic multiplicities are equal.

(d) $\sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i) = \dim V$, where $\lambda_1, \ldots, \lambda_k \in K$ are all the distinct eigenvalues of T.

(e) $V \cong \bigoplus_{i=1}^{k} \mathrm{Eig}_T(\lambda_i)$.

Proof:

(a) $\iff$ **(b)** has already been proved.

(a) $\implies$ **(c)**: Let $\mathcal{B}$ be a basis in which $[T]_{\mathcal{B}}^{\mathcal{B}}$ has diagonal form. By changing the order of the elements in $\mathcal{B}$, we can arrange that $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the following block-diagonal shape:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|c|c} \begin{matrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_1 \end{matrix} & 0 & 0 \\ \hline 0 & \ddots & 0 \\ \hline 0 & 0 & \begin{matrix} \lambda_k & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_k \end{matrix} \end{array}\right) \tag{21.11}$$

where $\mathcal{B} = (v_1^{(1)}, \ldots, v_{\ell_1}^{(1)}, v_1^{(2)}, \ldots, v_{\ell_2}^{(2)}, \ldots, v_1^{(k)}, \ldots, v_{\ell_k}^{(k)})$. Here, $\lambda_1, \ldots, \lambda_k$ are the eigenvalues of T (with $\lambda_i \neq \lambda_j$ for all $i \neq j$) and $v_1^{(i)}, \ldots, v_{\ell_i}^{(i)}$ are eigenvectors of λ_i for all i. The characteristic polynomial is:

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = (\lambda_1 - x)^{\ell_1} \cdot \dots \cdot (\lambda_k - x)^{\ell_k}. \tag{21.12}$$

This shows that $P_T(x)$ splits as a product of linear factors. It also shows that $\lambda_1, \ldots, \lambda_k$ are all the eigenvalues of T. It remains to show that $m_g(T; \lambda_i) = m_a(T; \lambda_i)$ for all i. Clearly, $v_1^{(i)}, \ldots, v_{\ell_i}^{(i)} \in \mathrm{Eig}_T(\lambda_i)$ are linearly independent; hence $\ell_i \leq \dim \mathrm{Eig}_T(\lambda_i) = m_g(T; \lambda_i)$ by definition. But from (21.12), we also have $\ell_i = m_a(T; \lambda_i)$. So $m_a(T; \lambda_i) \leq m_g(T; \lambda_i)$ for all i. At the same time, we always have the inequality $m_g(T; \lambda_i) \leq m_a(T; \lambda_i)$. Therefore, $m_g(T; \lambda_i) = m_a(T; \lambda_i)$ for all i. (The proof also shows that $\ell_i = m_g(T; \lambda_i) = m_a(T; \lambda_i)$).

(c) $\implies$ **(d)**: We assume that $P_T(x) = (\lambda_1 - x)^{\ell_1} \dots (\lambda_k - x)^{\ell_k}$ and $m_g(T; \lambda_i) = m_a(T; \lambda_i)$. Then $\lambda_1, \ldots, \lambda_k$ are exactly the eigenvalues of T. By definition, $\ell_i = m_a(T; \lambda_i)$. We have $n = \dim V = \sum_{i=1}^{k} m_a(T; \lambda_i)$ (because $\deg P_T(x) = n$). By assumption, this equals

$$\sum_{i=1}^{k} m_g(T; \lambda_i) = \sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i).$$

(d) $\implies$ **(e)**: Recall from Proposition 21.a.13 that $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k)$ is a direct sum. Consequently,

$$\dim(\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k)) = \dim \mathrm{Eig}_T(\lambda_1) + \cdots + \dim \mathrm{Eig}_T(\lambda_k) = \dim V$$

by assumption. Since $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) \subseteq V$ is a linear subspace, we have $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) = V$. This implies that $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) \cong \mathrm{Eig}_T(\lambda_1) \oplus \cdots \oplus \mathrm{Eig}_T(\lambda_k)$.

(e) $\implies$ **(a)**: Suppose $V \cong \mathrm{Eig}_T(\lambda_1) \oplus \cdots \oplus \mathrm{Eig}_T(\lambda_k)$. Isomorphic spaces have equal dimension, by Corollary 15.b.11, and the dimension of an external direct sum is the sum of the dimensions, so $\dim V = \sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i)$. Put $W := \mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) \subseteq V$. The sum is direct by Proposition 21.a.13, so concatenating bases of the summands gives a basis of W by Exercise 21.1, whence $\dim W = \sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i) = \dim V$. A subspace of the same finite dimension as the whole space is the whole space, by Proposition 12.3, so $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) = V$. If we choose a basis $v_1^{(i)}, \ldots, v_{\ell_i}^{(i)}$ for $\mathrm{Eig}_T(\lambda_i)$ for all i (where $\ell_i = m_g(T; \lambda_i)$), then the union $(v_1^{(1)}, \ldots, v_{\ell_1}^{(1)}, \ldots, v_1^{(k)}, \ldots, v_{\ell_k}^{(k)})$ is a basis of V and all the vectors in it are eigenvectors of T.

Q.E.D.

Corollary 21.c.2 (*Sufficient Condition for Diagonalizability*): If $P_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity 1, then T is diagonalizable.

Proof:

Let λ be an eigenvalue of T. Then $\mathrm{Eig}_T(\lambda) \neq \{0_V\}$, since it contains an eigenvector, so $m_g(T; \lambda) \geq 1$; and $m_g(T; \lambda) \leq m_a(T; \lambda) = 1$ by Proposition 21.b.4. Hence $m_g(T; \lambda) = 1 = m_a(T; \lambda)$. Now use point **(c)** of Theorem 21.c.1. **Q.E.D.**

Trigonalization

(= bringing a matrix/endomorphism to upper triangular form)

Question 21.1: Let $T \in \mathrm{End}(V)$. We want to find a basis $\mathcal{B}$ for T such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ has as simple a form as possible.

- If T is diagonalizable, we are okay. But we have seen that this is not always the case. For example, $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is not diagonalizable (we saw: $m_a(A; 1) = 2, m_g(A; 1) = 1$).
- A weaker (hence more general) condition is trigonalizability.

Definition 21.c.3 (*Trigonalizable Endomorphism*): We say that $T \in \mathrm{End}(V)$ is "*trigonalizable*" if there exists a basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular, i.e.,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \ldots & * \\ 0 & \ddots & \vdots \\ 0 & \ldots & \lambda_n \end{pmatrix} \quad (n = \dim V). \tag{21.13}$$

Theorem 21.c.4 (*Trigonalizability Condition*): Let V be a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$. Then T is trigonalizable if and only if $P_T(x)$ splits as a product of linear factors, i.e., $P_T(x) = (\lambda_1 - x)^{\ell_1} \ldots (\lambda_k - x)^{\ell_k}$.

Short Digression

Theorem 21.c.5 (*Fundamental Theorem of Algebra*): Every polynomial $p(x) \in \mathbb{C}[x]$ with complex coefficients splits as a product of linear factors and a constant, i.e., $p(x) = C \cdot (x - a_1)^{n_1} \ldots (x - a_r)^{n_r}$, with $C \in \mathbb{C}$, $r \geq 0$, $a_i \in \mathbb{C}$, and $n_j \in \mathbb{N}_{>0}$.

Remark 21.4: The proof is typically covered in an analysis course or a complex analysis course in the third semester.

Corollary 21.c.6 (*Trigonalizability over* $\mathbb{C}$): Let V be a finite-dimensional vector space over $\mathbb{C}$. Then every $T \in \mathrm{End}(V)$ is trigonalizable.

Proof:

The characteristic polynomial $P_T(x)$ lies in $\mathbb{C}[x]$, so by Theorem 21.c.5 it splits as a product of linear factors. Now apply Theorem 21.c.4. **Q.E.D.**

Exercise 21.5 (*Splitting Passes to the Cofactor*): Let $\lambda \in K$ and let $p(x), q(x) \in K[x]$ be non-zero polynomials with $p(x) = (\lambda - x) \cdot q(x)$. Show that if $p(x)$ splits into a product of linear factors over K, then so does $q(x)$. (This is the step Prof. Biran leaves as an exercise in both proofs of Theorem 21.c.4, applied to $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$.)

Proof of the Theorem on Trigonalizability

Proof of Theorem 21.c.4:

$\Longrightarrow$: If there exists a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular as in (21.13), then:

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = \det \begin{pmatrix} \lambda_1 - x & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n - x \end{pmatrix} = (\lambda_1 - x) \dots (\lambda_n - x).$$

$\Longleftarrow$: We will use induction on $n = \dim V$.

Base case $n = 1$: In this case, V is 1-dimensional and $[T]_{\mathcal{B}}^{\mathcal{B}} = (\, \lambda \,)$, so T is even diagonalizable.

Inductive step: Let $n \geq 1$. Assume that the theorem holds for all vector spaces of dimension $\leq n$ and every endomorphism of such a space. Let V be an $(n+1)$-dimensional vector space and $T \in \operatorname{End}(V)$ such that $P_T(x)$ splits as a product of linear factors.

Let λ_1 be a zero of $P_T(x)$. This implies that there exists an eigenvector $u_1 \in \ker(T - \lambda_1 \cdot \mathrm{id}_V)$. Extend u_1 to a basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$ of V. Then the matrix of T in basis $\mathcal{B}'$ has the block form:

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \left(\begin{array}{c|c} \lambda_1 & * \cdots * \\ \hline 0 & \\ \vdots & A \\ 0 & \end{array}\right) =: M \tag{21.14}$$

where A is an $n \times n$ matrix in $\mathcal{M}_{n \times n}(K)$. By block determinant properties (Proposition 20.c.4), we have $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$. Since $P_T(x)$ splits as a product of linear factors, so does $P_A(x)$ (Exercise 21.5).

Consider the linear map $T_A : K^n \to K^n$ defined by $T_A(u) = A \cdot u$. By the induction hypothesis, there exists a basis $\mathcal{C}$ of K^n such that:

$$[T_A]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}. \tag{21.15}$$

Denote by $\mathcal{E}$ the standard basis of K^n, and put $Q := [\mathrm{id}]_{\mathcal{E}}^{\mathcal{C}}$, the matrix whose columns are the vectors of $\mathcal{C}$ written in the standard basis, so that $Q^{-1} = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{E}}$. We know that $[T_A]_{\mathcal{C}}^{\mathcal{C}} = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{E}} \cdot [T_A]_{\mathcal{E}}^{\mathcal{E}} \cdot [\mathrm{id}]_{\mathcal{E}}^{\mathcal{C}} = Q^{-1}AQ$. Consequently, $Q^{-1}AQ = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}$.

Now, define the $(n+1) \times (n+1)$ block matrix:

$$\underline{P} := \left(\begin{array}{c|c} 1 & 0 \dots 0 \\ \hline 0 & \\ \vdots & Q \\ 0 & \end{array}\right) \in \mathcal{M}_{(n+1)\times(n+1)}(K). \tag{21.16}$$

Claim: $\underline{P}$ is invertible with $\underline{P}^{-1} = \left(\begin{array}{c|c} 1 & 0\dots0 \\ \hline 0 & \\ \vdots & Q^{-1} \\ 0 & \end{array}\right)$, and $\underline{P}^{-1}M\underline{P} = \left(\begin{array}{c|c} \lambda_1 & *\cdots* \\ \hline 0 & \\ \vdots & Q^{-1}AQ \\ 0 & \end{array}\right)$.

Proof of Claim:

This follows from direct block matrix multiplication. First, we verify the product $\underline{P}\cdot\underline{P}^{-1}$ to confirm the inverse:

$$\underline{P}\cdot\underline{P}^{-1} = \left(\begin{array}{c|c} 1\cdot1+0\cdot0 & 1\cdot0+0\cdot Q^{-1} \\ \hline 0\cdot1+Q\cdot0 & 0\cdot0+Q\cdot Q^{-1} \end{array}\right) = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & I_n \end{array}\right).$$

Next, we compute the similarity transformation $\underline{P}^{-1}M\underline{P}$ in two steps. Let us first multiply $\underline{P}^{-1}$ and M:

$$\begin{aligned}\underline{P}^{-1}M &= \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q^{-1} \end{array}\right)\left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & A \end{array}\right) \\ &= \left(\begin{array}{c|c} 1\cdot\lambda_1+0\cdot0 & 1\cdot*+0\cdot A \\ \hline 0\cdot\lambda_1+Q^{-1}\cdot0 & 0\cdot*+Q^{-1}\cdot A \end{array}\right) = \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & Q^{-1}A \end{array}\right).\end{aligned}$$

Finally, we multiply the resulting matrix by $\underline{P}$ from the right:

$$\begin{aligned}(\underline{P}^{-1}M)\underline{P} &= \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & Q^{-1}A \end{array}\right)\left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q \end{array}\right) \\ &= \left(\begin{array}{c|c} \lambda_1\cdot1+*\cdot0 & \lambda_1\cdot0+*\cdot Q \\ \hline 0\cdot1+Q^{-1}A\cdot0 & 0\cdot0+Q^{-1}A\cdot Q \end{array}\right) \\ &= \left(\begin{array}{c|c} \lambda_1 & *' \\ \hline 0 & Q^{-1}AQ \end{array}\right),\end{aligned}$$

where $*'$ denotes the new row vector $*Q$. Since $Q^{-1}AQ$ is upper triangular with diagonal entries $\lambda_2,\dots,\lambda_{n+1}$ (as guaranteed by the induction hypothesis), the block structure ensures the entire matrix $\underline{P}^{-1}M\underline{P}$ is upper triangular. This proves the claim. **Q.E.D.**

We continue with the proof of the Theorem. Recall we have the basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$, where $[T]_{\mathcal{B}'}^{\mathcal{B}'} = M$.

Claim: There exists a basis $\mathcal{B}$ for V such that $[\mathrm{id}]_{\mathcal{B}'}^{\mathcal{B}} = \underline{P}$.

Proof of Claim:

Write the entries of $\underline{P}$ as P_{ij}. Define $m := n+1$ and set vectors $v_1,\dots,v_m$ as linear combinations of $\mathcal{B}'$ using the columns of $\underline{P}$:

$$v_k := \sum_{j=1}^{m} P_{jk}u_j \quad \text{for } k = 1,\dots,m.$$

The set $\mathcal{B} := (v_1,\dots,v_m)$ is a basis for V. To see this, assume $c_1v_1+\cdots+c_mv_m = 0$. Substituting the definitions of v_k and using the linear independence of $(u_1,\dots,u_m)$ yields the matrix equation $\underline{P}\cdot(c_1,\dots,c_m)^\mathsf{T} = 0$. Because $\underline{P}$ is invertible, $(c_1,\dots,c_m)^\mathsf{T} = 0$, proving linear independence.
Q.E.D.

We now have the basis $\mathcal{B}$. Computing the matrix of T in this new basis yields:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}]_{\mathcal{B}}^{\mathcal{B}'}\cdot[T]_{\mathcal{B}'}^{\mathcal{B}'}\cdot[\mathrm{id}]_{\mathcal{B}'}^{\mathcal{B}} = \underline{P}^{-1}\cdot M\cdot\underline{P}. \tag{21.17}$$

As proven above, $\underline{P}^{-1}M\underline{P}$ is upper triangular, with $\lambda_1, \lambda_2, \ldots, \lambda_{n+1}$ along its diagonal. Therefore, T is trigonalizable. The induction is now complete. **Q.E.D.**

An Alternative (And More Elegant) Proof

Exercise 21.6 (*Bases of a Quotient Space*): Let V be a finite-dimensional vector space over K, let $U \subseteq V$ be a linear subspace, and let $(u_1, \ldots, u_r)$ be a basis for U, extended to a basis $(u_1, \ldots, u_r, w_{r+1}, \ldots, w_n)$ of V. Show that the equivalence classes $([w_{r+1}], \ldots, [w_n])$ form a basis for the quotient space V/U. (In particular, $\dim V/U = \dim V - \dim U$.)

Alternative proof of the direction $\Longleftarrow$ of Theorem 21.c.4:

$\Longleftarrow$: We will use induction on $n = \dim V$, and also quotient spaces.

Let us first recall the latter. If $U \subseteq V$ is a linear subspace, we have the quotient space V/U, whose elements are the equivalence classes $[v]$ of the vectors $v \in V$, where $[v] = [v']$ if and only if $v - v' \in U$. Recall also that if V is finite-dimensional, then so is V/U, and

$$\dim V/U = \dim V - \dim U.$$

Now to the proof.

Base case $n = 1$: In this case, V is 1-dimensional and $[T]_{\mathcal{B}}^{\mathcal{B}} = (\,\lambda\,)$, so T is even diagonalizable.

Inductive step: Let $n \geq 1$. Assume that the direction $\Longleftarrow$ of the theorem holds for all vector spaces of dimension $\leq n$ and every endomorphism of such a space. Let V be an $(n+1)$-dimensional vector space and $T \in \mathrm{End}(V)$ such that $P_T(x)$ splits as a product of linear factors.

Let $\lambda_1 \in K$ be a zero of $P_T(x)$. This implies that there exists an eigenvector $v_1 \in \ker(T - \lambda_1 \cdot \mathrm{id}_V)$. Define the 1-dimensional subspace $U := \mathrm{Sp}(v_1)$ and consider the quotient space V/U. Since v_1 is an eigenvector, $T(U) \subseteq U$. This invariance property allows us to obtain a well-defined induced endomorphism on the quotient space, $\overline{T} : V/U \to V/U$, defined by $\overline{T}([v]) := [Tv]$.

Extend v_1 to a basis $\mathcal{B}_1 = (v_1, w_2, \ldots, w_{n+1})$ of V. The equivalence classes $[w_2], \ldots, [w_{n+1}]$ form a basis for V/U (Exercise 21.6, applied with $r = 1$ and the basis (v_1) of U). In this basis, $\overline{T}$ is represented by an $n \times n$ matrix A, and the full transformation T in basis $\mathcal{B}_1$ takes the block form:

$$[T]_{\mathcal{B}_1}^{\mathcal{B}_1} = \left(\begin{array}{c|ccc} \lambda_1 & * & \ldots & * \\ \hline 0 & & & \\ \vdots & & A & \\ 0 & & & \end{array}\right). \tag{21.18}$$

By the properties of block-triangular determinants (Proposition 20.c.4), we have $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$. Because $\overline{T}$ has the matrix representation A, we know $P_{\overline{T}}(x) = P_A(x)$, which gives us:

$$P_T(x) = (\lambda_1 - x) \cdot P_{\overline{T}}(x).$$

Since $P_T(x)$ splits into a product of linear factors by our initial assumption, $P_{\overline{T}}(x)$ must also split into linear factors (Exercise 21.5).

Because $\dim(V/U) = n$, we can apply the induction hypothesis to $\overline{T}$. There exists a basis $\mathcal{C} := (z_2, \ldots, z_{n+1})$ for V/U such that its matrix is upper triangular:

$$[\overline{T}]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} \lambda_2 & \ldots & * \\ 0 & \ddots & \vdots \\ 0 & \ldots & \lambda_{n+1} \end{pmatrix}. \tag{21.19}$$

Now, each element in $\mathcal{C}$ is represented by some vector $v_i \in V$, meaning $z_i = [v_i]$. (Such a v_i is not unique, but we pick one specific representative for every i). Define the list of vectors $\mathcal{B} := (v_1, v_2, \ldots, v_{n+1})$. We claim that $\mathcal{B}$ is a basis for V.

To show this, we check for linear independence. Assume:

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{n+1} v_{n+1} = 0.$$

Applying the quotient map to this equation (where $[v_1] = 0$), we get:

$$\alpha_2 [v_2] + \cdots + \alpha_{n+1} [v_{n+1}] = 0.$$

This implies that $\alpha_2 z_2 + \cdots + \alpha_{n+1} z_{n+1} = 0$. Since the z_i vectors form a basis for V/U, they are linearly independent, which forces $\alpha_2 = \cdots = \alpha_{n+1} = 0$. Plugging this back into our original sum leaves $\alpha_1 v_1 = 0$. Since $v_1 \neq 0$, we must have $\alpha_1 = 0$. Because the $n+1$ vectors are linearly independent and $\dim V = n+1$, $\mathcal{B}$ is indeed a basis for V.

Finally, we claim that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular. For the first column, we already know $T(v_1) = \lambda_1 v_1$. For the remaining columns ($2 \leq i \leq n+1$), we read from the upper-triangular matrix (21.19) that:

$$[\overline{T} z_i]_{\mathcal{C}} = \alpha_{2i} z_2 + \cdots + \alpha_{i-1,i} z_{i-1} + \lambda_i z_i$$

for some scalars $\alpha \in K$. Translating this back to the equivalence classes:

$$[\overline{T} v_i] = \alpha_{2i} [v_2] + \cdots + \alpha_{i-1,i} [v_{i-1}] + \lambda_i [v_i].$$

This implies that the difference between Tv_i and the linear combination of the v's must belong to the kernel of the quotient map, which is U:

$$Tv_i = \alpha_{2i} v_2 + \cdots + \alpha_{i-1,i} v_{i-1} + \lambda_i v_i + u \quad \text{for some } u \in U.$$

Since $U = \mathrm{Sp}(v_1)$, we can write $u = \beta_i v_1$ for some scalar β_i. Thus,

$$Tv_i = \beta_i v_1 + \alpha_{2i} v_2 + \cdots + \alpha_{i-1,i} v_{i-1} + \lambda_i v_i.$$

Because Tv_i can be expressed as a linear combination of $v_1, \ldots, v_i$ (and not $v_{i+1}, \ldots, v_{n+1}$), $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \beta_2 & \ldots & \beta_{n+1} \\ 0 & \lambda_2 & \ldots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda_{n+1} \end{pmatrix}.$$

This completes the inductive step, showing that T is trigonalizable. **Q.E.D.**

Solutions to the Exercises

Proof of Exercise 21.5:

If $q(x)$ is a non-zero constant there is nothing to prove, so assume $\deg q(x) \geq 1$ and hence $\deg p(x) \geq 2$. By hypothesis

$$p(x) = c \cdot (\mu_1 - x) \cdot \cdots \cdot (\mu_m - x)$$

for some $c \in K \setminus \{0\}$, some $\mu_1, \ldots, \mu_m \in K$ and $m := \deg p(x)$.

First, λ occurs among the μ_i. Indeed, substituting $x = \lambda$ into $p(x) = (\lambda - x) q(x)$ gives $p(\lambda) = 0$, so

$$c \cdot (\mu_1 - \lambda) \cdot \cdots \cdot (\mu_m - \lambda) = 0.$$

A field has no zero divisors and $c \neq 0$, so $\mu_j = \lambda$ for at least one index j; after renumbering we may assume $\mu_1 = \lambda$.

Consequently,

$$(\lambda - x) \cdot q(x) = p(x) = (\lambda - x) \cdot \big(c \cdot (\mu_2 - x) \cdot \cdots \cdot (\mu_m - x)\big).$$

The polynomial ring $K[x]$ over a field has no zero divisors, the degree of a product is the sum of the degrees, so we may cancel the non-zero factor $(\lambda - x)$ and obtain

$$q(x) = c \cdot (\mu_2 - x) \cdot \cdots \cdot (\mu_m - x),$$

which is a product of linear factors and a constant, as claimed. **Q.E.D.**

Proof of Exercise 21.6:

Recall that the quotient map $\pi : V \to V/U$, $\pi(v) := [v]$, is linear and satisfies $[u] = 0$ exactly for $u \in U$.

Spanning. Let $[v] \in V/U$ be arbitrary and expand v in the basis of V:

$$v = a_1u_1 + \cdots + a_ru_r + b_{r+1}w_{r+1} + \cdots + b_nw_n.$$

Applying π and using $[u_i] = 0$ for $i \leq r$ leaves $[v] = b_{r+1}[w_{r+1}] + \cdots + b_n[w_n]$. Hence $([w_{r+1}], \ldots, [w_n])$ spans V/U.

Linear independence. Suppose $b_{r+1}[w_{r+1}] + \cdots + b_n[w_n] = 0$, that is, $b_{r+1}w_{r+1} + \cdots + b_nw_n \in U$. Since $(u_1, \ldots, u_r)$ is a basis for U, there are scalars $a_1, \ldots, a_r$ with

$$b_{r+1}w_{r+1} + \cdots + b_nw_n = a_1u_1 + \cdots + a_ru_r,$$

and therefore

$$a_1u_1 + \cdots + a_ru_r - b_{r+1}w_{r+1} - \cdots - b_nw_n = 0.$$

This is a vanishing linear combination of the basis $(u_1, \ldots, u_r, w_{r+1}, \ldots, w_n)$ of V, so all its coefficients vanish; in particular $b_{r+1} = \cdots = b_n = 0$.

Thus $([w_{r+1}], \ldots, [w_n])$ is a basis for V/U, and counting its members gives $\dim V/U = n - r = \dim V - \dim U$. **Q.E.D.**

The Minimal Polynomial & The Cayley-Hamilton Theorem (Section 21.d)

A powerful tool in linear algebra is the ability to evaluate polynomials not just at scalar numbers, but at matrices and endomorphisms. Let $A \in \mathcal{M}_{n\times n}(K)$. For any integer $k \geq 1$, the matrix power $A^k = \underbrace{A \cdot \cdots \cdot A}_{k}$ is well-defined. Therefore, for any formal polynomial $g(x) = a_dx^d + a_{d-1}x^{d-1} + \cdots + a_1x + a_0 \in K[x]$, we can define the matrix evaluation $g(A) \in \mathcal{M}_{n\times n}(K)$ naturally as:

$$g(A) := a_dA^d + a_{d-1}A^{d-1} + \cdots + a_1A + a_0I_n.$$

A natural question arises: can we find a polynomial that completely *"kills"* or annihilates a given matrix?

Claim (*Existence of an Annihilating Polynomial*): For every matrix $A \in \mathcal{M}_{n\times n}(K)$, there exists a non-zero polynomial $g \in K[x]$ such that $g(A) = 0$.

Proof:

Consider the sequence of successive matrix powers. Since the vector space $\mathcal{M}_{n\times n}(K)$ has dimension n^2, we can choose an integer $r \geq n^2$ to construct a sequence of length $r+1$:

$$\underbrace{I_n,\ A,\ A^2,\ A^3,\ \ldots,\ A^r}_{r+1 \text{ elements in a space of dimension } n^2} \tag{21.20}$$

Because the number of elements ($r+1$) strictly exceeds the dimension of the ambient space (n^2), the matrices in (21.20) cannot be linearly independent. This implies there exist scalars $a_0, a_1, \ldots, a_r \in K$, not all zero, such that:

$$a_0I_n + a_1A + a_2A^2 + \cdots + a_rA^r = 0.$$

By setting $g(x) := a_rx^r + \cdots + a_1x + a_0$, we obtain the desired non-zero annihilating polynomial. **Q.E.D.**

A similar logic applies seamlessly to any abstract operator $T \in \mathrm{End}(V)$, where V is a finite-dimensional vector space over K, writing $T^k := \underbrace{T \circ \cdots \circ T}_{k}$ and defining $g(T) := a_dT^d + \cdots +$

$a_1T + a_0\,\mathrm{id}_V$. Since annihilating polynomials exist, it is mathematically fruitful to look for the *"simplest"* one, which acts as a sort of algebraic DNA for the operator.

Definition 21.d.1 (*The Minimal Polynomial*): Let V be a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$. A "*minimal polynomial*" for T is a non-zero, monic polynomial $m_T(x) \in K[x]$ of the minimal possible degree such that $m_T(T) = 0$.

Remark 21.5 (*The Minimal Polynomial Exists and is Unique*): Two points are needed before the notation $m_T(x)$ is legitimate, and the hypothesis that V be finite-dimensional is what supplies the first of them.

Existence. Some non-zero polynomial annihilates T, by the Claim on Page 203 applied to $[T]_{\mathcal{B}}^{\mathcal{B}}$ for any basis $\mathcal{B}$; the set of degrees of such polynomials is a non-empty set of non-negative integers and so has a least element. Dividing by the leading coefficient makes a polynomial of that least degree monic without changing its degree or the fact that it annihilates T. Without finite-dimensionality this fails: the argument counts powers of T inside a space of dimension n^2, and there are endomorphisms of infinite-dimensional spaces that no non-zero polynomial annihilates.

Uniqueness. If f and g are both monic of that least degree d with $f(T) = g(T) = 0$, then $h := f - g$ satisfies $h(T) = 0$ and $\deg h < d$, the leading terms cancelling. Were $h \neq 0$, dividing it by its leading coefficient would produce a monic annihilating polynomial of degree less than d, contradicting minimality. So $h = 0$ and $f = g$. This is what licenses speaking of **the** minimal polynomial and writing $m_T(x)$.

While the minimal polynomial guarantees that an operator satisfies *some* polynomial equation, one of the most celebrated theorems in mathematics provides a canonical polynomial that always works: the characteristic polynomial.

Theorem 21.d.2 (*Cayley-Hamilton*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then

(a) $P_A(A) = 0$.

(b) Let V be a finite-dimensional vector space over K, and $T \in \mathrm{End}(V)$. Then $P_T(T) = 0$.

Exercise 21.7 (*Cayley-Hamilton for 2×2 Matrices*): Let $A := \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Its characteristic polynomial is

$$P_A(x) = \det\begin{pmatrix} a-x & b \\ c & d-x \end{pmatrix} = (a-x)(d-x) - bc = x^2 - \underbrace{(a+d)}_{\mathrm{Tr}(A)}\,x + \underbrace{(ad-bc)}_{\det(A)}.$$

Verify Theorem 21.d.2 in this case by calculating

$$P_A(A) = A^2 - (a+d)\cdot A + (ad-bc)\cdot I_2$$

directly, and obtain $\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.

Exercise 21.8 (*Expressing Matrix Inverses via Cayley-Hamilton*): Let $A \in \mathcal{M}_{n\times n}(K)$ be an invertible matrix. Use Theorem 21.d.2 to prove that the inverse matrix A^{-1} can be expressed as a polynomial in A of degree at most $n-1$.

Proof of Cayley-Hamilton Theorem (Analytic Approach)

Before we dive into a purely algebraic mechanism, it is highly instructive to view this theorem through the lens of topology. The core idea is elegant: the theorem is trivially true for diagonal matrices. If we can show that diagonalizable matrices are *"dense"* (meaning almost all matrices are diagonalizable), we can use the continuity of polynomials to force the theorem to hold for *every* matrix.

First, we note that it is enough to prove part **(a)** (for matrices). Suppose we know that **(a)** is true, and let $T \in \mathrm{End}(V)$. Pick a basis $\mathcal{B}$ for V, and assign $A := [T]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_{n\times n}(K)$.

Claim (*Matrix Representation of Polynomials*): $[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} = P_A(A)$.

Proof of Claim:

The map assigning an endomorphism to its matrix representation is an algebra isomorphism. Thus, $[R+S]_{\mathcal{B}}^{\mathcal{B}} = [R]_{\mathcal{B}}^{\mathcal{B}} + [S]_{\mathcal{B}}^{\mathcal{B}}$, and $[R \circ S]_{\mathcal{B}}^{\mathcal{B}} = [R]_{\mathcal{B}}^{\mathcal{B}} \cdot [S]_{\mathcal{B}}^{\mathcal{B}}$. Since $P_T(x) = P_A(x)$ by definition, we write $P_A(x) = c_n x^n + \cdots + c_0$. Then:

$$\begin{aligned}[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} &= [c_n T^n + \cdots + c_1 T + c_0 \operatorname{id}_V]_{\mathcal{B}}^{\mathcal{B}} \\ &= c_n [T^n]_{\mathcal{B}}^{\mathcal{B}} + \cdots + c_1 [T]_{\mathcal{B}}^{\mathcal{B}} + c_0 I_n \\ &= c_n A^n + \cdots + c_1 A + c_0 I_n = P_A(A).\end{aligned}$$

Q.E.D.

If Cayley-Hamilton holds for matrices, then $P_A(A) = 0$, meaning $[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} = 0$. Since the zero matrix uniquely represents the zero endomorphism, we conclude $P_T(T) = 0$. We now concentrate entirely on proving the theorem for matrices, beginning over the complex field $\mathbb{C}$.

Lemma 21.d.3 (*Cayley-Hamilton for Diagonalizable Matrices*): Suppose $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is diagonalizable. Then $P_A(A) = 0$.

Proof:

By assumption, there exists an invertible matrix Q such that

$$Q^{-1}AQ = \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Since the roots of the characteristic polynomial are exactly the eigenvalues, we know $P_A(\lambda_i) = 0$ for all i. Writing $P_A(x) = \sum c_j x^j$, we substitute A:

$$Q^{-1}P_A(A)Q = \sum c_j Q^{-1}A^j Q = \sum c_j (Q^{-1}AQ)^j = \sum c_j \Lambda^j = P_A(\Lambda).$$

Because Λ is diagonal, $P_A(\Lambda)$ is simply the diagonal matrix with entries $P_A(\lambda_1), \ldots, P_A(\lambda_n)$. Since every entry is zero, $Q^{-1}P_A(A)Q = 0$, which immediately implies $P_A(A) = 0$. **Q.E.D.**

Lemma 21.d.4 (*Density of Diagonalizable Matrices*): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be any arbitrary matrix. Then there exists a sequence of matrices $(A_k)_{k=1}^{\infty}$ such that:

(a) Each A_k is diagonalizable.

(b) $\lim_{k\to\infty} A_k = A$, in the entrywise sense that $\lim_{k\to\infty} A_k(i,j) = A(i,j)$ for all $1 \le i \le n$ and $1 \le j \le n$.

Proof:

Because $\mathbb{C}$ is algebraically closed, every matrix is trigonalizable (Corollary 21.c.6). Thus, there exists an invertible matrix $P \in \mathrm{GL}_n(\mathbb{C})$ such that

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

where $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ are *"the"* eigenvalues of A, or more precisely the zeros of $P_A(x)$, each of them appearing on the diagonal as many times as its algebraic multiplicity, and $b_{ij} \in \mathbb{C}$ for $1 \le i < j \le n$. Note that, equivalently,

$$A = P \begin{pmatrix} \lambda_1 & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} P^{-1}. \tag{21.21}$$

To ensure diagonalizability, we must prevent repeated eigenvalues, and we fix this by a tiny perturbation of the diagonal, one chosen afresh for each k. Let $k \in \mathbb{Z}_{\geq 1}$ and pick $\lambda_1(k), \dots, \lambda_n(k) \in \mathbb{C}$ such that

(a) $|\lambda_i(k) - \lambda_i| < \frac{1}{2023 \cdot k}$ for every i, and
(b) $\lambda_1(k), \dots, \lambda_n(k)$ are pairwise distinct.

This is always possible: distinct points may be chosen inside any collection of discs of positive radius, however small. Now define

$$A_k := P \begin{pmatrix} \lambda_1(k) & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n(k) \end{pmatrix} P^{-1},$$

keeping the very same P and the very same entries b_{ij} above the diagonal.

Clearly A_k has n distinct eigenvalues, namely $\lambda_1(k), \dots, \lambda_n(k)$, and is therefore diagonalizable. (Alternatively: $P_{A_k}(x)$ splits into the n distinct linear factors $(\lambda_1(k) - x) \cdot \dots \cdot (\lambda_n(k) - x)$, so $m_a(A_k; \lambda_i(k)) = 1$ for every i, and Corollary 21.c.2 applies.)

Finally, by condition **(a)** the middle matrix converges entrywise to the middle matrix of (21.21) as $k \to \infty$, and the map $X \mapsto PXP^{-1}$ is continuous (Exercise 21.9). Consequently,

$$\lim_{k\to\infty} A_k = P \left(\lim_{k\to\infty} \begin{pmatrix} \lambda_1(k) & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n(k) \end{pmatrix} \right) P^{-1} = P \begin{pmatrix} \lambda_1 & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} P^{-1} = A.$$

Q.E.D.

Exercise 21.9 (*Continuity of Two-Sided Multiplication*): Let $P, Q \in \mathcal{M}_{n\times n}(\mathbb{C})$ be given matrices. Show that the map

$$\mathcal{M}_{n\times n}(\mathbb{C}) \longrightarrow \mathcal{M}_{n\times n}(\mathbb{C}), \qquad X \longmapsto PXQ,$$

is continuous. (This is what licenses pulling the limit through P and P^{-1} at the end of the proof above.)

Lemma 21.d.5 (*Cayley-Hamilton over $\mathbb{C}$*): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be any complex matrix. The Cayley-Hamilton theorem holds for A.

Proof:

By our construction in Lemma 21.d.4, we can find a sequence of diagonalizable matrices $A_k \to A$. By Lemma 21.d.3, the theorem holds for all of them: $P_{A_k}(A_k) = 0$.

Notice that the mapping $\Phi : \mathcal{M}_{n\times n}(\mathbb{C}) \to \mathcal{M}_{n\times n}(\mathbb{C})$ defined by $\Phi(M) = P_M(M)$ is built entirely from taking determinants, adding matrices, and multiplying matrices. Therefore, Φ is a continuous polynomial function of the matrix entries. Because continuous functions commute with limits, we have:

$$P_A(A) = \Phi(A) = \Phi\left(\lim_{k\to\infty} A_k\right) = \lim_{k\to\infty} \Phi(A_k) = \lim_{k\to\infty} 0 = 0.$$

Q.E.D.

To generalize this result from $\mathbb{C}$ to any arbitrary field K, we rely on a formal algebraic identity regarding multi-variable polynomials.

Here $\mathbb{R}[x_1, \dots, x_r]$ denotes the set of all polynomials in the variables $x_1, \dots, x_r$ with coefficients in $\mathbb{R}$, that is, the finite sums

$$f(x_1, \dots, x_r) = \sum_{\underline{i} \in \mathcal{I}} c_{\underline{i}}\, x_1^{i_1} \cdot \dots \cdot x_r^{i_r}, \qquad c_{\underline{i}} \in \mathbb{R},$$

taken over a finite set $\mathcal{I} \subseteq (\mathbb{Z}_{\geq 0})^r$ of multi-indices $\underline{i} = (i_1, \dots, i_r)$.

Lemma 21.d.6 (*Formal Zero of Multi-Variable Polynomials*): Suppose $f(x_1, \dots, x_r) \in \mathbb{R}[x_1, \dots, x_r]$ satisfies $f(a_1, \dots, a_r) = 0$ for all $(a_1, \dots, a_r) \in \mathbb{R}^r$. Then $f \equiv 0$ as a formal polynomial (all coefficients are 0).

Proof:

Write $f(x_1, \dots, x_r) = \sum_{\underline{i} \in \mathcal{I}} c_{\underline{i}} x_1^{i_1} \dots x_r^{i_r}$, fix a multi-index $\underline{i} = (i_1, \dots, i_r) \in \mathcal{I}$, and put $m := i_1 + \dots + i_r$. We show that $c_{\underline{i}} = 0$.

Since $f(a_1, \dots, a_r) = 0$ for every $(a_1, \dots, a_r) \in \mathbb{R}^r$, the function f is identically zero on $\mathbb{R}^r$, and hence so are all of its partial derivatives, of any order, all over $\mathbb{R}^r$. In particular,

$$\left. \frac{\partial^m f}{\partial x_1^{i_1} \dots \partial x_r^{i_r}} \right|_{x_1 = 0, \dots, x_r = 0} = 0. \tag{21.22}$$

On the other hand, we can compute that same derivative monomial by monomial. For a multi-index $\underline{j} = (j_1, \dots, j_r)$,

$$\frac{\partial^m}{\partial x_1^{i_1} \dots \partial x_r^{i_r}} \left(x_1^{j_1} \cdot \dots \cdot x_r^{j_r} \right) = \begin{cases} 0, & \text{if } j_k < i_k \text{ for some } k, \\ \prod_{k=1}^{r} j_k (j_k - 1) \dots (j_k - i_k + 1) \, x_k^{j_k - i_k}, & \text{if } j_k \geq i_k \text{ for all } k. \end{cases}$$

Now evaluate at the origin. In the first case the result is 0. In the second case, if $\underline{j} \neq \underline{i}$, then $j_k > i_k$ for at least one k, so a positive power of x_k survives and the value at the origin is again 0. Only $\underline{j} = \underline{i}$ contributes, and it contributes

$$\prod_{k=1}^{r} i_k (i_k - 1) \dots 1 = i_1! \cdot \dots \cdot i_r!.$$

Summing over the monomials of f therefore gives

$$\left. \frac{\partial^m f}{\partial x_1^{i_1} \dots \partial x_r^{i_r}} \right|_{x_1 = 0, \dots, x_r = 0} = c_{\underline{i}} \cdot i_1! \cdot \dots \cdot i_r!.$$

Comparing with (21.22) and dividing by the non-zero real number $i_1! \dots i_r!$ yields $c_{\underline{i}} = 0$, as claimed. **Q.E.D.**

Exercise 21.10 (*Where the Argument Breaks over Other Fields*): What fails in the proof of Lemma 21.d.6 if we replace $\mathbb{R}[x_1, \dots, x_r]$ by $K[x_1, \dots, x_r]$, even for $r = 1$, with, for example, $K = \mathbb{Z}_2$, or some other field, finite or infinite?

Analytic Proof of Cayley-Hamilton over every field K:

The point of departure is that the determinant is given by one and the same formula over every field: for every $n \times n$ matrix B over every field K,

$$\det B = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot B_{1,\sigma 1} \cdot \dots \cdot B_{n,\sigma n}. \tag{21.23}$$

Consequently $P_A(x) = \det(A - x \cdot I_n)$ may be regarded as an element of $K[x, A_{11}, A_{12}, \dots, A_{nn}]$, a polynomial in the $n^2 + 1$ variables $x, A_{11}, \dots, A_{nn}$. But (21.23) shows more: the coefficients produced by that formula are sums of products of ± 1, so they lie in $\mathbb{Z}$, and hence in $\mathbb{R}$. Thus, $P_A(x) \in \mathbb{R}[x, A_{11}, \dots, A_{nn}]$, and the polynomial does not depend on K at all.

For an $n \times n$ matrix A, the entries of $P_A(A)$ are given by multi-variable polynomials in the n^2 variables $A_{11}, A_{12}, \dots, A_{nn}$. These polynomials have integer (and hence real) coefficients. Let the polynomial representing the (i, j)-th entry be $q_{ij}(A_{11}, \dots, A_{nn})$.

Since Cayley-Hamilton holds for $\mathbb{C}$ (and thus for any matrix in $\mathbb{R}$), we know $q_{ij}(A_{11},\dots)=0$ for all evaluations in $\mathbb{R}^{n^2}$. By Lemma 21.d.6, $q_{ij}\equiv 0$ as formal polynomials. Because they are formally the zero polynomial, evaluating them over *any* field K will always yield 0. Thus, $P_A(A)=0$ for any matrix over any field. **Q.E.D.**

Example 21.7 (*The Polynomials q_{ij} for $n=2$*): Let $n=2$ and $A:=\left(\begin{smallmatrix} a & b\\ c & d\end{smallmatrix}\right)$, so that $P_A(x)=(a-x)(d-x)-bc=x^2-(a+d)x+(ad-bc)$ as in Exercise 21.7. Then

$$\begin{aligned} P_A(A) &= \begin{pmatrix} a & b\\ c & d\end{pmatrix}^2-(a+d)\cdot\begin{pmatrix} a & b\\ c & d\end{pmatrix}+\begin{pmatrix} ad-bc & 0\\ 0 & ad-bc\end{pmatrix}\\ &= \begin{pmatrix} a^2+bc & ab+bd\\ ac+cd & bc+d^2\end{pmatrix}-(a+d)\cdot\begin{pmatrix} a & b\\ c & d\end{pmatrix}+\begin{pmatrix} ad-bc & 0\\ 0 & ad-bc\end{pmatrix}\\ &= \begin{pmatrix} q_{11} & q_{12}\\ q_{21} & q_{22}\end{pmatrix}, \qquad q_{ij}\in\mathbb{R}[a,b,c,d]. \end{aligned}$$

This displays concretely what the proof above uses: the four entries of $P_A(A)$ are polynomial expressions in the entries of A, with coefficients that are plain integers and owe nothing to the field the entries are drawn from.

A Purely Algebraic Proof of Cayley-Hamilton Thm

While the analytic proof is mathematically beautiful, it relies on topological machinery (limits and continuity) that feels slightly alien in pure algebra. Let us now construct a purely structural, algebraic proof using invariant subspaces.

Definition 21.d.7 (*T-Cyclic Subspace*): Let V be a vector space over K and $T\in\mathrm{End}(V)$. Let $v\in V$. We define the "*T-cyclic subspace*" generated by v as the span of the orbit of v under T:

$$\langle v\rangle_T:=\mathrm{Sp}\{T^k v\mid k\in\mathbb{Z}_{\geq 0}\}=\mathrm{Sp}(v,Tv,T^2v,T^3v,\dots)\subseteq V,$$

where, as usual, $T^0:=\mathrm{id}_V$.

Exercise 21.11 (*Minimality of the Cyclic Subspace*): Let V be a vector space over K, let $T\in\mathrm{End}(V)$ and let $v\in V$. Prove:

(a) $\langle v\rangle_T$ is invariant under T, that is, $T(\langle v\rangle_T)\subseteq\langle v\rangle_T$.

(b) $\langle v\rangle_T$ is the smallest linear subspace of V that contains v and is T-invariant. That is, if $U\subseteq V$ is a linear subspace with $v\in U$ and $T(U)\subseteq U$, then $\langle v\rangle_T\subseteq U$.

Lemma 21.d.8 (*Basis and Companion Matrix*): Suppose that $v,Tv,\dots,T^{d-1}v$ $(d\geq 1)$ are linearly independent, and $T^dv=c_0v+c_1Tv+\cdots+c_{d-1}T^{d-1}v$. Then:

(a) $\mathcal{B}=(v,Tv,\dots,T^{d-1}v)$ forms a basis for $\langle v\rangle_T$.

(b) Let $S=T|_{\langle v\rangle_T}$. Then $[S]^{\mathcal{B}}_{\mathcal{B}}$ is a companion matrix:

$$[S]^{\mathcal{B}}_{\mathcal{B}}=\begin{pmatrix} 0 & 0 & \dots & 0 & c_0\\ 1 & 0 & \dots & 0 & c_1\\ 0 & 1 & \dots & 0 & c_2\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ 0 & 0 & \dots & 1 & c_{d-1}\end{pmatrix}.$$

Proof:

(a) Let $U:=\mathrm{Sp}(\mathcal{B})$. Applying T to any basis vector T^kv yields either the next basis element $T^{k+1}v\in U$, or (in the case of $k=d-1$) it yields $T^dv\in U$ due to the given linear combination.

Thus $T(U) \subseteq U$, so U is a T-invariant subspace of V containing v. Clearly $U \subseteq \langle v\rangle_T$; and by part **(b)** of Exercise 21.11 (which identifies $\langle v\rangle_T$ as the smallest T-invariant subspace containing v), we also have $\langle v\rangle_T \subseteq U$. Hence $U = \langle v\rangle_T$. Since the elements of $\mathcal{B}$ are linearly independent by assumption and span this space, they form a basis for it.

(b) This follows directly from the definition of the matrix representation; each vector maps to the next standard basis vector, creating the 1s on the subdiagonal, and the last vector maps to the prescribed linear combination, populating the final column.

Q.E.D.

Lemma 21.d.9 (*Characteristic Polynomial of a Companion Matrix*): Under the assumptions of Lemma 21.d.8, the characteristic polynomial of the restriction S is given by:

$$P_S(x) = (-1)^d(x^d - c_{d-1}x^{d-1} - \cdots - c_1x - c_0).$$

For $d = 1$ this reads $P_S(x) = -(x - c_0)$.

Proof:

We compute $\det([S]_{\mathcal{B}}^{\mathcal{B}} - xI_d)$ using cofactor expansion along the first row:

$$P_S(x) = \begin{vmatrix} -x & 0 & \dots & 0 & c_0 \\ 1 & -x & \dots & 0 & c_1 \\ 0 & 1 & \dots & 0 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & c_{d-1} - x \end{vmatrix}$$

Expanding across the first row, we get two non-zero terms:

$$P_S(x) = (-x)\det(M_{11}) + (-1)^{1+d}c_0\det(M_{1d})$$

where M_{11} is the submatrix obtained by deleting the first row and first column, and M_{1d} is the submatrix obtained by deleting the first row and d-th column. Notice that M_{1d} is an upper triangular matrix with entirely 1s on its main diagonal. Thus, $\det(M_{1d}) = 1$. The matrix M_{11} is exactly the $(d-1) \times (d-1)$ matrix of the same shape, built from the constants $c_1, \dots, c_{d-1}$ in place of $c_0, \dots, c_{d-1}$.

We can now run the induction on d explicitly. For $d = 1$ the matrix is the 1×1 matrix $(c_0 - x)$, whose determinant is $c_0 - x = (-1)^1(x - c_0)$, which is the asserted formula. Let $d \geq 2$ and assume the formula for size $d - 1$. Applied to M_{11}, whose constants are $c_1, \dots, c_{d-1}$, it gives

$$\det(M_{11}) = (-1)^{d-1}\left(x^{d-1} - c_{d-1}x^{d-2} - \cdots - c_2x - c_1\right).$$

Substituting this and $\det(M_{1d}) = 1$ into the cofactor expansion above yields

$$\begin{aligned} P_S(x) &= (-x)\cdot(-1)^{d-1}\left(x^{d-1} - c_{d-1}x^{d-2} - \cdots - c_1\right) + (-1)^{d+1}c_0 \\ &= (-1)^d\left(x^d - c_{d-1}x^{d-1} - \cdots - c_1x\right) + (-1)^d(-c_0) \\ &= (-1)^d\left(x^d - c_{d-1}x^{d-1} - \cdots - c_1x - c_0\right), \end{aligned}$$

which is exactly the formula claimed. Here we used $(-1)^{d+1} = (-1)^{d-1} = -(-1)^d$ in both of the last two steps. **Q.E.D.**

Algebraic Proof of Cayley-Hamilton:

Let V be finite-dimensional, and $T \in \mathrm{End}(V)$. We want to show $P_T(T) = 0$, meaning $P_T(T)v = 0$ for all $v \in V$.

Let $v \in V$, $v \neq 0$. Consider the cyclic subspace $\langle v\rangle_T$. Take the maximal integer $d \geq 1$ such that $v, Tv, \dots, T^{d-1}v$ are linearly independent. Then $T^dv = c_0v + \cdots + c_{d-1}T^{d-1}v$ for some

$c_0, \ldots, c_{d-1} \in K$, and $\dim\langle v\rangle_T = d$ by Lemma 21.d.8. Put $n := \dim V$. If $d < n$, extend this basis $\mathcal{B}$ to a full basis $\mathcal{C} := (v, \ldots, T^{d-1}v, w_1, \ldots, w_{n-d})$ for V. (If $d = n$, take $\mathcal{C} := \mathcal{B}$; the block C below is then empty and $P_C(x) = 1$, so the argument goes through unchanged.) The matrix representation of T with respect to $\mathcal{C}$ is block upper-triangular:

$$[T]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} [S]_{\mathcal{B}}^{\mathcal{B}} & * \\ 0 & [C] \end{pmatrix}$$

Because the determinant of a block-triangular matrix is the product of the determinants of its diagonal blocks (Proposition 20.c.4), we have $P_T(x) = P_S(x) \cdot P_C(x)$. Consider the endomorphism $R := P_T(T) \in \operatorname{End}(V)$. Since polynomials evaluated at T commute with one another, we can factor R:

$$R = P_C(T) \circ P_S(T).$$

Therefore, $Rv = P_C(T)\Big(P_S(T)v\Big)$. However, applying Lemma 21.d.9 directly to v:

$$P_S(T)v = (-1)^d(T^d v - c_{d-1}T^{d-1}v - \cdots - c_0 v).$$

By the very definition of our constants c_i, this evaluates exactly to 0. Hence $Rv = P_C(T)(0) = 0$. Since $v \neq 0$ was chosen completely arbitrarily, $P_T(T)$ evaluates to zero on every vector in V, meaning $P_T(T) = 0$. **Q.E.D.**

Remark 21.6 (*The Lazy Proof*): A completely incorrect *"proof"* often presented by students is the following: $P_A(A) = \det(A - A \cdot I) = \det(0) = 0$. This is fundamentally invalid. The expression $P_A(x) = \det(A - xI)$ evaluates to a polynomial with scalar coefficients. Substituting matrices into a scalar expression *before* expanding the determinant leads to a severe category error. One must compute the scalar polynomial first, and only then evaluate it at the matrix A. Note also that the two sides do not even live in the same place: $\det(A - xI)$ is a **scalar**, whereas $P_A(A)$ is an $n \times n$ **matrix**, so the very first equality of the *"proof"* equates a scalar with a matrix.

Solutions to the Exercises

Solution of Exercise 21.7:

Write $A := \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ and compute the three summands separately:

$$A^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{pmatrix}, \qquad -(a+d) \cdot A = \begin{pmatrix} -a^2 - ad & -ab - bd \\ -ac - cd & -ad - d^2 \end{pmatrix},$$

$$(ad - bc) \cdot I_2 = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix}.$$

Adding them entry by entry:

$$\begin{aligned}
(1,1): &\quad (a^2 + bc) + (-a^2 - ad) + (ad - bc) = 0, \\
(1,2): &\quad (ab + bd) + (-ab - bd) + 0 = 0, \\
(2,1): &\quad (ac + cd) + (-ac - cd) + 0 = 0, \\
(2,2): &\quad (bc + d^2) + (-ad - d^2) + (ad - bc) = 0.
\end{aligned}$$

Hence $P_A(A) = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, confirming Theorem 21.d.2 for $n = 2$. Note that every cancellation above is an identity in the four symbols a, b, c, d with integer coefficients, it uses nothing about the field K, which is precisely the observation exploited in Example 21.7. **Q.E.D.**

Proof of Exercise 21.8:

Write $P_A(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0$. Its constant coefficient is

$$c_0 = P_A(0) = \det(A - 0 \cdot I_n) = \det(A),$$

which is non-zero precisely because A is invertible. By Theorem 21.d.2,

$$c_n A^n + c_{n-1} A^{n-1} + \cdots + c_1 A + c_0 I_n = 0,$$

so, moving the constant term to the other side and factoring out one copy of A (which is legitimate, since A commutes with every power of itself),

$$c_0 I_n = -\left(c_n A^n + \cdots + c_1 A\right) = A \cdot \left(-c_n A^{n-1} - c_{n-1} A^{n-2} - \cdots - c_1 I_n\right).$$

Dividing by the non-zero scalar $c_0 = \det(A)$ and multiplying by A^{-1} gives

$$A^{-1} = -\frac{1}{\det(A)} \left(c_n A^{n-1} + c_{n-1} A^{n-2} + \cdots + c_2 A + c_1 I_n\right),$$

which exhibits A^{-1} as a polynomial in A of degree at most $n-1$, as required.

For $n = 2$ this recovers the familiar formula: there $c_2 = 1$ and $c_1 = -(a+d) = -\operatorname{Tr}(A)$, so

$$A^{-1} = \frac{1}{\det(A)} \left(\operatorname{Tr}(A) \cdot I_2 - A\right) = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Q.E.D.

Proof of Exercise 21.9:

Identify $\mathcal{M}_{n \times n}(\mathbb{C})$ with $\mathbb{C}^{n^2}$ by listing the entries; under this identification, convergence of matrices is exactly entrywise convergence, which is what the statement of Lemma 21.d.4 asks for. A map into $\mathbb{C}^{n^2}$ is continuous if and only if each of its n^2 coordinate functions is continuous, so it suffices to look at one entry of PXQ at a time. By the definition of matrix multiplication,

$$(PXQ)_{ij} = \sum_{k=1}^{n} \sum_{\ell=1}^{n} P_{ik} \, X_{k\ell} \, Q_{\ell j}.$$

Here P_{ik} and $Q_{\ell j}$ are fixed complex numbers, so the right-hand side is a linear, in particular polynomial, expression in the n^2 coordinates $X_{k\ell}$ of X. Polynomial functions $\mathbb{C}^{n^2} \to \mathbb{C}$ are continuous, being built from the (continuous) coordinate projections by finitely many additions and multiplications. Hence every entry of PXQ depends continuously on X, and so does PXQ itself.

Taking $Q := P^{-1}$ gives the case used in Lemma 21.d.4: we have $\lim_{k \to \infty} P B_k P^{-1} = P \left(\lim_{k \to \infty} B_k\right) P^{-1}$ whenever the limit on the right exists. **Q.E.D.**

Solution of Exercise 21.10:

Two separate things go wrong, and it is worth keeping them apart.

The proof breaks immediately. The argument is analytic: it differentiates f as a **function** on $\mathbb{R}^r$, and differentiation rests on limits, which a bare field K does not provide. One can of course define the formal derivative of a polynomial without any analysis, but the step that made the proof work, *"f vanishes identically, therefore all its derivatives vanish identically"*, is a statement about functions and is no longer available. Even granting formal derivatives, the last step divides by

$i_1! \cdot \dots \cdot i_r!$, and in a field of characteristic p those factorials can be 0: for $K = \mathbb{Z}_2$ and $i = 2$ already $2! = 0$ in K.

For finite K the statement itself is false, so no repair of the proof is possible. Take $K = \mathbb{Z}_2$ and $r = 1$. Then $f(x) := x^2 + x$ satisfies $f(0) = 0$ and $f(1) = 1 + 1 = 0$, so f vanishes at every point of K, yet f is not the zero polynomial. This is exactly the phenomenon of Important remark 21.1, and Exercise 21.4 shows that it occurs over every finite field.

For infinite K the statement survives, but it must be proved algebraically rather than analytically. Induct on r. The case $r = 1$ is the content of Exercise 21.4: a non-zero polynomial in one variable has at most deg many zeros, so over an infinite field it cannot vanish everywhere. For $r \geq 2$, collect f by powers of the last variable,

$$f(x_1, \dots, x_r) = \sum_j g_j(x_1, \dots, x_{r-1}) \cdot x_r^j,$$

and fix any $(a_1, \dots, a_{r-1}) \in K^{r-1}$. The one-variable polynomial $\sum_j g_j(a_1, \dots, a_{r-1})\, x_r^j$ then vanishes at every point of K, so by the case $r = 1$ all of its coefficients $g_j(a_1, \dots, a_{r-1})$ are 0. As $(a_1, \dots, a_{r-1})$ was arbitrary, each g_j vanishes identically on K^{r-1}, hence $g_j \equiv 0$ by the inductive hypothesis, and therefore $f \equiv 0$.

This is why the proof of Cayley-Hamilton over an arbitrary field routes through $\mathbb{R}$: Lemma 21.d.6 is applied only over $\mathbb{R}$, where it is both true and easy, and its conclusion, that the q_{ij} are the zero polynomial **formally**, is then field-independent. **Q.E.D.**

Proof of Exercise 21.11:

(a) Let $u \in \langle v \rangle_T$. By the definition of the span, $u = a_0 v + a_1 T v + \dots + a_m T^m v$ for some $m \in \mathbb{Z}_{\geq 0}$ and $a_0, \dots, a_m \in K$. Applying the linear map T gives

$$Tu = a_0 T v + a_1 T^2 v + \dots + a_m T^{m+1} v,$$

which is again a linear combination of vectors of the form $T^k v$, and hence lies in $\langle v \rangle_T$. Therefore $T(\langle v \rangle_T) \subseteq \langle v \rangle_T$.

(b) Let $U \subseteq V$ be a linear subspace with $v \in U$ and $T(U) \subseteq U$. We show by induction on k that $T^k v \in U$ for every $k \in \mathbb{Z}_{\geq 0}$. For $k = 0$ this says $v \in U$, which holds by assumption. If $T^k v \in U$, then $T^{k+1} v = T(T^k v) \in T(U) \subseteq U$, which completes the induction.

Thus $\{T^k v \mid k \in \mathbb{Z}_{\geq 0}\} \subseteq U$. Since U is a linear subspace, it contains every linear combination of its elements, and therefore contains the span of that set, which is exactly $\langle v \rangle_T$. Hence $\langle v \rangle_T \subseteq U$.

Together, **(a)** and **(b)** say that $\langle v \rangle_T$ is itself a T-invariant subspace containing v and is contained in every other one, that is, it is the smallest such subspace, which is the property used in the proof of Lemma 21.d.8. **Q.E.D.**

AI-Note: The scan `21.eigenvectors.a` is absent from `source/`, so the material of §21.a could not be collated against Prof. Biran's notes; parts b, c and d were checked page by page. That gap decides one question below.

The rigour pass found one error of substance and several statements resting on a fact the book never states.

A change-of-basis matrix named as its own inverse. In the first proof of Theorem 21.c.4, the transition matrix was introduced as $Q = [\mathrm{id}]^{\mathcal{E}}_{\mathcal{C}}$ and the conjugation written $Q^{-1}AQ$. With the source basis on top, as this book writes it, $[\mathrm{id}]^{\mathcal{E}}_{\mathcal{C}}$ is the **left** factor of the displayed product, so that product is QAQ^{-1}; the matrix meant is $Q := [\mathrm{id}]^{\mathcal{C}}_{\mathcal{E}}$, whose columns are the vectors of $\mathcal{C}$ in standard coordinates. This is the failure mode of ch. 17b again, a product that composes perfectly while the letter naming one factor denotes its inverse, and `check-repmatrix.pl` cannot see it: the bases do meet correctly, only the abbreviation is wrong.

Eigenvalues and roots. The proof of Lemma 21.d.3 says *"the roots of the characteristic polynomial are exactly the eigenvalues"*, and both proofs of Theorem 21.c.4 open by drawing an eigenvector out of a zero of $P_T(x)$. That equivalence is the reason the characteristic polynomial is worth defining, and it appears nowhere in the book. It is now AI-Proposition 21.1. Because the scan for §21.a is missing, it cannot be checked whether Prof. Biran states it, and the rule keys on where a statement demonstrably appears: an `ai` environment is therefore the honest choice, and it leaves 21.a.1 to 21.a.9 untouched either way.

What $\det(A - xI_n)$ means. Definition 21.a.6 takes a determinant of a matrix whose entries lie in $K[x]$, whereas ch. 20 defines det over a field, and it defines P_T through a basis without saying that the result does not depend on it. Both are settled by material already in the book, the Leibniz formula being a polynomial identity valid over any commutative ring and Lemma 21.a.8 **(e)** giving the basis-independence, and Remark 21.2 now says so. The evaluation identity $P_A(\mu) = \det(A - \mu I_n)$ recorded there is the only bridge between the formal polynomial and the scalars, and it is used every time a root is turned into an eigenvector.

The minimal polynomial. Definition 21.d.1 placed no restriction on V, although the existence of an annihilating polynomial is proved by counting powers of T in a space of dimension n^2; and the notation $m_T(x)$, used ever after, presupposes a uniqueness never argued. Remark 21.5 supplies both, and the hypothesis is now finite-dimensional.

Smaller repairs. The proof of Proposition 21.a.5 argued an equivalence with two implications pointing the same way. Part **(e)** $\Rightarrow$ **(a)** of Theorem 21.c.1 concluded that a direct sum of eigenspaces filled V from its being a subspace, which needs the dimension count that an abstract isomorphism supplies. Corollary 21.c.2 passed from $m_a = 1$ to $m_g = 1$ without noting that $m_g \geq 1$ holds because an eigenvalue has an eigenvector. And ten `enumerate` environments printed their items as $1, 2, 3$ while the proofs and the surrounding prose refer to them as **(a)**, **(b)**, **(c)**.

Euclidean and Hermitian Spaces (Chapter 22)

Euclidean and Hermitian (Unitary) spaces (Section 22.a)

At last (!) we endow our spaces with a geometric structure. We work here with vector spaces V over $K = \mathbb{R}$ or $K = \mathbb{C}$. Over $\mathbb{R}$ we will have Euclidean spaces. Over $\mathbb{C}$ we will have Hermitian (a.k.a. unitary) spaces.

Definition 22.a.1 (*Scalar Product*): Let V be a vector space over $\mathbb{R}$. A "*scalar product*" (a.k.a. inner product) is a function $V \times V \longrightarrow \mathbb{R}$, denoted by $v, w \mapsto \langle v, w\rangle$ (or (v, w)), that has the following properties:

(a) Linearity in the 1st variable:

$$\langle v_1 + v_2, w\rangle = \langle v_1, w\rangle + \langle v_2, w\rangle \quad \forall v_1, v_2, w \in V$$
$$\langle \alpha v, w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{R},\ v, w \in V.$$

(b) Linearity in the 2nd variable:

$$\langle v, w_1 + w_2\rangle = \langle v, w_1\rangle + \langle v, w_2\rangle \quad \forall v, w_1, w_2 \in V$$
$$\langle v, \alpha w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{R},\ v, w \in V.$$

(c) Symmetry:

$$\langle v, w\rangle = \langle w, v\rangle \quad \forall v, w \in V.$$

(d) Positivity:

$$\forall 0 \neq v \in V, \quad \langle v, v\rangle > 0.$$

We call $(V, \langle\cdot,\cdot\rangle)$ a "*Euclidean space*", or a space endowed with a scalar product.

Definition 22.a.2 (*Hermitian Product*): Let V be a vector space over $\mathbb{C}$. A "*Hermitian product*" (or inner product) on V is a function $V \times V \longrightarrow \mathbb{C}$, denoted by $v, w \mapsto \langle v, w\rangle$, with the following properties:

(a) Linearity in the 1st variable:

$$\langle v_1 + v_2, w\rangle = \langle v_1, w\rangle + \langle v_2, w\rangle \quad \forall v_1, v_2, w \in V$$
$$\langle \alpha v, w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{C},\ v, w \in V.$$

(b) Sesquilinearity (complex anti-linearity) in the 2nd variable:

$$\langle v, w_1 + w_2\rangle = \langle v, w_1\rangle + \langle v, w_2\rangle \quad \forall v, w_1, w_2 \in V$$
$$\langle v, \alpha w\rangle = \overline{\alpha}\langle v, w\rangle \quad \forall \alpha \in \mathbb{C},\ v, w \in V.$$

(c) Hermitian property:

$$\langle v, w\rangle = \overline{\langle w, v\rangle} \quad \forall v, w \in V.$$

(d) Positivity:

$$\forall 0 \neq v \in V, \quad \langle v, v\rangle > 0.$$

(The inequality makes sense because $\langle v, v\rangle$ is real: property **(c)** applied with $w := v$ gives $\langle v, v\rangle = \overline{\langle v, v\rangle}$, and a complex number equal to its own conjugate has zero imaginary part. Without **(c)** the condition "$\langle v, v\rangle > 0$" would not even be well formed, there being no order on $\mathbb{C}$.)

We call $(V, \langle\cdot,\cdot\rangle)$ a "*Hermitian*" (or unitary) space.

Remark 22.1: Recall that if $\alpha = a + ib \in \mathbb{C}$, where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$, then $\overline{\alpha} := a - ib$ is the complex conjugate of α. Recall also that $\alpha \cdot \overline{\alpha} = a^2 + b^2 = |\alpha|^2 \geq 0$, with equality if and only if $\alpha = 0$.

The definition of a Hermitian product *"coincides"* with a scalar product if V is over $\mathbb{R}$, and we allow α only in $\mathbb{R}$.

Lemma 22.a.3: Let V be a Euclidean/Hermitian vector space. Then:

(a) $\forall v \in V$, $\langle 0, v\rangle = \langle v, 0\rangle = 0$.

(b) Let $w \in V$. If $\langle v, w\rangle = 0$ for all $v \in V$, then $w = 0$.

(c) Let $w_1, w_2 \in V$. If $\langle v, w_1\rangle = \langle v, w_2\rangle$ for all $v \in V$, then $w_1 = w_2$.

All three parts are left as an exercise.

Definition 22.a.4:

(a) Let V be a Euclidean/Hermitian vector space. For all $v \in V$, define $\|v\| := \sqrt{\langle v, v\rangle} \in \mathbb{R}_{\geq 0}$. The mapping $\|\cdot\|$ is called the "*norm*" on V induced by $\langle \cdot, \cdot \rangle$.

(b) A vector $u \in V$ is called a "*unit vector*" if $\|u\| = 1$. Note that if $0 \neq v \in V$ is any vector, then $\frac{1}{\|v\|}v$ is a unit vector (which is positively proportional to v, so it points in the same direction as v). We call $\frac{v}{\|v\|}$ the "*normalization*" of v.

(c) For $v, w \in V$, we define the "*distance*" between v and w by $d(v, w) := \|v - w\|$.

(d) Two vectors $u, v \in V$ are called "*orthogonal*" (or perpendicular one to the other) if $\langle u, v\rangle = 0$. If this is the case, we sometimes write $u \perp v$.

(e) A subset $S \subseteq V$ is called an "*orthogonal system*" if $0 \notin S$ and for every two distinct elements $u, v \in S$ we have $u \perp v$.

(f) An orthogonal system $S \subseteq V$ is called "*orthonormal*" if for all $u \in S$ we have $\|u\| = 1$.

Theorem 22.a.5 (*Pythagoras' Theorem*): Let V be a Euclidean/Hermitian vector space, and $u, v \in V$ with $u \perp v$. Then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Proof:

By the definition of the norm and the properties of the inner product, we have:

$$\begin{aligned}\|u+v\|^2 &= \langle u+v, u+v\rangle \\ &= \langle u, u\rangle + \underbrace{\langle u, v\rangle}_{=0} + \underbrace{\langle v, u\rangle}_{=0} + \langle v, v\rangle \\ &= \|u\|^2 + \|v\|^2.\end{aligned}$$

Note that in the Euclidean case, $\langle u, v\rangle = 0$ directly. In the Hermitian case, $\langle v, u\rangle = \overline{\langle u, v\rangle} = \overline{0} = 0$.

Q.E.D.

Exercise 22.1 (*Polarization Identities*):

(a) Let V be a Euclidean space. Show that for all $u, v \in V$ we have

$$\langle u, v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right). \quad (*)$$

This shows that $\|\cdot\|$ determines $\langle \cdot, \cdot \rangle$. **Warning:** There exist norms that do **not** come from scalar products.

(b) Find an analogous formula to (*) for a Hermitian vector space.

Examples

(1) The standard scalar product on $\mathbb{R}^n$. Let $\mathbb{R}^n = \mathbb{R}^n_{\text{col}}$. For all $u, v \in \mathbb{R}^n$, define $\langle u, v\rangle = u^\mathsf{T} \cdot v \in \mathcal{M}_{1\times 1}(\mathbb{R}) = \mathbb{R}$.

$$\left\langle \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \right\rangle = \sum_{i=1}^n u_i \cdot v_i$$

$$\left\| \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \right\| = \sqrt{\sum_{i=1}^n v_i^2}.$$

(2) The standard Hermitian inner product on $\mathbb{C}^n$:

$$\langle u, v\rangle := u^\mathsf{T} \cdot \overline{v} = \sum_{k=1}^n u_k \cdot \overline{v}_k \in \mathbb{C}$$

$$\|v\| = \sqrt{\sum_{k=1}^n v_k \cdot \overline{v}_k} = \sqrt{\sum_{k=1}^n |v_k|^2}.$$

(3) If $(V, \langle\cdot,\cdot\rangle)$ is a Euclidean/Hermitian space and $W \subseteq V$ is a subspace, we can restrict $\langle\cdot,\cdot\rangle$ to W and obtain a Euclidean/Hermitian structure on W.

Definition 22.a.6: A matrix $A \in \mathcal{M}_{n\times n}(K)$ (where K is any field) is called "*symmetric*" if $A^\mathsf{T} = A$ (i.e., $a_{ij} = a_{ji}$ for all i, j).

Definition 22.a.7:

(a) Let $A \in \mathcal{M}_{n\times m}(\mathbb{C})$. We write $\overline{A}$ for the matrix whose entry at i, j is $\overline{(A_{ij})}$ (complex conjugation). That is, $(\overline{A})_{ij} := \overline{(A_{ij})}$.

(b) Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$. Define $A^* := \overline{A}^\mathsf{T}$. The matrix A^* is called the "*adjoint matrix*" of A (this has nothing to do with the classical adjoint, $\operatorname{adj}(A)$).

(c) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is called "*Hermitian*" if $A^* = A$ ($\iff$ $A^\mathsf{T} = \overline{A}$).

Important example. Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$. Define $\langle\cdot,\cdot\rangle_A : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\langle v, w\rangle_A = v^\mathsf{T} \cdot A \cdot w.$$

Exercise 22.2 (*Bilinearity of $\langle\cdot,\cdot\rangle_A$*): $\langle\cdot,\cdot\rangle_A$ is bilinear (i.e., linear in both variables).

Assume now that A is symmetric. Then $\langle v, w\rangle_A = \langle w, v\rangle_A$ for all v, w. Indeed,

$$\langle w, v\rangle_A = w^\mathsf{T} \cdot A \cdot v = w^\mathsf{T} A^\mathsf{T} v = (v^\mathsf{T} A \cdot w)^\mathsf{T} = (\langle v, w\rangle_A)^\mathsf{T} = \langle v, w\rangle_A.$$

(Note that this is just a scalar, viewed as a 1×1 matrix).

Is $\langle\cdot,\cdot\rangle_A$ a scalar product? In general, it is not. For example, $A = \left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Then, $\langle \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)\rangle_A = -1$. But for a scalar product, we need $\langle v, v\rangle > 0$ for all $v \neq 0$. So we lack positivity here.

Definition 22.a.8: A symmetric matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is called "*positive definite*" if $v^\mathsf{T} A v > 0$ for all $0 \neq v \in \mathbb{R}^n$.

Exercise 22.3 (*Positive Definiteness Yields a Scalar Product*): If A is positive definite, then $\langle\cdot,\cdot\rangle_A$ is a scalar product.

Claim: If $\langle\cdot,\cdot\rangle_A$ is a scalar product, then A is positive definite. (Proof: a bit later).

Example 22.1:

(a) Let $a_1, \dots, a_n > 0$ (with $a_i \in \mathbb{R}$). Then the diagonal matrix $D = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}$ is positive definite. Indeed, if $v = (v_1, \dots, v_n)^\mathsf{T}$, then
$$\langle v, v\rangle_D = v^\mathsf{T} \cdot D \cdot v = a_1 v_1^2 + \cdots + a_n v_n^2 \geq 0,$$
with equality if and only if $v_i = 0$ for all i.

(b) Let $A = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(\mathbb{R})$. Then A is positive definite if and only if $a > 0$ and $\det(A) > 0$. (Exercise).

The situation for Hermitian products is similar.

Definition 22.a.9: Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be a Hermitian matrix. We say that A is "*positive definite*" if for all $0 \neq v \in \mathbb{C}^n$, we have $v^\mathsf{T} A \overline{v} > 0$. (Note that $v^\mathsf{T} A \overline{v} \in \mathbb{R}$, because $\overline{(v^\mathsf{T} A \overline{v})} = \overline{v}^\mathsf{T} \overline{A} v = (\overline{v}^\mathsf{T} \overline{A} v)^\mathsf{T} = v^\mathsf{T} \overline{A}^\mathsf{T} \overline{v} = v^\mathsf{T} A \overline{v}$. Here, we used $(\cdot)^\mathsf{T} = (\cdot)$ for a scalar/1×1 matrix, and the fact that A is Hermitian. Recall that if $v = (v_1, \dots, v_n)^\mathsf{T} \in \mathbb{C}^n$, then $\overline{v} := (\overline{v}_1, \dots, \overline{v}_n)^\mathsf{T}$).

Given a positive definite complex matrix A, we can define a Hermitian product on $\mathbb{C}^n$ via $\langle v, w\rangle_A = v^\mathsf{T} A \overline{w}$.

Claim: $\langle \cdot, \cdot \rangle_A$ is a Hermitian product if and only if A is Hermitian and positive definite. (Both halves of the hypothesis are needed on the right, since Definition 22.a.9 declares positive definiteness only for a Hermitian matrix; that A must be Hermitian is the first thing the proof below extracts.)

Exercise 22.4 (*Positive Definiteness of 2×2 Hermitian Matrices*): $A = \left(\begin{smallmatrix} a & b \\ \overline{b} & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(\mathbb{C})$ is positive definite if and only if $a \in \mathbb{R}$, $a > 0$, and $\det A \in \mathbb{R}$, $\det A > 0$.

Proof of the Claim:

We will prove it for complex matrices and Hermitian products (the case of symmetric matrices and scalar products is very similar and even simpler, see Exercise 22.5).

Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ and suppose that $\langle v, w\rangle_A := v^\mathsf{T} A \overline{w}$ is a Hermitian product. Then for all $v, w \in \mathbb{C}^n$ we have $\langle v, w\rangle_A = \overline{\langle w, v\rangle_A}$.
$$\implies v^\mathsf{T} A \overline{w} = \overline{w^\mathsf{T} A \overline{v}} \implies v^\mathsf{T} A \overline{w} = \overline{w}^\mathsf{T} \overline{A} v \implies v^\mathsf{T} A \overline{w} = (\overline{w}^\mathsf{T} \overline{A} v)^\mathsf{T} \implies v^\mathsf{T} A \overline{w} = v^\mathsf{T} \overline{A}^\mathsf{T} \overline{w}.$$

Take $v = e_i$, $w = e_j$ (standard basis). Write $A := \begin{pmatrix} a_{11} & \dots & a_{1j} & \dots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & a_{nj} & \dots & a_{nn} \end{pmatrix}$. We have
$$e_i^\mathsf{T} \cdot A \cdot e_j = \begin{pmatrix} - & e_i & - \end{pmatrix} A \begin{pmatrix} | \\ e_j \\ | \end{pmatrix} = \begin{pmatrix} - & e_i & - \end{pmatrix} \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix} = a_{ij}.$$

Similarly, $e_i^\mathsf{T} \overline{A}^\mathsf{T} e_j = \overline{a}_{ji}$. Thus, $\overline{a}_{ji} = a_{ij}$ for all i, j, which implies $\overline{A}^\mathsf{T} = A$.

The fact that A is positive definite is precisely the condition that $\langle v, v\rangle_A > 0$ for all $v \neq 0$, which is to say $v^\mathsf{T} A \overline{v} > 0$.

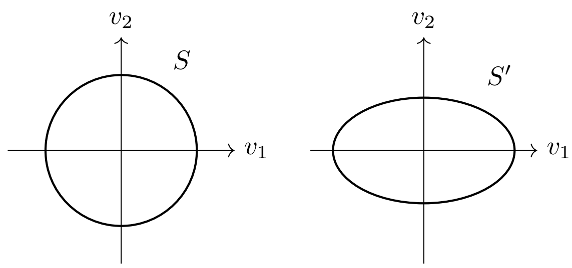
The opposite statement, that a positive definite matrix A makes $\langle \cdot, \cdot \rangle_A$ a Hermitian product, is Exercise 22.6. **Q.E.D.**

Exercise 22.5 (*The Real Case of the Claim*): Carry out the argument above for a real matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ and the form $\langle v, w\rangle_A = v^\mathsf{T} A w$, thereby proving the Claim on Page 216: if $\langle \cdot, \cdot \rangle_A$ is a scalar product, then A is symmetric and positive definite.

Exercise 22.6 (*Positive Definiteness Yields a Hermitian Product*): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be a positive definite Hermitian matrix. Show that $\langle v, w\rangle_A := v^{\mathsf{T}} A\overline{w}$ is a Hermitian product on $\mathbb{C}^n$.

The geometry on V depends on the choice of inner product

Example 22.2: Let $V = \mathbb{R}^2$, $\langle v, w\rangle = v_1w_1 + v_2w_2$ (the standard scalar product on $\mathbb{R}^2$). Let $\langle v, w\rangle' = a_1v_1w_1 + a_2v_2w_2$, where $a_1, a_2 > 0$. Note that $\langle v, w\rangle'$ is $\langle v, w\rangle_A$ for $A = \left(\begin{smallmatrix} a_1 & 0 \\ 0 & a_2 \end{smallmatrix}\right)$.
The unit sphere for $\langle\cdot,\cdot\rangle$: $S = \{v \in \mathbb{R}^2 \mid \|v\| = 1\} = \{(v_1, v_2) \mid v_1^2 + v_2^2 = 1\}$.
The unit sphere for $\langle\cdot,\cdot\rangle'$: $S' = \{v \in \mathbb{R}^2 \mid \|v\|' = 1\} = \{(v_1, v_2) \mid a_1v_1^2 + a_2v_2^2 = 1\}$.



Example 22.3: Let $A = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. It is positive definite (why?). Let $\langle\cdot,\cdot\rangle :=$ std. scalar prod, and $\langle\cdot,\cdot\rangle' := \langle\cdot,\cdot\rangle_A$. We have:

$$\left\langle \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix} \right\rangle = 1, \quad \left\langle \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix} \right\rangle' = \begin{pmatrix}1 & 0\end{pmatrix} \cdot \begin{pmatrix}1 & -1\\ -1 & 2\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1 & -1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = 0.$$

So $\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) \perp' \left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$ but $\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) \not\perp \left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$.

A different type of example

Let $C[a, b] =$ space of continuous functions $f : [a, b] \to \mathbb{R}$. It is a vector space over $\mathbb{R}$ with respect to the standard operations. We define a scalar product:

$$\langle f, g\rangle := \int_a^b f(x) \cdot g(x)\, dx.$$

(This is well-defined and finite. Why?)

Exercise 22.7 (*The Integral Pairing is Bilinear and Symmetric*): Show that $\langle\cdot,\cdot\rangle$ is linear in both variables and symmetric.

Let's check positivity: let $0 \neq f \in C[a, b]$. Then $\langle f, f\rangle = \int_a^b f(x)^2\, dx$. Since $f \neq 0$, there exists $x_0 \in (a, b)$ such that $f(x_0) \neq 0$. Since f is continuous, so is f^2. Hence, there exists $\delta > 0$ such that $[x_0 - \delta, x_0 + \delta] \subseteq [a, b]$ and $f(x) \neq 0$ for all $x \in [x_0 - \delta, x_0 + \delta]$. This implies $f(x)^2 > 0$ for all $x \in [x_0 - \delta, x_0 + \delta]$. Therefore,

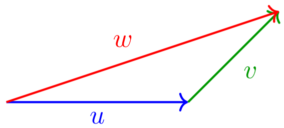
$$\langle f, f\rangle = \int_a^b f(x)^2\, dx \geq \int_{x_0-\delta}^{x_0+\delta} f(x)^2\, dx \geq 2\delta \cdot \min_{x\in[x_0-\delta, x_0+\delta]} f(x)^2 > 0.$$

Remark 22.2: By this scalar product on $C[-\pi, \pi]$, the functions $f(x) = \sin x$ and $g(x) \equiv 1$ are orthogonal ($f \perp g$).

Norms and angles

Definition 22.a.10: Let V be a vector space over K (where $K = \mathbb{R}$ or $\mathbb{C}$). A "*norm*" on V is a function $\|\cdot\| : V \to \mathbb{R}_{\geq 0}$ such that:

(a) $\|u+v\| \leq \|u\| + \|v\|$ for all $u, v \in V$. (Triangle inequality)
(b) $\|\alpha \cdot v\| = |\alpha| \cdot \|v\|$ for all $\alpha \in K$, $v \in V$. (Homogeneity)
(c) If $\|v\| = 0$, then $v = 0$. (Non-degeneracy)



Claim: If $\langle \cdot, \cdot \rangle$ is an inner product on V, then $\|v\| := \sqrt{\langle v, v\rangle}$ is a norm on V. (We'll prove this soon).

Exercise 22.8 (*Examples of Non-Inner-Product Norms*)**:** Define $\left\|\binom{a}{b}\right\|' := \max\{|a|, |b|\}$, and $\left\|\binom{a}{b}\right\|'' := |a| + |b|$ for all $\binom{a}{b} \in \mathbb{R}^2$. Show that $\|\cdot\|'$ and $\|\cdot\|''$ are norms, but they are not induced by any scalar product.
Hint: Recall that if the scalar product $\langle \cdot, \cdot \rangle$ induces $\|\cdot\|$, then $\langle u, v\rangle = \frac{1}{2}(\|u+v\|^2 - \|u\|^2 - \|v\|^2)$.

Theorem 22.a.11 (*Cauchy-Schwarz Inequality*)**:** Let $(V, \langle \cdot, \cdot \rangle)$ be a vector space endowed with an inner product $\langle \cdot, \cdot \rangle$. Then for all $u, v \in V$ we have

$$|\langle u, v\rangle| \leq \|u\| \cdot \|v\|,$$

with equality if and only if u and v are proportional (i.e., $u = \alpha \cdot v$, or $v = \alpha \cdot u$ for some $\alpha \in K \iff u, v$ are linearly dependent).

Proof:

If $u = 0$, the (in)equality holds, so assume $u \neq 0$. Put $w := v - \lambda \cdot u$, where $\lambda := \frac{\langle v, u\rangle}{\|u\|^2}$. (*)
We have $\langle w, u\rangle = \langle v, u\rangle - \lambda\langle u, u\rangle = 0$. Now:

$$\begin{aligned}
0 \leq \|w\|^2 &= \langle v - \lambda u, v - \lambda u\rangle \\
&= \langle v, v\rangle - \overline{\lambda}\langle v, u\rangle - \lambda\langle u, v\rangle + \lambda\overline{\lambda}\langle u, u\rangle \\
&= \|v\|^2 - \overline{\lambda} \cdot \lambda\|u\|^2 - \lambda \cdot \overline{\lambda}\|u\|^2 + \lambda \cdot \overline{\lambda}\|u\|^2 \quad \left(\text{since } \langle v, u\rangle = \lambda\|u\|^2 \text{ and } \langle u, v\rangle = \overline{\langle v, u\rangle} = \overline{\lambda}\|u\|^2\right) \\
&= \|v\|^2 - |\lambda|^2\|u\|^2 \\
&= \|v\|^2 - \frac{|\langle v, u\rangle|^2}{\|u\|^4} \cdot \|u\|^2 \\
&= \|v\|^2 - \frac{|\langle v, u\rangle|^2}{\|u\|^2}.
\end{aligned}$$

This implies $0 \leq \|v\|^2 - \frac{|\langle v, u\rangle|^2}{\|u\|^2}$, which implies $|\langle v, u\rangle| \leq \|u\| \cdot \|v\|$.
We have equality if and only if either $u = 0$, or $w = 0$. But $w = 0 \iff v = \lambda u$ where $\lambda = \frac{\langle v, u\rangle}{\|u\|^2}$. But for $u \neq 0$, the condition $v = \lambda u$ implies $\lambda = \frac{\langle v, u\rangle}{\|u\|^2}$. So, equality if and only if either $u = 0$ or $v = \lambda u$ for some $\lambda \in K \iff u, v$ are linearly dependent. $\boxed{\text{Q.E.D.}}$

Corollary 22.a.12: Let V be a vector space and $\langle \cdot, \cdot \rangle$ an inner product on V. Then $V \ni v \mapsto \|v\| := \sqrt{\langle v, v\rangle}$ is a norm on V.

Proof:

We assume $(V, \langle \cdot, \cdot \rangle)$ is a Hermitian space (the case of Euclidean space is very similar).
Triangle inequality: Let $u, v \in V$.

$$\begin{aligned}
\|u+v\|^2 &= \langle u+v, u+v\rangle = \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle \\
&= \|u\|^2 + 2\operatorname{Re}\langle u, v\rangle + \|v\|^2. \quad (*)
\end{aligned}$$

Note that for all $z \in \mathbb{C}$ we have $|\operatorname{Re} z| \le |z|$. (Indeed, if $z = x + iy$, $x, y \in \mathbb{R}$, then $|\operatorname{Re} z| = |x| \le \sqrt{x^2 + y^2} = |z|$). Applying this to our situation, we have $\operatorname{Re}\langle u, v\rangle \le |\operatorname{Re}\langle u, v\rangle| \le |\langle u, v\rangle|$. Going back to (*), we get:

$$\|u+v\|^2 \le \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2 \le \|u\|^2 + 2\|u\| \cdot \|v\| + \|v\|^2 = (\|u\| + \|v\|)^2,$$

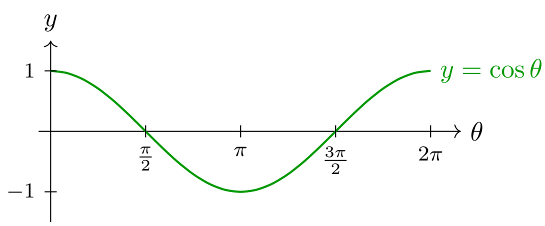
where the second inequality follows from the Cauchy-Schwarz inequality. Since both $\|u+v\|$ and $\|u\| + \|v\|$ are non-negative, the last inequality implies that $\|u+v\| \le \|u\| + \|v\|$.

The other properties of being a norm are Exercise 22.9. **Q.E.D.**

Exercise 22.9 (*The Remaining Two Norm Axioms*): Complete the proof of Corollary 22.a.12 by verifying the homogeneity $\|\alpha \cdot v\| = |\alpha| \cdot \|v\|$ and the non-degeneracy "$\|v\| = 0 \implies v = 0$" for $\|v\| := \sqrt{\langle v, v\rangle}$.

Definition 22.a.13: Let $(V, \langle\cdot,\cdot\rangle)$ be a Euclidean space. Let $u, v \in V$ be two non-zero vectors. By the Cauchy-Schwarz inequality, $-1 \le \frac{\langle u, v\rangle}{\|u\| \cdot \|v\|} \le 1$. We define the "*angle*" between u and v to be the unique $\alpha \in [0, \pi]$ such that:

$$\cos\alpha = \frac{\langle u, v\rangle}{\|u\| \cdot \|v\|} \quad \left(= \left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle\right).$$



The Gram-Schmidt orthogonalization process.

Theorem 22.a.14: Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space.

(a) If $S \subseteq V$ is an orthogonal system, then S is linearly independent.

(b) If $v_1, \dots, v_n$ is an orthogonal system and $v \in \operatorname{Sp}(v_1, \dots, v_n)$, write $v = a_1 v_1 + \cdots + a_n v_n$, with $a_j \in K$ (where $K = \mathbb{R}$ or $\mathbb{C}$). Then the coefficients a_j are given by

$$a_j = \frac{\langle v, v_j\rangle}{\langle v_j, v_j\rangle} \quad \forall\, 1 \le j \le n.$$

Proof:

(b) Write $v = a_1 v_1 + \cdots + a_n v_n$. Let $1 \le j \le n$. We have:

$$\langle v, v_j\rangle = \sum_{k=1}^{n} a_k \underbrace{\langle v_k, v_j\rangle}_{=0\ \forall k \ne j} = a_j \langle v_j, v_j\rangle,$$

which implies that

$$a_j = \frac{\langle v, v_j\rangle}{\langle v_j, v_j\rangle}.$$

(a) Suppose $\sum_{k=1}^{n} a_k v_k = 0$, where $n \ge 1$, $v_1, \dots, v_n \in S$ are distinct and $a_1, \dots, a_n \in K$. By **(b)**,

$$a_j = \frac{\langle 0, v_j\rangle}{\langle v_j, v_j\rangle} = 0 \quad \forall\, 1 \le j \le n.$$

Q.E.D.

Proposition 22.a.15: Let V, W be vector spaces over K $(=\mathbb{R}$ or $\mathbb{C})$ and let $\langle\cdot,\cdot\rangle$ be an inner product on W. Let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V and $\mathcal{C} = (w_1, \ldots, w_m)$ an **orthonormal** basis for W. Let $T : V \to W$ be a linear map. Write $[T]^{\mathcal{B}}_{\mathcal{C}} = (a_{ij})$. Then $a_{ij} = \langle T(v_j), w_i\rangle$ for all i, j. (If $\mathcal{C}$ is only an orthogonal basis, then $a_{ij} = \langle T(v_j), w_i\rangle / \langle w_i, w_i\rangle$). The proof is left as an exercise.

So, having orthogonal/orthonormal bases for our spaces is useful!

Theorem 22.a.16 (*Gram-Schmidt orthogonalization process*): Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over K $(=\mathbb{R}$ or $\mathbb{C})$ and let $n = \dim V$. Let $(v_1, \ldots, v_n)$ be a basis of V. Define a new list of vectors $w_1, \ldots, w_n$ by induction:

$$w_1 := v_1, \quad w_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i\rangle}{\langle w_i, w_i\rangle} \cdot w_i \quad \text{for } j = 2, \ldots, n.$$

Then:

(a) $(w_1, \ldots, w_n)$ is an orthogonal basis for V.

(b) $\left(\frac{w_1}{\|w_1\|}, \ldots, \frac{w_n}{\|w_n\|}\right)$ is an orthonormal basis for V.

(c) $\forall\, 1 \le j \le n$ we have $\mathrm{Sp}(v_1, \ldots, v_j) = \mathrm{Sp}(w_1, \ldots, w_j) = \mathrm{Sp}\left(\frac{w_1}{\|w_1\|}, \ldots, \frac{w_j}{\|w_j\|}\right)$.

Proof:

We claim that for all $1 \le j \le n$, w_j is well defined (i.e., $w_1, \ldots, w_{j-1} \ne 0$, recall that we need to divide by $\langle w_i, w_i\rangle$ in $i = 1, \ldots, j-1$ in order to define w_j), and moreover $(w_1, \ldots, w_j)$ is an orthogonal system with $\mathrm{Sp}(w_1, \ldots, w_j) = \mathrm{Sp}(v_1, \ldots, v_j)$.

We prove this by induction on j.

For $j = 1$, $w_1 = v_1$, so the claim is obvious ($v_1 \ne 0$ because it is a part of a basis).

Let $2 \le j \le n$. Assume the claim holds for $w_1, \ldots, w_{j-1}$. Recall that $w_j = v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i\rangle}{\langle w_i, w_i\rangle} \cdot w_i$. Note that w_j is well defined because $w_i \ne 0$ for all $1 \le i \le j-1$. We claim that $w_j \ne 0$. Indeed, if $w_j = 0$, then

$$v_j = \sum_{i=1}^{j-1} \frac{\langle v_j, w_i\rangle}{\langle w_i, w_i\rangle} w_i \implies v_j \in \mathrm{Sp}(w_1, \ldots, w_{j-1}) = \mathrm{Sp}(v_1, \ldots, v_{j-1})$$

(by inductive hypothesis), which is impossible because $(v_1, \ldots, v_j)$ are linearly independent.

This shows $w_j \ne 0$. By the inductive hypothesis, $w_1, \ldots, w_{j-1}$ is an orthogonal system. So we have to prove that $w_j \perp w_1, \ldots, w_{j-1}$. Indeed, let $1 \le k \le j-1$. Then

$$\langle w_j, w_k\rangle = \langle v_j, w_k\rangle - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i\rangle}{\langle w_i, w_i\rangle} \cdot \langle w_i, w_k\rangle = \langle v_j, w_k\rangle - \frac{\langle v_j, w_k\rangle}{\langle w_k, w_k\rangle} \cdot \langle w_k, w_k\rangle = 0.$$

This shows $(w_1, \ldots, w_j)$ is an orthogonal system.

To complete the induction, it remains to show that $\mathrm{Sp}(w_1, \ldots, w_j) = \mathrm{Sp}(v_1, \ldots, v_j)$. By the definition of w_j we have

$$w_j \in \mathrm{Sp}(w_1, \ldots, w_{j-1}, v_j) = \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_j) \quad \text{(inductive hypothesis)}.$$

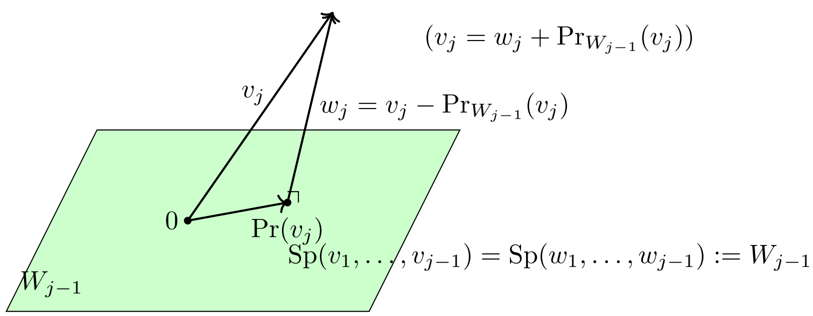
This implies $\mathrm{Sp}(w_1, \ldots, w_{j-1}, w_j) \subseteq \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_j)$. But both these vector spaces have dimension j (because $v_1, \ldots, v_j$ are linearly independent, and also $w_1, \ldots, w_j$ are linearly independent, being an orthogonal system, by Theorem 22.a.14 **(a)**). A subspace of the same finite dimension as the space containing it is that space, by Proposition 12.3, so $\mathrm{Sp}(w_1, \ldots, w_{j-1}, w_j) = \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_j)$. **Q.E.D.**

Corollary 22.a.17: Every finite-dimensional inner product space has an orthonormal basis.

Proof:

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space with $n := \dim V < \infty$. If $n = 0$ the empty family is an orthonormal basis, vacuously. Otherwise V has a basis $(v_1, \ldots, v_n)$ by Theorem 11.b.2; feeding it to Theorem 22.a.16 produces, by part **(b)** of that theorem, the orthonormal basis $\left(\frac{w_1}{\|w_1\|}, \ldots, \frac{w_n}{\|w_n\|}\right)$ of V. **Q.E.D.**

Geometric idea behind the Gram-Schmidt process.



Orthogonal projection:

$$\Pr_{W_{j-1}}(v_j) = \sum_{i=1}^{j-1} \frac{\langle v_j, w_i\rangle}{\langle w_i, w_i\rangle} \cdot w_i.$$

We'll see more about this soon.

Solutions to the Exercises

Proof of Lemma 22.a.3:

(a) Homogeneity in the first variable with $\alpha := 0$ gives

$$\langle 0, v\rangle = \langle 0 \cdot 0, v\rangle = 0 \cdot \langle 0, v\rangle = 0,$$

and homogeneity in the second variable gives $\langle v, 0\rangle = \langle v, 0{\cdot}0\rangle = \overline{0}{\cdot}\langle v, 0\rangle = 0$ (in the Euclidean case simply $0 \cdot \langle v, 0\rangle$, since there is no conjugation).

(b) Suppose $\langle v, w\rangle = 0$ for every $v \in V$. Taking in particular $v := w$ gives $\langle w, w\rangle = 0$. If w were non-zero, positivity would force $\langle w, w\rangle > 0$, a contradiction. Hence $w = 0$.

(c) Let $v \in V$ be arbitrary. Additivity in the second variable, together with $\overline{-1} = -1$, gives

$$\langle v, w_1 - w_2\rangle = \langle v, w_1\rangle + \overline{(-1)}\langle v, w_2\rangle = \langle v, w_1\rangle - \langle v, w_2\rangle = 0.$$

Since this holds for every $v \in V$, part **(b)** applied to $w := w_1 - w_2$ gives $w_1 - w_2 = 0$, that is, $w_1 = w_2$.

Q.E.D.

Proof of Exercise 22.1 (a):

Let V be a Euclidean space and $u, v \in V$. Expanding by bilinearity and then using symmetry,

$$\|u+v\|^2 = \langle u+v, u+v\rangle = \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2.$$

Rearranging gives exactly

$$\langle u, v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right).$$

The right-hand side involves nothing but the norm, so two scalar products on V that induce the same norm are equal; this is the sense in which $\|\cdot\|$ determines $\langle\cdot,\cdot\rangle$. The converse fails: not every norm arises this way, as Exercise 22.8 shows. **Q.E.D.**

Solution of Exercise 22.1 (b):

Let V be a Hermitian space. Here $\langle u,v\rangle$ is in general not real, so one formula must recover two real numbers, and we obtain them one at a time.

The real part. Expanding as above, but now with $\langle v,u\rangle = \overline{\langle u,v\rangle}$,

$$\|u+v\|^2 = \|u\|^2 + \langle u,v\rangle + \overline{\langle u,v\rangle} + \|v\|^2 = \|u\|^2 + \|v\|^2 + 2\,\mathrm{Re}\langle u,v\rangle, \tag{22.1}$$

so that $\mathrm{Re}\langle u,v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right)$, which is literally the Euclidean formula $(*)$ with $\langle u,v\rangle$ replaced by its real part.

The imaginary part. Apply (22.1) with iv in place of v. Since $\|iv\| = |i|\cdot\|v\| = \|v\|$ and $\langle u,iv\rangle = \bar{i}\langle u,v\rangle = -i\langle u,v\rangle$, we get

$$\|u+iv\|^2 = \|u\|^2 + \|v\|^2 + 2\,\mathrm{Re}\left(-i\langle u,v\rangle\right).$$

Writing $\langle u,v\rangle = x+iy$ with $x,y\in\mathbb{R}$ gives $-i(x+iy) = y - ix$, whose real part is $y = \mathrm{Im}\langle u,v\rangle$. Hence

$$\mathrm{Im}\langle u,v\rangle = \frac{1}{2}\left(\|u+iv\|^2 - \|u\|^2 - \|v\|^2\right).$$

The formula. Putting the two together,

$$\langle u,v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right) + \frac{i}{2}\left(\|u+iv\|^2 - \|u\|^2 - \|v\|^2\right),$$

which is the analogue of $(*)$ asked for. Subtracting the versions of (22.1) for v and for $-v$ (respectively for iv and $-iv$) eliminates $\|u\|^2+\|v\|^2$ and yields the more symmetric form usually called the polarization identity:

$$\langle u,v\rangle = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\right).$$

Q.E.D.

Proof of Exercise 22.2:

Everything follows from the distributivity and homogeneity of matrix multiplication. For the first variable, using $(v_1+v_2)^\mathsf{T} = v_1^\mathsf{T} + v_2^\mathsf{T}$ and $(\alpha v)^\mathsf{T} = \alpha v^\mathsf{T}$,

$$\langle v_1+v_2, w\rangle_A = (v_1+v_2)^\mathsf{T}Aw = v_1^\mathsf{T}Aw + v_2^\mathsf{T}Aw = \langle v_1,w\rangle_A + \langle v_2,w\rangle_A,$$
$$\langle \alpha v, w\rangle_A = (\alpha v)^\mathsf{T}Aw = \alpha\left(v^\mathsf{T}Aw\right) = \alpha\langle v,w\rangle_A.$$

For the second variable,

$$\langle v, w_1+w_2\rangle_A = v^\mathsf{T}A(w_1+w_2) = v^\mathsf{T}Aw_1 + v^\mathsf{T}Aw_2 = \langle v,w_1\rangle_A + \langle v,w_2\rangle_A,$$
$$\langle v, \alpha w\rangle_A = v^\mathsf{T}A(\alpha w) = \alpha\left(v^\mathsf{T}Aw\right) = \alpha\langle v,w\rangle_A.$$

Note that no hypothesis on A is needed here; symmetry and positivity are what will require one.

Q.E.D.

Proof of Exercise 22.3:

Let $A\in\mathcal{M}_{n\times n}(\mathbb{R})$ be positive definite; by Definition 22.a.8 this includes the requirement that A be symmetric. We check the four axioms of Definition 22.a.1 for $\langle v,w\rangle_A := v^\mathsf{T}Aw$.

Linearity in each of the two variables is Exercise 22.2. Symmetry is the computation carried out in the text above: since $A^\mathsf{T} = A$ and a 1×1 matrix equals its own transpose,

$$\langle w,v\rangle_A = w^\mathsf{T}Av = w^\mathsf{T}A^\mathsf{T}v = \left(v^\mathsf{T}Aw\right)^\mathsf{T} = v^\mathsf{T}Aw = \langle v,w\rangle_A.$$

Positivity is the definition of positive definiteness: $\langle v,v\rangle_A = v^\mathsf{T}Av > 0$ for every $0\neq v\in\mathbb{R}^n$.

Q.E.D.

Proof of Example 22.1 (b):

Let $A = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(\mathbb{R})$, which is symmetric, and write $v = \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$. Then

$$Q(v) := v^\mathsf{T} A v = ax^2 + 2bxy + dy^2.$$

Necessity. Suppose A is positive definite. Taking $v = \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ gives $Q(v) = a > 0$. Taking $v = \left(\begin{smallmatrix} -b/a \\ 1 \end{smallmatrix}\right)$, which is legitimate now that $a \neq 0$, and is non-zero, gives

$$Q(v) = a \cdot \frac{b^2}{a^2} - 2b \cdot \frac{b}{a} + d = \frac{b^2 - 2b^2 + ad}{a} = \frac{ad - b^2}{a} = \frac{\det(A)}{a} > 0,$$

and since $a > 0$ this forces $\det(A) > 0$.

Sufficiency. Suppose $a > 0$ and $\det(A) = ad - b^2 > 0$. Completing the square,

$$Q(v) = a\left(x + \frac{b}{a}y\right)^2 + \frac{ad - b^2}{a}\, y^2 = a\left(x + \frac{b}{a}y\right)^2 + \frac{\det(A)}{a}\, y^2. \tag{22.2}$$

Both coefficients are positive, so $Q(v) \geq 0$ always, and $Q(v) = 0$ forces first $y = 0$ and then $x + \frac{b}{a}y = x = 0$. Hence $Q(v) > 0$ for every $v \neq 0$, which is positive definiteness. **Q.E.D.**

Proof of Exercise 22.4:

Let $A = \left(\begin{smallmatrix} a & b \\ \overline{b} & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(\mathbb{C})$ be Hermitian. The two reality assertions are automatic: $A^* = A$ forces $a = \overline{a}$ and $d = \overline{d}$, so $a, d \in \mathbb{R}$, and consequently

$$\det A = ad - b\overline{b} = ad - |b|^2 \in \mathbb{R}.$$

So only the two inequalities are at issue. Write $v = \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \in \mathbb{C}^2$ and compute

$$Q(v) := v^\mathsf{T} A \overline{v} = a|x|^2 + bx\overline{y} + \overline{b}\,\overline{x}y + d|y|^2 = a|x|^2 + 2\,\mathrm{Re}\,(bx\overline{y}) + d|y|^2,$$

the last step because $\overline{b}\,\overline{x}y = \overline{bx\overline{y}}$, and $z + \overline{z} = 2\,\mathrm{Re}\,z$.

Necessity. If A is positive definite, then $v = \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ gives $Q(v) = a > 0$, and $v = \left(\begin{smallmatrix} -\overline{b}/a \\ 1 \end{smallmatrix}\right)$ gives, after the computation below, $Q(v) = \det(A)/a > 0$, whence $\det A > 0$.

Sufficiency. Assume $a > 0$ and $\det A > 0$. Completing the square exactly as in the real case, but taking the conjugations into account,

$$a\left|x + \frac{\overline{b}}{a}y\right|^2 = a\left(x + \frac{\overline{b}}{a}y\right)\overline{\left(x + \frac{\overline{b}}{a}y\right)} = a|x|^2 + bx\overline{y} + \overline{b}\,\overline{x}y + \frac{|b|^2}{a}|y|^2,$$

and therefore

$$Q(v) = a\left|x + \frac{\overline{b}}{a}y\right|^2 + \frac{ad - |b|^2}{a}|y|^2 = a\left|x + \frac{\overline{b}}{a}y\right|^2 + \frac{\det(A)}{a}|y|^2. \tag{22.3}$$

Both coefficients are positive real numbers and both squared moduli are non-negative, so $Q(v) \geq 0$; and $Q(v) = 0$ forces $y = 0$ and then $x = 0$. Hence $Q(v) > 0$ for all $v \neq 0$. Reading (22.3) at $x = -\overline{b}y/a$, $y = 1$ also supplies the value $\det(A)/a$ used in the necessity half. **Q.E.D.**

Proof of Exercise 22.5:

Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ and suppose $\langle v, w\rangle_A := v^\mathsf{T} A w$ is a scalar product on $\mathbb{R}^n$. Symmetry of the product says $v^\mathsf{T} A w = w^\mathsf{T} A v$ for all $v, w \in \mathbb{R}^n$. Take $v := e_i$ and $w := e_j$ from the standard basis. As in the complex computation of the text,

$$e_i^\mathsf{T} A e_j = a_{ij}, \qquad e_j^\mathsf{T} A e_i = a_{ji},$$

since Ae_j is the j-th column of A and left multiplication by e_i^T selects its i-th entry. Hence $a_{ij} = a_{ji}$ for all i, j, that is, $A^\mathsf{T} = A$: the matrix is symmetric.

Positivity of the scalar product says $\langle v, v\rangle_A = v^\mathsf{T} A v > 0$ for every $0 \neq v \in \mathbb{R}^n$, which is precisely the statement that the symmetric matrix A is positive definite (Definition 22.a.8). This proves the

Claim on Page 216, and, read together with Exercise 22.3, shows that $\langle \cdot, \cdot \rangle_A$ is a scalar product if and only if A is symmetric and positive definite.

It is indeed simpler than the complex case: no conjugation has to be tracked, and the Hermitian axiom collapses to plain symmetry. **Q.E.D.**

Proof of Exercise 22.6:

Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be Hermitian and positive definite, and put $\langle v, w\rangle_A := v^{\mathsf{T}} A\overline{w}$. We verify the four axioms of Definition 22.a.2.

Linearity in the first variable. As in Exercise 22.2, $(v_1+v_2)^{\mathsf{T}} A\overline{w} = v_1^{\mathsf{T}} A\overline{w} + v_2^{\mathsf{T}} A\overline{w}$ and $(\alpha v)^{\mathsf{T}} A\overline{w} = \alpha\left(v^{\mathsf{T}} A\overline{w}\right)$.

Sesquilinearity in the second variable. Complex conjugation is additive and satisfies $\overline{\alpha w} = \overline{\alpha}\,\overline{w}$, so

$$\langle v, w_1 + w_2\rangle_A = v^{\mathsf{T}} A\left(\overline{w_1} + \overline{w_2}\right) = \langle v, w_1\rangle_A + \langle v, w_2\rangle_A,$$
$$\langle v, \alpha w\rangle_A = v^{\mathsf{T}} A\left(\overline{\alpha}\,\overline{w}\right) = \overline{\alpha}\left(v^{\mathsf{T}} A\overline{w}\right) = \overline{\alpha}\langle v, w\rangle_A.$$

The Hermitian property. Using that a 1×1 matrix equals its own transpose, and then $\overline{A}^{\mathsf{T}} = A$:

$$\overline{\langle w, v\rangle_A} = \overline{w^{\mathsf{T}} A\overline{v}} = \overline{w}^{\mathsf{T}}\overline{A}v = \left(\overline{w}^{\mathsf{T}}\overline{A}v\right)^{\mathsf{T}} = v^{\mathsf{T}}\overline{A}^{\mathsf{T}}\overline{w} = v^{\mathsf{T}} A\overline{w} = \langle v, w\rangle_A.$$

Positivity. $\langle v, v\rangle_A = v^{\mathsf{T}} A\overline{v} > 0$ for every $0 \neq v \in \mathbb{C}^n$ is exactly Definition 22.a.9; that this quantity is real at all was checked there.

Together with the half proved in the text, this completes the Claim on Page 217. **Q.E.D.**

Proof of Exercise 22.7:

First, the pairing is well defined and finite, Prof. Biran's *"Why?"*. If $f, g \in C[a,b]$, then $f\cdot g$ is continuous on the closed bounded interval $[a,b]$, hence Riemann integrable there, and its integral is a real number.

Linearity in the first variable. By the linearity of the integral,

$$\langle f_1 + f_2, g\rangle = \int_a^b \left(f_1(x) + f_2(x)\right) g(x)\,dx = \int_a^b f_1 g\,dx + \int_a^b f_2 g\,dx = \langle f_1, g\rangle + \langle f_2, g\rangle,$$

and for $\alpha \in \mathbb{R}$,

$$\langle \alpha f, g\rangle = \int_a^b \alpha f(x)g(x)\,dx = \alpha\int_a^b f(x)g(x)\,dx = \alpha\langle f, g\rangle.$$

Symmetry. Pointwise multiplication of real numbers is commutative, so $f(x)g(x) = g(x)f(x)$ for every x, and therefore $\langle f, g\rangle = \langle g, f\rangle$.

Linearity in the second variable now follows from the first variable together with symmetry, without any further computation.

Positivity is the part checked in the text, so $\langle\cdot,\cdot\rangle$ is indeed a scalar product on $C[a,b]$. **Q.E.D.**

Proof of Exercise 22.8:

Write $v = \left(\begin{smallmatrix} a \\ b \end{smallmatrix}\right)$ and $v' = \left(\begin{smallmatrix} a' \\ b' \end{smallmatrix}\right)$.

Both are norms. For $\|\cdot\|''$ the three axioms are immediate from the corresponding properties of the absolute value on $\mathbb{R}$:

$$\|v + v'\|'' = |a + a'| + |b + b'| \leq |a| + |a'| + |b| + |b'| = \|v\|'' + \|v'\|'',$$

$\|\alpha v\|'' = |\alpha a| + |\alpha b| = |\alpha|\cdot\|v\|''$, and $\|v\|'' = 0$ forces $|a| = |b| = 0$, so $v = 0$. For $\|\cdot\|'$,

$$\|v + v'\|' = \max\{|a + a'|, |b + b'|\} \leq \max\{|a| + |a'|, |b| + |b'|\} \leq \max\{|a|, |b|\} + \max\{|a'|, |b'|\},$$

since each of $|a| + |a'|$ and $|b| + |b'|$ is bounded by the right-hand side; homogeneity is $\max\{|\alpha a|, |\alpha b|\} = |\alpha| \max\{|a|, |b|\}$, and $\|v\|' = 0$ again forces $v = 0$.

Neither comes from a scalar product. Suppose a norm $\|\cdot\|$ on a real vector space is induced by a scalar product. Expanding as in Exercise 22.1 **(a)** for both $u+v$ and $u-v$ and adding, the cross terms cancel and we obtain the "*parallelogram law*"

$$\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2 \qquad \forall u, v. \tag{22.4}$$

(Equivalently: the formula in the hint would have to define a scalar product, and (22.4) is what its additivity in the first variable amounts to.) So it suffices to exhibit two vectors violating (22.4). Take $u := \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ and $v := \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$, so that $u+v = \left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$ and $u - v = \left(\begin{smallmatrix}1\\-1\end{smallmatrix}\right)$.

For $\|\cdot\|'$ we get $\|u\|' = \|v\|' = \|u+v\|' = \|u-v\|' = 1$, so the left-hand side of (22.4) is $1+1=2$ while the right-hand side is $2+2=4$.

For $\|\cdot\|''$ we get $\|u\|'' = \|v\|'' = 1$ and $\|u+v\|'' = \|u-v\|'' = 2$, so the left-hand side is $4+4=8$ while the right-hand side is $2+2=4$.

In both cases (22.4) fails, so neither norm is induced by a scalar product. This is the warning attached to Exercise 22.1 **(a)**: a scalar product determines its norm, but a norm need not come from one. **Q.E.D.**

Proof of Exercise 22.9:

First note that $\|v\| := \sqrt{\langle v, v\rangle}$ makes sense: in the Hermitian case $\langle v, v\rangle = \overline{\langle v, v\rangle}$ by the Hermitian property, so $\langle v, v\rangle$ is real; it is 0 for $v = 0$ by Lemma 22.a.3 **(a)** and positive otherwise, so the square root is defined and lies in $\mathbb{R}_{\geq 0}$.

Homogeneity. For $\alpha \in K$ and $v \in V$,

$$\|\alpha v\|^2 = \langle \alpha v, \alpha v\rangle = \alpha\overline{\alpha}\langle v, v\rangle = |\alpha|^2\|v\|^2,$$

using homogeneity in the first variable and sesquilinearity in the second (in the Euclidean case $\alpha \cdot \alpha = \alpha^2 = |\alpha|^2$). Both $\|\alpha v\|$ and $|\alpha| \cdot \|v\|$ are non-negative, so taking square roots gives $\|\alpha v\| = |\alpha| \cdot \|v\|$.

Non-degeneracy. If $\|v\| = 0$, then $\langle v, v\rangle = 0$. Were $v \neq 0$, positivity would give $\langle v, v\rangle > 0$; hence $v = 0$. **Q.E.D.**

Proof of Proposition 22.a.15:

By the definition of the representation matrix, the j-th column of $[T]^{\mathcal{B}}_{\mathcal{C}} = (a_{ij})$ holds the coordinates of $T(v_j)$ in the basis $\mathcal{C}$, that is,

$$T(v_j) = \sum_{i=1}^{m} a_{ij}\, w_i.$$

Fix $1 \leq k \leq m$ and pair both sides with w_k. Using linearity in the first variable and the orthogonality $\langle w_i, w_k\rangle = 0$ for $i \neq k$,

$$\langle T(v_j), w_k\rangle = \sum_{i=1}^{m} a_{ij}\langle w_i, w_k\rangle = a_{kj}\langle w_k, w_k\rangle.$$

If $\mathcal{C}$ is orthonormal, then $\langle w_k, w_k\rangle = \|w_k\|^2 = 1$ and we get $a_{kj} = \langle T(v_j), w_k\rangle$, which is the claim after renaming k to i. If $\mathcal{C}$ is merely orthogonal, then $\langle w_k, w_k\rangle \neq 0$ (an orthogonal system does not contain 0) and dividing gives

$$a_{ij} = \frac{\langle T(v_j), w_i\rangle}{\langle w_i, w_i\rangle}.$$

This is Theorem 22.a.14 **(b)** applied to the vector $T(v_j) \in \mathrm{Sp}(w_1, \ldots, w_m) = W$. **Q.E.D.**

AI-Note: The rigour pass found this chapter sound and its exercises already fully solved. Four things were added.

Corollary 22.a.17 stood without a proof, although it is the result the whole Gram-Schmidt section exists to produce, and chs. 22b, 24 and 25 all begin from it. Two lines suffice, the only point worth stating being that $\dim V = 0$ is covered by the empty family.

The positivity axiom of Definition 22.a.2 writes "$\langle v, v\rangle > 0$" for a quantity whose values lie in $\mathbb{C}$, where there is no order. What makes the condition well formed is the Hermitian axiom immediately above it, which forces $\langle v, v\rangle$ to equal its own conjugate and hence to be real. The definition now says so. Definition 22.a.9 already made the corresponding remark for matrices, so the omission here was a local one.

The Claim on Page 217 asked for A "*positive definite*", a property Definition 22.a.9 declares only for Hermitian matrices; the statement now carries both halves, which is also what its proof extracts first and what Exercise 22.5 says in the real case.

And the last step of the proof of Theorem 22.a.16, that two spans of equal dimension coincide, now names Proposition 12.3 and Theorem 22.a.14 **(a)**.

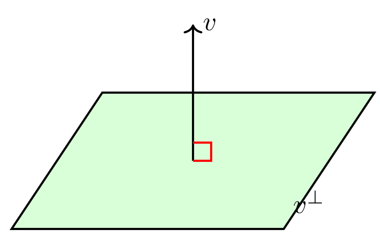
The Orthogonal Complement (Section 22.b)

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$).

Definition 22.b.1 (*The Orthogonal Complement*): Let $\emptyset \neq S \subseteq V$ be a subset. We define the "*orthogonal complement*" of S as:

$$S^\perp := \{v \in V \mid v \perp s \ \forall s \in S\} = \{v \in V \mid \langle v, s\rangle = 0 \ \forall s \in S\}.$$

When $S = \{v\}$, we write $v^\perp := \{v\}^\perp$.



Lemma 22.b.2 (*Properties of the Orthogonal Complement*): Let $\emptyset \neq S, T \subseteq V$. Then:

(a) $S^\perp \subseteq V$ is a linear subspace.
(b) $0^\perp = V$, and $V^\perp = \{0\}$.
(c) $S \cap S^\perp$ is either $\emptyset$ or equal to $\{0\}$.
(d) If $S \subseteq T$, then $S^\perp \supseteq T^\perp$.
(e) $(\mathrm{Sp}(S))^\perp = S^\perp$.
(f) $S \subseteq (S^\perp)^\perp$.

Proof:

(a) We have $0 \in S^\perp$ because $\langle 0, s\rangle = 0$ for all $s \in S$. If $\alpha, \beta \in K$ and $v, w \in S^\perp$, then for all $s \in S$ we have:

$$\langle \alpha v + \beta w, s\rangle = \alpha\langle v, s\rangle + \beta\langle w, s\rangle = 0.$$

This implies that $\alpha v + \beta w \in S^\perp$. Thus, $S^\perp$ is a linear subspace.

(b) Showing $0^\perp = V$ is left as an exercise. Let $v \in V^\perp$. Then $\langle v, w\rangle = 0$ for all $w \in V$. In particular, $\langle v, v\rangle = 0$, which implies $v = 0$. This shows $V^\perp \subseteq \{0\}$. Since it is clear that $\{0\} \subseteq V^\perp$, we conclude $V^\perp = \{0\}$.

(c) If $S \cap S^\perp = \emptyset$, we are done. Otherwise, assume that $s \in S \cap S^\perp$. This implies that $\langle s, s\rangle = 0$, which yields $s = 0$.

(d) (Exercise).

(e) We have $S \subseteq \mathrm{Sp}(S)$. By statement **(d)**, this yields $(\mathrm{Sp}(S))^\perp \subseteq S^\perp$. Next, we will show that $S^\perp \subseteq (\mathrm{Sp}(S))^\perp$. Indeed, let $v \in S^\perp$, and let $u \in \mathrm{Sp}(S)$. We can write $u = \sum_{i=1}^k \alpha_i u_i$ with $\alpha_i \in K$ and $u_i \in S$ for all $1 \le i \le k$. We compute:

$$\langle v, u\rangle = \sum_{i=1}^k \overline{\alpha_i}\, \underbrace{\langle v, u_i\rangle}_{=0} = 0,$$

where we used the fact that $\langle v, u_i\rangle = 0$ because $v \in S^\perp$ and $u_i \in S$. Thus, $v \perp u$. This holds for all $u \in \mathrm{Sp}(S)$, and therefore $v \in (\mathrm{Sp}(S))^\perp$. This shows that $S^\perp \subseteq (\mathrm{Sp}(S))^\perp$.

(f) Let $s \in S$. Then for all $v \in S^\perp$ we have $\langle v, s\rangle = 0$, which implies $\langle s, v\rangle = 0$. Therefore, $s \perp v$ for all $v \in S^\perp$. Consequently, $s \in (S^\perp)^\perp$.

Q.E.D.

Theorem 22.b.3 (*Orthogonal Complement in Finite Dimensions*): Let V be an inner product space (possibly of infinite dimension), and let $U \subseteq V$ be a finite-dimensional subspace. Then $U^\perp$ is a complement of U in V (i.e., $U + U^\perp = V$ and $U \cap U^\perp = \{0\}$).

Proof:

By Lemma 22.b.2 **(c)**, the intersection $U \cap U^\perp$ is either empty or $\{0\}$; it is not empty, since U is a subspace and so contains 0, which lies in $U^\perp$ as well. Hence $U \cap U^\perp = \{0\}$, and it is enough to show that $U + U^\perp = V$. Put $r := \dim U$ and pick an orthonormal basis $(e_1, \ldots, e_r)$ for U, which exists by Corollary 22.a.17 because U is finite-dimensional. (Note: this is not necessarily the standard basis or anything similar). Let $v \in V$. We introduce the temporary notation $\widetilde{P}_U(v) := \langle v, e_1\rangle e_1 + \cdots + \langle v, e_r\rangle e_r \in U$. (This is the orthogonal projection of v onto U.) We have

$$v = \underbrace{\widetilde{P}_U(v)}_{\in U} + (v - \widetilde{P}_U(v)). \tag{22.5}$$

We claim that $v - \widetilde{P}_U(v) \in U^\perp$. Indeed, for any $1 \le i \le r$,

$$\begin{aligned}\langle v - \widetilde{P}_U(v), e_i\rangle &= \langle v, e_i\rangle - \sum_{k=1}^r \langle v, e_k\rangle\langle e_k, e_i\rangle \\ &= \langle v, e_i\rangle - \langle v, e_i\rangle = 0.\end{aligned}$$

This holds for all $1 \le i \le r$, hence $(v - \widetilde{P}_U(v)) \perp \mathrm{Sp}(e_1, \ldots, e_r) = U$, meaning $v - \widetilde{P}_U(v) \in U^\perp$. Going back to (22.5), we see that

$$v = \underbrace{\widetilde{P}_U(v)}_{\in U} + \underbrace{(v - \widetilde{P}_U(v))}_{\in U^\perp}.$$

This holds for all $v \in V$, which implies $V = U + U^\perp$. **Q.E.D.**

Remark 22.3: If $U \subseteq V$ is a finite-dimensional subspace, then since $U^\perp$ is a complement of U in V, we have a canonical isomorphism $V \cong U \oplus U^\perp$.

Definition 22.b.4 (*Orthogonal Projection*): Let V be an inner product space and $U \subseteq V$ a finite-dimensional subspace. We have seen that $U^\perp$ is a complement of U in V (which implies that for all $v \in V$, there exist unique vectors $u \in U$ and $w \in U^\perp$ such that $v = u + w$). Define a map $P_U \colon V \to U$ as follows: for all $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp$ in a unique way. We define $P_U(v) := u$. The map P_U is called the "*orthogonal projection*" on U.

Proposition 22.b.5 (*Properties of Orthogonal Projection*): The map P_U has the following properties:

(a) P_U is a linear map.
(b) $\operatorname{Im} P_U = U$ and $\ker P_U = U^\perp$.
(c) For all $v \in V$, $v - P_U(v) \in U^\perp$.
(d) For all $u \in U$, $P_U(u) = u$.
(e) If $(e_1, \dots, e_r)$ is any orthonormal basis for U, then $P_U(v) = \sum_{i=1}^r \langle v, e_i\rangle \cdot e_i$ (which exactly equals $\widetilde{P}_U(v)$ from the previous proof).

Proof:

Properties **(a)–(d)** follow from the general fact that if W_1, W_2 are vector subspaces of W and $W_1 \oplus W_2 = W$, then the map $P_{W_1} : W \to W_1$, defined by $P_{W_1}(w) = w_1$, where we write $w = w_1 + w_2$ in a unique way with $w_1 \in W_1, w_2 \in W_2$, is a linear map. (That map is well defined precisely because the decomposition is unique, which is Corollary 14.3 **(d)**. It is linear because if $w = w_1 + w_2$ and $w' = w_1' + w_2'$ are the decompositions of w and w', then $\alpha w + \beta w' = (\alpha w_1 + \beta w_1') + (\alpha w_2 + \beta w_2')$ exhibits a decomposition of $\alpha w + \beta w'$ with the first summand in W_1 and the second in W_2; by uniqueness it **is** the decomposition, so $P_{W_1}(\alpha w + \beta w') = \alpha w_1 + \beta w_1' = \alpha P_{W_1}(w) + \beta P_{W_1}(w')$.) Furthermore, $\operatorname{Im} P_{W_1} = W_1$, $\ker P_{W_1} = W_2$, and for all $w \in W$ we have $w - P_{W_1}(w) \in W_2$. (Thus, $w = P_{W_1}(w) + (w - P_{W_1}(w))$ is the unique decomposition of w as $w = w_1 + w_2$). We also have $P_{W_1}(w) = w$ for all $w \in W_1$.

For statement **(e)**, we can write

$$v = \underbrace{\left(\sum_{i=1}^r \langle v, e_i\rangle \cdot e_i\right)}_{\text{(I)}} + \underbrace{\left(v - \sum_{i=1}^r \langle v, e_i\rangle e_i\right)}_{\text{(II)}}. \tag{22.6}$$

Now, part (I) is exactly $\widetilde{P}_U(v)$ from the previous proof, and we have also seen in that proof that part (II) belongs to $U^\perp$. This implies that (22.6) is the unique decomposition of v with respect to $V \cong U \oplus U^\perp$. Hence, $P_U(v) = \text{(I)}$. **Q.E.D.**

Corollary 22.b.6 (*Bidual Orthogonal Complement*): Let V be an inner product space and let $U \subseteq V$ be a finite-dimensional subspace. Then $(U^\perp)^\perp = U$.

Proof:

By Lemma 22.b.2, we have $U \subseteq (U^\perp)^\perp$. We will show that $(U^\perp)^\perp \subseteq U$ also holds. Indeed, let $v \in (U^\perp)^\perp$. Recall that $U^\perp$ is a complement of U in V. Write $v = u + w$ with $u \in U$ and $w \in U^\perp$. Since $u \in U \subseteq (U^\perp)^\perp$ and also $v \in (U^\perp)^\perp$, we have that $w = v - u \in (U^\perp)^\perp$. But $w \in U^\perp$, hence $w \in U^\perp \cap (U^\perp)^\perp = \{0\}$. This shows that $w = 0$, and therefore, $v = u \in U$. We have proven that $(U^\perp)^\perp \subseteq U$. **Q.E.D.**

Corollary 22.b.7 (*Dimension Formula for the Orthogonal Complement*): If V is a finite-dimensional inner product space and $U \subseteq V$ is a subspace, then $\dim U + \dim U^\perp = \dim V$.

Proof:

Since V is finite-dimensional, so is its subspace U, and Theorem 22.b.3 applies: $U + U^\perp = V$ and $U \cap U^\perp = \{0\}$, so the sum $U + U^\perp$ is direct and equals V. Choosing a basis for U and a basis for $U^\perp$, both of which exist because U and $U^\perp$ are subspaces of the finite-dimensional V (Proposition 12.3), and concatenating them therefore produces a basis for V, by the basis criterion of Exercise 21.1, whence

$$\dim U + \dim U^\perp = \dim\left(U \oplus U^\perp\right) = \dim V.$$

Q.E.D.

Orthogonal and Unitary Matrices

Definition 22.b.8 (*Orthogonal and Unitary Matrices*):

(a) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is called "*orthogonal*" if its columns form an orthonormal system in $\mathbb{R}^n$, with respect to the standard scalar product of $\mathbb{R}^n$. That is, writing A in terms of its columns,

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix},$$

the condition means that $\langle v_i, v_j\rangle = \delta_{ij}$ for all $1 \le i, j \le n$ (which in particular implies that the columns form an orthonormal basis for $\mathbb{R}^n$). We denote the set of orthogonal matrices of size $n \times n$ by $\mathrm{O}(n) \subseteq \mathcal{M}_{n\times n}(\mathbb{R})$.

(b) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is called "*unitary*" if its columns form an orthonormal system in $\mathbb{C}^n$, with respect to the standard Hermitian product on $\mathbb{C}^n$. We denote the set of all unitary matrices of size $n \times n$ by $\mathrm{U}(n) \subseteq \mathcal{M}_{n\times n}(\mathbb{C})$.

Remark 22.4: Why are they called *"orthogonal matrices"* and not *"orthonormal matrices"*? This is simply a historical convention in the terminology!

Lemma 22.b.9 (*Characterization of Orthogonal Matrices*): For a matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$, the following statements are equivalent:

(a) A is an orthogonal matrix.
(b) $A^{\mathsf{T}}A = I_n$.
(c) $AA^{\mathsf{T}} = I_n$.
(d) $A^{-1} = A^{\mathsf{T}}$.

Lemma 22.b.10 (*Characterization of Unitary Matrices*): For a matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$, the following statements are equivalent:

(a) A is a unitary matrix.
(b) $A^*A = I_n$ (which is equivalent to $\overline{A}^{\mathsf{T}}A = I_n$).
(c) $AA^* = I_n$ (which is equivalent to $A\overline{A}^{\mathsf{T}} = I_n$).
(d) $A^{-1} = A^*$.

Proof of Lemma 22.b.10:

(The proof of Lemma 22.b.9 is extremely similar). Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix},$$

where $v_k \in \mathbb{C}^n_{\mathrm{col}}$ for $1 \le k \le n$. We compute the product:

$$\begin{aligned} A^*A &= \begin{pmatrix} — & \overline{v_1}^{\mathsf{T}} & — \\ & \vdots & \\ — & \overline{v_n}^{\mathsf{T}} & — \end{pmatrix} \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix} \\ &= \left(\overline{v_i}^{\mathsf{T}} \cdot v_j\right)_{1\le i,j\le n} \\ &= \left(\langle v_j, v_i\rangle\right)_{1\le i,j\le n}. \end{aligned} \tag{22.7}$$

Now, A is unitary if and only if $v_1, \dots, v_n$ form an orthonormal system, which is equivalent to saying that the resulting matrix in (22.7) is exactly I_n. This proves **(a)** $\iff$ **(b)**.

For the remaining equivalences, suppose $A^*A = I_n$. Taking determinants and using $\det(A^*) = \det(\overline{A}^{\mathsf{T}}) = \det(\overline{A}) = \overline{\det(A)}$, we obtain $|\det(A)|^2 = 1$, so $\det(A) \neq 0$ and A is invertible. Multiplying $A^*A = I_n$ on the right by A^{-1} gives $A^* = A^{-1}$, which is **(d)**; and then $AA^* = AA^{-1} = I_n$, which is **(c)**. Conversely, **(c)** gives by the same argument $A^* = A^{-1}$ and hence **(b)**, so all four statements are equivalent. **Q.E.D.**

Exercise 22.10 (*The Orthogonal Case of the Characterization*): Carry out the argument above for a real matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ and the standard scalar product on $\mathbb{R}^n$, thereby proving Lemma 22.b.9. (Prof. Biran remarks that it is *"similar"*; the point of writing it out is to see exactly where the complex conjugations of the Hermitian case disappear.)

Corollary 22.b.11 (*Rows of Orthogonal and Unitary Matrices*):

- **(a)** Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$. The matrix $A \in \mathrm{O}(n)$ if and only if $A^{\mathsf{T}} \in \mathrm{O}(n)$, which holds if and only if the rows of A form an orthonormal basis for $\mathbb{R}^n$.
- **(b)** Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$. The matrix $A \in \mathrm{U}(n)$ if and only if $A^* \in \mathrm{U}(n)$, which holds if and only if the rows of A form an orthonormal basis for $\mathbb{C}^n$.

Proof:

- **(a)** By Lemma 22.b.9, $A \in \mathrm{O}(n)$ if and only if $A^{\mathsf{T}}A = I_n$, and equally if and only if $AA^{\mathsf{T}} = I_n$. Applying the same characterization to the matrix A^{T}, whose transpose is A, the condition $A^{\mathsf{T}} \in \mathrm{O}(n)$ reads $(A^{\mathsf{T}})^{\mathsf{T}}A^{\mathsf{T}} = AA^{\mathsf{T}} = I_n$, the very same equation. Hence $A \in \mathrm{O}(n) \iff A^{\mathsf{T}} \in \mathrm{O}(n)$. Finally, the columns of A^{T} are exactly the rows of A, so $A^{\mathsf{T}} \in \mathrm{O}(n)$ says precisely that the rows of A form an orthonormal system, and an orthonormal system of n vectors in $\mathbb{R}^n$ is a basis by Theorem 22.a.14 **(a)**.
- **(b)** Identical, with Lemma 22.b.10 in place of Lemma 22.b.9 and A^* in place of A^{T}: the condition $A^* \in \mathrm{U}(n)$ reads $(A^*)^*A^* = AA^* = I_n$, which is one of the equivalent forms of $A \in \mathrm{U}(n)$; and the columns of A^* are the conjugates of the rows of A, which form an orthonormal system exactly when the rows do.

Q.E.D.

QR-Decomposition

Theorem 22.b.12 (*QR-Decomposition for Invertible Matrices*):

- **(a)** Let $A \in \mathrm{GL}_n(\mathbb{R})$. Then there exists $Q \in \mathrm{O}(n)$ and an upper triangular matrix $R \in \mathcal{M}_{n\times n}(\mathbb{R})$ such that $A = Q \cdot R$.
- **(b)** Let $A \in \mathrm{GL}_n(\mathbb{C})$. Then there exists $Q \in \mathrm{U}(n)$ and an upper triangular matrix $R \in \mathcal{M}_{n\times n}(\mathbb{C})$ such that $A = Q \cdot R$.

Proof of Theorem 22.b.12:

We prove parts **(a)** and **(b)** together, writing $K = \mathbb{R}$ or $K = \mathbb{C}$ throughout; the only difference between the two cases is whether the resulting Q is called orthogonal or unitary.

Let $A \in \mathrm{GL}_n(K)$ and write

$$A := \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

with $v_k \in K^n_{\text{col}}$. Since $A \in \mathrm{GL}_n(K)$, the vectors $v_1, \dots, v_n$ form a basis for K^n. Apply the Gram-Schmidt orthogonalization process (Theorem 22.a.16) to $v_1, \dots, v_n$ to obtain an orthonormal basis $e_1, \dots, e_n$ for K^n, and recall from part **(c)** of that theorem that

$$\mathrm{Sp}(v_1, \dots, v_j) = \mathrm{Sp}(e_1, \dots, e_j) \qquad \forall\, 1 \le j \le n. \tag{22.8}$$

Fix $1 \le j \le n$. By (22.8) the vector v_j lies in $\mathrm{Sp}(e_1, \dots, e_j)$, and since $(e_1, \dots, e_j)$ is orthonormal, Theorem 22.a.14 **(b)** identifies its coefficients: they are $\langle v_j, e_i\rangle / \langle e_i, e_i\rangle = \langle v_j, e_i\rangle$. So the whole list expands as

$$\begin{aligned}
v_1 &= \langle v_1, e_1\rangle e_1, \\
v_2 &= \langle v_2, e_1\rangle e_1 + \langle v_2, e_2\rangle e_2, \\
&\ \vdots \\
v_j &= \langle v_j, e_1\rangle e_1 + \dots + \langle v_j, e_j\rangle e_j, \\
&\ \vdots \\
v_n &= \langle v_n, e_1\rangle e_1 + \dots + \langle v_n, e_n\rangle e_n,
\end{aligned}$$

with no e_i of index $i > j$ ever appearing in v_j. Read column by column, these n identities are exactly one matrix product:

$$\underbrace{\begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} | & & | \\ e_1 & \dots & e_n \\ | & & | \end{pmatrix}}_{Q} \underbrace{\begin{pmatrix} \langle v_1, e_1\rangle & \langle v_2, e_1\rangle & \dots & \langle v_n, e_1\rangle \\ 0 & \langle v_2, e_2\rangle & \dots & \langle v_n, e_2\rangle \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \langle v_n, e_n\rangle \end{pmatrix}}_{R},$$

that is, $R_{ij} = \langle v_j, e_i\rangle$ for $i \le j$ and $R_{ij} = 0$ for $i > j$.

The columns of Q are $e_1, \dots, e_n$, an orthonormal basis of K^n, so $Q \in \mathrm{O}(n)$ when $K = \mathbb{R}$ and $Q \in \mathrm{U}(n)$ when $K = \mathbb{C}$ (Definition 22.b.8); and R is upper triangular by construction. Hence $A = Q \cdot R$ as required. **Q.E.D.**

Theorem (*Extension of QR-Decomposition for All Matrices*): Let $A \in \mathcal{M}_{n\times n}(K)$, and let $r := \mathrm{rank}(A)$. Then there exist matrices Q and R with $Q \in \mathrm{O}(n)$ if $K = \mathbb{R}$ (and $Q \in \mathrm{U}(n)$ if $K = \mathbb{C}$), and $R = \left(\begin{array}{c|c} C & * \\ \hline 0 & 0 \end{array}\right) \in \mathcal{M}_{n\times n}(K)$, where $C \in \mathcal{M}_{r\times r}(K)$ is an upper triangular matrix, such that $A = Q \cdot R$.

Proof:

Let

$$A := \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

Note that $r = \dim \mathrm{Sp}(v_1, \dots, v_n)$. We may assume $r \ge 1$, because otherwise $r = 0$, meaning $A = 0$, in which case we can take $R = 0$ and Q to be any arbitrary orthogonal/unitary matrix. Note that there exist indices $1 \le i_1 < \dots < i_r \le n$ such that:

(a) $v_{i_1}, \dots, v_{i_r}$ are linearly independent.

(b) $v_{i_1}, \dots, v_{i_k}$ is a basis for $\mathrm{Sp}(v_1, v_2, \dots, v_{i_k})$ (and $k = \dim \mathrm{Sp}(v_1, v_2, \dots, v_{i_k}) \le i_k$).

To see this, let $1 \le i_1$ be the minimal index j such that $v_j \ne 0$. (Since $r \ge 1$, there exists such an index.) Clearly, $\mathrm{Sp}(v_1, \dots, v_{i_1}) = \mathrm{Sp}(v_{i_1})$. Suppose we have defined $1 \le i_1 < \dots < i_k < n$ as above. Consider now the minimal index ℓ, with $i_k < \ell \le n$ and such that $v_\ell \notin \mathrm{Sp}(v_{i_1}, \dots, v_{i_k})$.

If there is no such ℓ, then $k = r$ and we are done. Otherwise, take $i_{k+1} := \ell$. Continue the construction by induction.

Now, apply the Gram-Schmidt process to the list of vectors $v_{i_1}, \ldots, v_{i_r}$ and obtain an orthonormal system $e_1, \ldots, e_r$ such that $(e_1, \ldots, e_k)$ is a basis for $\mathrm{Sp}(v_{i_1}, \ldots, v_{i_k}) = \mathrm{Sp}(v_1, v_2, \ldots, v_{i_k})$. Extend $e_1, \ldots, e_r$ to an orthonormal basis $(e_1, \ldots, e_r, e_{r+1}, \ldots, e_n)$ of K^n (how? See Exercise 22.11). Take

$$Q := \begin{pmatrix} | & & | \\ e_1 & \ldots & e_n \\ | & & | \end{pmatrix},$$

and let R be built exactly as in the previous proof, by $R_{ij} := \langle v_j, e_i \rangle$.

The block shape of R is now forced. Every v_j lies in $\mathrm{Sp}(v_1, \ldots, v_j) = \mathrm{Sp}(e_1, \ldots, e_{k(j)})$, where $k(j) \leq r$ counts how many of the indices $i_1 < \cdots < i_r$ do not exceed j; in particular no e_i with $i > r$ ever occurs, so the last $n - r$ rows of R vanish. Among the first r rows, the columns $j = i_1, \ldots, i_r$ carry the upper triangular block $C := \left(\langle v_{i_j}, e_i \rangle\right)_{1 \leq i,j \leq r}$, upper triangular for the same reason as before, while the remaining columns are the arbitrary entries denoted by $*$. Hence $A = Q \cdot R$ with R of the stated form. **Q.E.D.**

Exercise 22.11 (*Extending an Orthonormal System to an Orthonormal Basis*): Let V be a finite-dimensional inner product space and let $(e_1, \ldots, e_r)$ be an orthonormal system in V with $r \leq n := \dim V$. Show that it can be extended to an orthonormal basis $(e_1, \ldots, e_r, e_{r+1}, \ldots, e_n)$ of V. (This is the step marked *"how?"* in the proof above.)

Exercise 22.12 (*Gram-Schmidt on Linearly Dependent Vectors*): Let V be an inner product space. Suppose $v_1, \ldots, v_m \in V$ are not necessarily linearly independent, but not all of them are 0. What happens if we apply the Gram-Schmidt orthogonalization process?

Suppose that at stage j we already have an orthogonal system $e_1, \ldots, e_k$ that spans $\mathrm{Sp}(v_1, \ldots, v_{i_k})$ and we encounter a vector $v_{i_{k+1}} \in \mathrm{Sp}(v_1, v_2, \ldots, v_{i_k})$. How will the next vector in the Gram-Schmidt process look like?

Solutions to the Exercises

Proof of Lemma 22.b.2 (b), the identity $0^\perp = V$:

Here 0 denotes the subset $\{0\} \subseteq V$, so by Definition 22.b.1

$$0^\perp = \{v \in V \mid \langle v, 0 \rangle = 0\}.$$

By Lemma 22.a.3 **(a)** we have $\langle v, 0 \rangle = 0$ for **every** $v \in V$, so the defining condition is satisfied by every vector of V and $0^\perp = V$.

The companion statement $V^\perp = \{0\}$, proved in the text, is the mirror image: the larger the set, the more orthogonality conditions, and the smaller its complement. **Q.E.D.**

Proof of Lemma 22.b.2 (d):

Assume $S \subseteq T$ and let $v \in T^\perp$, so that $\langle v, t \rangle = 0$ for every $t \in T$. Every $s \in S$ is in particular an element of T, so $\langle v, s \rangle = 0$ for every $s \in S$, that is, $v \in S^\perp$. Hence $T^\perp \subseteq S^\perp$, which is the assertion $S^\perp \supseteq T^\perp$.

Note that the inclusion reverses: passing to a larger set imposes more conditions on v, so fewer vectors survive. **Q.E.D.**

Proof of Exercise 22.10, that is, of Lemma 22.b.9:

Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}, \qquad v_k \in \mathbb{R}^n_{\mathrm{col}},$$

and recall that the standard scalar product on $\mathbb{R}^n$ is $\langle u, v\rangle = u^\mathsf{T} v$.

(a) $\iff$ **(b)**. The rows of A^T are the vectors $v_1^\mathsf{T}, \dots, v_n^\mathsf{T}$, so

$$A^\mathsf{T} A = \begin{pmatrix} — & v_1^\mathsf{T} & — \\ & \vdots & \\ — & v_n^\mathsf{T} & — \end{pmatrix} \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix} = \left(v_i^\mathsf{T} v_j\right)_{1\le i,j\le n} = \left(\langle v_i, v_j\rangle\right)_{1\le i,j\le n}.$$

By Definition 22.b.8, A is orthogonal exactly when $\langle v_i, v_j\rangle = \delta_{ij}$ for all i, j, which says precisely that the matrix just computed is I_n.

This is where the conjugations of the Hermitian case have gone. There the left factor was $\overline{A}^\mathsf{T}$, so the (i,j) entry came out as $\overline{v_i}^\mathsf{T} v_j = \langle v_j, v_i\rangle$, the arguments in the opposite order. Over $\mathbb{R}$ there is no conjugation, the scalar product is symmetric, and the entry is $\langle v_i, v_j\rangle$. Since the condition imposed is the symmetric one δ_{ij}, the difference is invisible in the conclusion, but it is real in the computation.

(b) $\implies$ **(d)**. Suppose $A^\mathsf{T} A = I_n$. Taking determinants and using $\det(A^\mathsf{T}) = \det(A)$ gives $\det(A)^2 = 1$, so $\det(A) = \pm 1 \neq 0$ and A is invertible. Multiplying $A^\mathsf{T} A = I_n$ on the right by A^{-1} yields $A^\mathsf{T} = A^{-1}$.

(d) $\implies$ **(c)**. If $A^\mathsf{T} = A^{-1}$, then $AA^\mathsf{T} = AA^{-1} = I_n$.

(c) $\implies$ **(b)**. Suppose $AA^\mathsf{T} = I_n$. Determinants again give $\det(A)^2 = 1$, so A is invertible; multiplying on the left by A^{-1} gives $A^\mathsf{T} = A^{-1}$, and hence $A^\mathsf{T} A = A^{-1} A = I_n$.

The four statements are therefore equivalent. **Q.E.D.**

Proof of Exercise 22.11:

An orthonormal system is in particular an orthogonal system, hence linearly independent by Theorem 22.a.14 **(a)**. So $(e_1, \dots, e_r)$ can be extended to a basis $(e_1, \dots, e_r, u_{r+1}, \dots, u_n)$ of V.

Now apply the Gram-Schmidt process (Theorem 22.a.16) to that basis, producing $w_1, \dots, w_n$. The point is that Gram-Schmidt does not disturb an initial segment that is already orthonormal. Indeed $w_1 = e_1$, and if $2 \le j \le r$ and $w_i = e_i$ for all $i < j$, then

$$w_j = e_j - \sum_{i=1}^{j-1} \frac{\langle e_j, w_i\rangle}{\langle w_i, w_i\rangle} w_i = e_j - \sum_{i=1}^{j-1} \frac{\langle e_j, e_i\rangle}{1} e_i = e_j,$$

because $\langle e_j, e_i\rangle = 0$ for $i \neq j$ and $\langle e_i, e_i\rangle = 1$. By induction $w_i = e_i$ for all $1 \le i \le r$.

Therefore $(w_1, \dots, w_n) = (e_1, \dots, e_r, w_{r+1}, \dots, w_n)$ is an orthogonal basis of V, and normalizing only the tail (the first r vectors already have norm 1) gives the orthonormal basis

$$\left(e_1, \dots, e_r, \frac{w_{r+1}}{\|w_{r+1}\|}, \dots, \frac{w_n}{\|w_n\|}\right)$$

extending the given system.

There is a second route, worth recording because it is the one the geometry suggests: put $U := \mathrm{Sp}(e_1, \dots, e_r)$. By Corollary 22.b.7, $\dim U^\perp = n - r$; choose an orthonormal basis $(f_{r+1}, \dots, f_n)$ of $U^\perp$, which exists by Corollary 22.a.17. Every f_i is orthogonal to every e_j by the definition of $U^\perp$, so the concatenation $(e_1, \dots, e_r, f_{r+1}, \dots, f_n)$ is an orthonormal system of n vectors, hence linearly independent, hence a basis of V. **Q.E.D.**

Solution of Exercise 22.12:

The next vector is 0. Suppose that after k successful steps we have an orthogonal system $e_1, \ldots, e_k$ with

$$U := \mathrm{Sp}(e_1, \ldots, e_k) = \mathrm{Sp}(v_1, \ldots, v_{i_k}),$$

and that the next vector to be processed satisfies $v_{i_{k+1}} \in \mathrm{Sp}(v_1, \ldots, v_{i_k}) = U$. The Gram-Schmidt formula prescribes

$$w := v_{i_{k+1}} - \sum_{i=1}^{k} \frac{\langle v_{i_{k+1}}, e_i\rangle}{\langle e_i, e_i\rangle} e_i.$$

But $(e_1, \ldots, e_k)$ is an orthogonal system spanning U, and $v_{i_{k+1}}$ lies in U, so Theorem 22.a.14 **(b)** tells us the coefficients of $v_{i_{k+1}}$ in that system exactly:

$$v_{i_{k+1}} = \sum_{i=1}^{k} \frac{\langle v_{i_{k+1}}, e_i\rangle}{\langle e_i, e_i\rangle} e_i.$$

The subtracted sum is therefore $v_{i_{k+1}}$ itself; it is the orthogonal projection $P_U(v_{i_{k+1}})$, which fixes U pointwise by Proposition 22.b.5 **(d)**, and consequently

$$w = v_{i_{k+1}} - v_{i_{k+1}} = 0.$$

What this means for the process. The construction then stalls: the next step would divide by $\langle w, w\rangle = 0$. Geometrically this is exactly right, $v_{i_{k+1}}$ contributes no new direction, so there is nothing left after removing its component along U.

The remedy. Discard every vector that produces 0 and carry on with the next one. The vectors that survive are precisely a maximal linearly independent sublist $v_{i_1}, \ldots, v_{i_r}$ of $v_1, \ldots, v_m$, where $r = \dim \mathrm{Sp}(v_1, \ldots, v_m)$, and the output $e_1, \ldots, e_r$ is an orthogonal basis of $\mathrm{Sp}(v_1, \ldots, v_m)$ satisfying $\mathrm{Sp}(e_1, \ldots, e_k) = \mathrm{Sp}(v_1, \ldots, v_{i_k})$ for every k.

This is what the proof of the extension of the QR-decomposition on Page 232 arranges in advance: rather than letting the process stall, it selects the indices $i_1 < \cdots < i_r$ first and applies Gram-Schmidt only to $v_{i_1}, \ldots, v_{i_r}$, which are linearly independent by construction.

Q.E.D.

AI-Note: Nothing in this section is false, and the source-verification pass had already restored the QR construction and unnumbered the QR extension, which is why Page 232 is a `theorem*` and 22.b.12 lands where Prof. Biran puts it.

The rigour pass named four steps. The proof of Theorem 22.b.3 reads $U \cap U^{\perp} = \{0\}$ off Lemma 22.b.2 **(c)**, which offers a choice between $\emptyset$ and $\{0\}$; what excludes the first is that U, being a subspace, contains 0. That proof also picks an orthonormal basis of U without saying where one comes from, which is Corollary 22.a.17 of the previous section.

Proposition 22.b.5 rests its first four parts on a general fact about projections along a direct sum, asserted in one sentence and proved nowhere in the book. The well-definedness of that projection is the uniqueness in Corollary 14.3 **(d)**, and its linearity follows from the same uniqueness in two lines; both are now written out. Finally, Corollary 22.b.7 concatenates two bases into one without naming the basis criterion of Exercise 21.1.

The two places where the text says *"exercise"* in the middle of a proof, the identity $0^{\perp} = V$ and the monotonicity Lemma 22.b.2 **(d)**, already had their solutions at the end of the section, and **(d)** is used in the proof of **(e)** directly above it.

Dual Spaces and Inner Products (Chapter 23)

Dual spaces and inner products (Section 23.a)

Let V be a vector space over K (any field). Recall that we have the dual space $V^* := \mathrm{Hom}(V, K)$, which is the space of linear maps $\varphi\colon V \to K$ (also called "*functionals*"). The dual space V^* is also a vector space over K.

Warm up 23.1: Assume $V = K^n$ and let $\varphi\colon K^n \to K$ be a functional. Then there exists a unique matrix $A = (a_1, \dots, a_n) \in \mathcal{M}_{1\times n}(K)$ such that $\varphi(v) = A \cdot v$ for all $v \in K^n$.

Indeed, let $e_1, \dots, e_n$ be the standard basis for K^n, and define $a_1 := \varphi(e_1), \dots, a_n := \varphi(e_n)$. Let $v = (v_1, \dots, v_n)^\mathsf{T} \in K^n$. Then

$$\varphi(v) = \varphi(v_1 e_1 + \cdots + v_n e_n) = v_1\varphi(e_1) + \cdots + v_n\varphi(e_n) = v_1 a_1 + \cdots + v_n a_n = A \cdot v.$$

This also settles the uniqueness of A: any row matrix $A' = (a'_1, \dots, a'_n)$ with $\varphi(v) = A' \cdot v$ for all $v \in K^n$ satisfies $a'_i = A' \cdot e_i = \varphi(e_i) = a_i$, so its entries are already forced by φ.

Consider now an inner product space $(V, \langle\cdot,\cdot\rangle)$ over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $u \in V$. We define a map $\varphi_u\colon V \to K$ by mapping $v \mapsto \langle v, u\rangle \in K$.

Claim: The map φ_u is linear.

(This is left as an exercise).

Theorem 23.a.1: Let $(V, \langle\cdot,\cdot\rangle)$ be a finite-dimensional inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $\varphi \in V^*$. Then there exists a unique vector $u \in V$ such that $\varphi = \varphi_u$, that is, $\varphi(v) = \langle v, u\rangle$ for all $v \in V$.

In fact, for $K = \mathbb{R}$, the map $\Phi\colon V \to V^*$, given by $\Phi(u) := \varphi_u$, is an isomorphism. For $K = \mathbb{C}$, this map is injective and surjective, but it is only $\mathbb{C}$-antilinear, meaning that $\Phi(u_1+u_2) = \Phi(u_1)+\Phi(u_2)$ and $\Phi(c \cdot u) = \overline{c} \cdot \Phi(u)$ for all $c \in \mathbb{C}$ and $u \in V$.

So, at least for $K = \mathbb{R}$, a scalar product on V gives us a canonical isomorphism $V \cong V^*$. (However, this isomorphism depends on the scalar product!)

Proof:

First, we prove that for all $\varphi \in V^*$ there exists a unique $u \in V$ such that $\varphi = \varphi_u$. Fix an orthonormal basis $\mathcal{B} := (e_1, \dots, e_n)$ for V, where $n = \dim V$; one exists by Corollary 22.a.17. Let $\varphi \in V^*$, and let $v \in V$. We have $v = \sum_{i=1}^n \langle v, e_i\rangle \cdot e_i$, the coefficients of a vector in an orthonormal system being given by Theorem 22.a.14 **(b)**, with $\langle e_i, e_i\rangle = 1$. This implies:

$$\varphi(v) = \varphi\left(\sum_{i=1}^n \langle v, e_i\rangle \cdot e_i\right) = \sum_{i=1}^n \langle v, e_i\rangle \cdot \varphi(e_i) = \left\langle v, \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i\right\rangle.$$

(Note that the conjugation is needed here only if $K = \mathbb{C}$). Define $u := \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i$. It follows that $\varphi(v) = \langle v, u\rangle$, so $\varphi = \varphi_u$.

Uniqueness of u: Suppose that $\varphi_{u_1} = \varphi_{u_2}$ for $u_1, u_2 \in V$. This implies that $\langle v, u_1\rangle = \langle v, u_2\rangle$ for all $v \in V$. By a previous result (Lemma 22.a.3 **(c)**), this implies that $u_1 = u_2$.

The statements about the linearity of Φ are left as an exercise. **Q.E.D.**

The adjoint map

Let V, W be vector spaces over K. Let $T\colon V \to W$ be a linear map. We have seen in Linear Algebra I that T induces a linear map $T^*\colon W^* \to V^*$ defined by $T^*(\varphi) := \varphi \circ T$.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[dr, "T^*(\varphi)=\varphi\circ T"'] & W \arrow[d, "\varphi"] \\ & K \end{tikzcd}$$

Assume now that $(V, \langle\cdot,\cdot\rangle_V)$ and $(W, \langle\cdot,\cdot\rangle_W)$ are inner product spaces over K (where $K = \mathbb{R}$ or $K = \mathbb{C}$). Assume that $\dim V < \infty$. Given T as above, we can define a map $T'\colon W \to V$ using our *"canonical"* identifications $W \cong W^*$ and $V \cong V^*$. (This is straightforward for $K = \mathbb{R}$, but we have to be careful with $K = \mathbb{C}$.)

$$\begin{tikzcd} W^* \arrow[r, "T^*"] & V^* \arrow[d, shift left, "\Phi_V^{-1}"] \\ W \arrow[u, "\Phi_W"] \arrow[r, dashed, "T'"'] & V \arrow[u, shift left, "\Phi_V"] \end{tikzcd}$$

Here, we define $T' := \Phi_V^{-1} \circ T^* \circ \Phi_W$. (Note that here we are using the fact that $\dim V < \infty$ so that Φ_V^{-1} exists).

What properties does T' have? Note that T' is uniquely defined by $\Phi_V \circ T' = T^* \circ \Phi_W$. We can unpack this relation as follows:

$$\begin{aligned} &\Phi_V \circ T' = T^* \circ \Phi_W \\ &\qquad\iff \forall w \in W, \quad (\Phi_V \circ T')(w) = (T^* \circ \Phi_W)(w) \\ &\qquad\iff \varphi_{T'(w)} = T^*(\varphi_w) \quad (\text{in } V^*) \\ &\qquad\iff \forall v \in V, \quad \varphi_{T'(w)}(v) = (T^*(\varphi_w))(v) \\ &\qquad\iff \langle v, T'(w)\rangle_V = \varphi_w(T(v)) = \langle T(v), w\rangle_W. \end{aligned}$$

In other words,

$$\langle T(v), w\rangle_W = \langle v, T'(w)\rangle_V \quad \forall v \in V, \forall w \in W. \tag{23.1}$$

Claim: The map T' is a linear map.

Proof:

For $K = \mathbb{R}$, the maps Φ_W, Φ_V, and T^* are linear, and therefore T' is also linear. For $K = \mathbb{C}$, the maps Φ_W and Φ_V are additive but only $\mathbb{C}$-antilinear. This implies that Φ_V^{-1} is also additive and $\mathbb{C}$-antilinear (this is an exercise to check!). The map T^* remains linear. Consequently, $T'(w_1 + w_2) = T'(w_1) + T'(w_2)$ (exercise). Now, let $c \in \mathbb{C}$ and $w \in W$. We compute:

$$\begin{aligned} T'(cw) &= (\Phi_V^{-1} \circ T^* \circ \Phi_W)(cw) \\ &= \Phi_V^{-1}(T^*(\bar{c} \cdot \Phi_W(w))) \\ &= \Phi_V^{-1}(\bar{c} \cdot T^*(\Phi_W(w))) \\ &= c \cdot \Phi_V^{-1}(T^*(\Phi_W(w))) \\ &= c \cdot T'(w). \end{aligned}$$

This proves that T' is linear. **Q.E.D.**

Notation 23.1: The linear map $T'\colon W \to V$ is denoted many times by $T^*\colon W \to V$. This makes sense because of the *"canonical"* identifications:

$$T\colon V \to W \quad \rightsquigarrow \quad \begin{tikzcd} W^* \arrow[r, "T^*"] \arrow[d, phantom, "\wr\|"] & V^* \arrow[d, phantom, "\wr\|"] \\ W \arrow[r, "T'"'] & V \end{tikzcd}$$

(Here we also assume $\dim W < \infty$ to have the canonical identification on the left). The map $T^*\colon W \to V$ is called the "*adjoint*" of T.

Basic properties of adjoint maps

Lemma 23.a.2 (*Properties of the Adjoint Map*): Let V, W, U be inner product spaces over K (where $K = \mathbb{R}$ or $\mathbb{C}$), all finite-dimensional. Let $S, T\colon V \to W$ and $R\colon W \to U$ be linear maps. Then:

(a) T^* is a linear map.
(b) $(S+T)^* = S^* + T^*$.
(c) $(\lambda T)^* = \overline{\lambda} T^*$ for all $\lambda \in K$.
(d) $(T^*)^* = T$.
(e) $(\mathrm{id}_V)^* = \mathrm{id}_V$.
(f) $(R \circ T)^* = T^* \circ R^*$.

Proof:

(a) This was proved already. (We will use repeatedly that if $\langle x, y_1\rangle = \langle x, y_2\rangle$ for all x, then $y_1 = y_2$; this is Lemma 22.a.3 **(c)**).

(b) Let $w \in W$ and $v \in V$. We have:

$$\begin{aligned} \langle v, (S+T)^*(w)\rangle &= \langle (S+T)(v), w\rangle \\ &= \langle S(v), w\rangle + \langle T(v), w\rangle \\ &= \langle v, S^*(w)\rangle + \langle v, T^*(w)\rangle \\ &= \langle v, (S^* + T^*)(w)\rangle. \end{aligned}$$

This holds for all $v \in V$, which implies $(S+T)^*(w) = (S^*+T^*)(w)$. Since this holds for all $w \in W$, we conclude that $(S+T)^* = S^* + T^*$.

(c) This is left as an exercise.

(d) Let $v \in V$ and $w \in W$. We have:

$$\begin{aligned} \langle w, (T^*)^*(v)\rangle &= \langle T^*(w), v\rangle \\ &= \overline{\langle v, T^*(w)\rangle} \\ &= \overline{\langle T(v), w\rangle} \\ &= \langle w, T(v)\rangle. \end{aligned}$$

This holds for all $w \in W$, which implies $(T^*)^*(v) = T(v)$. Since this holds for all $v \in V$, we conclude that $(T^*)^* = T$.

(e) This is left as an exercise.

(f) Let $v \in V$ and $u \in U$. We have:

$$\begin{aligned} \langle v, (R\circ T)^*(u)\rangle &= \langle (R\circ T)(v), u\rangle \\ &= \langle R(T(v)), u\rangle \\ &= \langle T(v), R^*(u)\rangle \\ &= \langle v, T^*(R^*(u))\rangle \\ &= \langle v, (T^*\circ R^*)(u)\rangle. \end{aligned}$$

This holds for all $v \in V$, which implies $(R \circ T)^*(u) = (T^* \circ R^*)(u)$. Since this holds for all $u \in U$, we conclude that $(R \circ T)^* = T^* \circ R^*$.

Q.E.D.

Lemma 23.a.3 (*Orthogonal Relations of Adjoint Maps*): Let V, W be finite-dimensional inner product spaces. Let $T\colon V \to W$ and let $T^*\colon W \to V$ be the adjoint of T. Then:

(a) $\ker(T^*) = (\operatorname{Im} T)^{\perp}$.
(b) $\operatorname{Im} T^* = (\ker T)^{\perp}$.
(c) $\ker T = (\operatorname{Im} T^*)^{\perp}$.
(d) $\operatorname{Im} T = (\ker T^*)^{\perp}$.

Proof:

Only **(a)** is established from scratch; the notes then deduce the other three from it in the order **(d)**, **(b)**, **(c)**, each step relying only on the ones already obtained. The forward references below are therefore not circular. The passages from a subspace to its double orthogonal complement all invoke Corollary 22.b.6, which requires the subspace in question to be finite-dimensional; we point out each time why it is.

(a) Let $w \in W$. Then:

$$\begin{aligned} w \in \ker(T^*) &\iff T^*(w) = 0 \\ &\iff \forall v \in V, \quad \langle v, T^*(w)\rangle = 0 \\ &\iff \forall v \in V, \quad \langle T(v), w\rangle = 0 \\ &\iff w \perp T(v) \quad \forall v \in V \\ &\iff w \perp \operatorname{Im}(T) \\ &\iff w \in (\operatorname{Im} T)^{\perp}. \end{aligned}$$

(b) Use statement **(d)** with T^* instead of T:

$$\operatorname{Im} T^* = (\ker(T^*)^*)^{\perp} = (\ker T)^{\perp}.$$

(c) Apply the orthogonal complement $(-)^{\perp}$ to statement **(b)**:

$$(\operatorname{Im} T^*)^{\perp} = ((\ker T)^{\perp})^{\perp} = \ker T.$$

The second equality is Corollary 22.b.6 applied to the subspace $\ker T \subseteq V$, which is finite-dimensional because V is.

(d) Apply the orthogonal complement $(-)^{\perp}$ to both sides of **(a)**:

$$(\ker T^*)^{\perp} = ((\operatorname{Im} T)^{\perp})^{\perp} = \operatorname{Im} T.$$

Here the second equality is Corollary 22.b.6 applied to $\operatorname{Im} T \subseteq W$; recall that $\operatorname{Im} T$ is finite-dimensional, being the image of the finite-dimensional space V.

Q.E.D.

Matrix representation of adjoint maps

Proposition 23.a.4 (*Matrix Representation of the Adjoint*): Let V, W be finite-dimensional inner product spaces and $T\colon V \to W$ a linear map. Let $\mathcal{B} := (e_1, \ldots, e_n)$ be an orthonormal basis for V, and $\mathcal{C} := (f_1, \ldots, f_m)$ an orthonormal basis for W. Then:

$$[T^*]^{\mathcal{C}}_{\mathcal{B}} = \left(\overline{[T]^{\mathcal{B}}_{\mathcal{C}}}\right)^{\mathsf{T}} = \left([T]^{\mathcal{B}}_{\mathcal{C}}\right)^* \quad (\text{“\textit{adjoint matrix}”})$$

Proof:

Write $A := (a_{ij})_{\substack{1\leq i\leq m\\ 1\leq j\leq n}} := [T]^{\mathcal{B}}_{\mathcal{C}}$ and $B := (b_{ij})_{\substack{1\leq i\leq n\\ 1\leq j\leq m}} := [T^*]^{\mathcal{C}}_{\mathcal{B}}$. Both bases being orthonormal, Proposition 22.a.15 reads the entries of a representation matrix off the inner product: $b_{ij} = \langle T^*(f_j), e_i\rangle$ and $a_{ji} = \langle T(e_i), f_j\rangle$. Hence

$$b_{ij} = \langle T^*(f_j), e_i\rangle = \overline{\langle e_i, T^*(f_j)\rangle} = \overline{\langle T(e_i), f_j\rangle} = \overline{a_{ji}},$$

the middle step by the Hermitian axiom and the next by the defining property (23.1) of the adjoint. Therefore, $B = (\overline{A})^{\mathsf{T}} = A^*$. **Q.E.D.**

Corollary 23.a.5 (*Adjoint of a Matrix Map*): Let $A \in \mathcal{M}_{m\times n}(K)$ (where $K = \mathbb{R}$ or $\mathbb{C}$), and let $T_A\colon K^n \to K^m$ be the linear map defined by $T_A(v) = A\cdot v$. Endow K^n and K^m with their standard inner products. Then $(T_A)^* = T_{A^*}$.

Proof:

Let $\mathcal{E} := (e_1,\dots,e_n)$ and $\mathcal{F} := (f_1,\dots,f_m)$ be the standard bases of K^n and of K^m. With respect to the standard inner products both are orthonormal, so Proposition 23.a.4 applies to them. The j-th column of $[T_A]^{\mathcal{E}}_{\mathcal{F}}$ is $T_A(e_j) = A\cdot e_j$, the j-th column of A, and therefore $[T_A]^{\mathcal{E}}_{\mathcal{F}} = A$. Consequently,

$$[(T_A)^*]^{\mathcal{F}}_{\mathcal{E}} = \left([T_A]^{\mathcal{E}}_{\mathcal{F}}\right)^* = A^*.$$

The same computation applied to A^* gives $[T_{A^*}]^{\mathcal{F}}_{\mathcal{E}} = A^*$. Two linear maps $K^m \to K^n$ with the same representation matrix with respect to the same pair of bases are equal, and hence $(T_A)^* = T_{A^*}$. **Q.E.D.**

Properties of the adjoint matrix

Proposition 23.a.6 (*Properties of the Adjoint Matrix*): Let $K = \mathbb{R},\mathbb{C}$, and let $A, B \in \mathcal{M}_{m\times n}(K)$ and $C \in \mathcal{M}_{n\times p}(K)$. Then:

- **(a)** $\overline{(A+B)}^{\mathsf{T}} = \overline{A}^{\mathsf{T}} + \overline{B}^{\mathsf{T}}$.
- **(b)** $\overline{(\lambda A)}^{\mathsf{T}} = \overline{\lambda A}^{\mathsf{T}}$ for all $\lambda \in K$.
- **(c)** $\overline{(\overline{A}^{\mathsf{T}})}^{\mathsf{T}} = A$.
- **(d)** $\overline{I_n}^{\mathsf{T}} = I_n$.
- **(e)** $\overline{(A\cdot C)}^{\mathsf{T}} = \overline{C}^{\mathsf{T}}\cdot\overline{A}^{\mathsf{T}}$.

Proof:

This is left as an exercise. (The verification is carried out in the solutions at the end of this section.) **Q.E.D.**

Solutions to the Exercises

Proof of Linearity of φ_u (on Page 236):

Let $v_1, v_2 \in V$ and $c \in K$. We have:

$$\varphi_u(v_1+v_2) = \langle v_1+v_2, u\rangle = \langle v_1, u\rangle + \langle v_2, u\rangle = \varphi_u(v_1) + \varphi_u(v_2),$$

and

$$\varphi_u(cv_1) = \langle cv_1, u\rangle = c\langle v_1, u\rangle = c\varphi_u(v_1).$$

Thus, φ_u is a linear map. **Q.E.D.**

Proof of Theorem 23.a.1:

For $K=\mathbb{R}$, we have $\Phi(u)=\varphi_u$. Then:

$$\Phi(u_1+u_2)(v)=\langle v,u_1+u_2\rangle=\langle v,u_1\rangle+\langle v,u_2\rangle=\Phi(u_1)(v)+\Phi(u_2)(v),$$

and

$$\Phi(cu)(v)=\langle v,cu\rangle=c\langle v,u\rangle=c\Phi(u)(v).$$

(Here we used the fact that for $K=\mathbb{R}$, $c=\overline{c}$). Thus Φ is linear.

For $K=\mathbb{C}$, the additivity follows identically. For scalar multiplication:

$$\Phi(cu)(v)=\langle v,cu\rangle=\overline{c}\langle v,u\rangle=\overline{c}\Phi(u)(v).$$

Thus $\Phi(cu)=\overline{c}\Phi(u)$, making it $\mathbb{C}$-antilinear.

For the inverse Φ^{-1}: since Φ is additive, taking $\varphi_1=\Phi(u_1)$ and $\varphi_2=\Phi(u_2)$ gives $\Phi^{-1}(\varphi_1+\varphi_2)=\Phi^{-1}(\varphi_1)+\Phi^{-1}(\varphi_2)$, so Φ^{-1} is additive. For scalar multiplication, let $\varphi=\Phi(u)$ and $\lambda\in\mathbb{C}$. Then $\overline{\lambda}\varphi=\overline{\lambda}\Phi(u)=\Phi(\lambda u)$. Applying Φ^{-1} to both sides gives $\Phi^{-1}(\overline{\lambda}\varphi)=\lambda u=\lambda\Phi^{-1}(\varphi)$. Setting $c=\overline{\lambda}$ (so $\lambda=\overline{c}$) gives $\Phi^{-1}(c\varphi)=\overline{c}\Phi^{-1}(\varphi)$, proving Φ^{-1} is $\mathbb{C}$-antilinear.

This last paragraph is also the verification Prof. Biran marks with *"check!"* in the proof that T' is linear: it is exactly the assertion that Φ_V^{-1} is additive and $\mathbb{C}$-antilinear. **Q.E.D.**

Proof of Additivity of T' (on Page 237):

We have:

$$\begin{aligned}
T'(w_1+w_2)&=(\Phi_V^{-1}\circ T^*\circ\Phi_W)(w_1+w_2)\\
&=(\Phi_V^{-1}\circ T^*)(\Phi_W(w_1)+\Phi_W(w_2)) && \text{(since } \Phi_W \text{ is additive)}\\
&=\Phi_V^{-1}(T^*(\Phi_W(w_1))+T^*(\Phi_W(w_2))) && \text{(since } T^* \text{ is linear)}\\
&=\Phi_V^{-1}(T^*(\Phi_W(w_1)))+\Phi_V^{-1}(T^*(\Phi_W(w_2))) && \text{(since } \Phi_V^{-1} \text{ is additive)}\\
&=T'(w_1)+T'(w_2).
\end{aligned}$$

Q.E.D.

Proof of Lemma 23.a.2 (c):

Let $v\in V$ and $w\in W$. We have:

$$\langle v,(\lambda T)^*(w)\rangle=\langle(\lambda T)(v),w\rangle=\langle\lambda T(v),w\rangle=\lambda\langle T(v),w\rangle=\lambda\langle v,T^*(w)\rangle=\langle v,\overline{\lambda}T^*(w)\rangle.$$

This holds for all $v\in V$, so $(\lambda T)^*(w)=\overline{\lambda}T^*(w)$. Thus $(\lambda T)^*=\overline{\lambda}T^*$. **Q.E.D.**

Proof of Lemma 23.a.2 (e):

Let $v,w\in V$. We have:

$$\langle v,(\mathrm{id}_V)^*(w)\rangle=\langle\mathrm{id}_V(v),w\rangle=\langle v,w\rangle.$$

This holds for all $v\in V$, so $(\mathrm{id}_V)^*(w)=w=\mathrm{id}_V(w)$. Thus $(\mathrm{id}_V)^*=\mathrm{id}_V$. **Q.E.D.**

Proof of Proposition 23.a.6:

These statements follow directly from the properties of matrix transpose and complex conjugation:

- **(a)** $\overline{(A+B)}^{\mathsf{T}}=(\overline{A}+\overline{B})^{\mathsf{T}}=\overline{A}^{\mathsf{T}}+\overline{B}^{\mathsf{T}}$.
- **(b)** $\overline{(\lambda A)}^{\mathsf{T}}=(\overline{\lambda A})^{\mathsf{T}}=\overline{\lambda A}^{\mathsf{T}}$.
- **(c)** $\overline{(\overline{A}^{\mathsf{T}})}^{\mathsf{T}}=(\overline{\overline{A}}^{\mathsf{T}})^{\mathsf{T}}=(A^{\mathsf{T}})^{\mathsf{T}}=A$.
- **(d)** $\overline{I_n}^{\mathsf{T}}=I_n^{\mathsf{T}}=I_n$ (since I_n is a real matrix).

(e) $\overline{(A \cdot C)}^{\mathsf{T}} = (\overline{A} \cdot \overline{C})^{\mathsf{T}} = \overline{C}^{\mathsf{T}} \cdot \overline{A}^{\mathsf{T}}$.

Q.E.D.

AI-Note: The source-verification pass had already put this section right: it restored the subsection headings the transcription had dropped, corrected the direction of Φ_V^{-1} in the commutative square, and supplied the solutions to every step Prof. Biran leaves to the reader, including the *"check!"* that Φ_V^{-1} is additive and antilinear.

The rigour pass found nothing false here and named three steps. The proof of Theorem 23.a.1 fixes an orthonormal basis without saying where it comes from (Corollary 22.a.17) and expands v in it without naming the coefficient formula (Theorem 22.a.14 **(b)**). And the proof of Proposition 23.a.4 opens with $b_{ij} = \langle T^*(f_j), e_i \rangle$, which is Proposition 22.a.15 of the previous chapter and is exactly where the orthonormality of the two bases enters; the chain then runs through the Hermitian axiom and (23.1).

Worth recording as correct: the proof of Lemma 23.a.3 deduces its four parts in the order **(a)**, **(d)**, **(b)**, **(c)**, so that its forward references are not circular, and it checks each time that the subspace it feeds to Corollary 22.b.6 is finite-dimensional. That is the sort of bookkeeping the pass usually has to add.

Spectral Theorems (Chapter 24)

AI-Note: The professor left a note to *"explain the name"* of the Spectral Theorems. In linear algebra, the "*spectrum*" of a linear operator is its set of eigenvalues. The Spectral Theorems provide conditions under which an operator can be completely described by its spectrum, specifically, when there exists a basis for the entire space consisting solely of the operator's eigenvectors (making the operator diagonalizable).

Definition 24.a.1: Let V be an inner product space and $T\colon V \to V$ an endomorphism. We say that T is "*orthogonally diagonalizable*" if there exists an orthonormal basis for V consisting of eigenvectors of T.

Lemma 24.a.2: If T is orthogonally diagonalizable, then $T \circ T^* = T^* \circ T$. (Here, and wherever an adjoint appears in this chapter, V is understood to be finite-dimensional: the construction of T^* in the previous chapter inverts the map Φ_V, and every statement about adjoints, Lemma 23.a.2 included, assumes as much.)

Proof:

Let $\mathcal{B}$ be an orthonormal basis for V such that all the vectors in $\mathcal{B}$ are eigenvectors of T. This implies that the representation matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$ is a diagonal matrix.

Since $\mathcal{B}$ is an orthonormal basis, we also know that the matrix of the adjoint map is the conjugate transpose of the matrix of the map, meaning $[T^*]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$ (this is Proposition 23.a.4, applied with $\mathcal{C} := \mathcal{B}$). Since the conjugate transpose of a diagonal matrix is still a diagonal matrix, $[T^*]^{\mathcal{B}}_{\mathcal{B}}$ is also a diagonal matrix.

Recall that if M and N are diagonal matrices of the form

$$\begin{pmatrix} * & & 0 \\ & \ddots & \\ 0 & & * \end{pmatrix},$$

then they must commute, meaning $M \cdot N = N \cdot M$. Therefore, we have:

$$[T]^{\mathcal{B}}_{\mathcal{B}} \cdot [T^*]^{\mathcal{B}}_{\mathcal{B}} = [T^*]^{\mathcal{B}}_{\mathcal{B}} \cdot [T]^{\mathcal{B}}_{\mathcal{B}}.$$

By the properties of representation matrices (namely the composition rule, the Proposition on Page 116), this implies that

$$[T \circ T^*]^{\mathcal{B}}_{\mathcal{B}} = [T^* \circ T]^{\mathcal{B}}_{\mathcal{B}},$$

and therefore, $T \circ T^* = T^* \circ T$ (two endomorphisms with the same representation matrix in the same basis are equal, since $\Psi^{\mathcal{B}}_{\mathcal{B}}$ is an isomorphism by Theorem 18.a.2). **Q.E.D.**

Definition 24.a.3: Let V be an inner product space. An endomorphism $T\colon V \to V$ is called "*normal*" if $T \circ T^* = T^* \circ T$.

Similarly, a matrix $A \in \mathcal{M}_{n\times n}(K)$ (where $K = \mathbb{R}$ or $\mathbb{C}$) is called "*normal*" if $A \cdot \overline{A}^{\mathsf{T}} = \overline{A}^{\mathsf{T}} \cdot A$. (Note that A is a normal matrix if and only if the map $T_A\colon K^n \to K^n$ is a normal endomorphism, provided we endow K^n with its standard inner product).

We have seen in the previous lemma (Lemma 24.a.2) that if $T\colon V \to V$ is orthogonally diagonalizable, then T is normal.

Question 24.1: Is the converse statement true?

Over $K = \mathbb{R}$, the answer is NO! For example, consider the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The linear map $T_A \colon \mathbb{R}^2 \to \mathbb{R}^2$ is a clockwise rotation by $90°$, so there are no eigenvalues (in $\mathbb{R}$) and no eigenvectors (this is the transpose of the rotation of Example 21.5, and has the same characteristic polynomial $P_A(x) = x^2 + 1$, which has no root in $\mathbb{R}$; now apply AI-Proposition 21.1). Thus, it cannot be diagonalizable. However, $\overline{A}^{\mathsf{T}} = A^{\mathsf{T}} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and we can easily check that $A \cdot A^{\mathsf{T}} = A^{\mathsf{T}} \cdot A = I_2$, meaning A is indeed normal.

What about over $\mathbb{C}$?

Theorem 24.a.4 (*Spectral Theorem over* $\mathbb{C}$): Let V be a finite-dimensional inner product space over $\mathbb{C}$, and let $T \colon V \to V$ be a linear map. Then T is orthogonally diagonalizable if and only if T is normal.

For the proof, we need a lemma.

Lemma 24.a.5 (*Properties of Normal Endomorphisms*): Let V be a finite-dimensional inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $T \colon V \to V$ be a normal endomorphism. Then:

(a) For all $v \in V$, $\|Tv\| = \|T^* v\|$.

(b) For all $\lambda \in K$, the endomorphism $T - \lambda\,\mathrm{id}_V$ is also normal.

(c) If v is an eigenvector of T with eigenvalue λ, then v is also an eigenvector of T^* with eigenvalue $\overline{\lambda}$.

(d) Let v_1, v_2 be eigenvectors of T with eigenvalues λ_1, λ_2 respectively. If $\lambda_1 \neq \lambda_2$, then $\langle v_1, v_2 \rangle = 0$ (i.e., $v_1 \perp v_2$).

Proof:

(a) We compute the squared norm:

$$\|Tv\|^2 = \langle Tv, Tv \rangle = \langle v, T^*Tv \rangle = \langle v, TT^*v \rangle = \langle T^*v, T^*v \rangle = \|T^*v\|^2,$$

where we used the defining property (23.1) of the adjoint twice, in the first and in the last step, and the fact that T is normal ($T \circ T^* = T^* \circ T$) in the middle.

(b) We expand the compositions, using $(T - \lambda\,\mathrm{id}_V)^* = T^* - \overline{\lambda}\,\mathrm{id}_V$, which is Lemma 23.a.2 **(b)**, **(c)** and **(e)** taken together:

$$\begin{aligned}(T - \lambda\,\mathrm{id}_V) \circ (T - \lambda\,\mathrm{id}_V)^* &= (T - \lambda\,\mathrm{id}_V) \circ (T^* - \overline{\lambda}\,\mathrm{id}_V) \\ &= T \circ T^* - \overline{\lambda}T - \lambda T^* + \lambda\overline{\lambda}\,\mathrm{id}_V . \quad (*)\end{aligned}$$

On the other hand:

$$\begin{aligned}(T - \lambda\,\mathrm{id}_V)^* \circ (T - \lambda\,\mathrm{id}_V) &= (T^* - \overline{\lambda}\,\mathrm{id}_V) \circ (T - \lambda\,\mathrm{id}_V) \\ &= T^* \circ T - \lambda T^* - \overline{\lambda}T + \overline{\lambda}\lambda\,\mathrm{id}_V . \quad (**)\end{aligned}$$

Because $T \circ T^* = T^* \circ T$, the expressions $(*)$ and $(**)$ are identically equal.

(c) If v is an eigenvector of T with eigenvalue λ, then $(T - \lambda\,\mathrm{id}_V)v = 0$. By statement **(b)**, $T - \lambda\,\mathrm{id}_V$ is normal, hence by statement **(a)**, we have:

$$\|(T^* - \overline{\lambda}\,\mathrm{id}_V)v\| = \|(T - \lambda\,\mathrm{id}_V)v\| = \|0\| = 0.$$

This implies that $(T^* - \overline{\lambda}\,\mathrm{id}_V)v = 0$, and therefore $T^*v = \overline{\lambda}v$.

(d) We compute the inner product in two ways:

$$\lambda_1 \langle v_1, v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle = \langle Tv_1, v_2 \rangle = \langle v_1, T^*v_2 \rangle.$$

By statement **(c)**, $T^*v_2 = \overline{\lambda_2}v_2$. Thus:

$$\langle v_1, T^*v_2 \rangle = \langle v_1, \overline{\lambda_2}v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle.$$

Equating the two results gives $\lambda_1 \langle v_1, v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle$, which implies $(\lambda_1 - \lambda_2)\langle v_1, v_2 \rangle = 0$. Since $\lambda_1 \neq \lambda_2$, we must have $\langle v_1, v_2 \rangle = 0$.

Q.E.D.

AI-Lemma 24.1 (*Norm of a Vector in an Orthonormal Basis*): Let V be a finite-dimensional inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$), and let $\mathcal{B} := (v_1, \ldots, v_n)$ be an orthonormal basis for V. Then:

(a) For every $v = \sum_{i=1}^n a_i v_i \in V$ we have $\|v\|^2 = \sum_{i=1}^n |a_i|^2$. That is, $\Phi_{\mathcal{B}}$ carries the norm of V to the standard norm of K^n.

(b) Let $T \in \mathrm{End}(V)$ and write $[T]_{\mathcal{B}}^{\mathcal{B}} = (a_{ij})$. Then, for every $1 \le j \le n$,
$$\|Tv_j\|^2 = \sum_{i=1}^n |a_{ij}|^2,$$
the norm of Tv_j being read off the j-th column of $[T]_{\mathcal{B}}^{\mathcal{B}}$.

Proof:

(a) Expand in both arguments, using linearity in the first variable and anti-linearity in the second (Definition 22.a.2):
$$\|v\|^2 = \left\langle \sum_{i=1}^n a_i v_i, \sum_{j=1}^n a_j v_j \right\rangle = \sum_{i=1}^n \sum_{j=1}^n a_i \overline{a_j} \langle v_i, v_j \rangle = \sum_{i=1}^n a_i \overline{a_i} = \sum_{i=1}^n |a_i|^2,$$
where the third equality holds because $\langle v_i, v_j \rangle = \delta_{ij}$, the basis being orthonormal, so that only the diagonal terms survive. Over $K = \mathbb{R}$ the conjugation is trivial and the same computation reads $\|v\|^2 = \sum_i a_i^2$.

(b) By the column description of the representation matrix (the Proposition on Page 113), the entries $a_{1j}, \ldots, a_{nj}$ of the j-th column are exactly the coefficients in $Tv_j = \sum_{i=1}^n a_{ij} v_i$. Now apply **(a)** to $v := Tv_j$.

Q.E.D.

Proof of the Spectral Theorem over $\mathbb{C}$ (Theorem 24.a.4):

We only need to prove the direction "*T is normal $\implies$ T is orthogonally diagonalizable*", as the reverse direction was already established in Lemma 24.a.2.

By a previous theorem (Corollary 21.c.6), T is trigonalizable (here we explicitly use the fact that $K = \mathbb{C}$ and the Fundamental Theorem of Algebra). That is, there exists a basis $\mathcal{C}$ for V such that $[T]_{\mathcal{C}}^{\mathcal{C}}$ is an upper triangular matrix. By applying the Gram-Schmidt orthogonalization process to the basis $\mathcal{C}$, we obtain a new basis $\mathcal{B} := (v_1, \ldots, v_n)$ which is orthonormal, and such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is still upper triangular. (This is left as an exercise to check, see Exercise 24.1. The main point is that if $\mathcal{C} = (u_1, \ldots, u_n)$, then Gram-Schmidt gives an orthonormal basis $\mathcal{B} = (v_1, \ldots, v_n)$ with $\mathrm{Sp}(v_1, \ldots, v_k) = \mathrm{Sp}(u_1, \ldots, u_k)$ for all $1 \le k \le n$, which is Theorem 22.a.16 **(c)**).

We claim that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is actually diagonal. To see this, let us write the matrix explicitly as $[T]_{\mathcal{B}}^{\mathcal{B}} = (a_{ij})_{1 \le i \le n, 1 \le j \le n}$:
$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ 0 & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & a_{nn} \end{pmatrix}.$$

Since $\mathcal{B}$ is orthonormal, the norm of Tv_1 is simply the norm of the first column of the matrix (AI-Lemma 24.1 **(b)**), which means $\|Tv_1\|^2 = |a_{11}|^2$, the entries below a_{11} in that column vanishing by upper triangularity. We also know that $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$ (again Proposition 23.a.4, which applies because $\mathcal{B}$ is orthonormal), which means taking the conjugate transpose:

$$\left(\overline{[T]_{\mathcal{B}}^{\mathcal{B}}}\right)^{\mathsf{T}} = \begin{pmatrix} \overline{a_{11}} & 0 & \dots & 0 \\ \overline{a_{12}} & \overline{a_{22}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \overline{a_{1n}} & \overline{a_{2n}} & \dots & \overline{a_{nn}} \end{pmatrix}.$$

Thus, the first column of $[T^*]_{\mathcal{B}}^{\mathcal{B}}$ is the conjugate transpose of the first row of $[T]_{\mathcal{B}}^{\mathcal{B}}$. Therefore, we have (by AI-Lemma 24.1 **(b)** applied to T^*, together with $|\overline{a}| = |a|$):

$$\|T^* v_1\|^2 = \sum_{k=1}^{n} |a_{1k}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2.$$

As T is normal, Lemma 24.a.5 **(a)** tells us that $\|Tv_1\| = \|T^* v_1\|$. This implies that

$$|a_{11}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2,$$

hence $a_{12} = \dots = a_{1n} = 0$. This shows that the matrix now looks like:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & * & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & * & \dots & * \end{pmatrix}.$$

Now, consider v_2. We have $\|Tv_2\|^2 = |a_{22}|^2$ and $\|T^* v_2\|^2 = |a_{22}|^2 + |a_{23}|^2 + \dots + |a_{2n}|^2$. As $\|Tv_2\| = \|T^* v_2\|$, we similarly obtain $a_{23} = \dots = a_{2n} = 0$.

We can continue this process by induction, and show that for all $1 \le k < n$ we have $a_{k,k+1} = \dots = a_{k,n} = 0$. (The induction hypothesis is what makes each step legitimate. Suppose rows $1, \dots, k-1$ have already been cleared beyond the diagonal. The k-th column of $[T]_{\mathcal{B}}^{\mathcal{B}}$ then has $a_{ik} = 0$ for $i > k$ by upper triangularity and $a_{ik} = 0$ for $i < k$ by those earlier steps, so $\|Tv_k\|^2 = |a_{kk}|^2$; the k-th column of $[T^*]_{\mathcal{B}}^{\mathcal{B}}$ is the conjugated k-th row of $[T]_{\mathcal{B}}^{\mathcal{B}}$, untouched so far, so $\|T^* v_k\|^2 = \sum_{j \ge k} |a_{kj}|^2$. Equating the two clears row k in turn.) Hence, $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal. Since $\mathcal{B}$ is an orthonormal basis that diagonalizes T, it consists of eigenvectors (the j-th column of a diagonal $[T]_{\mathcal{B}}^{\mathcal{B}}$ says $Tv_j = a_{jj} v_j$, by the Proposition on Page 113), proving that T is orthogonally diagonalizable.

Q.E.D.

Exercise 24.1 (*Gram-Schmidt Preserves Upper Triangularity*): This is the step marked *"exc.: check this"* in the proof above. Let $\mathcal{C} := (u_1, \dots, u_n)$ be a basis for V with $[T]_{\mathcal{C}}^{\mathcal{C}}$ upper triangular, and let $\mathcal{B} := (v_1, \dots, v_n)$ be the orthonormal basis produced from it by Gram-Schmidt. Show that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is again upper triangular. (Prof. Biran's hint: the point is that $\operatorname{Sp}(v_1, \dots, v_k) = \operatorname{Sp}(u_1, \dots, u_k)$ for all $1 \le k \le n$.)

Remark 24.1: In the proof above, we used $K = \mathbb{C}$ **only** at one place: when we stated that T is trigonalizable. Therefore, we immediately obtain the following corollary for the real case.

Corollary 24.a.6: Let V be a finite-dimensional inner product space over $\mathbb{R}$, and let $T\colon V \to V$ be an endomorphism which is normal. If T is trigonalizable (which by Theorem 21.c.4 is true if and only if the characteristic polynomial $P_T(x)$ factors as a product of linear factors over $\mathbb{R}$), then T is orthogonally diagonalizable.

Proof:

Read the proof of Theorem 24.a.4 again with $K = \mathbb{R}$. Trigonalizability is now a hypothesis instead of a consequence of Corollary 21.c.6, and every step after it is available over $\mathbb{R}$: Theorem 22.a.16, AI-Lemma 24.1, Proposition 23.a.4 and Lemma 24.a.5 are all stated for $K = \mathbb{R}$ as well as for $K = \mathbb{C}$, and over $\mathbb{R}$ the conjugations appearing there are the identity. **Q.E.D.**

The Spectral Theorem over $\mathbb{R}$

We've seen that over $\mathbb{R}$, the statement *"$T \in \operatorname{End}(V)$ is normal $\implies$ T is orthogonally diagonalizable"* does not hold in general. We need a stronger condition.

Lemma 24.a.7: Let V be a finite-dimensional inner product space over $\mathbb{R}$, and let $T \in \operatorname{End}(V)$ be orthogonally diagonalizable. Then $T^* = T$.

Proof:

Let $\mathcal{B}$ be an orthonormal basis such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal matrix. Since we are over $\mathbb{R}$, the conjugate transpose is simply the standard transpose. Then:

$$[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^{\mathsf{T}} = [T]_{\mathcal{B}}^{\mathcal{B}},$$

where the first equality is Proposition 23.a.4, whose conjugation drops away because we are over $\mathbb{R}$, and the second equality holds because $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal matrix. Because their representation matrices with respect to the same basis are equal, it follows that $T^* = T$ (by Theorem 18.a.2 once more). **Q.E.D.**

Definition 24.a.8: Let V be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$), and $T \in \operatorname{End}(V)$. T is called "*self adjoint*" if $T^* = T$.

Similarly, a matrix $A \in \mathcal{M}_{n\times n}(K)$ is called "*self adjoint*" if $A^* = A$, where $A^* := \overline{A}^{\mathsf{T}}$ is the adjoint matrix of Definition 22.a.7. (Note that over $K = \mathbb{R}$, a self adjoint matrix is precisely a symmetric matrix, $A^{\mathsf{T}} = A$).

Exercise 24.2 (*Matrix Representation of Self-Adjoint Maps*): Show that $T\colon V \to V$ is self adjoint if and only if for every orthonormal basis $\mathcal{B}$ of V, the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is self adjoint.

Theorem 24.a.9 (*Spectral Theorem over $\mathbb{R}$*): Let V be a Euclidean space (i.e., an inner product space over $\mathbb{R}$) of finite dimension, and let $T \in \operatorname{End}(V)$. Then T is orthogonally diagonalizable if and only if T is self adjoint.

For the proof, we need the following lemma:

Lemma 24.a.10: Let V be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$), and let $T \in \operatorname{End}(V)$ be self adjoint. Then:

(a) All eigenvalues of T are real.

(b) The characteristic polynomial $P_T(x)$ factors into a product of linear factors over K.

Proof:

(a) If $K = \mathbb{R}$, there is nothing to show. So assume $K = \mathbb{C}$. Let $\lambda \in \mathbb{C}$ be an eigenvalue and $v \in V$ a corresponding non-zero eigenvector. We've seen in Lemma 24.a.5 **(c)** that v is also an eigenvector of T^* with eigenvalue $\overline{\lambda}$ (that lemma applies because a self adjoint endomorphism is in particular normal: $T \circ T^* = T \circ T = T^* \circ T$). But $T^* = T$. This implies:

$$\lambda v = Tv = T^* v = \overline{\lambda} v.$$

Since $v \neq 0$, we must have $\lambda = \overline{\lambda}$, which implies $\lambda \in \mathbb{R}$.

(b) For $K = \mathbb{C}$, this follows immediately from the Fundamental Theorem of Algebra (Theorem 21.c.5). So assume $K = \mathbb{R}$. Let $\mathcal{B}$ be an orthonormal basis for V (one exists by Corollary 22.a.17) and consider the matrix $A := [T]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_{n\times n}(\mathbb{R})$, where $n := \dim V$. The statement would follow if we show that the characteristic polynomial $P_A(x)$ factors as a product of linear terms with real coefficients, since $P_T = P_A$ for the representation matrix A of T in any basis (the second half of Remark 21.2).

Now, $T^* = T$, hence $A = [T]^{\mathcal{B}}_{\mathcal{B}} = [T^*]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^* = A^* = A^{\mathsf{T}}$, the third equality by Proposition 23.a.4 and the last because A has real entries. Consider $\mathbb{C}^n$ endowed with its standard inner product, and view A as a matrix in $\mathcal{M}_{n\times n}(\mathbb{C})$. Let $T_A\colon \mathbb{C}^n \to \mathbb{C}^n$ be the corresponding linear map. The standard basis $\mathcal{E}_n = (e_1, \dots, e_n)$ is orthonormal, and $[T_A]^{\mathcal{E}_n}_{\mathcal{E}_n} = A$.

We know that $P_{T_A}(x) = P_A(x)$ (once more Remark 21.2, the standard basis representing T_A by A). When viewed as a polynomial over $\mathbb{C}$ (i.e., in $\mathbb{C}[x]$), it factors as a product of linear terms:

$$P_A(x) = \prod_{k=1}^{n}(x - \lambda_k) \in \mathbb{C}[x], \quad (*)$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of T_A. But T_A is self adjoint (even when viewed over $\mathbb{C}$ since $A^{\mathsf{T}} = A = A^*$), so by part **(a)**, all the eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{R}$. Therefore, the factorization $(*)$ actually holds in $\mathbb{R}[x]$.

Q.E.D.

Proof of the Spectral Theorem over $\mathbb{R}$ (Theorem 24.a.9):

The forward direction was proven in Lemma 24.a.7. For the reverse direction, if T is self adjoint, then by Lemma 24.a.10 **(b)**, its characteristic polynomial splits over $\mathbb{R}$. This implies T is trigonalizable over $\mathbb{R}$, by Theorem 21.c.4. Furthermore, since $T = T^*$, T trivially commutes with its adjoint, making it normal. By Corollary 24.a.6, a normal map that is trigonalizable over $\mathbb{R}$ is orthogonally diagonalizable. **Q.E.D.**

Spectral Theorems for Matrices

Let $K = \mathbb{R}$ or $\mathbb{C}$. Recall that a matrix $A \in \mathcal{M}_{n\times n}(K)$ is called normal if $A \cdot A^* = A^* \cdot A$ (the second half of Definition 24.a.3, written there as $A \cdot \overline{A}^{\mathsf{T}} = \overline{A}^{\mathsf{T}} \cdot A$).

Example 24.1 (*Classes of Normal Matrices*):

(a) For $K = \mathbb{R}$: Symmetric matrices and orthogonal matrices $\mathrm{O}(n)$ are normal.

(b) For $K = \mathbb{C}$: Self adjoint matrices and unitary matrices $\mathrm{U}(n)$ are normal.

In each case the verification is immediate: a self adjoint A has $A^* = A$, so both products read $A \cdot A$; and a unitary A has $AA^* = A^*A = I_n$ by Lemma 22.b.10. The real statements are the special cases $A^{\mathsf{T}} = A$ and $AA^{\mathsf{T}} = A^{\mathsf{T}}A = I_n$ of these, since over $\mathbb{R}$ the adjoint is the transpose.

Remark 24.2: There exist symmetric matrices in $\mathcal{M}_{n\times n}(\mathbb{C})$ that are not normal. For example, let $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$. We have $A^{\mathsf{T}} = A$, so it is symmetric. However, $A^* = \left(\begin{smallmatrix} -i & -i \\ -i & 1 \end{smallmatrix}\right)$. A quick calculation shows that $A \cdot A^* \neq A^* \cdot A$, so A is not normal.

Exercise 24.3 (*The Symmetric Matrix that is Not Normal*): Carry out the calculation Prof. Biran marks with *"check that $AA^* \neq A^*A$"*: verify for $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$ that the two products really do differ.

Theorem 24.a.11 (*Matrix Spectral Theorems*):

(a) If $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is normal, then there exists a unitary matrix $U \in \mathrm{U}(n)$ such that $U^{-1}AU$ is diagonal. (Note that $U^{-1} = U^*$, by Lemma 22.b.10 **(d)**).

(b) If $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is a symmetric matrix, then there exists an orthogonal matrix $O \in \mathrm{O}(n)$ such that $O^{-1}AO$ is diagonal. (Note that $O^{-1} = O^{\mathsf{T}}$, by Lemma 22.b.9 **(d)**).

Proof Sketch:

The proof is left as an exercise, based on the following key points translating between maps and matrices:

(a) $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is normal if and only if $T_A\colon \mathbb{C}^n \to \mathbb{C}^n$ is normal (when we endow $\mathbb{C}^n$ with its standard inner product).
(b) If $\mathcal{B}$ is an orthonormal basis for $\mathbb{C}^n$, then the transition matrix $U := [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ is unitary.
(c) $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is symmetric if and only if $T_A\colon \mathbb{R}^n \to \mathbb{R}^n$ is self adjoint, when we endow $\mathbb{R}^n$ with its standard inner product.
(d) If $\mathcal{B}$ is an orthonormal basis for $\mathbb{R}^n$, then the transition matrix $O := [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ is orthogonal.

None of the four is hard, and each rests on a result already available. Points **(a)** and **(c)** follow from $(T_A)^* = T_{A^*}$, which is Corollary 23.a.5: composing gives $T_A \circ (T_A)^* = T_{AA^*}$ and $(T_A)^* \circ T_A = T_{A^*A}$, and $B \mapsto T_B$ is injective, so the two compositions agree exactly when $AA^* = A^*A$; for **(c)** read the same identity with $A^* = A$ and recall that over $\mathbb{R}$ the adjoint matrix is the transpose. Points **(b)** and **(d)** are the definition of a unitary and of an orthogonal matrix, Definition 22.b.8: the columns of $[\mathrm{id}]^{\mathcal{B}}_{\mathcal{E}_n}$ are the coordinate vectors of the members of $\mathcal{B}$ in the standard basis, by the Proposition on Page 113, hence the members of $\mathcal{B}$ themselves, and those form an orthonormal system by hypothesis. **Q.E.D.**

Solutions to the Exercises

Proof of Exercise 24.1:

Let $\mathcal{C} := (u_1, \ldots, u_n)$ be a basis for V such that the representation matrix $[T]^{\mathcal{C}}_{\mathcal{C}}$ is upper triangular. This means that for any $1 \le k \le n$, the vector $T(u_k)$ is a linear combination of $u_1, \ldots, u_k$. In other words, $T(u_k) \in \mathrm{Sp}(u_1, \ldots, u_k)$. Consequently, the subspace $W_k := \mathrm{Sp}(u_1, \ldots, u_k)$ is T-invariant, meaning $T(W_k) \subseteq W_k$.

Applying the Gram-Schmidt orthogonalization process to the basis $\mathcal{C}$ produces an orthonormal basis $\mathcal{B} := (v_1, \ldots, v_n)$ with the property that for every $1 \le k \le n$, the spans are preserved: $\mathrm{Sp}(v_1, \ldots, v_k) = \mathrm{Sp}(u_1, \ldots, u_k) = W_k$.

Now, consider the matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$. The k-th column of this matrix consists of the coordinates of $T(v_k)$ with respect to the basis $\mathcal{B}$. Since $v_k \in W_k$ and W_k is T-invariant, we must have $T(v_k) \in W_k = \mathrm{Sp}(v_1, \ldots, v_k)$. This means $T(v_k)$ can be written strictly as a linear combination of $v_1, \ldots, v_k$, meaning all coefficients for vectors v_j with $j > k$ are exactly zero. Therefore, $[T]^{\mathcal{B}}_{\mathcal{B}}$ is an upper triangular matrix. **Q.E.D.**

Proof of Exercise 24.2:

Let $T \in \mathrm{End}(V)$. First, suppose T is self adjoint, so $T = T^*$. Let $\mathcal{B}$ be an arbitrary orthonormal basis for V. By the properties of representation matrices for adjoint maps in orthonormal bases (Proposition 23.a.4), we have $[T^*]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$. Since $T = T^*$, we substitute to get $[T]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$, which means the matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$ is self adjoint.

Conversely, suppose that for any orthonormal basis $\mathcal{B}$ of V, the matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$ is self adjoint, i.e., $[T]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$. One such basis exists, by Corollary 22.a.17, and one is all the argument needs. Again, using the identity $[T^*]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$, we see that $[T]^{\mathcal{B}}_{\mathcal{B}} = [T^*]^{\mathcal{B}}_{\mathcal{B}}$. Because the map sending an endomorphism to its representation matrix with respect to a fixed basis is an isomorphism (Theorem 18.a.2), this equality of matrices implies the equality of the maps, hence $T = T^*$, meaning T is self adjoint. **Q.E.D.**

Solution of Exercise 24.3:

With $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$ we have $A^* = \overline{A}^{\mathsf{T}} = \left(\begin{smallmatrix} -i & -i \\ -i & 1 \end{smallmatrix}\right)$, and, using $i \cdot (-i) = -i^2 = 1$ throughout,

$$A \cdot A^* = \begin{pmatrix} i & i \\ i & 1 \end{pmatrix} \begin{pmatrix} -i & -i \\ -i & 1 \end{pmatrix} = \begin{pmatrix} 1+1 & 1+i \\ 1-i & 1+1 \end{pmatrix} = \begin{pmatrix} 2 & 1+i \\ 1-i & 2 \end{pmatrix},$$

while

$$A^* \cdot A = \begin{pmatrix} -i & -i \\ -i & 1 \end{pmatrix} \begin{pmatrix} i & i \\ i & 1 \end{pmatrix} = \begin{pmatrix} 1+1 & 1-i \\ 1+i & 1+1 \end{pmatrix} = \begin{pmatrix} 2 & 1-i \\ 1+i & 2 \end{pmatrix}.$$

The diagonals agree, but the off-diagonal entries are conjugate to one another and $1+i \neq 1-i$, so $A \cdot A^* \neq A^* \cdot A$ and A is not normal, even though $A^{\mathsf{T}} = A$.

The example is not an accident: over $\mathbb{C}$ it is $A^* = A$, and not $A^{\mathsf{T}} = A$, that forces normality, and the two conditions are unrelated as soon as A has non-real entries. **Q.E.D.**

Proof of Theorem 24.a.11:

(a) Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be a normal matrix. By point **(a)** of the proof sketch above, the linear map $T_A \colon \mathbb{C}^n \to \mathbb{C}^n$ is a normal endomorphism with respect to the standard inner product. By the Spectral Theorem over $\mathbb{C}$ (Theorem 24.a.4), T_A is orthogonally diagonalizable. Thus, there exists an orthonormal basis $\mathcal{B} := (v_1, \ldots, v_n)$ for $\mathbb{C}^n$ consisting of eigenvectors of T_A.

The representation matrix of T_A with respect to $\mathcal{B}$, which is $[T_A]^{\mathcal{B}}_{\mathcal{B}}$, is a diagonal matrix D. Let $\mathcal{E}_n$ be the standard basis for $\mathbb{C}^n$. The transition matrix from $\mathcal{B}$ to $\mathcal{E}_n$ is $U := [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$, whose columns are precisely the vectors $v_1, \ldots, v_n$. Since $\mathcal{B}$ is an orthonormal basis, point **(b)** of that sketch guarantees that U is a unitary matrix.

Using the change of basis formula, we have:

$$D = [T_A]^{\mathcal{B}}_{\mathcal{B}} = [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{E}_n}_{\mathcal{B}} \cdot [T_A]^{\mathcal{E}_n}_{\mathcal{E}_n} \cdot [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{B}}_{\mathcal{E}_n} = U^{-1} \cdot A \cdot U.$$

Thus, $U^{-1}AU$ is diagonal.

(b) Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ be a symmetric matrix. By point **(c)** of the proof sketch above, the linear map $T_A \colon \mathbb{R}^n \to \mathbb{R}^n$ is self adjoint. By the Spectral Theorem over $\mathbb{R}$ (Theorem 24.a.9), T_A is orthogonally diagonalizable. Thus, there exists an orthonormal basis $\mathcal{B}$ for $\mathbb{R}^n$ consisting of eigenvectors of T_A, making $[T_A]^{\mathcal{B}}_{\mathcal{B}}$ a diagonal matrix D.

Let $O := [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ be the transition matrix. Since $\mathcal{B}$ is an orthonormal basis in $\mathbb{R}^n$, point **(d)** of that sketch guarantees that O is an orthogonal matrix. By the change of basis formula:

$$D = [T_A]^{\mathcal{B}}_{\mathcal{B}} = [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{E}_n}_{\mathcal{B}} \cdot [T_A]^{\mathcal{E}_n}_{\mathcal{E}_n} \cdot [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{B}}_{\mathcal{E}_n} = O^{-1} \cdot A \cdot O.$$

Thus, $O^{-1}AO$ is diagonal.

Q.E.D.

AI-Note: The rigour pass found nothing false in this chapter, and one thing genuinely missing: the identity that makes the proof of Theorem 24.a.4 work at all. That proof reads $\|Tv_1\|^2 = |a_{11}|^2$, then $\|T^*v_1\|^2 = \sum_k |a_{1k}|^2$, then the same twice more for v_2, each time on the strength of *"$\mathcal{B}$ is orthonormal"*. The fact behind it, that coordinates taken in an orthonormal basis compute the norm, is stated nowhere in the book: chs. 22a and 22b construct orthonormal bases at length, prove Pythagoras for two vectors, and never write down the n-term version. It is now AI-Lemma 24.1, lifted out of the proof so that the four steps can point at it. An `ai` environment, since the statement appears nowhere in the notes, only inside this one argument.

Named, and previously silent: the identity $[T^*]^{\mathcal{B}}_{\mathcal{B}} = ([T]^{\mathcal{B}}_{\mathcal{B}})^*$, which this chapter uses five separate times and never once credits, is Proposition 23.a.4 of the previous chapter, and it is where orthonormality enters each of those five arguments. Passing from an equality of representation matrices to an equality of maps, done three times, is Theorem 18.a.2. The four points of the proof sketch of Theorem 24.a.11 were asserted without any reason at all, although the solution below them leans on every one; they now have their two citations, Corollary 23.a.5 and Definition 22.b.8. Corollary 24.a.6 had no proof, only the remark above it observing that $K = \mathbb{C}$ was used once; the proof now checks that claim result by result.

Two places where the argument was right but stopped a step short. The induction at the end of the proof of Theorem 24.a.4 is not a repetition of the first step: at stage k it needs the entries above the k-th diagonal entry to have been cleared by stages $1, \ldots, k-1$, which is exactly what makes $\|Tv_k\|^2 = |a_{kk}|^2$; the parenthetical there spells that out. And the proof of Lemma 24.a.10 **(b)** passes from P_A to P_T twice without naming Remark 21.2, which is the result that licenses it.

Finally, a standing convention was made explicit in Lemma 24.a.2: the chapter says *"inner product space"* throughout while using T^*, and the adjoint of ch. 23a exists only in finite dimension.

Checked and found correct, so that nobody re-derives them: the rotation $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ really is clockwise and really is normal; the two products AA^* and A^*A for $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$ differ exactly in the off-diagonal conjugate pair $1 \pm i$; and the change-of-basis chain $D = [\mathrm{id}]^{\mathcal{E}_n}_{\mathcal{B}} \cdot A \cdot [\mathrm{id}]^{\mathcal{B}}_{\mathcal{E}_n}$ composes correctly in the house convention, with the source basis on top throughout.

Orthogonal and Unitary Maps / Isometries (Chapter 25)

Linear Isometries (Section 25.a)

Definition 25.a.1 (*Linear Isometry*): Let V, W be inner product spaces over K (where $K = \mathbb{R}$ or $\mathbb{C}$). A map $T\colon V \to W$ is called a "*linear isometry*" if T is linear and for all $v_1, v_2 \in V$,

$$\langle Tv_1, Tv_2 \rangle_W = \langle v_1, v_2 \rangle_V.$$

(In other words, T preserves the inner product).

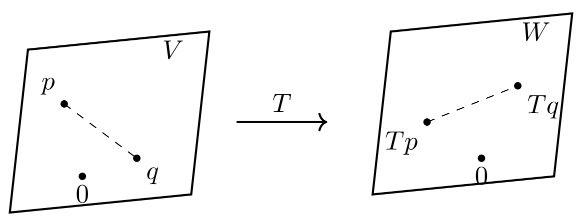
Remark 25.1: If $T\colon V \to W$ is a linear isometry, then:

(a) For all $v \in V$, $\|Tv\| = \|v\|$.

(b) For all $p, q \in V$, the distance is preserved:

$$\mathrm{dist}_W(Tp, Tq) = \|Tp - Tq\|_W = \|p - q\|_V = \mathrm{dist}_V(p, q).$$

(c) T is injective. Indeed, if $v \in \ker T$, then $0 = \|Tv\| = \|v\|$, which implies $v = 0$. So, $\ker T = \{0\}$.



From now on, we will concentrate on the case where $W = V$, meaning $T \in \mathrm{End}(V)$.

Definition (*Orthogonal and Unitary Endomorphism*): Let V be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$) and let $T \in \mathrm{End}(V)$. If such a map T is an isometry, we also call it an "*orthogonal endomorphism*" if $K = \mathbb{R}$, or a "*unitary endomorphism*" if $K = \mathbb{C}$.

Theorem 25.a.2 (*Characterization of Isometries*): Let V be a finite-dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let $T \in \mathrm{End}(V)$. The following statements are equivalent:

(a) T is an isometry.

(b) For all $v \in V$, $\|Tv\| = \|v\|$.

(c) For every orthonormal system $e_1, \ldots, e_k$ of V, the list of vectors $Te_1, \ldots, Te_k$ is also an orthonormal system.

(d) There exists an orthonormal **basis** $\mathcal{B} := (e_1, \ldots, e_n)$ of V such that $Te_1, \ldots, Te_n$ is also an orthonormal basis.

(e) $T \circ T^* = \mathrm{id}_V$.

(f) $T^* \circ T = \mathrm{id}_V$.

(g) $T^* = T^{-1}$.

(h) T^* is an isometry.

Proof:

We will prove some of the equivalences here; the rest are left as exercises (and for the exercise sheet).

(b) $\implies$ **(a)**: Using the polarization identity for the real case (Exercise 22.1 **(a)**), we have:

$$\begin{aligned}\langle Tv, Tw\rangle &= \frac{1}{2}\left(\|Tv+Tw\|^2 - \|Tv\|^2 - \|Tw\|^2\right)\\ &= \frac{1}{2}\left(\|T(v+w)\|^2 - \|Tv\|^2 - \|Tw\|^2\right)\\ &= \frac{1}{2}\left(\|v+w\|^2 - \|v\|^2 - \|w\|^2\right)\\ &= \langle v, w\rangle.\end{aligned}$$

The identity used here is stated for a Euclidean space, so this settles only the case $K = \mathbb{R}$, whereas the theorem is asserted over $\mathbb{R}$ and over $\mathbb{C}$ alike. The complex case runs identically once the Hermitian polarization identity of Exercise 22.1 **(b)** is put in place of the real one: that identity, worked out in the solutions to ch. 22a, writes $\langle u, v\rangle$ in terms of $\|u\|$, $\|v\|$, $\|u+v\|$ and $\|u+iv\|$ alone, and a T satisfying **(b)** leaves all four unchanged.

(a) $\implies$ **(c)**: Let $e_1, \dots, e_k$ be an orthonormal system. Then, for all $1 \le i, j \le k$, we have:

$$\langle Te_i, Te_j\rangle = \langle e_i, e_j\rangle = \delta_{ij}.$$

This implies that $Te_1, \dots, Te_k$ is also an orthonormal system.

(d) $\implies$ **(e)**: Assume there exists an orthonormal basis $\mathcal{B} = (e_1, \dots, e_n)$ of V such that $Te_1, \dots, Te_n$ is also an orthonormal basis. Let $1 \le j \le n$. Then, for all $1 \le i \le n$, the defining property (23.1) of the adjoint map gives

$$\langle e_i, e_j\rangle = \delta_{ij} = \langle Te_i, Te_j\rangle = \langle e_i, T^*Te_j\rangle. \tag{25.1}$$

Since $(e_1, \dots, e_n)$ is a basis for V, the equality (25.1) extends by linearity in the first argument to

$$\langle v, e_j\rangle = \langle v, T^*Te_j\rangle \quad \text{for all } v \in V,$$

and therefore, $T^*Te_j = e_j$ (by Lemma 22.a.3 **(c)**). This last equality holds for all $1 \le j \le n$, so the linear map $T^* \circ T$ agrees with id_V on the basis $\mathcal{B}$. Thus, $T^* \circ T = \mathrm{id}_V$ (by the uniqueness half of Theorem 15.a.3), which proves **(d)** $\implies$ **(f)**.

Finally, $T^* \circ T = \mathrm{id}_V$ shows that T is injective, and since V is finite-dimensional, T is an isomorphism by Corollary 15.b.10. Consequently, $T^* = T^* \circ T \circ T^{-1} = T^{-1}$, and therefore, $T \circ T^* = \mathrm{id}_V$, which proves **(e)**.

The remaining implications are left as exercises, as Prof. Biran indicates. **Q.E.D.**

AI-Note: In the handwritten notes, this proof is labeled as proving **(d)** $\implies$ **(e)**. However, the first part actually proves **(d)** $\implies$ **(f)** (which is $T^* \circ T = \mathrm{id}_V$), from which **(e)** is then deduced. We have clarified the steps here.

Orthogonal and Unitary Matrices (Section 25.b)

Back to matrices. Recall that:

- $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is orthogonal if $A^{\mathsf{T}}A = I \iff AA^{\mathsf{T}} = I \iff A^{-1} = A^{\mathsf{T}}$. The set of such matrices is $\mathrm{O}(n)$.
- $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is unitary if $A^*A = I \iff AA^* = I \iff A^{-1} = A^*$. The set of such matrices is $\mathrm{U}(n)$.

Proposition 25.a.3: Let V be an n-dimensional inner product space over $\mathbb{R}$ (resp. $\mathbb{C}$). Let $\mathcal{B}$ be an orthonormal basis for V, and let $T \in \mathrm{End}(V)$. Then, T is orthogonal (resp. unitary) if and only if $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathrm{O}(n)$ (resp. $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathrm{U}(n)$).

Proof:

This is left as an exercise, see Exercise 25.1. **Q.E.D.**

Exercise 25.1 (*Isometries versus Orthogonal and Unitary Matrices*): This is the step Prof. Biran marks with *"Exc."* in the proof of Proposition 25.a.3. Let $\mathcal{B}$ be an orthonormal basis for V and let $T \in \operatorname{End}(V)$. Show that T is orthogonal (resp. unitary) if and only if $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathrm{O}(n)$ (resp. $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathrm{U}(n)$). (The point is that $\mathcal{B}$ is orthonormal, so that Proposition 23.a.4 applies and turns the adjoint map T^* into the conjugate transpose of the representation matrix.)

Proposition (*Determinant of an Orthogonal or Unitary Matrix*): Note that for all $A \in \mathrm{O}(n)$, we have $\det(A) = \pm 1$. Similarly, for all $A \in \mathrm{U}(n)$, we have $|\det(A)| = 1$. (Note that $\det(A) \in \mathbb{C}$).

Proof:

Indeed, by Theorems 20.c.1 and 20.b.5,

$$1 = \det(I) = \det(AA^{\mathsf{T}}) = \det(A)\cdot\det(A^{\mathsf{T}}) = \det(A)\cdot\det(A) = \det(A)^2.$$

Similarly, by the same two results,

$$1 = \det(AA^*) = \det(A)\cdot\det(\overline{A}^{\mathsf{T}}) = \det(A)\cdot\det(\overline{A}) = \det(A)\cdot\overline{\det(A)} = |\det(A)|^2.$$

Both chains open with $AA^{\mathsf{T}} = I_n$, resp. $AA^* = I_n$, which is Lemma 22.b.9 **(c)**, resp. Lemma 22.b.10 **(c)**. The fourth equality of the second chain is $\det(\overline{A}) = \overline{\det(A)}$, which holds because the Leibniz formula of Theorem 20.b.2 writes det as a sum of products of entries and complex conjugation respects both sums and products. **Q.E.D.**

Definition 25.a.4 (*Special Orthogonal and Unitary Groups*): We define the following subsets:

$$\mathrm{SO}(n) := \{A \in \mathrm{O}(n) \mid \det(A) = 1\} \subseteq \mathrm{O}(n) \subseteq \mathrm{GL}_n(\mathbb{R}),$$
$$\mathrm{SU}(n) := \{A \in \mathrm{U}(n) \mid \det(A) = 1\} \subseteq \mathrm{U}(n) \subseteq \mathrm{GL}_n(\mathbb{C}).$$

Proposition 25.a.5 (*Group Properties*): Let G be any of the sets $\mathrm{O}(n), \mathrm{SO}(n), \mathrm{U}(n)$, or $\mathrm{SU}(n)$. Then:

- **(a)** $I_n \in G$.
- **(b)** For all $A, B \in G$, $A \cdot B \in G$.
- **(c)** For all $A \in G$, $A^{-1} \in G$.

Proof:

This is left as an exercise, see Exercise 25.2. **Q.E.D.**

Exercise 25.2 (*The Four Sets are Groups*): This is the claim Prof. Biran circles with *"Exc."* Prove the three statements of Proposition 25.a.5 for each of $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, and $\mathrm{SU}(n)$. Together with the associativity of matrix multiplication, they say precisely that each of these four sets is a group, and hence a subgroup of $\mathrm{GL}_n(\mathbb{R})$ resp. $\mathrm{GL}_n(\mathbb{C})$.

Classification of the Elements of $\mathrm{O}(2)$, $\mathrm{O}(3)$, etc. (Section 25.c)

Lemma 25.a.6: Let $A \in \mathrm{O}(n)$. If λ is a real eigenvalue of A, then $\lambda = \pm 1$.

Proof:

Suppose $Av = \lambda v$, where $0 \neq v \in \mathbb{R}^n$. Since the standard basis $\mathcal{E}_n$ of $\mathbb{R}^n$ is orthonormal and represents T_A by A itself ($[T_A]_{\mathcal{E}_n}^{\mathcal{E}_n} = A$, by the Proposition on Page 113), Proposition 25.a.3 shows that T_A is an isometry, and therefore, it preserves the standard scalar product. Then:

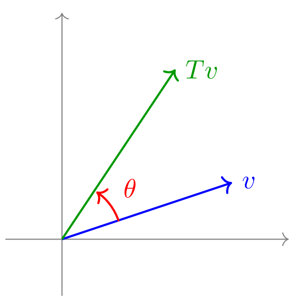
$$\langle v, v\rangle = \langle Av, Av\rangle = \langle \lambda v, \lambda v\rangle = \lambda^2 \langle v, v\rangle.$$

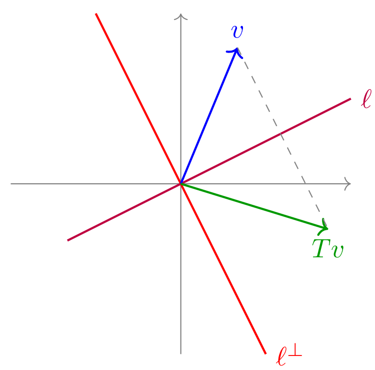
As $v \neq 0$, we have $\langle v, v\rangle \neq 0$, which implies $\lambda^2 = 1$, and therefore $\lambda = \pm 1$. **Q.E.D.**

Proposition 25.a.7 (*Classification of* $\mathrm{O}(2)$): Let $A \in \mathrm{O}(2)$.

(a) If $\det A = 1$, then there exists $\theta \in [0, 2\pi)$ such that
$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} := R_\theta.$$
The map $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ is an anti-clockwise rotation by an angle θ.

(b) If $\det A = -1$, then there exists a basis $\mathcal{B} := (v_1, v_2)$ for $\mathbb{R}^2$ such that $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ has the following representation matrix in the basis $\mathcal{B}$:
$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{25.2}$$
In other words, T_A performs a reflection with respect to the line $\ell := \mathrm{Sp}(v_1)$. Moreover, there exists $\theta \in [0, 2\pi)$ such that
$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} = R_\theta \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$
The basis $\mathcal{B}$ for which (25.2) holds is given by
$$v_1 = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) \end{pmatrix}, \quad v_2 = \begin{pmatrix} \cos\left(\frac{\theta}{2} + \frac{\pi}{2}\right) \\ \sin\left(\frac{\theta}{2} + \frac{\pi}{2}\right) \end{pmatrix}.$$



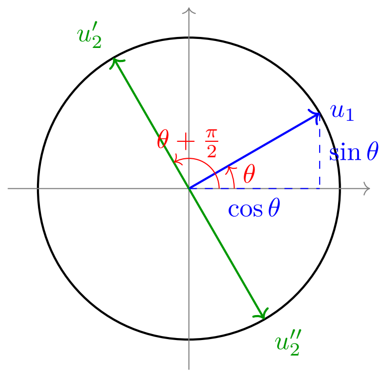


Proof:

Note that if $v \in \mathbb{R}^2$ has $\|v\| = 1$, then there exists $\theta \in [0, 2\pi)$ such that $v = \left(\begin{smallmatrix} \cos\theta \\ \sin\theta \end{smallmatrix}\right)$.

Now, let
$$A = \begin{pmatrix} | & | \\ u_1 & u_2 \\ | & | \end{pmatrix} \in \mathrm{O}(2).$$
Since A is orthogonal, its columns form an orthonormal basis by Definition 22.b.8. Thus, $\|u_1\| = 1$, which implies $u_1 = \left(\begin{smallmatrix} \cos\theta \\ \sin\theta \end{smallmatrix}\right)$ for some $\theta \in [0, 2\pi)$. We also have $u_1 \perp u_2$ and $\|u_2\| = 1$.

By Corollary 22.b.7, the orthogonal complement of u_1 is a line, and the vector displayed below does lie on it, since $\langle \left(\begin{smallmatrix} \cos\theta \\ \sin\theta \end{smallmatrix}\right), \left(\begin{smallmatrix} -\sin\theta \\ \cos\theta \end{smallmatrix}\right) \rangle = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$ and it is non-zero. Hence:
$$\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}^{\perp} = \mathrm{Sp}\begin{pmatrix} \cos(\theta + \frac{\pi}{2}) \\ \sin(\theta + \frac{\pi}{2}) \end{pmatrix} = \mathrm{Sp}\begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}.$$
Within the space $\mathrm{Sp}\left(\begin{smallmatrix} -\sin\theta \\ \cos\theta \end{smallmatrix}\right)$, there are precisely two elements with norm 1, namely:
$$u_2' = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}, \quad u_2'' = \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix}.$$



If $u_2 = u_2'$, then $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = R_\theta$, and $\det A = 1$.
If $u_2 = u_2''$, then $A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} = R_\theta \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and $\det A = -1$.

In the second case, a direct calculation shows that $v_1 = \begin{pmatrix} \cos(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) \end{pmatrix}$ and $v_2 = \begin{pmatrix} \cos(\frac{\theta}{2}+\frac{\pi}{2}) \\ \sin(\frac{\theta}{2}+\frac{\pi}{2}) \end{pmatrix}$ are eigenvectors of A for the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -1$, respectively.

Let us carry that calculation out. For any $\varphi \in \mathbb{R}$, the angle-subtraction identities give

$$A \begin{pmatrix} \cos\varphi \\ \sin\varphi \end{pmatrix} = \begin{pmatrix} \cos\theta\cos\varphi + \sin\theta\sin\varphi \\ \sin\theta\cos\varphi - \cos\theta\sin\varphi \end{pmatrix} = \begin{pmatrix} \cos(\theta-\varphi) \\ \sin(\theta-\varphi) \end{pmatrix}.$$

For v_1 we have $\varphi = \frac{\theta}{2}$, hence $\theta - \varphi = \frac{\theta}{2} = \varphi$, and therefore, $Av_1 = v_1$. For v_2 we have $\varphi = \frac{\theta}{2} + \frac{\pi}{2}$, hence $\theta - \varphi = \frac{\theta}{2} - \frac{\pi}{2} = \varphi - \pi$, and since $\cos(\varphi - \pi) = -\cos\varphi$ and $\sin(\varphi-\pi) = -\sin\varphi$, this yields $Av_2 = -v_2$.

Finally, $\|v_1\| = \|v_2\| = 1$, and

$$\langle v_1, v_2 \rangle = \cos\left(\tfrac{\theta}{2}\right)\cos\left(\tfrac{\theta}{2}+\tfrac{\pi}{2}\right) + \sin\left(\tfrac{\theta}{2}\right)\sin\left(\tfrac{\theta}{2}+\tfrac{\pi}{2}\right) = \cos\left(-\tfrac{\pi}{2}\right) = 0,$$

so the basis $\mathcal{B} = (v_1, v_2)$ appearing in statement **(b)** is in fact an **orthonormal** basis. This sharpening is what Corollary 25.a.8 below relies on. **Q.E.D.**

Summary 25.1: $\mathrm{SO}(2) =$ rotations.
$\mathrm{O}(2) \setminus \mathrm{SO}(2) =$ reflections.

Corollary 25.a.8: Let V be a 2-dimensional Euclidean space and $T\colon V \to V$ a linear isometry. Then $\det(T) = \pm 1$. Moreover, there exists an orthonormal basis $\mathcal{B}$ for V such that:

- **(a)** If $\det(T) = 1$, then $[T]_{\mathcal{B}}^{\mathcal{B}} = R_\theta$.
- **(b)** If $\det(T) = -1$, then $[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Proof:

The notes state this corollary without proof, so we supply the short argument; it is nothing more than Proposition 25.a.7 transported from $\mathbb{R}^2$ to V along a coordinate map.

By Corollary 22.a.17, V admits an orthonormal basis $\mathcal{C} := (c_1, c_2)$. Put $A := [T]_{\mathcal{C}}^{\mathcal{C}}$. Since T is an isometry and $\mathcal{C}$ is orthonormal, Proposition 25.a.3 gives $A \in \mathrm{O}(2)$, and by Corollary 20.c.8 together with the Proposition on Page 254, we obtain $\det(T) = \det(A) = \pm 1$.

Consider the coordinate map $\Phi\colon V \to \mathbb{R}^2$, $\Phi(w) := [w]_{\mathcal{C}}$. Because $\mathcal{C}$ is orthonormal, Φ is itself a linear isometry: it preserves the norm by AI-Lemma 24.1 **(a)**, and the polarization argument proving **(b)** $\implies$ **(a)** in Theorem 25.a.2 never uses that the two spaces coincide, so it recovers the scalar product here as well. By construction $\Phi \circ T = T_A \circ \Phi$. Consequently, if (v_1, v_2) is an orthonormal basis for $\mathbb{R}^2$, then $\mathcal{B} := (\Phi^{-1}(v_1), \Phi^{-1}(v_2))$ is an orthonormal basis for V with $[T]_{\mathcal{B}}^{\mathcal{B}} = [T_A]_{(v_1,v_2)}^{(v_1,v_2)}$.

(a) If $\det(T) = 1$, then $\det A = 1$, so $A = R_\theta$ for some $\theta \in [0, 2\pi)$ by Proposition 25.a.7. Here we may take (v_1, v_2) to be the standard basis of $\mathbb{R}^2$, that is, $\mathcal{B} := \mathcal{C}$, and then $[T]_{\mathcal{B}}^{\mathcal{B}} = A = R_\theta$.

(b) If $\det(T) = -1$, then $\det A = -1$, so Proposition 25.a.7 supplies a basis (v_1, v_2) for $\mathbb{R}^2$ with $[T_A]_{(v_1,v_2)}^{(v_1,v_2)} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$; by the calculation at the end of its proof, this basis is orthonormal. Transporting it back gives an orthonormal basis $\mathcal{B}$ for V with $[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$.

Q.E.D.

Proposition 25.a.9 (*Classification of* $\mathrm{SO}(3)$): Let $A \in \mathrm{SO}(3)$. Then 1 is an eigenvalue of A. Moreover, there exists an orthonormal basis $\mathcal{B} := (v_1, v_2, v_3)$ and $\theta \in [0, 2\pi)$ such that

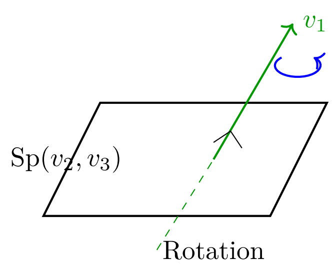
$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & R_\theta \end{pmatrix}.$$

So, T_A is a rotation with angle θ around the axis $\mathrm{Sp}(v_1)$. Moreover:

(a) If -1 is an eigenvalue of A, then $\theta = \pi$, hence

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(b) If -1 is **not** an eigenvalue of A (which is true if and only if 1 is the only real eigenvalue), then $\theta \neq \pi$.



Proof:

The characteristic polynomial $P_A(x)$ of A is a polynomial of degree 3 with real coefficients. By a theorem from analysis (the Intermediate Value Theorem), every polynomial of odd degree with real coefficients has a real zero, so there exists $\lambda \in \mathbb{R}$ with $P_A(\lambda) = 0$. This implies that A has at least one real eigenvalue, by AI-Proposition 21.1. By Lemma 25.a.6, $\lambda = \pm 1$.

Let $v_1 \in \mathbb{R}^3$ be an eigenvector of λ. By normalizing it if necessary, we may assume $\|v_1\| = 1$. Consider the subspace

$$L := v_1^\perp = \{v \in \mathbb{R}^3 \mid v \perp v_1\}.$$

Since A is orthogonal, we have $A \cdot v \in L$ for all $v \in L$ (i.e., $T_A(L) \subseteq L$, meaning L is invariant under T_A). Indeed, if $v \in L$, this implies $0 = \langle v, v_1 \rangle$. Because $A \in \mathrm{O}(3)$, the map T_A preserves the scalar product by Proposition 25.a.3, and therefore,

$$0 = \langle v, v_1 \rangle = \langle Av, Av_1 \rangle = \langle Av, \pm v_1 \rangle = \pm \langle Av, v_1 \rangle,$$

which implies that $Av \in L$.

By Theorem 22.b.3, we also have $\mathbb{R}^3 = \mathrm{Sp}(v_1) \oplus L$, hence $\dim L = 2$ by Corollary 22.b.7. Now, $T_A|_L \colon L \to L$ is a linear isometry. (Indeed, T_A preserves the standard scalar product of $\mathbb{R}^3$ by

Proposition 25.a.3, hence so does its restriction to L, and that restriction is well defined as a map $L \to L$ precisely because of the invariance just established). Let $\mathcal{L} = (w_1, w_2)$ be an orthonormal basis for L. By Proposition 25.a.3, the representation matrix is $Q := [T_A|_L]_{\mathcal{L}}^{\mathcal{L}} \in \mathrm{O}(2)$.

Case (a): $\lambda = 1$. Take the basis $\mathcal{B} := (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & Q \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}.$$

Now, by Corollary 20.c.8 and Proposition 20.c.4, $1 = \det A = \det(T_A) = 1 \cdot \det(Q)$, which implies $\det Q = 1$. So, $Q \in \mathrm{SO}(2)$. By Proposition 25.a.7, there exists $\theta \in [0, 2\pi)$ such that $Q = R_\theta$. Thus,

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & R_\theta \end{pmatrix}.$$

Case (b): $\lambda = -1$. By the same argument as in case **(a)**, we get that $1 = \det A = (-1) \cdot \det(T_A|_L)$, which implies $\det(T_A|_L) = -1$. By Corollary 25.a.8, there exists an orthonormal basis $\mathcal{L} = (w_1, w_2)$ for L such that $[T_A|_L]_{\mathcal{L}}^{\mathcal{L}} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Take $\mathcal{B}' := (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}'}^{\mathcal{B}'} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

And for the reordered basis $\mathcal{B} := (w_1, v_1, w_2)$, we have

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & R_\pi \end{pmatrix}.$$

Note that in case **(b)** too, 1 is an eigenvalue of A (corresponding to the eigenvector w_1).

It remains to read off the two supplementary statements from these cases. For **(a)**, suppose -1 is an eigenvalue of A. Then we are free to take $\lambda := -1$ as the real eigenvalue chosen at the start of the proof, which puts us in case **(b)**; that case produced an orthonormal basis with $[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & R_\pi \end{smallmatrix}\right)$, that is, $\theta = \pi$. For **(b)**, we argue by contraposition: if $\theta = \pi$, then

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

which is a diagonal matrix carrying -1 on its diagonal. Since A and $[T_A]_{\mathcal{B}}^{\mathcal{B}}$ are similar, they have the same eigenvalues, so -1 is an eigenvalue of A. **Q.E.D.**

Exercise 25.3 (*Why the Two Conditions in Proposition 25.a.9 (b) Agree*): Prof. Biran annotates the parenthesis in Proposition 25.a.9 **(b)** with a *"why?"*. Justify it: for $A \in \mathrm{SO}(3)$, show that -1 is not an eigenvalue of A if and only if 1 is the only real eigenvalue of A.

Proposition 25.a.10 (*Classification of $\mathrm{O}(3) \setminus \mathrm{SO}(3)$*): Let $A \in \mathrm{O}(3) \setminus \mathrm{SO}(3)$. Then there exists an orthonormal basis $\mathcal{B} := (v_1, v_2, v_3)$ for $\mathbb{R}^3$ such that

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & R_\theta \end{pmatrix}$$

for some $\theta \in [0, 2\pi)$. (So T_A does a rotation by θ around $\mathrm{Sp}(v_1)$, followed by a reflection with respect to $\mathrm{Sp}(v_2, v_3)$).

Proof:

Let $A' := -A = (-I) \cdot A$. Since $-I \in \mathrm{O}(3)$, Proposition 25.a.5 gives $A' \in \mathrm{O}(3)$, and by Theorem 20.b.5, $\det A' = \det(-I) \cdot \det A = (-1)^3 \cdot \det A = (-1) \cdot (-1) = 1$ (the last step because $\det A = \pm 1$ by the Proposition on Page 254, while $\det A \neq 1$ since $A \notin \mathrm{SO}(3)$). This implies $A' \in \mathrm{SO}(3)$. By Proposition 25.a.9, there exists an orthonormal basis $\mathcal{B}$ for $\mathbb{R}^3$, and $\theta' \in [0, 2\pi)$, such that $[T_{A'}]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & R_{\theta'} \end{smallmatrix}\right)$. This implies:

$$\begin{aligned} [T_A]_{\mathcal{B}}^{\mathcal{B}} &= [-\mathrm{id}_{\mathbb{R}^3}]_{\mathcal{B}}^{\mathcal{B}} \cdot [T_{A'}]_{\mathcal{B}}^{\mathcal{B}} \\ &= -I \cdot \begin{pmatrix} 1 & 0 \\ 0 & R_{\theta'} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -R_{\theta'} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & R_{\theta} \end{pmatrix}, \end{aligned}$$

where $\theta = (\theta' + \pi) \pmod{2\pi}$. (Note that $-R_{\theta'} = R_\pi \circ R_{\theta'} = R_{\theta' + \pi}$). **Q.E.D.**

Solutions to the Exercises

Proof of Proposition 25.a.3:

Let $\mathcal{B} = (e_1, \ldots, e_n)$ be an orthonormal basis for V. First, suppose T is an orthogonal (resp. unitary) endomorphism. By Theorem 25.a.2, $T \circ T^* = \mathrm{id}_V$ and $T^* \circ T = \mathrm{id}_V$. Taking the representation matrix with respect to $\mathcal{B}$, we have:

$$[T^* \circ T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n.$$

By the properties of representation matrices (the composition rule, the Proposition on Page 116), this splits into $[T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$. Since $\mathcal{B}$ is an orthonormal basis, Proposition 23.a.4 gives $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$. Thus, $\left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^* \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$, which by Lemma 22.b.9 (resp. Lemma 22.b.10) means that the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is orthogonal (resp. unitary).

Conversely, assume that $A := [T]_{\mathcal{B}}^{\mathcal{B}}$ is orthogonal (resp. unitary). This means $A^* A = I_n$. Again because $\mathcal{B}$ is an orthonormal basis, Proposition 23.a.4 yields $A^* = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^* = [T^*]_{\mathcal{B}}^{\mathcal{B}}$. Substituting this back yields $[T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$, and therefore $[T^* \circ T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}}$. This implies the maps are equal (by Theorem 18.a.2), so $T^* \circ T = \mathrm{id}_V$. By Theorem 25.a.2, this guarantees T is an isometry. **Q.E.D.**

Proof of Proposition 25.a.5:

We will prove this generically for $G = \mathrm{U}(n)$ and $G = \mathrm{SU}(n)$ (the real cases are identical, replacing the conjugate transpose * with the transpose $^\mathsf{T}$).

- **(a)** The identity matrix I_n satisfies $I_n^* \cdot I_n = I_n \cdot I_n = I_n$, so $I_n \in \mathrm{U}(n)$. Furthermore, $\det(I_n) = 1$, so $I_n \in \mathrm{SU}(n)$.
- **(b)** Let $A, B \in \mathrm{U}(n)$. We evaluate $(AB)^*(AB) = (B^* A^*)(AB) = B^*(A^* A)B$. Since $A \in \mathrm{U}(n)$, $A^* A = I_n$, so this reduces to $B^* I_n B = B^* B$. Since $B \in \mathrm{U}(n)$, $B^* B = I_n$. Thus, $AB \in \mathrm{U}(n)$. If additionally $A, B \in \mathrm{SU}(n)$, then by Theorem 20.b.5, $\det(AB) = \det(A)\det(B) = 1 \cdot 1 = 1$, so $AB \in \mathrm{SU}(n)$.
- **(c)** Let $A \in \mathrm{U}(n)$. This implies $A^* A = I_n$, meaning $A^{-1} = A^*$. We check if A^{-1} is unitary: $(A^{-1})^* A^{-1} = (A^*)^* A^* = AA^* = I_n$. Thus, $A^{-1} \in \mathrm{U}(n)$. If additionally $A \in \mathrm{SU}(n)$, then by Corollary 20.b.6, $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{1} = 1$, so $A^{-1} \in \mathrm{SU}(n)$.

Q.E.D.

Proof of Exercise 25.3:

Let $A \in \mathrm{SO}(3)$. Two facts about the real eigenvalues of A are already available. First, $\mathrm{SO}(3) \subseteq \mathrm{O}(3)$, so by Lemma 25.a.6 every real eigenvalue of A equals 1 or -1. Second, by the first assertion of Proposition 25.a.9, the value 1 **is** an eigenvalue of A. Taken together, the set of real eigenvalues of A is either $\{1\}$ or $\{1,-1\}$, and no other possibility occurs.

Now suppose -1 is not an eigenvalue of A. Then the set of real eigenvalues cannot be $\{1,-1\}$, so it equals $\{1\}$, which says exactly that 1 is the only real eigenvalue of A.

Conversely, suppose 1 is the only real eigenvalue of A. Since -1 is a real number different from 1, it cannot be an eigenvalue of A.

Note that the first assertion of Proposition 25.a.9 is what makes the equivalence work: without it, an orthogonal matrix could a priori have -1 as its only real eigenvalue, and then both conditions would fail simultaneously for the wrong reason. **Q.E.D.**

AI-Note: Two of Prof. Biran's statements were sitting in running prose with nothing to cite. The names "*orthogonal endomorphism*" and "*unitary endomorphism*", which Proposition 25.a.3, Lemma 25.a.6, Corollary 25.a.8 and both exercises all use, were introduced in a single sentence between 25.a.2 and 25.a.3; they are now the Definition on Page 252. And $\det A = \pm 1$ for $A \in \mathrm{O}(n)$, with $|\det A| = 1$ for $A \in \mathrm{U}(n)$, was a paragraph of running prose that Corollary 25.a.8 had to cite in words, as *"the computation following Proposition 25.a.3"*; it is now the Proposition on Page 254, with that paragraph as its proof. Both unnumbered, so that 25.a.1 to 25.a.10 do not move.

One gap in a proof. The implication **(b)** $\implies$ **(a)** of Theorem 25.a.2 is proved with the polarization identity of Exercise 22.1 **(a)**, which is stated for a Euclidean space, while the theorem is asserted over $\mathbb{R}$ and over $\mathbb{C}$ alike. The complex half needed the Hermitian identity of Exercise 22.1 **(b)**, which the solutions to ch. 22a do supply; the proof now says so.

Three silent steps in the same proof and the ones after it now name their reason: passing from $\langle v, e_j\rangle = \langle v, T^*Te_j\rangle$ for all v to $T^*Te_j = e_j$ is Lemma 22.a.3 **(c)**; concluding $T^* \circ T = \mathrm{id}_V$ from agreement on a basis is the uniqueness half of Theorem 15.a.3; and $\det(\overline{A}) = \overline{\det(A)}$, the fourth equality of the unitary determinant computation, holds because the Leibniz formula of Theorem 20.b.2 is a sum of products and conjugation respects both.

Also: the odd-degree polynomial argument opening the proof of Proposition 25.a.9 produces a root of P_A, and turning that into an eigenvalue is AI-Proposition 21.1. The coordinate map Φ in the proof of Corollary 25.a.8 was called an isometry without argument; it preserves the norm by AI-Lemma 24.1 **(a)**, and the polarization step of Theorem 25.a.2 never uses that the two spaces coincide.

Checked and found correct: the two unit vectors orthogonal to $(\cos\theta, \sin\theta)^{\mathsf{T}}$ really are $\pm(-\sin\theta, \cos\theta)^{\mathsf{T}}$; $R_\theta \cdot \mathrm{diag}(1,-1)$ really is $\left(\begin{smallmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{smallmatrix}\right)$; the reordering $(v_1, w_1, w_2) \to (w_1, v_1, w_2)$ in case **(b)** of Proposition 25.a.9 does turn $\mathrm{diag}(-1,1,-1)$ into $\mathrm{diag}(1,-1,-1)$; and $-R_{\theta'} = R_{\theta'+\pi}$.

Singular Value Decomposition (Chapter 26)

Theorem 26.a.1 (*Singular Value Decomposition*): Let $K = \mathbb{R}$ or $\mathbb{C}$, and let $A \in \mathcal{M}_{m\times n}(K)$. Then there exist $P \in \mathrm{O}(m)$ (if $K = \mathbb{R}$) or $P \in \mathrm{U}(m)$ (if $K = \mathbb{C}$), and $Q \in \mathrm{O}(n)$ (if $K = \mathbb{R}$) or $Q \in \mathrm{U}(n)$ (if $K = \mathbb{C}$), and a *"diagonal"* matrix

$$D = \left(\begin{array}{ccc} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_k \\ \hline & 0\dots 0 & \end{array}\right) \in \mathcal{M}_{m\times n}(\mathbb{R})$$

with $\lambda_1, \dots, \lambda_k > 0$, where $k = \operatorname{rank} A$, such that $A = PDQ^{-1}$.

Remark 26.1: Recall that we previously defined the equivalence of matrices. Here, the factorization $A = PDQ^{-1}$ demonstrates that we have an orthogonal (or unitary) equivalence. The definition in question is Definition 17.b.5 **(a)**: two matrices $A, B \in \mathcal{M}_{m\times n}(K)$ are equivalent when $B = PAQ$ for some $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$. The theorem sharpens this considerably, because the two invertible matrices may be taken to be orthogonal (resp. unitary) rather than merely invertible.

Proof:

We will prove only a special case: we assume $\operatorname{rank} A = n$ (which is true if and only if T_A is injective. Hence, we also assume $n \le m$). Both of these follow from Theorem 15.b.9: $\operatorname{rank} A = n$ says $\dim\ker T_A = 0$, and the image of the injective T_A is then an n-dimensional subspace of K^m, so $n \le m$ by Proposition 12.3.

Define $B := A^* \cdot A \in \mathcal{M}_{n\times n}(K)$. We claim that the linear map $T_B\colon K^n \to K^n$ is injective. Indeed, for all $0 \ne v \in K^n$, we have:

$$\langle T_B v, v\rangle = \langle Bv, v\rangle = \langle A^*Av, v\rangle = \langle Av, Av\rangle > 0. \tag{26.1}$$

This implies that $T_B v \ne 0$. Therefore, $\ker T_B = \{0\}$, hence T_B is injective.

Two steps in the chain (26.1) deserve a word. The third equality is the defining property (23.1) of the adjoint, which is available for T_A because the standard basis of K^n is orthonormal, so that $(T_A)^* = T_{A^*}$ by Proposition 23.a.4. The final inequality is strict precisely because of the standing assumption: $\operatorname{rank} A = n$ says that T_A is injective by Theorem 15.b.9, so $v \ne 0$ forces $Av \ne 0$, and positive definiteness of the inner product then yields $\langle Av, Av\rangle > 0$.

Note also that if $\eta \in K$ is an eigenvalue of B, then $\eta \in \mathbb{R}$ (even if $K = \mathbb{C}$) and $\eta > 0$. Indeed, B is self adjoint ($B^* = (A^*A)^* = A^*(A^*)^* = A^*A = B$, by Proposition 23.a.6 **(e)** and **(c)**), hence $\eta \in \mathbb{R}$. (This last step is Lemma 24.a.10 **(a)**, applied to the self-adjoint endomorphism T_B of K^n, which is self adjoint because $(T_B)^* = T_{B^*} = T_B$ by Corollary 23.a.5). To see that $\eta > 0$, let v be an eigenvector for η, so that $v \ne 0$ and $Bv = \eta v$. Substituting this into (26.1) gives

$$0 < \langle Bv, v\rangle = \langle \eta v, v\rangle = \eta \cdot \langle v, v\rangle,$$

where the last step uses linearity in the first variable together with $\eta \in \mathbb{R}$. Since $v \ne 0$, positivity of the inner product gives $\langle v, v\rangle > 0$, and therefore, $\eta > 0$.

By the Spectral Theorems for matrices (Theorem 24.a.11), B is orthogonally diagonalizable, hence there exists $Q \in \mathrm{O}(n)$ (or $\mathrm{U}(n)$) such that

$$Q^{-1}A^*AQ = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix},$$

where $\eta_j \in \mathbb{R}$, and $\eta_j > 0$ for all j. (For $K = \mathbb{R}$ this is part **(b)** of that theorem, because $B = A^\mathsf{T} A$ is then a real symmetric matrix; for $K = \mathbb{C}$ it is part **(a)**, because a self-adjoint matrix is in particular normal, $BB^* = BB = B^*B$. That the η_j are positive reals is the previous paragraph applied to them, which is legitimate because they are eigenvalues of B: the diagonal matrix $Q^{-1}BQ$ is similar to B, so the two have the same characteristic polynomial by Lemma 21.a.8 **(e)**, and the roots of that polynomial are the eigenvalues by AI-Proposition 21.1).

Let $\lambda_j := +\sqrt{\eta_j} > 0$ for $j = 1, \dots, n$. Define

$$C := AQ \cdot \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} \in \mathcal{M}_{m\times n}(K).$$

We have:

$$C^*C = \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} \underbrace{Q^{-1}A^*AQ}_{= \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix}} \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix}$$

$$= I_n.$$

(Two adjoints are taken silently here. By Proposition 23.a.6 **(e)**, $C^* = \big(\operatorname{diag}(\lambda_j^{-1})\big)^* Q^* A^*$, and the two factors on the left simplify: the diagonal matrix has real entries, so it is its own adjoint, while $Q^* = Q^{-1}$ because Q is orthogonal, resp. unitary, by Lemma 22.b.9 **(d)**, resp. Lemma 22.b.10 **(d)**. The final equality is then entry by entry: the product of the three diagonal matrices is $\operatorname{diag}(\eta_j/\lambda_j^2) = I_n$, since $\lambda_j^2 = \eta_j$).

It follows that the columns of C form an orthonormal system of vectors in K^m. To see this explicitly, let $c_1, \dots, c_n$ be the columns of C. The identity $C^*C = I_n$ means:

$$C^*C = \begin{pmatrix} - & \overline{c_1}^\mathsf{T} & - \\ & \vdots & \\ - & \overline{c_n}^\mathsf{T} & - \end{pmatrix} \begin{pmatrix} | & & | \\ c_1 & \dots & c_n \\ | & & | \end{pmatrix} = I_n.$$

Reading off the entries of this product, the identity $C^*C = I_n$ says precisely that $\langle c_j, c_i\rangle = \overline{c_i}^\mathsf{T} c_j = \delta_{ij}$ for all $1 \le i, j \le n$, which is the assertion that $c_1, \dots, c_n$ is an orthonormal system. Extend $c_1, \dots, c_n$ to an orthonormal basis $c_1, \dots, c_n, f_{n+1}, \dots, f_m$ of K^m and define:

$$P := \begin{pmatrix} | & & | & | & & | \\ c_1 & \dots & c_n & f_{n+1} & \dots & f_m \\ | & & | & | & & | \end{pmatrix} \in \mathcal{M}_{m\times m}(K).$$

Clearly, $P \in \mathrm{O}(m)$ (or $\mathrm{U}(m)$). (The extension is possible by Exercise 22.11, and P is orthogonal, resp. unitary, directly by Definition 22.b.8, since its columns are by construction an orthonormal basis of K^m).

We have:

$$AQ \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} = C = P \cdot \begin{array}{c} \left(\begin{array}{ccc} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \\ \hline & 0 & \end{array} \right) \end{array} \begin{array}{l} \left.\vphantom{\begin{array}{c}1\\ \ddots\\ 1\end{array}}\right\} n \\ \left.\vphantom{0}\right\} m-n \end{array}$$

(The right-hand equality is the definition of P read column by column: multiplying P by the $m\times n$ block $\binom{I_n}{0}$ selects its first n columns, which are $c_1, \dots, c_n$, that is, exactly C). Multiplying from the right by the diagonal matrix of λ_j's and then by Q^{-1}, we obtain:

$$A = P \cdot \left(\begin{array}{ccc} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \\ \hline & 0 & \end{array} \right) \cdot Q^{-1}.$$

This proves the theorem (under the assumption that $\operatorname{rank} A = n$).

For the general case, see the script, chapter 7.5. **Q.E.D.**

AI-Note: Nothing in this chapter is false, and the source-verification pass had already supplied most of what the argument leaves implicit: the definition of equivalence the Remark alludes to, the two steps of (26.1), the reading of $C^*C = I_n$ as an orthonormality statement, and the extension of an orthonormal system to a basis.

The rigour pass named four more. That $B := A^*A$ is self adjoint is Proposition 23.a.6 **(e)** and **(c)**, and passing from the matrix B to the endomorphism T_B, which is what Lemma 24.a.10 actually speaks about, is Corollary 23.a.5. The computation of C^*C takes two adjoints silently, of a real diagonal matrix and of Q, the second being $Q^* = Q^{-1}$ by Lemma 22.b.9 **(d)**, resp. Lemma 22.b.10 **(d)**. The identity $C = P \cdot \binom{I_n}{0}$ is the definition of P read column by column. And the reduction $n \leq m$ announced in the first line of the proof is Theorem 15.b.9 together with Proposition 12.3.

The one step that was doing real work unannounced: the proof calls the diagonal entries η_j of $Q^{-1}BQ$ positive reals, which is the paragraph above applied to them, and that paragraph speaks about *eigenvalues* of B. The two are the same list because $Q^{-1}BQ$ and B are similar, hence share a characteristic polynomial by Lemma 21.a.8 **(e)**, whose roots are the eigenvalues by AI-Proposition 21.1. Without that, $\lambda_j := +\sqrt{\eta_j}$ is not known to be defined.

On the numbering: Prof. Biran numbers nothing in this two-page section, and the file's own header explains why printing the single theorem as 26.a.1 shifts nothing. That decision is left as it stands.

Bilinear and Quadratic Forms (Chapter 27)

Definition 27.a.1 (*Bilinear Form*): Let V be a vector space over K (a general field). A function $B\colon V \times V \to K$ is called a "*bilinear form*" if the following two conditions are satisfied:

(a) For all $w \in V$, the map $B(-, w)\colon V \to K$, $v \mapsto B(v, w)$ is linear.

(b) For all $v \in V$, the map $B(v, -)\colon V \to K$, $w \mapsto B(v, w)$ is linear.

Lemma 27.a.2 (*Matrix Representation of a Bilinear Form*): Let V be a vector space over K and $\mathcal{C} := (e_1, \ldots, e_n)$ a basis for V. Let $B\colon V \times V \to K$ be a bilinear form. Then there exists a matrix $[B]_{\mathcal{C}} \in \mathcal{M}_{n\times n}(K)$ such that

$$B\left(\sum_{i=1}^{n} c_i e_i, \sum_{j=1}^{n} d_j e_j\right) = (c_1, \ldots, c_n) \cdot [B]_{\mathcal{C}} \cdot (d_1, \ldots, d_n)^{\mathsf{T}} \quad \forall c_i, d_j \in K.$$

The matrix $[B]_{\mathcal{C}}$ is called the "*matrix representation*" of B in the basis $\mathcal{C}$. (It is unique, which is what entitles us to the definite article; see the end of the proof).

Proof:

Define $[B]_{\mathcal{C}} := (a_{ij})$, where $a_{ij} := B(e_i, e_j)$. Then we compute:

$$\begin{aligned}
B\left(\sum_{i=1}^{n} c_i e_i, \sum_{j=1}^{n} d_j e_j\right) &= \sum_{i=1}^{n} c_i B\left(e_i, \sum_{j=1}^{n} d_j e_j\right) \\
&= \sum_{i=1}^{n} c_i \left(\sum_{j=1}^{n} d_j B(e_i, e_j)\right) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} c_i d_j B(e_i, e_j) \\
&= (c_1, \ldots, c_n) \cdot [B]_{\mathcal{C}} \cdot \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix}.
\end{aligned}$$

For uniqueness, suppose $M \in \mathcal{M}_{n\times n}(K)$ also satisfies the displayed identity. Fix i and j and choose the coefficients $c_1, \ldots, c_n$ to be 1 at the index i and 0 elsewhere, and $d_1, \ldots, d_n$ to be 1 at the index j and 0 elsewhere. The left-hand side is then $B(e_i, e_j)$ and the right-hand side is the single entry M_{ij}, so $M_{ij} = B(e_i, e_j) = ([B]_{\mathcal{C}})_{ij}$ for all i and j, that is, $M = [B]_{\mathcal{C}}$. **Q.E.D.**

We saw that also for $T \in \operatorname{End}(V)$ and $\mathcal{C}$ a basis for V, we have a matrix representation $[T]_{\mathcal{C}}^{\mathcal{C}} \in \mathcal{M}_{n\times n}(K)$, and that if $\mathcal{D}$ is another basis for V, then

$$[T]_{\mathcal{D}}^{\mathcal{D}} = [\mathrm{id}_V]_{\mathcal{D}}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{C}} \cdot [\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}, \tag{27.1}$$

where $[\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}} = ([\mathrm{id}_V]_{\mathcal{D}}^{\mathcal{C}})^{-1}$ is the transition matrix from basis $\mathcal{D}$ to $\mathcal{C}$.

Question 27.1: What happens for bilinear forms?

Lemma 27.a.3 (*Change of Basis for Bilinear Forms*): Let V be a finite-dimensional vector space over K and $\mathcal{C}, \mathcal{D}$ two bases of V. Let $B\colon V \times V \to K$ be a bilinear form. Then

$$[B]_{\mathcal{D}} = \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)^{\mathsf{T}} \cdot [B]_{\mathcal{C}} \cdot [\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}.$$

(Note that this is quite different from (27.1)).

Proof:

Let $\mathcal{C} := (v_1, \ldots, v_n)$ and $\mathcal{D} := (w_1, \ldots, w_n)$. We have $([B]_{\mathcal{C}})_{j,k} = B(v_j, v_k)$, and $([B]_{\mathcal{D}})_{j,k} = B(w_j, w_k)$ (this is how the previous proof constructed those matrices, and by the uniqueness just established there is no other reading). Recall that column j in $[\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}$ contains the coordinates of w_j in the basis $\mathcal{C}$, by the Proposition on Page 113 applied to id_V:

$$w_j = \sum_{p=1}^{n} \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)_{p,j} \cdot v_p.$$

This implies:

$$\begin{aligned} ([B]_{\mathcal{D}})_{j,k} &= B(w_j, w_k) \\ &= B\left(\sum_{p=1}^{n} \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)_{p,j} v_p, \sum_{q=1}^{n} \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)_{q,k} v_q\right) \\ &= \sum_{p,q=1}^{n} \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)_{p,j} \cdot B(v_p, v_q) \cdot \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)_{q,k} \\ &= \left(\left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)^{\mathsf{T}} \cdot [B]_{\mathcal{C}} \cdot [\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)_{j,k}. \end{aligned}$$

Q.E.D.

Definition 27.a.4 (*Symmetric, Positive Definite, and Quadratic Forms*):

(a) A bilinear form $B \colon V \times V \to K$ is called "*symmetric*" if $B(v, w) = B(w, v)$ for all $v, w \in V$.

(b) Assume $K = \mathbb{R}$. A bilinear form $B \colon V \times V \to \mathbb{R}$ is called "*positive definite*" if $B(v, v) > 0$ for all $0 \neq v \in V$.

(c) For a bilinear form $B \colon V \times V \to K$, we define $q_B \colon V \to K$ by $q_B(v) := B(v, v)$. We call q_B the "*quadratic form*" associated to B. Sometimes we just denote it by q.

Example 27.1 (*Standard Bilinear and Quadratic Forms*):

(a) Let $(V, \langle\cdot,\cdot\rangle)$ be a Euclidean vector space. Then $B(v, w) := \langle v, w\rangle$ is a symmetric, positive definite bilinear form. Its associated quadratic form is $q(v) = \|v\|^2$.

(b) Let $B \in \mathcal{M}_{n\times n}(K)$. Then $B \colon K^n \times K^n \to K$ defined by $B(v, w) := v^{\mathsf{T}} B w$ is a bilinear form. (The letter B is doing double duty here, standing both for the matrix and for the form it defines; throughout this item, B written with two arguments is the form and B written alone is the matrix). If $B^{\mathsf{T}} = B$ (i.e., B is a symmetric matrix), then B is a symmetric bilinear form (left as an exercise, see Exercise 27.1). The associated quadratic form is $q(v) = v^{\mathsf{T}} B v$.

Exercise 27.1 (*A Symmetric Matrix Gives a Symmetric Bilinear Form*): This is the step Prof. Biran marks with *"exc."* in the example above. Let $B \in \mathcal{M}_{n\times n}(K)$ with $B^{\mathsf{T}} = B$, and let $B \colon K^n \times K^n \to K$ be the bilinear form $B(v, w) := v^{\mathsf{T}} B w$. Show that it is symmetric, that is, $B(v, w) = B(w, v)$ for all $v, w \in K^n$.

Remark 27.1: Note that $B \in \mathcal{M}_{n\times n}(K)$ and $B^{\mathsf{T}} \in \mathcal{M}_{n\times n}(K)$ give the same quadratic form $q \colon K^n \to K$. Indeed, for all $v \in K^n$, we have:

$$q_B(v) = v^{\mathsf{T}} B v = (v^{\mathsf{T}} B v)^{\mathsf{T}} = v^{\mathsf{T}} B^{\mathsf{T}} v = q_{B^{\mathsf{T}}}(v),$$

where we used the fact that $v^{\mathsf{T}} B v$ is a scalar, and thus equals its own transpose.

Assume now that $2 \neq 0$ in K (i.e., $1 + 1 \neq 0$). Then $\frac{1}{2}(B + B^{\mathsf{T}})$ also gives the same quadratic form as B. Indeed, for all $v \in K^n$,

$$q_{\frac{1}{2}(B+B^{\mathsf{T}})}(v) = \tfrac{1}{2}\left(v^{\mathsf{T}} B v + v^{\mathsf{T}} B^{\mathsf{T}} v\right) = \tfrac{1}{2}\left(q_B(v) + q_{B^{\mathsf{T}}}(v)\right) = q_B(v),$$

the last step by the identity just proved. What makes this worth recording is not the equality itself but the symmetry of the new matrix: $\left(\frac{1}{2}(B+B^{\mathsf{T}})\right)^{\mathsf{T}} = \frac{1}{2}(B^{\mathsf{T}}+B)$, so every quadratic form on K^n is q_A for a **symmetric** A, which is exactly what the Conclusion below asserts.

Conclusion 27.1: If $2 \neq 0$ in K, then for the discussion on quadratic forms $q\colon K^n \to K$, we can assume they all come from symmetric matrices (or symmetric bilinear forms).

Theorem (*Polarization Identity*): Assume $2 \neq 0$ in K. (Note that $2 \neq 0 \implies 4 \neq 0$. Why? This is left as an exercise, see Exercise 27.2). If B is symmetric and defines $q := q_B$, then for all $v, w \in V$ we have:

(a) $B(v,w) = \frac{1}{4}q(v+w) - \frac{1}{4}q(v-w)$.
(b) $B(v,w) = \frac{1}{2}(q(v+w) - q(v) - q(w))$.

Proof:

Bilinearity and the symmetry of B give, for all $v, w \in V$,

$$q(v \pm w) = B(v \pm w, v \pm w) = B(v,v) \pm B(v,w) \pm B(w,v) + B(w,w) = q(v) \pm 2B(v,w) + q(w).$$

Subtracting the two versions of this identity gives $q(v+w) - q(v-w) = 4B(v,w)$, which is **(a)**, division by 4 being legitimate by Exercise 27.2. Taking only the version with the plus sign and moving $q(v)$ and $q(w)$ to the left gives $q(v+w)-q(v)-q(w) = 2B(v,w)$, which is **(b)**. **Q.E.D.**

Exercise 27.2 (*Why* $2 \neq 0$ *Forces* $4 \neq 0$): This is the question Prof. Biran annotates the Polarization Identity with. Let K be a field in which $2 \neq 0$. Show that then also $4 \neq 0$, so that the divisions by 4 in part **(a)** really are legitimate.

Theorem 27.a.5 (*Sylvester's Law of Inertia*): Let $B\colon V \times V \to \mathbb{R}$ be a **symmetric** bilinear form on a vector space V over $\mathbb{R}$, of dimension n. Then there exists a basis $\mathcal{C}$ for V such that

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_k & 0 & 0 \\ 0 & -I_\ell & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

for some $k, \ell \in \mathbb{Z}_{\geq 0}$ with $k + \ell \leq n$. (The zero block in the bottom right has size $n - (k+\ell)$).
Moreover, k and ℓ do **not** depend on the basis $\mathcal{C}$ (they depend only on B). In fact:

$$k = \max\{\dim W \mid W \subseteq V \text{ is a subspace and } B|_{W\times W} \text{ is positive definite}\},$$
$$\ell = \max\{\dim U \mid U \subseteq V \text{ is a subspace and } -B|_{U\times U} \text{ is positive definite}\},$$
$$n-(k+\ell) = \max\{\dim Z \mid Z \subseteq V \text{ is a subspace and } B|_{Z\times Z} \equiv 0\}.$$

Remark 27.2:

(a) In the basis $\mathcal{C}$, the quadratic form q associated to B looks like
$$q(x_1, \ldots, x_n) = x_1^2 + \cdots + x_k^2 - x_{k+1}^2 - \cdots - x_{k+\ell}^2.$$
(b) k is called the "*positive index*" of B, ℓ is the "*negative index*", and $n-(k+\ell)$ is the "*nullity*". The value $\sigma(B) := \operatorname{sgn}(B) := k - \ell$ is called the "*signature*" of B. (This is sometimes referred to as type $(k, \ell, 0)$).

Proof:

Choose any basis $\mathcal{C}'$ for V and let $A := [B]_{\mathcal{C}'} \in \mathcal{M}_{n\times n}(\mathbb{R})$. Note that A is a symmetric matrix (since B is symmetric: $A_{ij} = B(v_i', v_j') = B(v_j', v_i') = A_{ji}$). By the Spectral Theorem for matrices (Theorem 24.a.11 **(b)**), this implies that A is orthogonally diagonalizable, i.e., there exists $P \in \mathrm{O}(n)$ such that $P^{-1}AP$ is a diagonal matrix. But $P^{-1} = P^{\mathsf{T}}$ (by Lemma 22.b.9 **(d)**), so $P^{\mathsf{T}}AP$ is diagonal.

Let $\mathcal{C}''$ be the (unique) basis of V such that $[\mathrm{id}_V]_{\mathcal{C}'}^{\mathcal{C}''} = P$. (That is, if $\mathcal{C}' = (v_1', \dots, v_n')$ and $\mathcal{C}'' = (v_1'', \dots, v_n'')$, then $v_j'' = \sum_{i=1}^n P_{ij} v_i'$ for all j). We have, by Lemma 27.a.3 applied to the pair $\mathcal{C}'$, $\mathcal{C}''$,

$$[B]_{\mathcal{C}''} = P^{\mathsf{T}} A P = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix},$$

where $\eta_j \in \mathbb{R}$ for all j.

By changing the order of the elements in the basis $\mathcal{C}''$, we may assume $\eta_1, \dots, \eta_k > 0$, $\eta_{k+1}, \dots, \eta_{k+\ell} < 0$, and $\eta_{k+\ell+1} = \dots = \eta_n = 0$ for some $k, \ell \in \mathbb{Z}_{\geq 0}$ with $k + \ell \leq n$.

We can write:

$$\begin{aligned} &\eta_i = \lambda_i^2, \text{ with } \lambda_i > 0, \quad \forall 1 \leq i \leq k \quad (\text{so } B(v_i'', v_i'') = \lambda_i^2), \\ &\eta_i = -\lambda_i^2, \text{ with } \lambda_i > 0, \quad \forall k+1 \leq i \leq k+\ell \quad (\text{so } B(v_i'', v_i'') = -\lambda_i^2), \\ &\eta_i = 0, \quad \forall k+\ell+1 \leq i \leq n \quad (\text{so } B(v_i'', v_i'') = 0). \end{aligned}$$

Define $\mathcal{C} := (v_1, \dots, v_k, v_{k+1}, \dots, v_{k+\ell}, v_{k+\ell+1}, \dots, v_n)$, where

$$v_i = \frac{1}{\lambda_i} v_i'' \quad \forall 1 \leq i \leq k+\ell, \qquad v_i = v_i'' \quad \forall k+\ell+1 \leq i \leq n.$$

In other words:

$$\mathcal{C} = \left(\frac{1}{\lambda_1} v_1'', \dots, \frac{1}{\lambda_k} v_k'', \frac{1}{\lambda_{k+1}} v_{k+1}'', \dots, \frac{1}{\lambda_{k+\ell}} v_{k+\ell}'', v_{k+\ell+1}'', \dots, v_n'' \right).$$

Note that $v_i = c_i v_i''$ for some scalars $c_i \neq 0$, for all i. It is left as an exercise to show that $\mathcal{C}$ is indeed a basis, see Exercise 27.3.

We have $([B]_{\mathcal{C}})_{i,j} = B(v_i, v_j) = B(c_i v_i'', c_j v_j'') = 0$ for all $i \neq j$. And for $i = j$:

- if $1 \leq i \leq k$, then $([B]_{\mathcal{C}})_{i,i} = B(v_i, v_i) = B\left(\frac{1}{\lambda_i} v_i'', \frac{1}{\lambda_i} v_i''\right) = \frac{1}{\lambda_i^2} \lambda_i^2 = 1$,
- if $k+1 \leq i \leq k+\ell$, then $([B]_{\mathcal{C}})_{i,i} = B(v_i, v_i) = B\left(\frac{1}{\lambda_i} v_i'', \frac{1}{\lambda_i} v_i''\right) = \frac{1}{\lambda_i^2}(-\lambda_i^2) = -1$,
- if $k+\ell+1 \leq i \leq n$, then $([B]_{\mathcal{C}})_{i,i} = B(v_i, v_i) = B(v_i'', v_i'') = 0$.

This proves the existence of the basis $\mathcal{C}$ claimed in the theorem.

We will prove now that k is independent of $\mathcal{C}$. Put

$$k' := \max\{\dim W \mid W \subseteq V \text{ is a subspace such that } B|_{W \times W} \text{ is positive definite}\}.$$

We will show $k = k'$. Indeed, $k \leq k'$, because $B|_{U \times U}$ is positive definite on $U := \mathrm{Sp}(v_1, \dots, v_k)$. This is because:

$$\begin{aligned} &B(a_1 v_1 + \dots + a_k v_k, a_1 v_1 + \dots + a_k v_k) \\ &= (a_1, \dots, a_k, 0, \dots, 0) \cdot \begin{pmatrix} I_k & 0 & 0 \\ 0 & -I_\ell & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} \\ &= a_1^2 + \dots + a_k^2 \\ &> 0, \end{aligned}$$

if not all $a_i = 0$.

Suppose by contradiction that $k' \geq k+1$. This implies there exists $W \subseteq V$ of dimension $\geq k+1$ such that $B|_{W \times W}$ is positive definite. Take $X := \mathrm{Sp}(v_{k+1}, \dots, v_n)$. Then, by the dimension

formula (Proposition 14.2 **(c)**):
$\dim(W \cap X) = \dim W + \dim X - \dim(W + X) \geq (k+1) + (n-k) - \dim(W+X) \geq n + 1 - n = 1$.
This implies $W \cap X \neq \{0\}$. Take $0 \neq w \in W \cap X$. We can write $w = a_{k+1}v_{k+1} + \cdots + a_n v_n$. Then $B(w,w) = -a_{k+1}^2 - \cdots - a_{k+\ell}^2 \leq 0$. But, by the assumption that W is positive definite, $B(w,w) > 0$. This is a contradiction, which implies $k' \leq k$. This proves $k = k'$.
The proofs for ℓ and $n - (k+\ell)$ are similar. **Q.E.D.**

Exercise 27.3 (*Rescaling a Basis Leaves a Basis*): This is the step marked *"Exc."* in the proof above. Let $\mathcal{C}'' := (v_1'', \ldots, v_n'')$ be a basis for V, let $c_1, \ldots, c_n \in K \setminus \{0\}$, and put $v_i := c_i v_i''$. Show that $\mathcal{C} := (v_1, \ldots, v_n)$ is again a basis for V.

Exercise 27.4 (*Why Symmetry Cannot Be Dropped in Theorem 27.a.5*): Prof. Biran annotates the word *"symmetric"* in the statement of Theorem 27.a.5 with *"explain"*. Explain why the hypothesis cannot simply be dropped: show that if $[B]_{\mathcal{C}}$ has the form displayed in the theorem for **some** basis $\mathcal{C}$, then B is necessarily symmetric. (Hint: that displayed matrix is symmetric, and Lemma 27.a.3 tells you what happens to it under a change of basis). Then give a bilinear form on $\mathbb{R}^2$ for which the conclusion of the theorem fails.

Theorem 27.a.6 (*Sylvester's Law for Euclidean Spaces*): Let V be a finite-dimensional Euclidean space of dimension n. Let $B\colon V \times V \to \mathbb{R}$ be a symmetric bilinear form. Then there exists an **orthonormal** basis $\mathcal{C}$ for V in which

$$[B]_{\mathcal{C}} = \begin{pmatrix} \eta_1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & \eta_k & & & & & & \\ & & & \eta_{k+1} & & & & & \\ & & & & \ddots & & & & \\ & & & & & \eta_{k+\ell} & & & \\ & & & & & & 0 & & \\ & & & & & & & \ddots & \\ & & & & & & & & 0 \end{pmatrix},$$

where $k + \ell \leq n$, with $\eta_1, \ldots, \eta_k > 0$ and $\eta_{k+1}, \ldots, \eta_{k+\ell} < 0$, and where the remaining $n - (k+\ell)$ diagonal entries vanish. As before, k, ℓ are the positive and negative indices of B. Moreover, the scalars $\eta_1, \ldots, \eta_k, \eta_{k+1}, \ldots, \eta_{k+\ell}, 0, \ldots, 0$ are independent of $\mathcal{C}$. They depend only on B (and on the scalar product on V). They are the eigenvalues of $[B]_{\mathcal{D}}$ for every **orthonormal** basis $\mathcal{D}$ of V (independent of $\mathcal{D}$ and on whether or not $[B]_{\mathcal{D}}$ is diagonal).

Remark 27.3: This theorem has a geometric meaning. If $q\colon \mathbb{R}^n \to \mathbb{R}$ is a quadratic form $q(x_1, \ldots, x_n) = \sum a_{ij} x_i x_j$ for some $a_{ij} \in \mathbb{R}$, then the set

$$Q = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid q(x_1, \ldots, x_n) = C\} \quad \text{(for some constant } C \in \mathbb{R})$$

is called a "*quadric*". (Examples include an ellipse/ellipsoid, parabola/paraboloid, hyperbola/hyperboloid). The theorem tells us that there exists a linear isometry $T\colon \mathbb{R}^n \to \mathbb{R}^n$ such that

$$T(Q) = \left\{ (y_1, \ldots, y_n) \in \mathbb{R}^n \;\middle|\; \underbrace{\eta_1 y_1^2 + \cdots + \eta_k y_k^2}_{\text{positive coeffs.}} + \underbrace{\eta_{k+1} y_{k+1}^2 + \cdots + \eta_{k+\ell} y_{k+\ell}^2}_{\text{negative coeffs.}} = C \right\}.$$

Proof Outline:

Pick any orthonormal basis $\mathcal{C}'$ for V (similar to the previous proof, only now we take $\mathcal{C}'$ to be orthonormal). Define $A := [B]_{\mathcal{C}'}$, and proceed exactly as in the previous proof, but instead of defining $\mathcal{C}$ with the scaling factors λ_i, simply take $\mathcal{C} = \mathcal{C}''$.

To see why $\eta_1, \dots, \eta_{k+\ell}, 0, \dots, 0$ depend only on B etc., note that if $\mathcal{D}$ and $\mathcal{D}'$ are two orthonormal bases for V, then $[\mathrm{id}_V]^{\mathcal{D}}_{\mathcal{D}'} \in \mathrm{O}(n)$ (left as an exercise, see Exercise 27.5). Put $Q := [\mathrm{id}_V]^{\mathcal{D}}_{\mathcal{D}'}$. We have

$$[B]_{\mathcal{D}} = Q^{\mathsf{T}} \cdot [B]_{\mathcal{D}'} \cdot Q = Q^{-1}[B]_{\mathcal{D}'}Q,$$

since $Q \in \mathrm{O}(n)$, so that $Q^{\mathsf{T}} = Q^{-1}$ by Lemma 22.b.9 **(d)**. Hence, the matrices $[B]_{\mathcal{D}}$ and $[B]_{\mathcal{D}'}$ are similar, which means their eigenvalues are exactly the same: similar matrices share a characteristic polynomial by Lemma 21.a.8 **(e)**, and the eigenvalues are its roots by AI-Proposition 21.1. This is precisely where the congruence $Q^{\mathsf{T}}[B]_{\mathcal{D}'}Q$ of Lemma 27.a.3, which in general does **not** preserve eigenvalues, is upgraded to a similarity, and it is the reason the theorem must insist on orthonormal bases. $\boxed{\text{Q.E.D.}}$

Exercise 27.5 (*The Transition Matrix Between Two Orthonormal Bases*): This is the step marked *"exc."* in the proof above, and it is what makes the whole argument work: it converts the congruence $Q^{\mathsf{T}}[B]_{\mathcal{D}'}Q$ of Lemma 27.a.3 into a similarity, so that eigenvalues are preserved. Let V be a Euclidean space and let $\mathcal{D}, \mathcal{D}'$ be two orthonormal bases for V. Show that $[\mathrm{id}_V]^{\mathcal{D}}_{\mathcal{D}'} \in \mathrm{O}(n)$.

Theorem 27.a.7 (*Diagonalization of Symmetric Bilinear Forms*): Let V be a finite-dimensional vector space over a field K of characteristic 0. Let $B\colon V \times V \to K$ be a symmetric bilinear form. Then there exists a basis $\mathcal{C}$ of V such that $[B]_{\mathcal{C}}$ is a diagonal matrix.

Proof:

We proceed by induction on $n := \dim V$. We need to show there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ such that $B(v_i, v_j) = 0$ for all $i \neq j$.

If $B = 0$, or if $n = 1$, there is nothing to prove.

Let $n \geq 2$. Assume the statement of the theorem holds for all vector spaces of dimension $\leq n-1$. Let V be an n-dimensional vector space. If $B(v,v) = 0$ for all $v \in V$, then the polarization formula (on Page 266) shows $B = 0$ and we are done. (That identity is available here because K has characteristic 0, so in particular $2 \neq 0$ in K). So assume that there exists $v_1 \in V$ such that $B(v_1, v_1) \neq 0$.

Put $W := \mathrm{Sp}(v_1)$. The subspace W is 1-dimensional (because $v_1 \neq 0$). Define the orthogonal complement with respect to B as:

$$W^{\perp_B} := \{w \in V \mid B(v_1, w) = 0\}.$$

It is left as an exercise to show that $W^{\perp_B} \subseteq V$ is a subspace, see Exercise 27.6.

We claim that $W \cap W^{\perp_B} = \{0\}$. Indeed, if $w \in W \cap W^{\perp_B}$, then $w = c \cdot v_1$ and $B(v_1, w) = 0$. But $B(v_1, w) = B(v_1, c \cdot v_1) = c \cdot B(v_1, v_1)$. Since $B(v_1, v_1) \neq 0$, this implies $c = 0$, which means $w = 0$.

We claim that $V = W + W^{\perp_B}$. Indeed, if $v \in V$, then the vector

$$u := v - \frac{B(v_1, v)}{B(v_1, v_1)} \cdot v_1$$

belongs to $W^{\perp_B}$. To see this, we calculate:

$$B(v_1, u) = B(v_1, v) - \frac{B(v_1, v)}{B(v_1, v_1)} \cdot B(v_1, v_1) = 0.$$

This implies $u \in W^{\perp_B}$. So we can write v as:

$$v = \underbrace{\frac{B(v_1, v)}{B(v_1, v_1)} \cdot v_1}_{\in W} + \underbrace{u}_{\in W^{\perp_B}} \in W + W^{\perp_B}.$$

It follows that $V = W \oplus W^{\perp_B}$. This implies $\dim W^{\perp_B} = n - 1$.

Now, $B|_{W^{\perp_B} \times W^{\perp_B}}$ is a symmetric bilinear form on an $(n-1)$-dimensional vector space ($W^{\perp_B}$). By the induction hypothesis, there exists a basis $\mathcal{C}' = (v_2, \dots, v_n)$ for $W^{\perp_B}$ such that $B(v_i, v_j) = 0$ for all $i \neq j$ with $i, j \geq 2$.

Take $\mathcal{C} := (v_1, \dots, v_n)$. Since $v_1 \in W$ and $v_2, \dots, v_n \in W^{\perp_B}$, we obtain that $B(v_i, v_j) = 0$ for all $i \neq j$. **Q.E.D.**

Exercise 27.6 (*The Orthogonal Complement with Respect to a Bilinear Form*): This is the step marked *"Exc."* in the proof above. With B, V and v_1 as there, show that

$$W^{\perp_B} := \{w \in V \mid B(v_1, w) = 0\}$$

is a linear subspace of V.

Question 27.2: What happens over $\mathbb{C}$?

Theorem 27.a.8 (*Complex Symmetric Bilinear Forms*): Let $B\colon V \times V \to \mathbb{C}$ be a symmetric bilinear form on a vector space V over $\mathbb{C}$, of dimension n. Then there exists a basis $\mathcal{C}$ of V such that

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

for some $0 \leq r \leq n$. (The zero block in the bottom right has size $n - r$).

The number r is independent of the basis $\mathcal{C}$: it depends only on B. In fact,

$$n - r = \max\{\dim W \mid W \subseteq V \text{ is a subspace such that } B|_{W \times W} \equiv 0\}.$$

The number r is sometimes called the "*rank*" of B.

Proof:

By the previous theorem (Theorem 27.a.7), there exists a basis $\mathcal{C}' = (v'_1, \dots, v'_n)$ of V such that

$$[B]_{\mathcal{C}'} = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix},$$

where, after reordering the vectors of $\mathcal{C}'$ if necessary, $\eta_j \neq 0$ for all $1 \leq j \leq r$ and $\eta_j = 0$ for all $r < j \leq n$. Here r is simply the number of non-zero entries on the diagonal, so $0 \leq r \leq n$. (The reordering matters: the vectors v'_j with $\eta_j \neq 0$ must come first, since we are about to divide by λ_j).

Choose $\lambda_j \in \mathbb{C}$ such that $\lambda_j^2 = \eta_j$ for $j = 1, \dots, r$. (This is always possible over $\mathbb{C}$, and $\lambda_j \neq 0$ because $\eta_j \neq 0$). Define $v_j := \frac{1}{\lambda_j} v'_j$ for $j = 1, \dots, r$, and $v_q := v'_q$ for all $r < q \leq n$.

Take $\mathcal{C} := (v_1, \dots, v_n)$. Then $B(v_i, v_j) = 0$ for all $i \neq j$. Furthermore, for $1 \leq j \leq r$, we have:

$$B(v_j, v_j) = B\left(\frac{1}{\lambda_j} v'_j, \frac{1}{\lambda_j} v'_j\right) = \frac{1}{\lambda_j^2} \cdot \eta_j = 1,$$

and for all $r + 1 \leq j \leq n$, we have $B(v_j, v_j) = 0$. **Q.E.D.**

More on Positive Definite Bilinear Forms

Recall: A symmetric bilinear form $B\colon V\times V\to\mathbb{R}$ is called "*positive definite*" if $B(v,v)>0$ for all $0\neq v\in V$.

Theorem 27.a.9 (*Sylvester's Criterion*): Let V be a finite-dimensional vector space over $\mathbb{R}$ and $B\colon V\times V\to\mathbb{R}$ a symmetric bilinear form. Then the following conditions are equivalent:

(a) B is positive definite.

(b) There exists a basis $\mathcal{C}:=(e_1,\dots,e_n)$ for V such that the principal minors

$$\det(B(e_j,e_k))_{j,k=1,\dots,\ell}$$

for all $1\leq\ell\leq n$, are positive.

(c) Like **(b)**, but instead of *"there exists a basis…"*, it holds for *"all bases…"*.

Proof:

(b) $\implies$ **(a)**: We proceed by induction on $n=\dim V$.

For $n=1$, the assumption in **(b)** says $B(e_1,e_1)>0$. This implies that for all $0\neq c\in\mathbb{R}$, $B(ce_1,ce_1)=c^2B(e_1,e_1)>0$.

Suppose the implication holds for all spaces of dimension $\leq n$. Let V be an $(n+1)$-dimensional vector space, and $B\colon V\times V\to\mathbb{R}$ a symmetric bilinear form. Let $\mathcal{C}:=(e_1,\dots,e_{n+1})$ be a basis for V such that all the principal minors of $[B]_{\mathcal{C}}$ are positive.

Let $W:=\mathrm{Sp}(e_1,\dots,e_n)$ with the basis $\mathcal{C}_W:=(e_1,\dots,e_n)$. The principal minors of $[B|_{W\times W}]_{\mathcal{C}_W}$ are exactly the same as the first n principal minors of $[B]_{\mathcal{C}}$. Since these are positive, the induction hypothesis implies that $B|_{W\times W}$ is positive definite.

This implies that if we define $\langle w,w'\rangle_B:=B(w,w')$, then $\langle\cdot,\cdot\rangle_B$ is a valid scalar product on W.

Consider the linear functional $L\colon W\to\mathbb{R}$ defined by $L(w)=B(w,e_{n+1})$. By a previous result (Riesz Representation Theorem, Theorem 23.a.1), there exists $w'\in W$ such that $L(w)=\langle w,w'\rangle_B$ for all $w\in W$. This implies:

$$B(w,w')=B(w,e_{n+1})\quad\forall w\in W. \tag{27.2}$$

Define a new basis for V, $\mathcal{C}':=(e_1,\dots,e_n,e_{n+1}-w')$. (It is left as an exercise to show this is a basis, see Exercise 27.7. Hint: $w'\in W$). By (27.2), we have $B(e_j,e_{n+1}-w')=0$ for all $1\leq j\leq n$.

This implies that the matrix representation in the new basis is block diagonal:

$$[B]_{\mathcal{C}'}=\begin{pmatrix}C&0\\0&d\end{pmatrix}=S^{\mathsf{T}}[B]_{\mathcal{C}}S,$$

where $d=B(e_{n+1}-w',e_{n+1}-w')$, $S=[\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{C}'}$, and $C=[B|_{W\times W}]_{\mathcal{C}_W}$.

Taking the determinant, we get:

$$\det([B]_{\mathcal{C}'})=\det(C)\cdot d=\det(S)^2\cdot\det([B]_{\mathcal{C}}).$$

We know $\det(C)>0$ (it is a principal minor), $\det(S)^2>0$ (since S is invertible), and $\det([B]_{\mathcal{C}})>0$ (by assumption). This implies that $d>0$.

Let $v\in V$. We can write $v=w+\alpha\cdot(e_{n+1}-w')$, with $w\in W$ and $\alpha\in\mathbb{R}$. Recall that $\langle w,e_{n+1}-w'\rangle_B=0$. This implies:

$$B(v,v)=B(w,w)+\alpha^2B(e_{n+1}-w',e_{n+1}-w')=B(w,w)+\alpha^2d>0,$$

whenever either $w\neq0$ or $\alpha\neq0$. This implies that B is positive definite. The proof of *"**(b)** $\implies$ **(a)**"* follows now by induction.

(a) $\implies$ **(b)**: Left as an exercise, see Exercise 27.8. Similar ideas apply. **Q.E.D.**

Exercise 27.7 (*Shifting the Last Basis Vector Leaves a Basis*): This is the step Prof. Biran marks with *"exc. show this is a basis. Hint: $w' \in W$ "* in the proof above. Let $\mathcal{C} := (e_1, \ldots, e_n, e_{n+1})$ be a basis for V and let $w' \in W := \mathrm{Sp}(e_1, \ldots, e_n)$. Show that $\mathcal{C}' := (e_1, \ldots, e_n, e_{n+1} - w')$ is again a basis for V.

Exercise 27.8 (*The Forward Direction of Theorem 27.a.9*): This is the implication Prof. Biran leaves with *"Exc. Similar ideas."* Prove that **(a)** $\implies$ **(b)** in Theorem 27.a.9: if B is positive definite, then the principal minors $\det(B(e_j, e_k))_{j,k=1,\ldots,\ell}$ are positive for $1 \le \ell \le n$.

Solutions to the Exercises

Proof of Exercise 27.1:

Let $v, w \in K^n$. The product $w^\mathsf{T} B v$ is a 1×1 matrix, that is, a scalar, and therefore, it equals its own transpose. Using this together with $B^\mathsf{T} = B$, we compute

$$B(w, v) = w^\mathsf{T} B v = \left(w^\mathsf{T} B v\right)^\mathsf{T} = v^\mathsf{T} B^\mathsf{T} w = v^\mathsf{T} B w = B(v, w),$$

which is precisely the symmetry of the bilinear form. **Q.E.D.**

Proof of Exercise 27.2:

Assume $2 \neq 0$ in a field K. This means $1+1 \neq 0$. If $4 = 0$, then $2+2 = 0$, which means $2 \cdot 2 = 0$. Since K is a field, it has no zero divisors. Thus, $2 \cdot 2 = 0$ implies $2 = 0$, which contradicts our assumption. Therefore, $4 \neq 0$. **Q.E.D.**

Proof of Exercise 27.6:

Let $w_1, w_2 \in W^{\perp_B}$ and $\alpha, \beta \in K$. We need to show that $\alpha w_1 + \beta w_2 \in W^{\perp_B}$. By definition of $W^{\perp_B}$, we have $B(v_1, w_1) = 0$ and $B(v_1, w_2) = 0$. Using the linearity of B in the second argument, we compute:

$$B(v_1, \alpha w_1 + \beta w_2) = \alpha B(v_1, w_1) + \beta B(v_1, w_2) = \alpha \cdot 0 + \beta \cdot 0 = 0.$$

Thus, $\alpha w_1 + \beta w_2 \in W^{\perp_B}$, proving it is a subspace. **Q.E.D.**

Proof of Exercise 27.3:

We are given a basis $\mathcal{C}'' := (v_1'', \ldots, v_n'')$. The new set $\mathcal{C} := (v_1, \ldots, v_n)$ is formed by scaling each vector in $\mathcal{C}''$ by a non-zero scalar c_i (where $c_i = 1/\lambda_i$ or $c_i = 1$). Since scaling basis vectors by non-zero scalars preserves linear independence and the span, $\mathcal{C}$ remains a basis for V. **Q.E.D.**

Proof of Exercise 27.4:

Suppose that $\mathcal{C}$ is a basis for V in which $M := [B]_\mathcal{C}$ has the form displayed in Theorem 27.a.5. That matrix is symmetric, $M^\mathsf{T} = M$. Now let $v, w \in V$ be arbitrary and let $c, d \in \mathbb{R}^n$ be their coordinate vectors with respect to $\mathcal{C}$. By Lemma 27.a.2, $B(v, w) = c^\mathsf{T} M d$, and since a 1×1 matrix equals its own transpose,

$$B(w, v) = d^\mathsf{T} M c = \left(d^\mathsf{T} M c\right)^\mathsf{T} = c^\mathsf{T} M^\mathsf{T} d = c^\mathsf{T} M d = B(v, w).$$

So B is symmetric. In other words, symmetry is not an artefact of the proof but a genuine necessary condition: no non-symmetric form can be brought to that shape in any basis.

For a form where the conclusion fails, take $V := \mathbb{R}^2$ and

$$B(v, w) := v_1 w_2 - v_2 w_1,$$

which is bilinear with $B(e_1, e_2) = 1$ and $B(e_2, e_1) = -1$, hence not symmetric. By the argument just given, $[B]_\mathcal{C}$ never has the form of Theorem 27.a.5. One can also see the failure directly:

$q_B(v) = B(v,v) = v_1v_2 - v_2v_1 = 0$ for every v, so the associated quadratic form vanishes identically even though $B \neq 0$, and the numbers k and ℓ could not possibly describe B. **Q.E.D.**

Proof of Exercise 27.5:

Write $\mathcal{D} = (d_1, \dots, d_n)$ and $\mathcal{D}' = (d'_1, \dots, d'_n)$, and put $Q := [\mathrm{id}_V]^{\mathcal{D}}_{\mathcal{D}'}$. By the definition of a transition matrix, column j of Q holds the coordinates of d_j in the basis $\mathcal{D}'$, that is,

$$d_j = \sum_{p=1}^{n} Q_{p,j}\, d'_p \qquad \text{for all } 1 \le j \le n.$$

Since $\mathcal{D}$ is orthonormal, $\langle d_j, d_k \rangle = \delta_{jk}$, and since $\mathcal{D}'$ is orthonormal, $\langle d'_p, d'_q \rangle = \delta_{pq}$. Expanding by bilinearity,

$$\delta_{jk} = \langle d_j, d_k \rangle = \left\langle \sum_{p=1}^{n} Q_{p,j} d'_p,\ \sum_{q=1}^{n} Q_{q,k} d'_q \right\rangle = \sum_{p,q=1}^{n} Q_{p,j} Q_{q,k}\, \delta_{pq} = \sum_{p=1}^{n} Q_{p,j} Q_{p,k} = \left(Q^{\mathsf{T}} Q\right)_{j,k}.$$

Hence $Q^{\mathsf{T}}Q = I_n$, which is exactly the statement that $Q \in \mathrm{O}(n)$. (Equivalently, the displayed identity says that the columns of Q form an orthonormal system, which is Definition 22.b.8).

Q.E.D.

Proof of Exercise 27.7:

We are given the basis $\mathcal{C} = (e_1, \dots, e_n, e_{n+1})$. We define $\mathcal{C}' := (e_1, \dots, e_n, e_{n+1} - w')$, where $w' \in W = \mathrm{Sp}(e_1, \dots, e_n)$. We can write $w' = \sum_{i=1}^{n} \alpha_i e_i$. Suppose a linear combination of $\mathcal{C}'$ equals zero:

$$\sum_{i=1}^{n} c_i e_i + c_{n+1}(e_{n+1} - w') = 0 \implies \sum_{i=1}^{n} (c_i - c_{n+1}\alpha_i) e_i + c_{n+1} e_{n+1} = 0.$$

Because $\mathcal{C}$ is a basis, all coefficients must be zero. In particular, $c_{n+1} = 0$. Substituting this back into the other coefficients yields $c_i = 0$ for all $1 \le i \le n$. Thus, the vectors in $\mathcal{C}'$ are linearly independent and form a basis. **Q.E.D.**

Proof of Exercise 27.8:

Assume B is positive definite. Let $\mathcal{C} = (e_1, \dots, e_n)$ be any basis for V. For any $1 \le \ell \le n$, consider the subspace $W_\ell := \mathrm{Sp}(e_1, \dots, e_\ell)$. Since B is positive definite on V, its restriction $B|_{W_\ell \times W_\ell}$ is also positive definite. The matrix representation of this restriction in the basis $\mathcal{E}_\ell := (e_1, \dots, e_\ell)$ is exactly the $\ell \times \ell$ principal submatrix of $[B]_{\mathcal{C}}$, so it is the ℓ-th principal minor that we must show is positive.

Apply Theorem 27.a.5 to the symmetric bilinear form $B|_{W_\ell \times W_\ell}$ on W_ℓ. It supplies a basis $\mathcal{F}$ of W_ℓ in which the matrix of the restriction is $I_k \oplus (-I_m) \oplus 0$ with $k + m \le \ell$. Here k is the largest dimension of a subspace on which the form is positive definite; since the form is positive definite on all of W_ℓ, which has dimension ℓ, we get $k = \ell$, and therefore, $m = 0$ and there is no zero block. In other words,

$$[B|_{W_\ell \times W_\ell}]_{\mathcal{F}} = I_\ell, \qquad \text{so} \qquad \det [B|_{W_\ell \times W_\ell}]_{\mathcal{F}} = 1 > 0.$$

Now pass back to the basis $\mathcal{E}_\ell$. By Lemma 27.a.3, with $S := [\mathrm{id}_{W_\ell}]^{\mathcal{E}_\ell}_{\mathcal{F}}$,

$$[B|_{W_\ell \times W_\ell}]_{\mathcal{E}_\ell} = S^{\mathsf{T}} \cdot [B|_{W_\ell \times W_\ell}]_{\mathcal{F}} \cdot S,$$

and taking determinants gives $\det [B|_{W_\ell \times W_\ell}]_{\mathcal{E}_\ell} = \det(S)^2 \cdot 1 > 0$, because S is invertible. This is the ℓ-th principal minor of $[B]_{\mathcal{C}}$, and ℓ was arbitrary, which proves **(b)**.

Note that the basis $\mathcal{C}$ was arbitrary throughout, so what we have actually proved is the stronger implication **(a)** $\implies$ **(c)**. Since **(c)** $\implies$ **(b)** is immediate, the cycle **(a)** $\implies$ **(c)** $\implies$ **(b)**

$\Longrightarrow$ **(a)** closes, and all three conditions of Theorem 27.a.9 are equivalent. The notes state the three conditions but only ever discuss the passage between **(a)** and **(b)**, leaving **(c)** unproved; this is what closes that gap. **Q.E.D.**

AI-Note: Two inverted representation-matrix indices, the error class this pass exists to catch, both in transition matrices and both invisible to 'tools/check-repmatrix.pl' because neither stands inside a product.

In the proof of Theorem 27.a.5, the orthogonal P produced by the Spectral Theorem was introduced as $[\mathrm{id}_V]^{\mathcal{C}'}_{\mathcal{C}''} = P$, with the basis of the target on top. The parenthetical immediately after it says $v''_j = \sum_i P_{ij} v'_i$, that is, column j of P holds the coordinates of v''_j in $\mathcal{C}'$, which is $[\mathrm{id}_V]^{\mathcal{C}''}_{\mathcal{C}'}$; and $[B]_{\mathcal{C}''} = P^{\mathsf{T}} A P$ is what Lemma 27.a.3 yields for that reading and no other. The same swap occurs in the proof of Theorem 27.a.9, where $S = [\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{C}'}$ was written for the matrix that the identity $[B]_{\mathcal{C}'} = S^{\mathsf{T}}[B]_{\mathcal{C}} S$ forces to be $[\mathrm{id}_V]^{\mathcal{C}'}_{\mathcal{C}}$. Both are corrected; Lemma 27.a.3 itself, the proof outline of Theorem 27.a.6 and the solution to Exercise 27.8 were all checked and are right.

Two results asserted without proof. The Polarization Identity on Page 266 carried none, although it is what starts the induction of Theorem 27.a.7; it is two expansions and now has them. And Lemma 27.a.2 proves only that a matrix representation **exists**, then calls it *"the"* matrix representation; uniqueness is what entitles the later chapters to read $([B]_{\mathcal{C}})_{jk} = B(v_j, v_k)$ off the name alone, as the proof of Lemma 27.a.3 does in its first line, and it too is now proved.

One step whose point was the missing half. The Remark before the Conclusion shows that $\frac{1}{2}(B + B^{\mathsf{T}})$ gives the same quadratic form as B, without computing it and, more to the point, without saying that this matrix is **symmetric**, which is the whole content of the Conclusion that follows. Both are supplied.

Steps that now name their reason: Lemma 22.b.9 **(d)** for the three appearances of $P^{\mathsf{T}} = P^{-1}$; the Proposition on Page 113 for what a column of a transition matrix contains; Lemma 21.a.8 **(e)** with AI-Proposition 21.1 for *"similar, hence the same eigenvalues"* in Theorem 27.a.6, which is the exact point at which the congruence of Lemma 27.a.3 has to be upgraded to a similarity, and so the reason that theorem insists on orthonormal bases.

Flagged rather than changed: in the second item of the Example of standard forms the letter B denotes both the matrix and the bilinear form it defines, so that the defining line reads $B(v, w) := v^{\mathsf{T}} B w$. Nothing is wrong, and the notes do it too, but it is a trap for the eye of the same kind as the P of ch. 17c; a parenthetical now says which B is which.

The Jordan Normal Form (Chapter 28)

Introduction and Motivation (Section 28.a)

Let V be a finite-dimensional vector space over K, and let $T\colon V \to V$ be a linear map.

The easiest situation to understand is when T is diagonalizable. That is, there exists a basis $\mathcal{B}$ for V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of T.

Using the basis $\mathcal{B}$, it is also easy to calculate powers of T. For example,

$$[T^k]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{pmatrix} \quad \forall k \in \mathbb{Z}_{\geq 1}.$$

But **not all** $T \in \operatorname{End}(V)$ are diagonalizable! Why? There are two possible reasons:

(a) The characteristic polynomial $P_T(x)$ does not split as a product of linear factors. This is equivalent to saying that T is not (even) trigonalizable (hence, all the more so not diagonalizable!). In other words, $P_T(x)$ doesn't have enough zeroes in K (or even no zeroes at all).

For example, consider $T_A\colon \mathbb{R}^2 \to \mathbb{R}^2$, where $A = R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, with $\theta \neq \pi k$ for $k \in \mathbb{Z}$. The map T_A has **no** eigenvalues (in $\mathbb{R}$). It is a rotation, but not by $0°$ or $180°$.

Sometimes we can overcome this *"problem"* by enlarging (extending) K. For example, $K = \mathbb{R}$ can be extended to $K = \mathbb{C}$. Then the above matrix A has two eigenvalues when viewed as $A \in \mathcal{M}_{2\times 2}(\mathbb{C})$, namely $\lambda_1 = e^{i\theta}$ and $\lambda_2 = e^{-i\theta}$ (Exercise: check this, see Exercise 28.1).

(b) T is trigonalizable, or equivalently $P_T(x)$ splits as a product of linear factors, but for some eigenvalues λ of T, the geometric multiplicity is strictly less than the algebraic multiplicity:

$$m_g(\lambda) < m_a(\lambda).$$

Exercise 28.1 (*The Complex Eigenvalues of a Rotation*): This is the step Prof. Biran marks with *"exc. check this"*. Let $\theta \in \mathbb{R}$ and let $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, regarded as an element of $\mathcal{M}_{2\times 2}(\mathbb{C})$. Show that its eigenvalues are $e^{i\theta}$ and $e^{-i\theta}$.

Let's try to understand case **(b)** better.

Definition 28.a.1 (*Jordan Matrix*): Let $\lambda \in K$ and $n \in \mathbb{Z}_{\geq 1}$. Define:

$$J_{\lambda,n} = \begin{pmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix} \in \mathcal{M}_{n\times n}(K).$$

This matrix has λ on the main diagonal, 1's on the diagonal immediately above the main diagonal, and 0's everywhere else.

For $n = 1$, $J_{\lambda,1} = (\lambda)$. For $n = 2$, $J_{\lambda,2} = \left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$. For $n = 3$, $J_{\lambda,3} = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$, etc.

The matrix $J_{\lambda,n}$ is called the *"n-dimensional Jordan matrix with eigenvalue λ"*.

Lemma 28.a.2 (*Properties of the Jordan Matrix*):

(a) λ is the only eigenvalue of $J_{\lambda,n}$.

(b) $P_{J_{\lambda,n}}(x) = (\lambda - x)^n$, and $m_a(J_{\lambda,n};\lambda) = n$.

(c) $m_g(J_{\lambda,n};\lambda) = 1$. In fact, $\mathrm{Eig}_{J_{\lambda,n}}(\lambda) = \mathrm{Sp}(e_1)$, where $e_1 = (1,0,\ldots,0)^{\mathsf{T}}$.

Proof:

(a) The characteristic polynomial is the determinant of an upper triangular matrix:

$$P_{J_{\lambda,n}}(x) = \det \begin{pmatrix} \lambda - x & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda - x \end{pmatrix} = (\lambda - x)^n,$$

which is Lemma 21.a.8 **(c)**, the case of an upper triangular matrix with all diagonal entries equal to λ. The only root of $P_{J_{\lambda,n}}(x)$ is λ, so λ is the only eigenvalue of $J_{\lambda,n}$, by AI-Proposition 21.1.

(b) From part **(a)**, the root λ has multiplicity n, so $m_a(J_{\lambda,n};\lambda) = n$ (by Definition 21.b.3, the algebraic multiplicity being the multiplicity of λ as a root of the characteristic polynomial).

(c) To find $\mathrm{Eig}_{J_{\lambda,n}}(\lambda)$, we need to solve the equation $(J_{\lambda,n} - \lambda I_n)x = 0$:

$$(J_{\lambda,n} - \lambda I_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Subtracting λI_n clears the main diagonal and leaves only the superdiagonal of 1's, so this reads

$$\begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

This immediately implies the system of equations:

$$\begin{cases} x_2 = 0 \\ x_3 = 0 \\ \vdots \\ x_n = 0 \\ 0 = 0 \end{cases}$$

The only free variable is x_1. Therefore, $\mathrm{Eig}_{J_{\lambda,n}}(\lambda) = \mathrm{Sp}\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \mathrm{Sp}(e_1)$, which means the geometric multiplicity is 1.

Q.E.D.

We've seen that the statement *"$P_T(x)$ splits as a product of linear factors"* does **not** imply that *"T is diagonalizable"*, and $J_{\lambda,n}$ is a clear counterexample.

Question 28.1: Is $J_{\lambda,n}$ the *"only reason"* for this non-implication?

Theorem 28.a.3 (*Jordan Normal Form*): Let V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$ such that $P_T(x)$ splits as a product of linear factors in $K[x]$ (e.g., if $K = \mathbb{C}$, this always happens). Then there exists a basis $\mathcal{B}$ for V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1,n_1} & & & \\ & J_{\lambda_2,n_2} & & \\ & & \ddots & \\ & & & J_{\lambda_k,n_k} \end{pmatrix},$$

where $k \geq 1$, $\lambda_1, \ldots, \lambda_k \in K$, $n_1, \ldots, n_k \geq 1$, and $n_1 + \cdots + n_k = n$.

Moreover, up to permutation, the blocks $J_{\lambda_1,n_1}, \ldots, J_{\lambda_k,n_k}$ are unique.

Exercise 28.2 (*What the Uniqueness in Theorem 28.a.3 Says*): Prof. Biran annotates the uniqueness claim with *"explain"*. Make the statement precise: say exactly which data attached to T is being claimed to be independent of the choice of Jordan basis, and why *"up to permutation"* is the right qualification. Then verify two concrete instances of that independence directly from Corollary 28.a.5: for each eigenvalue λ, both the **number** of blocks carrying λ and the **size of the largest** such block are determined by T alone.

Remark 28.1: The basis $\mathcal{B}$ is called a "*Jordan basis*" for T. The resulting block diagonal matrix is called the "*Jordan Normal Form*" (JNF) of T. A similar theorem holds for matrices $A \in \mathcal{M}_{n\times n}(K)$: there exists an invertible matrix P such that $P^{-1}AP$ is in Jordan Normal Form.

Example 28.1 (*Jordan Normal Form Block Structure*): Consider a matrix $A \in \mathcal{M}_{13\times 13}(K)$ in Jordan Normal Form:

$$A = \begin{pmatrix}
\lambda_1 & 1 & 0 & & & & & & & & & & \\
0 & \lambda_1 & 1 & & & & & & & & & & \\
0 & 0 & \lambda_1 & & & & & & & & & & \\
& & & \lambda_1 & 1 & 0 & & & & & & & \\
& & & 0 & \lambda_1 & 1 & & & & & & & \\
& & & 0 & 0 & \lambda_1 & & & & & & & \\
& & & & & & \lambda_2 & 1 & & & & & \\
& & & & & & 0 & \lambda_2 & & & & & \\
& & & & & & & & \lambda_2 & & & & \\
& & & & & & & & & \lambda_2 & & & \\
& & & & & & & & & & \lambda_3 & 1 & 0 \\
& & & & & & & & & & 0 & \lambda_3 & 1 \\
& & & & & & & & & & 0 & 0 & \lambda_3
\end{pmatrix}$$

Here, $n = 13$. Let's assume the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ are distinct.

- **Characteristic polynomial:** $P_A(x) = (\lambda_1 - x)^6 \cdot (\lambda_2 - x)^4 \cdot (\lambda_3 - x)^3$.
- **Algebraic multiplicity:** $m_a(\lambda_1) = 6$, $m_a(\lambda_2) = 4$, $m_a(\lambda_3) = 3$.
- **Geometric multiplicity:** $m_g(\lambda)$ is exactly the number of blocks with eigenvalue λ. So, $m_g(\lambda_1) = 2$, $m_g(\lambda_2) = 3$, and $m_g(\lambda_3) = 1$.
- **Minimal polynomial:** $\mu_A(x) = (x-\lambda_1)^{\ell_1} \cdot (x-\lambda_2)^{\ell_2} \cdot (x-\lambda_3)^{\ell_3}$, where $1 \leq \ell_i \leq m_a(\lambda_i)$ (why? See Exercise 28.3). We will see that ℓ_i is precisely the size of the **biggest** block with eigenvalue λ_i. Thus,
$$\mu_A(x) = (x-\lambda_1)^3 \cdot (x-\lambda_2)^2 \cdot (x-\lambda_3)^3.$$

Remark 28.2: Prof. Biran writes *"Remind"* beside the minimal polynomial in this example, so we recall the definition. By Definition 21.d.1, the minimal polynomial of T is the monic polynomial of least degree that annihilates T, that is, with $\mu_T(T) = 0$. Note that chapter 21 writes it $m_T(x)$, whereas the notes for the present chapter write $\mu_T(x)$; the two symbols denote the same polynomial, and we keep μ here to stay with Prof. Biran's notation for the Jordan chapters.

Exercise 28.3 (*The Bounds on the Exponents of the Minimal Polynomial*): These are the two steps Prof. Biran marks with *"why?"* in the example above. Let $T \in \mathrm{End}(V)$ with $P_T(x)$ splitting into linear factors, and write $\mu_T(x) = \prod_i (x - \lambda_i)^{\ell_i}$ over the distinct eigenvalues λ_i of T. Show that $1 \leq \ell_i \leq m_a(\lambda_i)$ for every i. (Hint for the lower bound: apply $\mu_T(T)$ to an eigenvector. Hint for the upper bound: Theorem 21.d.2).

Powers of Jordan Matrices and Minimal Polynomials (Section 28.b)

Let's go back to the Jordan matrices $J_{\lambda,n}$. Recall that $J_{\lambda,n} \in \mathcal{M}_{n\times n}(K)$ for $n \in \mathbb{Z}_{\geq 1}$ and $\lambda \in K$. We can write $J_{\lambda,n} = \lambda \cdot I_n + N$, where

$$N = \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix} = J_{0,n}.$$

Lemma 28.a.4 (*Powers of the Nilpotent Matrix N and Jordan Blocks*):

(a) The powers of N shift the diagonal of 1's further up:

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & & 0 \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ 0 & & & & 0 \end{pmatrix}, \quad N^3 = \begin{pmatrix} 0 & 0 & 0 & 1 & \dots \\ & \ddots & \ddots & \ddots & \ddots \\ & & \ddots & \ddots & \ddots \end{pmatrix}, \quad \dots,$$

$$N^{n-1} = \begin{pmatrix} 0 & \dots & 0 & 1 \\ \vdots & & & 0 \\ \vdots & & & \vdots \\ 0 & \dots & \dots & 0 \end{pmatrix}.$$

Furthermore, $N^n = 0$, and $N^k = 0$ for all $k \geq n$. In other words, N^k has 1's in the k-th *"diagonal"* above the *"central diagonal"* and 0's everywhere else.

(b) For all $k \geq 1$, we have:

$$J_{\lambda,n}^k = \begin{pmatrix} \lambda^k & \binom{k}{1}\lambda^{k-1} & \binom{k}{2}\lambda^{k-2} & \dots \\ & \lambda^k & \binom{k}{1}\lambda^{k-1} & \dots \\ & & \ddots & \dots \\ & & & \binom{k}{1}\lambda^{k-1} \\ 0 & & & \lambda^k \end{pmatrix}.$$

Written differently, we have $J_{\lambda,n}^k = \sum_{j=0}^k \binom{k}{j}\lambda^{k-j}N^j$ for all $k \geq 1$.

Proof:

(a) This can be shown by direct calculation (exercise, see Exercise 28.4). Alternatively, denote by $e_1, \dots, e_n$ the standard basis for K^n. Then the linear map $T_N \colon K^n \to K^n$ acts as a shift operator:

$$T_N e_i = e_{i-1} \quad \forall 2 \le i \le n, \quad \text{and} \quad T_N e_1 = 0.$$

In other words:

$$e_n \xrightarrow{T_N} e_{n-1} \xrightarrow{T_N} \dots \xrightarrow{T_N} e_1 \xrightarrow{T_N} 0.$$

The map $T_{N^2} (= T_N \circ T_N = T_N^2)$ applies the shift twice:

$$e_n \xrightarrow{T_N^2} e_{n-2} \xrightarrow{T_N^2} e_{n-4} \cdots \to 0$$

$$e_{n-1} \xrightarrow{T_N^2} e_{n-3} \xrightarrow{T_N^2} \cdots \to 0$$

By induction on k, it follows that for all $k \ge 1$:

$$T_{N^k}(e_j) = \begin{cases} e_{j-k} & \text{if } k+1 \le j \le n \\ 0 & \text{if } j \le k \end{cases}.$$

Since $N^k = [T_{N^k}]_{\mathcal{E}_n}^{\mathcal{E}_n}$, the matrix form follows immediately.

(b) Since the scalar matrix λI_n and the matrix N commute, we can use the standard binomial theorem:

$$J_{\lambda,n}^k = (\lambda I_n + N)^k = \sum_{j=0}^{k} \binom{k}{j} (\lambda I_n)^{k-j} N^j = \sum_{j=0}^{k} \binom{k}{j} \lambda^{k-j} N^j.$$

Using the structure of N^j from part **(a)**, this yields the upper triangular matrix with binomial coefficients along the diagonals.

Q.E.D.

Exercise 28.4 (*The Powers of N by Direct Calculation*): This is the route Prof. Biran marks with *"Direct calculation (exc.)"*, offered as an alternative to the shift-operator argument given in the proof. With $N = J_{0,n}$, so that $N_{i,j} = 1$ when $j = i+1$ and $N_{i,j} = 0$ otherwise, show by induction on k that

$$\left(N^k\right)_{i,j} = \begin{cases} 1 & \text{if } j = i+k, \\ 0 & \text{otherwise,} \end{cases}$$

and deduce that $N^k = 0$ for all $k \ge n$.

Corollary 28.a.5 (*Multiplicities and the Minimal Polynomial from JNF*): Suppose $T \in \operatorname{End}(V)$ admits a matrix representation in Jordan Normal Form in some basis $\mathcal{B}$:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1,n_1} & & \\ & \ddots & \\ & & J_{\lambda_k,n_k} \end{pmatrix}. \tag{28.1}$$

Then for every eigenvalue λ of T, the geometric multiplicity $m_g(\lambda)$ equals the number of blocks in (28.1) labeled with λ (i.e., blocks of the type $J_{\lambda,\dots}$). Note that blocks of size 1×1 also count. In other words,

$$m_g(\lambda) = \#\{j \mid 1 \le j \le k, \lambda_j = \lambda\}.$$

Furthermore, the minimal polynomial $\mu_T(x)$ is given by:

$$\mu_T(x) = \prod_{\lambda \in \operatorname{Spec}(T)} (x - \lambda)^{s(\lambda)},$$

where $\operatorname{Spec}(T) := \{\lambda_1, \dots, \lambda_k\}$ is the "*spectrum*" of T, that is, its set of eigenvalues, each counted once, and where $s(\lambda)$ is the size of the largest block labeled by λ, i.e., $s(\lambda) = \max\{n_j \mid 1 \le j \le k, \lambda_j = \lambda\}$.

Proof:

Geometric Multiplicity: Suppose that $A \in \mathcal{M}_{n\times n}(K)$ has the block diagonal shape $A = \left(\begin{smallmatrix} B & 0 \\ 0 & C \end{smallmatrix}\right)$ with $B \in \mathcal{M}_{\ell\times\ell}(K)$ and $C \in \mathcal{M}_{m\times m}(K)$. Let $0 \neq u = \binom{v}{w}$ with $v \in K^\ell$ and $w \in K^m$. Then u is an eigenvector of A for the eigenvalue λ if and only if $Bv = \lambda v$ and $Cw = \lambda w$.

Moreover, the eigenspace decomposes as $\operatorname{Eig}_A(\lambda) = \operatorname{Eig}_B(\lambda) \oplus \operatorname{Eig}_C(\lambda)$, which is the criterion just proved, read as a statement about sets rather than about a single u. In particular, $m_g(A,\lambda) = m_g(B,\lambda) + m_g(C,\lambda)$. (We use the convention here that if λ is not an eigenvalue of a matrix D, then $m_g(D,\lambda) = 0$).

Now recall from Lemma 28.a.2 that for a single Jordan block, $m_g(J_{\lambda,r};\lambda') = \begin{cases} 1 & \text{if } \lambda' = \lambda \\ 0 & \text{if } \lambda' \neq \lambda \end{cases}$. Applying this additively to $A := [T]^{\mathcal{B}}_{\mathcal{B}}$, by induction on the number of blocks, we obtain:

$$m_g(T,\lambda) = m_g(A,\lambda) = \sum_{i=1}^{k} m_g(J_{\lambda_i,n_i};\lambda) = \#\{i \mid \lambda_i = \lambda\}.$$

(The first equality is the passage from T to its representation matrix. It holds because the coordinate map $\Phi_{\mathcal{B}}$ carries $\operatorname{Eig}_T(\lambda)$ onto $\operatorname{Eig}_A(\lambda)$: for $v \in V$, $Tv = \lambda v$ if and only if $A \cdot [v]_{\mathcal{B}} = \lambda [v]_{\mathcal{B}}$, by Proposition 16.4, and $\Phi_{\mathcal{B}}$ is an isomorphism by Proposition 16.2, so the two eigenspaces have the same dimension).

Minimal Polynomial: Throughout this half we work with $A := [T]^{\mathcal{B}}_{\mathcal{B}}$ in place of T, which is legitimate because T and A have the same minimal polynomial: $q(A) = [q(T)]^{\mathcal{B}}_{\mathcal{B}}$ for every $q(x) \in K[x]$, since $S \mapsto [S]^{\mathcal{B}}_{\mathcal{B}}$ respects sums, scalar multiples and compositions, and it is injective by Theorem 18.a.2; hence $q(T) = 0$ if and only if $q(A) = 0$.

Let $A := \left(\begin{smallmatrix} B & 0 \\ 0 & C \end{smallmatrix}\right)$ as above. We have $A^k = \begin{pmatrix} B^k & 0 \\ 0 & C^k \end{pmatrix}$ (check! See Exercise 28.5). This implies that for every polynomial $q(x) \in K[x]$, we have $q(A) = \begin{pmatrix} q(B) & 0 \\ 0 & q(C) \end{pmatrix}$.

Consider a single Jordan matrix $J_{\lambda,n}$. For every $q(x) \in K[x]$, we have:

$$q(J_{\lambda,n}) = \begin{pmatrix} q(\lambda) & & * \\ & \ddots & \\ 0 & & q(\lambda) \end{pmatrix} \quad \text{(exercise, see Exercise 28.6).}$$

Note the following key properties:

(a) If $q(\lambda) \neq 0$, then $q(J_{\lambda,n})$ is an upper triangular matrix with non-zero diagonal entries, hence it is invertible.

(b) If $q(x) = (x-\lambda)^s$, then $q(J_{\lambda,n}) = (J_{\lambda,n} - \lambda I_n)^s = N^s$. So, for $q(x) = (x-\lambda)^s$, $q(J_{\lambda,n}) = 0$ if and only if $s \geq n$.

(c) Now, take a general polynomial $q(x) \in K[x]$ of positive degree (i.e., $q(x) \neq$ const.) and let λ be a zero of $q(x)$. We can write $q(x) = (x-\lambda)^s \cdot p(x)$, where $s \geq 1$ and $p(\lambda) \neq 0$. This implies $q(J_{\lambda,n}) = N^s \cdot p(J_{\lambda,n})$. Since $p(\lambda) \neq 0$, $p(J_{\lambda,n})$ is invertible by **(a)**. Therefore, $q(J_{\lambda,n}) = 0$ if and only if $N^s = 0$, which occurs if and only if $s \geq n$.

Let A be a matrix in Jordan Normal Form, and let $q(x) \in K[x]$ be a polynomial of positive degree such that $q(A) = 0$. We must have $q(\lambda_j) = 0$ for all $1 \leq j \leq k$, because if $q(\lambda_j) \neq 0$, then $q(J_{\lambda_j,n_j})$ would be invertible (and thus non-zero) by **(a)**, which would imply that $q(A) \neq 0$. This proves that $\lambda_1, \ldots, \lambda_k$ are all zeroes of $q(x)$.

Let $\lambda \in \{\lambda_1, \ldots, \lambda_k\}$. Write $q(x) = (x-\lambda)^s \cdot p(x)$ where $p(\lambda) \neq 0$. By **(c)**, we must have $s \geq n_j$ for all j with $\lambda_j = \lambda$. That is, $s \geq \max\{n_j \mid \lambda_j = \lambda\} = s(\lambda)$. This implies that any annihilating polynomial $q(x)$ must be of the form:

$$q(x) = \prod_{\lambda \in \{\lambda_1,\ldots,\lambda_k\}} (x-\lambda)^{s(\lambda)} \cdot r(x) \quad \text{for some } r(x) \in K[x]. \tag{28.2}$$

On the other hand, we claim that the specific polynomial $\tilde{q}(x) := \prod_{\lambda\in\{\lambda_1,\ldots,\lambda_k\}}(x-\lambda)^{s(\lambda)}$ satisfies $\tilde{q}(A) = 0$. If we prove this, then together with (28.2), it establishes that $\tilde{q}(x)$ is the monic polynomial of least degree annihilating A, meaning $\mu_T(x) = \tilde{q}(x)$.

Indeed, let $A := \text{blockdiag}(J_{\lambda_1,n_1},\ldots,J_{\lambda_k,n_k})$. We have:

$$\tilde{q}(A) = \text{blockdiag}(\tilde{q}(J_{\lambda_1,n_1}),\ldots,\tilde{q}(J_{\lambda_k,n_k})). \tag{28.3}$$

Let $1 \le j \le k$, and consider the block $\tilde{q}(J_{\lambda_j,n_j})$. We can write $\tilde{q}(x) = (x-\lambda_j)^{s(\lambda_j)} \cdot p_j(x)$ where $p_j(x) \in K[x]$. Explicitly, $p_j(x) = \prod_{\lambda\neq\lambda_j}(x-\lambda)^{s(\lambda)}$, the product running over the **distinct** eigenvalues other than λ_j, and therefore, $p_j(\lambda_j) \neq 0$. This is what property **(c)** requires. By the definition of $s(\lambda_j)$, we have $s(\lambda_j) \ge n_j$. By property **(c)** above, $\tilde{q}(J_{\lambda_j,n_j}) = 0$. This holds for all $1 \le j \le k$, hence the entire block diagonal matrix in (28.3) is zero. This proves the last claim, and hence concludes the proof of the corollary. **Q.E.D.**

Exercise 28.5 (*Powers of a Block Diagonal Matrix*): This is the step Prof. Biran marks with *"check!"* in the proof above. Let $A = \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix}$ with $B \in \mathcal{M}_{\ell\times\ell}(K)$ and $C \in \mathcal{M}_{m\times m}(K)$. Show that $A^k = \begin{pmatrix} B^k & 0 \\ 0 & C^k \end{pmatrix}$ for every $k \ge 1$, and deduce that $q(A) = \begin{pmatrix} q(B) & 0 \\ 0 & q(C) \end{pmatrix}$ for every polynomial $q(x) \in K[x]$.

Exercise 28.6 (*A Polynomial Evaluated on a Jordan Block*): This is the step Prof. Biran marks with *"exc."* in the proof above. Let $q(x) \in K[x]$ and let $J_{\lambda,n}$ be a Jordan matrix. Show that $q(J_{\lambda,n})$ is upper triangular with every diagonal entry equal to $q(\lambda)$. (Hint: write $J_{\lambda,n} = \lambda I_n + N$ and expand using Lemma 28.a.4; the terms involving N^j with $j \ge 1$ are strictly upper triangular).

Solutions to the Exercises

Proof of Exercise 28.1:

We compute the characteristic polynomial directly:

$$P_{R_\theta}(x) = \det\begin{pmatrix} \cos\theta - x & -\sin\theta \\ \sin\theta & \cos\theta - x \end{pmatrix} = (\cos\theta - x)^2 + \sin^2\theta = x^2 - 2\cos\theta\cdot x + 1,$$

where the last step uses $\cos^2\theta + \sin^2\theta = 1$. Over $\mathbb{C}$ the roots are

$$x = \cos\theta \pm \sqrt{\cos^2\theta - 1} = \cos\theta \pm \sqrt{-\sin^2\theta} = \cos\theta \pm i\sin\theta,$$

which by Euler's formula are exactly $e^{i\theta}$ and $e^{-i\theta}$.

This also makes the point of the surrounding discussion visible: these two roots are real if and only if $\sin\theta = 0$, that is, if and only if $\theta = \pi k$ for some $k \in \mathbb{Z}$, which is precisely the case excluded above. **Q.E.D.**

Proof of Exercise 28.2:

The data claimed to be independent of the choice of Jordan basis is the **multiset** of pairs

$$\{(\lambda_1, n_1), (\lambda_2, n_2), \ldots, (\lambda_k, n_k)\},$$

that is, the unordered list, counted with repetitions, of the eigenvalues labelling the blocks together with the sizes of those blocks. Equivalently: for every $\lambda \in K$ and every $r \ge 1$, the number of indices j with $\lambda_j = \lambda$ and $n_j = r$ depends only on T.

The qualification *"up to permutation"* is needed because no ordering of the blocks can possibly be canonical. If $\mathcal{B}$ is a Jordan basis, then reordering the basis vectors block by block is again a basis, and it produces the same blocks arranged along the diagonal in a different order. So the ordered list of blocks is not an invariant of T, while the multiset is.

Two instances of this independence follow immediately from Corollary 28.a.5.

(a) The number of blocks carrying λ equals $m_g(\lambda) = \dim \mathrm{Eig}_T(\lambda)$. The eigenspace $\mathrm{Eig}_T(\lambda) = \ker(T - \lambda \, \mathrm{id}_V)$ is defined without reference to any basis, so its dimension is an invariant of T, and hence so is the number of blocks.

(b) The size of the largest block carrying λ is the exponent $s(\lambda)$ of $(x-\lambda)$ in the minimal polynomial $\mu_T(x)$. The minimal polynomial is defined by Definition 21.d.1 purely in terms of T, with no basis involved, so $s(\lambda)$ is an invariant of T too.

These two invariants do not by themselves pin down the whole multiset when λ labels three or more blocks, which is why the theorem asserts more than **(a)** and **(b)** together give. **Q.E.D.**

Proof of Exercise 28.3:

The lower bound $\ell_i \geq 1$. Let λ_i be an eigenvalue of T and pick an eigenvector $v \neq 0$, so $Tv = \lambda_i v$. Then $T^m v = \lambda_i^m v$ for every $m \geq 0$, and consequently $g(T)v = g(\lambda_i)v$ for every polynomial $g(x) \in K[x]$. Applying this to $g = \mu_T$ gives

$$0 = \mu_T(T)v = \mu_T(\lambda_i) \cdot v.$$

Since $v \neq 0$, this forces $\mu_T(\lambda_i) = 0$, so $(x - \lambda_i)$ divides $\mu_T(x)$, that is, $\ell_i \geq 1$.

The upper bound $\ell_i \leq m_a(\lambda_i)$. First note that μ_T divides every annihilating polynomial. Indeed, if $g(T) = 0$, divide with remainder, $g = q \cdot \mu_T + r$ with $r = 0$ or $\deg r < \deg \mu_T$. Evaluating at T gives $r(T) = g(T) - q(T)\mu_T(T) = 0$, and if $r \neq 0$ this would contradict the minimality of $\deg \mu_T$. Hence $r = 0$ and $\mu_T \mid g$.

By Theorem 21.d.2 we have $P_T(T) = 0$, so $\mu_T(x)$ divides $P_T(x)$. Since $P_T(x)$ splits, the multiplicity of the factor $(x - \lambda_i)$ in $P_T(x)$ is by definition $m_a(\lambda_i)$, and a divisor cannot contain that factor more often than P_T does. Therefore, $\ell_i \leq m_a(\lambda_i)$. **Q.E.D.**

Proof of Exercise 28.4:

By definition $N_{i,j} = 1$ if $j = i+1$ and $N_{i,j} = 0$ otherwise, which is the case $k=1$ of the claim. Assume the claim for some $k \geq 1$. Then

$$\left(N^{k+1}\right)_{i,j} = \sum_{p=1}^{n} \left(N^k\right)_{i,p} \cdot N_{p,j}.$$

By the induction hypothesis the factor $\left(N^k\right)_{i,p}$ vanishes unless $p = i+k$, and $N_{p,j}$ vanishes unless $j = p+1$. So the whole sum is 1 exactly when $j = (i+k)+1 = i+(k+1)$, and 0 otherwise, which is the claim for $k+1$.

For $k \geq n$ there is no pair of indices $1 \leq i, j \leq n$ with $j = i+k$ at all, since that would force $j \geq 1+n > n$. Hence every entry of N^k vanishes, that is, $N^k = 0$. **Q.E.D.**

Proof of Exercise 28.5:

We argue by induction on k. For $k=1$ there is nothing to prove. Assuming the claim for k, block multiplication gives

$$A^{k+1} = A^k \cdot A = \begin{pmatrix} B^k & 0 \\ 0 & C^k \end{pmatrix} \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix} = \begin{pmatrix} B^k B & 0 \\ 0 & C^k C \end{pmatrix} = \begin{pmatrix} B^{k+1} & 0 \\ 0 & C^{k+1} \end{pmatrix}.$$

For the second statement, write $q(x) = \sum_{m=0}^{d} a_m x^m$. The identity also holds for $m = 0$, because $A^0 = I_n = \left(\begin{smallmatrix} I_\ell & 0 \\ 0 & I_m \end{smallmatrix}\right)$. Hence

$$q(A) = \sum_{m=0}^{d} a_m A^m = \sum_{m=0}^{d} a_m \begin{pmatrix} B^m & 0 \\ 0 & C^m \end{pmatrix} = \begin{pmatrix} \sum_m a_m B^m & 0 \\ 0 & \sum_m a_m C^m \end{pmatrix} = \begin{pmatrix} q(B) & 0 \\ 0 & q(C) \end{pmatrix}.$$

Q.E.D.

Proof of Exercise 28.6:

Write $q(x) = \sum_{m=0}^{d} a_m x^m$ and $J_{\lambda,n} = \lambda I_n + N$ as in the text. By Lemma 28.a.4 **(b)**,

$$J_{\lambda,n}^m = \sum_{j=0}^{m} \binom{m}{j} \lambda^{m-j} N^j \qquad \text{for all } m \geq 1,$$

and this also holds for $m = 0$, both sides being I_n. Substituting and collecting the terms according to the power of N,

$$q(J_{\lambda,n}) = \sum_{m=0}^{d} a_m \sum_{j=0}^{m} \binom{m}{j} \lambda^{m-j} N^j = \underbrace{\left(\sum_{m=0}^{d} a_m \lambda^m \right)}_{= q(\lambda)} N^0 + \sum_{j \geq 1} c_j N^j,$$

where $c_j := \sum_{m \geq j} a_m \binom{m}{j} \lambda^{m-j}$ and $N^0 = I_n$.

By Lemma 28.a.4 **(a)**, for every $j \geq 1$ the matrix N^j has its only non-zero entries in positions $(i, i+j)$, so it is strictly upper triangular. A sum of strictly upper triangular matrices is again strictly upper triangular, and therefore,

$$q(J_{\lambda,n}) = q(\lambda) \cdot I_n + (\text{strictly upper triangular}).$$

This is exactly the assertion: $q(J_{\lambda,n})$ is upper triangular, and each of its diagonal entries equals $q(\lambda)$. **Q.E.D.**

AI-Note: Nothing false, and the source-verification pass had already written up all six of Prof. Biran's *"exc."* marks, including the two *"why?"*s on the exponents of the minimal polynomial and the *"explain"* on the uniqueness claim.

A notation used once and defined nowhere. Corollary 28.a.5 writes its product over $\lambda \in \operatorname{Spec}(T)$, the only occurrence of Spec in the book: the operator is declared in the preamble, the word *"spectrum"* is explained only in the AI-Note opening ch. 24, and the corollary's own proof works throughout with $\{\lambda_1, \ldots, \lambda_k\}$ instead. The statement now says which set that is.

Two passages between T and its matrix, both silent. The proof computes $m_g(T, \lambda) = m_g(A, \lambda)$ for $A := [T]_{\mathcal{B}}^{\mathcal{B}}$ without argument; the reason is that $\Phi_{\mathcal{B}}$ carries $\operatorname{Eig}_T(\lambda)$ onto $\operatorname{Eig}_A(\lambda)$, which is Proposition 16.4 together with Proposition 16.2. The whole second half then proves a statement about the minimal polynomial of the **matrix** A and concludes one about μ_T; these agree because $q(A) = [q(T)]_{\mathcal{B}}^{\mathcal{B}}$ for every polynomial q, the representation map being a ring homomorphism and injective by Theorem 18.a.2. This is the more serious of the two, since without it the corollary computes the minimal polynomial of the wrong object.

Smaller ones: the determinant of the triangular $J_{\lambda,n} - xI_n$ is Lemma 21.a.8 **(c)**; passing from *"the only root"* to *"the only eigenvalue"* is AI-Proposition 21.1; the algebraic multiplicity is read off Definition 21.b.3; and the two-block additivity of m_g reaches k blocks by an induction that is now named.

The Remark introducing the terms "*Jordan basis*" and "*Jordan Normal Form*" keeps its Remark and simply carries a label now, so later chapters can cite it where it stands.

Checked and found correct: the shift description $T_N e_i = e_{i-1}$ agrees with $N = J_{0,n}$; the binomial expansion of $(\lambda I_n + N)^k$ is legitimate because the two commute; and every one of the six exercise solutions was re-verified, including the two roots $\cos\theta \pm i \sin\theta$ of $x^2 - 2\cos\theta \cdot x + 1$.

Proof of the Existence of the Jordan Normal Form (Section 28.c)

Preparations

Lemma 28.b.1 (*The Generalized Eigenspace*): Let V be a finite-dimensional vector space over K, and put $n := \dim V$. Let $T \in \operatorname{End}(V)$ and let $\lambda \in K$ be an eigenvalue of T. Consider the "*generalized eigenspace*"

$$\widetilde{\operatorname{Eig}}_T(\lambda) := \bigcup_{j=1}^{\infty} \ker\left(T - \lambda \operatorname{id}_V\right)^j .$$

Then:

(a) $\widetilde{\operatorname{Eig}}_T(\lambda)$ is a subspace of V. In fact, $\widetilde{\operatorname{Eig}}_T(\lambda) = \ker\left(T - \lambda \operatorname{id}_V\right)^n$.

(b) If $v \in \widetilde{\operatorname{Eig}}_T(\lambda)$ is an eigenvector of T corresponding to some eigenvalue μ of T, then $\mu = \lambda$. In other words, the only eigenvectors of T that belong to $\widetilde{\operatorname{Eig}}_T(\lambda)$ are the eigenvectors of λ.

Proof:

(a) We have the chain of nested subspaces

$$\{0\} \subsetneq \ker(T - \lambda \operatorname{id}_V) \subseteq \ker(T - \lambda \operatorname{id}_V)^2 \subseteq \cdots \subseteq \ker(T - \lambda \operatorname{id}_V)^n \subseteq \ker(T - \lambda \operatorname{id}_V)^{n+1} \subseteq V,$$

where the first inclusion is strict because λ is an eigenvalue of T, so that $\ker(T - \lambda \operatorname{id}_V)$ is at least 1-dimensional, and where V is n-dimensional.

We therefore have a sequence of $n+1$ nested subspaces of V, inside a space of dimension n, whose first member is already at least 1-dimensional. The dimensions cannot strictly increase at every step, and consequently, there exists $1 \le k \le n$ such that

$$\ker(T - \lambda \operatorname{id}_V)^k = \ker(T - \lambda \operatorname{id}_V)^{k+1}. \tag{28.4}$$

We claim that

$$\ker(T - \lambda \operatorname{id}_V)^k = \ker(T - \lambda \operatorname{id}_V)^{k+1} = \ker(T - \lambda \operatorname{id}_V)^{k+2} = \cdots = \ker(T - \lambda \operatorname{id}_V)^{k+\ell}$$

for all $\ell \ge 1$. In particular, $\ker(T - \lambda \operatorname{id}_V)^k = \ker(T - \lambda \operatorname{id}_V)^n$.

Proof of the claim. Let us first show that $\ker(T - \lambda \operatorname{id}_V)^{k+2} = \ker(T - \lambda \operatorname{id}_V)^k$. It is enough to show the inclusion $\ker(T - \lambda \operatorname{id}_V)^{k+2} \subseteq \ker(T - \lambda \operatorname{id}_V)^k$, because the reverse inclusion is obvious.

Let $v \in \ker(T - \lambda \operatorname{id}_V)^{k+2}$, so that $(T - \lambda \operatorname{id}_V)^{k+2} v = 0$. Put $w := (T - \lambda \operatorname{id}_V) v$. Then $(T - \lambda \operatorname{id}_V)^{k+1} w = 0$, which means that

$$w \in \ker(T - \lambda \operatorname{id}_V)^{k+1} = \ker(T - \lambda \operatorname{id}_V)^k,$$

where the equality is (28.4). Consequently, $(T - \lambda \operatorname{id}_V)^k w = 0$, that is, $(T - \lambda \operatorname{id}_V)^{k+1} v = 0$, and therefore, $v \in \ker(T - \lambda \operatorname{id}_V)^{k+1}$. This proves that

$$\ker(T - \lambda \operatorname{id}_V)^{k+2} \subseteq \ker(T - \lambda \operatorname{id}_V)^{k+1} = \ker(T - \lambda \operatorname{id}_V)^k,$$

and hence $\ker(T - \lambda \operatorname{id}_V)^{k+2} = \ker(T - \lambda \operatorname{id}_V)^k$.

Applying the same argument again and again, we obtain $\ker(T - \lambda \operatorname{id}_V)^k = \ker(T - \lambda \operatorname{id}_V)^{k+\ell}$ for all $\ell \ge 1$, which proves the claim.

It follows that

$$\widetilde{\operatorname{Eig}}_T(\lambda) = \bigcup_{j=1}^{\infty} \ker(T - \lambda \operatorname{id}_V)^j = \ker(T - \lambda \operatorname{id}_V)^k = \ker(T - \lambda \operatorname{id}_V)^n.$$

Being a kernel of a linear map, this is a subspace of V.

(b) Let v be an eigenvector of T with eigenvalue μ, and assume that $v \in \widetilde{\operatorname{Eig}}_T(\lambda)$. We have $Tv = \mu v$, hence $T^j v = \mu^j v$ for all $j \ge 0$, and consequently, $q(T)v = q(\mu) \cdot v$ for every polynomial $q(x) \in K[x]$.

Apply this to $q(x) := (x - \lambda)^n$. We obtain

$$q(T)v = (T - \lambda \operatorname{id}_V)^n v = q(\mu) \cdot v = (\mu - \lambda)^n v.$$

The left-hand side vanishes by part **(a)**, since $v \in \widetilde{\mathrm{Eig}}_T(\lambda) = \ker(T - \lambda\,\mathrm{id}_V)^n$. Therefore, $(\mu-\lambda)^n v = 0$. As $v \neq 0$, this forces $(\mu-\lambda)^n = 0$, and hence $\mu = \lambda$.

Q.E.D.

Definition 28.b.2 (*Invariant Subspace*): Let W be a vector space over K and $S \in \mathrm{End}(W)$. A subspace $\mathcal{U} \subseteq W$ is called "*S-invariant*" if $S(\mathcal{U}) \subseteq \mathcal{U}$, that is, if $Su \in \mathcal{U}$ for all $u \in \mathcal{U}$. If $\mathcal{U}$ is S-invariant, then S defines an endomorphism $S|_{\mathcal{U}} \colon \mathcal{U} \to \mathcal{U}$.

Exercise 28.7 (*The Characteristic Polynomial of a Restriction*): Let $S \colon W \to W$ be an endomorphism and let $\mathcal{U} \subseteq W$ be a subspace invariant under S. Denote by $S_U := S|_{\mathcal{U}} \colon \mathcal{U} \to \mathcal{U}$ the induced endomorphism. Show that $P_{S_U}(x)$ divides $P_S(x)$.

(Prof. Biran's hint: choose a basis $\mathcal{B}$ for $\mathcal{U}$ and extend it to a basis $\mathcal{C}$ for W. Then

$$[S]_{\mathcal{C}}^{\mathcal{C}} = \left(\begin{array}{c|c} [S_U]_{\mathcal{B}}^{\mathcal{B}} & M \\ \hline 0 & N \end{array}\right),$$

which implies $P_S(x) = P_{S_U}(x) \cdot P_N(x)$).

Lemma 28.b.3 (*Kernels of Polynomials in T are Invariant*): Let V be a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$. Let $f(x) \in K[x]$. Then $\ker f(T)$ is a T-invariant subspace of V. In particular, if λ is an eigenvalue of T, then $\widetilde{\mathrm{Eig}}_T(\lambda)$ is a T-invariant subspace.

Proof:

Let $v \in \ker f(T)$. Since T and $f(T)$ commute,

$$f(T)\,Tv = Tf(T)v = T \cdot 0 = 0,$$

and therefore, $Tv \in \ker f(T)$. This shows that $\ker f(T)$ is T-invariant.

Now apply this to $f(x) := (x-\lambda)^n$, where $n := \dim V$. We obtain that

$$\ker f(T) = \ker(T - \lambda\,\mathrm{id}_V)^n = \widetilde{\mathrm{Eig}}_T(\lambda)$$

is T-invariant, the last equality being Lemma 28.b.1 **(a)**. **Q.E.D.**

Notation 28.1 (*Splitting into Linear Factors*): Prof. Biran introduces the abbreviation *"splf"* for *"splits as a product of linear factors (in $K[x]$)"*, noting that it is a non-standard abbreviation to be used only in these lectures. We keep the full phrase in the prose below, but record the abbreviation here so that the notes can be read alongside this text.

Lemma 28.b.4 (*The Characteristic Polynomial on a Generalized Eigenspace*): Let V be a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$ such that $P_T(x)$ splits as a product of linear factors. Let $\lambda \in K$ be an eigenvalue of T and denote

$$T_\lambda := T|_{\widetilde{\mathrm{Eig}}_T(\lambda)} \in \mathrm{End}\left(\widetilde{\mathrm{Eig}}_T(\lambda)\right),$$

which is well defined by Lemma 28.b.3. Then $P_{T_\lambda}(x) = (\lambda - x)^m$, where $m \leq m_a(\lambda; T)$, and $m_a(\lambda;T)$ denotes the algebraic multiplicity of λ for T.

Moreover, if μ is another eigenvalue of T with $\mu \neq \lambda$, then

$$\widetilde{\mathrm{Eig}}_T(\lambda) \cap \widetilde{\mathrm{Eig}}_T(\mu) = \{0\}.$$

Proof:

Write $P_T(x) = (\lambda - x)^{m_a(\lambda)} \cdot q(x)$, where $q(\lambda) \neq 0$ and $m_a(\lambda) := m_a(\lambda; T)$. Since $P_T(x)$ splits as a product of linear factors, so does $q(x)$.

By Exercise 28.7, $P_{T_\lambda}(x)$ divides $P_T(x)$. It is therefore enough to show that $P_{T_\lambda}(x)$ has no zeroes other than λ, because this will imply that $P_{T_\lambda}(x)$ divides $(\lambda-x)^{m_a(\lambda)}$, which is exactly the assertion. (A divisor of a polynomial that splits into linear factors splits itself, by unique factorization in $K[x]$; so $P_{T_\lambda}(x)$ is a product of factors $(\lambda' - x)$ with each λ' a zero of it, and if λ is the only such zero, then $P_{T_\lambda}(x) = (\lambda - x)^m$ with $m \leq m_a(\lambda)$, the leading coefficient being right because $P_S(x) = \det(S - x \cdot \mathrm{id})$ always has leading term $(-x)^{\deg}$).

So let λ' be a zero of $P_{T_\lambda}(x)$. Then $T_\lambda \in \mathrm{End}\left(\widetilde{\mathrm{Eig}}_T(\lambda)\right)$ has λ' as an eigenvalue, and an eigenvector for it lies in $\widetilde{\mathrm{Eig}}_T(\lambda)$ by construction. By Lemma 28.b.1 **(b)**, $\lambda' = \lambda$.

For the second assertion, suppose $\mu \neq \lambda$ is another eigenvalue of T and put

$$\mathcal{U} := \widetilde{\mathrm{Eig}}_T(\lambda) \cap \widetilde{\mathrm{Eig}}_T(\mu).$$

Assume by contradiction that $\mathcal{U} \neq \{0\}$, so that $\dim \mathcal{U} \geq 1$. The subspace $\mathcal{U}$ is T-invariant, being an intersection of two T-invariant subspaces, so $T|_{\mathcal{U}}$ is an endomorphism of $\mathcal{U}$ and $P_{T|_{\mathcal{U}}}(x)$ is a polynomial of degree $\dim \mathcal{U} \geq 1$.

Since $\mathcal{U} \subseteq \widetilde{\mathrm{Eig}}_T(\lambda)$ and $\mathcal{U} \subseteq \widetilde{\mathrm{Eig}}_T(\mu)$, the polynomial $P_{T|_{\mathcal{U}}}(x)$ divides both

$$P_{T_\lambda}(x) = (\lambda - x)^{m_a(\lambda)} \qquad \text{and} \qquad P_{T_\mu}(x) = (\mu - x)^{m_a(\mu)},$$

again by Exercise 28.7. This is impossible, because $\lambda \neq \mu$ and a non-constant polynomial cannot divide two coprime polynomials. Consequently, $\dim \mathcal{U} = 0$, that is, $\mathcal{U} = \{0\}$. **Q.E.D.**

Lemma 28.b.5 (*Generalized Eigenspaces Form a Direct Sum*): Under the same assumptions as in Lemma 28.b.4, if $\lambda_1, \ldots, \lambda_r$ are distinct eigenvalues of T, then the sum

$$\widetilde{\mathrm{Eig}}_T(\lambda_1) + \cdots + \widetilde{\mathrm{Eig}}_T(\lambda_r)$$

forms a direct sum.

Proof:

We proceed by induction on r. For $r = 1$ there is nothing to prove, and for $r = 2$ the statement follows from Lemma 28.b.4.

Assume the lemma holds for $r - 1$ distinct eigenvalues, so that

$$\widetilde{\mathrm{Eig}}_T(\lambda_1) + \cdots + \widetilde{\mathrm{Eig}}_T(\lambda_{r-1}) = \widetilde{\mathrm{Eig}}_T(\lambda_1) \oplus \cdots \oplus \widetilde{\mathrm{Eig}}_T(\lambda_{r-1}) =: W_1.$$

Put $W_2 := \widetilde{\mathrm{Eig}}_T(\lambda_r)$, and write $T_{W_1} := T|_{W_1}$ and $T_{W_2} := T|_{W_2}$.

Assume by contradiction that $\mathcal{U} := W_1 \cap W_2 \neq \{0\}$, so that $\dim \mathcal{U} \geq 1$. Each $\widetilde{\mathrm{Eig}}_T(\lambda_i)$ is T-invariant by Lemma 28.b.3, and hence so are W_1, W_2 and $\mathcal{U}$. Put $T_{\mathcal{U}} := T|_{\mathcal{U}}$.

By Lemma 28.b.4,

$$P_{T_{W_1}}(x) = (\lambda_1 - x)^{m_1} \cdots (\lambda_{r-1} - x)^{m_{r-1}}, \qquad P_{T_{W_2}}(x) = (\lambda_r - x)^{m_r}.$$

By Exercise 28.7, $P_{T_{\mathcal{U}}}(x)$ divides both $P_{T_{W_1}}(x)$ and $P_{T_{W_2}}(x)$, which is impossible because $\deg P_{T_{\mathcal{U}}}(x) \geq 1$ and the two polynomials have no common zero, the eigenvalues $\lambda_1, \ldots, \lambda_r$ being distinct. This is a contradiction, and therefore, $\mathcal{U} = \{0\}$.

Consequently, $\widetilde{\mathrm{Eig}}_T(\lambda_1) + \cdots + \widetilde{\mathrm{Eig}}_T(\lambda_r)$ forms a direct sum. (What the induction has just supplied is exactly criterion **(b)** of Exercise 21.1, one index at a time: at step r it gives $W_2 \cap W_1 = \{0\}$, that is, $\widetilde{\mathrm{Eig}}_T(\lambda_r) \cap \left(\widetilde{\mathrm{Eig}}_T(\lambda_1) + \cdots + \widetilde{\mathrm{Eig}}_T(\lambda_{r-1})\right) = \{0\}$, while the earlier steps give the same for every smaller index). **Q.E.D.**

Proposition 28.b.6 (*Decomposition into Generalized Eigenspaces*): Let V be a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$. Assume that $P_T(x)$ splits into a product of linear factors in $K[x]$. Then

$$V = \bigoplus_{\lambda \text{ eigenvalue of } T} \widetilde{\mathrm{Eig}}_T(\lambda).$$

Proof:

We proceed by induction on the number of different eigenvalues of T, equivalently the number of distinct zeroes of $P_T(x)$. Denote this number by z_T, and put $n := \dim V$.

Base of the induction, $z_T = 1$. We have $P_T(x) = (\lambda - x)^n$. By the Cayley-Hamilton theorem (Theorem 21.d.2), $P_T(T) = 0$, so $(\lambda \operatorname{id}_V - T)^n = 0$. Consequently,

$$V = \ker(T - \lambda \operatorname{id}_V)^n = \widetilde{\operatorname{Eig}}_T(\lambda),$$

the last equality by Lemma 28.b.1 **(a)**.

The induction step. Let $\ell \geq 2$ and assume the proposition holds for all endomorphisms T' satisfying the assumptions of the proposition with $z_{T'} \leq \ell - 1$. Let $T \in \operatorname{End}(V)$ be such that $P_T(x)$ splits into a product of linear factors and $z_T = \ell$. Let λ be an eigenvalue of T and write

$$P_T(x) = (x - \lambda)^k q(x), \qquad k \geq 1, \quad q(x) \in K[x], \quad q(\lambda) \neq 0.$$

We claim that

$$V = \operatorname{Im} q(T) \oplus \ker q(T) \tag{28.5}$$

and moreover that

$$V = \widetilde{\operatorname{Eig}}_T(\lambda) \oplus \ker q(T). \tag{28.6}$$

To prove (28.5), note that $\dim V = \dim \operatorname{Im} q(T) + \dim \ker q(T)$ by Theorem 15.b.9, and that $\operatorname{Im} q(T) + \ker q(T) \subseteq V$. It is therefore enough to show that

$$\operatorname{Im} q(T) \cap \ker q(T) = \{0\}. \tag{28.7}$$

We will show the following two statements:

(a) $\operatorname{Im} q(T) \subseteq \widetilde{\operatorname{Eig}}_T(\lambda)$, and
(b) $\widetilde{\operatorname{Eig}}_T(\lambda) \cap \ker q(T) = \{0\}$.

These imply that

$$\operatorname{Im} q(T) \cap \ker q(T) \subseteq \widetilde{\operatorname{Eig}}_T(\lambda) \cap \ker q(T) = \{0\},$$

which is (28.7), and hence (28.5). Also, **(b)** says that $\widetilde{\operatorname{Eig}}_T(\lambda) + \ker q(T)$ is a direct sum, so together with **(a)** and (28.5) we obtain (28.6), and also that the inclusion in **(a)** is in fact an equality, $\operatorname{Im} q(T) = \widetilde{\operatorname{Eig}}_T(\lambda)$. So instead of proving the claim, we prove **(a)** and **(b)**.

*Proof of **(a)**.* By the Cayley-Hamilton theorem (Theorem 21.d.2),

$$0 = P_T(T) = (T - \lambda \operatorname{id}_V)^k \circ q(T),$$

and therefore, $\operatorname{Im} q(T) \subseteq \ker(T - \lambda \operatorname{id}_V)^k \subseteq \widetilde{\operatorname{Eig}}_T(\lambda)$.

*Proof of **(b)**.* By Lemma 28.b.3, both $\ker q(T)$ and $\widetilde{\operatorname{Eig}}_T(\lambda)$ are T-invariant, and consequently so is

$$Y := \widetilde{\operatorname{Eig}}_T(\lambda) \cap \ker q(T).$$

We want to show that $Y = \{0\}$. Assume by contradiction that $Y \neq \{0\}$, that is, $\dim Y \geq 1$. Since Y is invariant under T, the restriction $T|_Y$ defines an endomorphism $T_Y : Y \to Y$.

By assumption $P_T(x)$ splits into a product of linear factors. As $P_{T_Y}(x)$ divides $P_T(x)$ by Exercise 28.7, and $\deg P_{T_Y}(x) \geq 1$ because $\dim Y \geq 1$, the polynomial $P_{T_Y}(x)$ also splits as a product of linear factors. Consequently, $P_{T_Y}(x)$ has a zero in K, so T_Y has at least one eigenvalue and an eigenvector $v \in Y$.

But we have seen in Lemma 28.b.1 **(b)** that the only eigenvectors of T lying inside $\widetilde{\operatorname{Eig}}_T(\lambda)$ are the eigenvectors of λ. So $Tv = \lambda v$, and therefore,

$$q(T)v = q(\lambda) \cdot v \neq 0,$$

because $q(\lambda) \neq 0$ and $v \neq 0$. At the same time $v \in Y = \widetilde{\operatorname{Eig}}_T(\lambda) \cap \ker q(T) \subseteq \ker q(T)$, so $q(T)v = 0$. This is a contradiction, and hence $Y = \{0\}$. This proves **(b)**.

Now put $W := \ker q(T)$, which we recall is T-invariant, and define $S := T|_W \colon W \to W$. By Exercise 28.7, $P_S(x)$ divides $P_T(x)$. Also, λ is not a zero of $P_S(x)$, because λ cannot be an eigenvalue of S: if $Tv = \lambda v$ for some $0 \neq v \in W$, then

$$\underbrace{q(T)v}_{=0} = \underbrace{q(\lambda)v}_{\neq 0},$$

which is impossible. Consequently, the number z_S of different zeroes of $P_S(x)$ is strictly smaller than the number $z_T = \ell$ of different zeroes of $P_T(x)$.

By the induction hypothesis,

$$W = \bigoplus_{\mu \text{ eigenvalue of } S} \widetilde{\operatorname{Eig}}_S(\mu).$$

Recall from (28.6) that $V = \widetilde{\operatorname{Eig}}_T(\lambda) \oplus W$, and therefore,

$$V = \widetilde{\operatorname{Eig}}_T(\lambda) \oplus \bigoplus_{\mu \text{ eigenvalue of } S} \widetilde{\operatorname{Eig}}_S(\mu). \tag{28.8}$$

But for every eigenvalue μ of S we have $\widetilde{\operatorname{Eig}}_S(\mu) \subseteq \widetilde{\operatorname{Eig}}_T(\mu)$. By Lemma 28.b.5, the sum

$$V' := \sum_{\eta \text{ eigenvalue of } T} \widetilde{\operatorname{Eig}}_T(\eta)$$

forms a direct sum, and V' contains the right-hand side of (28.8). Consequently, $V' = V$, which is the assertion. **Q.E.D.**

Proof of the Existence

Definition 28.b.7 (*Nilpotent Endomorphisms and Matrices*):

- **(a)** Let V be a finite-dimensional vector space over K and $N \in \operatorname{End}(V)$. We call N "*nilpotent*" if there exists $k \in \mathbb{Z}_{\geq 1}$ such that $N^k = 0$.
- **(b)** A matrix $A \in \mathcal{M}_{n\times n}(K)$ is called "*nilpotent*" if there exists k such that $A^k = 0$.

The minimal $k \in \mathbb{Z}_{\geq 1}$ such that $N^k = 0$ (resp. $A^k = 0$) is called the "*nilpotency index*" of N (resp. of A).

Example 28.2: The Jordan matrix $J_{0,n} \in \mathcal{M}_{n\times n}(K)$ is nilpotent of index n, by Lemma 28.a.4 **(a)**.

Exercise 28.8 (*Basic Properties of Nilpotent Endomorphisms*): Prof. Biran collects the following five statements as an exercise. Throughout, V is a finite-dimensional vector space over K.

- **(a)** If $N \in \operatorname{End}(V)$ is nilpotent, then 0 is an eigenvalue of N, and it is the only eigenvalue of N.
- **(b)** If $N \in \operatorname{End}(V)$ is nilpotent, then $N^{\dim V} = 0$. Hence the nilpotency index of N is at most $\dim V$. (Prof. Biran's hint: by the definition of the generalized eigenspace we have $\widetilde{\operatorname{Eig}}_N(0) = \bigcup_{j=1}^{\infty} \ker(N^j) = V$. By Lemma 28.b.1, $\widetilde{\operatorname{Eig}}_N(0) = \ker\left(N^{\dim V}\right)$, and therefore, $\ker\left(N^{\dim V}\right) = V$, which gives $N^{\dim V} = 0$).
- **(c)** Suppose that $N \in \operatorname{End}(V)$ is nilpotent and that $P_N(x)$ splits as a product of linear factors in $K[x]$. Then $P_N(x) = (-x)^{\dim V}$. (Prof. Biran notes that this assumption is not really necessary, but assumes it for simplicity).
- **(d)** Let $T \in \operatorname{End}(V)$ and assume $P_T(x)$ splits as a product of linear factors over $K[x]$. Let $\eta \in K$ and define $S := T - \eta\,\mathrm{id}_V$. Then $P_S(x)$ also splits as a product of linear factors over $K[x]$. In fact, $P_S(x) = P_T(x+\eta)$.
- **(e)** Let $T \in \operatorname{End}(V)$ and assume that $\lambda \in K$ is the only eigenvalue of T and that $P_T(x)$ splits as a product of linear factors in $K[x]$. Define $N := T - \lambda\,\mathrm{id}_V$. Then $P_N(x) = (-x)^{\dim V}$ and $N^{\dim V} = 0$.

Corollary 28.b.8 (*Nilpotency Index on a Generalized Eigenspace*): Let $T \in \operatorname{End}(V)$ and let $\lambda \in K$ be an eigenvalue of T. We have seen that $\widetilde{\operatorname{Eig}}_T(\lambda) = \ker(T - \lambda \operatorname{id}_V)^{\dim V}$, so that

$$(T - \lambda \operatorname{id}_V)^{\dim V}|_{\widetilde{\operatorname{Eig}}_T(\lambda)} = 0.$$

In fact we have the sharper statement

$$(T - \lambda \operatorname{id}_V)^{m}|_{\widetilde{\operatorname{Eig}}_T(\lambda)} = 0, \qquad \text{where } m := \dim \widetilde{\operatorname{Eig}}_T(\lambda),$$

and note that $m \leq \dim V$. Moreover, if $P_T(x)$ splits as a product of linear factors, then

$$\dim \widetilde{\operatorname{Eig}}_T(\lambda) = m_a(\lambda; T).$$

Proof:

The first statement follows from Exercise 28.8 **(b)**, applied to the restriction of $T - \lambda \operatorname{id}_V$ to $\widetilde{\operatorname{Eig}}_T(\lambda)$, which is an endomorphism of that subspace because $\widetilde{\operatorname{Eig}}_T(\lambda)$ is T-invariant by Lemma 28.b.3, and is nilpotent on it because $\widetilde{\operatorname{Eig}}_T(\lambda) = \ker(T - \lambda \operatorname{id}_V)^{\dim V}$.

For the statement about $\dim \widetilde{\operatorname{Eig}}_T(\lambda)$, use Proposition 28.b.6, which gives $V = \bigoplus_\lambda \widetilde{\operatorname{Eig}}_T(\lambda)$, together with the fact that

$$P_{T|_{\widetilde{\operatorname{Eig}}_T(\lambda)}}(x) = (\lambda - x)^{\dim \widetilde{\operatorname{Eig}}_T(\lambda)}$$

from Lemma 28.b.4. Multiplying these characteristic polynomials over all eigenvalues recovers $P_T(x)$, and comparing the exponent of $(\lambda - x)$ on both sides gives $\dim \widetilde{\operatorname{Eig}}_T(\lambda) = m_a(\lambda; T)$. (The multiplication is legitimate because taking a basis of each summand and concatenating them, which is a basis of V by Exercise 21.1, represents T by a block diagonal matrix whose blocks are the $[T|_{\widetilde{\operatorname{Eig}}_T(\lambda)}]$; Proposition 20.c.4 then splits the determinant, and Remark 21.2 says that the polynomial so computed is P_T). **Q.E.D.**

Back to the proof of existence of the Jordan Normal Form

Claim 28.1 (*Reduction to the Nilpotent Case*): In order to prove the existence of the Jordan Normal Form, it is enough to prove it for $N \in \operatorname{End}(\mathcal{U})$ with $N^m = 0$, where $m = \dim \mathcal{U}$.

Reason:

Let $T \in \operatorname{End}(V)$ with $P_T(x)$ splitting as a product of linear factors. We have seen in Proposition 28.b.6 that

$$V = \bigoplus_\lambda \widetilde{\operatorname{Eig}}_T(\lambda).$$

Since each subspace $\widetilde{\operatorname{Eig}}_T(\lambda)$ is T-invariant, we can consider

$$N_\lambda := (T - \lambda \operatorname{id}_V)|_{\widetilde{\operatorname{Eig}}_T(\lambda)},$$

and we have that N_λ sends $\widetilde{\operatorname{Eig}}_T(\lambda)$ to $\widetilde{\operatorname{Eig}}_T(\lambda)$, so $N_\lambda \in \operatorname{End}\left(\widetilde{\operatorname{Eig}}_T(\lambda)\right)$. Also, $P_{N_\lambda}(x)$ splits as a product of linear factors, because $P_T(x)$ has this property: the restriction $T|_{\widetilde{\operatorname{Eig}}_T(\lambda)}$ inherits the property by Exercise 28.7, and subtracting $\lambda \operatorname{id}$ preserves it by Exercise 28.8 **(d)**. Consequently,

$$N_\lambda^{m_\lambda} = (T - \lambda \operatorname{id}_V)^{m_\lambda}|_{\widetilde{\operatorname{Eig}}_T(\lambda)} = 0, \qquad \text{where } m_\lambda := \dim \widetilde{\operatorname{Eig}}_T(\lambda),$$

by Corollary 28.b.8.

If we obtain a Jordan Normal Form for N_λ, that is, a basis $\mathcal{B}_\lambda$ for $\widetilde{\operatorname{Eig}}_T(\lambda)$ such that $[N_\lambda]^{\mathcal{B}_\lambda}_{\mathcal{B}_\lambda}$ is in Jordan Normal Form, then we also get a Jordan Normal Form for $T|_{\widetilde{\operatorname{Eig}}_T(\lambda)}$, because

$$\left[T|_{\widetilde{\operatorname{Eig}}_T(\lambda)}\right]^{\mathcal{B}_\lambda}_{\mathcal{B}_\lambda} = [N_\lambda]^{\mathcal{B}_\lambda}_{\mathcal{B}_\lambda} + \lambda \cdot I_{m_\lambda}.$$

Doing this on each $\widetilde{\operatorname{Eig}}_T(\lambda)$ separately, where λ runs over the eigenvalues of T, we obtain a basis $\mathcal{B} = (\mathcal{B}_{\lambda_1}, \ldots, \mathcal{B}_{\lambda_r})$ for V such that $[T]^{\mathcal{B}}_{\mathcal{B}}$ is in Jordan Normal Form.

In conclusion, it is enough to show the existence of a Jordan Normal Form for each N_λ.

Q.E.D.

Remark 28.3 (*Change in Notation*): From here on, V is a finite-dimensional vector space, $n = \dim V$, and $N \in \mathrm{End}(V)$ is nilpotent with $P_N(x)$ splitting as a product of linear factors, so that $N^n = 0$. Previously this V was $\widetilde{\mathrm{Eig}_T}(\lambda)$, this N was N_λ, and this n was m_λ.

Proof of the existence of the Jordan Normal Form for a nilpotent N:

We do induction on $n = \dim V$.

The case $n = 1$. We have $N = 0$, and consequently $[N]_{\mathcal{B}}^{\mathcal{B}} = 0$ for every basis $\mathcal{B}$, which is already in Jordan Normal Form.

The induction step. Let $n > 1$, and assume we have already proved the existence of a Jordan Normal Form for all N as above on vector spaces of dimension at most $n-1$. Let V be of dimension n and $N \in \mathrm{End}(V)$ nilpotent with $P_N(x)$ splitting as a product of linear factors. Define

$$k := \min\{j \geq 1 \mid N^j = 0\} \leq n.$$

If $k = 1$, then $N = 0$ and $[N]_{\mathcal{B}}^{\mathcal{B}} = 0$ for every basis $\mathcal{B}$. So assume $k > 1$. Then $N^{k-1} \neq 0$, because k was minimal, and consequently there exists $e_k \in V$ such that $N^{k-1}e_k \neq 0$, while $N^k e_k = 0$ because $N^k = 0$. Define

$$e_j := N^{k-j}e_k, \qquad j = 1, \dots, k-1,$$

so that the list $e_1, e_2, \dots, e_{k-1}, e_k$ is precisely $N^{k-1}e_k, N^{k-2}e_k, \dots, Ne_k, e_k$.

Claim: The vectors $e_1, \dots, e_k$ are linearly independent.

Proof of the claim:

Suppose that

$$c_1 e_1 + \cdots + c_k e_k = 0. \tag{28.9}$$

Apply N^{k-1} to (28.9). For all $1 \leq j \leq k-1$ we have

$$N^{k-1}e_j = N^{k-1}N^{k-j}e_k = N^{2k-(j+1)}e_k = 0,$$

because $2k - (j+1) \geq k$ and $N^k = 0$. What remains is

$$0 + \cdots + 0 + c_k \underbrace{N^{k-1}e_k}_{=e_1 \neq 0} = 0,$$

and therefore, $c_k = 0$.

Now apply N^{k-2} to (28.9). By the same reasoning all terms with $j \leq k-2$ vanish, while the term with $j = k$ has already been removed, its coefficient c_k being zero by the step just carried out. What is left is the term involving e_{k-1}, giving

$$0 + \cdots + 0 + c_{k-1} \underbrace{N^{k-2}e_{k-1}}_{=N^{k-1}e_k = e_1 \neq 0} + 0 = 0,$$

and therefore, $c_{k-1} = 0$. Continuing in this way, we obtain $c_{k-2} = \cdots = c_1 = 0$. This proves the claim.

Q.E.D.

Note that $W := \mathrm{Sp}(e_1, \dots, e_k)$ is N-invariant, so $N|_W \in \mathrm{End}(W)$. Moreover, with $\mathcal{B}_1 := (e_1, \dots, e_k)$,

$$[N|_W]_{\mathcal{B}_1}^{\mathcal{B}_1} = \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix} = J_{0,k},$$

which is a Jordan matrix, and in fact the biggest block of N.

Since W is N-invariant, N induces a map

$$N'\colon V/W \to V/W, \qquad N'[v] := [Nv],$$

on the quotient space of Definition 18.c.1. Clearly N' is nilpotent (see Exercise 28.9). Moreover $P_{N'}(x)$ divides $P_N(x)$, and hence $P_{N'}(x)$ splits as a product of linear factors.

To see why $P_{N'}(x)$ divides $P_N(x)$, extend $e_1, \dots, e_k$ to a basis $\mathcal{C} = (e_1, \dots, e_k, v_{k+1}, \dots, v_n)$ of V. Let $[v_j] \in V/W$, for $j = k+1, \dots, n$, be the images of v_j in the quotient. Then $\mathcal{C}' = ([v_{k+1}], \dots, [v_n])$ is a basis for V/W, and we have

$$[N]_{\mathcal{C}}^{\mathcal{C}} = \left(\begin{array}{c|c} [N|_W]_{\mathcal{B}_1}^{\mathcal{B}_1} & * \\ \hline 0 & [N']_{\mathcal{C}'}^{\mathcal{C}'} \end{array}\right),$$

where the top left block has size $k \times k$, which implies $P_N(x) = P_{N|_W}(x) \cdot P_{N'}(x)$.

Now, $\dim V/W = \dim V - \dim W = n - k < n$ by Proposition 18.c.4. We may therefore apply the induction hypothesis to $N' \in \operatorname{End}(V/W)$. We obtain a basis $\mathcal{B}' = (u_1, \dots, u_\ell)$ of V/W, where $\ell = n - k$, such that $[N']_{\mathcal{B}'}^{\mathcal{B}'}$ is in Jordan Normal Form. We can write $u_i = [f_i]$ with $f_i \in V$, for $i = 1, \dots, \ell$. Note that

$$\mathcal{B}_2 := (e_1, \dots, e_k, f_1, \dots, f_\ell)$$

is a basis for V, and we have

$$[N]_{\mathcal{B}_2}^{\mathcal{B}_2} = \left(\begin{array}{c|c} J_{0,k} & * \\ \hline 0 & [N']_{\mathcal{B}'}^{\mathcal{B}'} \end{array}\right) = \left(\begin{array}{c|cc} J_{0,k} & * & * \\ \hline 0 & J_{0,m} & \\ 0 & & \ddots \end{array}\right), \tag{28.10}$$

where $J_{0,m}$ is the first Jordan block of $[N']_{\mathcal{B}'}^{\mathcal{B}'}$.

Note that $m \leq k$, and in fact every Jordan block in $[N']_{\mathcal{B}'}^{\mathcal{B}'}$ has size at most k. Indeed, if $m \geq k+1$, then $J_{0,m}^{m-1} \neq 0$, and consequently $J_{0,m}^k \neq 0$ because $k \leq m-1$. This would give $\left([N]_{\mathcal{B}_2}^{\mathcal{B}_2}\right)^k \neq 0$. But k was defined so that $N^k = 0$, a contradiction.

We will now modify the elements $f_1, \dots, f_m$ into $e_{k+1}, \dots, e_{k+m}$ such that $[f_i] = [e_{k+i}]$ and such that, in the basis

$$\mathcal{B}_3 := (e_1, \dots, e_k, e_{k+1}, \dots, e_{k+m}, f_{m+1}, \dots, f_\ell),$$

the matrix $[N]_{\mathcal{B}_3}^{\mathcal{B}_3}$ has no $*$ block above the second, $m \times m$, block, that is,

$$[N]_{\mathcal{B}_3}^{\mathcal{B}_3} = \left(\begin{array}{c|c|c} J_{0,k} & 0 & * \\ \hline 0 & J_{0,m} & \\ 0 & & \ddots \end{array}\right).$$

To prove this, note that by (28.10), for all $1 \leq j \leq m$ we have

$$N^{m-j} f_m = f_j + (\text{something from } W),$$

because $J_{0,m}^{m-j} \varepsilon_m = \varepsilon_j$, where $\varepsilon_1, \dots, \varepsilon_m$ is the standard basis for K^m. Also $N^m f_m \in W$, because $J_{0,m}^m = 0$. Consequently,

$$N^m f_m = c_1 e_1 + \dots + c_k e_k \qquad \text{for some } c_1, \dots, c_k \in K.$$

Claim: We have $c_{k-m+1} = \dots = c_k = 0$.

Proof of the claim:

Recall that $m \leq k$, and therefore,

$$0 = N^k f_m = N^{k-m} \circ N^m f_m = N^{k-m}\left(c_1 e_1 + \dots + c_k e_k\right) = c_{k-m+1} e_1 + \dots + c_k e_m,$$

where the first equality holds because $N^k = 0$, the third step needs $k - m \geq 0$, and the last uses that

$$J_{0,k}^r \cdot \varepsilon_j = \begin{cases} 0 & \text{if } j \leq r, \\ \varepsilon_{j-r} & \text{if } r+1 \leq j \leq k. \end{cases}$$

Since $e_1, \ldots, e_m$ are linearly independent, we obtain $c_{k-m+1} = \cdots = c_k = 0$, which proves the claim. **Q.E.D.**

We now define

$$e_{k+m} := f_m - (c_1 e_{m+1} + c_2 e_{m+2} + \cdots + c_{k-m} e_k),$$

where, if $m = k$, we simply define $e_{k+m} := f_m$. We then set

$$e_{k+m-1} := N e_{k+m}, \quad e_{k+m-2} := N^2 e_{k+m}, \quad \ldots, \quad e_{k+1} := N^{m-1} e_{k+m}.$$

Note that

$$e_{k+m} \xrightarrow{N} e_{k+m-1} \xrightarrow{N} e_{k+m-2} \longmapsto \ldots \xrightarrow{N} e_{k+1},$$

and

$$\begin{aligned} N e_{k+1} &= N \circ N^{m-1} e_{k+m} = N^m \left(f_m - c_1 e_{m+1} - c_2 e_{m+2} - \cdots - c_{k-m} e_k \right) \\ &= \underbrace{(c_1 e_1 + \cdots + c_{k-m} e_{k-m})}_{\text{by the previous claim}} - \underbrace{(c_1 e_1 + \cdots + c_{k-m} e_{k-m})}_{\text{because } [N|_W]_{\mathcal{B}_1}^{\mathcal{B}_1} = J_{0,k}} = 0. \end{aligned}$$

So $e_{k+1} \overset{N}{\longmapsto} 0$, and the vectors $e_{k+1}, \ldots, e_{k+m}$ form a Jordan chain of their own.

The elements of $\mathcal{B}_3$ are linearly independent, and hence form a basis for V; this is left as an exercise, see Exercise 28.10. It is easy to see now that

$$[N]_{\mathcal{B}_3}^{\mathcal{B}_3} = \left(\begin{array}{c|c|c} J_{0,k} & 0 & * \\ \hline 0 & J_{0,m} & \\ 0 & & \ddots \end{array} \right).$$

Now continue in the same way with the next blocks after $J_{0,m}$, if there are any, until we obtain a basis $\mathcal{B}$ such that $[N]_{\mathcal{B}}^{\mathcal{B}}$ is in Jordan Normal Form. **Q.E.D.**

Exercise 28.9 (*The Induced Map on the Quotient is Nilpotent*): This is the step Prof. Biran marks with *"check!"*. Let $N \in \operatorname{End}(V)$ be nilpotent and let $W \subseteq V$ be an N-invariant subspace. Show that the induced map $N' \colon V/W \to V/W$, $N'[v] := [Nv]$, is well defined and nilpotent.

Exercise 28.10 (*The Modified Family is a Basis*): This is the step Prof. Biran marks with *"Exc."* in the proof above. Show that the elements of

$$\mathcal{B}_3 = (e_1, \ldots, e_k, e_{k+1}, \ldots, e_{k+m}, f_{m+1}, \ldots, f_\ell)$$

are linearly independent, and hence form a basis for V. (Prof. Biran's hint: $e_{k+j} = f_{k+j} +$ (something from W)).

Proof of the Uniqueness

Lemma 28.b.9 (*Counting Jordan Blocks by Kernel Dimensions*): Let V be a finite-dimensional vector space over K, and let $N \in \operatorname{End}(V)$ be nilpotent. Let A be a matrix in Jordan Normal Form such that $A = [N]_{\mathcal{B}}^{\mathcal{B}}$ in some basis $\mathcal{B}$ for V. Then:

(a) $\dim \ker(N) = \#\{\text{Jordan blocks in } A\}$.

(b) $\dim \ker(N^2) = \#\{\text{Jordan blocks in } A\} + \#\{\text{Jordan blocks of dimension} \geq 2 \text{ in } A\}$.

(c) $\dim \ker(N^3) = \#\{\text{Jordan blocks in } A\} + \#\{\text{Jordan blocks of dimension} \geq 2 \text{ in } A\} + \#\{\text{Jordan blocks of dimension} \geq 3 \text{ in } A\}$.

(d) And so on: for every $r \geq 1$,
$$\dim\ker(N^r) = \sum_{i=1}^{r} \#\{\text{Jordan blocks of dimension} \geq i \text{ in } A\}.$$

Proof:

Here and below we identify matrices M with the linear maps $T_M(v) := M \cdot v$ they define.

For one Jordan block $J_{0,n}$ we have
$$\dim\ker\left(J_{0,n}^r\right) = r \quad \text{for all } r \leq n, \qquad \text{and} \qquad \dim\ker\left(J_{0,n}^r\right) = n \quad \text{for } r > n,$$
which follows from Lemma 28.a.4 **(a)**: the matrix $J_{0,n}^r$ has 1's on the r-th diagonal above the main diagonal and 0's elsewhere, so its rank is $n-r$ for $r \leq n$, and 0 for $r > n$.

If
$$A = \begin{pmatrix} J_{0,n_1} & & & \\ & J_{0,n_2} & & \\ & & \ddots & \\ & & & J_{0,n_\ell} \end{pmatrix}, \qquad \text{then} \qquad A^r = \begin{pmatrix} J_{0,n_1}^r & & & \\ & J_{0,n_2}^r & & \\ & & \ddots & \\ & & & J_{0,n_\ell}^r \end{pmatrix}$$
by Exercise 28.5, and the kernel of a block diagonal matrix is the direct sum of the kernels of its blocks.

(a) Each Jordan block in A contributes exactly 1 to the dimension of $\ker(A)$. This proves **(a)**.

(b) For $r = 2$, each Jordan block in A of dimension 1 contributes 1 to $\ker(A^2)$, and each Jordan block of dimension ≥ 2 in A contributes 2 to $\ker(A^2)$. So in total we have
$$\begin{aligned} \dim\ker(A^2) &= \#\{\text{Jordan blocks of dimension 1 in } A\} \\ &\quad + 2 \cdot \#\{\text{Jordan blocks of dimension} \geq 2 \text{ in } A\} \\ &= \#\{\text{Jordan blocks in } A\} \\ &\quad + \#\{\text{Jordan blocks of dimension} \geq 2 \text{ in } A\}. \end{aligned}$$

(c) Statements **(c)** and **(d)** are proved in the same way.

Q.E.D.

Proof of the uniqueness of the Jordan Normal Form:

Assume that A and A' are two matrices in Jordan Normal Form that both represent $T \in \operatorname{End}(V)$. Consequently, $P_A(x) = P_T(x) = P_{A'}(x)$, by Remark 21.2, which says that the characteristic polynomial computed from a representation matrix does not depend on the basis.

Now, A being upper triangular, Lemma 21.a.8 **(c)** gives
$$P_A(x) = \prod_{\lambda} (\lambda - x)^{\#\{\lambda\text{'s in the diagonal of } A\}},$$
and similarly for $P_{A'}(x)$. This shows that every λ appears the same number of times in the diagonals of both A and A'.

Let λ be an eigenvalue of T. By permuting the Jordan blocks in each of A and A', we obtain two matrices B and B', also in Jordan Normal Form, that both represent T in some bases, and such that
$$B = \left(\begin{array}{cccc|c} J_{\lambda,n_1} & & & & \\ & J_{\lambda,n_2} & & & \\ & & \ddots & & \\ & & & J_{\lambda,n_\ell} & \\ \hline & & & & \text{JNF with eigenvalues} \neq \lambda \end{array}\right)$$

and

$$B' = \left(\begin{array}{cccc|c} J_{\lambda,m_1} & & & & \\ & J_{\lambda,m_2} & & & \\ & & \ddots & & \\ & & & J_{\lambda,m_{\ell'}} & \\ \hline & & & & \text{JNF with eigenvalues} \neq \lambda \end{array}\right).$$

We have seen above that $n_1 + \cdots + n_\ell = m_1 + \cdots + m_{\ell'}$; denote this common value by r.

Consider $S := T - \lambda\,\mathrm{id}_V$. Note that both $C := B - \lambda I$ and $C' := B' - \lambda I$ are matrices that represent S, and

$$C = \left(\begin{array}{cccc|c} J_{0,n_1} & & & & \\ & J_{0,n_2} & & & \\ & & \ddots & & \\ & & & J_{0,n_\ell} & \\ \hline & & & & \text{JNF with eigenvalues} \neq 0 \end{array}\right),$$

and similarly for C', with the blocks $J_{0,m_1}, \ldots, J_{0,m_{\ell'}}$.

Define

$$D := \begin{pmatrix} J_{0,n_1} & & \\ & \ddots & \\ & & J_{0,n_\ell} \end{pmatrix}, \qquad D' := \begin{pmatrix} J_{0,m_1} & & \\ & \ddots & \\ & & J_{0,m_{\ell'}} \end{pmatrix}.$$

Both D and D' are nilpotent matrices of size $r \times r$.

Note that for every k,

$$\dim\ker S^k = \dim\ker C^k = \dim\ker D^k,$$

because the complementary part of C, namely the Jordan Normal Form with all eigenvalues different from 0, is invertible and so contributes nothing to the kernel. (It is invertible because it is upper triangular with all diagonal entries non-zero, so its determinant is a product of non-zero scalars; a power of it is invertible too, and the kernel of a block diagonal matrix is the direct sum of the kernels of its blocks. The first equality, $\dim\ker S^k = \dim\ker C^k$, is the passage from an endomorphism to a representation matrix, exactly as in the proof of Corollary 28.a.5: the coordinate map carries $\ker S^k$ isomorphically onto $\ker C^k$). Similarly, for every k,

$$\dim\ker S^k = \dim\ker(C')^k = \dim\ker(D')^k.$$

Consequently,

$$\dim\ker D^k = \dim\ker(D')^k \qquad \text{for all } k \geq 1. \tag{28.11}$$

We now apply Lemma 28.b.9 to (28.11) for the various values of k.

$k = 1$. We obtain $\ell = \ell'$, that is, the number of Jordan blocks in D and in D' coincide.

$k = 2$. We obtain $\#\{i \mid n_i \geq 2\} = \#\{j \mid m_j \geq 2\}$, which implies $\#\{i \mid n_i = 1\} = \#\{j \mid m_j = 1\}$.

$k = 3$. We obtain $\#\{i \mid n_i \geq 3\} = \#\{j \mid m_j \geq 3\}$, which implies $\#\{i \mid n_i = 2\} = \#\{j \mid m_j = 2\}$.

Continuing in this way, we obtain that for every k,

$$\#\{i \mid n_i = k\} = \#\{j \mid m_j = k\}.$$

Consequently, up to permutation of blocks, D and D' coincide, and therefore, the blocks in B and B' that correspond to λ are the same, up to permutation.

This holds for all eigenvalues λ of T, and consequently, A and A' are the same up to permutation of the Jordan blocks. **Q.E.D.**

Solutions to the Exercises

Proof of Exercise 28.7:

Choose a basis $\mathcal{B} := (u_1, \ldots, u_d)$ for $\mathcal{U}$ and extend it to a basis $\mathcal{C} := (u_1, \ldots, u_d, w_{d+1}, \ldots, w_p)$ for W, which is possible by the basis extension theorem.

Because $\mathcal{U}$ is S-invariant, each Su_i lies in $\mathcal{U}$, so its coordinate vector with respect to $\mathcal{C}$ has zeroes in the last $p-d$ entries. The first d columns of $[S]^{\mathcal{C}}_{\mathcal{C}}$ therefore have the shape claimed, and consequently

$$[S]^{\mathcal{C}}_{\mathcal{C}} = \left(\begin{array}{c|c} [S_U]^{\mathcal{B}}_{\mathcal{B}} & M \\ \hline 0 & N \end{array}\right)$$

for some M and N of the appropriate sizes.

Subtracting x from the diagonal preserves this block upper triangular shape, so by Proposition 20.c.4,

$$P_S(x) = \det\left([S]^{\mathcal{C}}_{\mathcal{C}} - xI_p\right) = \det\left([S_U]^{\mathcal{B}}_{\mathcal{B}} - xI_d\right) \cdot \det\left(N - xI_{p-d}\right) = P_{S_U}(x) \cdot P_N(x).$$

Hence $P_{S_U}(x)$ divides $P_S(x)$. **Q.E.D.**

Proof of Exercise 28.8:

(a) Let k be the nilpotency index of N, so $N^k = 0$ and $N^{k-1} \neq 0$ (with the convention $N^0 = \mathrm{id}_V$). Pick v with $N^{k-1}v \neq 0$ and put $u := N^{k-1}v$. Then $u \neq 0$ and $Nu = N^k v = 0$, so u is an eigenvector for the eigenvalue 0. Hence 0 is an eigenvalue of N.

Conversely, let μ be any eigenvalue of N with eigenvector $w \neq 0$. Then $N^k w = \mu^k w$, and since $N^k = 0$ this gives $\mu^k w = 0$, hence $\mu^k = 0$ and therefore $\mu = 0$. So 0 is the only eigenvalue.

(b) By the definition of the generalized eigenspace and the fact that some power of N vanishes,

$$\widetilde{\mathrm{Eig}}_N(0) = \bigcup_{j=1}^{\infty} \ker\left(N^j\right) = V.$$

By Lemma 28.b.1 **(a)** applied with $T := N$ and $\lambda := 0$, we also have $\widetilde{\mathrm{Eig}}_N(0) = \ker\left(N^{\dim V}\right)$. Combining the two gives $\ker\left(N^{\dim V}\right) = V$, that is, $N^{\dim V} = 0$. In particular the nilpotency index, being the least such exponent, is at most $\dim V$.

(c) By part **(a)**, the only eigenvalue of N is 0. Since $P_N(x)$ splits as a product of linear factors, it is a product of factors of the form $(\mu - x)$ with μ an eigenvalue of N, hence $\mu = 0$ in every factor. As $\deg P_N(x) = \dim V$ and P_N is normalized so that its leading term is $(-x)^{\dim V}$, we get $P_N(x) = (-x)^{\dim V}$.

(d) Let $\mathcal{B}$ be any basis for V and put $A := [T]^{\mathcal{B}}_{\mathcal{B}}$. Then $[S]^{\mathcal{B}}_{\mathcal{B}} = A - \eta I$, and consequently

$$P_S(x) = \det\left(A - \eta I - xI\right) = \det\left(A - (x+\eta)I\right) = P_T(x+\eta).$$

If $P_T(x) = \prod_i (\lambda_i - x)$, then $P_S(x) = \prod_i (\lambda_i - \eta - x)$, which is again a product of linear factors. So $P_S(x)$ splits as a product of linear factors.

(e) By part **(d)** with $\eta := \lambda$, the polynomial $P_N(x) = P_T(x+\lambda)$ splits as a product of linear factors. If μ were an eigenvalue of N, then $\mu + \lambda$ would be an eigenvalue of T, and since λ is the only eigenvalue of T this forces $\mu = 0$. So 0 is the only eigenvalue of N, and exactly as in part **(c)** we obtain $P_N(x) = (-x)^{\dim V}$.

By the Cayley-Hamilton theorem (Theorem 21.d.2), $P_N(N) = 0$, that is, $(-N)^{\dim V} = 0$, and therefore, $N^{\dim V} = 0$.

Q.E.D.

Proof of Exercise 28.9:

Well defined. Suppose $[v]=[v']$, that is, $v-v'\in W$. Since W is N-invariant, $N(v-v')\in W$, and consequently $[Nv]=[Nv']$. So N' does not depend on the representative chosen, and it is clearly linear because N is.

Nilpotent. Let k be such that $N^k=0$. An easy induction gives $(N')^j[v]=[N^jv]$ for all $j\geq 1$: the case $j=1$ is the definition, and

$$(N')^{j+1}[v]=N'\left((N')^j[v]\right)=N'\left[N^jv\right]=\left[N^{j+1}v\right].$$

Consequently,

$$(N')^k[v]=\left[N^kv\right]=[0]=0_{V/W}\qquad\text{for every } [v]\in V/W,$$

that is, $(N')^k=0$. So N' is nilpotent, with nilpotency index at most that of N. **Q.E.D.**

Proof of Exercise 28.10:

Recall that $\mathcal{B}_2=(e_1,\dots,e_k,f_1,\dots,f_\ell)$ is a basis for V, and that $\mathcal{B}_3$ is obtained from it by replacing $f_1,\dots,f_m$ with $e_{k+1},\dots,e_{k+m}$. Since $\mathcal{B}_3$ has $k+\ell=n$ elements, it is enough to prove linear independence.

The key point is Prof. Biran's hint: for each $1\leq j\leq m$ we have

$$e_{k+j}=f_j+w_j\qquad\text{for some } w_j\in W=\mathrm{Sp}(e_1,\dots,e_k). \tag{28.12}$$

For $j=m$ this is the definition of e_{k+m}, whose correction term $c_1e_{m+1}+\cdots+c_{k-m}e_k$ lies in W. For $j<m$ it follows because $e_{k+j}=N^{m-j}e_{k+m}$ and, as established in the proof, $N^{m-j}f_m=f_j+(\text{something from } W)$, while N^{m-j} maps W into W by N-invariance.

Now suppose

$$\sum_{i=1}^k a_ie_i+\sum_{j=1}^m b_je_{k+j}+\sum_{i=m+1}^{\ell} d_if_i=0.$$

Substituting (28.12) and collecting the terms lying in W, this becomes

$$\underbrace{\left(\sum_{i=1}^k a_ie_i+\sum_{j=1}^m b_jw_j\right)}_{\in W}+\sum_{j=1}^m b_jf_j+\sum_{i=m+1}^{\ell} d_if_i=0.$$

Pass to the quotient V/W. The first bracket dies, and we are left with

$$\sum_{j=1}^m b_j[f_j]+\sum_{i=m+1}^{\ell} d_i[f_i]=0\quad\text{in } V/W.$$

But $([f_1],\dots,[f_\ell])=\mathcal{B}'$ is a basis for V/W, so all b_j and all d_i vanish.

The relation therefore reduces to $\sum_{i=1}^k a_ie_i=0$, and $e_1,\dots,e_k$ are linearly independent by the first claim in the proof, so all a_i vanish as well. Hence $\mathcal{B}_3$ is linearly independent, and being of the right cardinality, it is a basis for V. **Q.E.D.**

AI-Note: It is worth stepping back from nineteen pages of induction to see the shape of the argument, because the two halves of this section are doing quite different things.

The **existence** proof is a reduction followed by a construction. The reduction is Proposition 28.b.6: once P_T splits, V falls apart into the generalized eigenspaces $\widetilde{\mathrm{Eig}}_T(\lambda)=\ker(T-\lambda\,\mathrm{id}_V)^n$, each of them T-invariant, so that T may be studied one eigenvalue at a time. On a single generalized eigenspace, $T-\lambda\,\mathrm{id}_V$ is nilpotent, and adding back the scalar λI turns a Jordan form for the nilpotent map into one for T; that is all Claim 28.1 says. What remains is a purely nilpotent question, and the construction answers it by chasing a single vector: take e_k with $N^{k-1}e_k\neq 0$, and the chain $e_k\mapsto Ne_k\mapsto\cdots\mapsto N^{k-1}e_k\mapsto 0$ is automatically independent and spans an invariant W on which N is exactly $J_{0,k}$. Pass to V/W, apply the induction hypothesis there, and the only real work left is the correction $e_{k+m} := f_m - (c_1 e_{m+1} + \cdots + c_{k-m} e_k)$, which shifts the lifted vectors by an element of W so as to kill the $*$ block that a naive lift leaves behind.

The **uniqueness** proof is of a completely different character: it is a counting argument, and it is the reason Lemma 28.b.9 exists. The numbers $\dim\ker(T - \lambda\,\mathrm{id}_V)^r$ are defined without reference to any basis, and the lemma says that this list of numbers determines, by successive differences, how many Jordan blocks of each size carry λ. Two Jordan forms of the same T therefore produce the same list, and hence the same blocks. Nothing about the construction above is used, which is why uniqueness holds even though the Jordan basis is very far from unique.

One warning about reading the notes alongside this text. Prof. Biran numbers his four preparatory lemmata *"Lemma 1"* to *"Lemma 4"* and leaves the rest unnumbered, so the labels 28.b.1 to 28.b.7 here are the book's own; a comment above each label records his numbering where he gives one. The abbreviation *"splf"* of Notation 28.1 is his and appears throughout the scans, but the prose here always writes the phrase out.

The rigour pass over this section found nothing false. Five steps were resting on a reason that was left unsaid, and now name it: that a divisor of a split polynomial splits, in the proof of Lemma 28.b.4, without which $P_{T_\lambda}(x) = (\lambda - x)^m$ does not follow from having no other zero; criterion **(b)** of Exercise 21.1, which is what the induction of Lemma 28.b.5 actually verifies; Proposition 20.c.4 with Remark 21.2 for multiplying the characteristic polynomials of the summands back together in Corollary 28.b.8; Exercise 28.8 **(d)** for the splitting of P_{N_λ} in Claim 28.1; and, in the uniqueness proof, Remark 21.2 and Lemma 21.a.8 **(c)** for the two identities the whole count starts from, together with the observation that $\dim\ker S^k = \dim\ker C^k$ is once again the passage from an endomorphism to a representation matrix.